GENERAL

PLATE	DRAWING NO.	CONTENT	DETAILS
1	1695-D-62526	DRAWING INDEX	
2	1695-529-60481	GENERAL MAP	
3	1695-529-60488	VICINITY MAP	SANOSTEE TURNOUT BUILDING
4	1695-529-60489	VICINITY MAP	NEWCOMB TURNOUT BUILDING
5	1695-529-60490	VICINITY MAP	NASCHITTI TURNOUT BUILDING
6	1695-529-60491	VICINITY MAP	BBN9 TURNOUT BUILDING
7	1695-529-60492	VICINITY MAP	COYOTE CANYON TURNOUT BUILDING
8	1695-529-60493	VICINITY MAP	ROCK SPRING TURNOUT BUILDING
9	1695-529-60494	VICINITY MAP	WINDOW ROCK TURNOUT BUILDING
10	1695-D-62527	SPECIAL INSPECTIONS	

CIVIL

PLATE	DRAWING NO.	CONTENT	DETAIL
11	1695-D-62528	SANOSTEE TURNOUT	SITE PLAN
12	1695-D-62529	SANOSTEE TURNOUT	GRADING PLAN
13	1695-D-62530	NEWCOMB TURNOUT	EXISTING CONDITIONS AND DEMOLITION PLAN
14	1695-D-62531	NEWCOMB TURNOUT	SITE PLAN
15	1695-D-62532	NEWCOMB TURNOUT	GRADING PLAN
16	1695-D-62533	NASCHITTI TURNOUT	SITE PLAN
17	1695-D-62534	NASCHITTI TURNOUT	GRADING PLAN
18	1695-D-62535	BBN9 TURNOUT	SITE PLAN
19	1695-D-62536	BBN9 TURNOUT	GRADING PLAN
20	1695-D-62537	COYOTE CANYON TURNOUT	SITE PLAN
21	1695-D-62538	COYOTE CANYON TURNOUT	GRADING PLAN
22	1695-D-62539	ROCK SPRINGS TURNOUT	SITE PLAN
23	1695-D-62540	ROCK SPRINGS TURNOUT	GRADING PLAN
24	1695-D-62541	WINDOW ROCK TURNOUT	SITE PLAN
25	1695-D-62542	WINDOW ROCK TURNOUT	GRADING PLAN
26	1695-D-62543	COYOTE CANYON ACCESS ROAD	PLAN & PROFILE
27	1695-D-62544	WINDOW ROCK ACCESS ROAD	PLAN & PROFILE
28	1695-D-62545	WINDOW ROCK ACCESS ROAD	PLAN & PROFILE
29	1695-D-62546	BBN9 TURNOUT	SITE PLAN - LOW WATER CROSSING
30	1695-D-62547	BBN9 TURNOUT	LOW WATER CROSSING - DETAIL AND SECTIONS

PIPELINE

PLATE	DRAWING NO.	CONTENT	DETAIL
31	1695-D-62548	BBN9 LINE PIPE TURNOUT	PLANS, SECTIONS AND DETAILS
32	1695-D-62549	BBN9 LINE PIPE - SECTIONALIZING VALVE INSTALLATION	PLANS, SECTIONS AND DETAILS
33	1695-D-62550	BBN9 LINE PIPE - COLLARS	DETAILS AND SECTIONS
34	1695-D-62551	BBN9 LINE PIPE - AIR VALVES WITH MANHOLES	SECTIONS AND DETAILS
35	1695-D-62552	BBN9 LINE PIPE - MANHOLE WITH BLOWOFF	SECTIONS AND DETAILS
36	1695-D-62553	BBN9 LINE PIPE - PRECAST CONCRETE MANHOLES	PLANS, SECTIONS AND DETAILS
37	1695-D-62554	PRESSURE PIPE TRENCH INSTALLATION	DETAILS AND SECTIONS

STRUCTURAL

PLATE	DRAWING NO.	CONTENT	DETAIL
38	1695-D-62555	STRUCTURAL DESIGN DATA & GENERAL ARRANGEMENT	PLAN AND SECTIONS
39	1695-D-62556	CONCRETE OUTLINE & REINFORCEMENT DESIGN	THRUST BLOCKS AND MISC. SITE CONCRETE

ARCHITECTURAL

PLATE	DRAWING NO.	CONTENT	DETAIL
40	1695-D-62557	TURNOUT BUILDING ARCHITECTURAL	CODE SUMMARY, PLANS, ELEVATIONS, SECTIONS

MECHANICAL - STEEL PIPE

PLATE	DRAWING NO.	CONTENT	DETAIL
41	1695-D-62558	COVER SHEET	DETAILS
42	1695-D-62559	6 INCH AND 8 INCH DIAMETER TURNOUT PIPING	PLAN AND SECTION
43	1695-D-62560	12 INCH DIAMETER TURNOUT PIPING	PLAN AND SECTION
44	1695-D-62561	SANOSTEE YARD PIPE	PLAN AND SECTION
45	1695-D-62562	NEWCOMB YARD PIPE	PLAN AND SECTION
46	1695-D-62563	NASCHITTI YARD PIPE	PLAN AND SECTION
47	1695-D-62564	BBN9 YARD PIPE	PLAN AND SECTION
48	1695-D-62565	COYOTE CANYON YARD PIPE	PLAN AND SECTION
49	1695-D-62566	ROCK SPRINGS YARD PIPE	PLAN AND SECTION
50	1695-D-62567	WINDOW ROCK YARD PIPE	PLAN AND SECTION
51	1695-D-62568	DETAILS	DETAILS

PIPING AND INSTRUMENTATION DIAGRAM

PLATE	DRAWING NO.	CONTENT	DETAIL
52	1695-D-62569	P&ID	LEGEND
53	1695-D-62570	P&ID	SYMBOLS AND ABBREVIATIONS
54	1695-D-62571	P&ID	SANOSTEE TURNOUT BUILDING
55	1695-D-62572	P&ID	NEWCOMB TURNOUT BUILDING
56	1695-D-62573	P&ID	NASCHITTI TURNOUT BUILDING
57	1695-D-62574	P&ID	BBN9 TURNOUT BUILDING
58	1695-D-62575	P&ID	COYOTE CANYON TURNOUT BUILDING
59	1695-D-62576	P&ID	ROCK SPRINGS TURNOUT BUILDING
60	1695-D-62577	P&ID	WINDOW ROCK TURNOUT BUILDING

ELECTRICAL

PLATE	DRAWING NO.	CONTENT	DETAIL
61	1695-D-62578	TYPICAL	SINGLE-LINE DIAGRAM
62	1695-D-62579	SANOSTEE TURNOUT	ELECTRICAL SITE PLAN
63	1695-D-62580	NEWCOMB	ELECTRICAL SITE PLAN
64	1695-D-62581	NASCHITTI	ELECTRICAL SITE PLAN
65	1695-D-62582	BBN9	ELECTRICAL SITE PLAN
66	1695-D-62583	COYOTE CANYON	ELECTRICAL SITE PLAN
67	1695-D-62584	ROCK SPRINGS	ELECTRICAL SITE PLAN
68	1695-D-62585	WINDOW ROCK	ELECTRICAL SITE PLAN
69	1695-D-62586	TYPICAL	GROUNDING PLAN
70	1695-D-62587	CABLE AND CONDUIT	SCHEDULE
71	1695-D-62588	NETWORK OVERVIEW	TURNOUT BUILDING
72	1695-D-62589	NOT INCLUDED IN PACKAGE	

CATHODIC PROTECTION

PLATE	DRAWING NO.	CONTENT	DETAIL
73	1695-D-62590	GALVANIC ANODE CATHODIC PROTECTION SYSTEM	JOINT BOND, CABLE TRENCH, IJFK, AND CABLE I.D. DETAILS
74	1695-D-62591	GALVANIC ANODE CATHODIC PROTECTION SYSTEM	TEST STATION AND ANODE BED DETAILS

INFORMATIONAL DRAWINGS

PLATE	DRAWING NO.	CONTENT	DETAIL
75	1695-D-60119	REACH 12.1 PLAN AND PROFILE	STA. 121300+00 TO STA. 121350+00
76	1695-D-60129	REACH 12.1 OUTLETS	PLAN, SECTIONS AND DETAILS
77	1695-D-60150	REACHES 12.2 & 12.2 2 OR 3-INCH AIR VALVE	PLAN, SECTION, AND DETAILS
78	1695-D-60157	REACHES 12.2 & 12.2 SECTIONIZING VALVE INSTALLATION	PLAN, SECTIONS, AND TABLE
79	1695-D-60253	REACHES 12.1 & 12.2 SECTIONALIZING VALVE INSTALLATIONS LAYOUT	PLAN, SECTION, AND TABLES
80	1695-D-60260	REACH 12.2 PLAN AND PROFILE	STA. 122280+00 TO END
81	1695-D-60297	REACH 9 PLAN AND PROFILE	STA. 90750+00 TO STA. 90800+00
82	1695-D-60309	REACH 11 PLAN AND PROFILE	BEGINNING TO STA. 11100+00
83	1695-D-60322	BLOCK 9-11 MANHOLE WITH BLOWOFF	SECTIONS, DETAILS, AND TABLES
84	1695-D-60324	BLOCK 9-11 SECTIONALIZING VALVE INSTALLATION	PLAN, SECTION, AND DETAILS
85	1695-D-60325	BLOCK 9-11 SECTIONALIZING VALVE INSTALLATION	TAPERS, ADAPTERS, AND COLLARS
86	1695-D-60326	BLOCK 9-11 SECTIONALIZING VALVE INSTALLATION	TYPICAL LAYOUT
87	1695-D-60327	BLOCK 9-11 PRECAST CONCRETE MANHOLES	DETAILS
88	1695-D-60684	REACH 4C PLAN AND PROFILE	STA. 42150+00 TO END
89	1695-D-60695	REACH 6 PLAN AND PROFILE	STA. 6500+00 TO END
90	1695-D-60713	REACH 8 PLAN AND PROFILE	STA. 8400+00 TO END
91	1695-D-60720	BLOCK 4C - 8 AIR VALVES	DATA TABLES

INFORMATIONAL DRAWINGS

PLATE	DRAWING NO.	CONTENT	DETAIL
92	1695-D-60722	BLOCK 4C - 8 AIR VALVES WITH MANHOLES	SECTIONS AND DETAILS
93	1695-D-60724	BLOCK 4C - 8 MANHOLE WITH BLOWOFF	SECTIONS, DETAILS, AND TABLES
94	1695-D-60727	BLOCK 4C - 8 TAPERS, ADAPTERS AND COLLARS	DETAILS, SECTIONS, AND DATA TABLES
95	1695-D-60728	BLOCK 4C - 8 TYPICAL TURNOUT	PLAN, SECTIONS AND DETAILS
96	1695-D-60731	BLOCK 4C - 8 PRECAST CONCRETE MANHOLES	PLAN, SECTIONS AND DETAILS
97	085-N-1110	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	SANOSTEE TURNOUT NIP LAYOUT
98	085-N-1111	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	SANOSTEE TURNOUT NIP POWER DISTRIBUTION
99	085-N-1115	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	NEWCOMB TURNOUT NIP LAYOUT
100	085-N-1116	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	NEWCOMB TURNOUT NIP POWER DISTRIBUTION
101	085-N-1120	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	NASCHITTI TURNOUT NIP LAYOUT
102	085-N-1121	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	NASCHITTI TURNOUT NIP POWER DISTRIBUTION
103	085-N-1125	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	BBN9 TURNOUT NIP LAYOUT
104	085-N-1126	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	BBN9 TURNOUT NIP POWER DISTRIBUTION
105	085-N-1130	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	COYOTE CANYON TURNOUT PP NIP LAYOUT
106	085-N-1131	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	COYOTE CANYON TURNOUT NIP POWER DISTRIBUTION
107	085-N-1135	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	ROCK SPRINGS TURNOUT NIP LAYOUT
108	085-N-1136	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	ROCK SPRINGS TURNOUT NIP POWER DISTRIBUTION
109	085-N-1150	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	WINDOW ROCK TURNOUT NIP LAYOUT
110	085-N-1151	SAN JUAN LATERAL WATER TREATMENT PLANT- I&C	WINDOW ROCK TURNOUT NIP POWER DISTRIBUTION
111	6928997 C-1	NACHITTI CHAPTER - SAN JUAN LATERAL CONNECTION	PLAN AND PROFILE STA 0+00 TO STA 28+00
112	6922268 C-1	NAVAJO GULLUP WATER SUPPLY PROJECT - BEACON BISTI N9 - REACH 10.1	PLAN AND PROFILE STA 0+35.5 TO STA 28+00
113	6928904 C-1	SANOSTEE CHAPTER - SAN JUAN LATERAL CONNECTION	PLAN AND PROFILE STA 1+03 TO STA 28+00
114	6922268 FIG-1	NAVAJO GULLUP WATER SUPPLY PROJECT - BEACON BISTI N9 - REACH 10.1.1	TURNOUT AS BUILT

STANDARD DRAWINGS

PLATE	DRAWING NO.	CONTENT	DETAIL
115	40-D-4334	Electrical Standards Electrical Standards - Typical Grounding Details	Typical Grounding Details
116	40-D-4335	Electrical Standards	Typical Grounding Details
117	40-D-4753	Electrical Installation - Typical Grounding Details - For Switchyards and Substations	
118	40-D-5246	Buildings and Other Structures	Method of Building Forms for Successive Lifts, Detail
119	40-D-5247	Buildings and Other Structures	Joints in Concrete Structures
120	40-D-6234	Standard Nameplates	
121	40-D-6551	Pressure Pipe Trench Installation	Select Material
122	40-D-6552	Pressure Pipe Trench Installation	Soil-Cement Slurry - 12-Inch and Larger Pipe
123	40-D-7102	Government Drawing Format	Drawing Border, Sample Title Block, and Signature Lines
124	40-D-60001	Electrical Equipment Support	Sections and Detail
125	40-D-60002	Typical Guard Posts	Non-Security Application
126	40-D-60003	General Notes for Concrete Outline and Reinforcement	
127	40-D-60004	Minimum Requirements for Detailing Reinforcement	
128	104-D-254	Box Termination	
129	104-D-757	Electrical Installations	Designations and Symbols
130	104-D-1150	Electrical Installation	Device Designations and Symbols to be Used on Single-Line and Schematic Diagrams
131	104-D-1165	Electrical Installation	Main Control Board CSA - Panel 7R - Wiring Diagram

NOTES

1. For the current drawing revision number, refer to Section 52 00 00 - Drawings.

BUREAU OF RECLAMATION

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Asya Mcdaniel
DESIGNED
Asya Mcdaniel
DRAWN
Sara Putnam
CHECKED
Sara Putnam
Digitally signed by SARA PUTNAM
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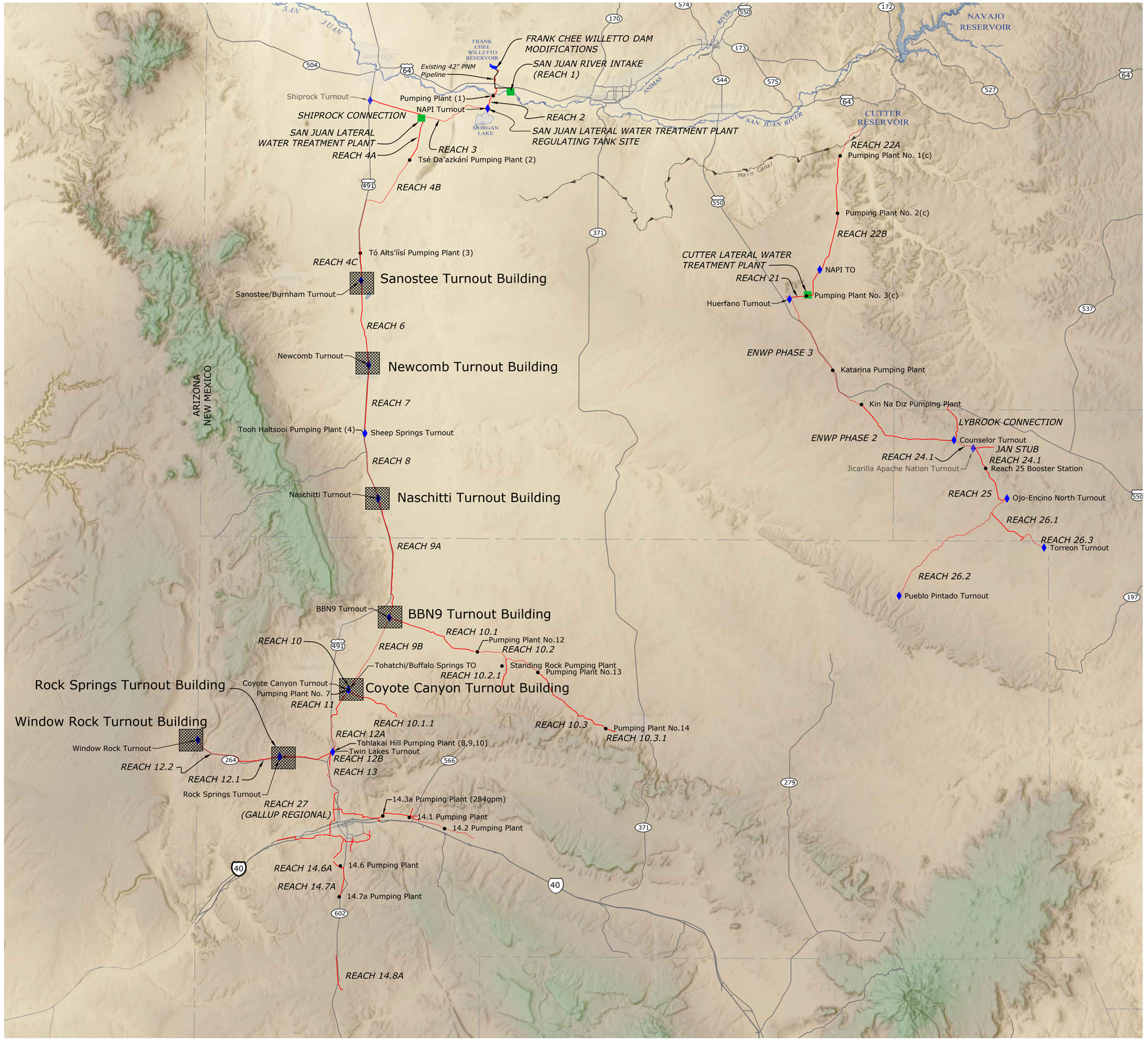
TECH. APPR.
Sara Putnam, P.E.
Lan Nguyen
APPROVED
Lan Nguyen, P.E.
Peer Reviewer - Plant Structures Group

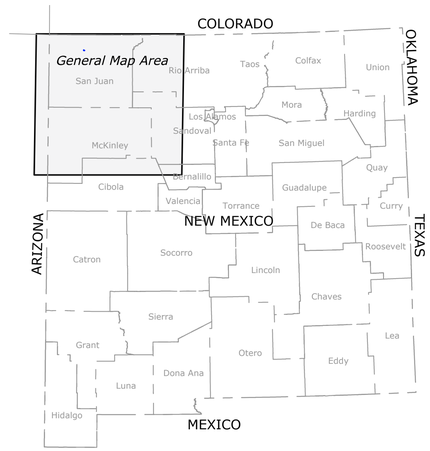
DENVER, CO 2026-06-19

DRAWING INDEX

1695-D-62526



KEY MAP



LEGEND

- *Pipeline (Varied by Reach)*
- *Water Treatment Plant*
- *Pumping Plant Location*
- *Turnout Location*
- *Highway (as designated)*
- *County Line*

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

D. Thornburg — DESIGNED
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Putnam, Sara K P.E. — TECH. APPR.
O`Shea, Kevin M P.E. — APPROVED Structural Engineer
PEER REVIEWER

FARMINGTON, NM 2026-06-05

GENERAL MAP

1695-529-60481

SHEET 1

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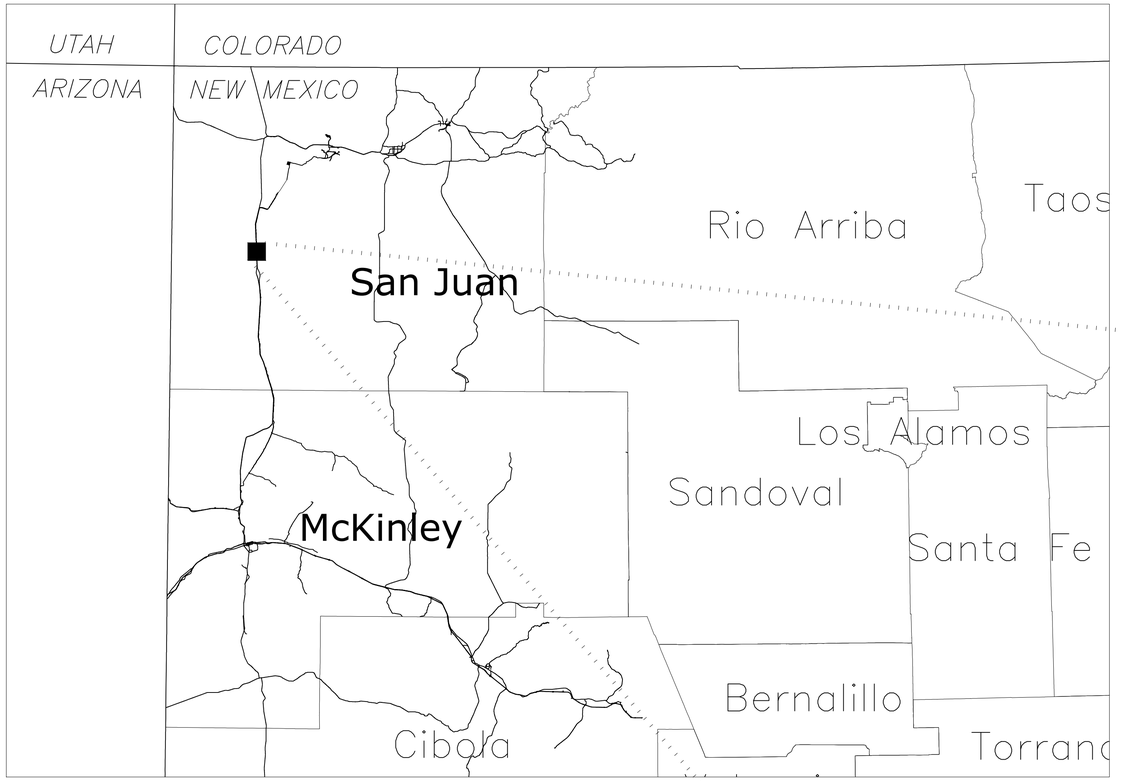

LEGEND

- State & Tribal Highway
- Pipeline Alignments
- Pipeline ROW

SANOSTEE TURNOUT BUILDING - LOCATION MAP
0.10 miles east from Indian Service Rt 34 and Hwy 491
intersection (SECTION 2, TOWNSHIP 25N, RANGE 18W)

50 25 0 50 100
SCALE OF FEET

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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ADMIN APPROVAL

FARMINGTON, NM 2026-06-16

VICINITY MAP

SANOSTEE TURNOUT BUILDING

1695-529-60488

SHEET 1

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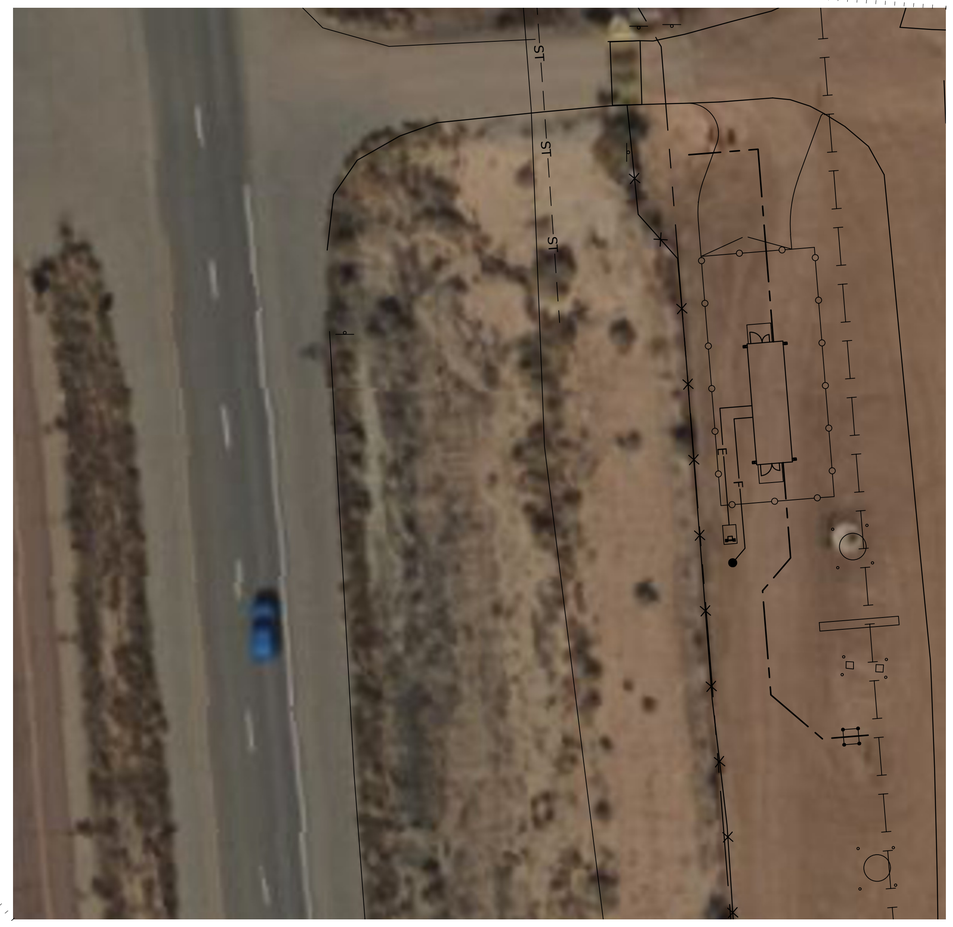
1695-529-60488-RD-1.DWG

LEGEND

- State & Tribal Highway
- Pipeline Alignments
- Pipeline ROW



NEWCOMB TURNOUT BUILDING - LOCATION MAP
0.11 miles east from Indian Service Rt 5022 and Hwy 491 intersection (SECTION 25, TOWNSHIP 24N, RANGE 18W)

20 10 0 20 40
SCALE OF FEET

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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AREEB HOSSAIN
DRAWN
S. Putnam
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Tihami, Moncef N P.E.
TECH. APPR.
Gladden, Ryan L P.E.
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ADMIN APPROVAL

FARMINGTON, NM 2026-06-16

VICINITY MAP

NEWCOMB TURNOUT BUILDING

1695-529-60489

SHEET 1

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2026-06-13
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CAD FILENAME
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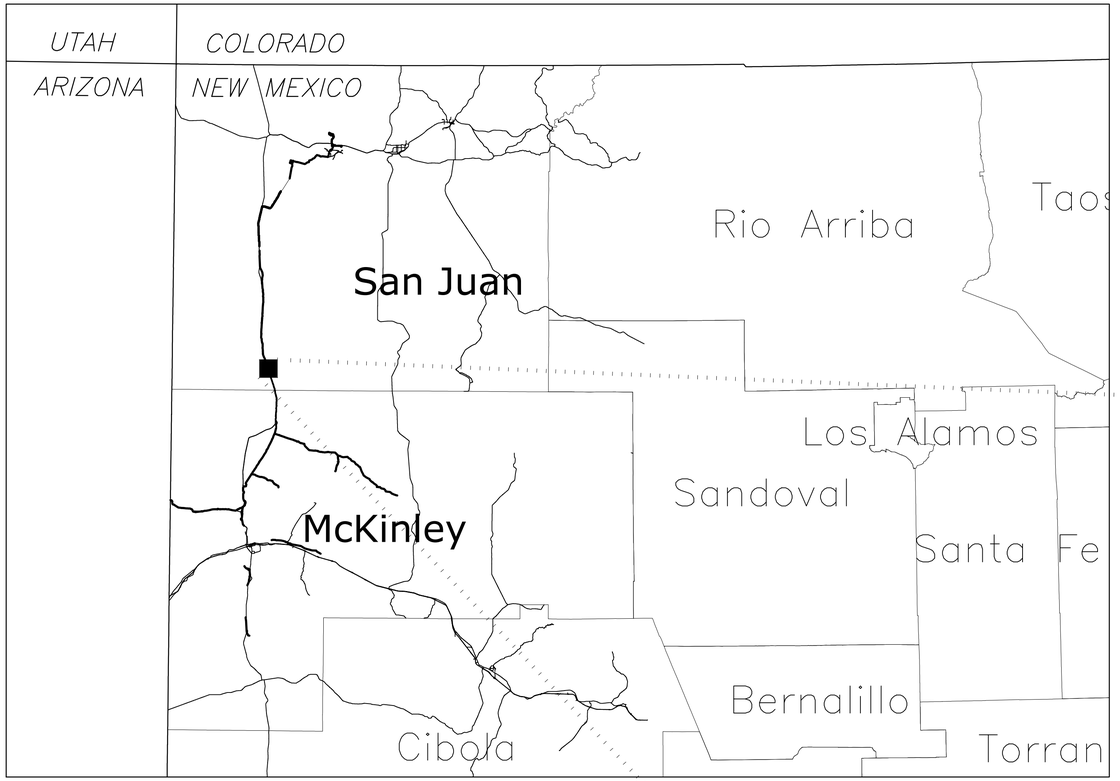
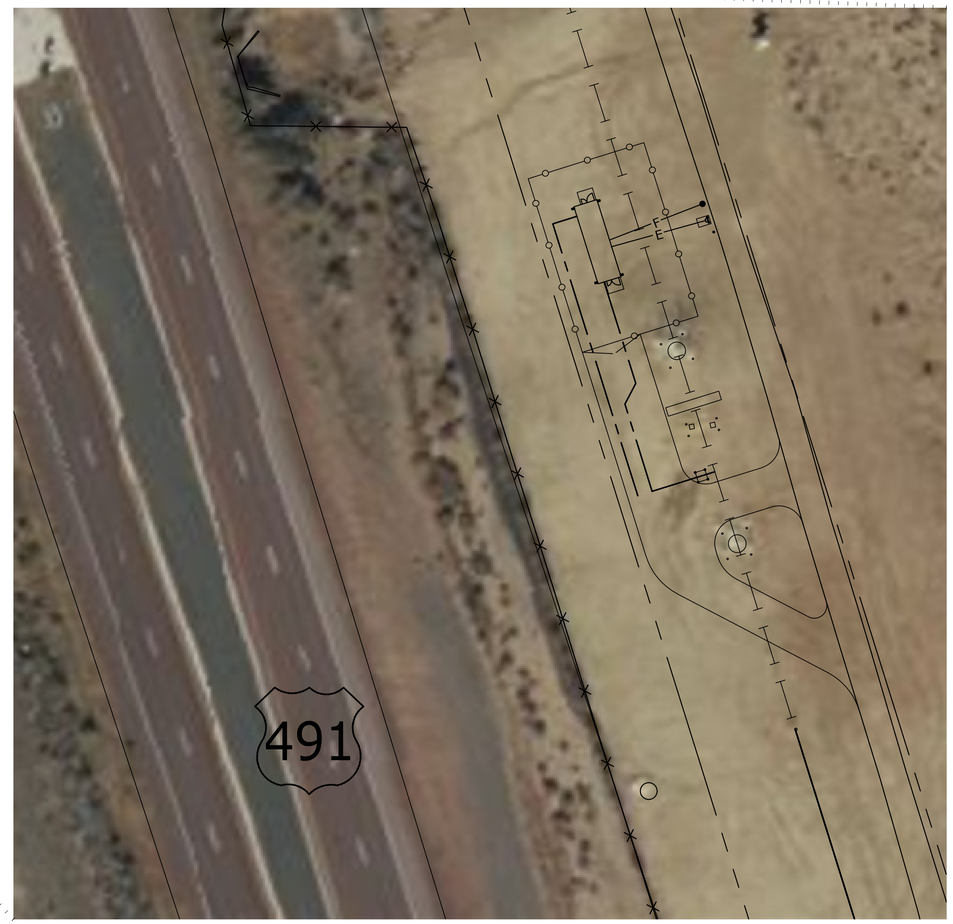


LEGEND

- State & Tribal Highway
- Pipeline Alignments
- Pipeline ROW



NASCHITTI TURNOUT BUILDING - LOCATION MAP
0.19 miles east from Indian Service Rt 69 and Hwy 491 intersection (SECTION 5, TOWNSHIP 21N, RANGE 17W)

30 15 0 30 60
SCALE OF FEET

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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APPROVED Deputy Construction Engineer
ADMIN APPROVAL

FARMINGTON, NM 2026-06-16

VICINITY MAP

NASCHITTI TURNOUT BUILDING

1695-529-60490

SHEET 1

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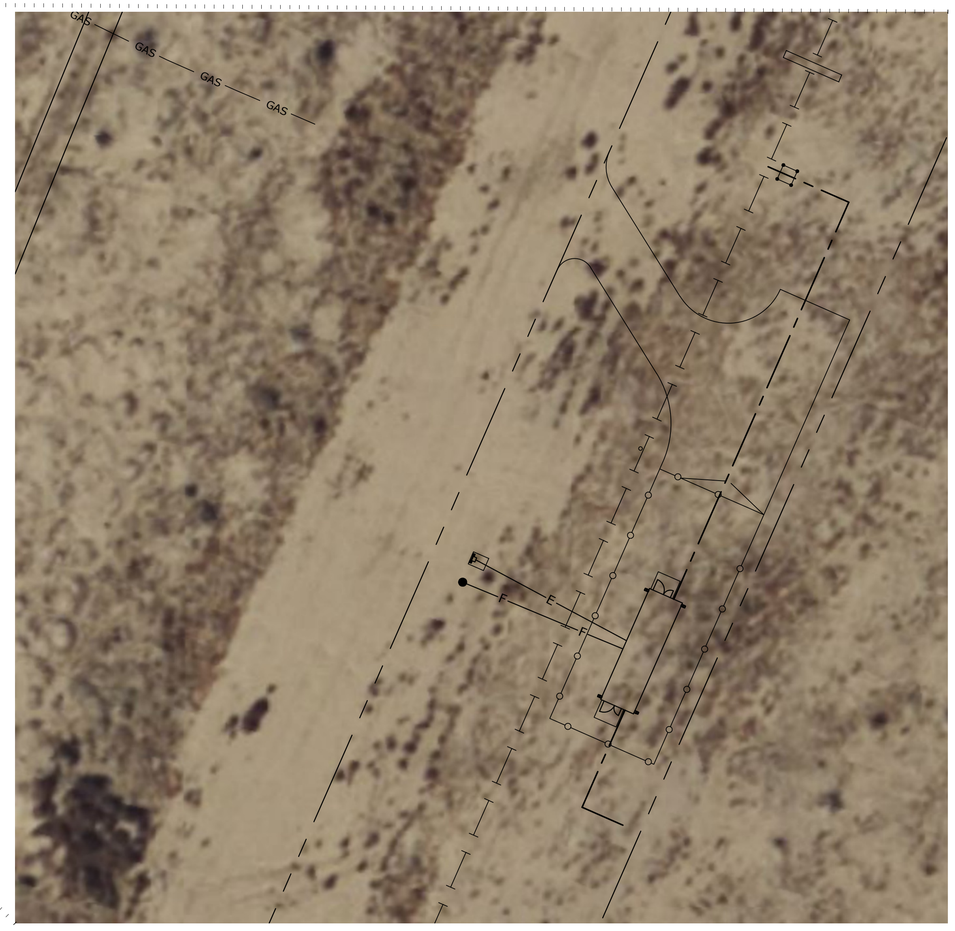
LEGEND

- State & Tribal Highway
- Pipeline Alignments
- Pipeline ROW



BBN9 TURNOUT BUILDING - LOCATION MAP
1.30 miles east from Indian Service Rt 11 and Hwy 491 intersection (SECTION 21, TOWNSHIP 19N, RANGE 17W)

20 10 0 20 40
SCALE OF FEET

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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Gladden, Ryan L P.E.
APPROVED Deputy Construction Engineer
ADMIN APPROVAL

FARMINGTON, NM 2026-06-16

VICINITY MAP

BBN9 TURNOUT BUILDING

1695-529-60491

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1695-529-60491-RD-1.DWG

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LEGEND

- State & Tribal Highway
- Pipeline Alignments
- Pipeline ROW

COYOTE CANYON TURNOUT BUILDING - LOCATION MAP
1.30 miles east from W Navajo 9 (SECTION 34, TOWNSHIP 18N, RANGE 18W)

30 15 0 30 60
SCALE OF FEET

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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AREEB HOSSAIN — DRAWN
S. Putnam — CHECKED
Tihani, Moncef N P.E. — TECH. APPR.
Gladden, Ryan L P.E. — APPROVED Deputy Construction Engineer — ADMIN APPROVAL

FARMINGTON, NM 2026-06-16

VICINITY MAP

COYOTE CANYON TURNOUT BUILDING

1695-529-60492

SHEET 1

LAST SAVED DATE
2026-06-15
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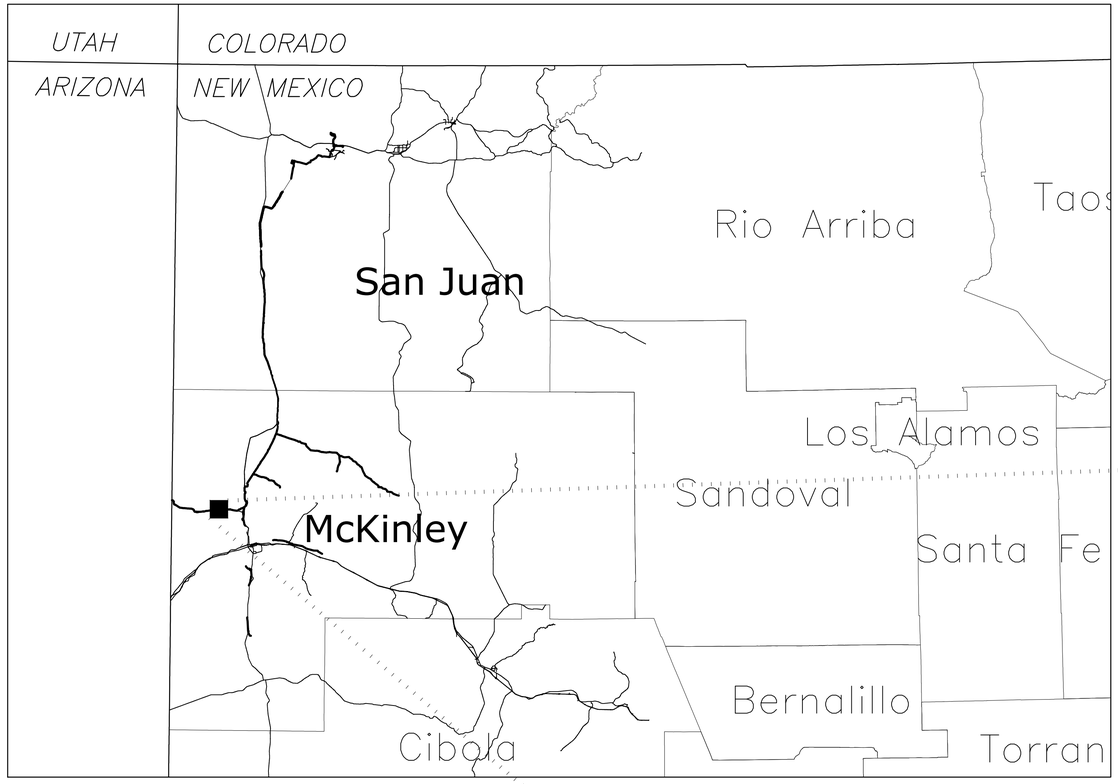

LEGEND

- State & Tribal Highway
- Pipeline Alignments
- Pipeline ROW

ROCK SPRINGS TURNOUT BUILDING - LOCATION MAP
0.05 miles east from intersection of HWY 264 and Pleasant View Rd (SECTION 8, TOWNSHIP 16N, RANGE 19W)

30 15 0 30 60
SCALE OF FEET

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RECLAMATION

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

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TURNOUT BUILDINGS

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FARMINGTON, NM 2026-06-16

VICINITY MAPS

ROCK SPRINGS
TURNOUT BUILDING

1695-529-60493

SHEET 1

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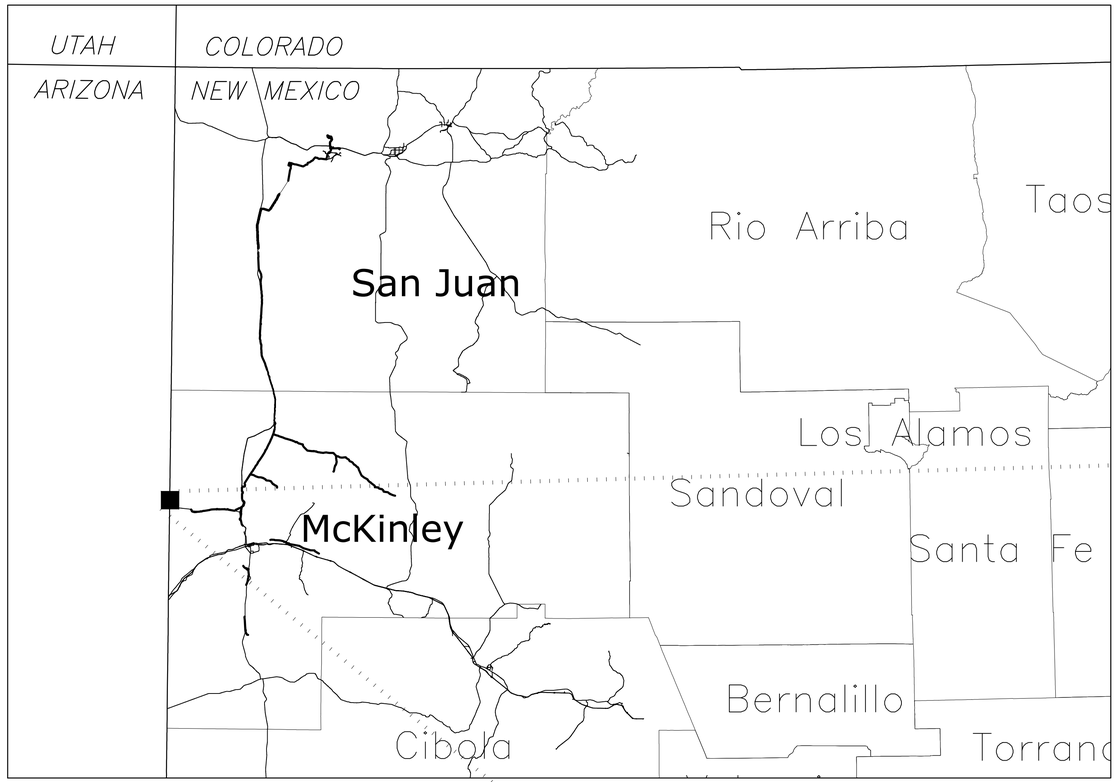

LEGEND

- State & Tribal Highway
- Pipeline Alignments
- Pipeline ROW

WINDOW ROCK` TURNOUT BUILDING - LOCATION MAP
0.08 miles east from HWY 264 (SECTION 36, TOWNSHIP 17N, RANGE 21W)

40 20 0 40 80
SCALE OF FEET

BUREAU OF RECLAMATION

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

AREEB HOSSAIN
DESIGNED
AREEB HOSSAIN
DRAWN
S. Putnam
CHECKED
Tihami, Moncef N P.E.
TECH. APPR.
Gladden, Ryan L P.E.
APPROVED Deputy Construction Engineer
ADMIN APPROVAL

FARMINGTON, NM 2026-06-16

VICINITY MAP

WINDOW ROCK
TURNOUT BUILDING

1695-529-60494

SHEET 1

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STATEMENT OF SPECIAL INSPECTIONS

SPECIAL INSPECTION REPORT REQUIREMENTS

Special Inspectors shall keep records of inspections. The Special Inspector shall furnish inspection reports to the COR, Architect, Engineer of Record, and Contractor. Reports shall indicate that the structural work inspected was done in conformance to approved contract documents. Discrepancies shall be brought to the immediate attention of the contractor for correction. If the discrepancies are not corrected, the discrepancies shall be brought to the attention of the COR, Architect and Engineer of Record prior to the completion of that phase of work. A final report of inspections documenting required special inspections of structural work and correction of any discrepancies noted in the inspections shall be submitted at a point in time agreed upon by the COR prior to the start of work. The Special Inspection program does not relieve the Contractor of his or her responsibilities.

REQUIRED SPECIAL INSPECTIONS AND TESTS

Tables 1,2 and 3 identify types of structural work that require special inspections. Refer to individual specification sections for specific testing and inspection requirements.

Continuous/Perform: The full-time observation of work by a qualified special inspector who is present in the area where the work is being performed.

Periodic/Observe: The part-time or intermittent observation of work by a qualified special inspector who is present in the area where the work has been or is being performed and at the completion of the work. Operations need not be delayed pending these inspections.

REFERENCE STANDARDS

See Structural Design Data on drawing 1695-D-62555 and specifications.

TABLE 1

REQUIRED SPECIAL INSPECTIONS AND TESTS OF CONCRETE CONSTRUCTION (IBC - Table 1705.3)

SYSTEM or MATERIAL	INSPECTION				REMARKS
	IBC CODE REFERENCE	CODE or STANDARD REFERENCE	FREQUENCY		
			CONTINUOUS	PERIODIC	
1. Inspect reinforcement, including prestressing tendons, and verify placement.		ACI 318 Ch. 20, 25.2, 25.3, 26.6.1-26.6.3		X	
2. Reinforcing bar welding: a. Verify weldability of reinforcing bars other than ASTM A706; b. Inspect single-pass fillet welds, maximum 5/16"; and c. Inspect all other welds.		AWS D1.4 ACI 318: 26.6.4	 X	 X X 	
3. Inspect anchors cast in concrete.		ACI 318: 17.8.2		X	
4. Inspect anchors post-installed in hardened concrete members. a. Adhesive anchors installed in horizontally or upwardly inclined orientations to resist sustained tension loads. b. Mechanical anchors and adhesive anchors not defined in 4.a.		 ICC Evaluation Report, ACI 318: 26.13.1.6/R26.13.1.6 ICC Evaluation Report, ACI 318: 26.13.1.5/R26.13.1.5 26.13.1.6/R26.13.1.6	 X 	 X	4.b: All anchors shall be visually inspected. Where periodic inspection is allowed in accordance with the anchor ICC evaluation report, inspections shall be as follows: • For all anchors, prior to concealment, verify anchor type, anchor dimensions, anchor spacing and edge distance. • For each anchor type and size, inspector shall be onsite to continuously inspect a minimum of the first 10 anchors installed by each installer for conformance with the ICC evaluation report. Provided all anchors are installed correctly per manufacturer's instructions, provide periodic inspections a minimum of the next 10% of the next 1000 anchors by each installer and a minimum of 5% of the remaining anchors by each installer. Inspections shall occur a minimum of once per week at a random time while anchor installation is ongoing. Any non-compliance issues shall reset the inspection requirements to ten (10) continuous inspections. Non-compliant anchors shall be brought to the attention of the engineer of record for review and shall be brought into compliance by either testing or re-installation. • Inspection reports shall identify names of installers. Special inspector shall provide documentation at the end of anchor installations stating that the minimum number of anchors were inspected.
5. Verify use of required design mix.	1904.1, 1904.2	ACI 318: Ch. 19, 26.4.3, 26.4.4		X	
6. Prior to concrete placement, fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete.		ASTM C172 ASTM C31 ACI 318: 26.4, 26.12	X		
7. Inspect concrete placement for proper application techniques.		ACI 318: 26.5	X		
8. Verify maintenance of specified curing temperature and techniques.		ACI 318: 26.5.3-26.5.5		X	
9. Inspect prestressed concrete for: a. Application of prestressing forces; and b. Grouting of bonded prestressing tendons.		ACI 318: 26.10	 X X		
10. Inspect erection of precast concrete members.		ACI 318: 26.9		X	
13. Verify in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.		ACI 318: 26.11.2		X	
14. Inspect formwork for shape, location and dimensions of the concrete member being formed.		ACI 318: 26.11.1.2(b)		X	

TABLE 2

REQUIRED SPECIAL INSPECTIONS AND TESTS OF SOILS (IBC - Table 1705.6)

SYSTEM or MATERIAL	INSPECTION				REMARKS
	IBC CODE REFERENCE	CODE or STANDARD REFERENCE	FREQUENCY		
			CONTINUOUS	PERIODIC	
1. Verify materials below shallow foundations are adequate to achieve the design bearing capacity.				X	
2. Verify excavations are extended to proper depth and have reached proper material.				X	
3. Perform classification and testing of compacted fill materials.				X	
4. During fill placement, verify use of proper materials and procedures in accordance with the provisions of the approved geotechnical report. Verify densities and lift thicknesses during placement and compaction of compacted fill.			X		
5. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly.				X	

TABLE 3

REQUIRED SPECIAL INSPECTIONS OF FABRICATED ITEMS (IBC - 1705.11)

SYSTEM or MATERIAL	INSPECTION				REMARKS
	IBC CODE REFERENCE	CODE or STANDARD REFERENCE	FREQUENCY		
			CONTINUOUS	PERIODIC	
1. Special inspections of fabricated items shall be performed in accordance with 1704.2.5.					Where fabrication of structural, load-bearing or lateral load-resisting members or assemblies is being conducted on the premises of a fabricator's shop, special inspections of the fabricated items shall be performed during fabrication, except where the fabricator has been approved to perform work without special inspections in accordance with IBC Section 1704.2.5.1.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

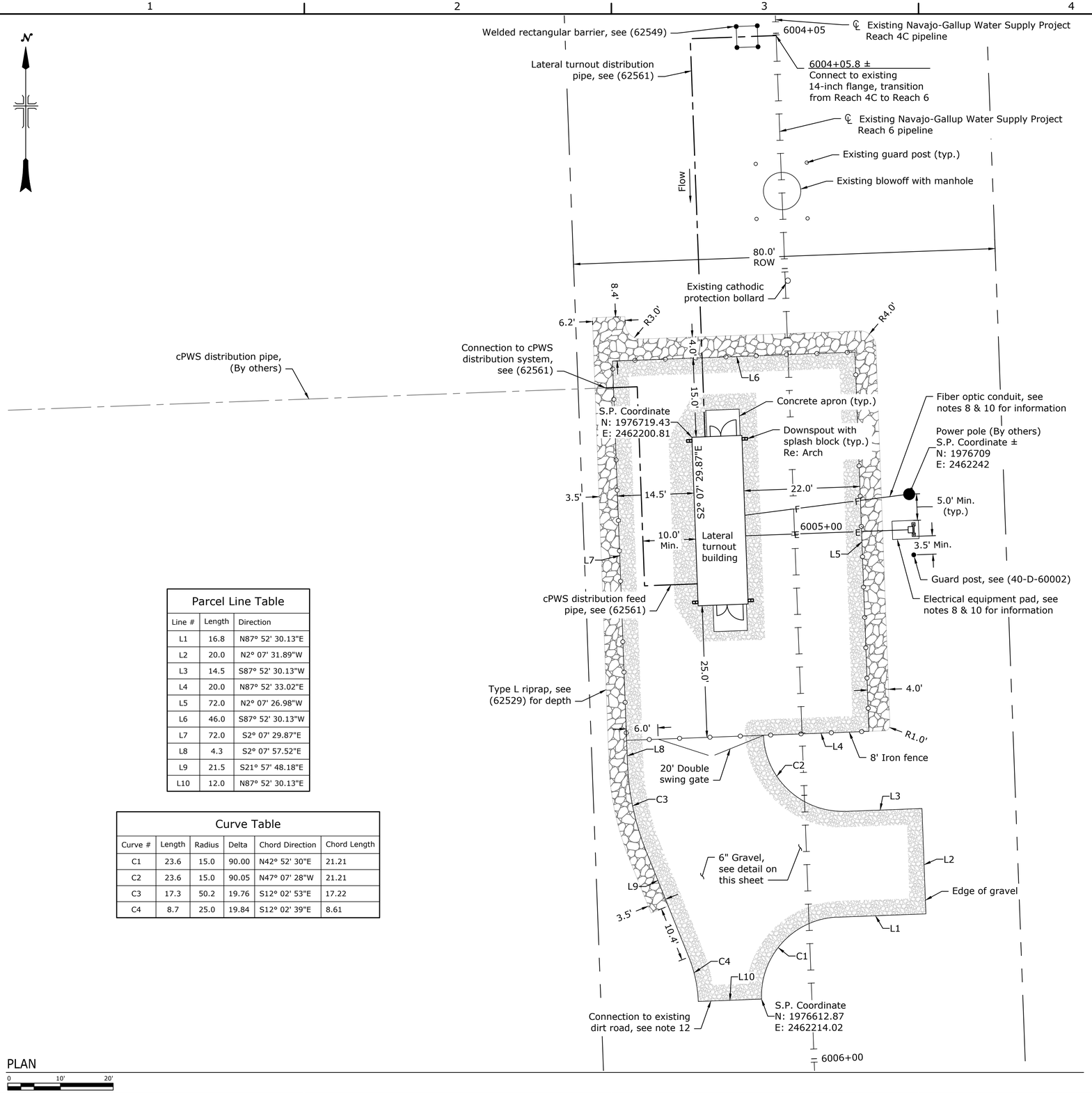
SAN JUAN LATERAL
TURNOUT BUILDINGS

SARA PUTNAM
DESIGNED
SARA PUTNAM
DRAWN
NICOLE M. SMITH
CHECKED
TECH. APPR. Kevin O'Shea, P.E. Digitally signed by LAN NGUYEN Date: 2026.06.05 12:54:14 -06'00'
APPROVED L. Nguyen, Ph.D, P.E.
Peer Reviewer - Plant Structures Group

DENVER, CO 2026-06-19

SPECIAL INSPECTIONS

1695-D-62527

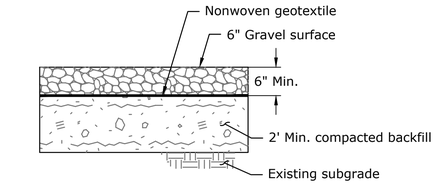


PLAN

0 10' 20'

Parcel Line Table		
Line #	Length	Direction
L1	16.8	N87° 52' 30.13"E
L2	20.0	N2° 07' 31.89"W
L3	14.5	S87° 52' 30.13"W
L4	20.0	N87° 52' 33.02"E
L5	72.0	N2° 07' 26.98"W
L6	46.0	S87° 52' 30.13"W
L7	72.0	S2° 07' 29.87"E
L8	4.3	S2° 07' 57.52"E
L9	21.5	S21° 57' 48.18"E
L10	12.0	N87° 52' 30.13"E

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	23.6	15.0	90.00	N42° 52' 30"E	21.21
C2	23.6	15.0	90.05	N47° 07' 28"W	21.21
C3	17.3	50.2	19.76	S12° 02' 53"E	17.22
C4	8.7	25.0	19.84	S12° 02' 39"E	8.61



DETAIL - YARD GRAVEL SURFACING SECTION

NOT TO SCALE

ABBREVIATIONS

S.P. - State Plane
℄ - Center line
ROW- Right-of-Way

LEGEND

- Permanent Right-of-Way
- ℄ Existing Navajo-Gallup Water Supply Project pipeline
- Lateral turnout distribution pipe/ cPWS distribution feed pipe
- cPWS distribution pipe (By others)
- E — Electrical conduit
- F — Fiber optic conduit
- Iron fence
- Concrete
- Riprap
- Gravel
- Compacted backfill
- Welded rectangular barrier
- Power pole (By others)
- Proposed air valve with manhole
- Proposed sectionalizing valve
- Proposed manhole with blowoff

GENERAL SITEWORK NOTES:

1. Reference Structural and Architectural drawings and specifications for concrete apron, downspout, and splash block information.
2. See 40-D-6551 and 40-D-6552 for pipeline trenching details.
3. Drawing numbers are 1695-D-XXXXX unless otherwise specified.
4. See specifications for sediment and erosion control information.
5. Reference Mechanical drawings and specifications for turnout yard piping information.
6. Clear and screen the turnout footprint and associated yard area for vegetation, large rocks, etc. before backfilling and grading. Protect in place existing infrastructure, see Specifications for additional information.
7. Power poles and overhead electric lines shall be installed by local utility provider.
8. Fiber and electrical conduits shown are for graphical purposes only and do not represent all conduits and vaults expected to be on site. See (62578-62588) for additional information. Coordinate with COR for final placement of electrical disconnect pad, see (62556) for pad information.
9. See specifications for fence and gate information.
10. See specifications for fiber and electrical conduit depths and minimum clearances at utility crossings.
11. See specifications for signage requirements and installation.
12. The location and elevation of the connection points to the existing dirt road are approximate and have not been field verified. Contractor to coordinate with the COR as needed for field adjustments for the road connection.
13. See (62558-62568) for yard pipe designs, (62555) for turnout building, and (62556) for thrust block information.
14. Existing infrastructure shall be protected in place in accordance with the specifications unless otherwise noted.
15. See (62556) for concrete apron details.

NOTES:

1. All control points are shown in New Mexico State Plan Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00026509926 (Reach 4) was used.

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DESIGNED	B. ZOOK
DRAWN	L. VOGT
CHECKED	C. HARMON
TECH. APPR.	Marner, Andrew (Andy) J P.E.
APPROVED Supervisory Civil Engineer PEER REVIEWER	Steen, Timothy M
DENVER, CO	2026-06-01

SANOSTEE TURNOUT

SITE PLAN

1695-D-62528

SHEET 1

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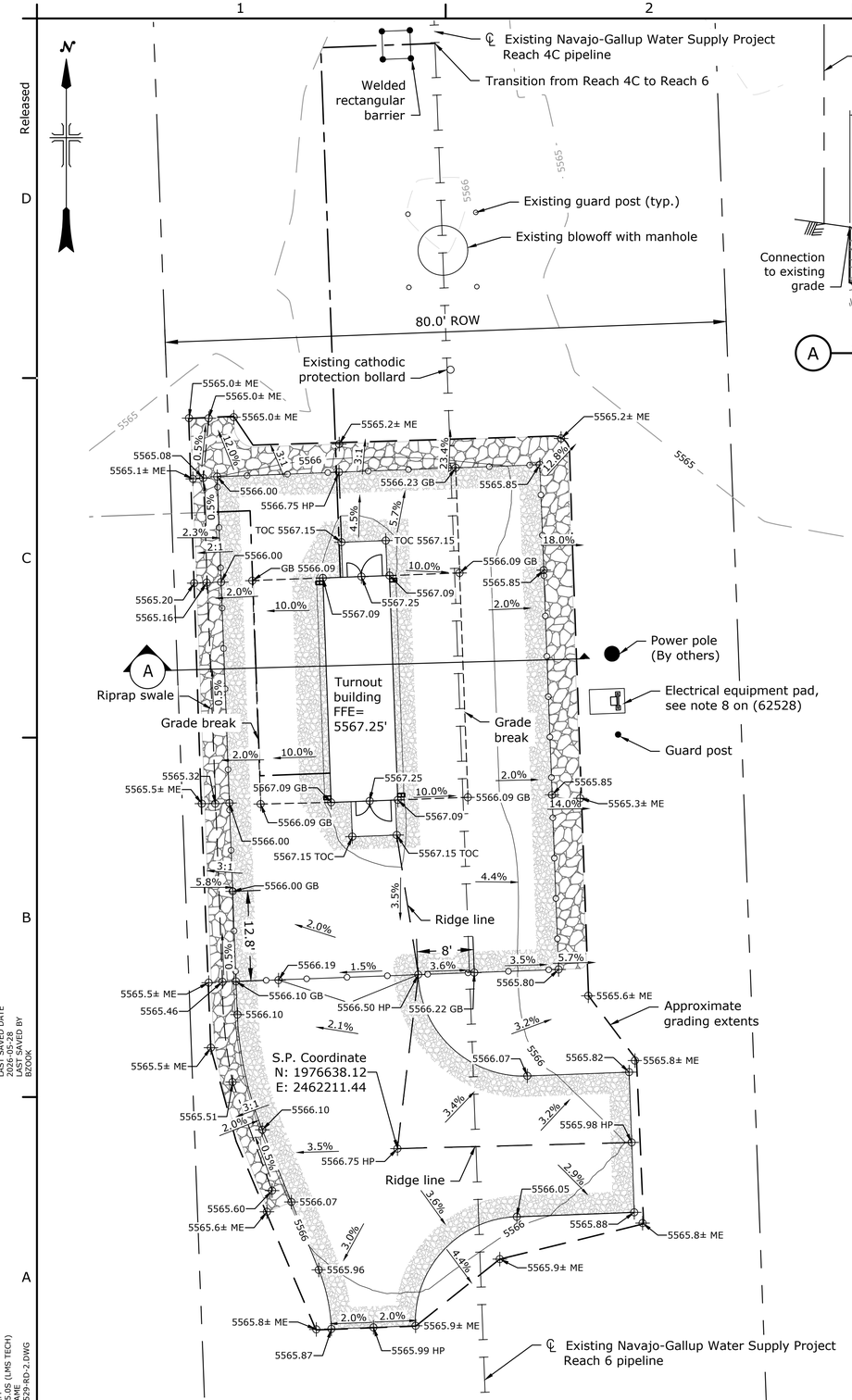

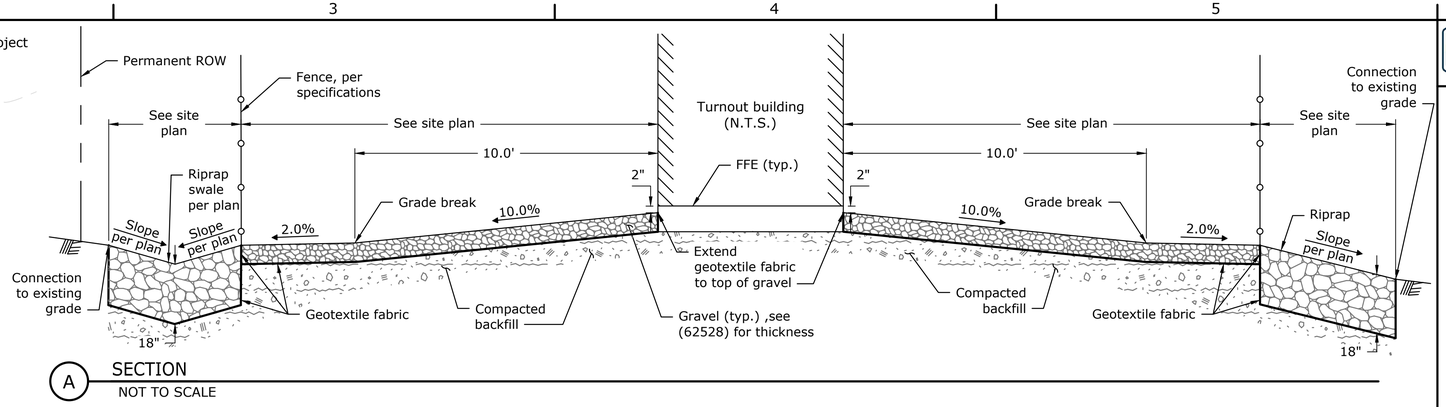

ABBREVIATIONS

ME - Match existing (±)
HP - High point
GB - Grade break
FFE - Finished floor elevation
TOC - Top of concrete

GENERAL SITEWORK GRADING NOTES:

1. Contractor to verify existing grades and utilities prior to construction, notify COR of any conflicts or discrepancies.
2. Yard elevations are shown to top of gravel or concrete unless otherwise specified.
3. Provide a minimum slope of 10% away from the building in gravel areas, extending 10' minimum, or continuing to the edge of the fence where the gravel width is less than 10'.
4. Slope concrete stoops and aprons away from the building at 2.0% slope.
5. All vertical elevations are based on the NAVD88 vertical datum.
6. Contours shown are based off of survey data provided by the Four Corners Construction Office.
7. Contours are shown in 1' intervals.
8. Slope riprap as shown on drawings, maximum slope for riprap shall not exceed 1.5(H):1(V).
9. See (62528) for legend, abbreviations, and general sitework notes.

NOTES:

1. All control points are shown in New Mexico State Plan Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00026509926 (Reach 4) was used.

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK	DESIGNED
L. VOGT	DRAWN
C. HARMON	CHECKED
Marner, Andrew (Andy) J P.E.	TECH. APPR.
Steen, Timothy M	APPROVED Supervisory Civil Engineer
	PEER REVIEWER
DENVER, CO	2026-06-01

SANOSTEE TURNOUT

GRADING PLAN

1695-D-62529

SHEET 2

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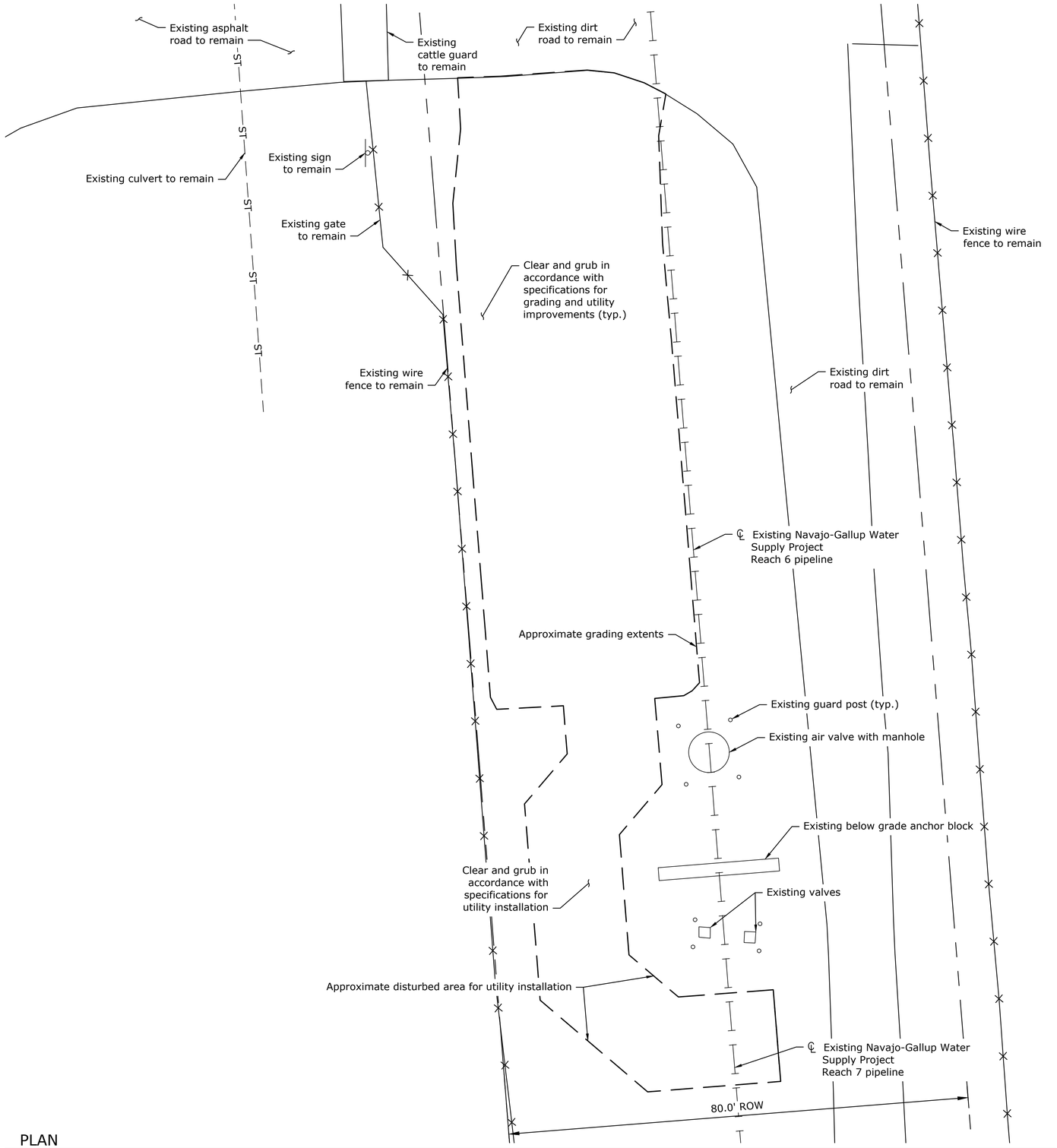

PLAN

GENERAL EXISTING CONDITIONS AND DEMOLITION NOTES:

1. Clear and screen the turnout footprint and associated yard area for vegetation, large rocks, etc. before backfilling and grading. Protect in place existing infrastructure, see specifications for additional information.
2. Contractor to verify existing grades, utilities, and horizontal & vertical tie in to existing roads prior to construction, notify COR of any conflicts or discrepancies.
3. Contractor to protect existing utilities and infrastructure in place in accordance with the specifications.

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027097141 (Reach 6) was used.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
DESIGNED
L. VOGT
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C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
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DENVER, CO 2026-06-01

NEWCOMB TURNOUT

EXISTING CONDITIONS AND DEMOLITION PLAN

1695-D-62530

SHEET 3

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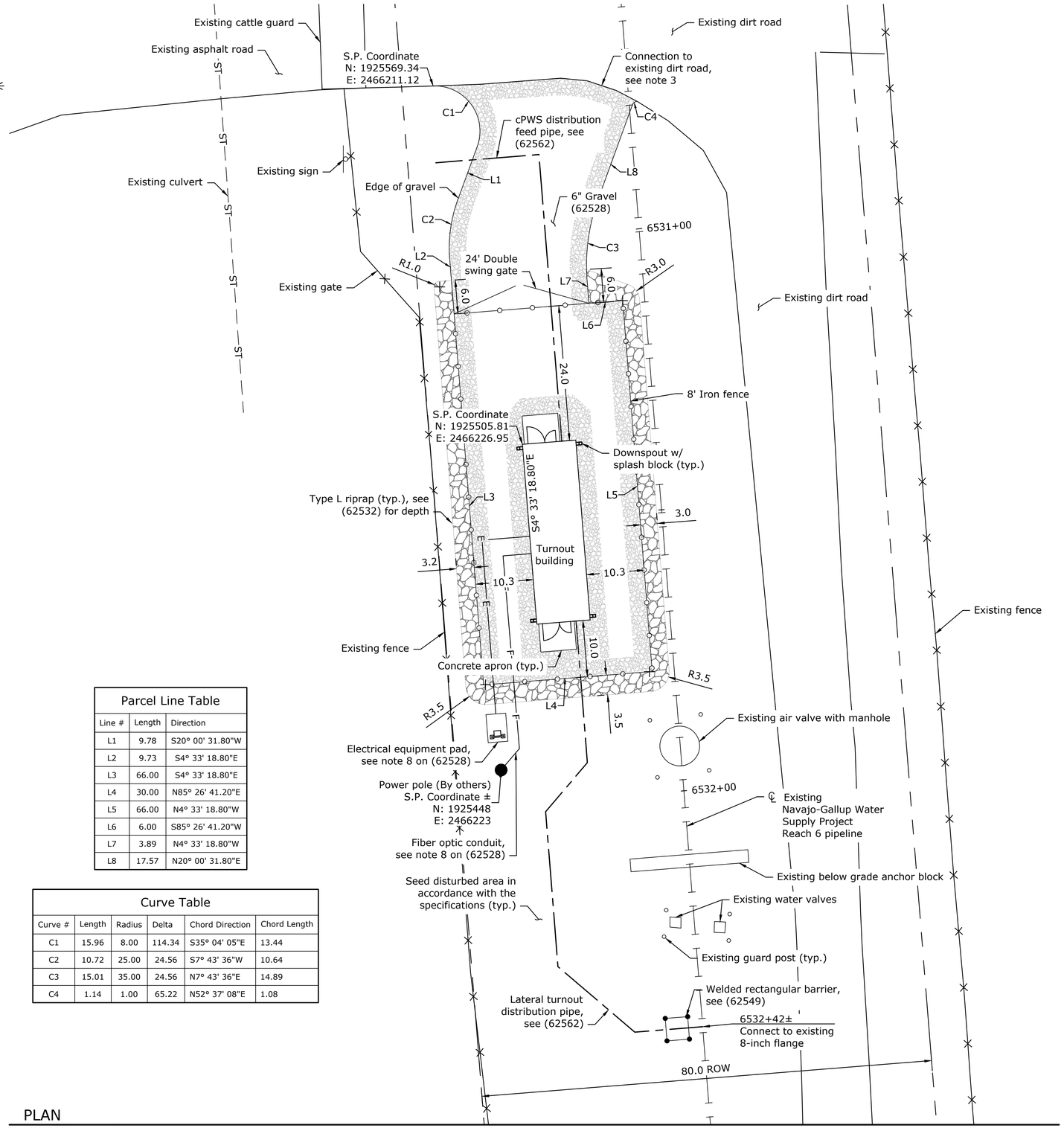

PLAN

0 10' 20'

Parcel Line Table		
Line #	Length	Direction
L1	9.78	S20° 00' 31.80"W
L2	9.73	S4° 33' 18.80"E
L3	66.00	S4° 33' 18.80"E
L4	30.00	N85° 26' 41.20"E
L5	66.00	N4° 33' 18.80"W
L6	6.00	S85° 26' 41.20"W
L7	3.89	N4° 33' 18.80"W
L8	17.57	N20° 00' 31.80"E

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	15.96	8.00	114.34	S35° 04' 05"E	13.44
C2	10.72	25.00	24.56	S7° 43' 36"W	10.64
C3	15.01	35.00	24.56	N7° 43' 36"E	14.89
C4	1.14	1.00	65.22	N52° 37' 08"E	1.08

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027097141 (Reach 6) was used.
3. The location and elevation of the connection points to the existing dirt road are approximate and have not been field verified. Contractor to coordinate with the COR as needed for field adjustments for the road connection.

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

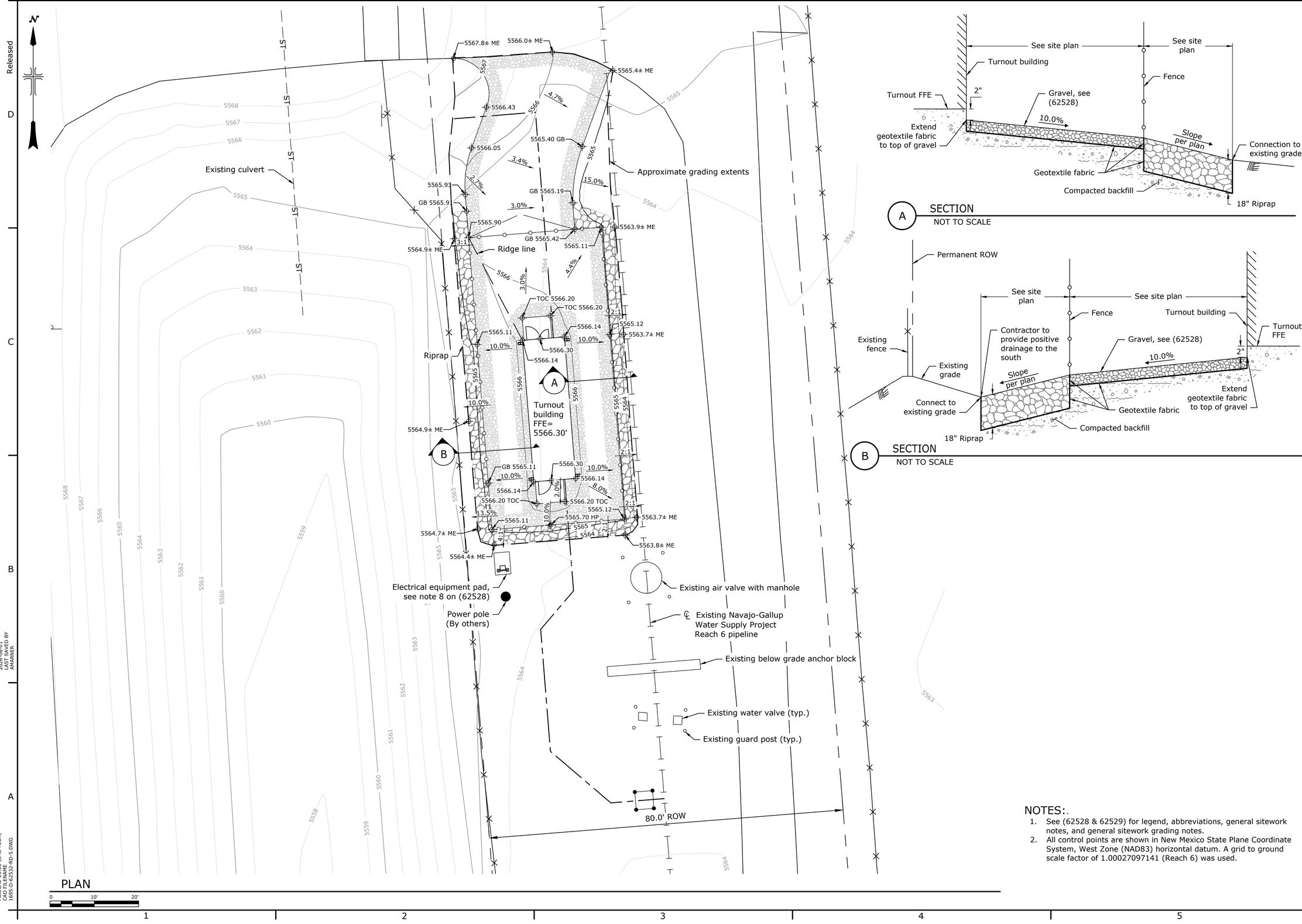
B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
PEER REVIEWER
DENVER, CO 2026-06-01

NEWCOMB TURNOUT

SITE PLAN

1695-D-62531

SHEET 4



PLAN

0 10' 20'

A SECTION
NOT TO SCALE

B SECTION
NOT TO SCALE

NOTES:.

1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general sitework grading notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027097141 (Reach 6) was used.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK	DESIGNED
L. VOGT	DRAWN
C. HARMON	CHECKED
Marner, Andrew (Andy) J P.E.	TECH. APPR.
Steen, Timothy M	APPROVED Supervisory Civil Engineer
	PEER REVIEWER
DENVER, CO	2026-06-01

NEWCOMB TURNOUT

GRADING PLAN

1695-D-62532

SHEET 5

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Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	19.79	25.00	45.35	N39° 39' 51"W	19.28
C2	19.63	25.00	45.00	N39° 50' 24"W	19.13
C3	16.58	10.04	94.57	S64° 44' 52"E	14.76
C4	13.99	10.00	80.17	N32° 34' 23"E	12.88
C5	15.98	10.00	91.55	N61° 33' 56"W	14.33
C6	15.71	10.00	90.00	S27° 39' 36"W	14.14
C7	11.78	15.00	45.00	S39° 50' 24"E	11.48
C8	11.72	5.00	134.34	N50° 29' 20"E	9.22

Parcel Line Table		
Line #	Length	Direction
L1	67.53	N62° 20' 23.50"W
L2	84.78	N17° 20' 23.50"W
L3	72.00	N17° 20' 23.50"W
L4	48.00	N72° 39' 36.50"E
L5	72.00	S17° 20' 23.50"E
L6	24.00	S72° 39' 36.50"W
L7	54.81	S17° 20' 23.50"E
L8	19.35	N72° 41' 33.87"E
L9	34.57	S15° 52' 51.59"E
L10	16.19	S72° 39' 36.50"W
L11	2.49	S17° 20' 23.50"E
L12	32.30	S62° 20' 23.50"E
L13	45.11	S16° 51' 39.56"E

Existing fence

Existing cPWS distribution system

Type L riprap swale, see (62534) for depth

S.P. Coordinate
N: 1845579.33
E: 2471950.19

Seed disturbed areas in accordance with the specifications (typ.)

See (62563) for pipe connection information

cPWS distribution feed pipe, see (62563)

Lateral turnout distribution pipe, see (62563)

Flow

S.P. Coordinate
N: 1845491.32
E: 2471989.67

N17° 20' 23.50"W

Edge of gravel

6" Gravel, see (62528)

24' Double swing gate

Lateral turnout building

Existing Navajo-Gallup Water Supply Project Reach 8 pipeline

Welded rectangular barrier, see (62549)

Roof drain downspout w/ splashblock (typ.)

Concrete apron (typ.)

Existing cathodic protection bollard

Existing blowoff with manhole

Existing valve (typ.)

Existing guard post (typ.)

Existing blowoff with manhole

Existing below grade anchor block

8' Iron fence

S.P. Coordinate
N: 1845322.48
E: 2472093.07

8414+00.6±
Connect to existing 10-inch flange

Electrical equipment pad, see note 8 on (62528)

Fiber optic cable, see note 8 on (62528)

Guard post, see (40-D-60002)

Power pole (By others)

S.P. Coordinate ±
N: 1845523
E: 2472031

Approximate connection to existing road

Approximate connection to existing road

Existing dirt access road

PLAN

0 10' 20'

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00028975894 (Reach 8) was used.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
PEER REVIEWER

DENVER, CO 2026-06-02

NASCHITTI TURNOUT

SITE PLAN

1695-D-62533

SHEET 6

LAST SAVED DATE
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1695-D-62533-RD-6.DWG

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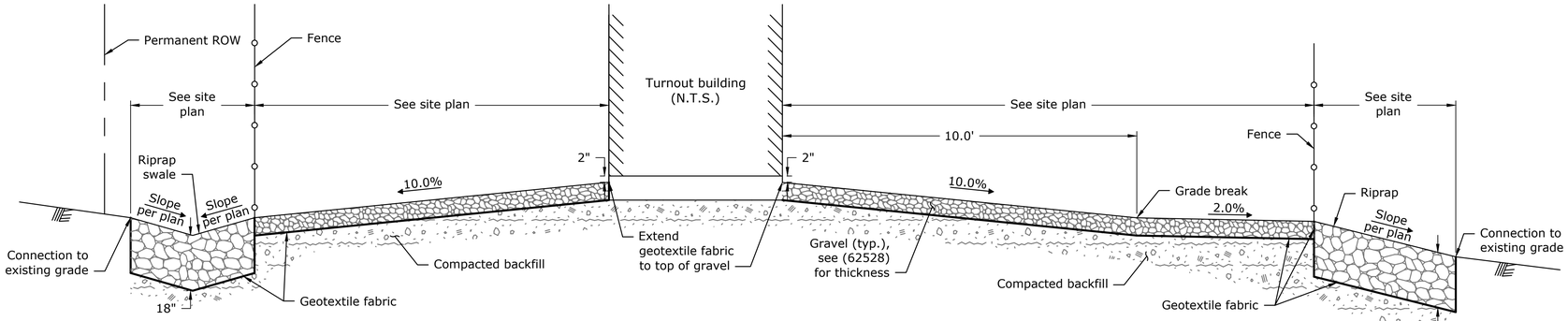
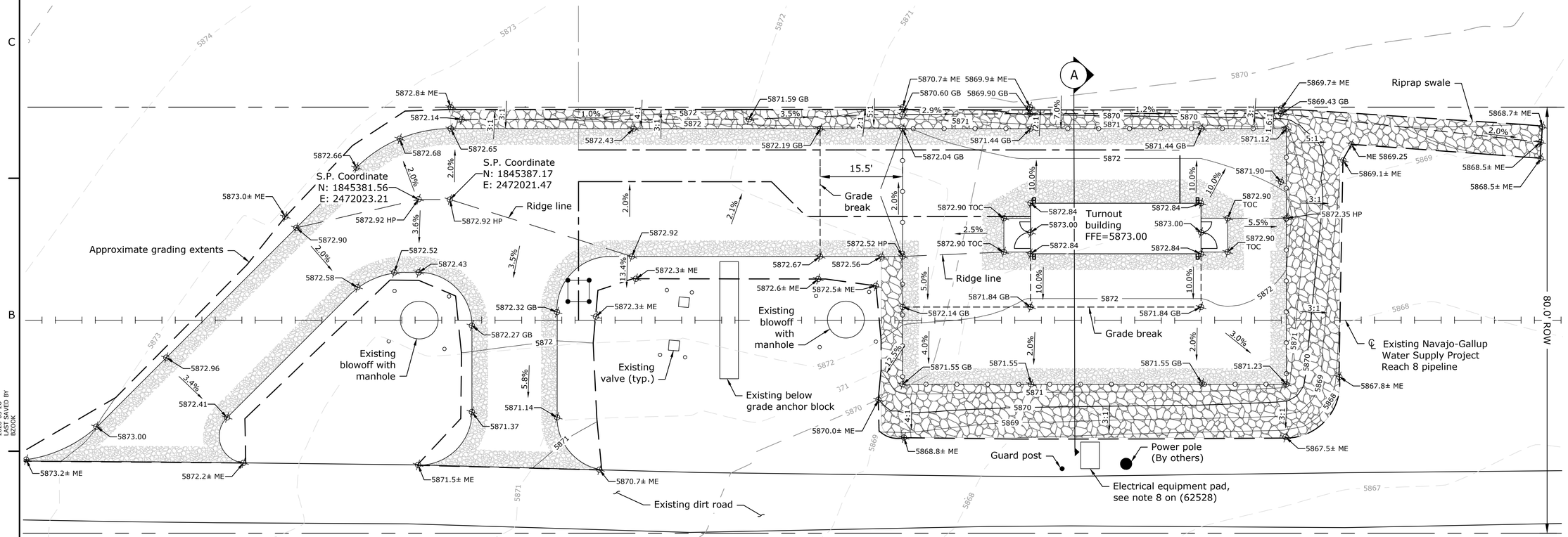


A SECTION
NOT TO SCALE



PLAN
0 10' 20'

NOTES:
1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general sitework grading notes.
2. All control points are shown in New Mexico State Plan Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00028975894 (Reach 8) was used.

LAST SAVED DATE
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CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62534-RD-7.DWG

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
ADMIN APPROVAL
DENVER, CO 2026-06-01

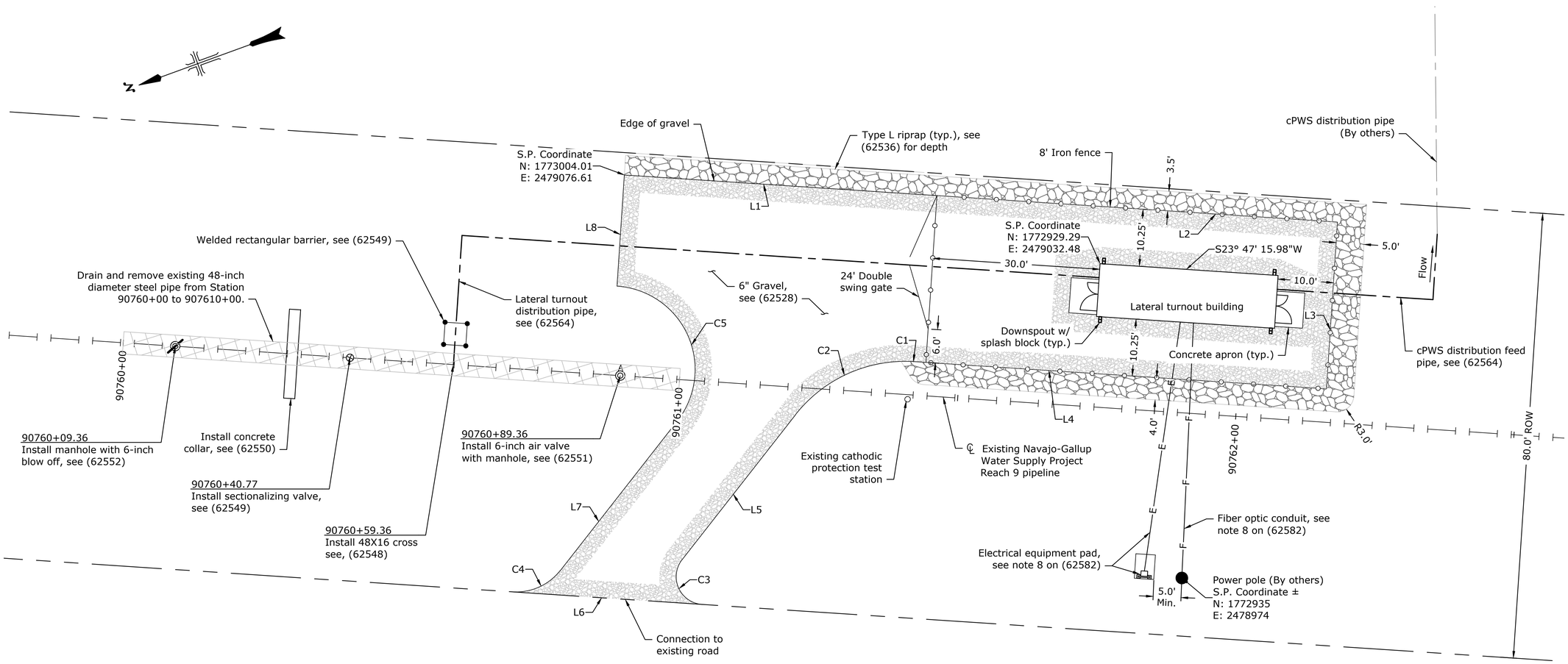
NASCHITTI TURNOUT

GRADING PLAN

1695-D-62534

SHEET 7

Released



PLAN

0 10' 20'

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	4.45	46.60	5.48	N22° 19' 10"E	4.45
C2	21.94	25.00	50.28	N6° 49' 31"W	21.24
C3	10.84	5.00	124.25	S85° 54' 48"W	8.84
C4	9.73	10.00	55.75	S4° 05' 18"E	9.35
C5	32.53	15.00	124.25	N85° 54' 45"E	26.52

Parcel Line Table		
Line #	Length	Direction
L1	56.19	S23° 47' 15.98"W
L2	72.00	S23° 47' 15.98"W
L3	30.00	N66° 12' 44.02"W
L4	72.00	N23° 47' 15.98"E
L5	32.97	N31° 57' 46.48"W
L6	34.10	N23° 47' 15.98"E
L7	34.10	S31° 57' 46.48"E
L8	20.00	S66° 12' 44.02"E

NOTES:
1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.000308263000 (Reach 9) was used.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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C. HARMON
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Steen, Timothy M
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PEER REVIEWER
DENVER, CO 2026-06-02

BBN9 TURNOUT

SITE PLAN

1695-D-62535

SHEET 8

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CAD SYSTEM
AUTODESK CIVIL 3DS (LMS TECH)
CAD FILENAME
1695-D-62535-RD-8.DWG

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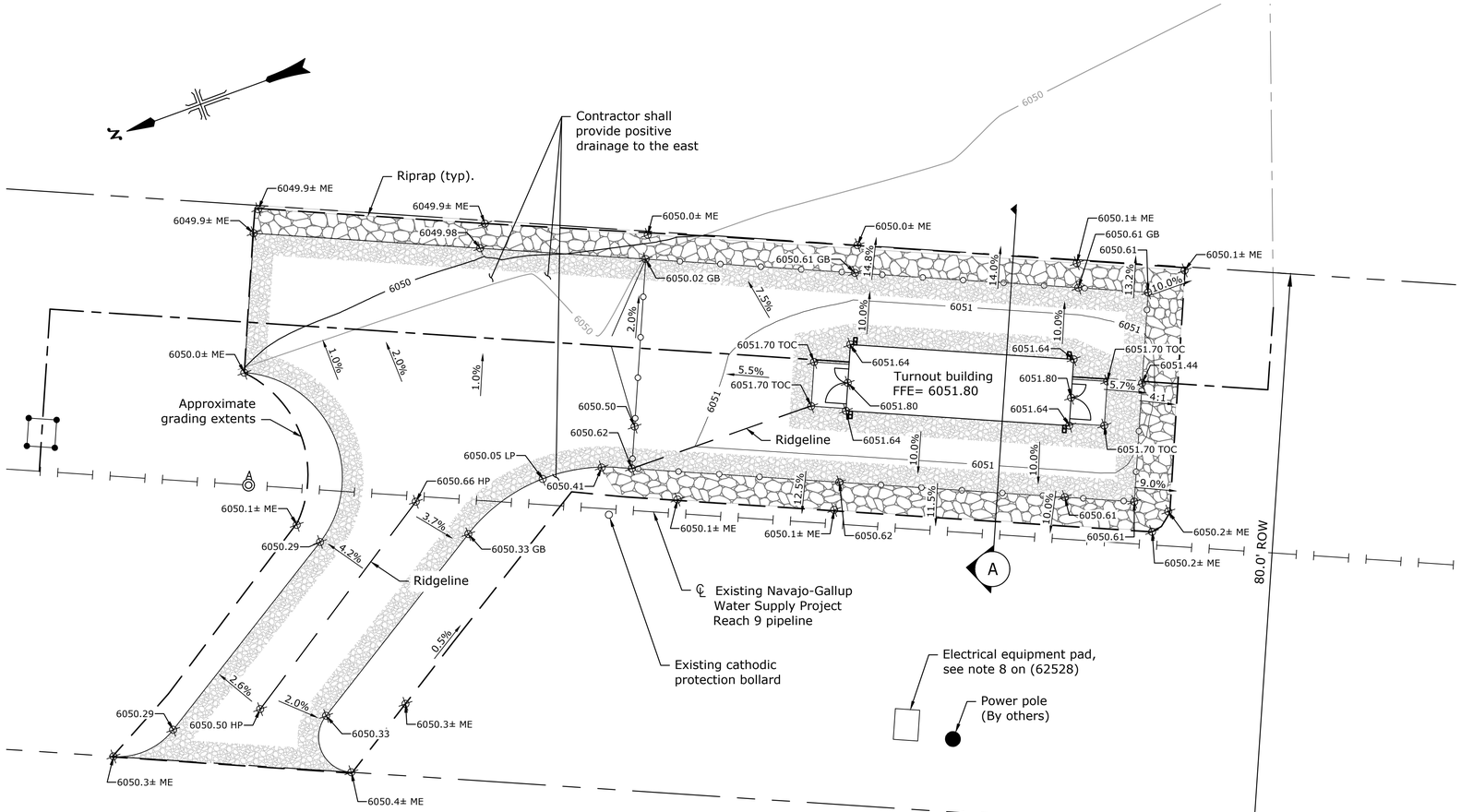
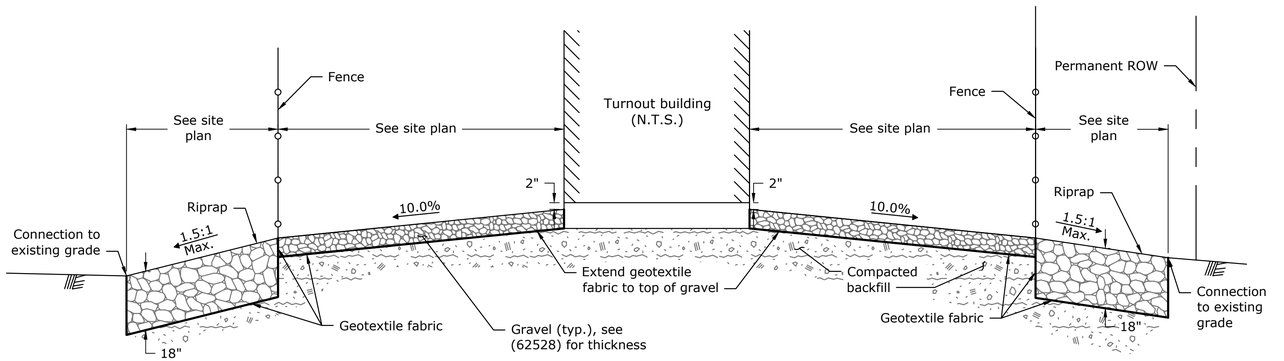


PLAN

0 10' 20'



A SECTION

NOT TO SCALE

NOTES:.

1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general sitework grading notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.000308263000 (Reach 9) was used.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
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DENVER, CO 2026-06-01

BBN9 TURNOUT

GRADING PLAN

1695-D-62536

SHEET 9

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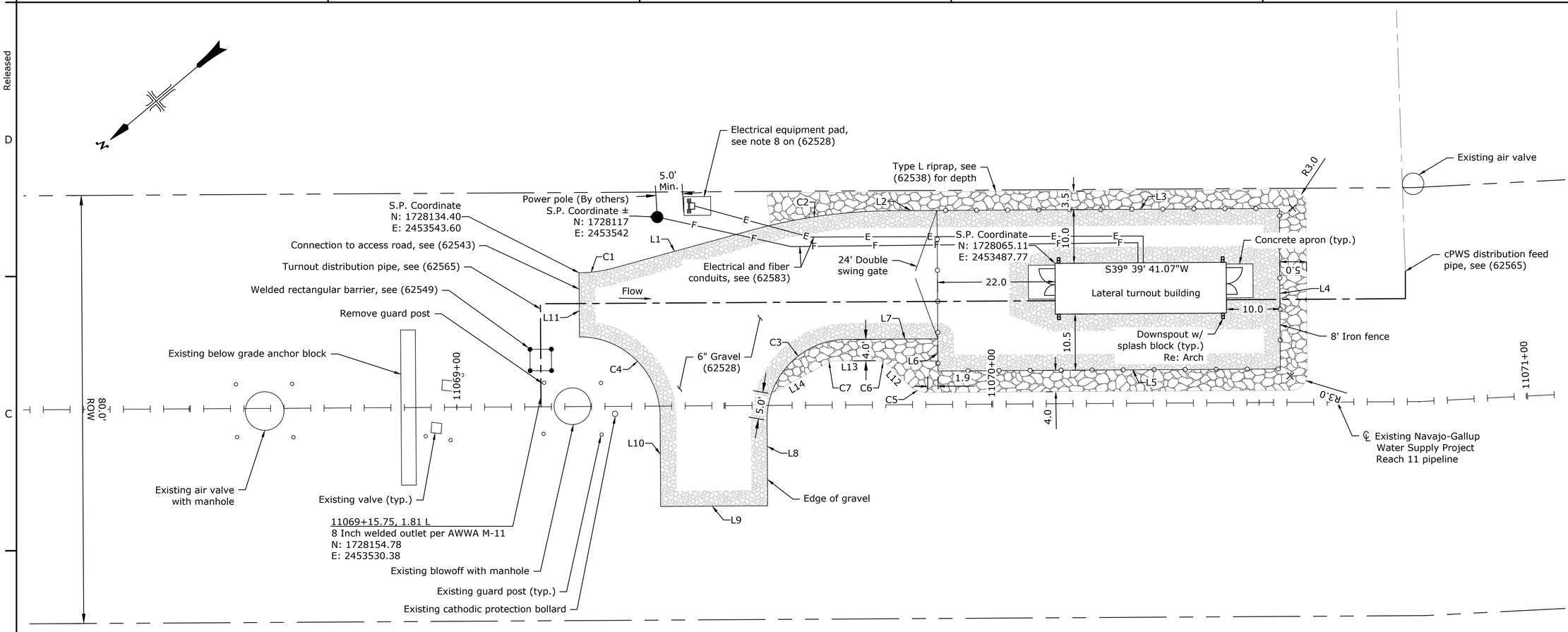
CAD SYSTEM
AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62536-RD-9.DWG



PLAN

0 10' 20'

Parcel Line Table		
Line #	Length	Direction
L1	27.01	S24° 39' 41.23"W
L2	8.56	S39° 39' 41.07"W
L3	64.00	S39° 39' 41.07"W
L4	30.00	S50° 20' 18.93"E
L5	64.00	N39° 39' 41.07"E
L6	6.00	S50° 20' 18.93"E
L7	17.00	N39° 39' 41.07"E
L8	16.00	N50° 20' 18.77"W
L9	20.00	N39° 39' 41.23"E
L10	16.75	S50° 20' 18.77"E
L11	12.00	S50° 20' 18.77"E
L12	4.34	N84° 39' 41.07"E
L13	6.67	N39° 39' 41.07"E
L14	11.41	N11° 09' 08.19"E

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	6.54	25.00	15.00	S32° 09' 41"W	6.53
C2	26.18	100.00	15.00	S32° 09' 41"W	26.11
C3	23.57	14.97	90.21	N5° 20' 19"W	21.21
C4	23.56	15.00	90.00	N84° 39' 41"E	21.21
C5	3.93	5.00	45.00	N62° 09' 41"E	3.83
C6	3.93	5.00	45.00	N62° 09' 41"E	3.83
C7	2.49	5.00	28.51	N25° 24' 25"E	2.46

NOTES:
1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00029693715 (Reach 11) was used.

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AutoCAD Civil 3D (LMS TECH)
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1695-D-62537-RD-10.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
PEER REVIEWER

DENVER, CO 2026-06-01

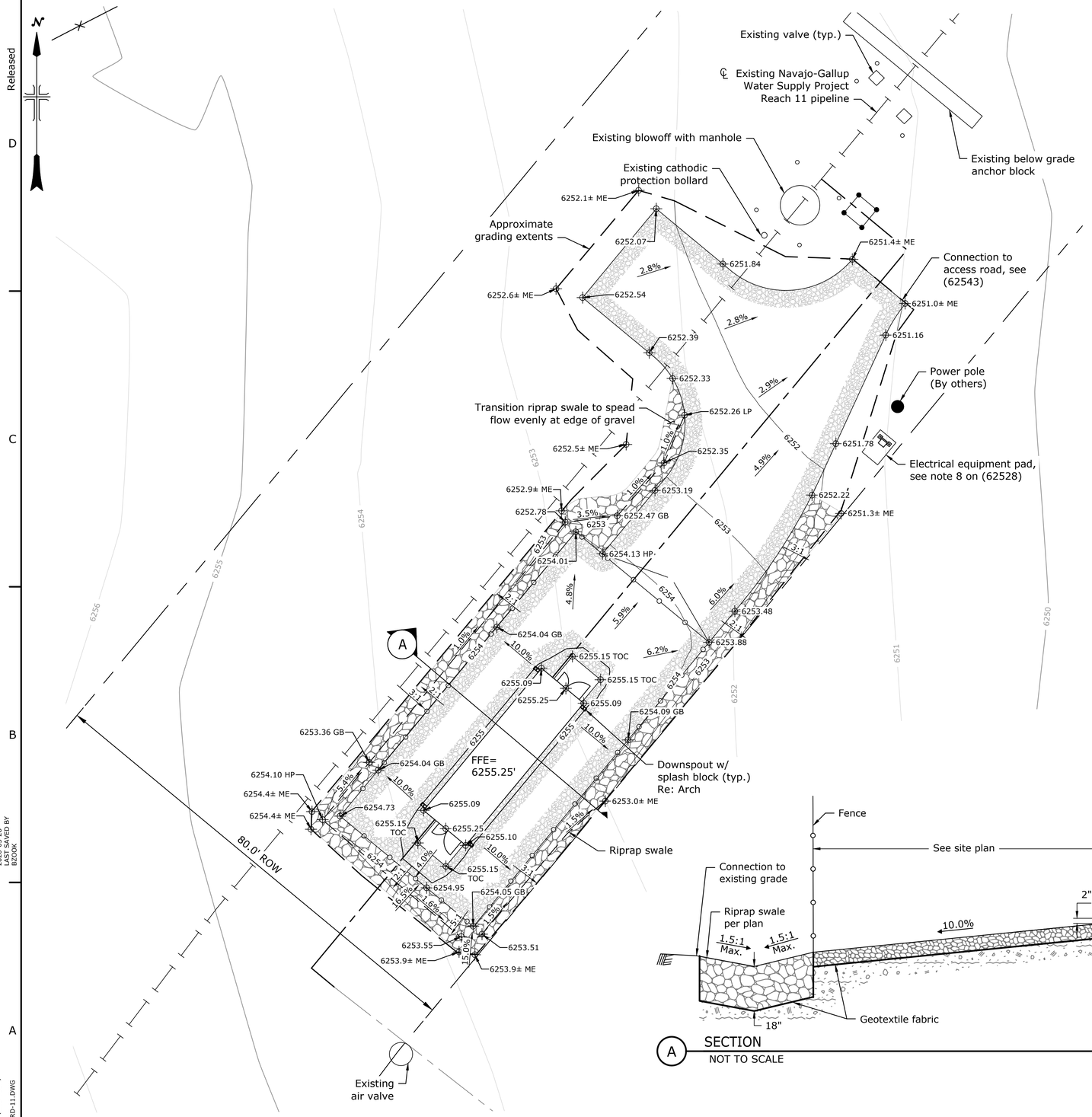
COYOTE CANYON TURNOUT

SITE PLAN

1695-D-62537

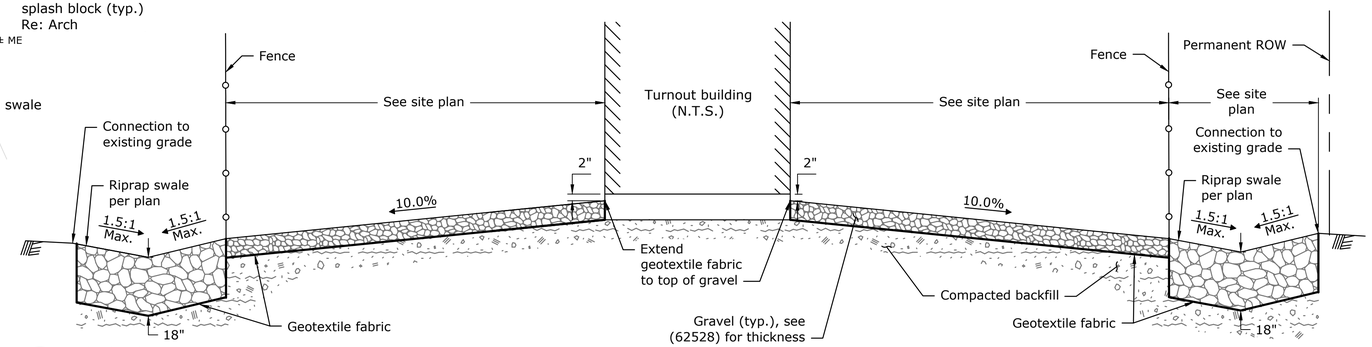
SHEET 10

Released



PLAN

0 10' 20'



A SECTION

NOT TO SCALE

NOTES:.

1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general grading notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00029693715 (Reach 11) was used.

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AUTOCAD OS (LMS TECH)
CAD FILENAME
1695-D-62538-RD-11.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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DENVER, CO 2026-06-01

COYOTE CANYON TURNOUT

GRADING PLAN

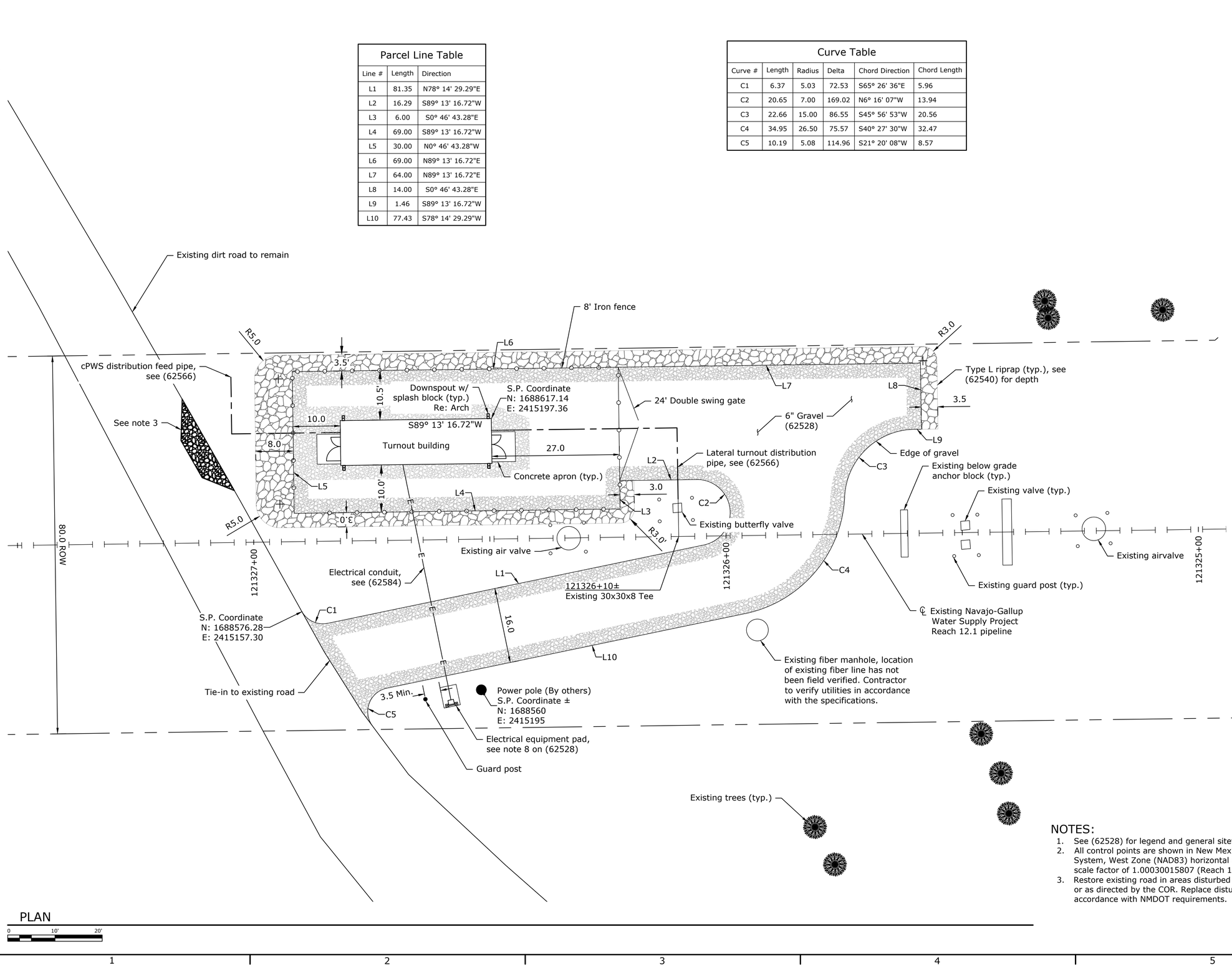
1695-D-62538

SHEET 11

Released

Parcel Line Table		
Line #	Length	Direction
L1	81.35	N78° 14' 29.29"E
L2	16.29	S89° 13' 16.72"W
L3	6.00	S0° 46' 43.28"E
L4	69.00	S89° 13' 16.72"W
L5	30.00	N0° 46' 43.28"W
L6	69.00	N89° 13' 16.72"E
L7	64.00	N89° 13' 16.72"E
L8	14.00	S0° 46' 43.28"E
L9	1.46	S89° 13' 16.72"W
L10	77.43	S78° 14' 29.29"W

Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	6.37	5.03	72.53	S65° 26' 36"E	5.96
C2	20.65	7.00	169.02	N6° 16' 07"W	13.94
C3	22.66	15.00	86.55	S45° 56' 53"W	20.56
C4	34.95	26.50	75.57	S40° 27' 30"W	32.47
C5	10.19	5.08	114.96	S21° 20' 08"W	8.57



PLAN

0 10' 20'

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00030015807 (Reach 12.1) was used.
3. Restore existing road in areas disturbed by construction activities or as directed by the COR. Replace disturbed road areas in accordance with NMDOT requirements.

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
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C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
PEER REVIEWER
DENVER, CO 2026-06-01

ROCK SPRINGS TURNOUT

SITE PLAN

1695-D-62539

SHEET 12

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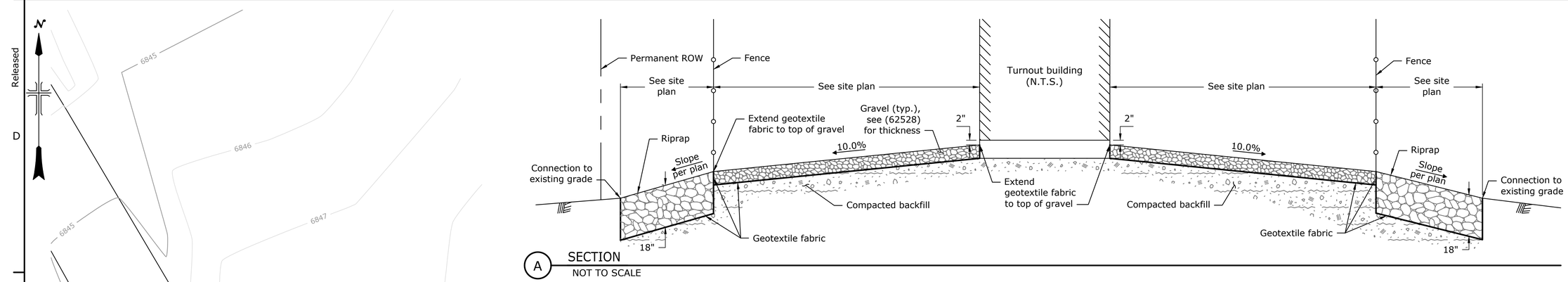
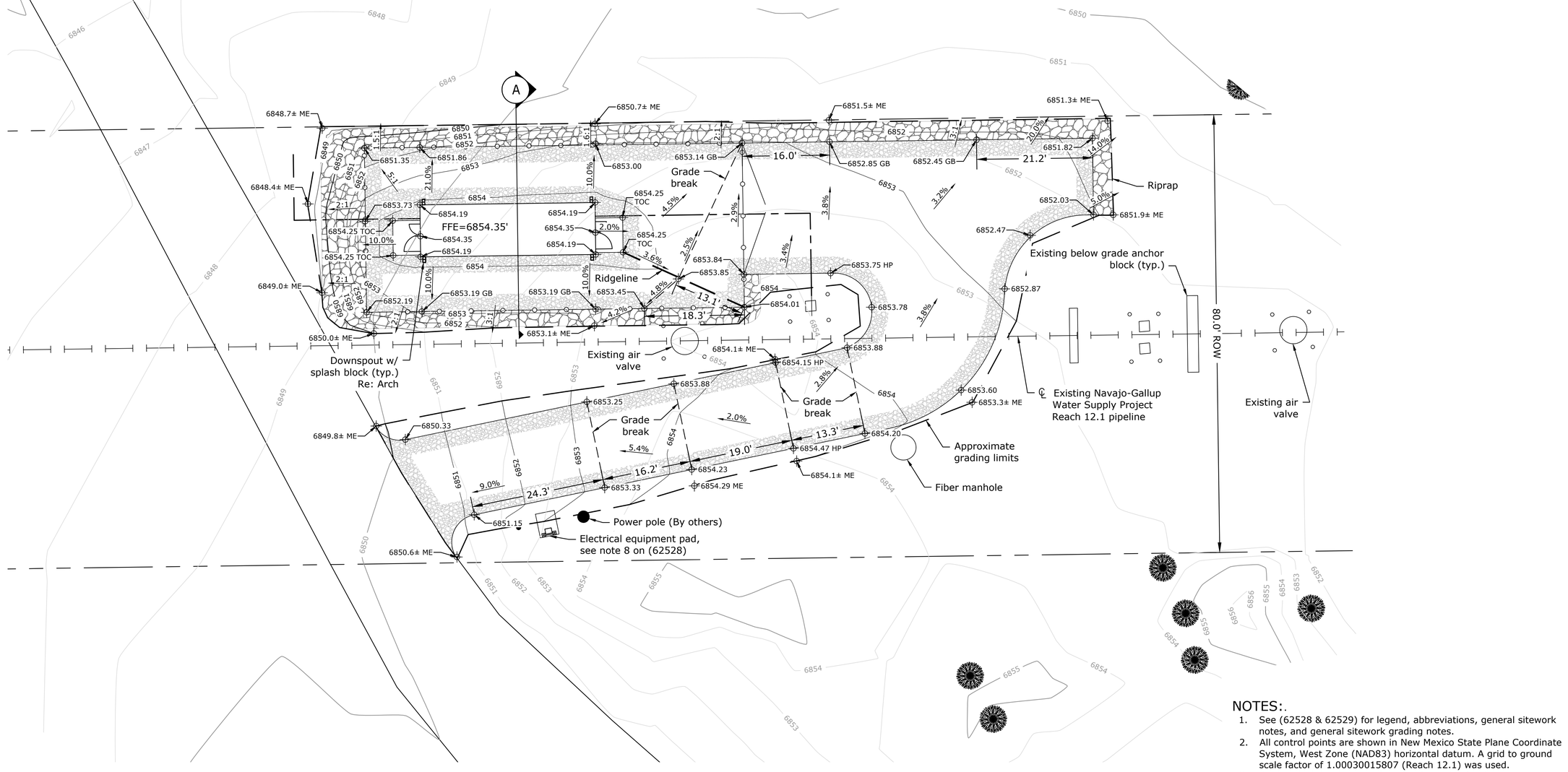
CAD SYSTEM
AUTOCAD OS (LMS TECH)
CAD FILENAME
1695-D-62539-RD-12.DWG



A SECTION
NOT TO SCALE



PLAN
0 10' 20'

NOTES:.

1. See (62528 & 62529) for legend, abbreviations, general sitework notes, and general sitework grading notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00030015807 (Reach 12.1) was used.
3. Angle splash blocks away from existing infrastructure.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
DESIGNED
L. VOGT
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
PEER REVIEWER
DENVER, CO 2026-06-01

ROCK SPRINGS TURNOUT

GRADING PLAN

1695-D-62540

SHEET 13

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CAD SYSTEM
AUTOCAD CIVIL 3D (LMS TECH)
CAD FILENAME
1695-D-62540-RD-13.DWG

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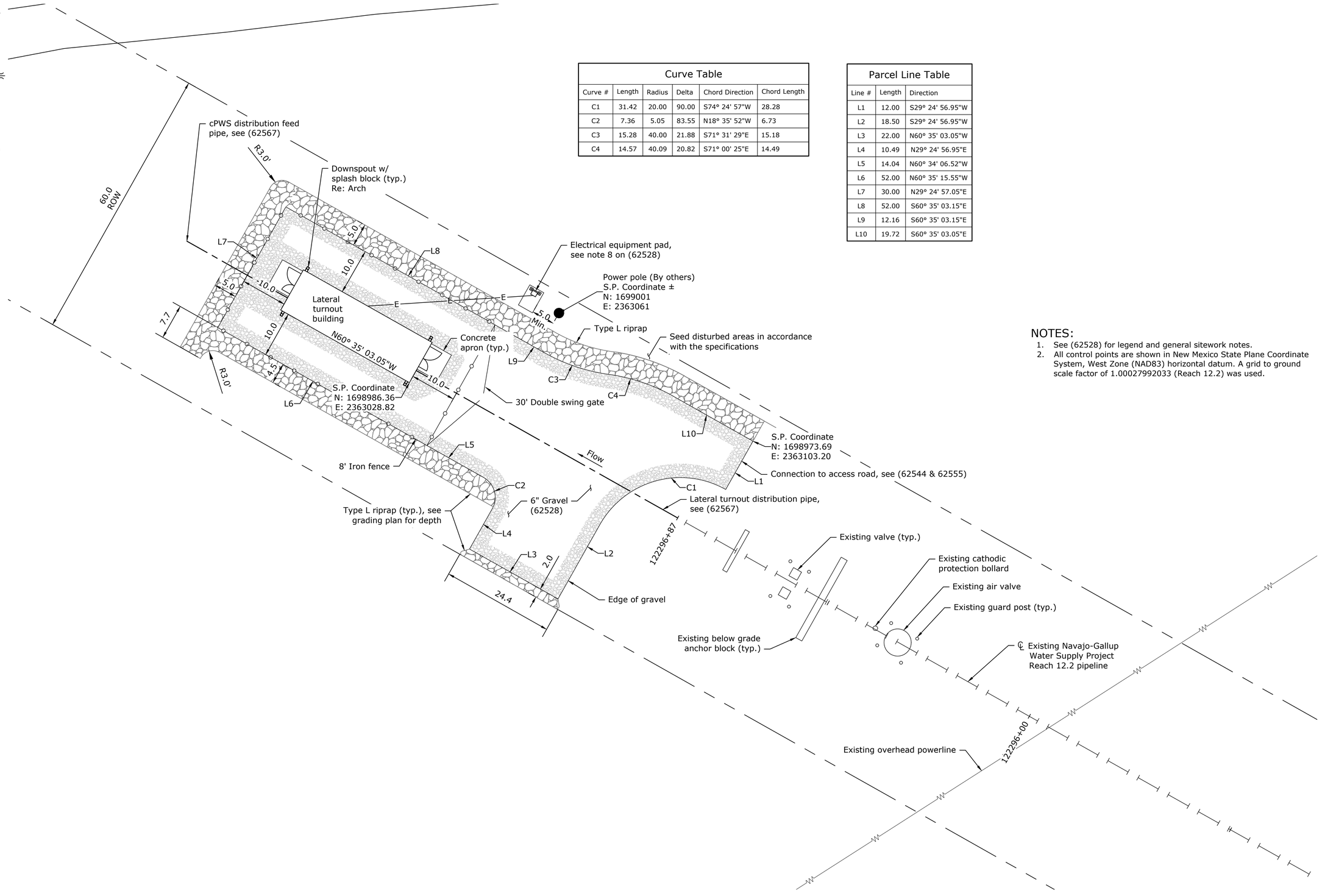
Curve Table					
Curve #	Length	Radius	Delta	Chord Direction	Chord Length
C1	31.42	20.00	90.00	S74° 24' 57"W	28.28
C2	7.36	5.05	83.55	N18° 35' 52"W	6.73
C3	15.28	40.00	21.88	S71° 31' 29"E	15.18
C4	14.57	40.09	20.82	S71° 00' 25"E	14.49

Parcel Line Table		
Line #	Length	Direction
L1	12.00	S29° 24' 56.95"W
L2	18.50	S29° 24' 56.95"W
L3	22.00	N60° 35' 03.05"W
L4	10.49	N29° 24' 56.95"E
L5	14.04	N60° 34' 06.52"W
L6	52.00	N60° 35' 15.55"W
L7	30.00	N29° 24' 57.05"E
L8	52.00	S60° 35' 03.15"E
L9	12.16	S60° 35' 03.15"E
L10	19.72	S60° 35' 03.05"E



PLAN

0 10' 20'

NOTES:

1. See (62528) for legend and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027992033 (Reach 12.2) was used.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
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DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
PEER REVIEWER

DENVER, CO 2026-06-01

WINDOW ROCK TURNOUT

SITE PLAN

1695-D-62541

SHEET 14

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AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-62541-RD-14.DWG

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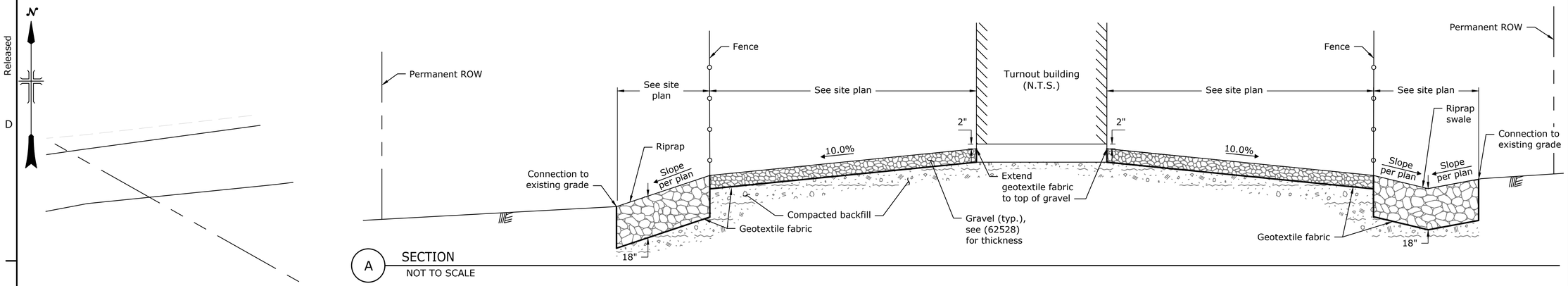
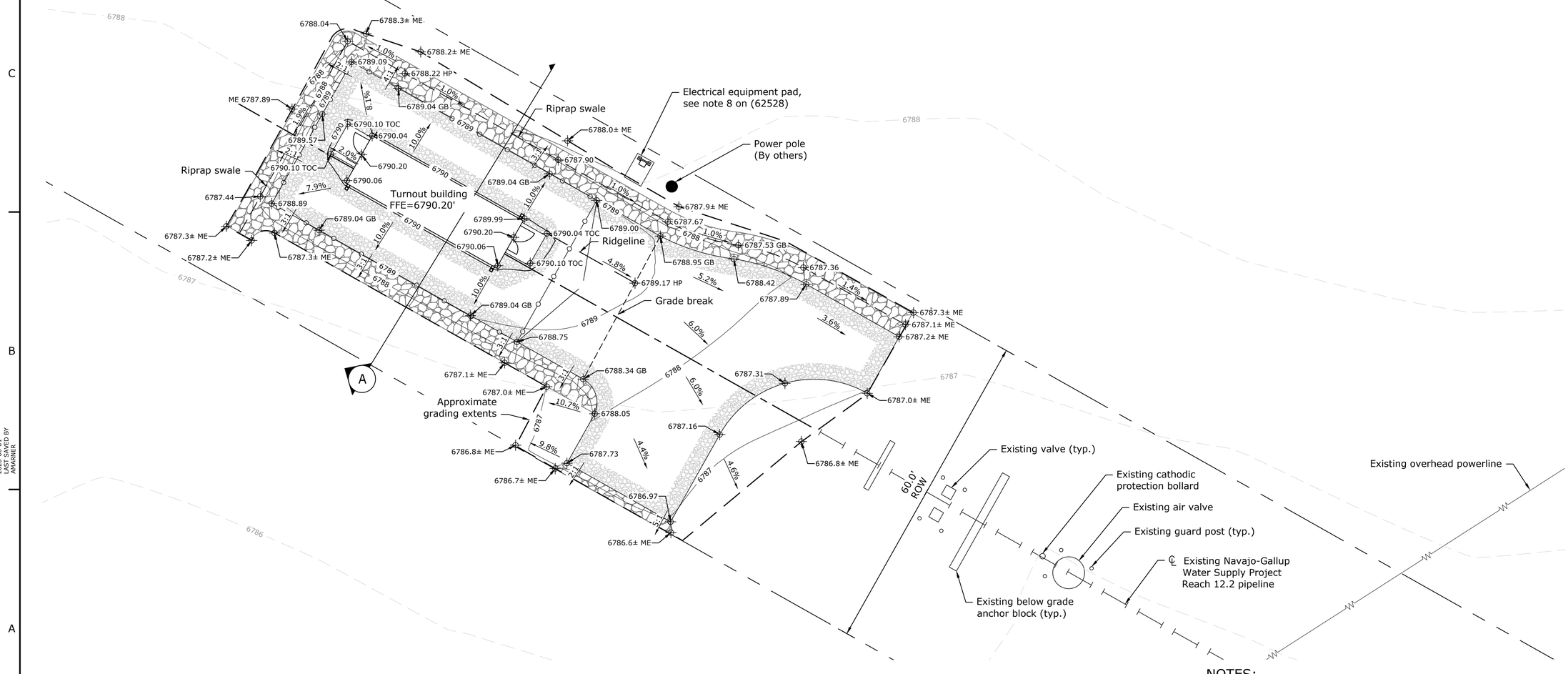


A SECTION
NOT TO SCALE



PLAN
0 10' 20'

NOTES:.
1. See (62528 & 62529) for legend, abbreviations, and general sitework notes.
2. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00027992033 (Reach 12.2) was used.

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MICROSTATION V8i (LMS TECH)
CAD FILENAME
1695-D-62542-RD-15.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

B. ZOOK
DESIGNED
L. VOGT
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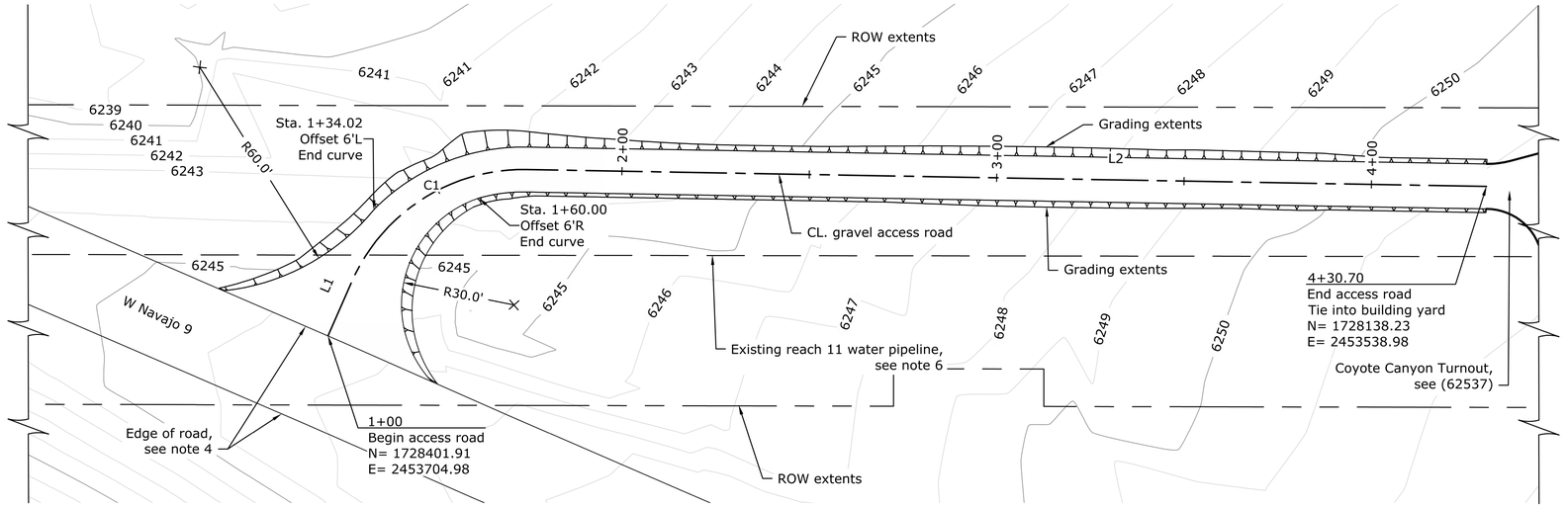
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
PEER REVIEWER
DENVER, CO 2026-06-01

WINDOW ROCK TURNOUT

GRADING PLAN

1695-D-62542

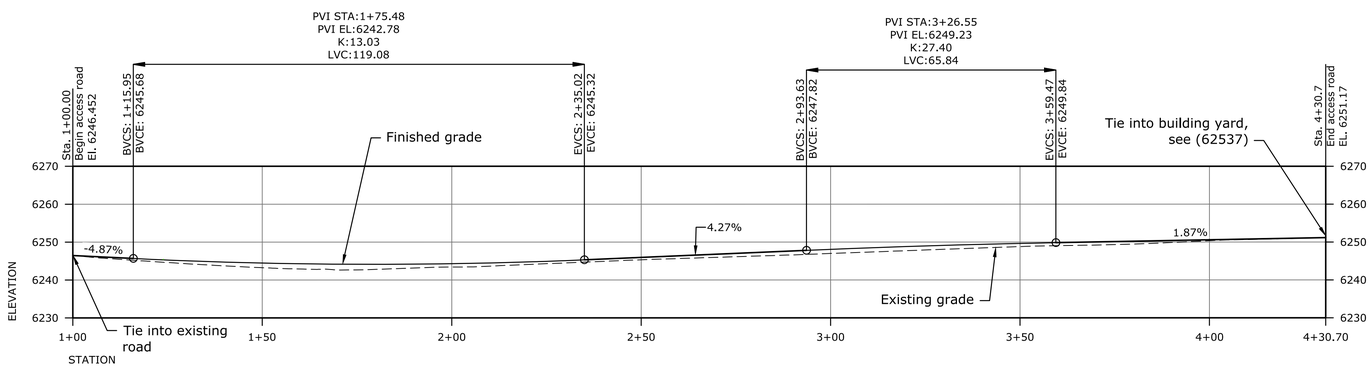
SHEET 15



PLAN

0 10' 20' 40'

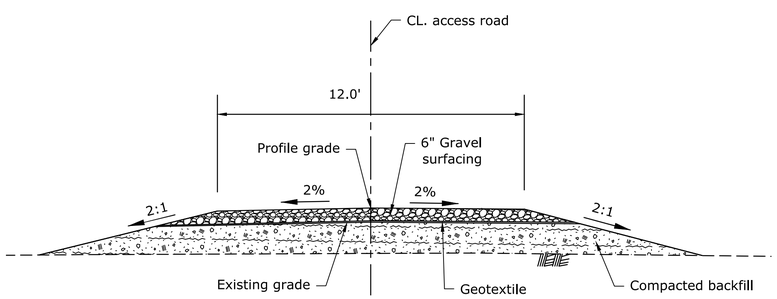
SCALE: 1" = 20'



PROFILE

0 10' 20' 40'

SCALE: 1" = 20'



TYPICAL CROSS SECTION

NOT TO SCALE

ALIGNMENT LINE DATA

Line #	START STA	END STA	LENGTH	BEARING
L1	1+00.00	1+15.84	15.84	S26° 45' 40.95"E
L2	1+74.59	4+30.70	256.11	S40° 33' 22.53"W

ALIGNMENT CURVE DATA

CURVE #	BC STA	PI STA	EC STA	LENGTH	RADIUS	DELTA	TANGENT	CHORD
C1	1+15.84	1+49.14	1+74.59	58.75	50.00	67.3176	33.29	55.42

NOTES

1. The design speed of the access road is 25 mph. The design vehicle is a passenger pickup truck.
2. Topography shown includes two sets of data. One prior to the pipe installation and the second post pipe installation construction completion.
3. Field verify the location and connection point from the access road to the existing road. If any major discrepancies exist, access road tie-in is to be constructed as directed by COR.
4. Boundaries shown for W. Navajo 9 are approximate.
5. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00029693715 (Reach 11) was used.
6. For Reach 11 pipeline crossing locations and elevation data, see 1695-D-60309.

ABBREVIATIONS

BVCE	Begin vertical curve elevation
BVCS	Begin vertical curve station
EVCE	End vertical curve elevation
EVCS	End vertical curve station
K	Rate of vertical curve curvature
LVC	Length of vertical curve
PI	Point of intersection
PVI	Point of vertical intersection
Sta.	Station
ROW	Right of way

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

SEAN MILLER
DESIGNED
SEAN MILLER
DRAWN
K HEWETT, P.E.
CHECKED
KYRA HEWETT Digitally signed by KYRA HEWETT Date: 2026.06.17 15:29:42 -06'00'
TECH. APP.
GABRIEL LEDEZMA Digitally signed by GABRIEL LEDEZMA Date: 2026.06.17 15:40:13 -06'00'
APPROVED
PEER REVIEWER -
DENVER, CO 2026-06-17

COYOTE CANYON
ACCESS ROAD

PLAN AND PROFILE

1695-D-62543

LAST SAVED DATE 2026-06-17
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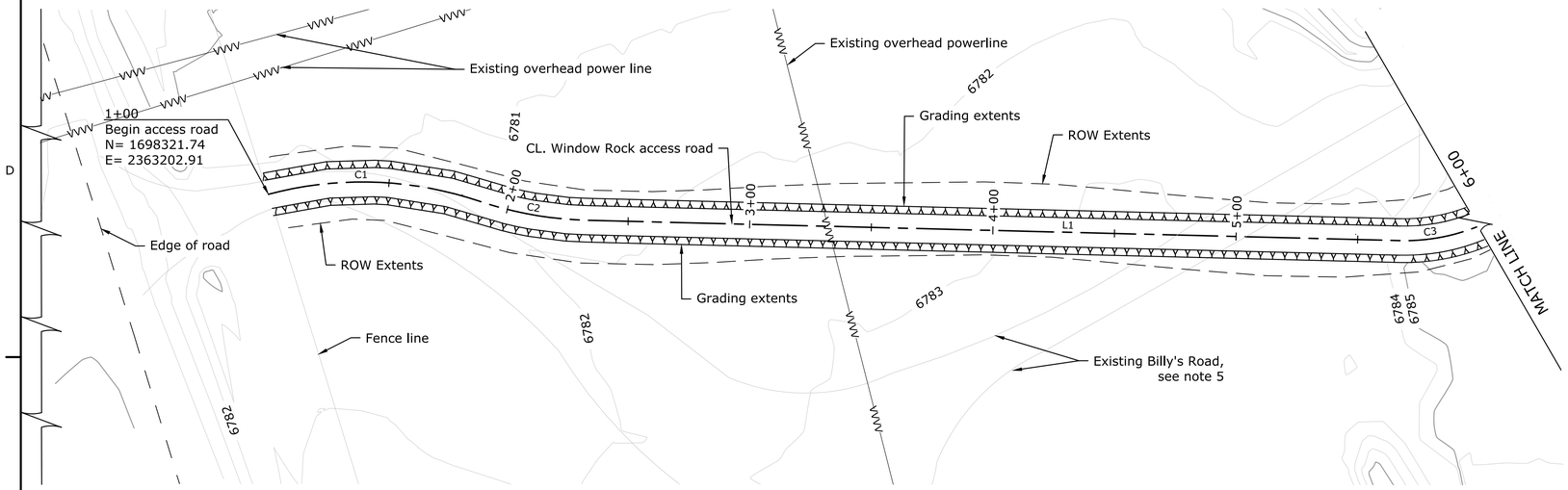
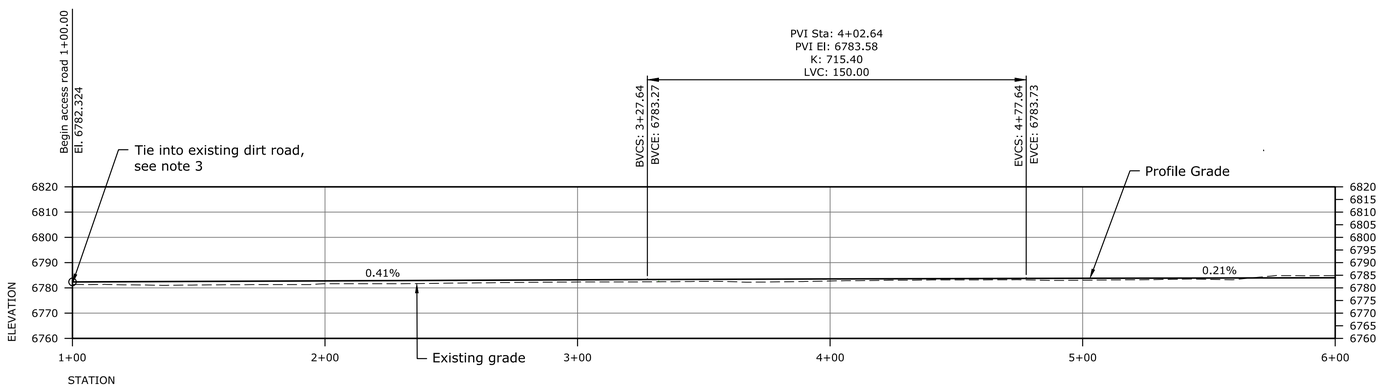
CAD SYSTEM AutoCAD Civil 3D
CAD FILENAME 1695-D-62543-RD-1.DWG



PLAN

0 30' 60'

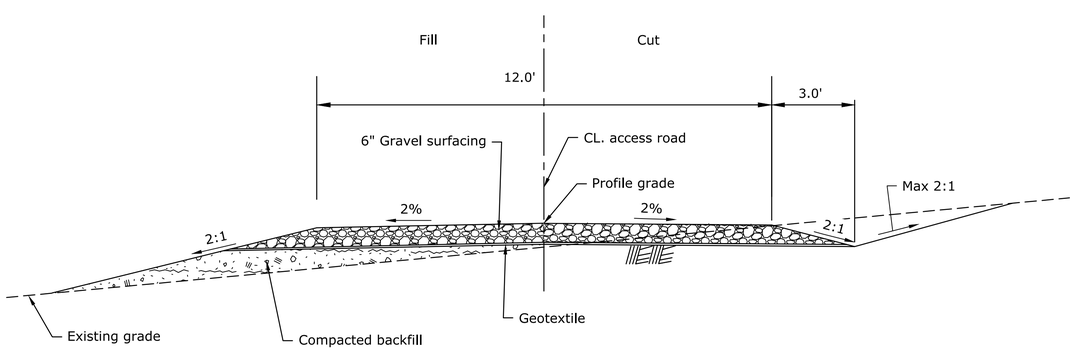
SCALE: 1" = 30'



PROFILE

0 30' 60'

SCALE: 1" = 30'



TYPICAL CROSS SECTION

NOT TO SCALE

ALIGNMENT LINE DATA				
Line #	START STA	END STA	LENGTH (ft)	Bearing
L1	2+33.02	5+67.40	334.37	N61° 57' 31.71"E

ALIGNMENT CURVE DATA								
Curve #	BC STA	PI STA	EC STA	LENGTH	RADIUS	DELTA	TANGENT	CHORD
C1	1+00.00	1+45.26	1+87.91	87.91	150.00	33.5800	45.26	86.66
C2	1+87.91	2+10.64	2+33.02	45.11	150.00	17.2310	22.73	44.94
C3	5+67.40	7+12.88	7+38.28	170.88	80.00	122.3864	145.48	140.20

ABBREVIATIONS

BVCE	Begin vertical curve elevation
BVCS	Begin vertical curve station
EVCE	End vertical curve elevation
EVCS	End vertical curve station
K	Rate of vertical curve curvature
LVC	Length of vertical curve
PVI	Point of vertical intersection
Sta.	Station

NOTES

1. The design speed of the access road is 25 mph. The design vehicle is a passenger pickup truck.
2. Topography shown includes two sets of data. One prior to the pipe installation and the second post pipe installation construction completion.
3. Field verify the location and connection point from the access road to the existing road. If any major discrepancies exist, access road tie-in is to be constructed as directed by COR.
4. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00029693715 (Reach 11) was used.
5. Contractor shall field verify the boundaries and location of Billy's Road. Billy's Road intersection design is approximate and is to be constructed as directed by COR. The portions of Billy's Road inside of ROW to be abandoned in place.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

SEAN MILLER
DESIGNED
SEAN MILLER
DRAWN
K HEWETT, P.E.
CHECKED KYRA HEWETT Digitally signed by KYRA HEWETT Date: 2026.06.17 15:10:54 -06'00'
TECH. APPR. GABRIEL LEDEZMA Digitally signed by GABRIEL LEDEZMA Date: 2026.06.17 15:42:29 -06'00'
APPROVED *PEER REVIEWER* -
DENVER, CO 2026-06-17

WINDOW ROCK ACCESS ROAD

PLAN AND PROFILE

1695-D-62544

SHEET 1 OF 2

LAST SAVED DATE
2026-06-17
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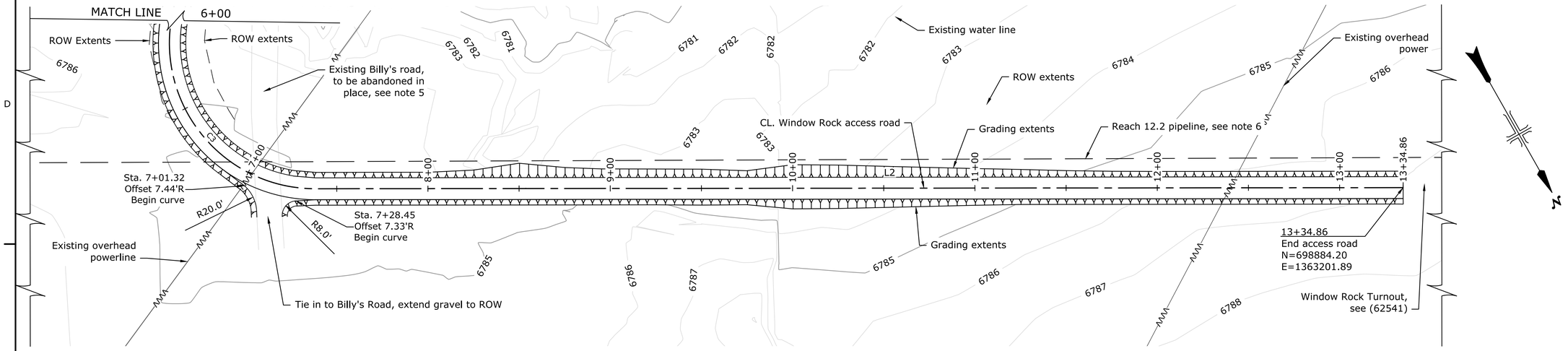
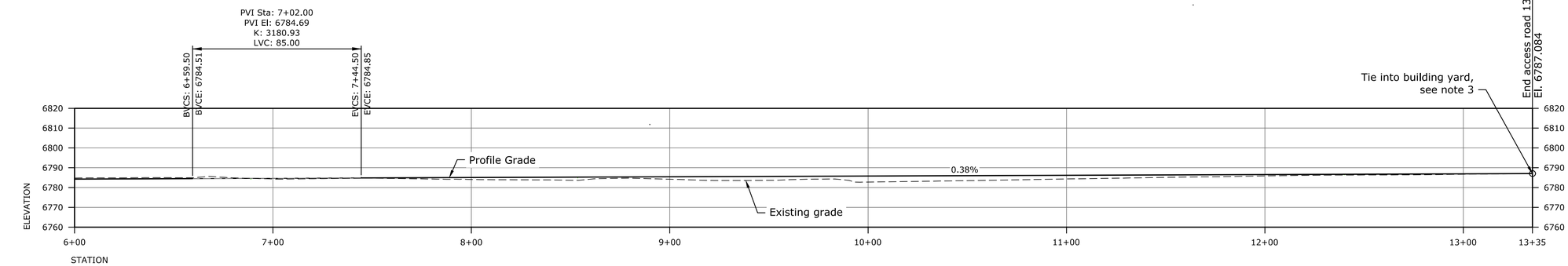
CAD SYSTEM
AutoCAD Civil 3D
CAD FILENAME
1695-D-62544-RD-1.DWG



PLAN

0 30' 60'

SCALE: 1" = 30'



PROFILE

0 30' 60'

SCALE: 1" = 30'

ALIGNMENT LINE DATA

LINE #	START STA	END STA	LENGTH (FT)	BEARING
L2	7+38.28	13+34.86	596.58	N60° 25' 39"W

ALIGNMENT CURVE DATA

Curve #	BC STA	PI STA	EC STA	LENGTH	RADIUS	DELTA	TANGENT	CHORD
C3	5+67.40	7+12.88	7+38.28	170.88	80.00	122.3864	145.48	140.20

NOTES

1. The design speed of the access road is 25 mph. The design vehicle is a passenger pickup truck.
2. Topography shown includes two sets of data. One prior to the pipe installation and the second post pipe installation construction completion.
3. Field verify the location and connection point from the access road to the existing road. If any major discrepancies exist, access road tie-in is to be constructed as directed by COR.
4. All control points are shown in New Mexico State Plane Coordinate System, West Zone (NAD83) horizontal datum. A grid to ground scale factor of 1.00029693715 (Reach 11) was used.
5. Contractor shall field verify the boundaries and location of Billy's road. Billy's Road intersection design is approximate and is to be constructed as directed by COR. The portions of Billy's road inside of ROW to be abandoned in place.
6. For Reach 12.2 pipeline crossing locations and elevation data, see 1695-D-60260.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DESIGNED	SEAN MILLER
DRAWN	SEAN MILLER
CHECKED	K HEWETT, P.E.
	KYRA HEWETT Digitally signed by KYRA HEWETT Date: 2020.06.17 16:05:29 -06'00'
TECH. APPR.	GABRIEL LEDEZMA Digitally signed by GABRIEL LEDEZMA Date: 2020.06.17 16:02:11 -06'00'
APPROVED	-
PEER REVIEWER	
DENVER, CO	1899-12-30

WINDOW ROCK ACCESS ROAD

PLAN AND PROFILE

1695-D-62545

SHEET 2 OF 2

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AutoCAD Civil 3D
CAD FILENAME
1695-D-62545-RD-1.DWG

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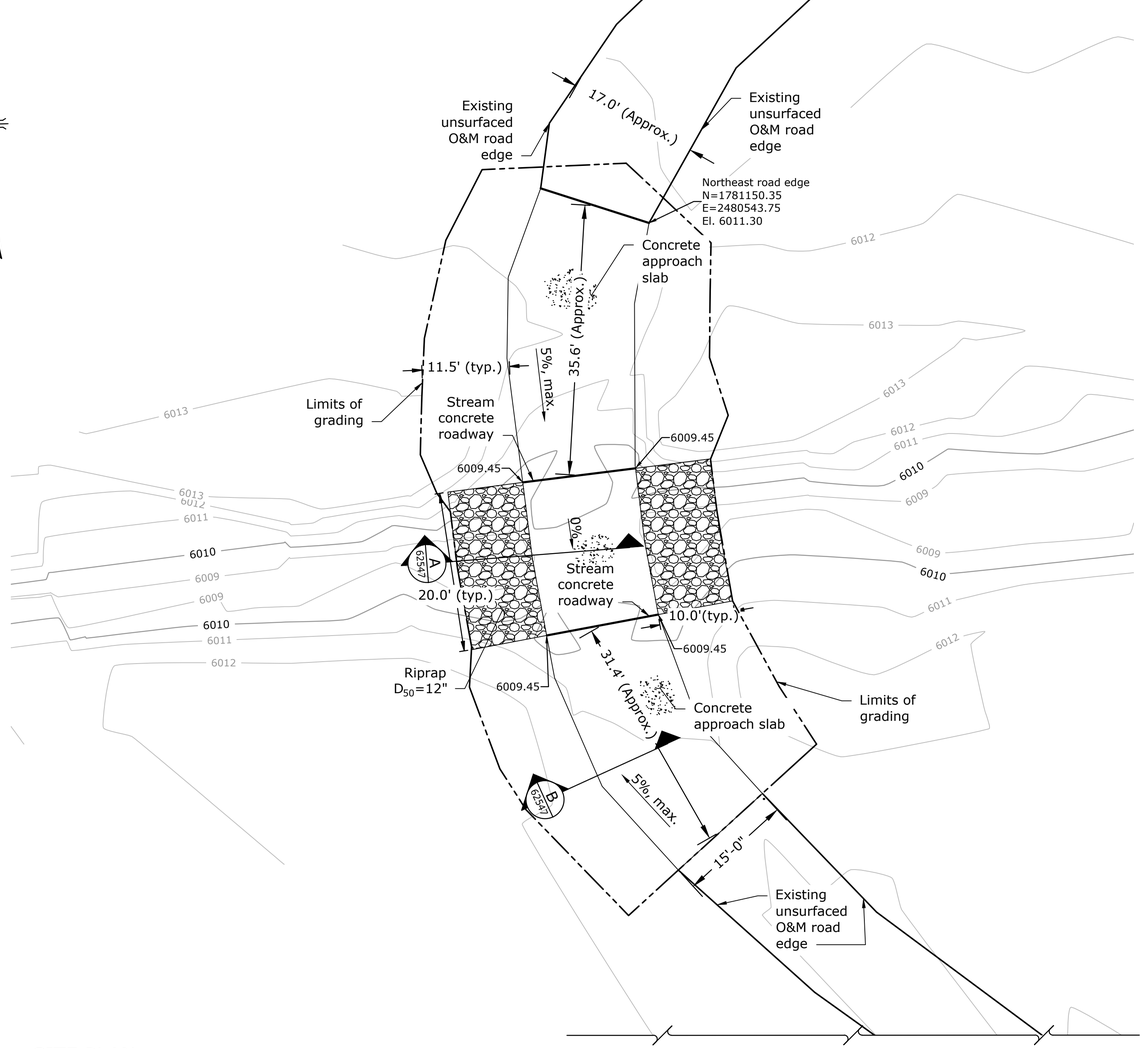
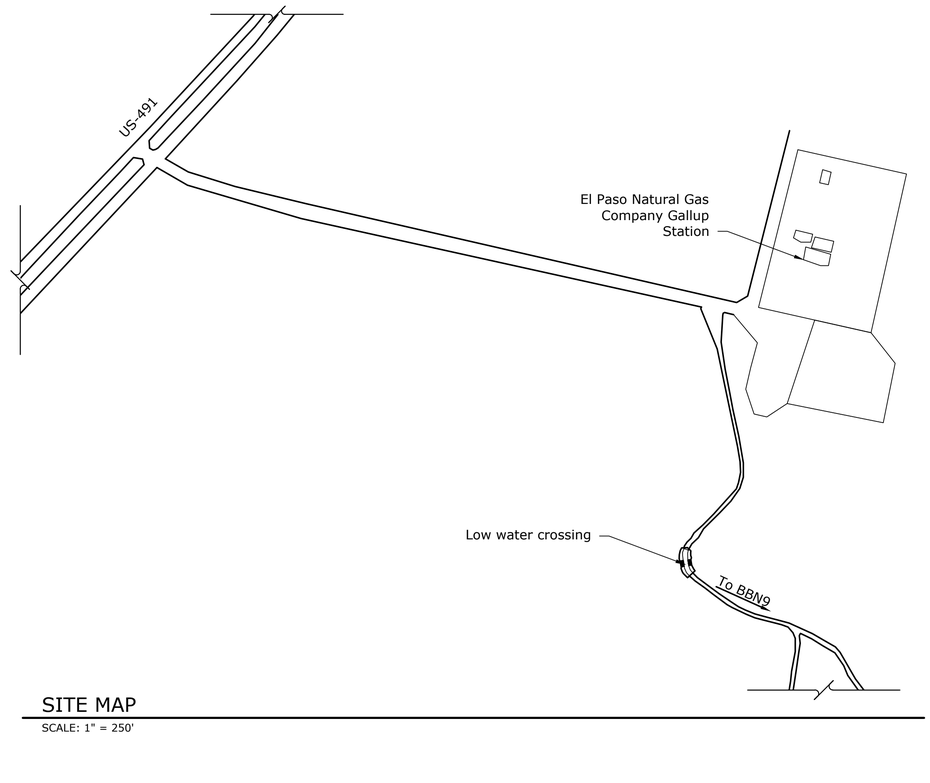


SITE PLAN

SCALE: 1" = 10'



SITE MAP

SCALE: 1" = 250'

NOTES

1. Field verify tie-in location and elevation of low water crossing road into existing O&M road.
2. Low water crossing roadway to follow centerline of existing O&M road.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DESIGNED	DOUGLAS RICHARDS
DRAWN	DOUGLAS RICHARDS
CHECKED	CASSIE HARMON
TECH. APPR.	Richards, Douglas A P.E.
APPROVED / PEER REVIEWER	Steen, Timothy M — Supervisory Civil Engineer
DENVER, CO	2026-05-29

BBN9 TURNOUT

SITE PLAN LOW WATER CROSSING

1695-D-62546

SHEET 1

LAST SAVED DATE
2026-05-29
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DARICHARDS

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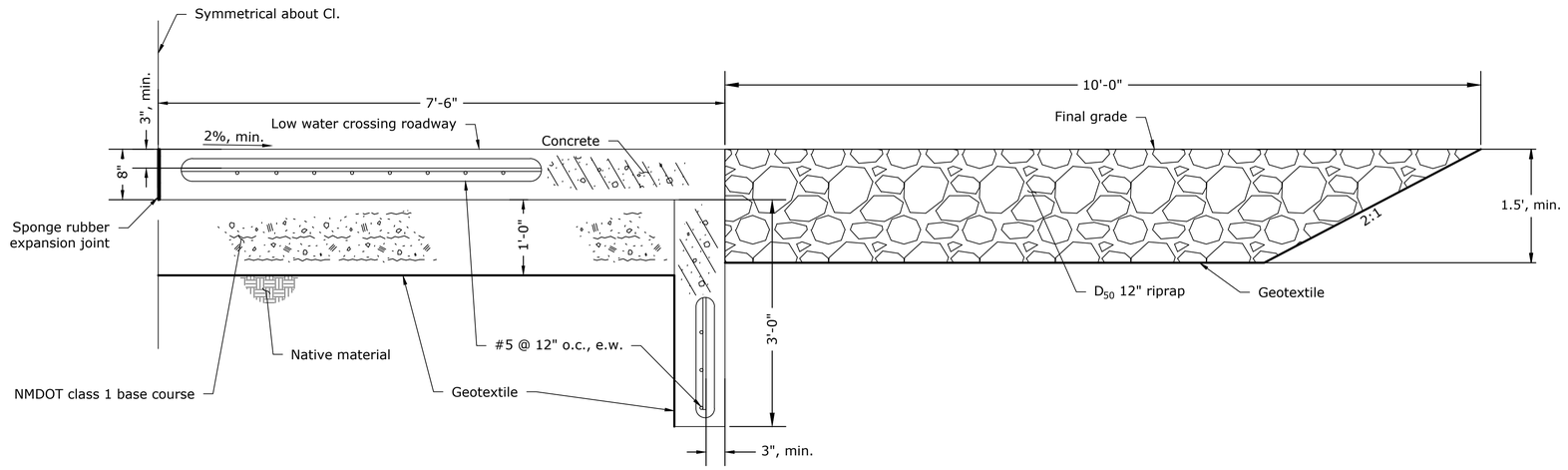
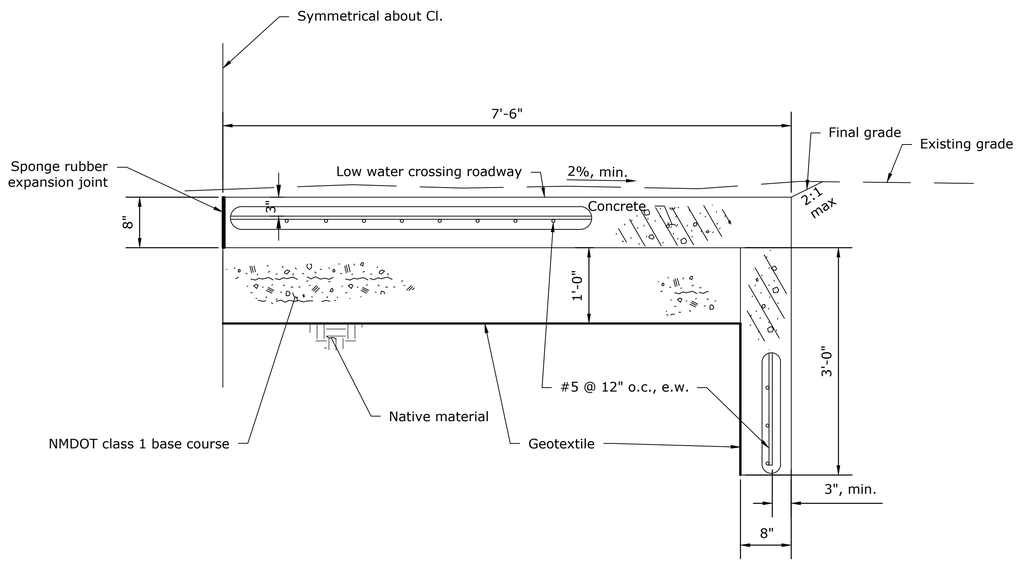
AutoCAD Civil 3DS (LMS TECH)
CAD FILENAME
1695-D-62546-RD-1.DWG



A TYPICAL SECTION - STREAM CHANNEL CROSSING

NOT TO SCALE



B TYPICAL SECTION - LOW WATER CROSSING ROADWAY

NOT TO SCALE

NOTES

1. Contraction joint placed every 15'-0" transverse to centerline.
2. See 40-D-5247 for details on joints in concrete structures.
3. See 40-D-6003 & 40-D-6004 for general notes and minimum requirements for reinforcement.
4. Geotextile placed below riprap for stabilization and drainage. Max drop height of 1 ft.
5. Geotextile placed on native material before placement of base course or concrete formwork.
6. Install and anchor geotextile per manufacturer requirements.
7. Sloping, shoring, and benching shall be in accordance with Specifications.
8. Welded wire reinforcement may be used in lieu of reinforcement bar, with approval of COR.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DOUGLAS RICHARDS
DESIGNED
DOUGLAS RICHARDS
DRAWN
CASSIE HARMON
CHECKED
Richards, Douglas A P.E.
TECH. APPR.
Steen, Timothy M
APPROVED Supervisory Civil Engineer
PEER REVIEWER
DENVER, CO 2026-03-20

BBN9 TURNOUT

LOW WATER CROSSING DETAIL AND SECTIONS

1695-D-62547

SHEET 2

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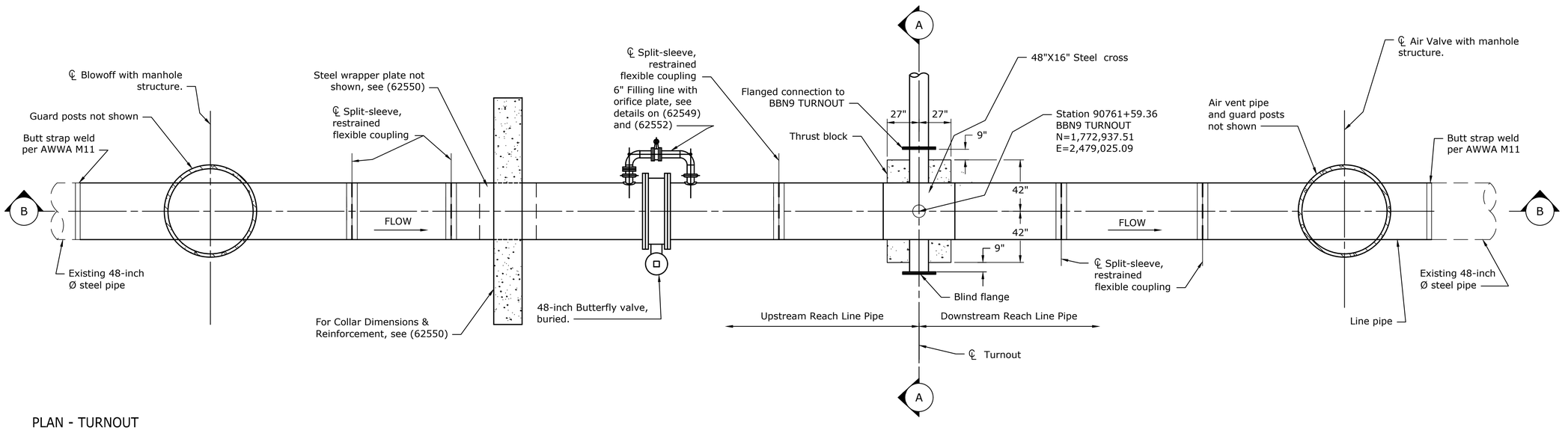
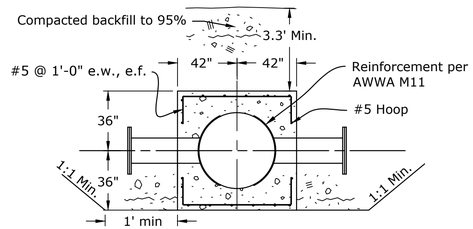
CAD FILENAME
1695-D-62547-RD-1.DWG



PLAN - TURNOUT

NOT TO SCALE - See (62535) for location



A SECTION

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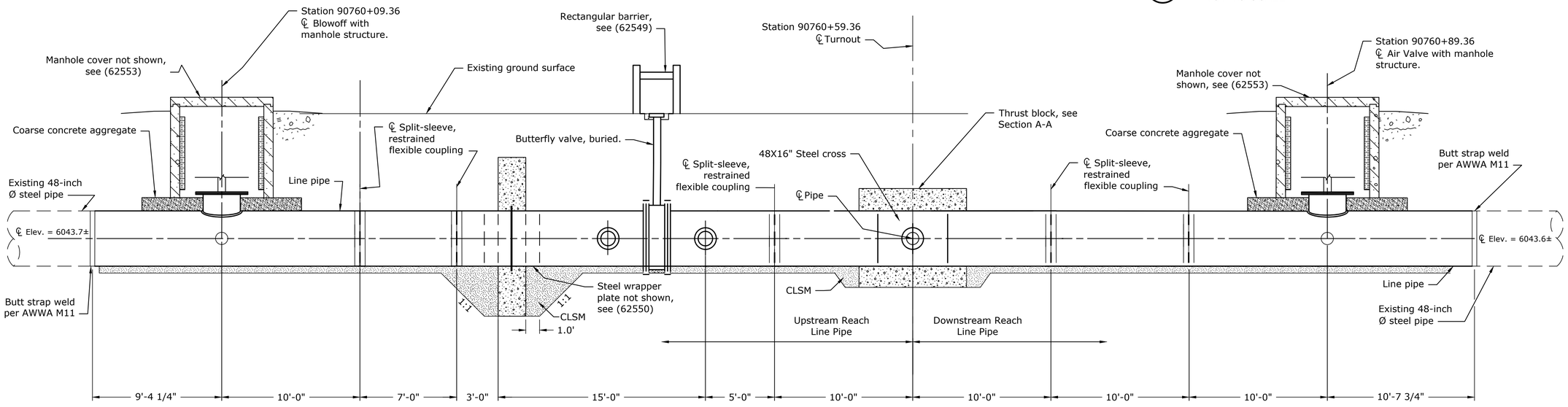

B SECTION

NOT TO SCALE - See (62535) for location

LAST SAVED DATE 2026-06-02
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CAD SYSTEM AutoCAD Civil 3D (LMS TECH)
CAD FILENAME 1695-D-62548-RD-1.DWG

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

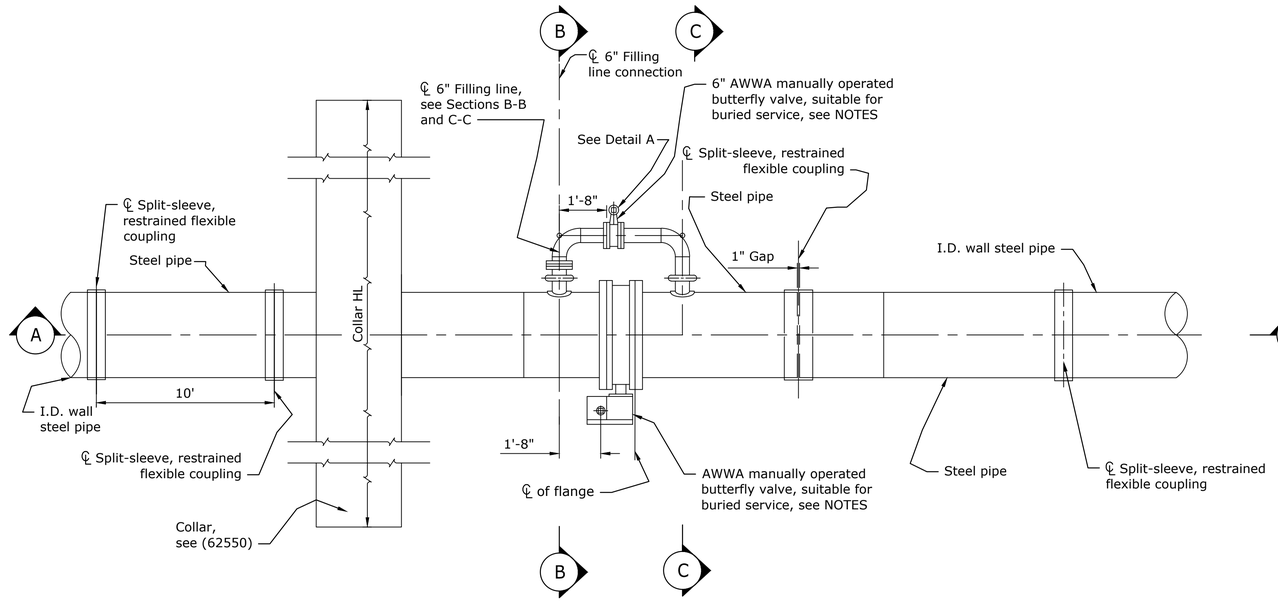
A. MARNER
DESIGNED
A. MARNER
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Pelzer, Kylie C P.E.
APPROVED Civil Engineer
PEER REVIEWER
DENVER, CO 2026-06-02

BBN9 LINE PIPE TURNOUT

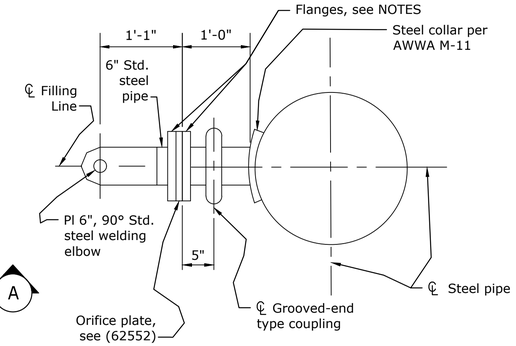
PLAN, SECTIONS AND DETAILS

1695-D-62548

SHEET 1



PLAN



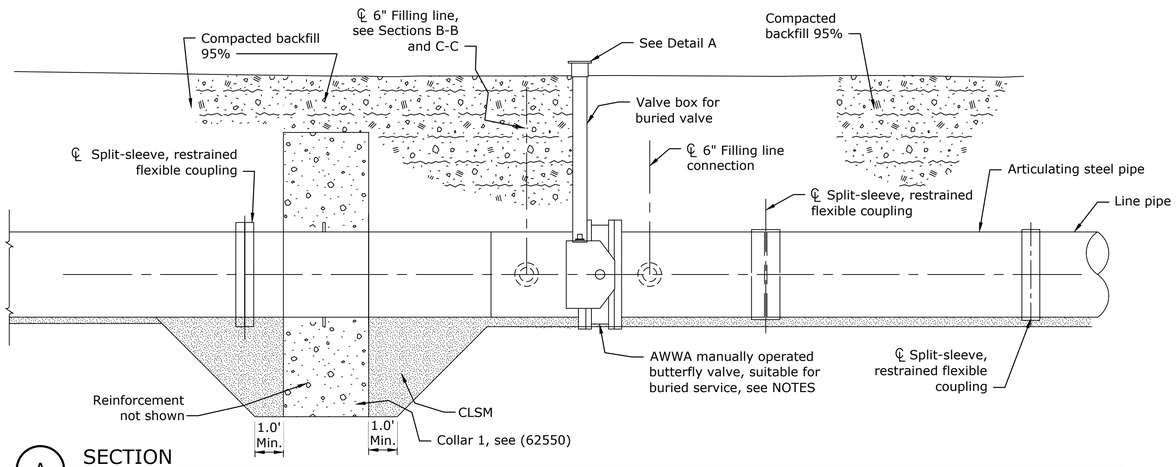
B SECTION

OPERATING CRITERIA

1. High performance Bypass butterfly valve shall be fully open when operated.
2. Excavation around blocking shall not occur when the pipeline is under hydrostatic pressure.

NOTES

1. Drawing not to scale.
2. See Specifications on exact label text required for each valve.
3. All sectionalizing valves will require a bypass.
4. For pipe trench earthwork details, see (62554).
5. The minimum steel cylinder thickness for the section of line pipe with the valve outlet shall be the thickness of the steel line pipe. For the steel line pipe thickness see the specification.
6. Piping next to butterfly valves may require 1" oversizing so that the applied interior lining will not reduce the inside dimension to a smaller size which could prevent the butterfly valve from opening. Valve bypass piping shall be epoxy lined.
7. For collar details, see (62550).
8. Flange class shall meet pressure requirement designated by adjacent line pipe. Flange drilling shall match mating flange.
9. For Flange and Valve ratings, see notes (62551).
10. Dissimilar metals require a dielectric coupling for electrical isolation.



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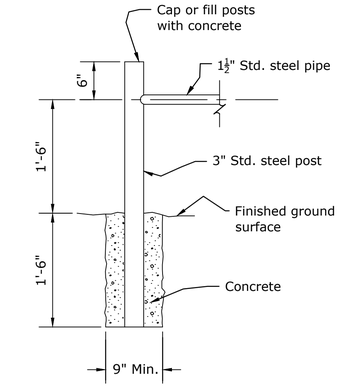

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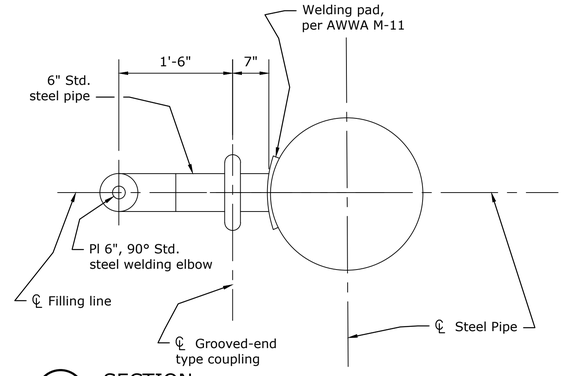
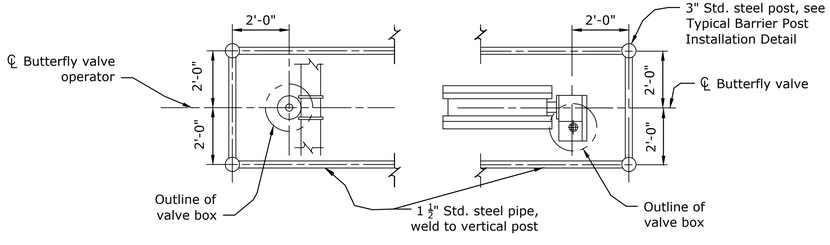
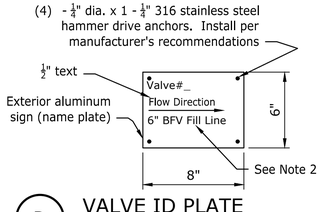
INSTALLATION DETAIL



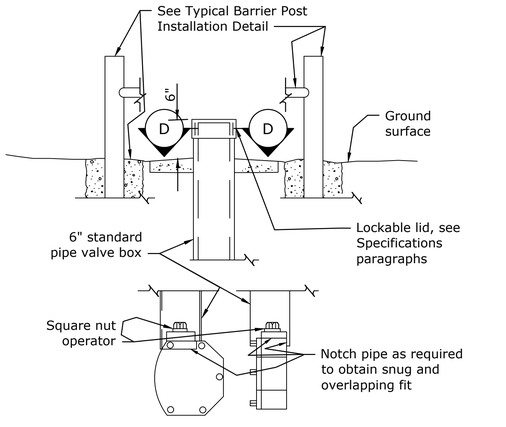
C SECTION



WELDED RECTANGULAR BARRIER



D VALVE ID PLATE



A DETAIL
(PROFILE VIEW)

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

A. MARNER	DESIGNED
A. MARNER	DRAWN
C. HARMON	CHECKED
Marner, Andrew (Andy) J P.E.	TECH. APPR.
Pelzer, Kylie C P.E.	APPROVED Civil Engineer PEER REVIEWER
DENVER, CO	2026-06-02

BBN9 LINE PIPE - SECTIONALIZING VALVE INSTALLATION

PLAN, SECTIONS AND DETAILS

1695-D-62549

SHEET 2

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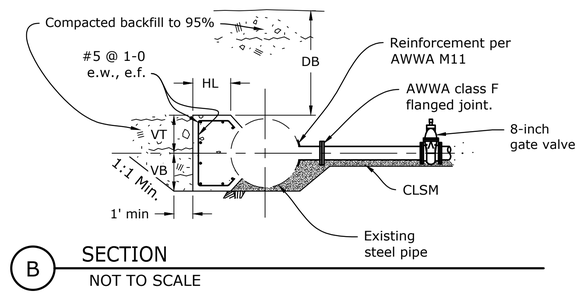
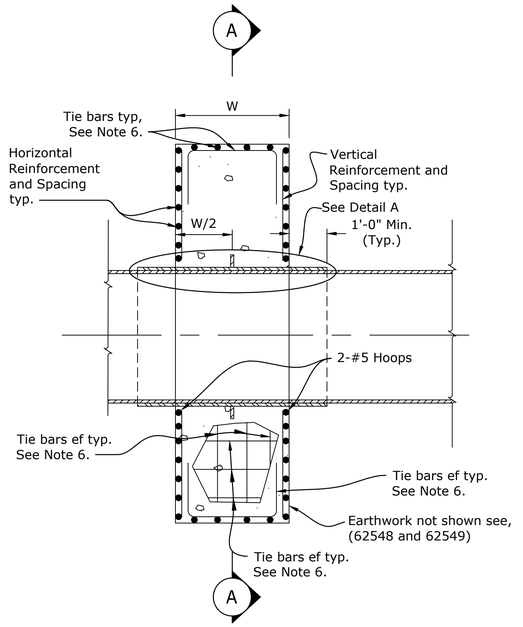
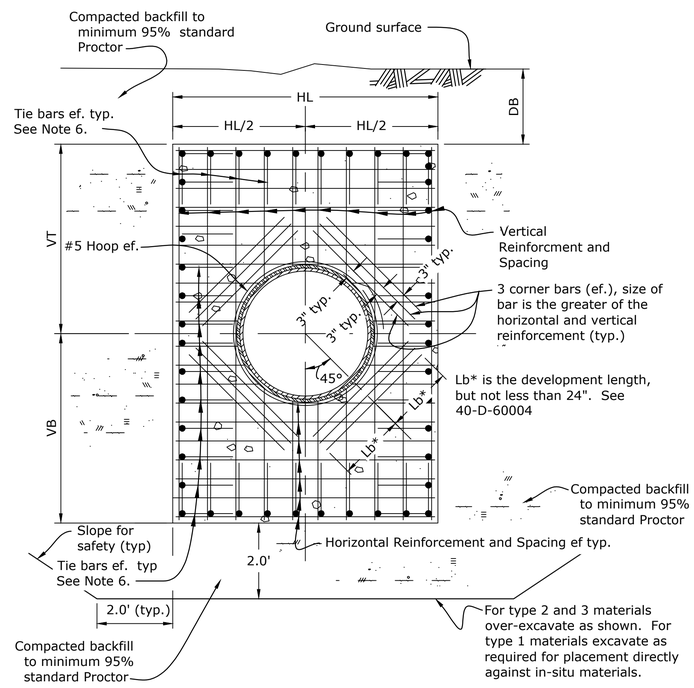
CAD SYSTEM
AutoCAD Civil 3DS (LMS TECH)
CAD FILENAME
1695-D-62549-RD-2.DWG

COLLAR / THRUST BLOCK TABLE									
STATION	DESIGN HEAD (FT)	PIPE DIAMETER (IN)	COVER (FT)	HL	VT	VB	W	DB (FT)	BAR SIZE
90761+59	300	48	3.3	22'-0"	4'-6"	4'-6"	2'-3"	2.2	#8@6" O.C.
11069+16	350	42	5.6	2'-0"	2'-0"	2'-0"	3'-0"	5.4	#5@12" O.C.

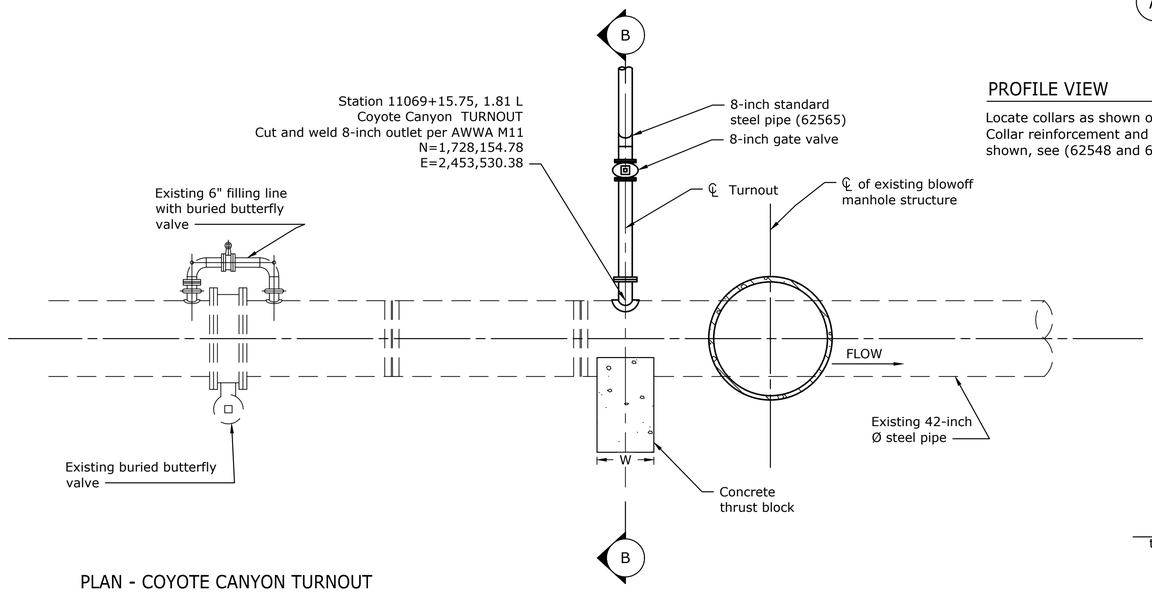


B SECTION
NOT TO SCALE





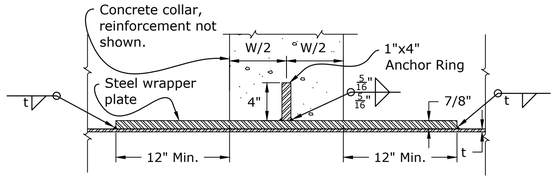
A SECTION
COLLAR DETAILS



PLAN - COYOTE CANYON TURNOUT
NOT TO SCALE

PROFILE VIEW

Locate collars as shown on (62548) for valve installation. For Collar reinforcement and thickness, see Collar Table. Earthwork not shown, see (62548 and 62549).



A DETAIL

OPERATING CRITERIA

1. Excavation around the blocking shall not occur when the pipeline is under hydrostatic pressure.

NOTES

1. Drawing not to scale.
2. Flange class shall meet pressure requirement designated by adjacent symbol line pipe. Flange drilling shall match mating flange. See Notes on (62551).
3. Steel plate minimum thickness, t, shall not be less than Specification requirements.
4. Concrete compressive strength, f'c, is 4,500 psi and specified yield point for reinforcement, fy, is 60,000 psi.
5. For general concrete outline notes and for minimum requirements for detailing reinforcement, see 40-D-60003, and 40-D-60004.
6. Tie bars as shown on the drawing are not required when for thickness of the collar "W" is 1'-6" or less. When placement is required tie bars are to be #6 bars spaced at a maximum 1'-0" and as detailed on 40-D-60004.
7. For coating and lining options, see Specification paragraphs.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

A. MARNER
DESIGNED
A. MARNER
DRAWN
C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Pelzer, Kylie C P.E.
APPROVED Civil Engineer
PEER REVIEWER
DENVER, CO 2026-06-02

BBN9 LINE PIPE - COLLARS

DETAILS AND SECTIONS

1695-D-62550
SHEET

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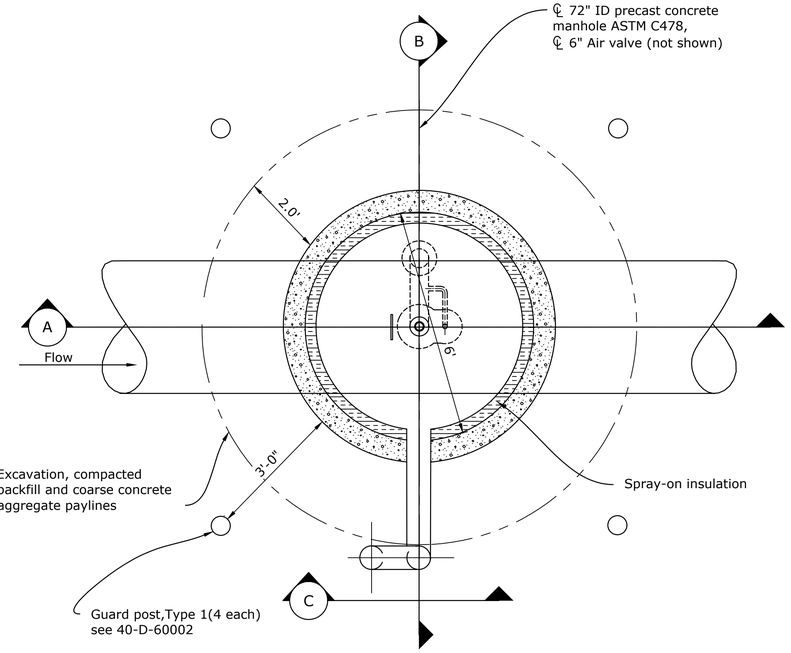
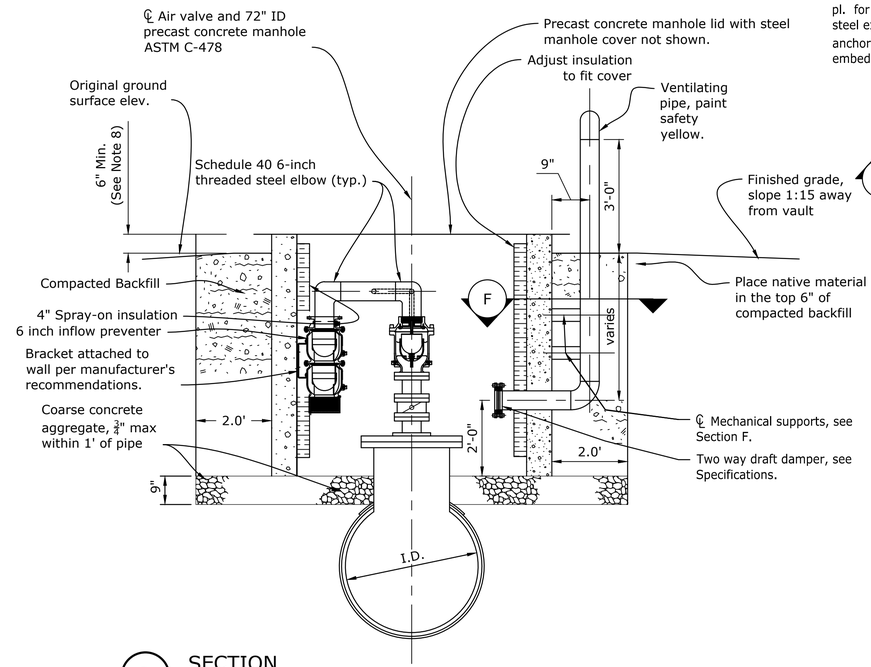
CAD FILENAME
1695-D-62550-RD-3.DWG



PLAN

WITH COVER REMOVED



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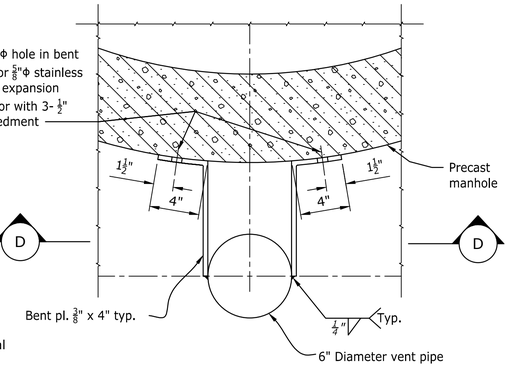

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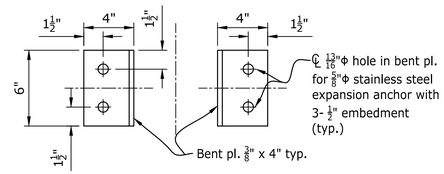
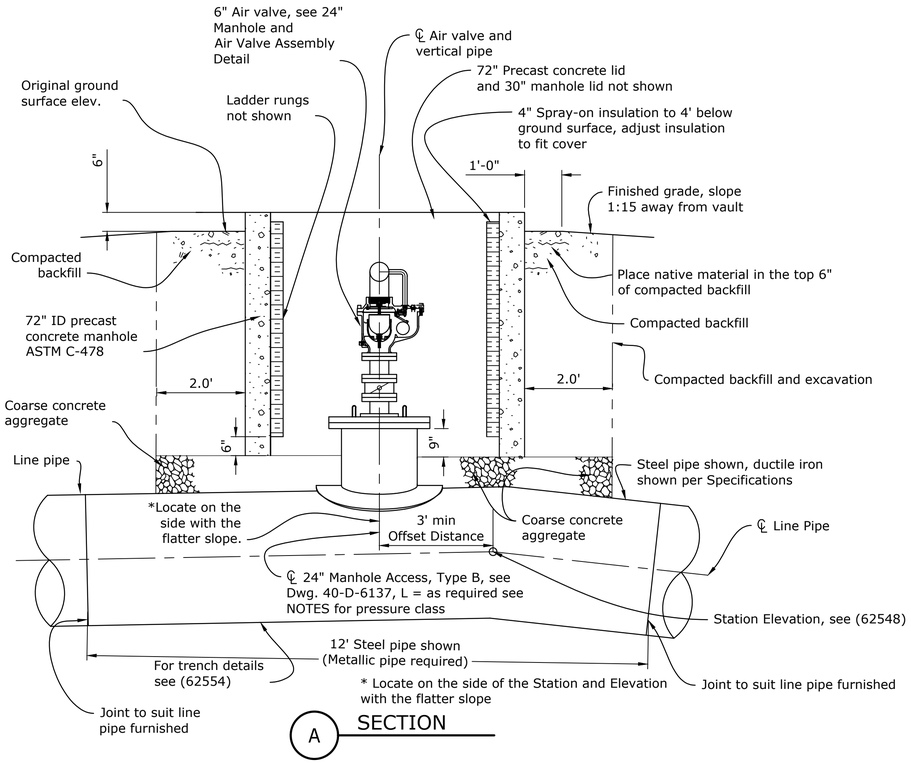


D MECHANICAL SUPPORTS FOR AIR VENT PIPING ON AIR VALVE VAULTS



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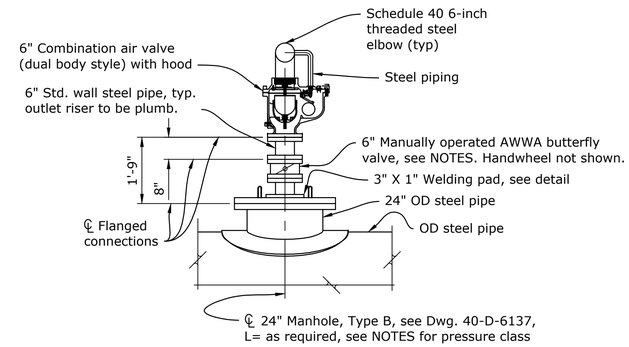
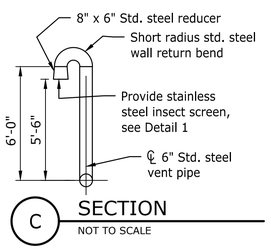


24" MANHOLE AND AIR VALVE ASSEMBLY



C SECTION

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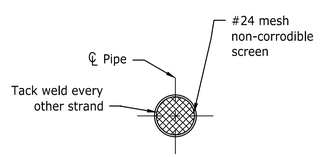

1 DETAIL

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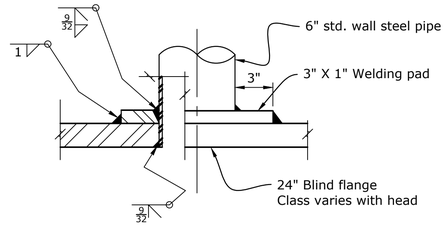

DETAIL - 3" X 1" WELDING PAD

NOTES

1. Flanges should be AWWA Class D when the head class is less than or equal to 346 ft, AWWA Class F when the head class is greater than 346 ft and is less than or equal to 692 ft.
2. The butterfly valve shall be AWWA Class 150B when the head class is less than or equal to 346 ft, AWWA Class 250B when the head class is less than or equal to 577 ft. See Specifications for butterfly valves with head classes greater than 577 ft.
3. The air valve inlet flange shall be ANSI B16.1 Class 125 or ANSI B16.5 Class 150 when the head class of the pipe is less than 346 ft and ANSI B16.1 Class 250 or ANSI B16.5 Class 350 for a head class of pipe greater than or equal to 692 ft.
4. The manhole class shall be Class 150 when the head class of the pipe is less than 346 feet and Class 300 for a head class of pipe less than or equal to 692 ft.
5. Drawing not to scale.
6. Construct air valve vertical.
7. Provide handwheel for all exposed valves.
8. The height of 72" precast concrete manhole shall allow enough clearance between the top of the air valve and the bottom of the insulation on the cover as per the air valve manufacturer's recommendations.
9. Match all mating flanges with highest rated flanges.
10. Outlet riser for air valve is to be plumb.
11. Dissimilar metals require a dielectric coupling for electrical isolation.

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

A. MARNER	DESIGNED
A. MARNER	DRAWN
C. HARMON	CHECKED
Marner, Andrew (Andy) J P.E.	TECH. APPR.
Pelzer, Kylie C P.E.	APPROVED Civil Engineer
	PEER REVIEWER
DENVER, CO	2026-06-02

BBN9 LINE PIPE - AIR VALVES WITH MANHOLES

SECTIONS AND DETAILS

1695-D-62551

SHEET 4

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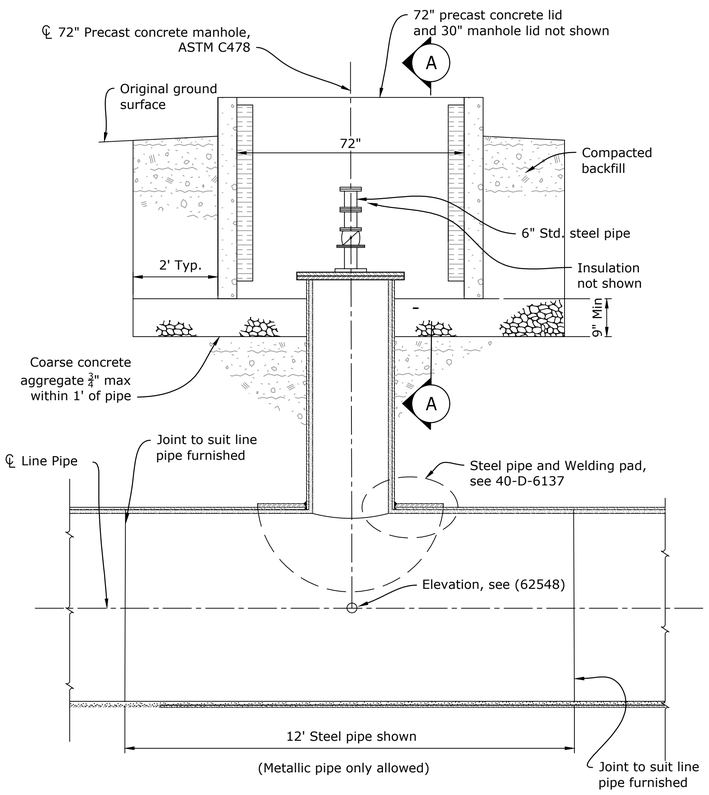

LONGITUDINAL SECTION

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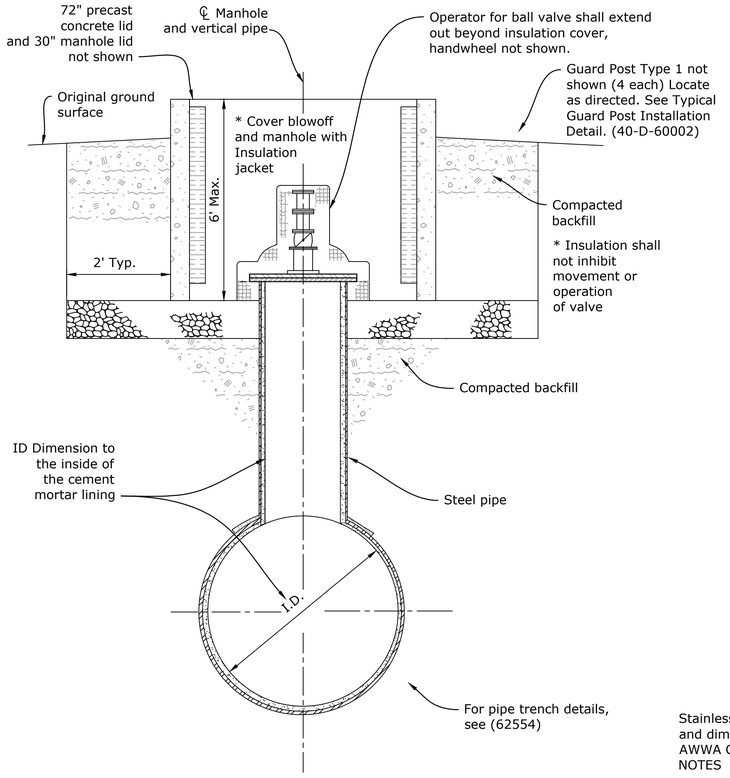

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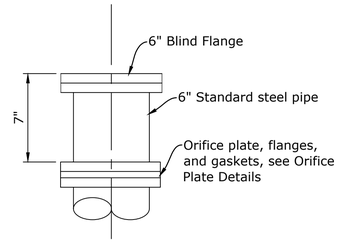

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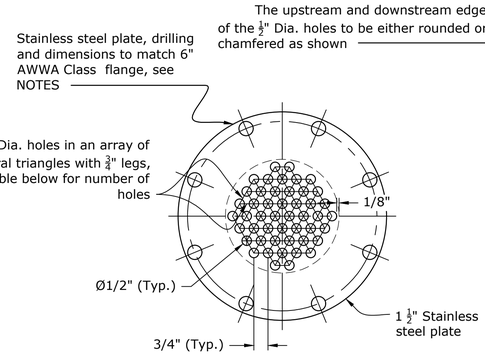
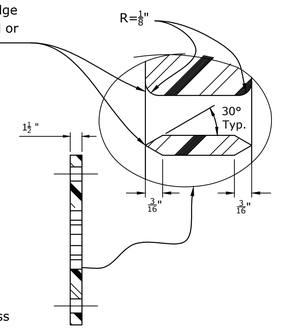
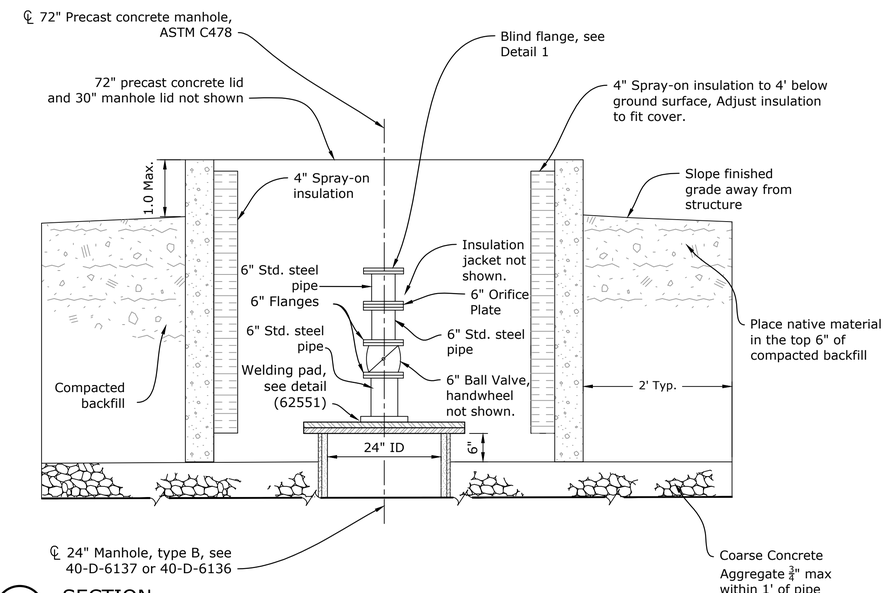




ORIFICE PLATE DETAILS

NOT TO SCALE

ORIFICE PLATES	
LOCATION	NUMBER OF HOLES
6-INCH BYPASS LINE	17
6-INCH BLOWOFF	48



A SECTION

NOT TO SCALE

NOTES

1. For blowoff location, see (62535 to 62548).
2. For coating and lining requirements for std. steel pipe and fittings, see Specification paragraphs.
3. Provide handwheel for all exposed valves.
4. Any welding to be completed prior to cement motor lining application.
5. The minimum steel cylinder thickness for the section of line pipe with the blowoff outlet shall be the thickness of the steel line pipe. For the steel line pipe thickness see the Specifications.
6. For coating and lining requirements, see Specification paragraphs.
7. Install four Guard posts Type 1 around each blowoff as directed by the COR. Install posts in accordance with 40-D-60002.
8. See Plan and Profile drawings for location of blowoff stations.
9. For Flange ratings, see Notes (62551).
10. Match all mating flanges with highest rated flange.
11. Dissimilar metals require a dielectric coupling for electrical isolation.

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

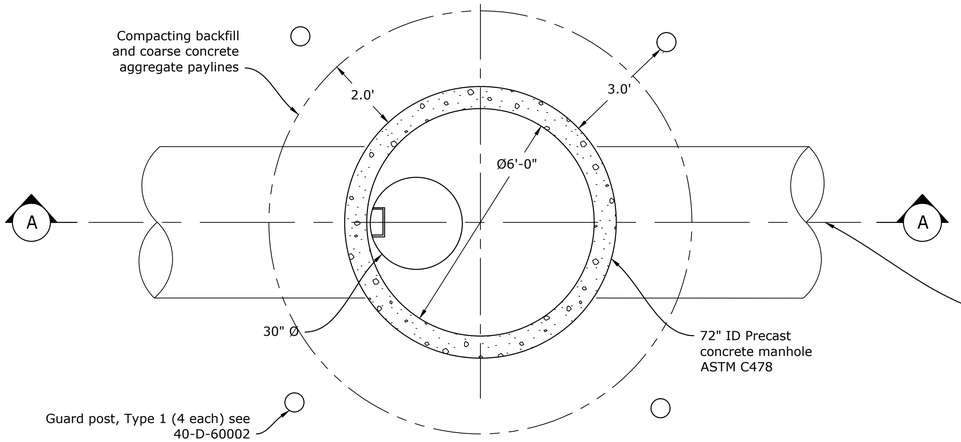
A. MARNER	DESIGNED
A. MARNER	DRAWN
C. HARMON	CHECKED
Marner, Andrew (Andy) J P.E.	TECH. APPR.
Pelzer, Kylie C P.E.	APPROVED Civil Engineer PEER REVIEWER
DENVER, CO	2026-06-02

BBN9 LINE PIPE - MANHOLE WITH BLOWOFF

SECTIONS AND DETAILS

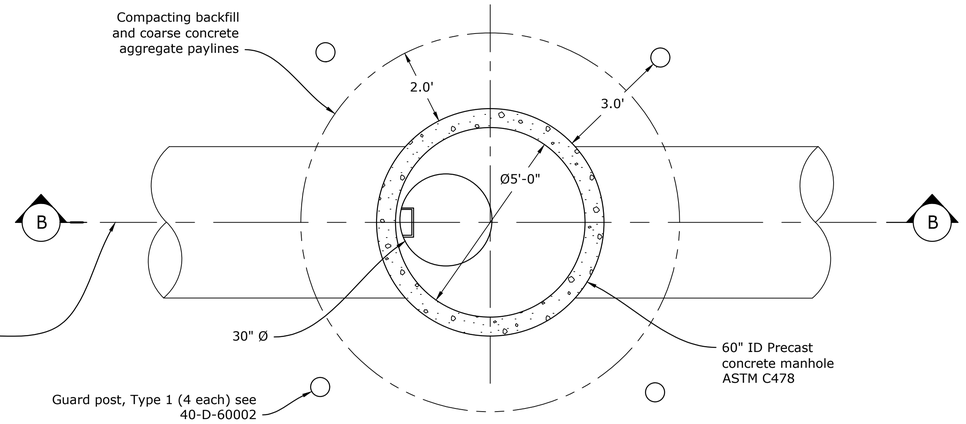
1695-D-62552

SHEET 5



PLAN - 72" Ø MANHOLE

NOT TO SCALE



PLAN - 60" Ø MANHOLE

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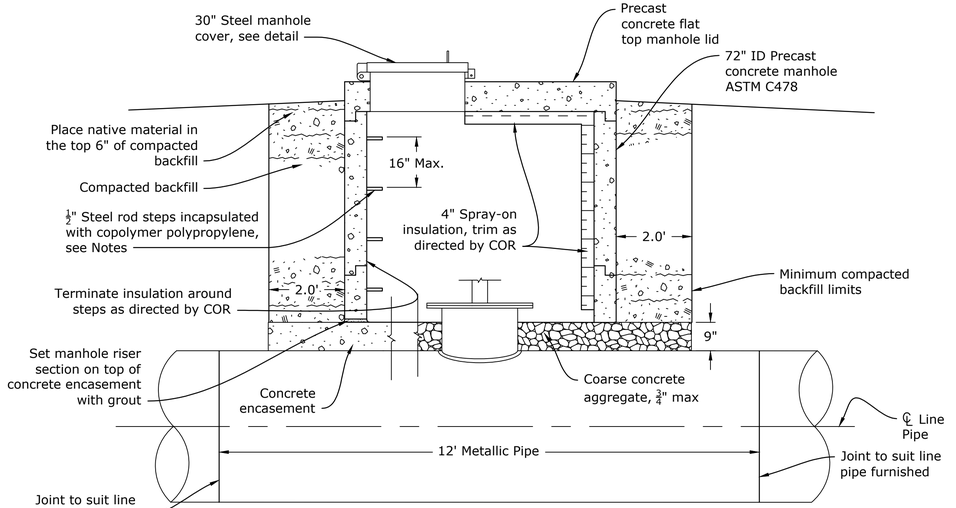

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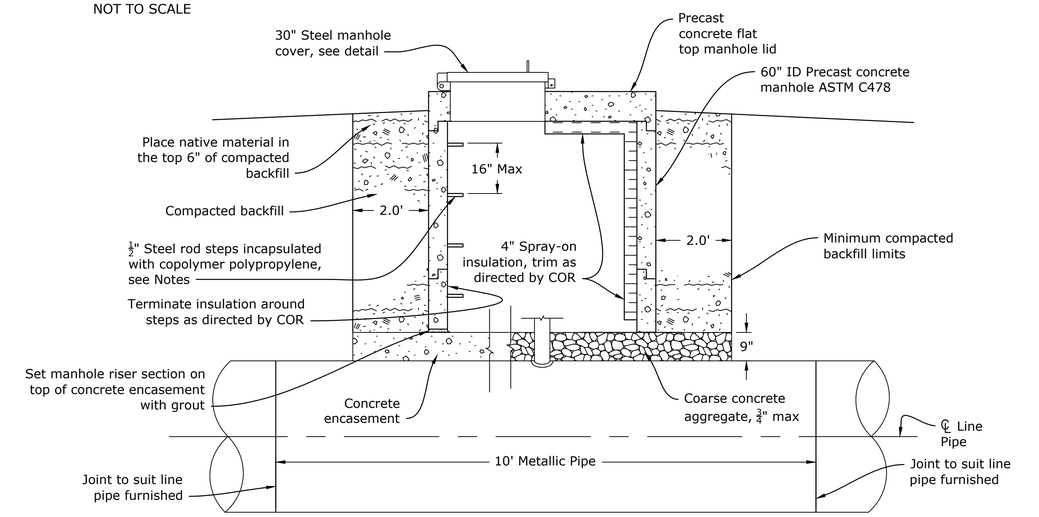

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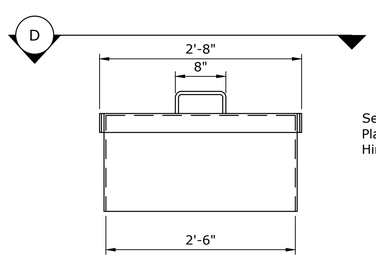

C SECTION

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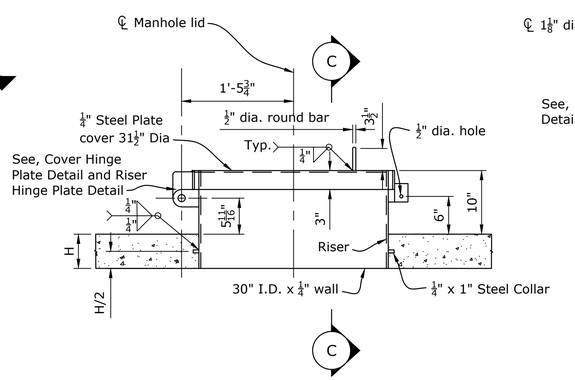

STEEL MANHOLE LID DETAIL

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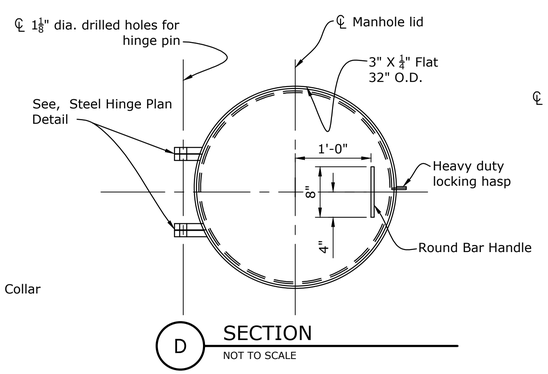

D SECTION

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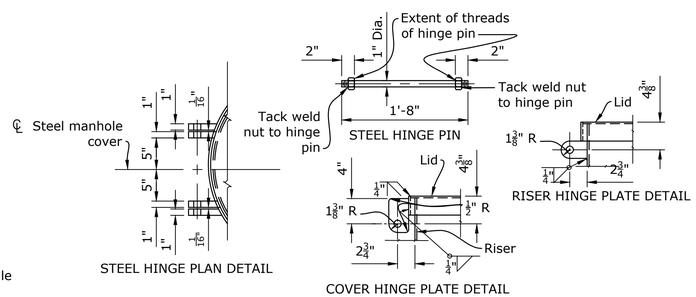

NOTES

1. Manhole steps shall be made of ½" steel rod encapsulated with copolymer polypropylene or approved equal and shall conform to ASTM C478. Steps shall have minimum projections of 4-inches, spaced no more than 16" apart, minimum overall widths of 14 inches and anchored into the walls at least 3½ inches deep.
2. Manhole cover shall be provided with a lockable lid.
3. Drawing not to scale.

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

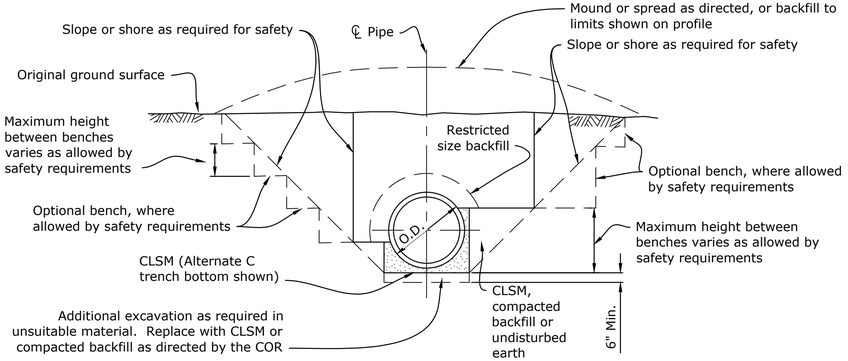
A. MARNER	DESIGNED
A. MARNER	DRAWN
C. HARMON	CHECKED
Marner, Andrew (Andy) J P.E.	TECH. APPR.
Pelzer, Kylie C P.E.	APPROVED Civil Engineer
	PEER REVIEWER
DENVER, CO	2026-06-02

BBN9 LINE PIPE - PRECAST CONCRETE MANHOLES

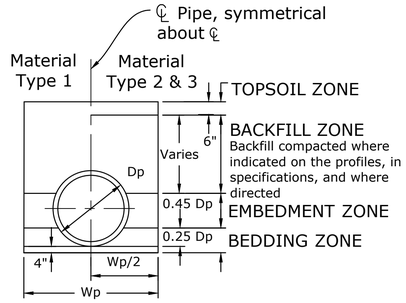
PLANS, SECTIONS, AND DETAILS

1695-D-62553

SHEET 6

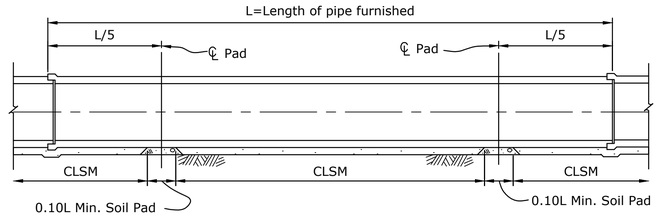


TYPICAL TRENCH DETAILS



TRENCH DEFINITION

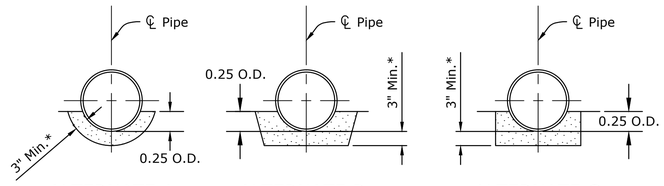
ALL TYPES OF PIPE



NOTE: Soil pads allow observation of the CLSM flowing from one side of the pipe to the other. See Note 8.

CLSM INSTALLATION WITH SOIL PADS

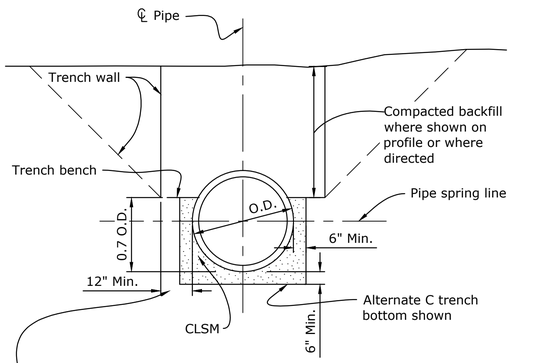
TYPICAL FOR ALL PIPE TYPES



ALTERNATE A ALTERNATE B ALTERNATE C

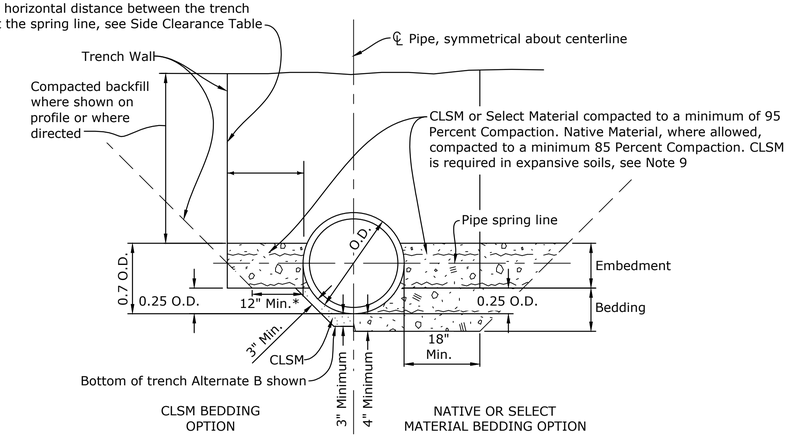
ACCEPTABLE TRENCH BOTTOMS FOR CLSM

* 6" Min required between bottom of pipe and rock surfaces



Minimum working/inspection clearance is the horizontal distance between the intersection of the trench wall with the trench bench and the pipe at spring line

MATERIAL TYPE 1 TRENCH



*12" Minimum bench not required using CLSM embedment, minimum side clearance still required

MATERIAL TYPE 2 AND 3 TRENCH

SELECT MATERIAL GRADATION LIMITS

Size *	Percent by Weight
Passing No. 200 sieve	12 or less
Passing No. 50 sieve	25 or less

* Maximum size shall not exceed ¾-inch

NATIVE MATERIAL PROPERTIES

MATERIAL TYPE 1 - Typical trench wall materials would be rock; materials described as claystone, mudstone, or siltstone; highly cemented soils even if low density; sands and gravels; and silty or clayey material with inplace densities 95% Standard Proctor or higher.

MATERIAL TYPE 2 - Typical trench wall materials would include silty or clayey material or sands and gravels with inplace densities between 85% Standard Proctor or higher, but less than 95%.

MATERIAL TYPE 3 - Typical trench wall materials would include elastic silts, low density silty or clayey soils, or low-density cohesionless soils with a density below 85% Standard Proctor.

Trench Material Type locations, based on the above Native Material Properties and geology investigation, have been identified and correspond to minimum side clearance widths listed in the Side Clearance Table. Material Type 1's inferred bedrock surface is shown in the line pipe profile on drawings. Material Types 2 and 3 locations are listed in the specifications.

1. Side clearance measurement is at pipe springline for all materials.
2. Backfill can not exceed 5' until CLSM embedment obtains an unconfined compressive strength of 30 psi.
3. Any dimension less than 12" must use CLSM embedment. Select and native material embedment requires minimum of 18" side clearance.

Steel - Mortar lined, flexible coated.
Ductile Iron - Special thickness class.
HDPE - Only DR ratios 21, 17, 13.5, 11, and 9 allowed.

CLSM/SELECT EMBEDMENT SIDE CLEARANCE

Height of Cover	STEEL					
	12"		16"		48"	
	Material Type 2	Material Type 3	Material Type 2	Material Type 3	Material Type 2	Material Type 3
<=10'	6	6	6	6	15	18
<=15'	8	8	8	8	21	27
<=20'	10	10	10	10	30	33

NATIVE MATERIAL EMBEDMENT

PIPE MATERIAL TYPE	MAX COVER DEPTH
Steel Pipe	Not Allowed

Screened excavated native material is allowed for the embedment zone for Material Types 2 and 3 based on the following conditions:

1. Particle sizes greater than ¾-inch are removed.
2. Compacted backfill to 95% is not required above pipe.
3. Native material embedment is compacted to 85%.
4. Side clearance is 18" minimum.
5. Restraint joints are not required for thrust.

NOTES

1. Drawing not to scale.
2. Dp and Wp are used for calculating pay quantities for all pipe and trench types. Calculations are based on vertical walls.
3. O.D is the outside diameter, in inches, of the pipe actually installed.
4. CLSM is Controlled Low Strength Material. See Specifications for details.
5. Sloping, shoring, and benching shall be in accordance with Reclamation Safety and Health Standards.
6. CLSM is required in the bedding for all pipe types.
7. Warning tape and tracer wire (for plastic pipes) not shown, but are required. See specifications for details.
8. Other material may be submitted in lieu of soil pads for approval. Soil pad materials must not exceed the compressive strength of the CLSM.
9. CLSM is required for the embedment when there is expansive soils and the native trench material's plastic index exceeds 25 as listed in Table 1.2 of Reclamation's Earth Manual.
10. CLSM is required for the embedment when saturated soil is encountered in the pipe trench.
11. Install pipe based on these Specifications or pipe manufacturer's, whichever is more stringent.

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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DESIGNED
A. MARNER
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C. HARMON
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Pelzer, Kylie C P.E.
APPROVED Civil Engineer
PEER REVIEWER

DENVER, CO 2026-06-02

PRESSURE PIPE TRENCH INSTALLATION

SECTIONS AND DETAILS

1695-D-62554

SHEET 7

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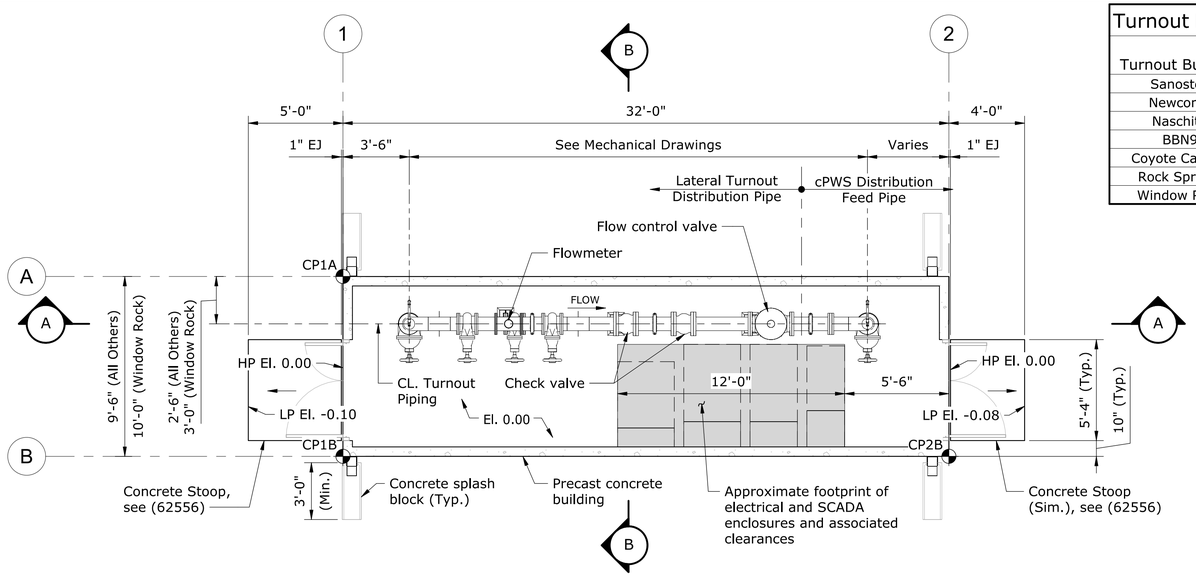
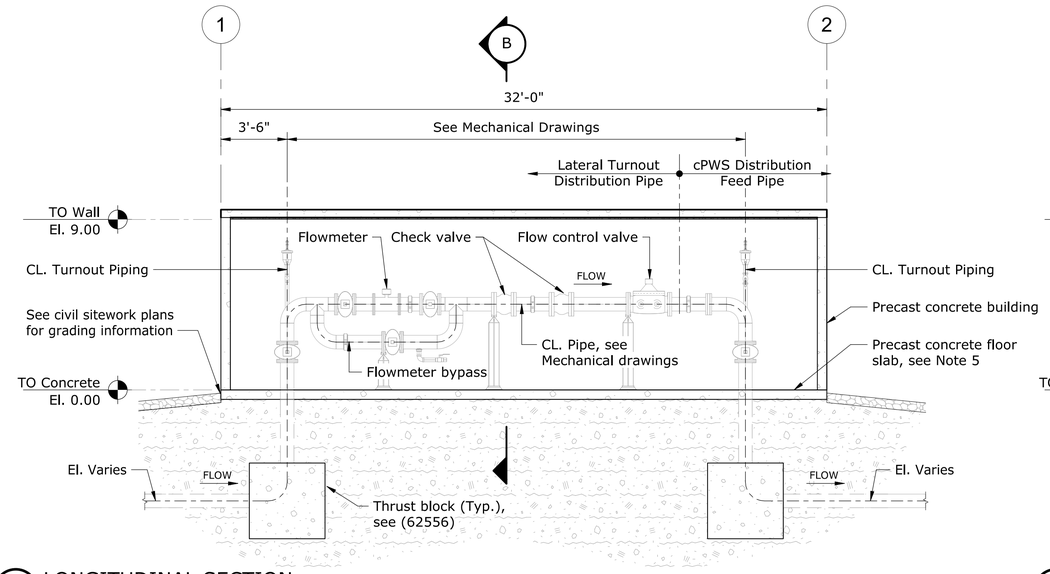
Turnout Building Control Points and Orientation			
Turnout Building	Control Point Identified	Building Plan	Site Plan Drawing
Sanostee	CP1A	Similar	(62528)
Newcomb	CP2B	Similar	(62531)
Naschitti	CP1A	As shown	(62533)
BBN9	CP1A	As shown	(62535)
Coyote Canyon	CP1B	Similar	(62537)
Rock Springs	CP1A	Similar	(62539)
Window Rock	CP1A	As shown	(62541)



PLAN
Scale: 1/4" = 1'-0"



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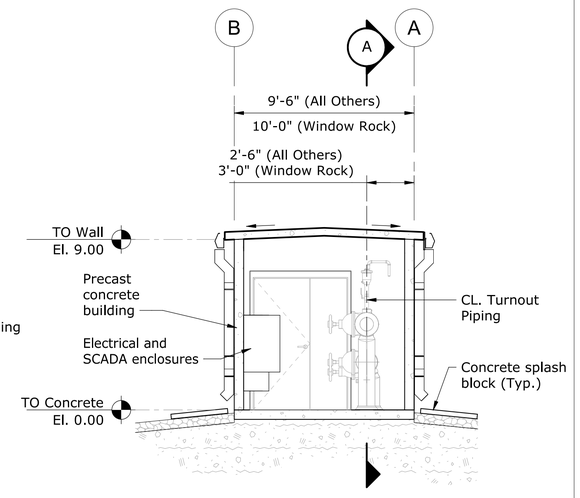
Scale: 1/4" = 1'-0"



B TRANSVERSE SECTION
Scale: 1/4" = 1'-0"

NOTES

1. For Drawing Index, see (62526).
2. For building control point coordinates and bearings, see civil sitework plans.
3. The finished floor elevation (FFE) identified on the civil sitework plans correspond to the El. 0.00 identified on this drawing. All elevations reported on the drawings are relative to the El. 0.00 elevation unless noted otherwise.
4. For structural excavation and backfill, see Specifications.
5. Building floor slab is shown as precast concrete. A cast-in-place concrete building floor slab is also permitted. See Specifications.
6. For Concrete Outline General Notes, see (62556).
7. For Concrete Reinforcement General Notes, see (62556).

ABBREVIATIONS

CL.	denotes centerline
cPWS	denotes consecutive public water system
EQ	denotes equal
HP	denotes high point
LP	denotes low point
NTS	denotes not to scale
Sim.	denotes similar
Typ.	denotes typical
TO	denotes top of

0 2' 4' 8'
Scale: 1/4" = 1'-0"

DESIGN CRITERIA & LOAD DESIGNATIONS

GENERAL DESIGN STANDARDS

DESIGN STANDARDS

- International Building Code (IBC) 2021.
- American Society of Civil Engineers (ASCE) 7-16, "Minimum Design Loads and Associated Criteria for Buildings and Other Structures," with Supplement 1.
- American Concrete Institute (ACI) 318-19, "Building Code Requirements for Structural Concrete"
- Bureau of Reclamation Design Standards No. 9, "Buildings and Other Structures."

DESIGN LOADS

(D) DEAD LOADS (IBC 1606)

See Specifications.

(L) FLOOR LIVE LOADS (IBC 1603-1.1)

Floor ---------- 150 lb/ft^2

(L_r) ROOF LIVE LOADS (IBC 1603-1.2)

Roof ---------- 20 lb/ft^2

(S) SNOW LOADS (IBC 1603-1.3)

The structure and its components shall be designed in accordance with the IBC and governing local codes for snow load criteria as follows.

Ground Snow Load (p_g) = 30.0 psf
Snow Importance Factor (I_s) = 1.10
Exposure Factor (C_e) = 0.9
Thermal Factor (C_t) = 1.0
Flat Roof Snow Load (p_f) = 20.8 psf

(W) WIND LOADS (IBC 1603-1.4)

The structure and its components shall be designed in accordance with the IBC and governing local codes for wind load criteria as follows:

Risk Category: III
Basic wind speed (3-second gust): 120 mph
Wind Exposure Category: C
Internal Pressure Coefficient: GC_{pi} = ±0.18

(E) SEISMIC LOADS (IBC 1603-1.5)

The seismic analysis and design of structures above grade shall be designed in accordance with the building code for seismic load parameters as follows:

Risk Category: III
Seismic importance factor (I_e): 1.25

Mapped (MCE_R) spectral response acceleration coefficients:
S_s = 0.149g
S_1 = 0.051g

Site class: D

Design spectral response acceleration coefficients:
S_{DS} = 0.159g
S_{D1} = 0.081g

Seismic design category: B

Spectral response acceleration coefficient adjusted for Site Class effects:
S_{MS} = 0.239g
S_{M1} = 0.122g

Non-Structural Components:
Unless noted otherwise, Mechanical-Electrical-Plumbing (MEP) and Architectural Components shall be anchored per ASCE 7, Chapter 13 - *Seismic Design Requirements for Nonstructural Components.*

Ground Motion:
Risk-Targeted Maximum Considered Earthquake (MCE_R) target risk of structural collapse equal to 1% in 50 years, design EQ = 2/3 MCE_R has an approximate 2% probability of exceedance in 50 years (return period of 2,475 years)

(H) LATERAL EARTH PRESSURES (IBC 1063-1.6)

See Specifications.

(F) FLUID LOADS (IBC 1063-1.7)

Includes all situations where horizontal water loads may exist and corresponding uplift pressures where applicable.

Water Pressure ---------- 62.4 lb/ft^2/ft

(R) RAIN LOADS (IBC 1063-1.9)

Apply in accordance with IBC, Section 1611 and ASCE 7, Chapter 8.

(T) TEMPERATURE VARIATIONS

All members above grade: ---------- ±50° F
All members below grade: ---------- ±20° F

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

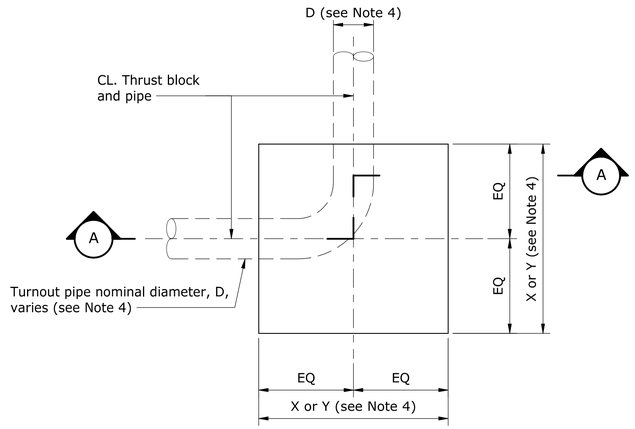
SARA PUTNAM	
DESIGNED	
SARA PUTNAM	
DRAWN	
NICOLE M. SMITH	
CHECKED	
TECH. APPR.	KEVIN O'SHEA, P.E.
APPROVED	L. NGUYEN, PH.D, P.E.
Peer Reviewer - PLANT STRUCTURES GROUP	

DENVER, CO 2026-06-19

STRUCTURAL DESIGN DATA & GENERAL ARRANGEMENT

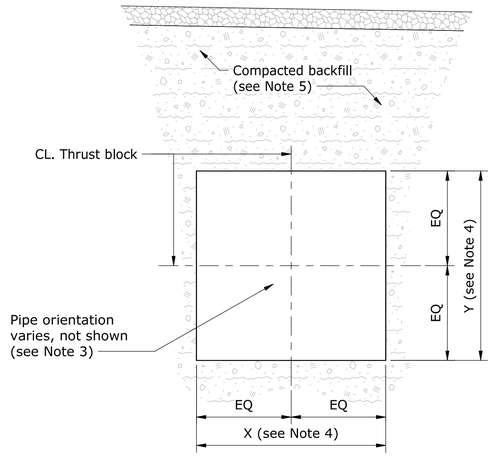
PLAN AND SECTIONS

1695-D-62555



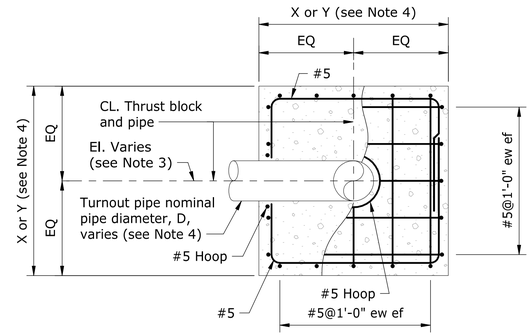
THRUST BLOCK ALONG PIPE BEND

NTS



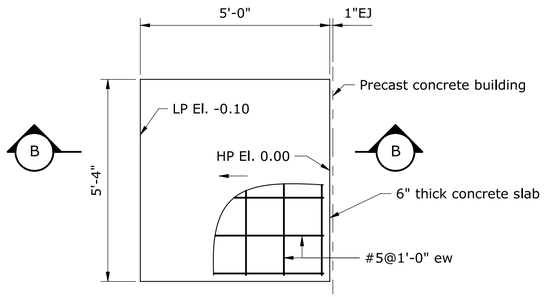
ELEVATION - THRUST BLOCK BACKFILL

NTS



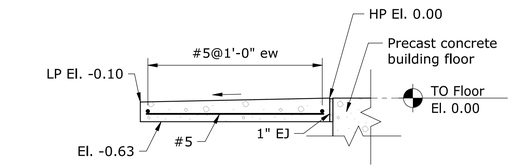
A SECTION - THRUST BLOCK

NTS



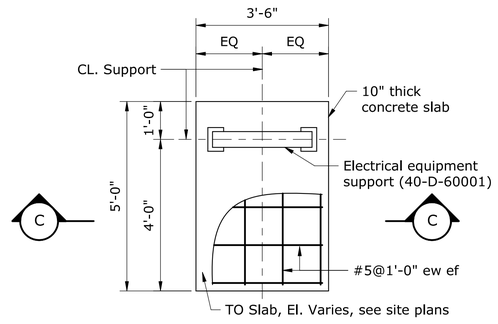
PLAN - BUILDING STOOP

Scale: 1/2" = 1'-0"



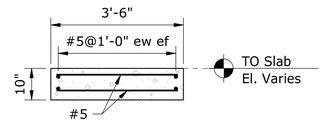
B SECTION - BUILDING STOOP

Scale: 1/2" = 1'-0"



PLAN - ELECTRICAL DISCONNECT PAD

Scale: 1/2" = 1'-0"



C SECTION - ELECTRICAL DISCONNECT PAD

Scale: 1/2" = 1'-0"

THRUST BLOCK DIMENSIONS

THRUST BLOCK NO.	PIPE NOMINAL DIAMETER, D	CONCRETE DIMENSION, X	CONCRETE DIMENSION, Y	MINIMUM SOIL COVER
SAN-1	6"	3' - 6"	3' - 6"	See (62529)
SAN-2	6"	3' - 6"	3' - 6"	See (62529)
SAN-3	14"	7' - 10"	7' - 10"	See (62529)
NEW-1	6"	3' - 6"	3' - 6"	See (62532)
NEW-2	6"	4' - 6"	4' - 6"	See (62532)
NEW-3	8"	6' - 2"	4' - 6"	See (62532)
NAS-1	6"	4' - 6"	4' - 6"	See (62534)
NAS-2	6"	4' - 6"	4' - 6"	See (62534)
NAS-3	10"	6' - 2"	6' - 2"	See (62534)
BBN9-1	8"	3' - 6"	3' - 6"	See (62536)
BBN9-2	8"	3' - 6"	3' - 6"	See (62536)
BBN9-3	16"	7' - 10"	7' - 10"	1.5' Minimum Compacted Backfill
CC-1	6"	3' - 6"	3' - 6"	See (62538)
CC-2	6"	4' - 6"	4' - 6"	See (62538)
CC-3	8"	6' - 2"	6' - 2"	1.5' Minimum Compacted Backfill
RS-1	6"	3' - 6"	3' - 6"	See (62540)
RS-2	6"	3' - 6"	3' - 6"	See (62540)
RS-3	8"	4' - 6"	4' - 6"	2.5' Minimum Compacted Backfill
WR-1	12"	6' - 2"	6' - 2"	See (62542)
WR-2	12"	6' - 2"	6' - 2"	See (62542)

CONCRETE OUTLINE GENERAL NOTES

For abbreviations, concrete symbols, and additional concrete outline general notes, see 40-D-60003.

Structural design for cast-in-place structural concrete is based on concrete with a compressive strength of 4,500 pounds per square inch at 28 days and reinforcement with a minimum yield strength of 60,000 pounds per square inch.

Dimensions not shown are the same dimensions for identical details and sections shown elsewhere on the drawings.

Where concrete with varying thickness is shown on the drawings, vary thickness of concrete uniformly between dimensions and elevations shown.

Pipe sleeves and embedded piping 12 inches in diameter and larger are shown on the outline drawings. Smaller diameter pipe sleeves or piping may or may not be shown.

Not all embedded items are shown on the outline drawings.

Before placing concrete on grade, ensure that all buried materials below the concrete are in place.

For details of method of building forms for successive lifts, see 40-D-5246.

For details of joints in concrete structures, see 40-D-5247.

For details of construction joint keys, see 40-D-5249.

All horizontal construction joints shall be thoroughly cleaned and roughened to a full amplitude of approximately 1/4 inch.

Backfill shall not be placed against the walls until concrete has attained a compressive strength of 4,500 pounds per square inch.

Conduit shall not be larger in outside diameter than one-third of the overall thickness of the slab, wall, or beam in which they are embedded.

Conduit shall not be spaced closer than 3 diameters or widths on center.

Conduit shall be located within the middle third of the cross sectional depth of the member, slab, or wall in which they are embedded. For conduit larger than 3-inches in diameter, locate as close as possible to the middle third of the cross sectional depth of the member.

REINFORCEMENT DESIGN GENERAL NOTES

For notes to designers and detailers, and for additional reinforcement general notes, see 40-D-60003.

For minimum requirements for detailing reinforcement, see 40-D-60004.

Structural design for cast-in-place structural concrete is based on concrete with a compressive strength of 4,500 pounds per square inch at 28 days with a minimum yield strength of 60,000 pounds per square inch.

"L" denotes length of reinforcement.

"L_s" denotes lap splice length, see 40-D-60004.

For add'l reinforcement around openings, see 40-D-60004.

Reinforcement not shown is similar to that shown elsewhere on the drawing for similar details.

Splices in reinforcement at construction joints may be shifted to agree with sequence of construction.

NOTES

1. For Drawing Index, see (62526).
2. For Abbreviations, see (62555).
3. For locations, elevations, and pipe bend orientations of thrust blocks, see (62561) through (62567).
4. See Thrust Block Dimensions Table for nominal pipe diameter and concrete dimensions. 'X' denotes length/width. 'Y' denotes height. Pipe orientations vary, details remain similar between pipe orientations.
5. Place compacted backfill, geotextile fabric, and gravel, per each applicable Turnout Grading Plan. For thrust blocks located outside of turnout building yard extents, place compacted backfill to a minimum depth specified in the Thrust Block Dimensions table (measured from top of concrete), or match existing grade elevation, whichever is greater.
6. For approximate location of electrical disconnect pad, see site plans. Coordinate with COR for final placement and location.
7. Contractor to prepare firm building surface prior to concrete placement. See Specifications for surface preparation and compaction requirements.

0 1' 2' 4'
Scale: 1/2" = 1'-0"

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

EDWARD DOWNS	
DESIGNED	
EDWARD DOWNS	
DRAWN	
NICOLE M. SMITH	
CHECKED	K.-C.S.L., P.E.
TECH. APPR.	Kevin O'Shea, P.E.
APPROVED	L. Nguyen, Ph.D, P.E.
Admin Approver - PLANT STRUCTURES GROUP	
DENVER, CO	2026-06-19

CONCRETE OUTLINE & REINFORCEMENT DESIGN

THRUST BLOCKS AND MISC. SITE CONCRETE

1695-D-62556

BUILDING CODE SUMMARY

All work to comply with the 2021 International Building Code (IBC)

Occupancy type:
Utility and Miscellaneous, U (IBC 312.1)

Construction type: V-B (IBC 602.5)

Allowable area: 5,500 sf (IBC Table 506.2)
Actual area: ≦320 sf (calculated as defined - IBC Section 506.2)

Allowable height: 40 feet, 1 stories (IBC Tables 504.3 and 504.4)
Actual height: 10 feet, 1 story (calculated as defined - IBC Section 502.1)

Occupant load (IBC Table 1004.5):
400 sf "Mechanical Equipment Room" / 300 gross = 2 total occupant load

Required number of exits: One - each side (IBC Table 1006.3.3)
Exits provided: One - each side

Travel distance (IBC Table 1017.2):
Total: 300'
Common path (IBC Table 1006.2.1): 100'

Fire resistance rating (IBC Table 601):
Primary structural frame: 0 hours
Bearing walls (exterior and interior): 0 hours
Nonbearing walls and partitions, exterior: 0 hours (IBC Table 602)
Nonbearing walls and partitions, interior: 0 hours
Floor construction and associated secondary members: 0 hours
Roof construction and associated secondary members: 0 hours

Plumbing fixtures (IBC 2902.1): not provided - facility designed for infrequent and intermittent occupancy.

Climate zone (ASHRAE 90.1): 5B
Insulation values:
Roof (attic/other): R-49
Walls, above grade (mass): R-11.4 C.I. min.
Foundation: R-15 to 24" below grade

TURNOUT SITE	BUILDING SIZE		SITE DRAWING
	L	W	
Rock Springs	32'-0"	9'-6"	1695-D-62539
Window Rock	32'-0"	10'-0"	1695-D-62541
Coyote Canyon	32'-0"	9'-6"	1695-D-62537
Crownpoint (BBN9)	32'-0"	9'-6"	1695-D-62535
Naschitti	32'-0"	9'-6"	1695-D-62533
Newcomb	32'-0"	9'-6"	1695-D-62531
Sanostee	32'-0"	9'-6"	1695-D-62528

SYMBOL LEGEND

1 — BUILDING ELEVATION SYMBOL
Indicated elevation number

1 — BUILDING SECTION SYMBOL
Indicates section number

ROOM NAME
101 — Room name and number

101 — Door symbol

T.O. Wall
ELEVATION — Vertical or spot elevation

GENERAL NOTES

1. Dimensions on drawings are taken to/from the following locations unless otherwise indicated: centerline of column, face of stud, face of concrete; rough opening edges of doors, windows, louvers.

2. Lightning protection system components are not shown on drawings. See specifications.

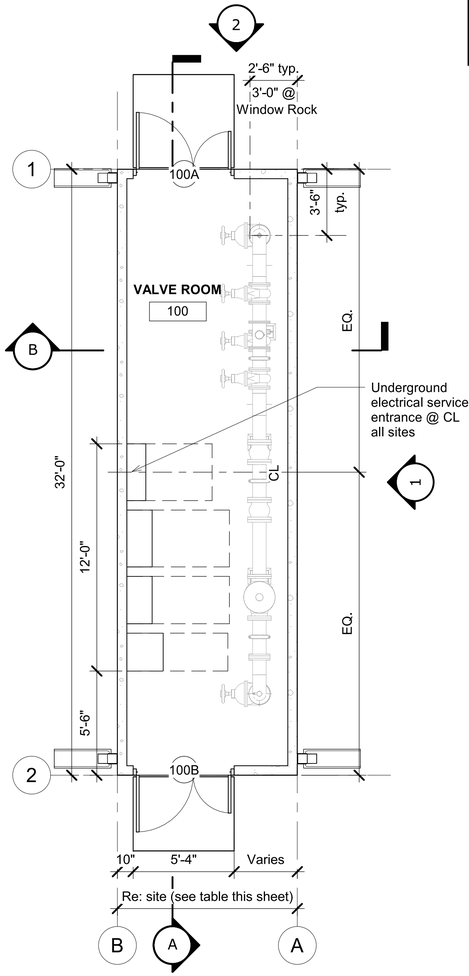
3. Mechanical, electrical, structural and site information shown on architectural drawings is provided for clarity and context only. See mechanical, electrical, structural and site work drawings for additional information.

4. Flashing and sealant color (may not be shown) to match adjacent material unless otherwise noted.

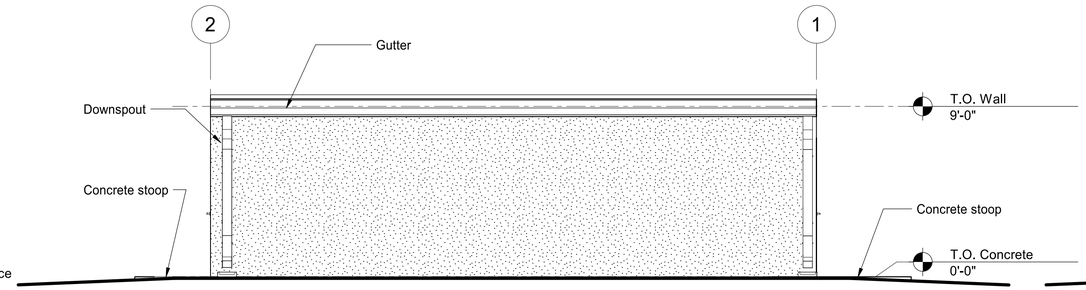
5. All sealant, gaskets and sealing materials may not be shown but shall be applied as required and/or recommended by building manufacturer for weatherproofing.

6. Room and door numbers shown on drawings are for construction purposes only.

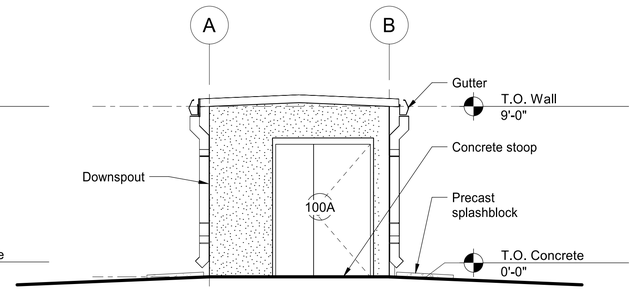
7. See site plans for flow direction and associated building location and orientation.



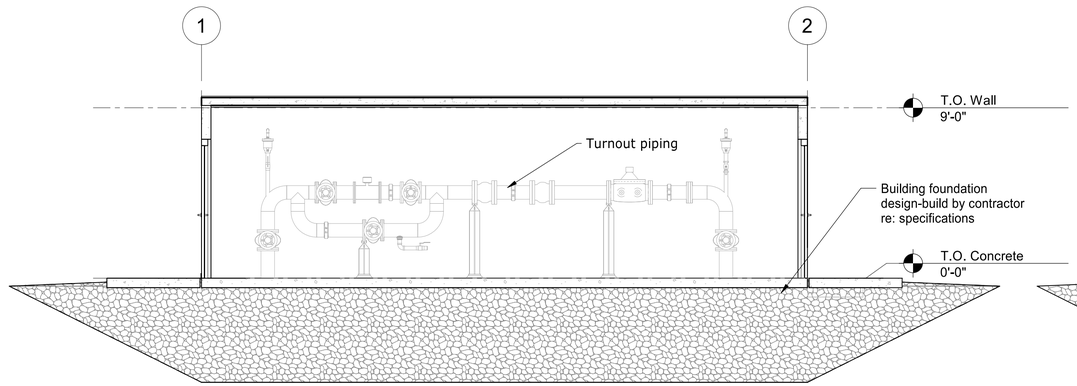
FLOOR PLAN
Scale: 1/4" = 1'-0"



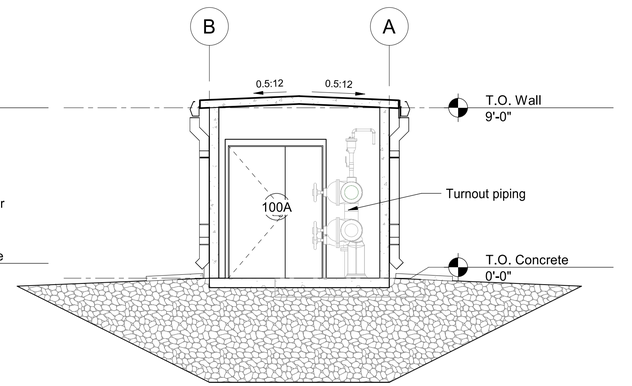
1 SIDE ELEVATION
Scale: 1/4" = 1'-0"



2 END ELEVATION
Scale: 1/4" = 1'-0"



A LONGITUDINAL SECTION.
Scale: 1/4" = 1'-0"



B CROSS SECTION
Scale: 1/4" = 1'-0"

0 2' 4' 8'
Scale: 1/4" = 1'-0"

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

A. MCDANIEL	
DESIGNED	
A. MCDANIEL	
DRAWN	
H. COOPER	
CHECKED	Digitally signed by BRUCE BARKEY Date: 2026.06.04 14:23:52 -06'00'
TECH. APPR.	B. BARKEY, R.A.
	Digitally signed by LAN NGUYEN Date: 2026.06.04 16:10:42 -06'00'
APPROVED	L. NGUYEN, P.E.
Peer Reviewer - PLANT STRUCTURES GROUP	
DENVER, CO	2026/06/04

TURNOUT BUILDING
ARCHITECTURAL

CODE SUMMARY, PLANS,
ELEVATIONS, SECTIONS

1695-D-62557

Flow Control Valve Sizes

Turnout Location	Flow Control Valve Size	Maximum Operating Pressure	Maximum Operating Flows	Yard Pipe Drawing	Interior Pipe Centerline Elevation	Site Plan
Sanostee	6 inch	233 psi	2.27 cfs	1695-D-62561	5571.75	1695-D-62528
Newcomb	8 inch	184 psi	1.52 cfs	1695-D-62562	5570.80	1695-D-62531
Naschitti	6 inch	210 psi	1.77 cfs	1695-D-62563	5877.50	1695-D-62533
BBN9	8 inch	123 psi	3.80 cfs	1695-D-62564	6056.30	1695-D-62535
Coyote Canyon	6 inch	230 psi	1.25 cfs	1695-D-62565	6259.75	1695-D-62537
Rock Springs	6 inch	148 psi	1.71 cfs	1695-D-62566	6858.85	1695-D-62539
Window Rock	12 inch	138 psi	11.51 cfs	1695-D-62567	6795.70	1695-D-62541

Location of Lateral Turnout

Lateral Turnout	Lateral Turnout Tie-In Location		Reference Drawing	Consecutive Distribution Entry Point	
	Northing[1]	Easting[1]		Northing[1]	Easting[1]
Sanostee	1976795.6044	2462216.8104	1695-D-60728	1976728.7800	2462184.4600
Newcomb	1925402.5175	2466258.7219	1695-D-60728	1925555.7300	2466211.6000
Naschitti	1845416.9037	2472035.7886	1695-D-60728	1845407.4200	2472005.3900
BBN9[2]	1773044.4500	2479055.1100	1695-D-62535	1772870.6700	2479016.7600
Coyote Canyon	1728154.7800	2453530.3800	1695-D-62537	1728015.9400	2453444.5400
Rock Springs	1688592.1811	2415237.0131	1695-D-60119	1688625.8900	2415142.2500
Window Rock	1698957.0600	2363086.8899	1695-D-60260	1699016.9700	2362980.6500

Notes:

1. Coordinate system code NM83-WF
2. Crownpoint is now BBN9

REFERENCE DRAWINGS

1695-D-60119 ------ STA. 121300+00 TO STA. 121350+00
1695-D-60260 ------ STA. 122280+00 TO END
1695-D-60326 ------ Sectionalizing Valve Installation
1695-D-60328 ------ Crownpoint Turnout
1695-D-60728 ------ Typical Turnout
1695-D-62535 ------ BBN9 Site Plan
1695-D-62537 ------ Coyote Canyon Site Plan
1695-D-62556 ------ Thrust Blocks and Misc. Site Concrete
1695-D-62590 ------ Galvanic Anode Cathodic Protection

GENERAL NOTES

1. Provide nuts, bolts and gaskets suitable for cold water service.
2. All flanges shall be AWWA Class F as shown unless otherwise noted.
3. Additional fittings, flanges, or welded joints added by the contractor to facilitate installation of the piping are subject to Government approval.
4. Set grooved coupling gap to half of manufacturer's maximum allowable pipe end separation.
5. Thead-O-Let type fittings may be substituted for the 3000 lb. steel threaded couplings and threaded half-couplings. Weld Thread-O-Let fittings to pipe as recommended by Thread-O-Let manufacturer.
6. Welding symbols apply to the joints of all members of similar identification.
7. Hydrostatic test pressure for the turnout building steel pipe: 1.5 times the maximum operating pressure, see table.
8. Dimensions marked with an asterisk (*) to be field verified.
9. All valves and equipment to be rated for a minimum 300 psi working pressure.
10. Provide control valve flange supports to match control valve flanges, round or flat bottomed.
11. All elbows shall be long radius unless otherwise noted.
12. At locations where fittings have a greater wall thickness than steel piping, machine to match lighter pipe wall and taper excess thickness internally to form transition zone.
13. All fillet welds shall be returned.

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

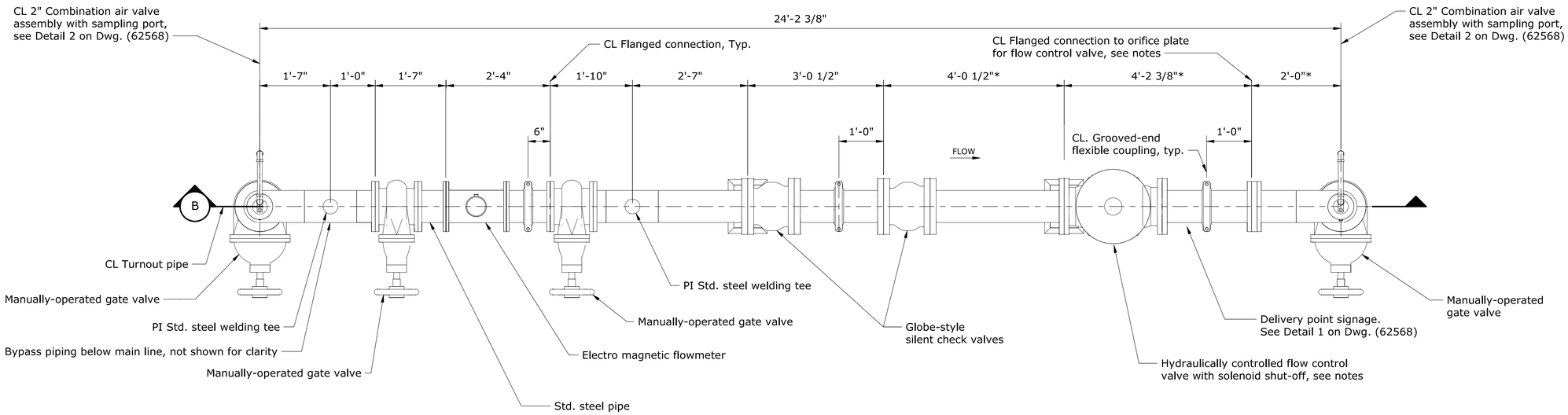
SAN JUAN LATERAL
TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E. Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

COVERSHEET

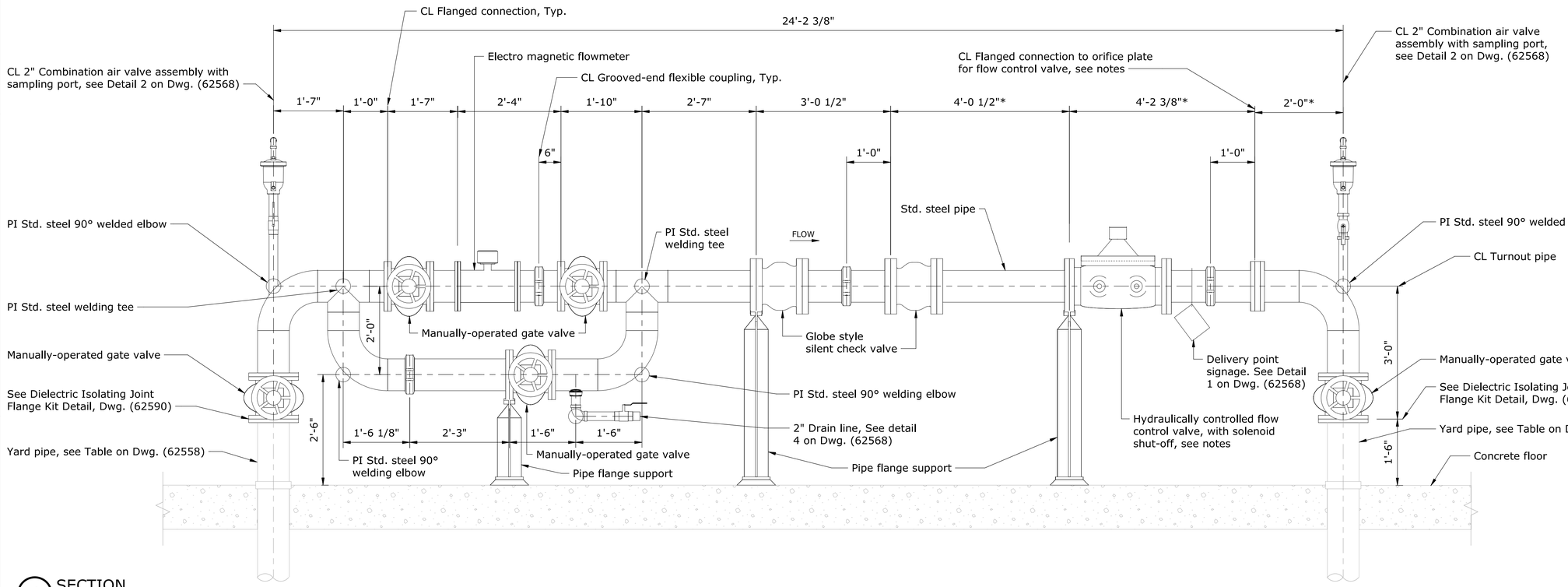
DETAILS

1695-D-62558



6 AND 8 INCH INTERIOR PIPE PLAN

Scale: 3/4" = 1'-0"



B SECTION

Scale: 3/4" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)
2. Locate and install flow control valve, orifice plate, and sensing line per flow control valve manufacturer's recommendations. It is understood that different control valve manufacturers have orifice plates in different locations. General layout is shown. Actual layout may vary.
3. This drawing depicts 8-inch diameter piping. The dimensions on this drawing indicates the layout for both 6-inch and 8-inch diameter piping.

0 1' 2' 3'
Scale: 3/4" = 1'-0"

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NEW MEXICO

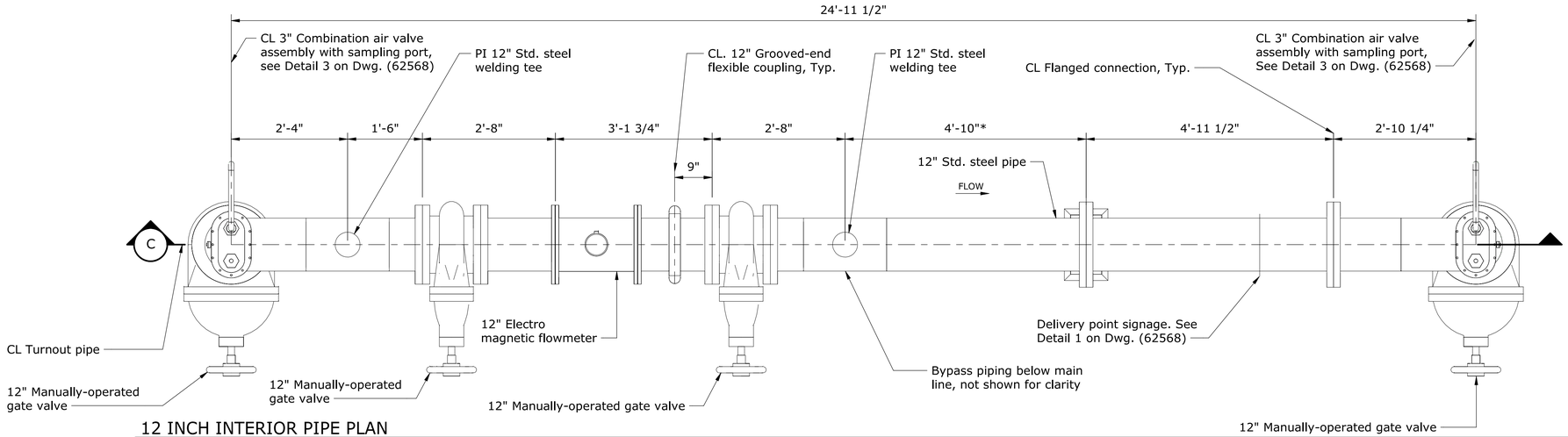
SAN JUAN LATERAL
TURNOUT BUILDINGS

CHRISTOPHER BELLECI	
DESIGNED	
CHRISTOPHER BELLECI	
DRAWN	
KYLE TAYLOR	
CHECKED	
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems	
DENVER, CO	2026-06-19

6 INCH AND 8 INCH DIAMETER TURNOUT PIPING

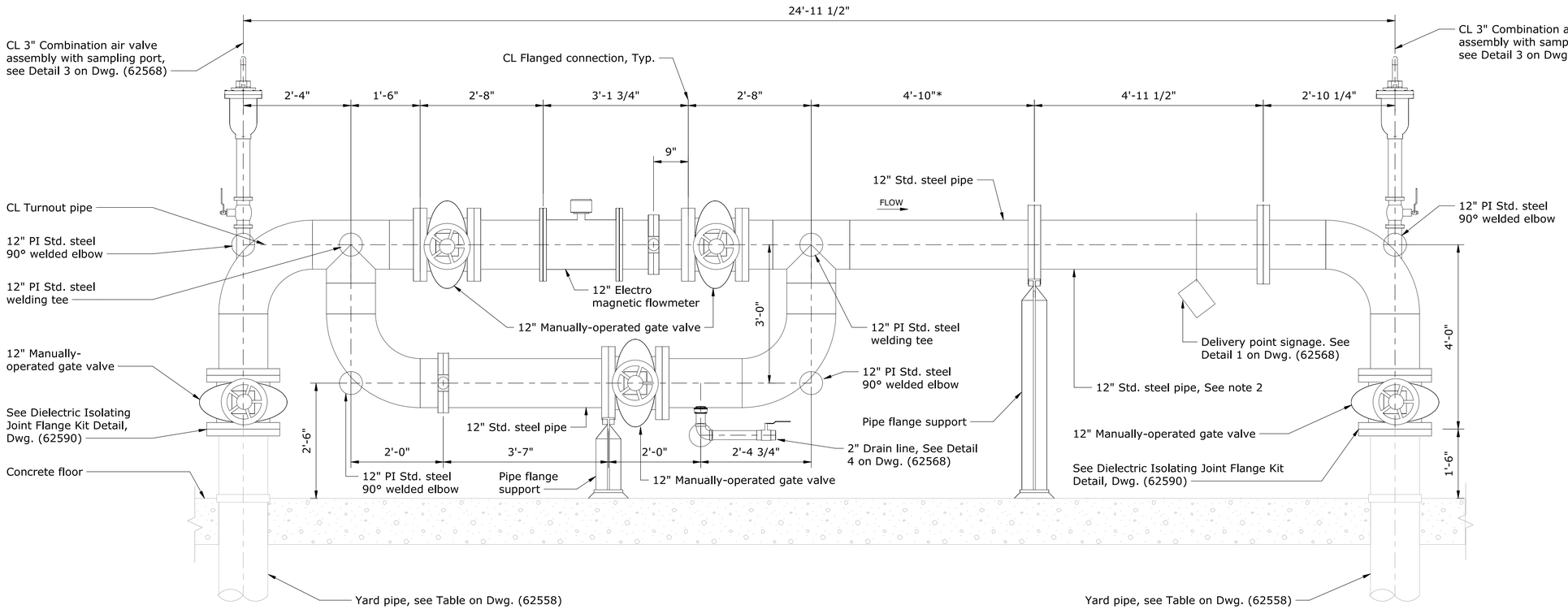
PLAN AND SECTION

1695-D-62559



12 INCH INTERIOR PIPE PLAN

Scale: 3/4" = 1'-0"



C SECTION

Scale: 3/4" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)
2. Install 12" Std. steel pipe spool piece in place of flow control valve and orifice plate.

0 1' 2' 3'

Scale: 3/4" = 1'-0"

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NEW MEXICO

SAN JUAN LATERAL

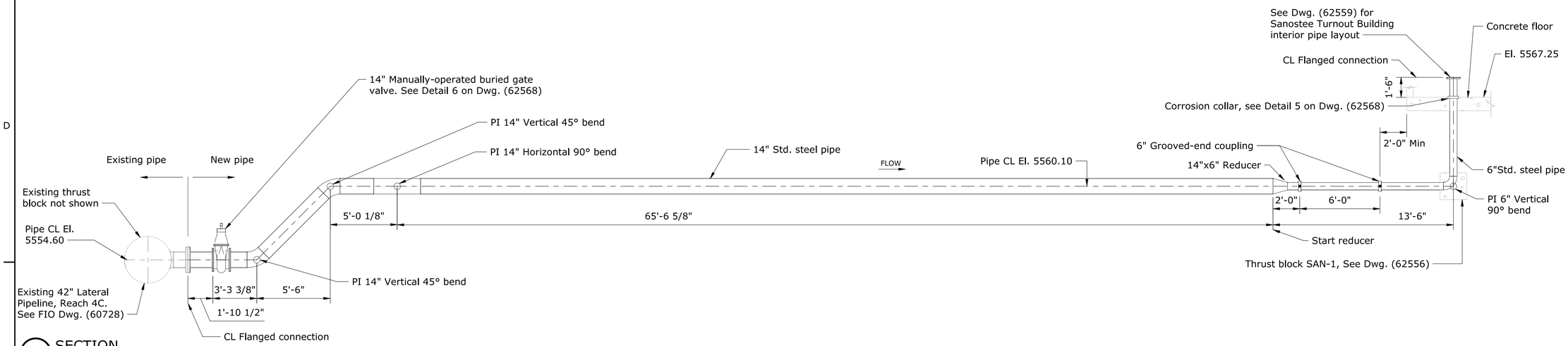
TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
	Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

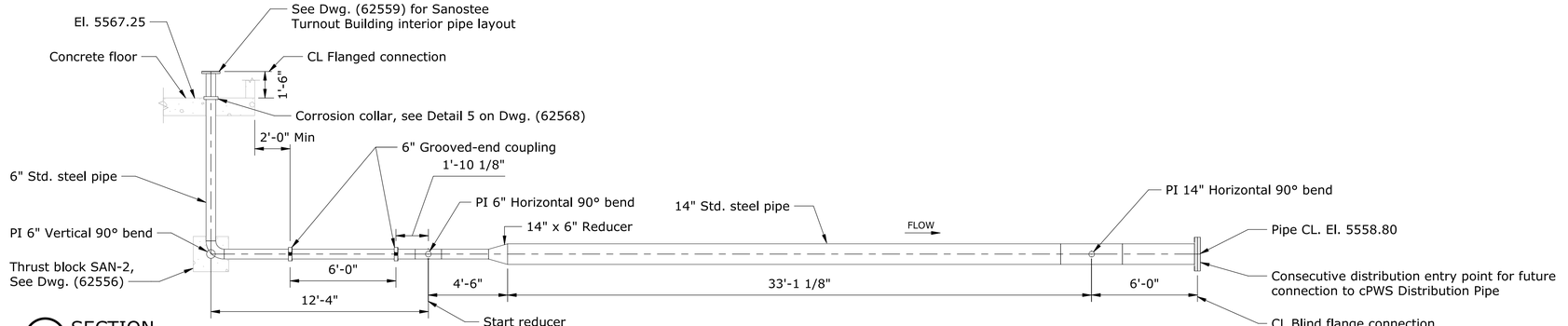
12 INCH DIAMETER TURNOUT PIPING

PLAN AND SECTION

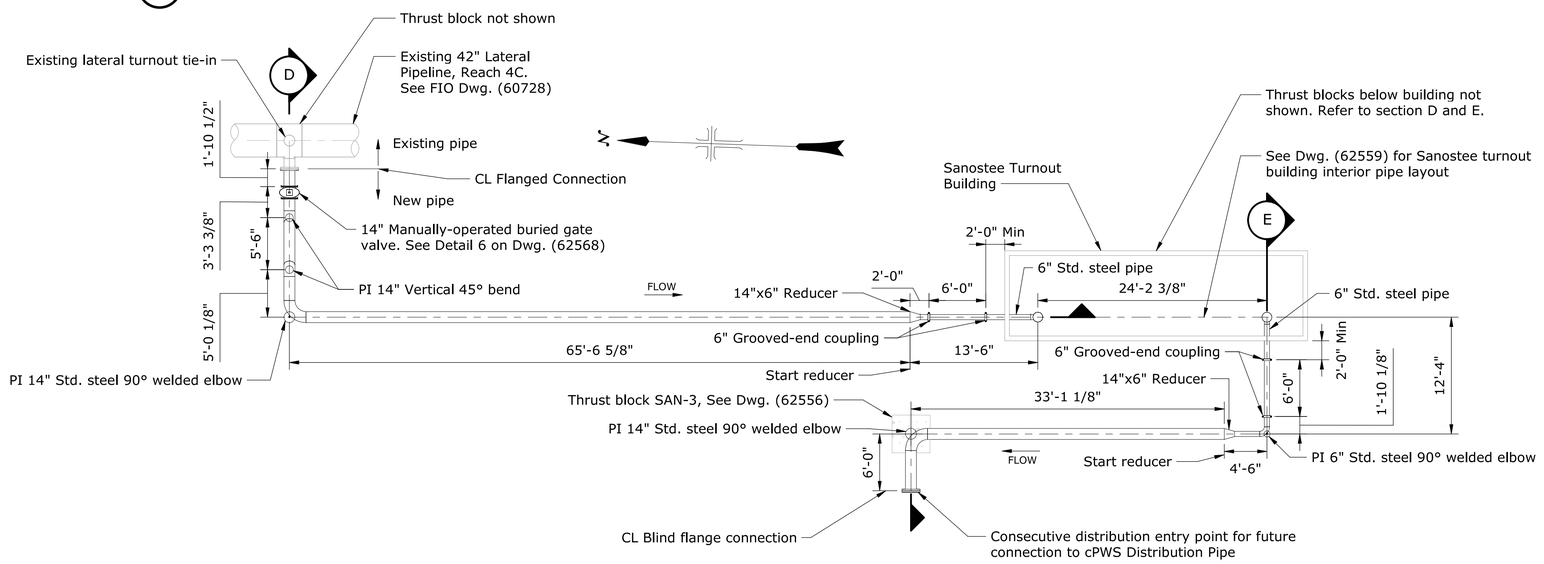
1695-D-62560



D SECTION
Scale: 1/4" = 1'-0"



E SECTION
Scale: 1/4" = 1'-0"



SANOSTEE YARD PIPE PLAN
Scale: 1/8" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 8'
Scale: 1/4" = 1'-0"

0 4' 8' 16'
Scale: 1/8" = 1'-0"

BUREAU OF RECLAMATION

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

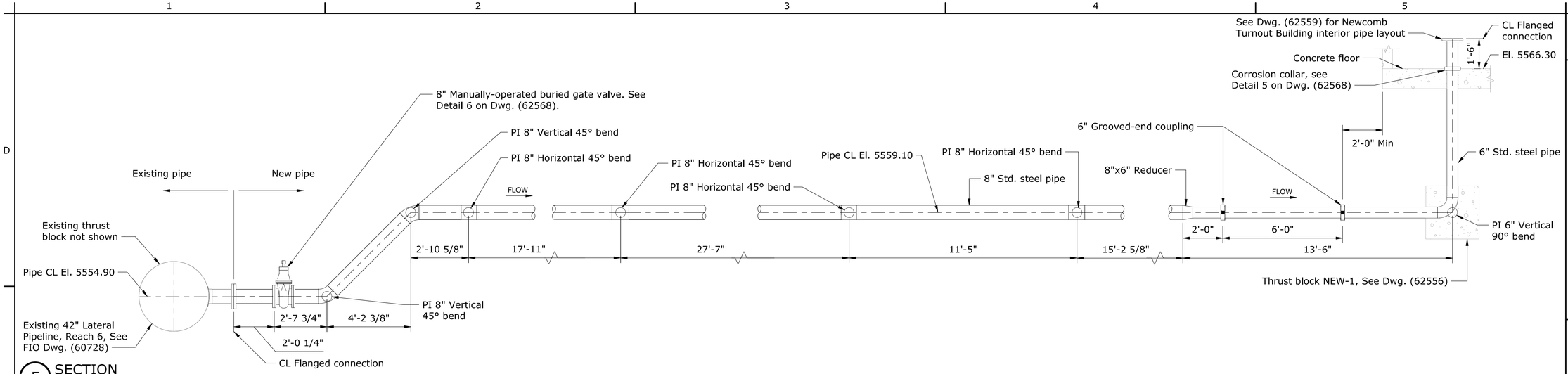
SAN JUAN LATERAL
TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
	Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

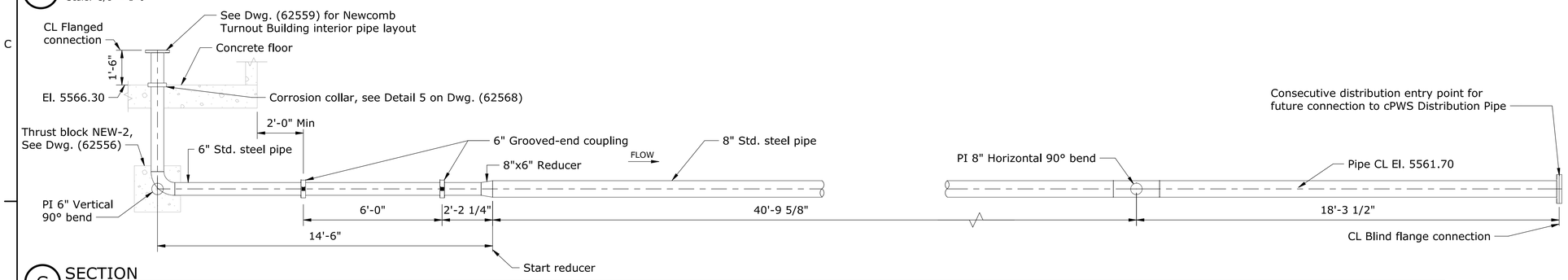
SANOSTEE YARD PIPE

PLAN AND SECTION

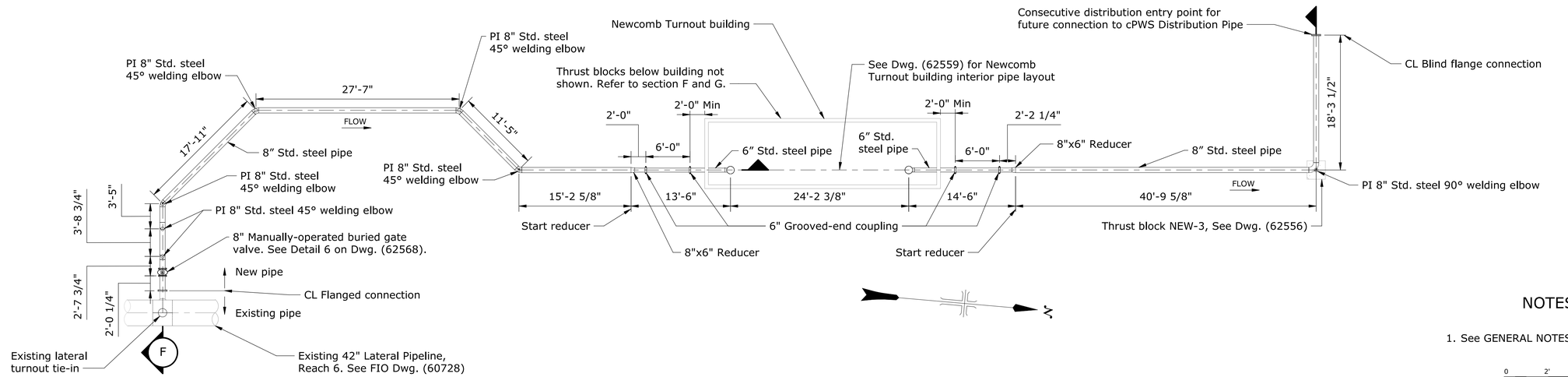
1695-D-62561



F SECTION
Scale: 3/8" = 1'-0"



G SECTION
Scale: 3/8" = 1'-0"



NEWCOMB YARD PIPE PLAN
Scale: 1/8" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

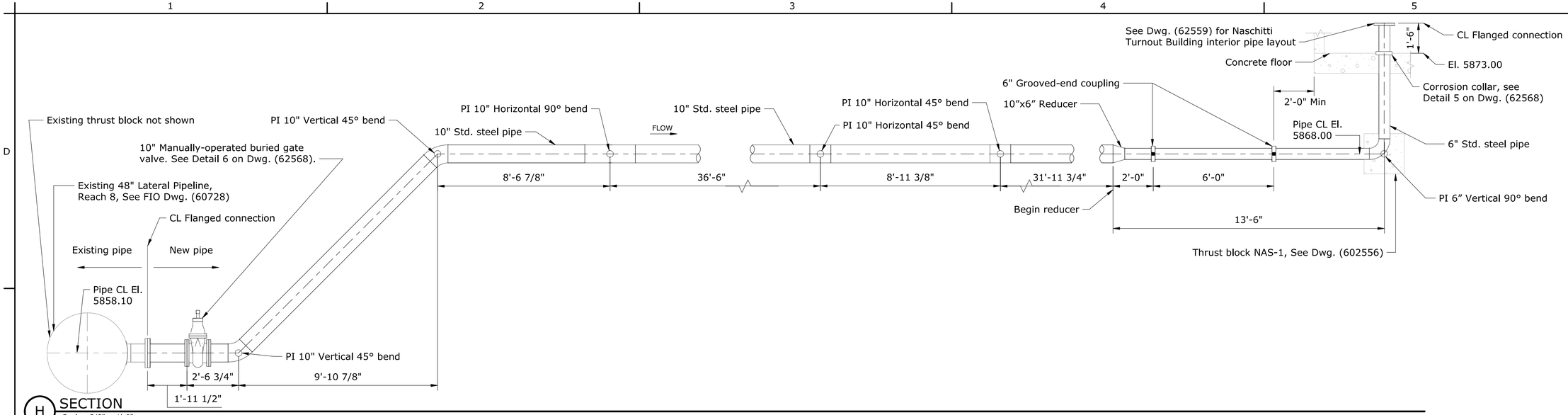
SAN JUAN LATERAL
TURNOUT BUILDINGS

CHRISTOPHER BELLECI	
DESIGNED	
CHRISTOPHER BELLECI	
DRAWN	
KYLE TAYLOR	
CHECKED	
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems	
DENVER, CO	2026-06-19

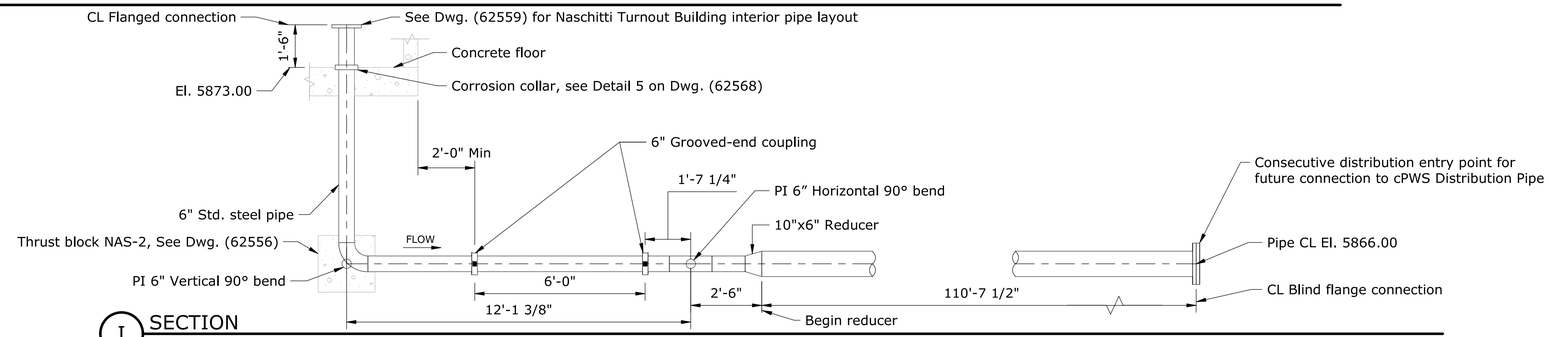
NEWCOMB YARD PIPE

PLAN AND SECTION

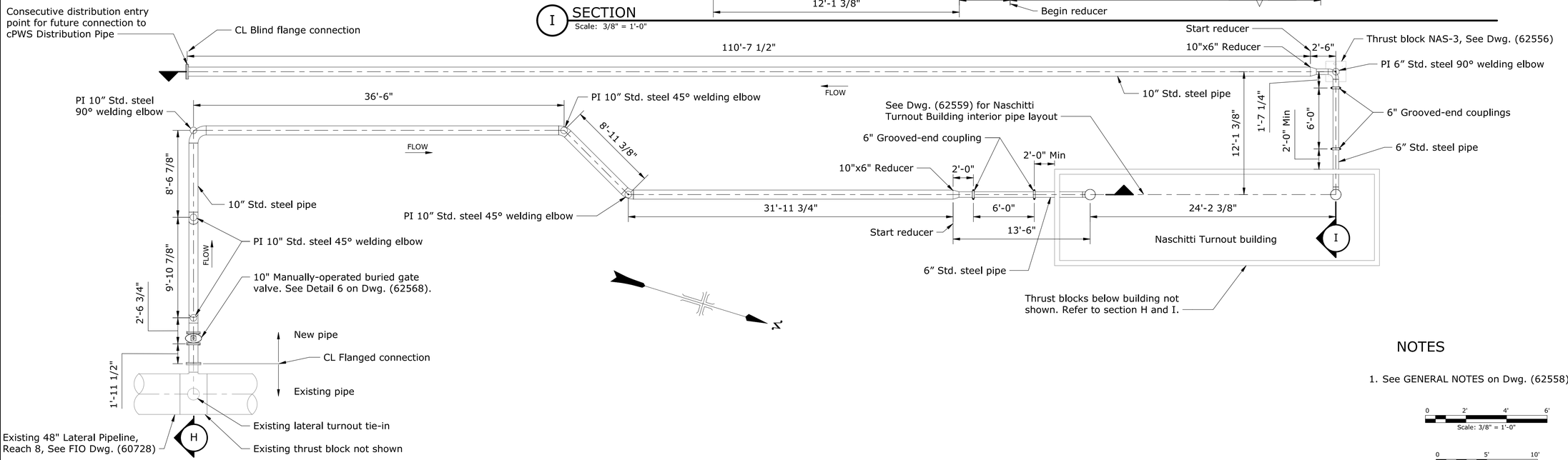
1695-D-62562



H SECTION
Scale: 3/8" = 1'-0"



I SECTION
Scale: 3/8" = 1'-0"



NASCHITTI YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

Scale: 3/8" = 1'-0"

Scale: 3/16" = 1'-0"

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

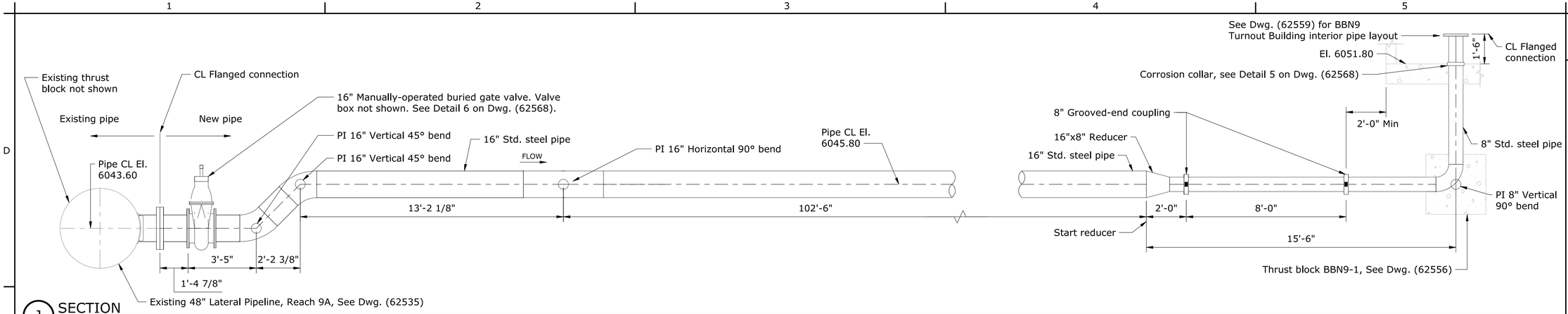
SAN JUAN LATERAL
TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems	
DENVER, CO	2026-06-19

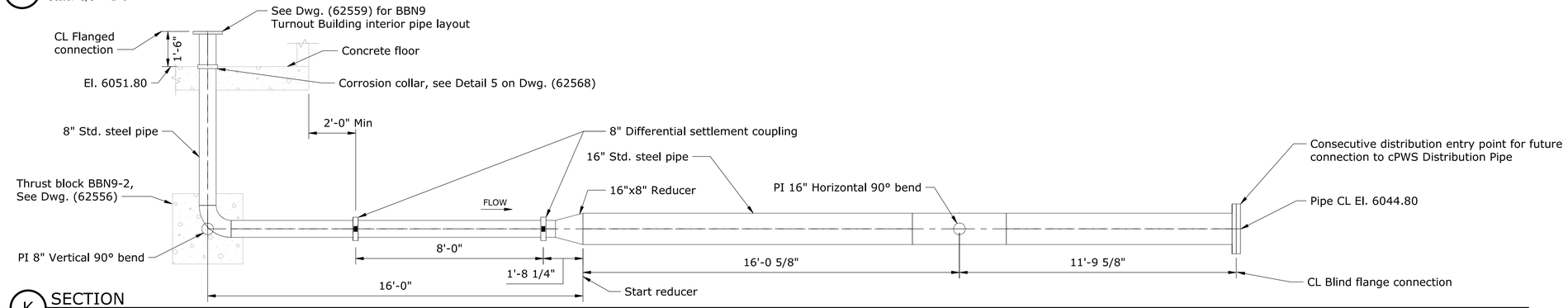
NASCHITTI YARD PIPE

PLAN AND SECTION

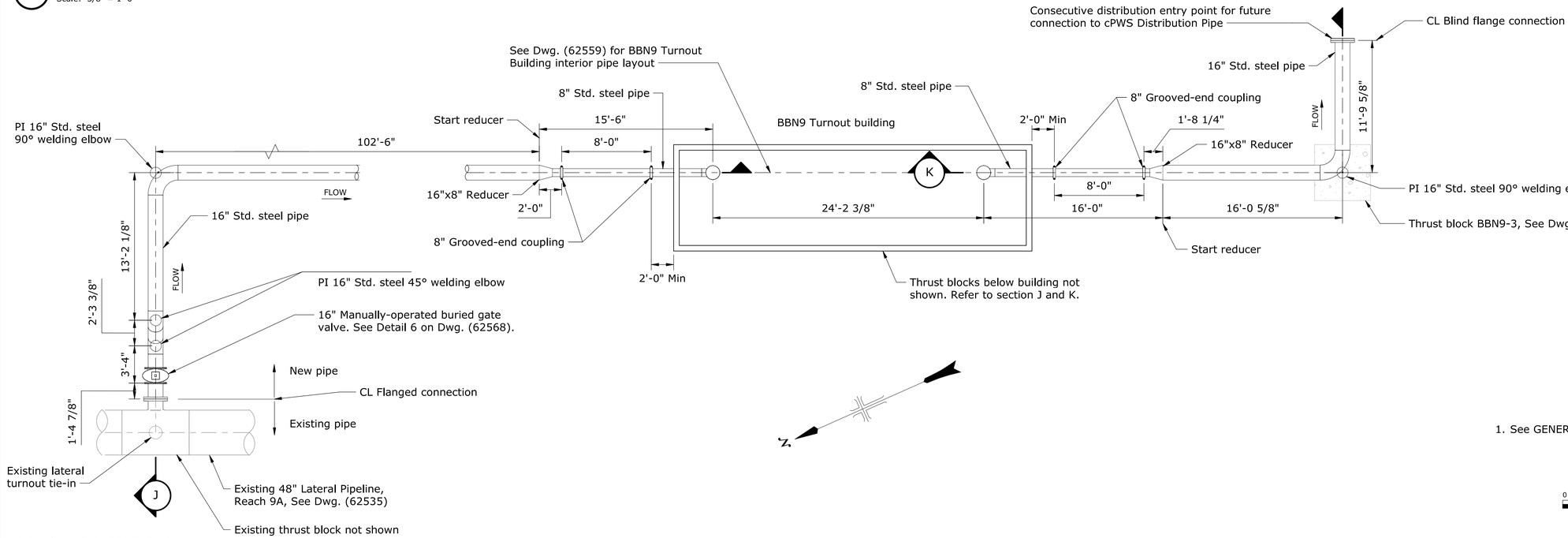
1695-D-62563



J SECTION
Scale: 3/8" = 1'-0"



K SECTION
Scale: 3/8" = 1'-0"



BBN9 YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

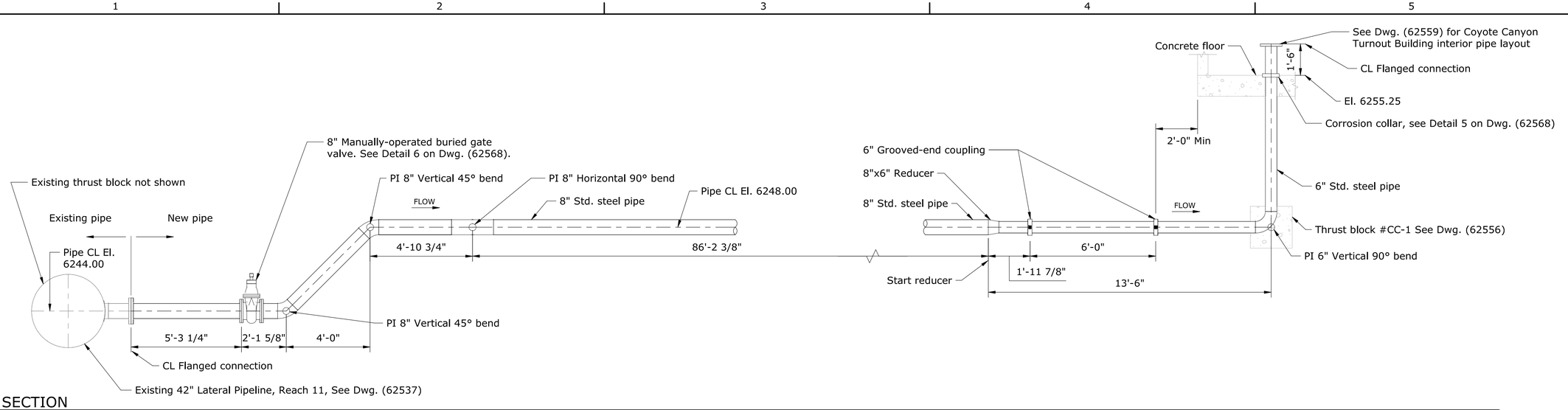
SAN JUAN LATERAL
TURNOUT BUILDINGS

CHRISTOPHER BELLECI	
DESIGNED	
CHRISTOPHER BELLECI	
DRAWN	
KYLE TAYLOR	
CHECKED	
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
Peer Reviewer - Supervisor, Mechanical Piping and Systems	
DENVER, CO	2026-06-19

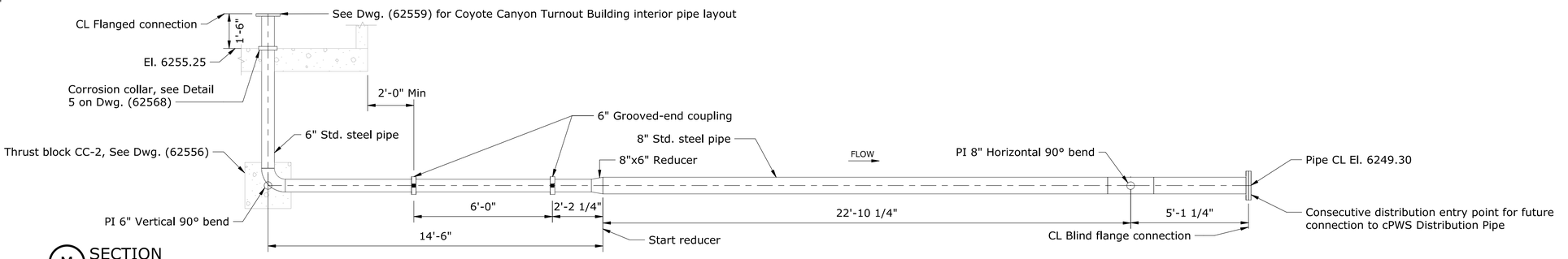
BBN9 YARD PIPE

PLAN AND SECTION

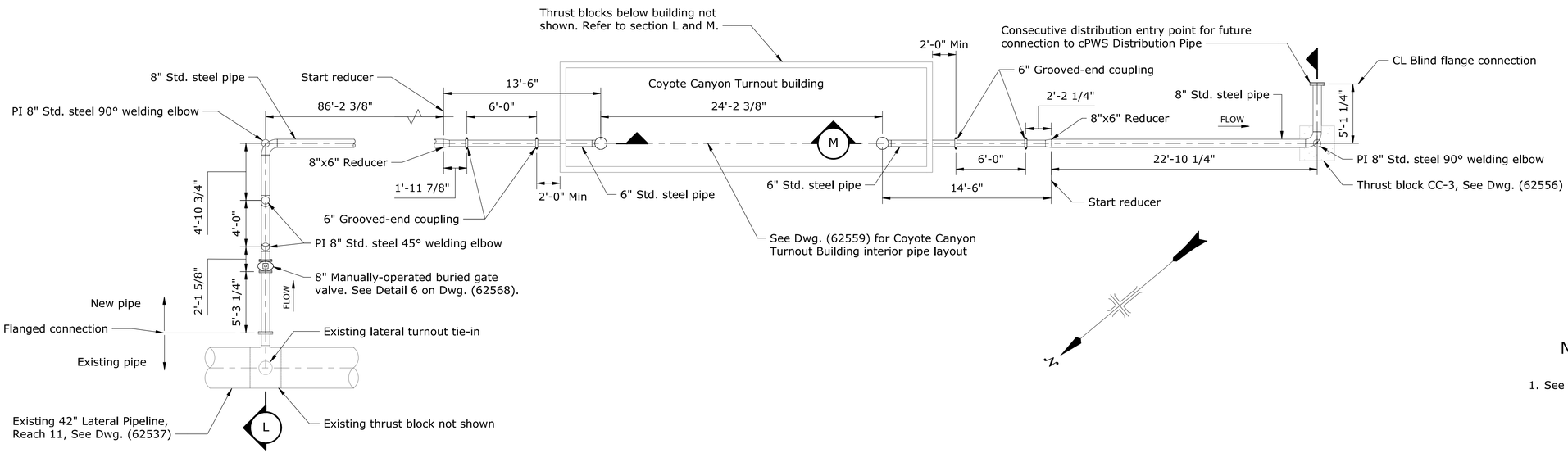
1695-D-62564



L SECTION
Scale: 3/8" = 1'-0"



M SECTION
Scale: 3/8" = 1'-0"



COYOTE CANYON YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
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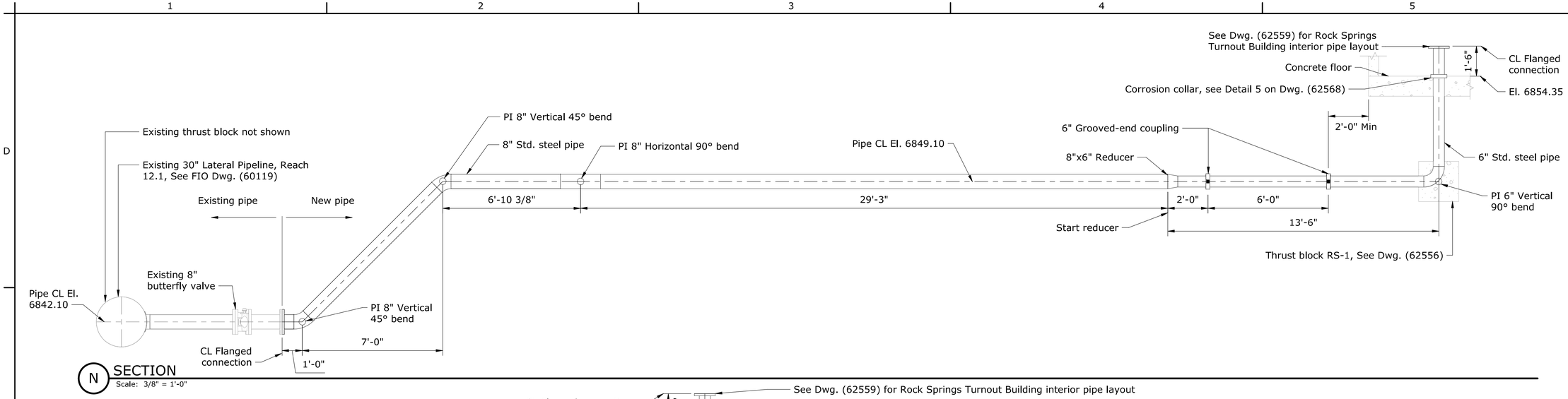
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E. Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

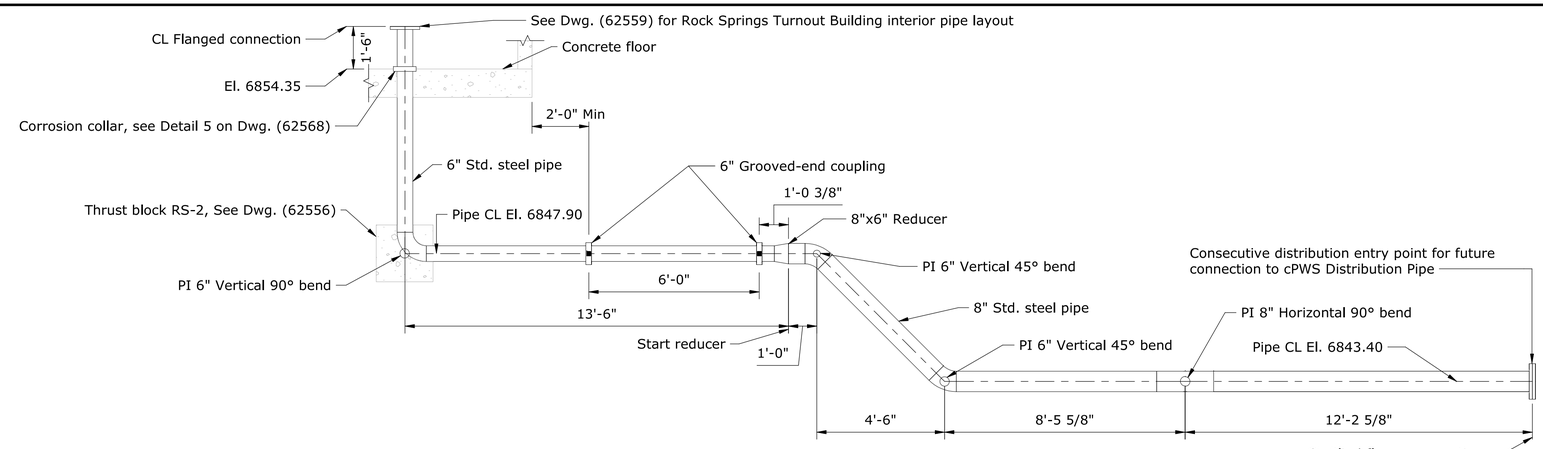
COYOTE CANYON YARD PIPE

PLAN AND SECTION

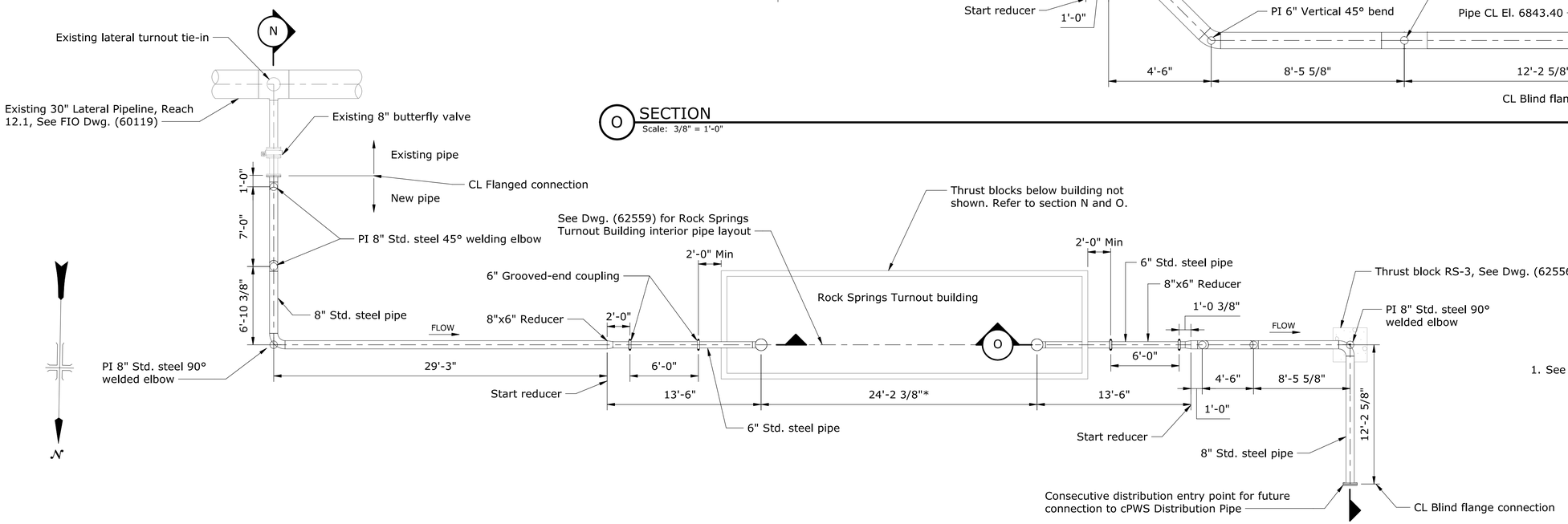
1695-D-62565



N SECTION
Scale: 3/8" = 1'-0"



O SECTION
Scale: 3/8" = 1'-0"



ROCK SPRINGS YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
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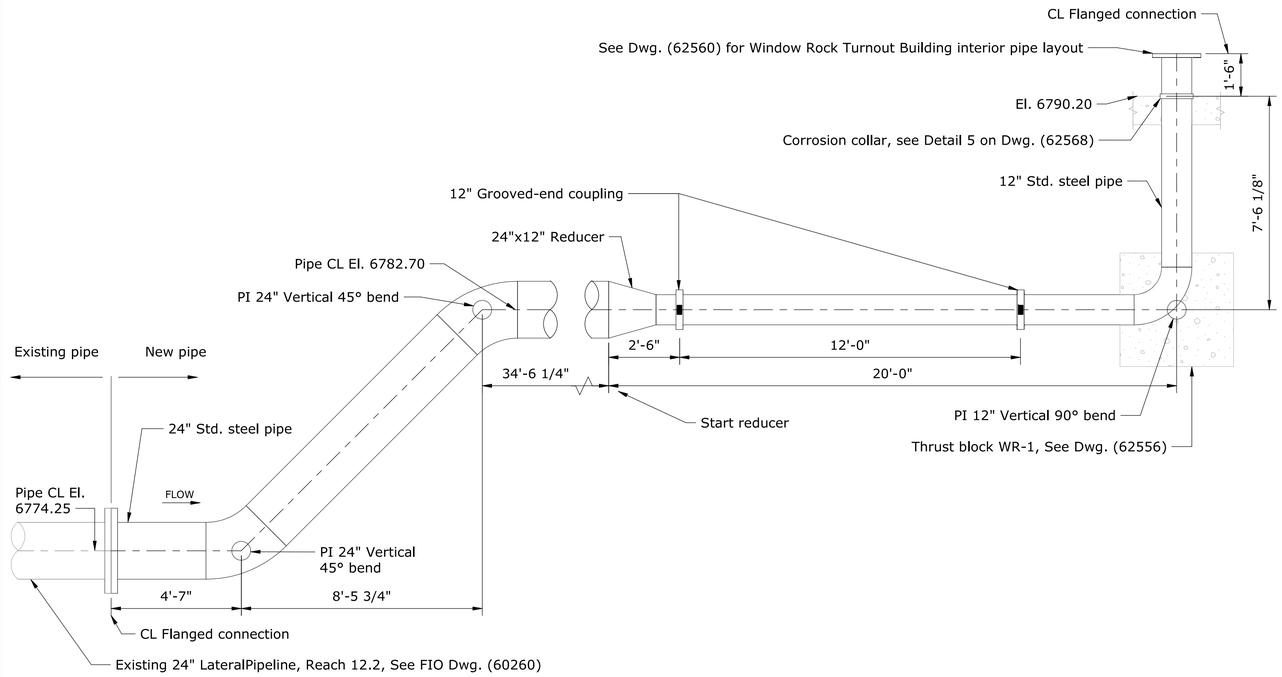
	Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

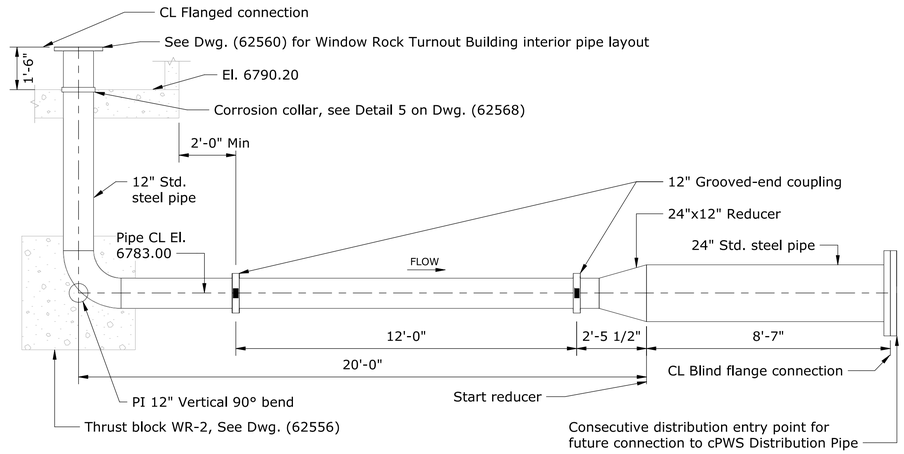
ROCK SPRINGS YARD PIPE

PLAN AND SECTION

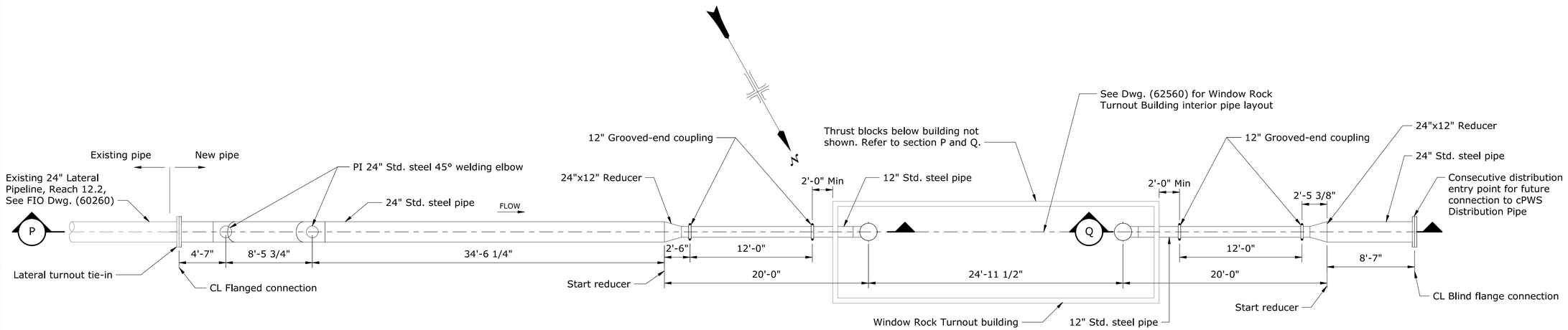
1695-D-62566



P SECTION
Scale: 3/8" = 1'-0"



Q SECTION
Scale: 3/8" = 1'-0"



WINDOW ROCK YARD PIPE PLAN
Scale: 3/16" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

0 2' 4' 6'
Scale: 3/8" = 1'-0"

0 5' 10'
Scale: 3/16" = 1'-0"

BUREAU OF RECLAMATION

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

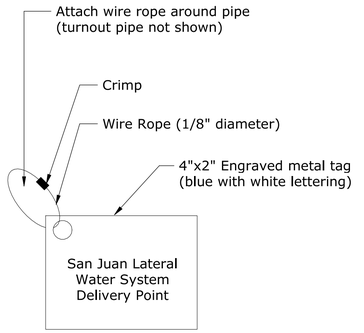
SAN JUAN LATERAL
TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
	Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

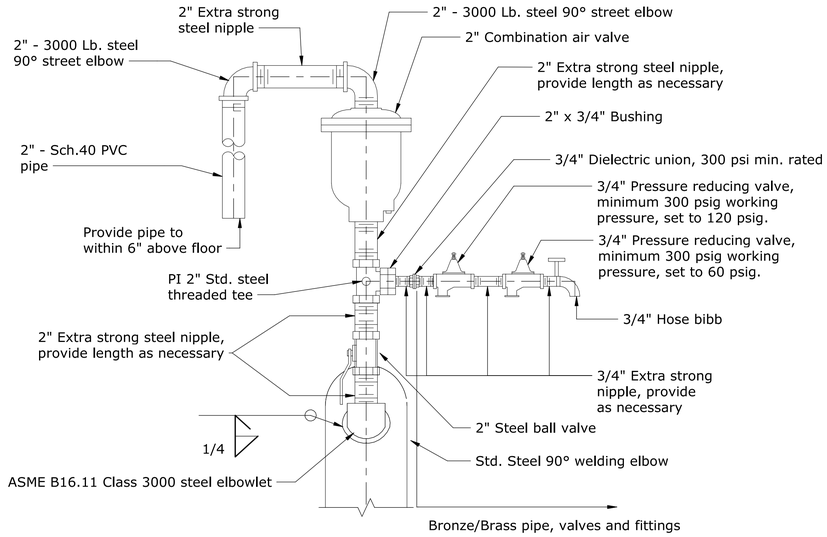
WINDOW ROCK YARD PIPE

PLAN AND SECTION

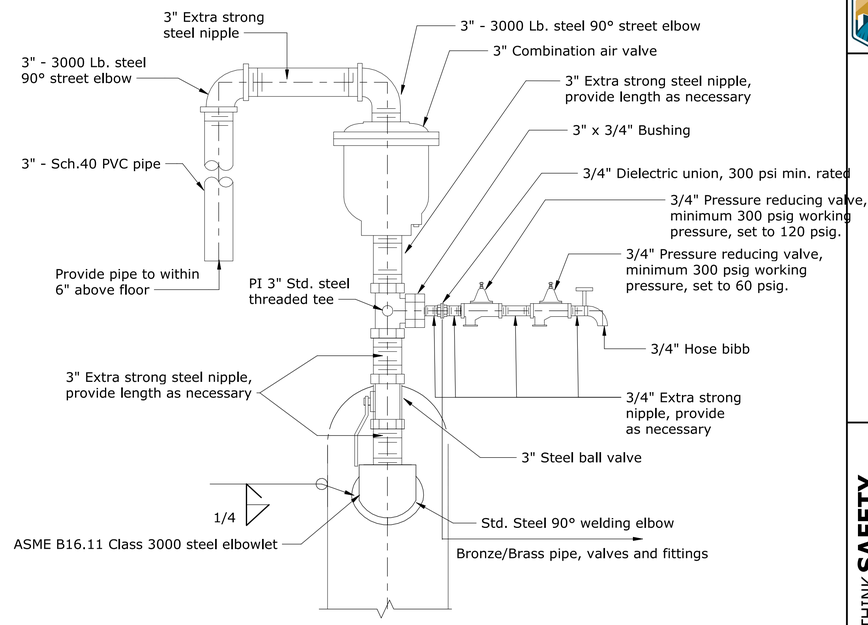
1695-D-62567



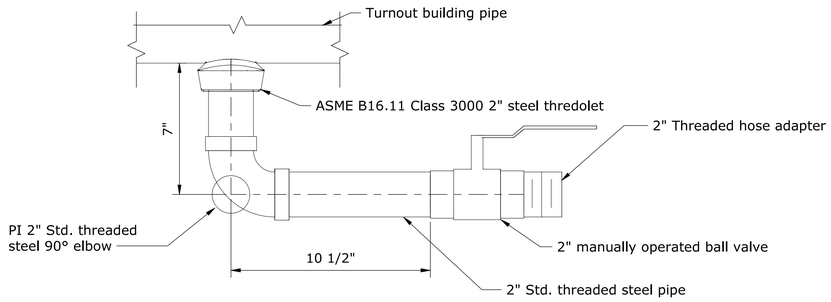
1 DELIVERY POINT SIGNAGE DETAIL
Scale: 6" = 1'-0"



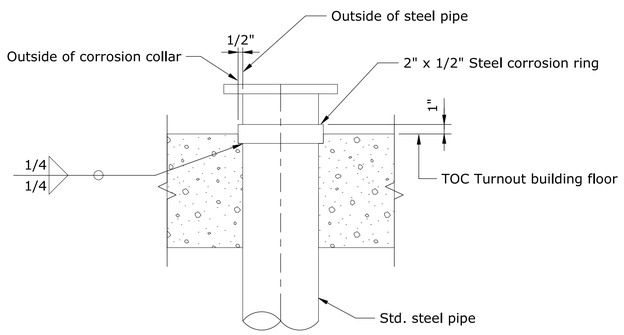
2 2-INCH COMBINATION AIR VALVE WITH SAMPLING PORT DETAIL
Scale: 1 1/2" = 1'-0"



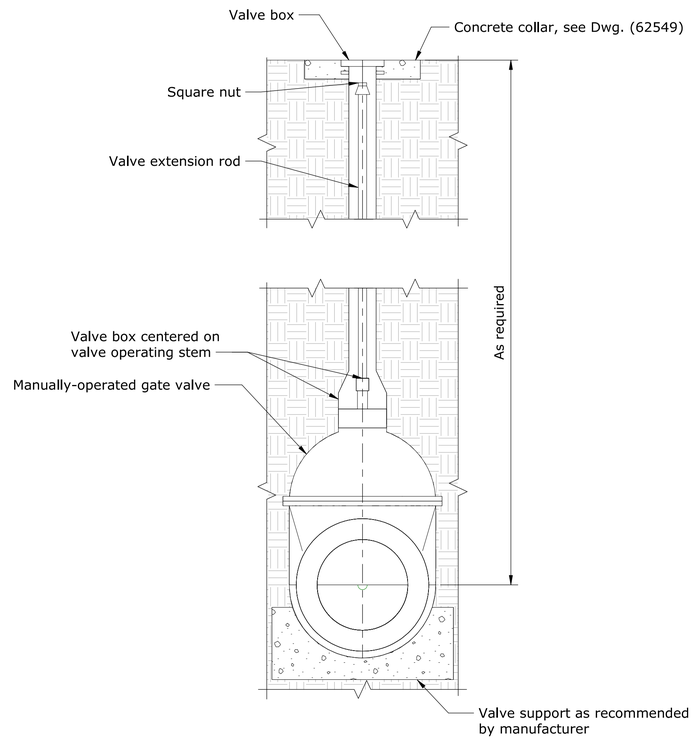
3 3 INCH COMBINATION AIR VALVE WITH SAMPLING PORT DETAIL
Scale: 1 1/2" = 1'-0"



4 INTERIOR PIPE DRAIN DETAIL
Scale: 3" = 1'-0"



5 Corrosion Collar
Scale: 1 1/2" = 1'-0"



6 BURIED VALVE BOX DETAIL
Scale: 1" = 1'-0"

0 1" 2" 3" 4"
Scale: 6" = 1'-0"

0 1/2' 1' 1 1/2'
Scale: 1 1/2" = 1'-0"

0 3" 6" 9"
Scale: 3" = 1'-0"

0 1' 2'
Scale: 1" = 1'-0"

NOTES

1. See GENERAL NOTES on Dwg. (62558)

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NEW MEXICO

SAN JUAN LATERAL
TURNOUT BUILDINGS

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DRAWN	CHRISTOPHER BELLECI
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TECH. APPR.	JEREMY HAMMOND, P.E.
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	Peer Reviewer - Supervisor, Mechanical Piping and Systems
DENVER, CO	2026-06-19

DETAILS

DETAILS

1695-D-62568

INSTRUMENTATION DEVICE AND FUNCTION SYMBOLS

Shared Display & Control System	Alternate Display, Shared Control	High Level Control System & Software	Standalone	Location and Accessibility
				Located in field Not panel, cabinet, or console mounted Visible at field location Normally operator accessible
				Located in or on front of central or main panel or console Visible on front of panel or on video display Normally operator accessible at panel front or console
				Located in rear of central or main panel Located in cabinet behind panel Not visible on front of panel or on video display Not normally operator accessible at panel or console
				Located in or on front of secondary or local panel or console Visible on front of panel or on video display Normally operator accessible at panel front or console
				Located in rear of secondary or local panel Located in field cabinet Not visible on front of panel or on video display Not normally operator accessible at panel or console
				Relay
				Undefined Interlock
				PLC Function
				Pilot Light or Gauge Glass Illuminator
				Dual Function or Instruments sharing common housing

FACILITY AREA DESIGNATION

Position	Description	Name		Number
1	Lateral	San Juan		S
		Cutter		C
2 & 3	Site	Lateral	Facility	
		San Juan	Intake & Pumping Plant	01
			Water Treatment Plant	02
			Pumping Plant No. 2	03
			Pumping Plant No. 3	04
			Pumping Plant No. 4	05
			Pumping Plant No. 7	06
			Pumping Plant No. 8	07
			Black Hat Tank	08
			McKinley Surge Tank	09
			NAPI Turnout	11
			Sanostee Turnout	12
			Newcomb Turnout	13
			Naschitti Turnout	14
			BBN9 (Crownpoint) Turnout	15
			Rock Springs Turnout	16
			Coyote Canyon Turnout	17
			Window Rock Turnout	18
		Cutter	Cutter Dam	01
			Pumping Plant No. 1	02
			Pumping Plant No. 2	03
			Regulating Tank No. 2	04
			NAPI Turnout	11
4 & 5	Physical Location	Yard		01
		Pump Room		02
		Air Chamber Room		03
		Chemical Building		04
		Turnout Building		05

NGWSP FACILITIES

Lateral	Facility	Formal Name
San Juan	Intake & Pumping Plant	Tsétaak'áán
San Juan	Water Treatment Plant	
San Juan	Pumping Plant 2	Tsé Da' azkání
San Juan	Pumping Plant 3	Tó Ałts'ísí
San Juan	Pumping Plant 4	Tooh Haltsooi
San Juan	Pumping Plant 7	Bahastł'ah
San Juan	Pumping Plant 8	Tohlakai
San Juan	Pumping Plant 12	
San Juan	Pumping Plant 13	Tsé Íí'áhí
San Juan	Pumping Plant 14.1	Tsé Daat'ees
San Juan	Pumping Plant 14.2	Iyanbito
San Juan	Pumping Plant 14.6	
San Juan	Pumping Plant 17.a	Tsé Łichii dah Azkání
Cutter	Cutter Dam	
Cutter	Pumping Plant No. 1	
Cutter	Pumping Plant No. 2	
Cutter	Regulating Tank No. 2	
Cutter	Water Treatment Plant	
Cutter	Regulating Tank No. 3	

NOTES

1. Equipment and instrumentation symbols are based on ANSI/ISA and PIP standards.

INSTRUMENT AND FUNCTION IDENTIFICATION

Letter	Column 1 Measuring/Initiating Variable	Column 2 Variable Modifier	Column 3 Readout/Passive Function	Column 4 Output/Active Function	Column 5 Function Modifier
A	Analysis		Alarm		
B	Burner, Combustion		Users Choice	Users Choice	Users Choice
C	Users Choice			Control	Close
D	Users Choice	Difference,Differential			Deviation
E	Voltage		Sensor, Primary Element		
F	Flow, Flow Rate	Ratio			
G	Users Choice		Glass, Gauge, Viewing Device		
H	Hand				High
I	Current		Indicate		
J	Power		Scan		
K	Time, Schedule	TIme Rate of Change		Control Station	
L	Level		Light		Low
M	Users Choice				Middle, Intermediate
N	Users Choice		Users Choice	Users Choice	Users Choice
O	Users Choice		Orifice, Restriction		Open
P	Pressure		Point (Test Connection)		
Q	Quantity	Integrate, Totalize	Integrate, Totalize		
R	Radiation		Record		Run
S	Speed, Frequency	Safety		Switch	Stop
T	Temperature			Transmit	
U	Multivariable		Multifunction	Multifunction	
V	Vibration, Mechanical Analysis			Valve, Damper, Louver	
W	Weight, Force		Well, Probe		
X	Unclassified	X-axis	Accessory Devices, Unclassified	Unclassified	Unclassified
Y	Event State, Presence	Y-axis		Auxiliary Device	
Z	Position, Dimension	Z-axis, Safety Instrument System		Driver, Actuator, Unclassified final control element	

TYPICAL TAG NUMBERS, TAGGING STANDARD & DESIGNATION

INSTRUMENTATION

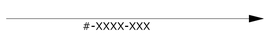
EQUIPMENT

X####-XX-###

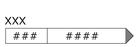
VALVES

X####-XXX-XX-###

PIPE



OFF-PAGE CONNECTOR



BUREAU OF RECLAMATION

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

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DESIGNED
CHRISTOPHER BELLECI
DRAWN
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CHECKED
TECH. APPR. JEREMY HAMMOND, P.E.
APPROVED CODY YOUNG, P.E.
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LEGEND

1695-D-62569

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VALVE SYMBOLS

Symbol	Description
	Butterfly Valve
	Ball Valve
	Globe Valve
	Gate Valve
	Plug Valve
	Needle Valve
	Pinch Valve
	Angled Valve
	Three Way Valve
	Four Way Valve
	Knife Valve
	Diaphragm Valve
	Rotary Valve
	Stop Check Valve
	Check Valve
	Control Valve
	Rupture Disc (Pressure Relief)
	Rupture Disc (Vacuum Relief)
	Pressure Relief Valve
	Vacuum Relief Valve
	Pressure and Vacuum Relief Valve
P	Pilot Operated Relief Valve

CONTROL VALVE ACTUATOR SYMBOLS

Symbol	Description
	Pressure Reducing Regulator
	Pressure Reducing Regulator with External Tap
	Back Pressure Regulator
	Back Pressure Regulator with External Tap
	Differential Pressure Reducing Regulator
	Pressure Reducing Regulator with Integral Outlet
	Piston/Cylinder Actuator
	Handwheel/Manual Actuator
	Diaphragm Actuator - Spring Return Actuator
	Pressure Balanced Diaphragm Actuator
	Diaphragm Actuator
	Throttling Actuator
	Spring Actuator
M	Motor Actuator
S	Solenoid Actuator
I/M	Electro-Hydraulic Actuator
D	Digital Actuator
S	Solenoid Actuator with Manual Reset

MECHANICAL COMPONENTS

Symbol	Description
	Pump
	Dome Roof Tank
	Electric Motor
	Agitator
	Compressor
	Vent Silencer
	Flange
	Orifice
	Duplex Strainer
	Y-Strainer
	Pulsation Damper
	Vertical Can Pump
M	Magnetic Flow Meter
	Pitot Tube Flow Meter
	Chlorine Residual Analyzer
	Peristaltic Pump
	Calibration Column

MECHANICAL COMPONENTS

Symbol	Description
	Heat Exchanger
	Vent
	Open Drain
	Expansion Joint

MECHANICAL COMPONENTS

Symbol	Description
	Flow Conditioning Device
	Conveyor Scraper
	Flexible Hose

GATES

Symbol	Description
	Weir Gate
	Fabricated Slide Gate
	Stop Log

SIGNAL PATHS

Linetype	Function
	Primary Fluid System
	Secondary Fluid System
	Instrument Supply
	Jacketed Pipe Segment
	Electrical Signal
	Pneumatic Signal
	Hydraulic Signal
	Data/Software Communication
	Capillary Tube
	Mechanical Link

EQUIPMENT TYPE

Designation	Description	Designation	Description
A	Agitator, Mixer	M	Motor
AC	Air Chamber	MBV	Motor Operated Ball Valve
ADV	Automatic Drain Valve	MBFV	Motor Operated Butterfly Valve
ARV	Air Release Valve	MCE	Motor Control Equipment
AVC	Combination Air Valve	P	Pump
BFP	Back Flow Preventer	PCV	Pressure Control Valve
BFV	Butterfly Valve	PRV	Pressure Relief Valve
BGT	Butterfly Gate	PSV	Pressure Safety Valve
BV	Ball Valve	SGT	Slide Gate
C	Compressor	SL	Stop Log
CP	Control Panel	SR	Strainer
CV	Check Valve	SRB	Bucket Strainer
FGT	Flap Gate	SRY	Y-Strainer
FIL	Filter	SRT	T-Strainer
GLV	Globe Valve	SV	Solenoid Valve
GV	Gate Valve	TK	Tank
HTR	Heater	VRV	Air Vacuum Valve
IMV	Integrated Manifold Valve	WGT	Weir Gate

LINE SERVICE CODES

AV	Atmospheric Vent	ME	Methanol
BD	Blowdown	N	Nitrogen
BRR	Brine Return	NA	Caustic
BRS	Brine Supply	NAS	Caustic Sewer
CHS	Chemical Sewer	NG	Natural Gas
CLF	Chlorine Feed	NH	Amonia
CLI	Chlorine Injection	OWS	Oily Water Sewer
CLS	Chlorine Sample	OX	Oxygen
CV	Cold Vent	P	General Process
CWR	Cooling Water Return	PA	Plant Air
CWS	Cooling Water Supply	PW	Process Water
DMW	Demineralized Water	RV	Relief Vent
DR	Drain	RW	Raw Water
DS	Dilution Steam	SO	Seal Oil
DW	Drinking Water	SS	Sanitary Sewer
FF	Flushing Fluid	STS	Storm Sewer
FG	Fuel Gas	SW	Service Water
FO	Fuel Oil	TWR	Tempered Water Return
FW	Fire Water	TWS	Tempered Water Supply
IA	Instrument Air	WW	Waste Water
LNG	Liquefied Natural Gas		
LO	Lube Oil		
LPG	Liquefied Petroleum Gas		

VALVE SYSTEM DESIGNATION

Designation	Description
AC#	Air Chamber
ACS	Air Chamber System
CS	Chlorine Sampling
CST	Chlorine Storage Tank
CF	Chlorine Feed
DH	Discharge Header
ES	Emergency Shower
ST	Storage Tank / Surge Tank
TO	Turnout
TS#	Tank Sampling
U#	Unit

PIPE TYPE

Designation	Description
CS-3/8" W.T.	Carbon Steel $\frac{3}{8}$" Wall Thickness
CS-1/4" W.T.	Carbon Steel $\frac{1}{4}$" Wall Thickness
CS Sch 10	Carbon Steel Schedule 10
CS Sch 20	Carbon Steel Schedule 20
CS-Std	Carbon Steel Standard
CU	Copper
PVC Sch 80	Polyvinyl Chloride Schedule 80
SS-Std	Stainless Steel Standard
SS 316L Sch 80	Stainless Steel 316L Schedule 80

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NAVAJO-GALLUP WATER SUPPLY PROJECT, NM
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

CHRISTOPHER BELLECI — DESIGNED
CHRISTOPHER BELLECI — DRAWN
KYLE TAYLOR — CHECKED
TECH. APPR. JEREMY HAMMOND, P.E.
APPROVED CODY YOUNG, P.E.
PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
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SYMBOLS AND ABBREVIATIONS

1695-D-62570

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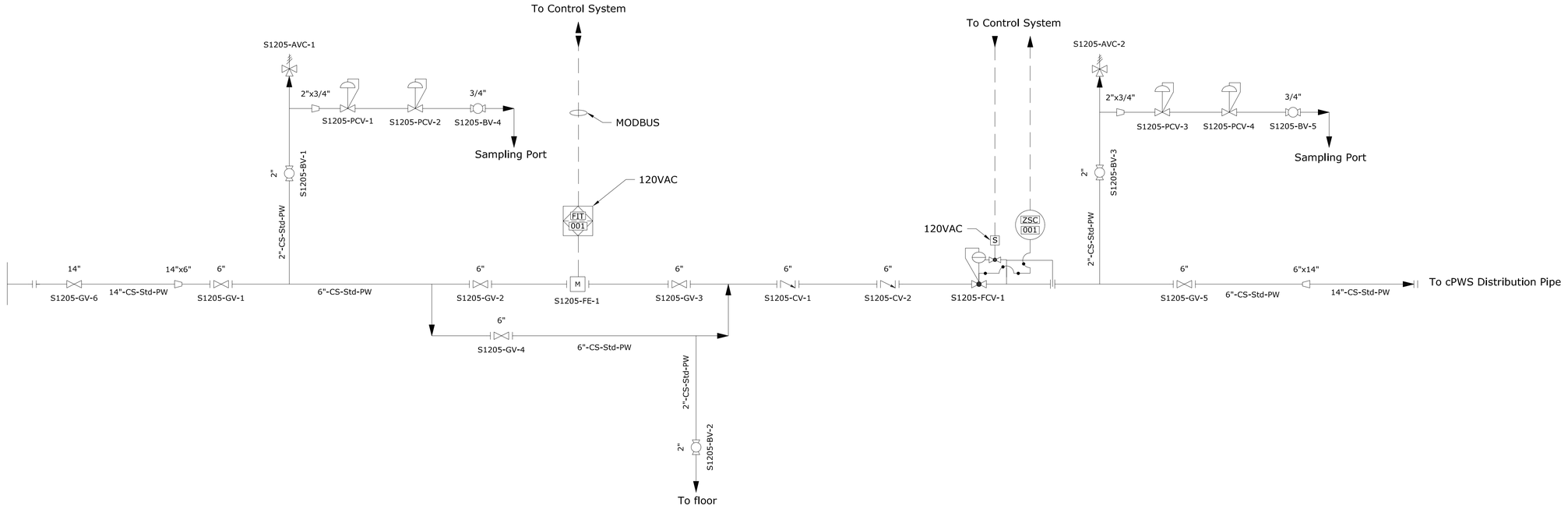

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NEW MEXICO

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TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E. PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
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SANOSTEE TURNOUT BUILDING

1695-D-62571

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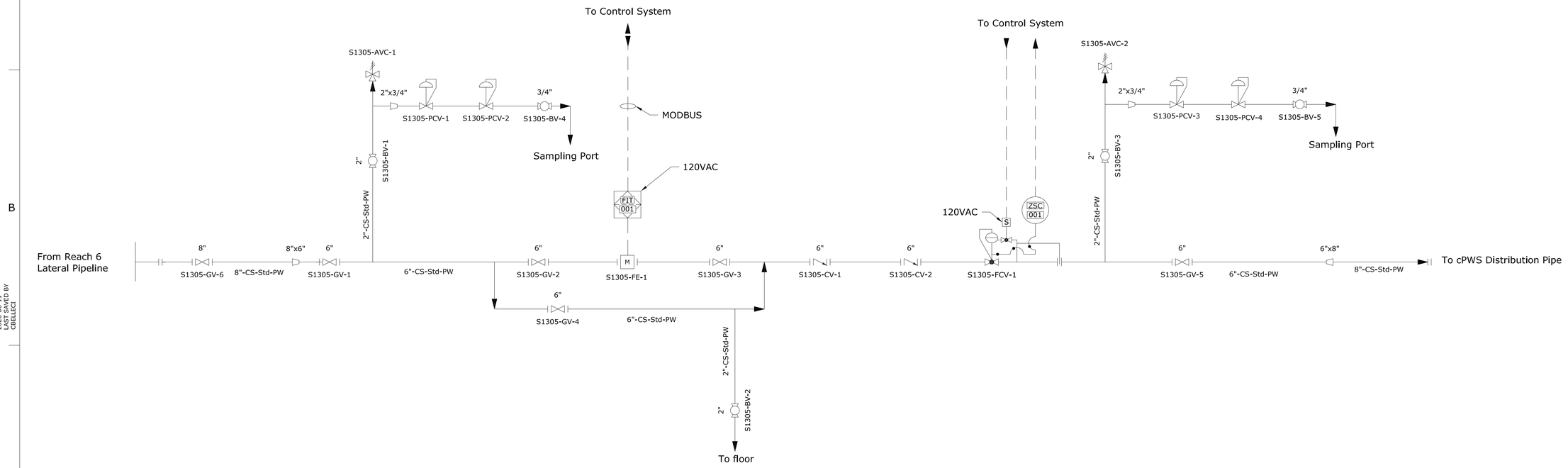

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E. PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
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NEWCOMB TURNOUT BUILDING

1695-D-62572

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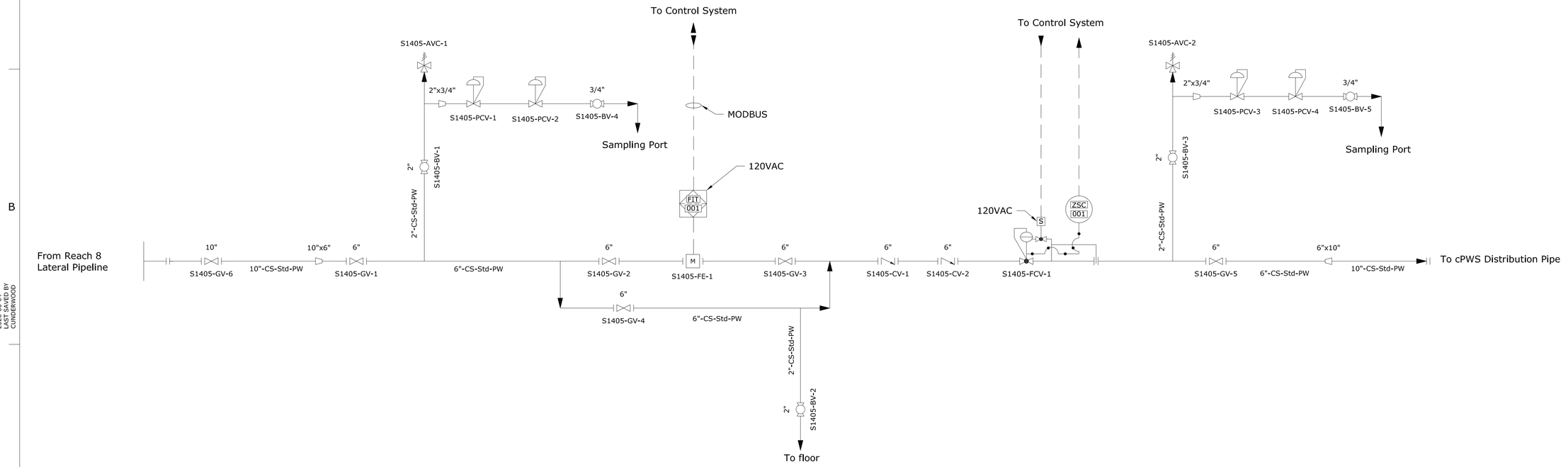

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TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
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CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E. PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
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NASCHITTI TURNOUT BUILDING

1695-D-62573

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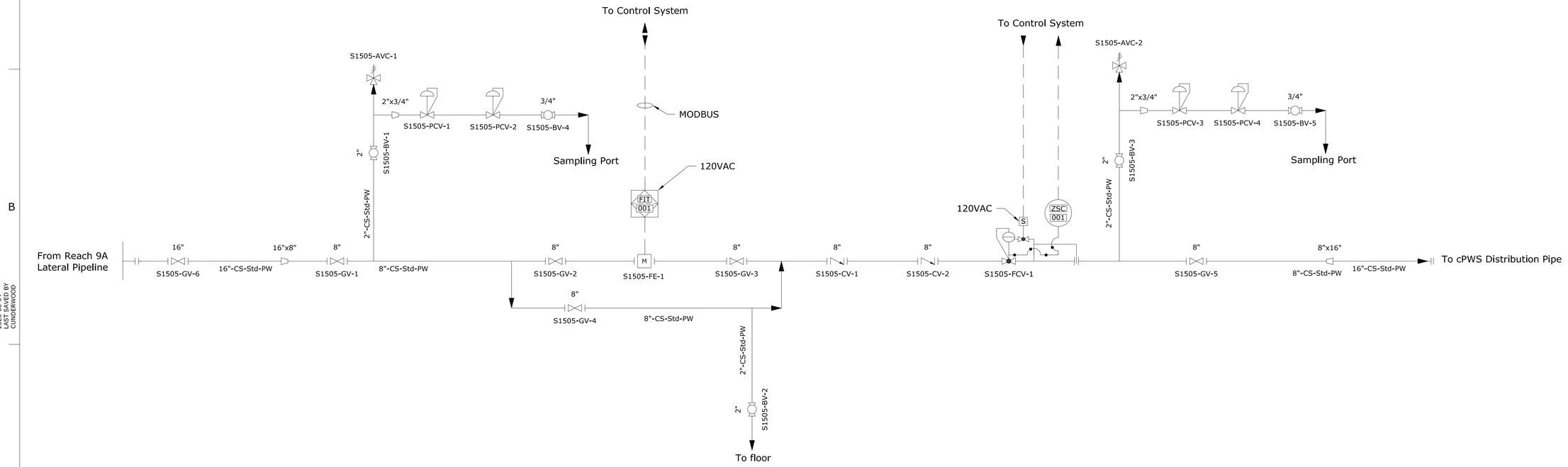

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CAD FILENAME
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TURNOUT BUILDINGS

DESIGNED	CHRISTOPHER BELLECI
DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E.
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BBN9 TURNOUT BUILDING

1695-D-62574

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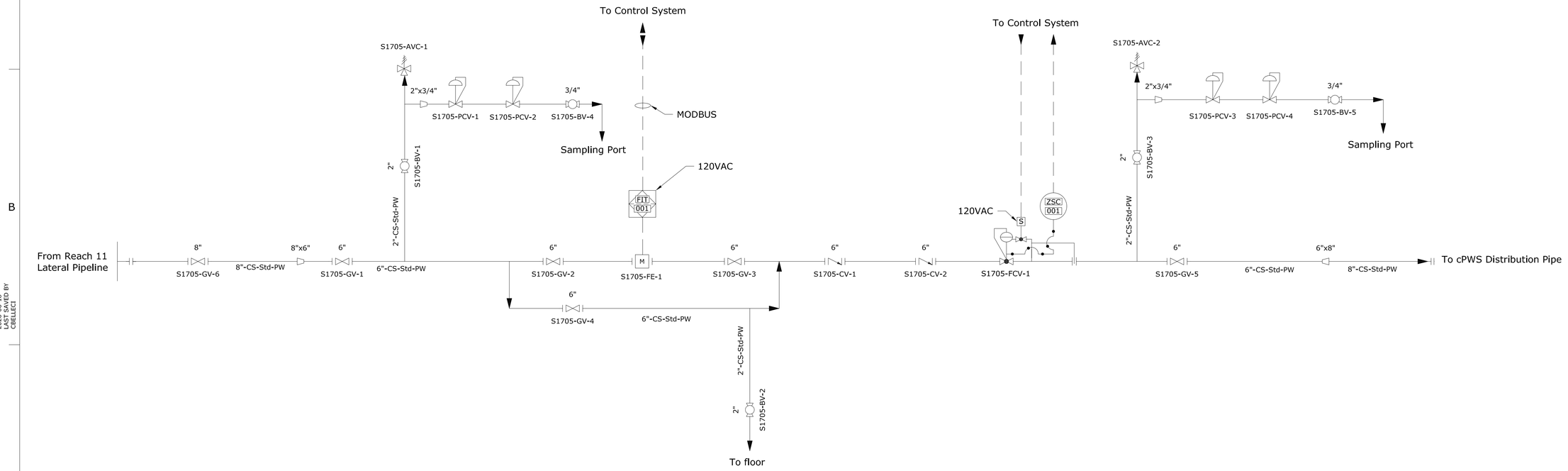

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CAD FILENAME
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TURNOUT BUILDINGS

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DRAWN	CHRISTOPHER BELLECI
CHECKED	KYLE TAYLOR
TECH. APPR.	JEREMY HAMMOND, P.E.
APPROVED	CODY YOUNG, P.E. PEER REVIEWER - SUP, MECH PIPING AND SYSTEMS
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COYOTE CANYON
TURNOUT BUILDING

1695-D-62575

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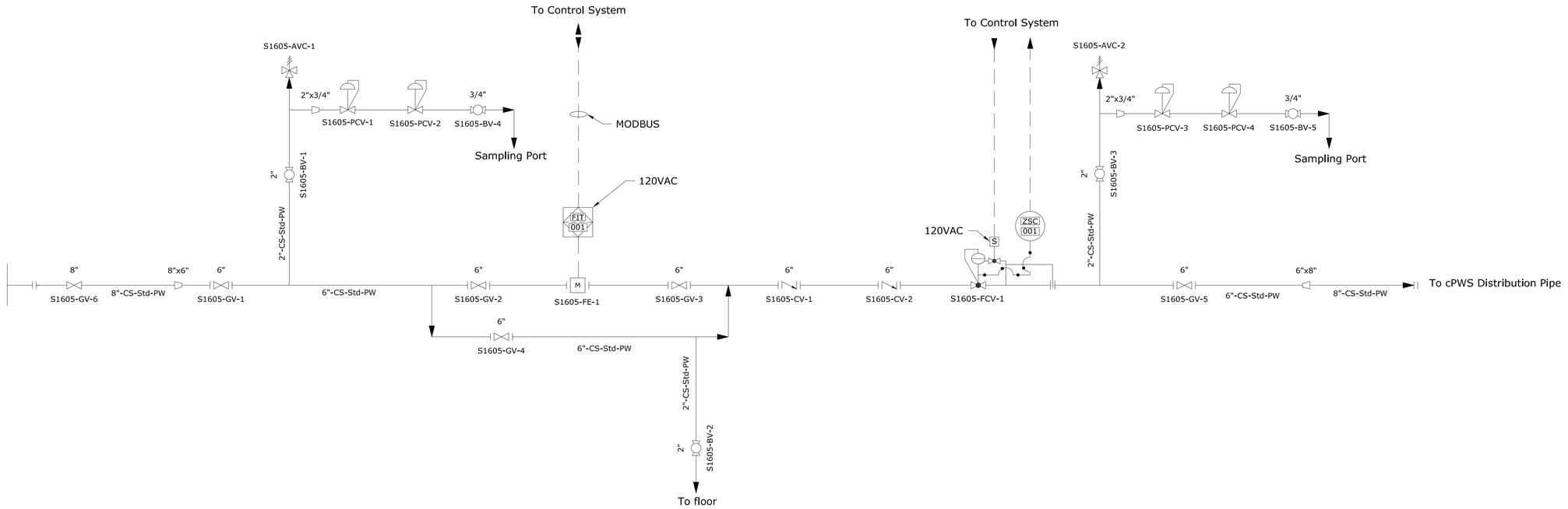

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CAD SYSTEM
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CAD FILENAME
ROCK SPRINGS TURNOUT.DWG

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TURNOUT BUILDINGS

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ROCK SPRINGS
TURNOUT BUILDING

1695-D-62576

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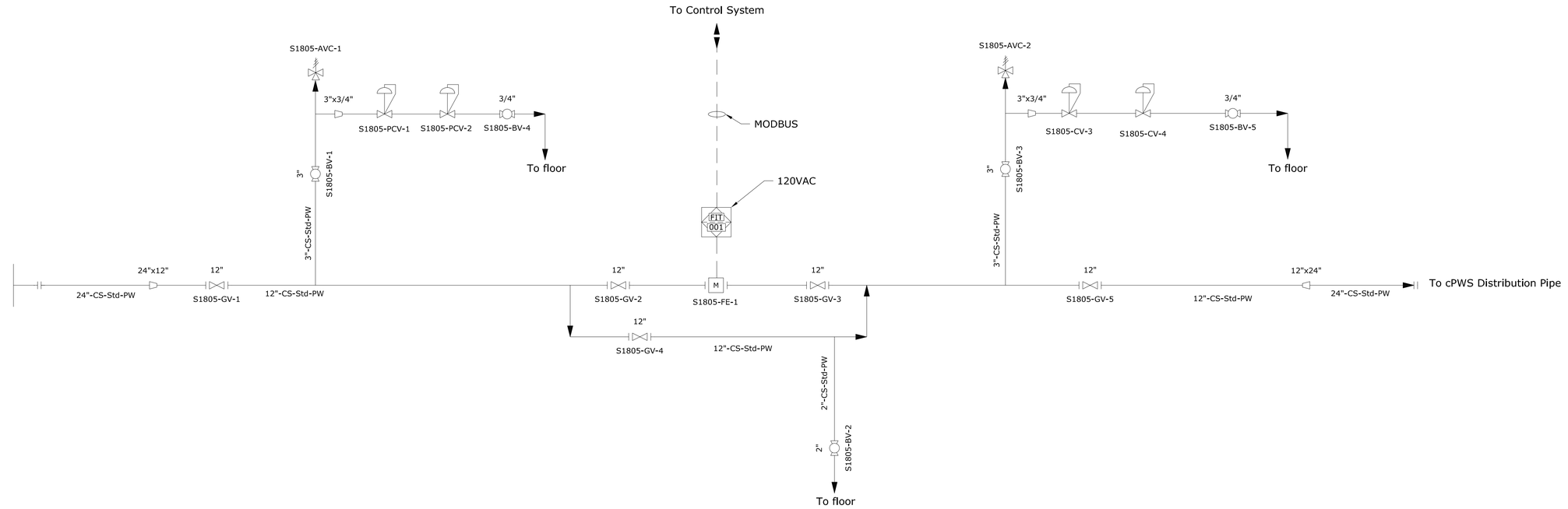

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WINDOW ROCK TURNOUT.DWG

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TURNOUT BUILDINGS

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CHRISTOPHER BELLECI	DRAWN
KYLE TAYLOR	CHECKED
JEREMY HAMMOND, P.E.	TECH. APPR.
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WINDOW ROCK
TURNOUT BUILDING

1695-D-62577

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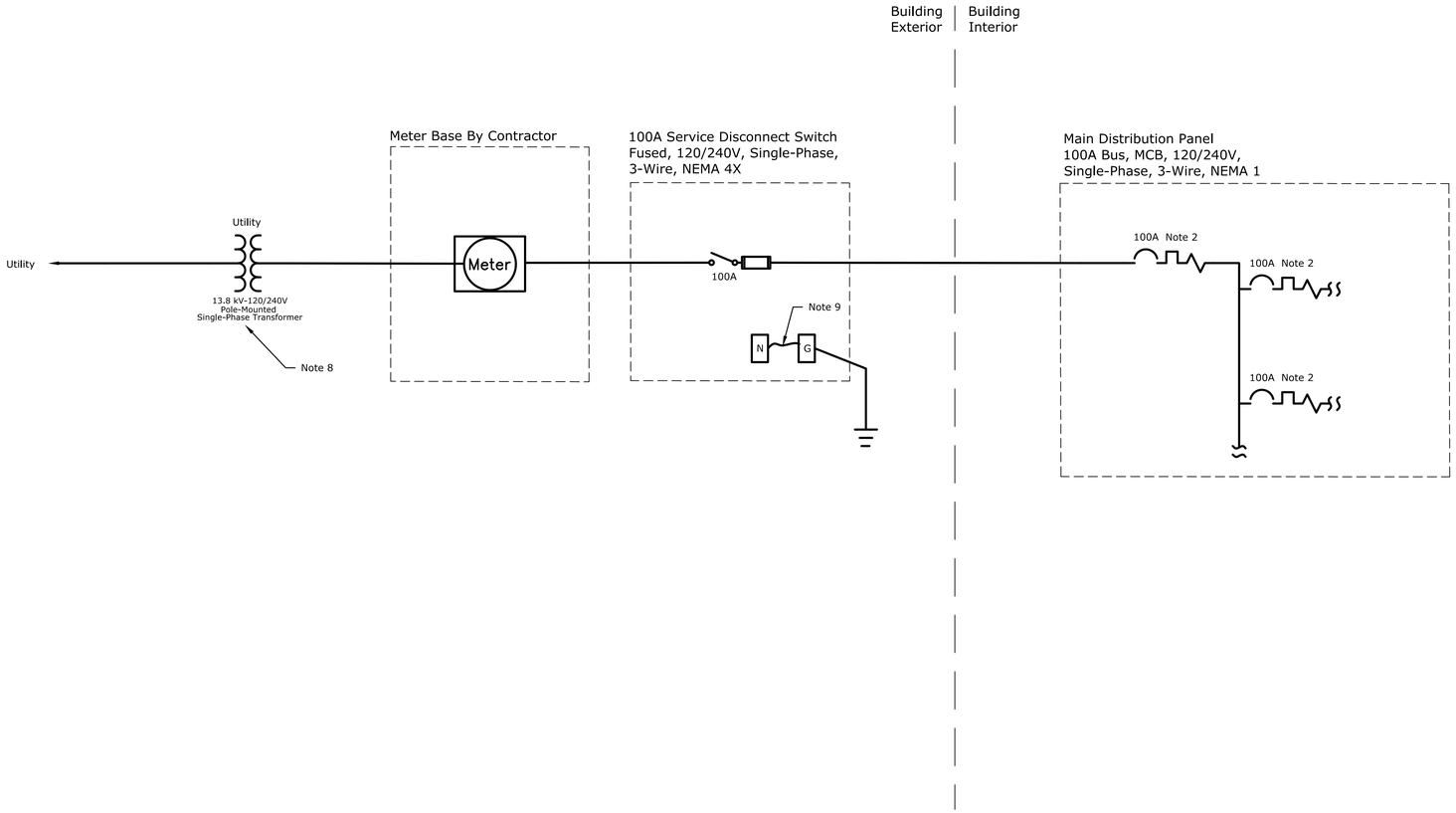

NOTES

1. Single-line is typical of all 7 Turnout Building Locations. See Conduit and Equipment Schedules on drawing 1695-D-62587 for more information on conduit and equipment designations.
2. Circuit Breaker ratings are estimated. Actual breaker sizes and ratings to be determined by contractor based on equipment furnished.
3. Equipment based on typical installations. Contractor to verify equipment sizes prior to construction.
4. Turnout building electrical system designed by contractor.
5. Cable and conduit from pole-mounted transformer to meter provided by the utility.
6. Contractor to make all electrical connections from the meter and downstream to ensure a properly functioning electrical system.
7. Meter Base shall be provided and installed by contractor in accordance with Utility standards and coordinated with the Utility as required.
8. Transformer primary voltage shown is an estimate. Transformer size will be determined by Utility.
9. The neutral-ground bounding jumper is to be size #6 American Wire Gauge (AWG) per NFPA 70 Table 250.102(C)(1).

REFERENCE DRAWINGS

Sanostee - Electrical Site Plan	1695-D-62579
Newcomb - Electrical Site Plan	1695-D-62580
Naschitti - Electrical Site Plan	1695-D-62581
BBN9 - Electrical Site Plan	1695-D-62582
Coyote Canyon - Electrical Site Plan	1695-D-62583
Rock Springs - Electrical Site Plan	1695-D-62584
Window Rock - Electrical Site Plan	1695-D-62585
Typical - Grounding Plan	1695-D-62586
Cable And Conduit - Schedules	1695-D-62587

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DENNIS O'DONNELL
DESIGNED
DENNIS O'DONNELL
DRAWN
FRANCIS AGBAGLO
CHECKED
JOHN KELLY, P.E.
TECH. APPR.
ERIC MENDLIN, P.E.
APPROVED
PEER REVIEWER

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TYPICAL

SINGLE-LINE DIAGRAM

1695-D-62578

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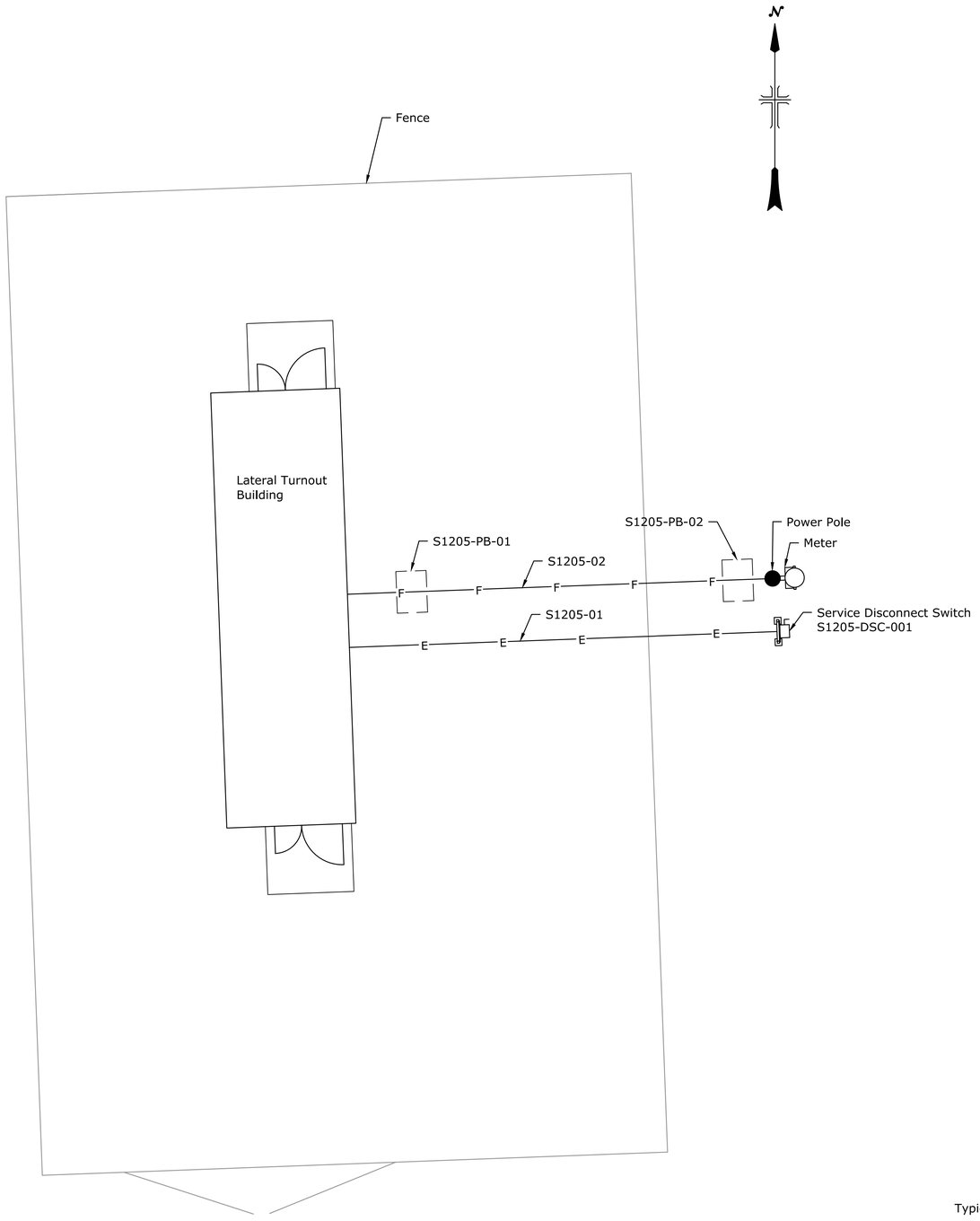
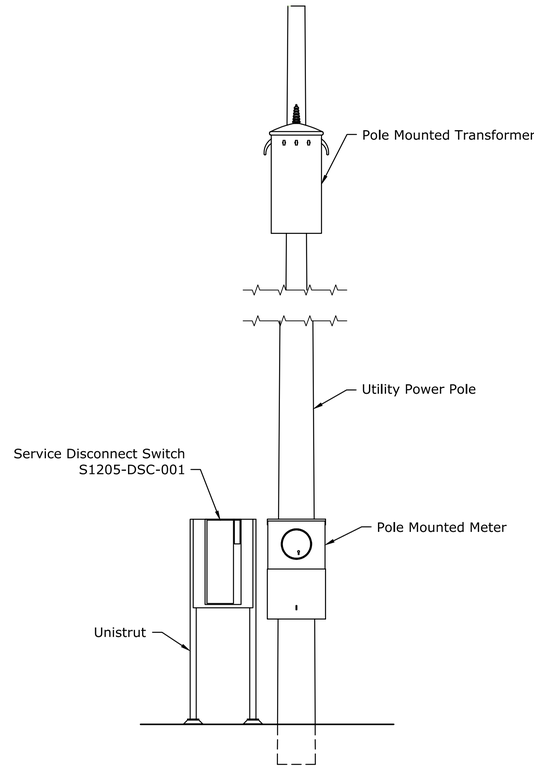


Electrical Plan View

1" = 5'



Power Pole Elevation View

Not To Scale

LEGEND

E E — Electrical Conduit

F F — Fiber optic Conduit

NOTES

1. Meter base shall be mounted on utility power pole.
2. Disconnect switch location is estimated, contractor to determine final disconnect location in accordance with NFPA 70 (NEC).
3. Utility pole, transformer, service disconnect, and meter equipment are not shown to scale.
4. Gate magnetic contact and associated conduit are not shown. Contractor to determine routing based on field conditions. Refer to section 28 10 01 for more details on the gate magnetic contacts.
5. Instrumentation and sensors not shown.
6. See Raceways and Boxes specification 26 05 33 for pull box installation details.
7. Disconnect concrete pad for strut mounting not shown. See Metal Fabrications section 05 50 00 for more details.

REFERENCE DRAWINGS

Typical - Single-Line Diagram	1695-D-62578
Typical - Grounding Plan	1695-D-62586
Cable and Conduit - Schedules	1695-D-62587

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DENNIS O'DONNELL
DESIGNED
ASHLEY WOODRUFF
DRAWN
FRANCIS AGBAGLO
CHECKED
JOHN KELLY, P.E.
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SANOSTEE TURNOUT

ELECTRICAL SITE PLAN

1695-D-62579

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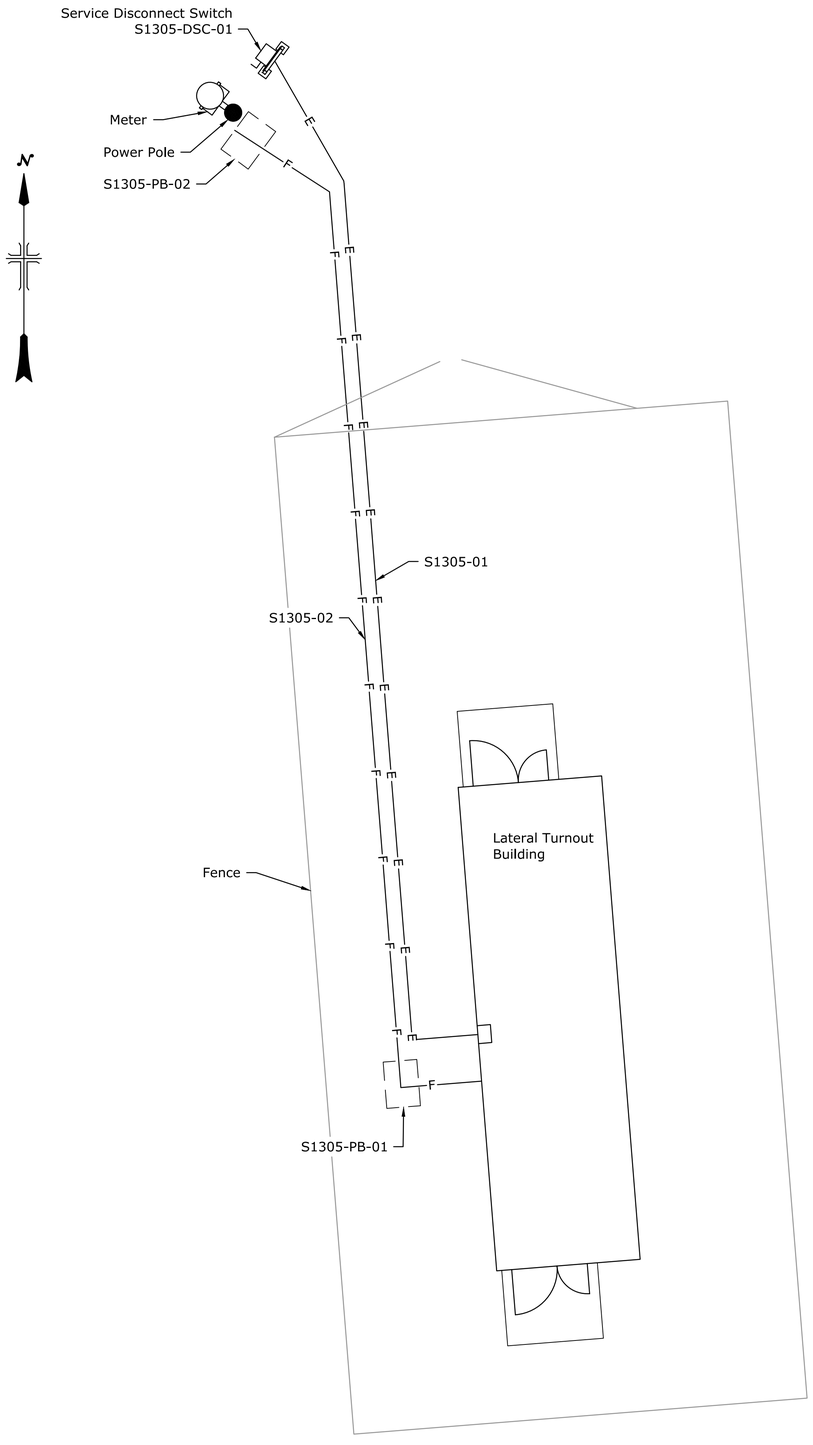
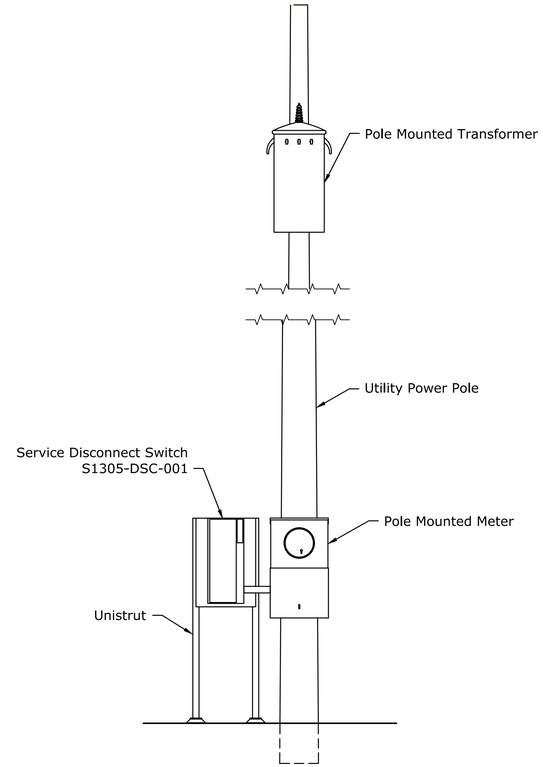


Electrical Plan View

1" = 5'



Power Pole Elevation View

Not To Scale

LEGEND

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Typical - Single-Line Diagram	1695-D-62578
Typical - Grounding Plan	1695-D-62586
Cable and Conduit - Schedules	1695-D-62587

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DESIGNED	DENNIS O'DONNELL
DRAWN	ASHLEY WOODRUFF
CHECKED	FRANCIS AGBAGLO
TECH. APPR.	JOHN KELLY, P.E.
APPROVED PEER REVIEWER	ERIC MENDLIN, P.E.
DENVER, CO	2026-05-29

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ELECTRICAL SITE PLAN

1695-D-62580

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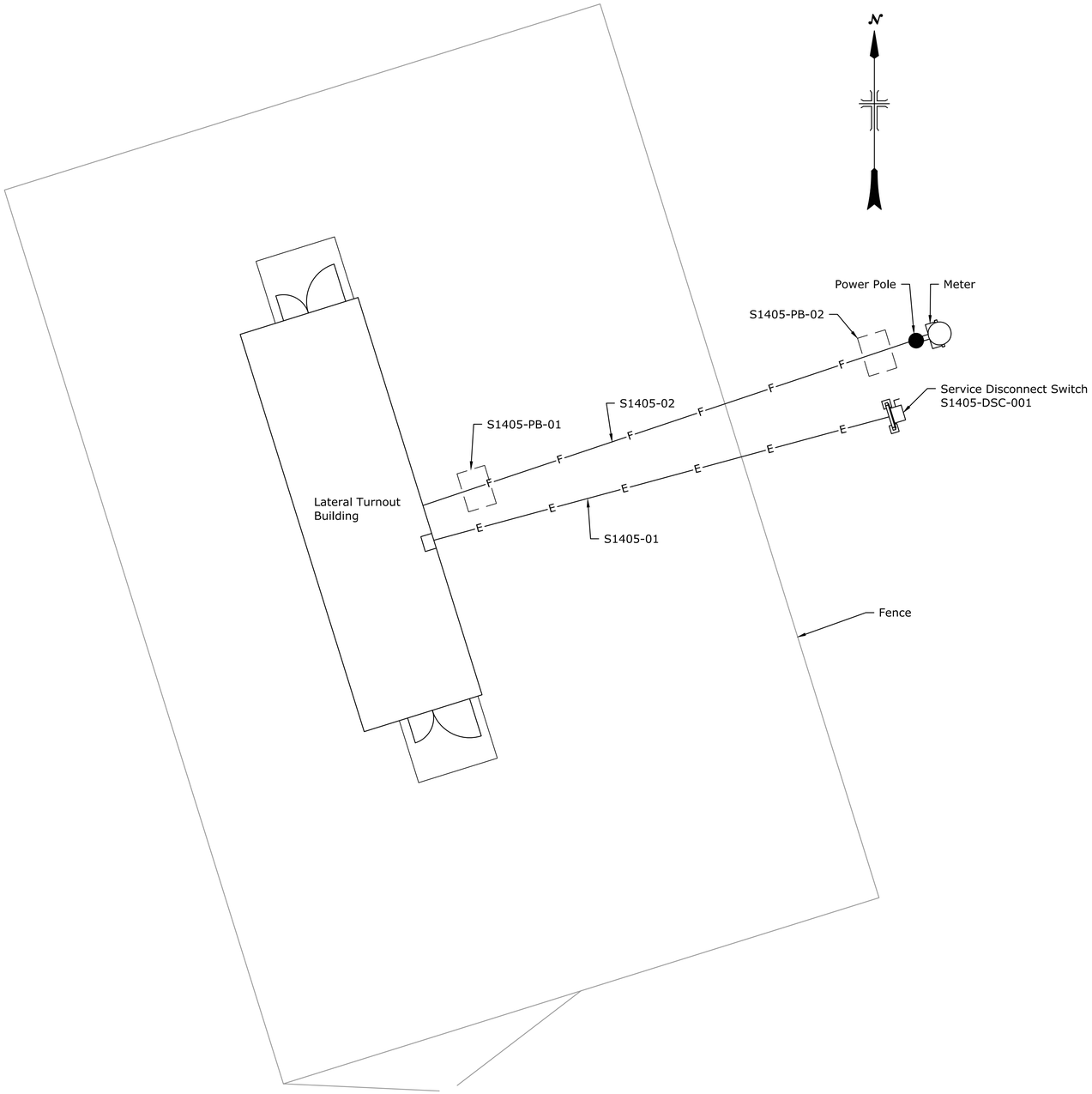
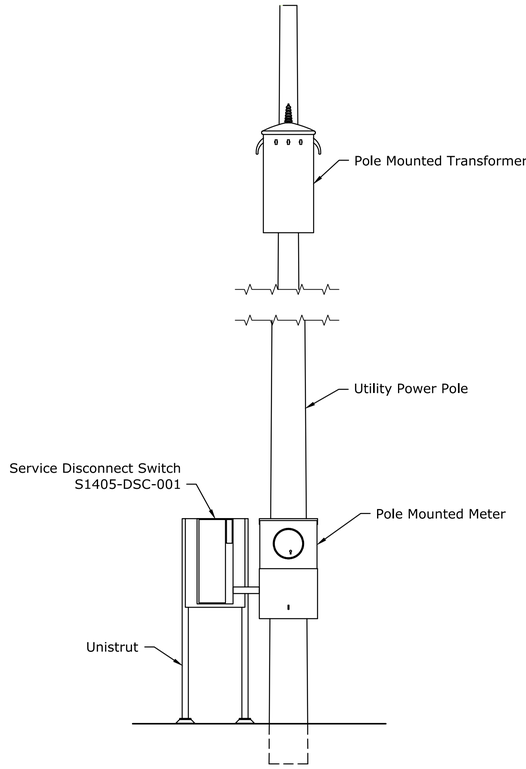


Electrical Plan View
1" = 5'



Power Pole Elevation View
Not To Scale

LEGEND

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F F — Fiber optic Conduit

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NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DENNIS O'DONNELL — DESIGNED
ASHLEY WOODRUFF — DRAWN
FRANCIS AGBAGLO — CHECKED
JOHN KELLY, P.E. — TECH. APPR.
ERIC MENDLIN, P.E. — APPROVED PEER REVIEWER
DENVER, CO — 2026-05-29

NASCHITTI

ELECTRICAL SITE PLAN

1695-D-62581

SHEET 1

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CAD SYSTEM
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CAD FILENAME
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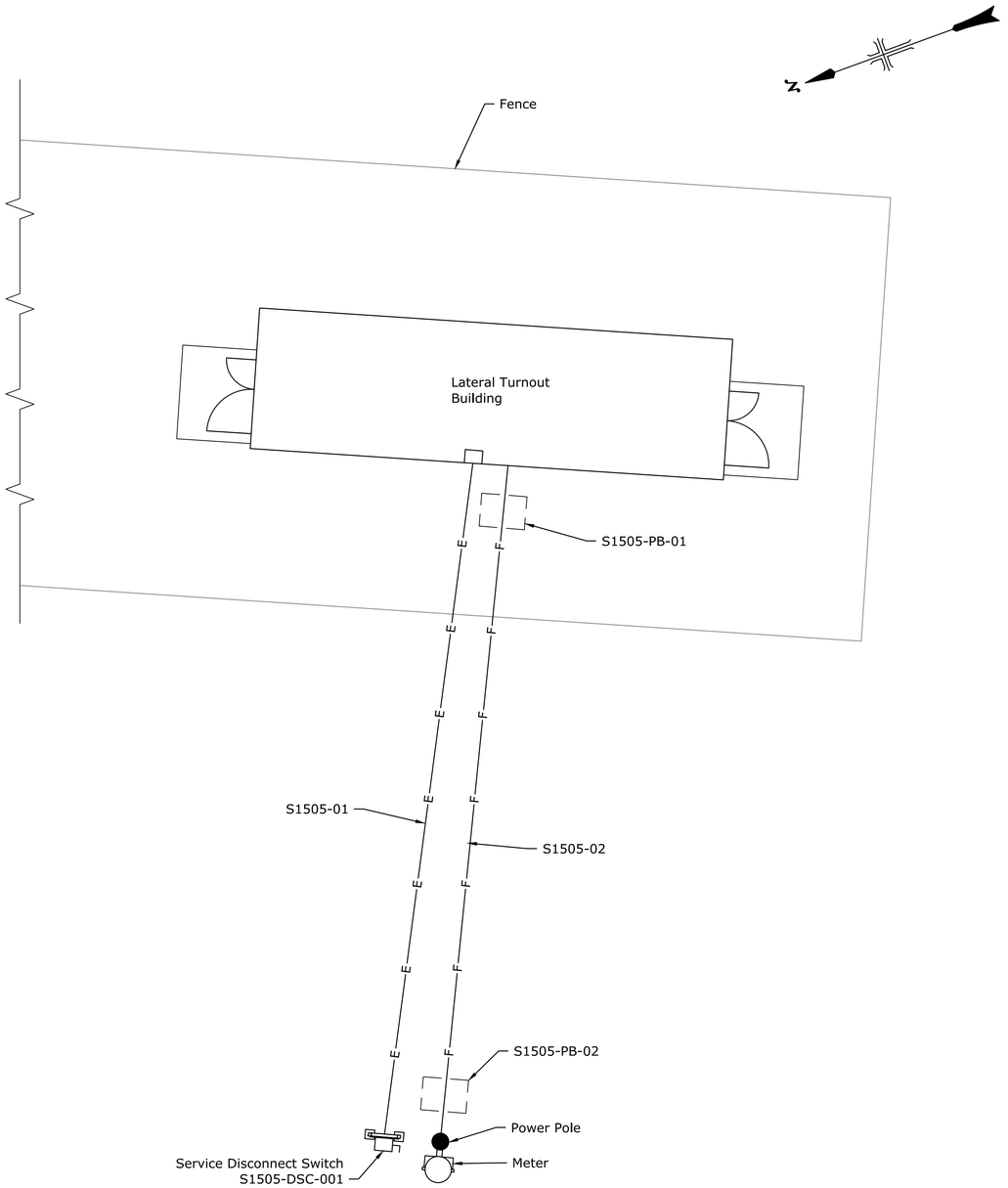
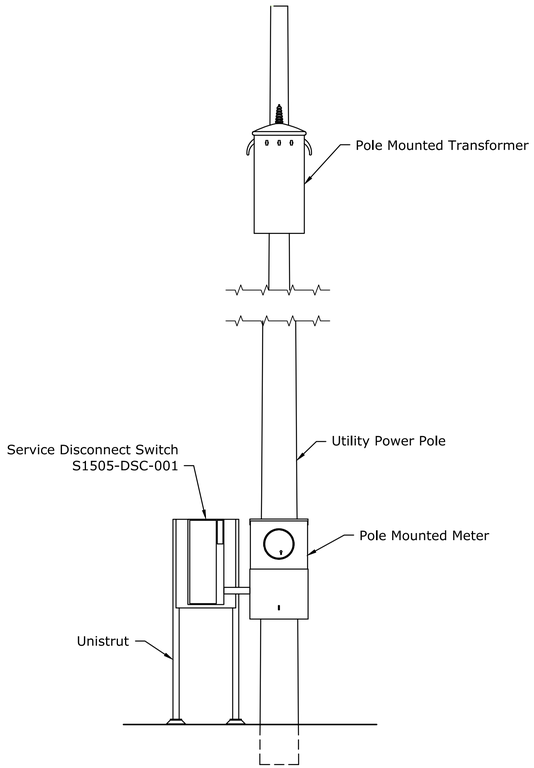


Electrical Plan View

1" = 5'



Power Pole Elevation View

Not To Scale

LEGEND

E E — Electrical Conduit

F F — Fiber optic Conduit

NOTES

1. Meter base shall be mounted on utility power pole.
2. Disconnect switch location is estimated, contractor to determine final disconnect location in accordance with NFPA 70 (NEC).
3. Utility pole, transformer, service disconnect, and meter equipment are not shown to scale.
4. Gate magnetic contact and associated conduit are not shown. Contractor to determine routing based on field conditions. Refer to section 28 10 01 for more details on the gate magnetic contacts.
5. Instrumentation and sensors not shown.
6. See Raceways and Boxes specification 26 05 33 for pull box installation details.
7. Fence gates not shown.
8. Disconnect concrete pad for strut mounting not shown. See Metal Fabrications section 05 50 00 for more details.

REFERENCE DRAWINGS

Typical - Single-Line Diagram	1695-D-62578
Typical - Grounding Plan	1695-D-62586
Cable and Conduit - Schedules	1695-D-62587

BUREAU OF RECLAMATION

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

JOHN KELLY, P.E.
DESIGNED
ASHLEY WOODRUFF
DRAWN
FRANCIS AGABGLO
CHECKED
JOHN KELLY, P.E.
TECH. APPR.
ERIC MENDLIN, P.E.
APPROVED
PEER REVIEWER

DENVER, CO 2026-05-29

BBN9

ELECTRICAL SITE PLAN

1695-D-62582

SHEET 1

LAST SAVED DATE
2026-05-27
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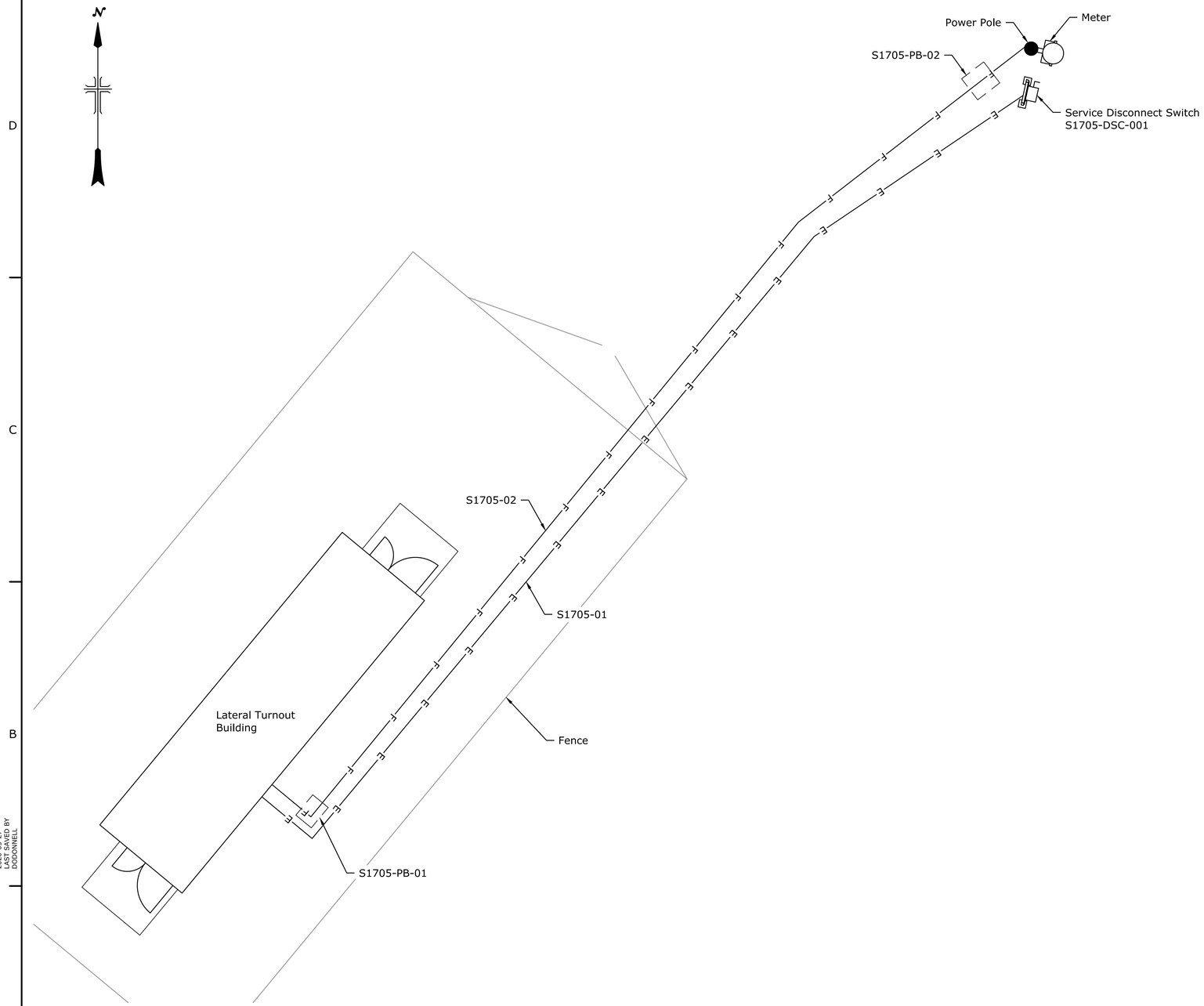
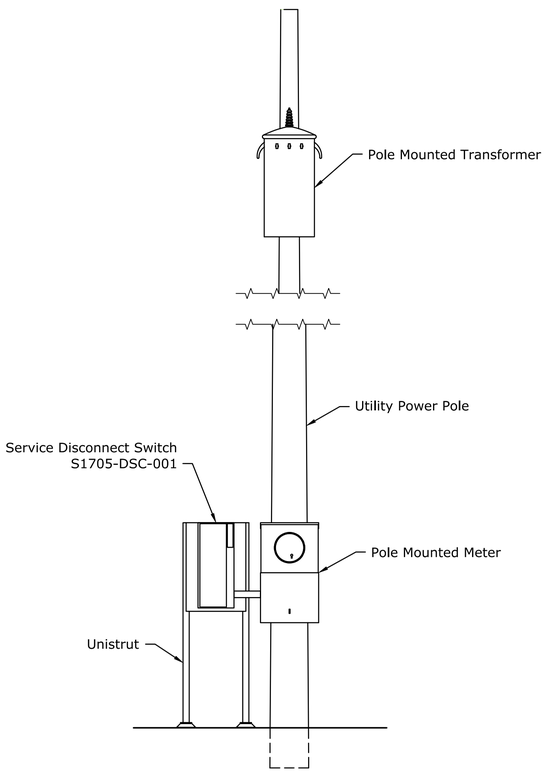
CAD SYSTEM
AUTODESK BIM 360 DOCS (LMS TECH)
CAD FILENAME
1695-D-T0171 - BBN9.DWG



Electrical Plan View

1" = 5'



Power Pole Elevation View

Not To Scale

LEGEND

E E — Electrical Conduit

F F — Fiber optic Conduit

NOTES

1. Meter base shall be mounted on utility power pole.
2. Disconnect switch location is estimated, contractor to determine final disconnect location in accordance with NFPA 70 (NEC).
3. Utility pole, transformer, service disconnect, and meter equipment are not shown to scale.
4. Gate magnetic contact and associated conduit are not shown. Contractor to determine routing based on field conditions. Refer to section 28 10 01 for more details.
5. Instrumentation and sensors not shown.
6. See Raceways and Boxes specification 26 05 33 for pull box installation details.
7. Disconnect concrete pad for strut mounting not shown. See Metal Fabrications section 05 50 00 for more details.

REFERENCE DRAWINGS

Typical - Single-Line Diagram	1695-D-62578
Typical - Grounding Plan	1695-D-62586
Cable and Conduit - Schedules	1695-D-62587

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DENNIS O'DONNELL — DESIGNED
ASHLEY WOODRUFF — DRAWN
FRANCIS AGBAGLO — CHECKED
JOHN KELLY, P.E. — TECH. APPR.
ERIC MENDLIN, P.E. — APPROVED, PEER REVIEWER

DENVER, CO 2026-05-29

COYOTE CANYON

ELECTRICAL SITE PLAN

1695-D-62583

SHEET 1

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2026-05-29
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DODONNELL

CAD SYSTEM
AUTOCAD 2025 (LMS TECH)
CAD FILENAME
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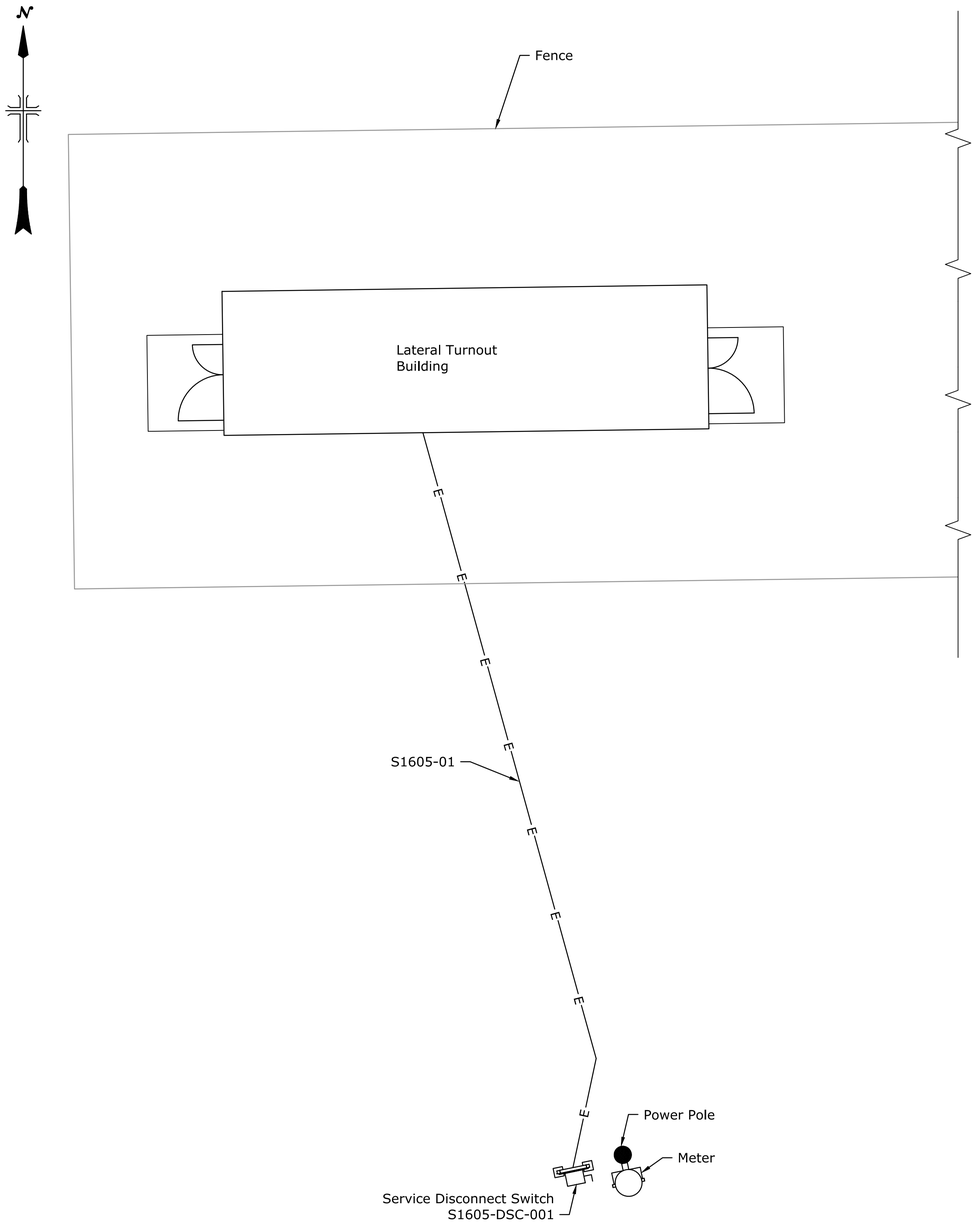
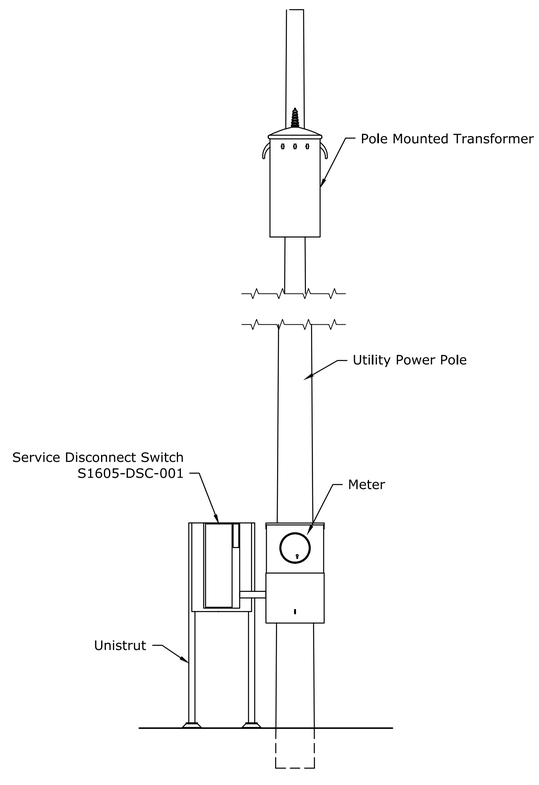


Electrical Plan View

1" = 5'



Power Pole Elevation View

Not To Scale

LEGEND

E E — Electrical Conduit

NOTES

1. Underground optical fiber utility connections for Rock Springs Turnout Building is not shown. Connect to existing underground fiber optic transmission system interface at the splice closure in the existing underground fiber optic pull box located at the site. Contractor to install a second pull box for fiber entrance into the building.
2. Utility pole, transformer, service disconnect, and meter equipment are not shown to scale.
3. Meter base shall be mounted on utility power pole.
4. Disconnect switch location is estimated, contractor to determine final disconnect location in accordance with NFPA 70 (NEC).
5. Gate magnetic contact and associated conduit are not shown. Contractor to determine routing based on field conditions. Refer to section 28 10 01 for more details on the gate magnetic contacts.
6. Instrumentation and sensors not shown.
7. See Raceways and Boxes specification 26 05 33 for pull box installation details.
8. Fence Gates not shown.
9. Disconnect concrete pad for strut mounting not shown. See Metal Fabrications section 05 50 00 for more details.

REFERENCE DRAWINGS

Typical - Single-Line Diagram	1695-D-62578
Typical - Grounding Plan	1695-D-62586
Cable and Conduit - Schedules	1695-D-62587

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DESIGNED	DENNIS O'DONNELL
DRAWN	ASHLEY WOODRUFF
CHECKED	FRANCIS AGBAGLO
TECH. APPR.	JOHN KELLY, P.E.
APPROVED PEER REVIEWER	ERIC MENDLIN, P.E.
DENVER, CO	2026-05-29

ROCK SPRINGS

ELECTRICAL SITE PLAN

1695-D-62584

SHEET 1

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2026-05-29
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CAD SYSTEM
AUTOCAD 2025 (LMS TECH)
CAD FILENAME
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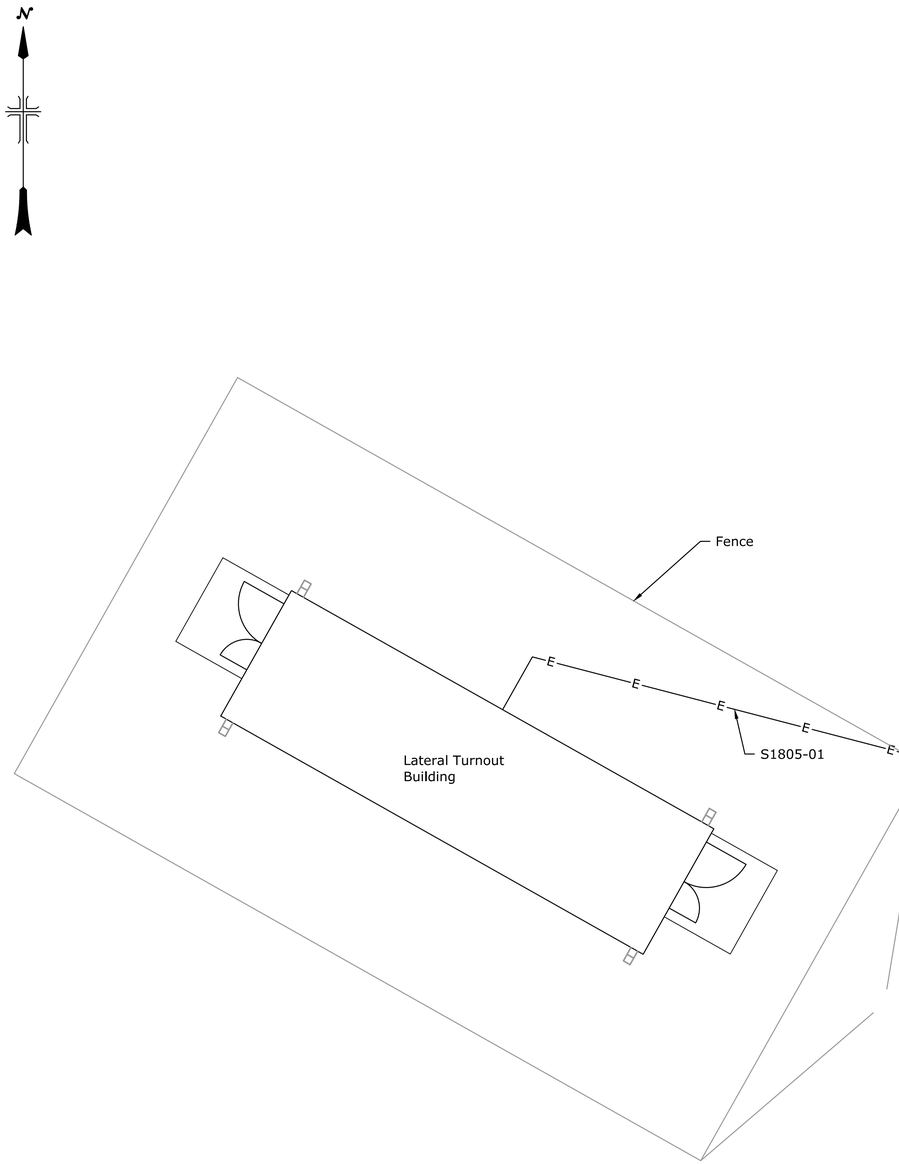
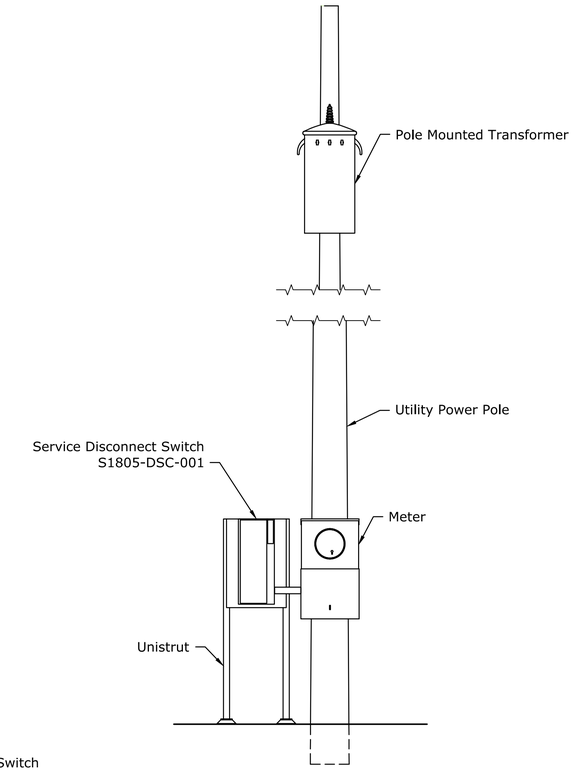


Electrical Plan View

1" = 5'



Power Pole Elevation View

Not To Scale

LEGEND

E E — Electrical Conduit

NOTES

1. Underground optical fiber utility connections for Window Rock Turnout Building is not shown. Connect to existing underground fiber optic transmission system interface at the splice closure in the existing underground fiber optic pull box located at the site. Contractor to install a second pull box for fiber entrance into the building.
2. Utility pole, transformer, service disconnect, and meter equipment are not shown to scale.
3. Meter base shall be mounted on utility power pole.
4. Disconnect switch location shown is estimated, contractor to determine final disconnect location in accordance with NFPA 70 (NEC).
5. Concrete Electrical Disconnect Pad for strut mounting not shown. See Metal Fabrications section 05 50 00 for more details.
6. Gate magnetic contact and associated conduit are not shown. Contractor to determine routing based on field conditions. Refer to section 28 10 01 for more details on the gate magnetic contacts.
7. Instrumentation and sensors not shown.
8. See Raceways and Boxes specification 26 05 33 for pull box installation details.

REFERENCE DRAWINGS

Typical - Single-Line Diagram	1695-D-62578
Typical - Grounding Plan	1695-D-62586
Cable and Conduit - Schedules	1695-D-62587

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DENNIS O'DONNELL
DESIGNED
ASHLEY WOODRUFF
DRAWN
FRANCIS AGBAGLO
CHECKED
JOHN KELLY, P.E.
TECH. APPR.
ERIC MENDLIN, P.E.
APPROVED
PEER REVIEWER

DENVER, CO 2026-05-29

WINDOW ROCK

ELECTRICAL SITE PLAN

1695-D-62585

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AUTOCAD CIVIL (LMS TECH)
CAD FILENAME
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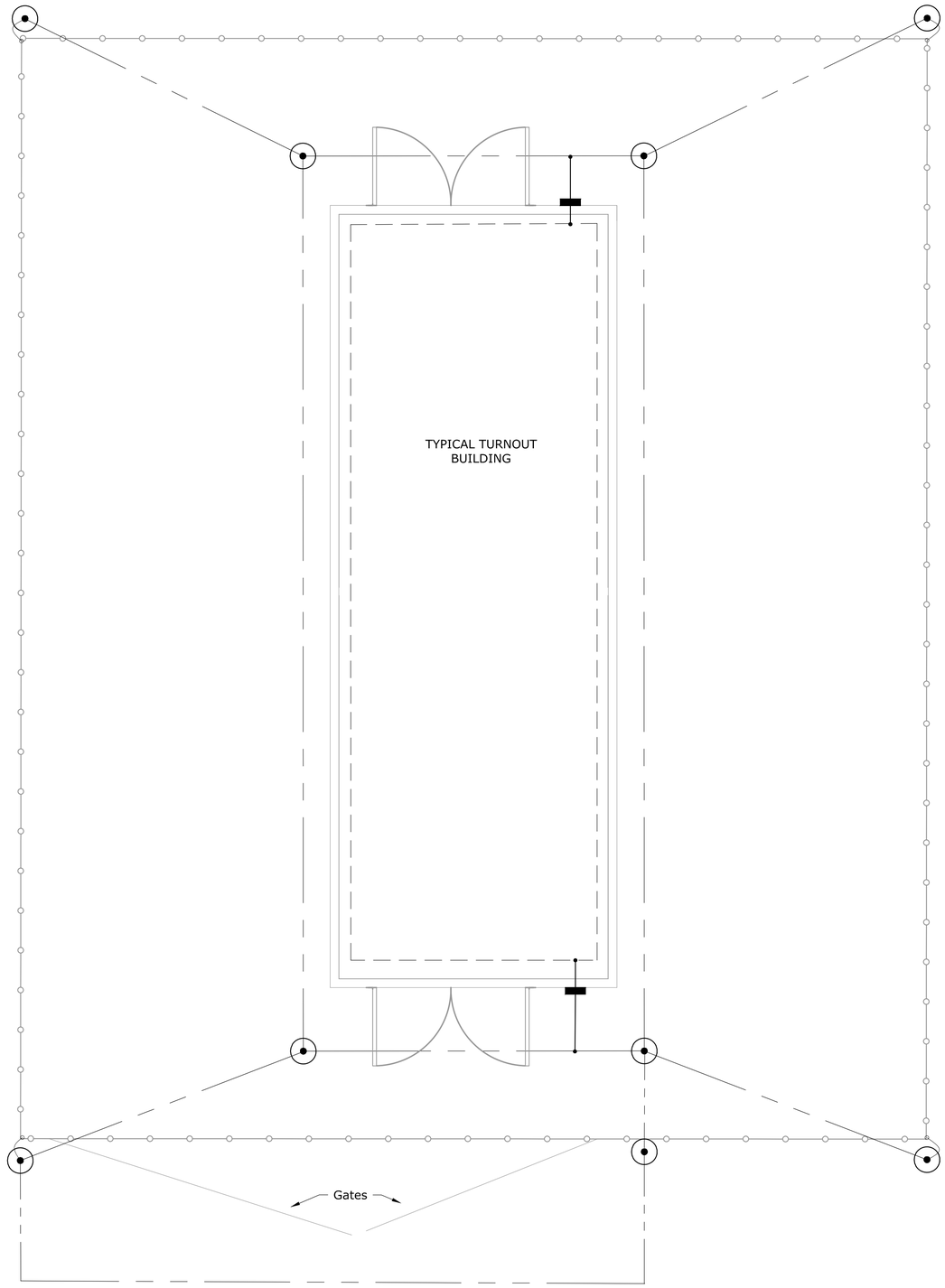

Typical Grounding Plan

Not To Scale

GROUNDING EQUIPMENT

- 4/0 AWG Ground Electrode Conductor - Ceiling
- 4/0 AWG Ground Electrode Conductor - Buried In Earth
- Fence
- Welded Or Bolted Ground Connection
- Ground Rod
- Wall-Mounted Ground Pad (Exterior)
- Connection To Fence Post

NOTES

1. This drawing is not to scale.
2. This drawing represents a typical grounding system for all seven Turnout Building Sites.
3. Coordinate grounding system installation with Turnout Building supplier.
4. Bond building ground ring to service entrance grounding (not shown) per NFPA 70.
5. Grounding system conductors are stranded bare copper.
6. Bond the grounding electrode conductor and ground pad to the Turnout Building's structural steel at a location selected by the contractor. Structural steel must be part of the Turnout Building's grounding electrode system and verified for electrical continuity.
7. Ground electrode conductor must be accessible for inspection and permanently secured.
8. Bond at locations that make direct contact with structural metal. Do not bond on decorative bar work.
9. All embedded ground terminations to be performed using Caldwell, Thermoweld, or an equivalent exothermic process.
10. Location of ground pad and ground electrode terminations to be determined by contractor.
11. Grounding layout shows typical location of grounding equipment.
12. Ground the structure and all equipment and components in accordance with Section 26 05 20 - Grounding and Bonding.
13. Ground loop is to be bonded to the grounding system.
14. Connect building ground electrode conductor to ground pads and connect ground pads to the buried ground electrode conductor.
15. All grounding electrodes present at a structure must be bonded together per NFPA 70 (NEC) 250.50.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DESIGNED	DENNIS O'DONNELL
DRAWN	ASHLEY WOODRUFF
CHECKED	FRANCIS AGBAGLO
TECH. APPR.	JOHN KELLY, P.E.
APPROVED PEER REVIEWER	ERIC MENDLIN, P.E.
DENVER, CO	2026-05-29

TYPICAL

GROUNDING PLAN

1695-D-62586

SHEET 1

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CAD SYSTEM
AUTOCAD CIVIL 3D (LMS TECH)
CAD FILENAME
1695-D-T0176 - TYPICAL GROUNDING.DWG

CABLE AND CONDUIT SCHEDULE

CONDUIT DESIGNATION	SIZE	CONDUIT CONTENTS	CONDUIT TYPE	FILL%	DESCRIPTION
S1205-01	3"	3-1 AWG + 1-8 AWG-G	PVC Sch 80 (Note 6)	7.8%	Power To Sanostee Turnout Building Main Distribution Panel
S1205-02	3"	FIBER	PVC Sch 80 (Note 6)	N/A	Sanostee Turnout Building SCADA And Controls
S1305-01	3"	3-1 AWG + 1-8 AWG-G	PVC Sch 80 (Note 6)	7.8%	Power To Newcomb Turnout Building Main Distribution Panel
S1305-02	3"	FIBER	PVC Sch 80 (Note 6)	N/A	Newcomb Turnout Building SCADA And Controls
S1405-01	3"	3-1 AWG + 1-8 AWG-G	PVC Sch 80 (Note 6)	7.8%	Power To Naschitti Turnout Building Main Distribution Panel
S1405-02	3"	FIBER	PVC Sch 80 (Note 6)	N/A	Naschitti Turnout Building SCADA And Controls
S1505-01	3"	3-1 AWG + 1-8 AWG-G	PVC Sch 80 (Note 6)	7.8%	Power To BBN9 Turnout Building Main Distribution Panel
S1505-02	3"	FIBER	PVC Sch 80 (Note 6)	N/A	BBN9 Turnout Building SCADA And Controls
S1605-01	3"	3-1 AWG + 1-8 AWG-G	PVC Sch 80 (Note 6)	7.8%	Power To Rock Springs Turnout Building Main Distribution Panel
S1605-02	3"	FIBER	PVC Sch 80 (Note 6)	N/A	Rock Springs Turnout Building SCADA And Controls
S1705-01	3"	3-1 AWG + 1-8 AWG-G	PVC Sch 80 (Note 6)	7.8%	Power To Coyote Canyon Turnout Building Main Distribution Panel
S1705-02	3"	FIBER	PVC Sch 80 (Note 6)	N/A	Coyote Canyon Turnout Building SCADA And Controls
S1805-01	3"	3-1 AWG + 1-8 AWG-G	PVC Sch 80 (Note 6)	7.8%	Power To Window Rock Turnout Building Main Distribution Panel
S1805-02	3"	FIBER	PVC Sch 80 (Note 6)	N/A	Window Rock Turnout Building SCADA And Controls

EXPLANATION

Device	Location	Description
METER	Mounted On Utility Pole	Utility Revenue Meter
S1205-DSC-001	Strut-Mounted Adjacent To Meter	Sanostee 120/240V, 100A Disconnect Switch
S1305-DSC-001	Strut-Mounted Adjacent To Meter	Newcomb 120/240V, 100A Disconnect Switch
S1405-DSC-001	Strut-Mounted Adjacent To Meter	Naschitti 120/240V, 100A Disconnect Switch
S1505-DSC-001	Strut-Mounted Adjacent To Meter	BBN9 120/240V, 100A Disconnect Switch
S1605-DSC-001	Strut-Mounted Adjacent To Meter	Rock Springs 120/240V, 100A Disconnect Switch
S1705-DSC-001	Strut-Mounted Adjacent To Meter	Coyote Canyon 120/240V, 100A Disconnect Switch
S1805-DSC-001	Strut-Mounted Adjacent To Meter	Window Rock 120/240V, 100A Disconnect Switch

NOTES

1. Conduit and disconnect switch designations reflect site location.
2. Cable and Conduit within the Turnout Buildings are not shown and will be provided and installed by others.
3. All work to be performed in accordance with the latest version of NFPA 70.
4. Contractor to verify cable and conduit size based on NFPA 70 prior to installation.
5. Refer to Site Plan drawings listed in the Reference Drawings list below for estimated conduit routing. Contractor to make final determination on conduit routing based on field conditions and buried pipe path.
6. Transition to RMC prior to conduit vertical transition. Conduit elbow, vertical segments, and above grade conduit shall be RMC.
7. Transition from RMC to LFMC for no more than the last 6 feet of conduit run to any equipment subject to movement, vibration, misalignment, or in need of flexibility.
8. Contractor to verify conductor ampacity, conduit fill, and jam ratio based on actual equipment provided.

REFERENCE DRAWINGS

Electrical Installation - Single Line Diagram	1695-D-62578
Sanostee - Electrical Site Plan	1695-D-62579
Newcomb - Electrical Site Plan	1695-D-62580
Naschitti - Electrical Site Plan	1695-D-62581
BBN9 - Electrical Site Plan	1695-D-62582
Coyote Canyon - Electrical Site Plan	1695-D-62583
Rock Springs - Electrical Site Plan	1695-D-62584
Window Rock - Electrical Site Plan	1695-D-62585

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

DENNIS O'DONNELL
DESIGNED
ASHLEY WOODRUFF
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FRANCIS AGBAGLO
CHECKED
JOHN KELLY, P.E.
TECH. APPR.
ERIC MENDLIN, P.E.
APPROVED
PEER REVIEWER

DENVER, CO 2026-05-29

CABLE AND CONDUIT

SCHEDULE

1695-D-62587

SHEET 1

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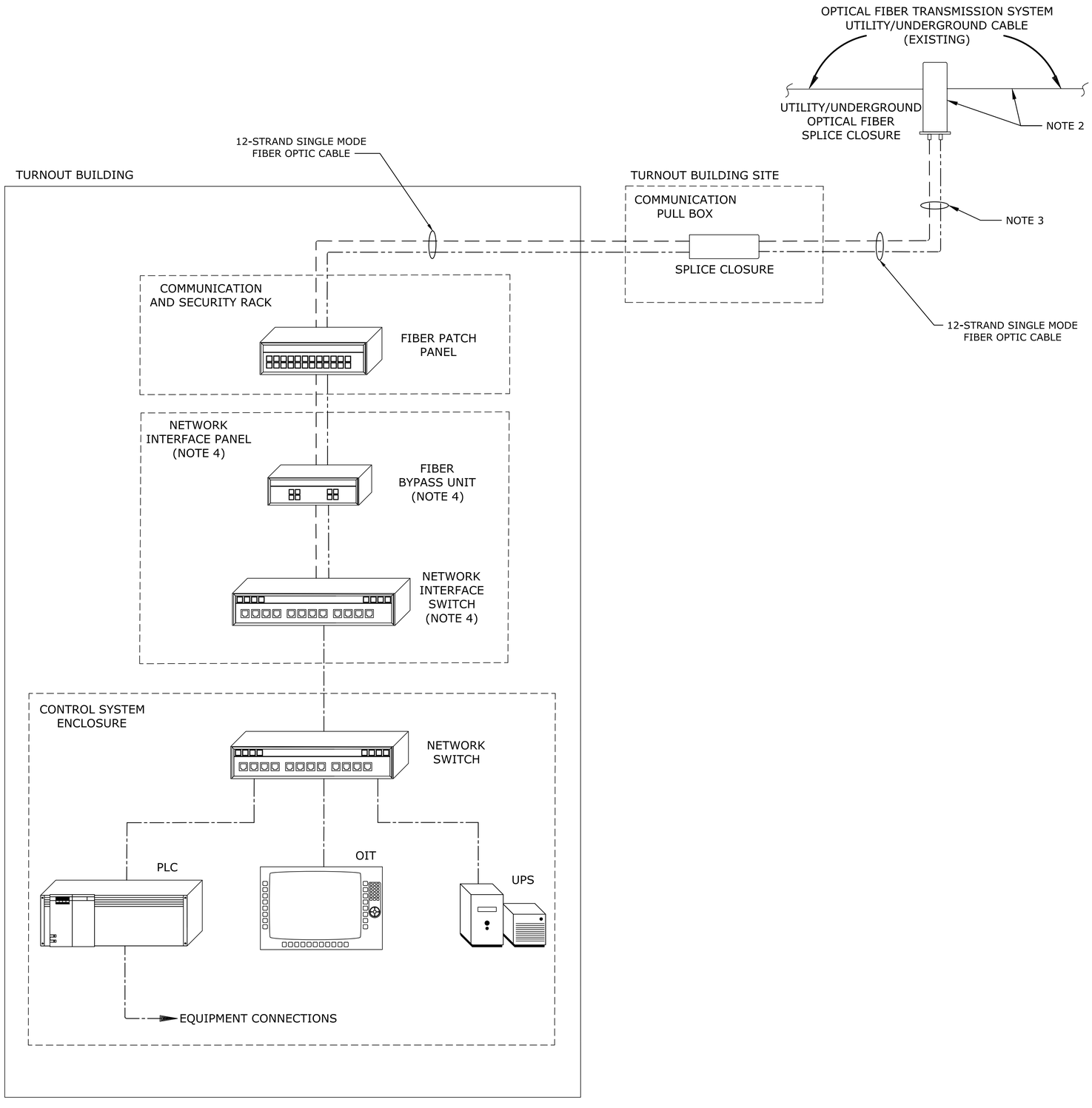

NOTES

1. NETWORK OVERVIEW IS TYPICAL FOR ALL TURNOUT BUILDING SITES EXCEPT AS NOTED.
2. UTILITY/UNDERGROUND OPTICAL FIBER CABLE AND UTILITY/UNDERGROUND SPLICE ENCLOSURE PROVIDED BY OTHERS.
3. OPTICAL FIBER UTILITY CONNECTIONS FOR EACH OF THE FOLLOWING SITES ARE AS FOLLOWS:
   - 3.1. SANOSTEE, NASCHITTI, NEWCOMB, BBN9, COYOTE CANYON: CONNECT TO UTILITY FIBER OPTIC TRANSMISSION SYSTEM INTERFACE AT THE UNDERGROUND CONTRACTOR SUPPLIED COMMUNICATION PULL BOX LOCATED CLOSEST TO UTILITY POLE.
   - 3.2. ROCK SPRINGS AND WINDOW ROCK: CONNECT TO EXISTING UNDERGROUND FIBER OPTIC TRANSMISSION SYSTEM INTERFACE AT THE SPLICE CLOSURE IN THE UNDERGROUND FIBER OPTIC PULL BOX LOCATED AT THE SITE.
4. NETWORK INTERFACE PANEL IS GFE TO BE PROVIDED TO CONTRACTOR FOR INSTALLATION. CONTRACTOR TO COMPLETE ALL POWER AND COMMUNICATION CONNECTIONS TO NETWORK INTERFACE PANEL.

ABBREVIATIONS

GFE	GOVERNMENT FURNISHED EQUIPMENT
OIT	OPERATOR INTERFACE TERMINAL
PLC	PROGRAMMABLE LOGIC CONTROLLER
SM	SINGLE MODE (FIBER OPTIC CABLE)
UPS	UNINTERRUPTIBLE POWER SUPPLY

CABLE LEGEND

- CAT 6 ETHERNET CABLE
- SM OPTICAL FIBER CABLE - PRIMARY PAIR
- SM OPTICAL FIBER CABLE - BACKUP PAIR

REFERENCE DRAWINGS

SINGLE LINE DIAGRAM - TYPICAL	1695-D-62578
ELECTRICAL SITE PLAN - SANOSTEE	1695-D-62579
ELECTRICAL SITE PLAN - NASCHITTI	1695-D-62581
ELECTRICAL SITE PLAN - NEWCOMB	1695-D-62580
ELECTRICAL SITE PLAN - BBN9	1695-D-62582
ELECTRICAL SITE PLAN - COYOTE CANYON	1695-D-62583
ELECTRICAL SITE PLAN - ROCK SPRINGS	1695-D-62584
ELECTRICAL SITE PLAN - WINDOW ROCK	1695-D-62585

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
1695, NAVAJO-GALLUP WATER SUPPLY PROJECT, NM

SAN JUAN LATERAL

TURNOUT BUILDINGS

Christian Underwood
DESIGNED
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Lisa Gentry
CHECKED
Lisa Gentry, P.E.
TECH. APPR.
Sean Kyer, F.P.E.
APPROVED
ADMIN APPROVAL
SEAN KYER
DENVER, CO
2026-05-29

NETWORK
OVERVIEW

TURNOUT BUILDING

1695-D-62588

SHEET 1

LAST SAVED DATE
2026-05-29
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CAD SYSTEM
MICROSTATION (LMS TECH)
CAD FILENAME
1695-D-62588.DWG

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

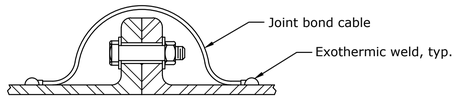
SAN JUAN LATERAL
TURNOUT BUILDINGS

NA
DESIGNED
NA
DRAWN
NA
CHECKED
TECH. APPR. NA
APPROVED Approver
Peer Reviewer - NA

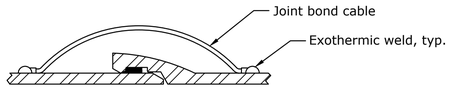
DENVER, CO 2026-06-19

NOT INCLUDED IN PACKAGE

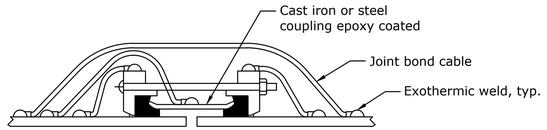
1695-D-62589



FLANGED JOINT (NON-ISOLATING)



PUSH-ON JOINT



FLEXIBLE COUPLING (MULTIPLE COMPONENT COUPLING)

JOINT BOND INSTALLATION

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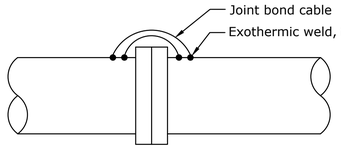
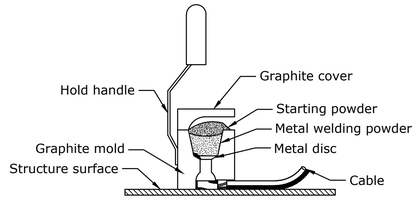
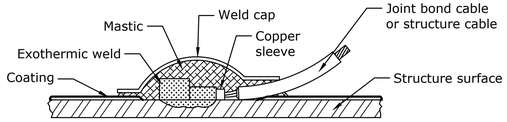


FLANGE-FLANGE BOND DETAIL

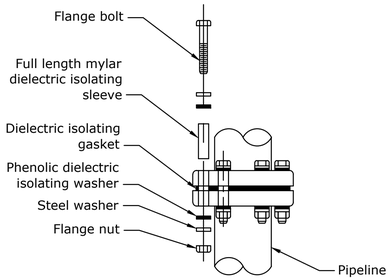
SCALE: NOT TO SCALE





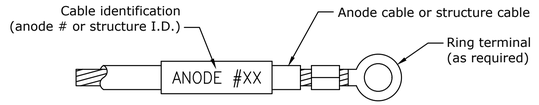
EXOTHERMIC WELD DETAIL

SCALE: NOT TO SCALE



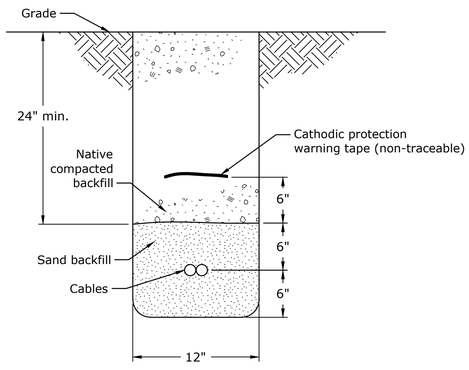
IJFK - DIELECTRIC ISOLATING JOINT FLANGE KIT

SCALE: NOT TO SCALE



CABLE IDENTIFICATION

SCALE: NOT TO SCALE



CABLE TRENCH DETAIL

SCALE: NOT TO SCALE

MATERIAL EXPLANATION	
Item	Description
Joint bond cables (6" to 8" dia. pipe)	#12 AWG/HMWPE (black); 2 per joint
Joint bond cables (12" dia. pipe)	#6 AWG/HMWPE (black); 2 per joint
Exothermic weld supplies	CS-32 type, M-102 mold, 15 CP weld metal (for steel), ductile iron requires different supplies, manufactured by thermOweld®, or equal
Weld cap	Prefabricated, integrated primer patch to repair coating surrounding exothermic weld, manufactured by thermOweld®, or equal
Warning tape	Polyethylene, 3" wide, "Caution-Cathodic Protection Cable Buried Below"
Sand	ASTM C33, fine aggregate
Isolation joint flange kit (IJFK)	Full face isolating gasket, isolation bolt sleeves, and one or more isolating washers per bolt

NOTES

1. Exothermic welding procedure is to be used as a general guide only. Consult manufacturer's literature for specific installation instructions.
2. All pipe joints are to be electrically bonded with cables, exothermically welded to the top portion of the pipe.
3. Do not install joint bond cables on welded pipe joints and/or pipe joints called out to be electrically isolated.
4. Any non-welded and non-isolating joints require bonding to achieve electrical continuity.
5. Coat exposed copper, weld nugget, and ferrous material at weld with approved dielectric coating.
6. Cover cable I.D. with clear heat shrink tubing.
7. Cable trench detail to be used for all horizontal segments of buried cables.
8. IJFK installation on pipe inside turnout building, first and last flanged Joints, see (-62559 and -62560).
9. Cathodic protection system components are not to scale for visualization.

REFERENCE DRAWINGS

STEEL PIPE INTERIOR 6 INCH AND 8 INCH - PLAN AND SECTION.....1695-D-62559
STEEL PIPE INTERIOR 12 INCH - PLAN AND SECTION.....1695-D-62560
TEST STATION AND ANODE BED DETAILS 1695-D-62591

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LAST SAVED DATE
2026-05-27
LAST SAVED BY
DLITTLE

CAD SYSTEM
AUTOCAD LT (LMS TECH)
CAD FILENAME
1695-D-62590-RD-1.DWG

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Little, Daryl
DESIGNED
Little, Daryl
DRAWN
Schilt, Avery
CHECKED
Little, Daryl A
TECH. APPR.
Plaseied, Atousa P.E.
APPROVED Civil Engineer
ADMIN APPROVAL

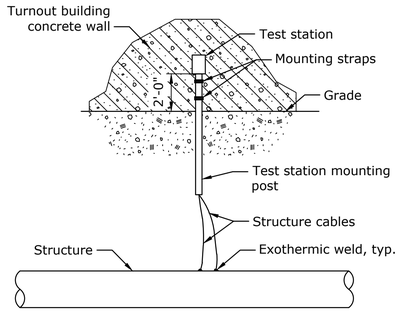
Denver, CO 2026-05-27

GALVANIC ANODE
CATHODIC PROTECTION

JOINT BOND, CABLE
TRENCH, IJFK, AND
CABLE I.D. DETAILS

1695-D-62590

SHEET 1



CORROSION MONITORING TEST STATION

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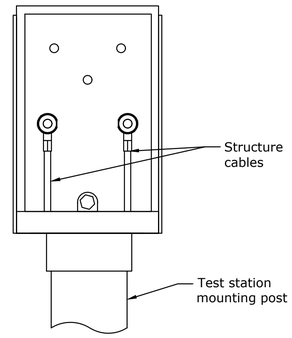
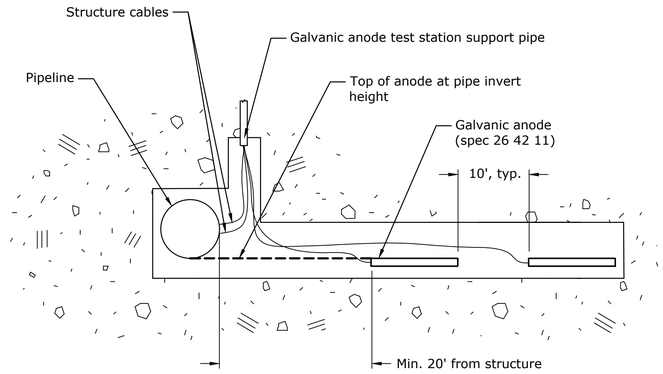


CORROSION MONITORING TEST BOARD

SCALE: NOT TO SCALE



PROFILE - GALVANIC ANODE BED DETAIL

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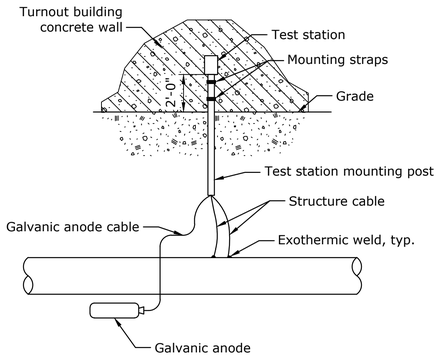

ANODE TEST STATION - SINGLE ANODE

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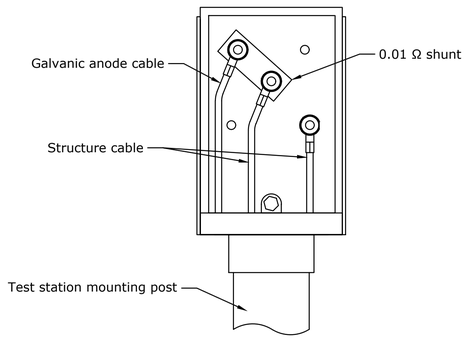


TEST BOARD - SINGLE ANODE

SCALE: NOT TO SCALE

TEST STATION AND CATHODIC PROTECTION REQUIREMENTS				
Turnout	Upstream side		Downstream side	
	Test station	Cathodic protection	Test station	Cathodic protection
Sanostee	Corrosion monitoring	Bond to Reach 4C pipeline and impressed current cathodic protection system	Single anode	Galvanic anode
Newcomb	Corrosion monitoring	Bond to Reach 6 pipeline and impressed current cathodic protection system	Single anode	Galvanic anode
Naschitti	Corrosion monitoring	Bond to Reach 8 pipeline and impressed current cathodic protection system	Single anode	Galvanic anode
BBN9	Corrosion monitoring	Bond to Reach 9A pipeline and impressed current cathodic protection system	Single anode	Galvanic anode
Coyote Canyon	Corrosion monitoring	Bond to Reach 11 pipeline and impressed current cathodic protection system	Single anode	Galvanic anode
Rock Springs	Single anode	Galvanic anode	Single anode	Galvanic anode
Window Rock	Single anode	Galvanic anode	Single anode	Galvanic anode

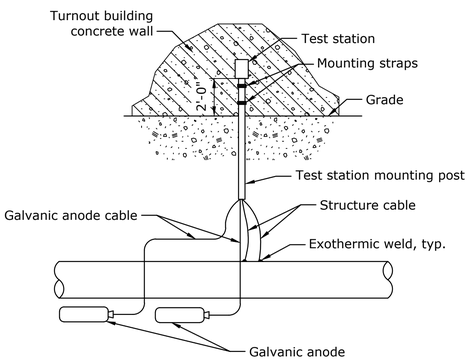
MATERIAL EXPLANATION	
Item	Description
Test station	Cast aluminum with 5 terminals, minimum, post mounted
Test station mounting post	2" Ø galvanized steel post
Shunt	0.01 Ohm calibrated resistor
Anode (galvanic)	Standard potential magnesium anodes (20-lb)
Galvanic anode cable	#12 AWG/HMWPE (red if available)
Structure cable (test station)	#12 AWG/HMWPE (black)
Exothermic weld	See drawing -62590

NOTES

1. Install test station against outside wall of turnout building and anchor support pole to concrete wall with metal straps.
2. Place anodes at the same depth as the bottom of the structure.
3. One shunt for each anode installed at location.
4. Install single anode at each location as determined in table. Install additional anode only if final test data determines it is required.
5. Galvanic anode bed details listed in Specification section 26 42 11.
6. Anodes are to be installed a minimum distance of 20 feet from the structure.
7. Minimum of one anode length from non-metallic pipe.
8. Exothermically weld cables to top center of pipe min. 12" apart.
9. Cathodic protection system components are not to scale for visualization.

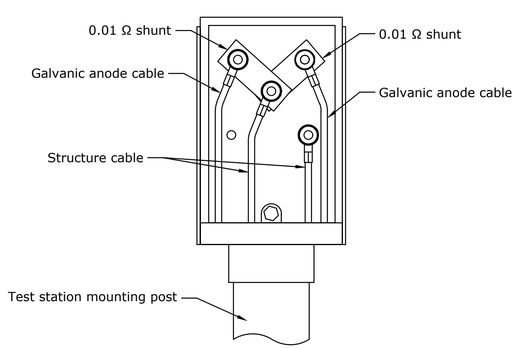
REFERENCE DRAWINGS

SANOSTEE YARD PIPE - PLAN AND SECTION.....1695-D-62561
NEWCOMB YARD PIPE - PLAN AND SECTION.....1695-D-62562
NASCHITTI YARD PIPE - PLAN AND SECTION.....1695-D-62563
BBN9 YARD PIPE - PLAN AND SECTION.....1695-D-62564
COYOTE CANYON YARD PIPE - PLAN AND SECTION.....1695-D-62565
ROCK SPRINGS YARD PIPE - PLAN AND SECTION.....1695-D-62566
WINDOW ROCK YARD PIPE - PLAN AND SECTION.....1695-D-62567
JOINT BOND, CABLE TRENCH, IJFK, AND CABLE I.D. DETAILS 1695-D-62590



ANODE TEST STATION - MULTIPLE ANODE

SCALE: NOT TO SCALE



TEST BOARD - MULTIPLE ANODE

SCALE: NOT TO SCALE

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

TURNOUT BUILDINGS

Little, Daryl
DESIGNED
Little, Daryl
DRAWN
Schilt, Avery
CHECKED
Little, Daryl A
TECH. APPR.
Plaseied, Atousa P.E.
APPROVED Civil Engineer
ADMIN APPROVAL

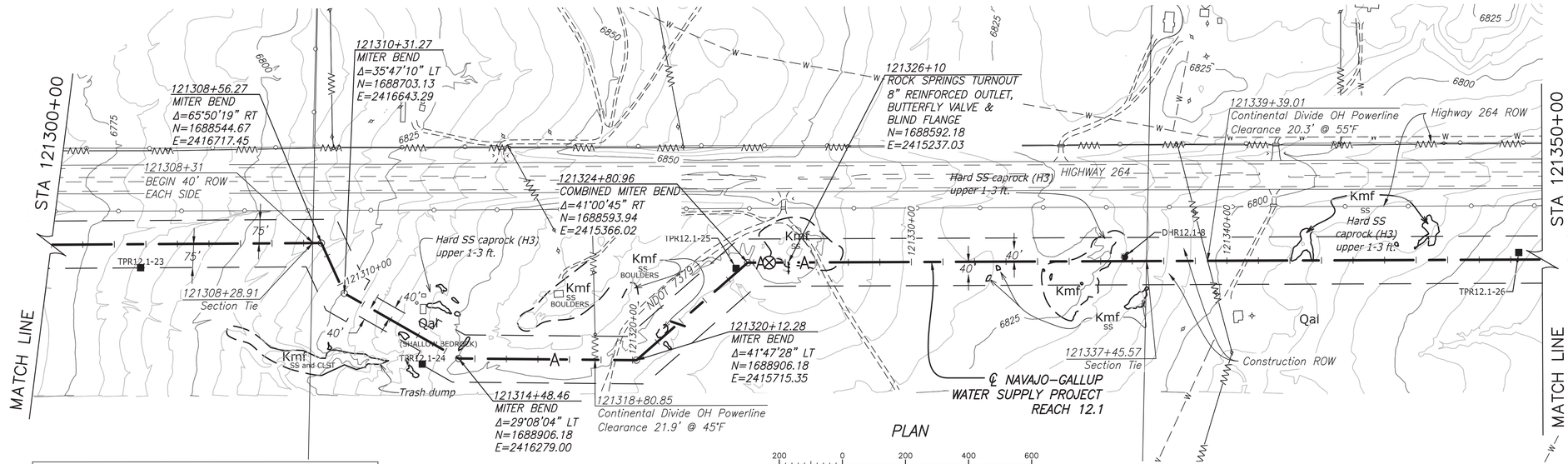
Denver, CO 2026-05-27

GALVANIC ANODE CATHODIC PROTECTION

TEST STATION AND ANODE BED DETAILS

1695-D-62591

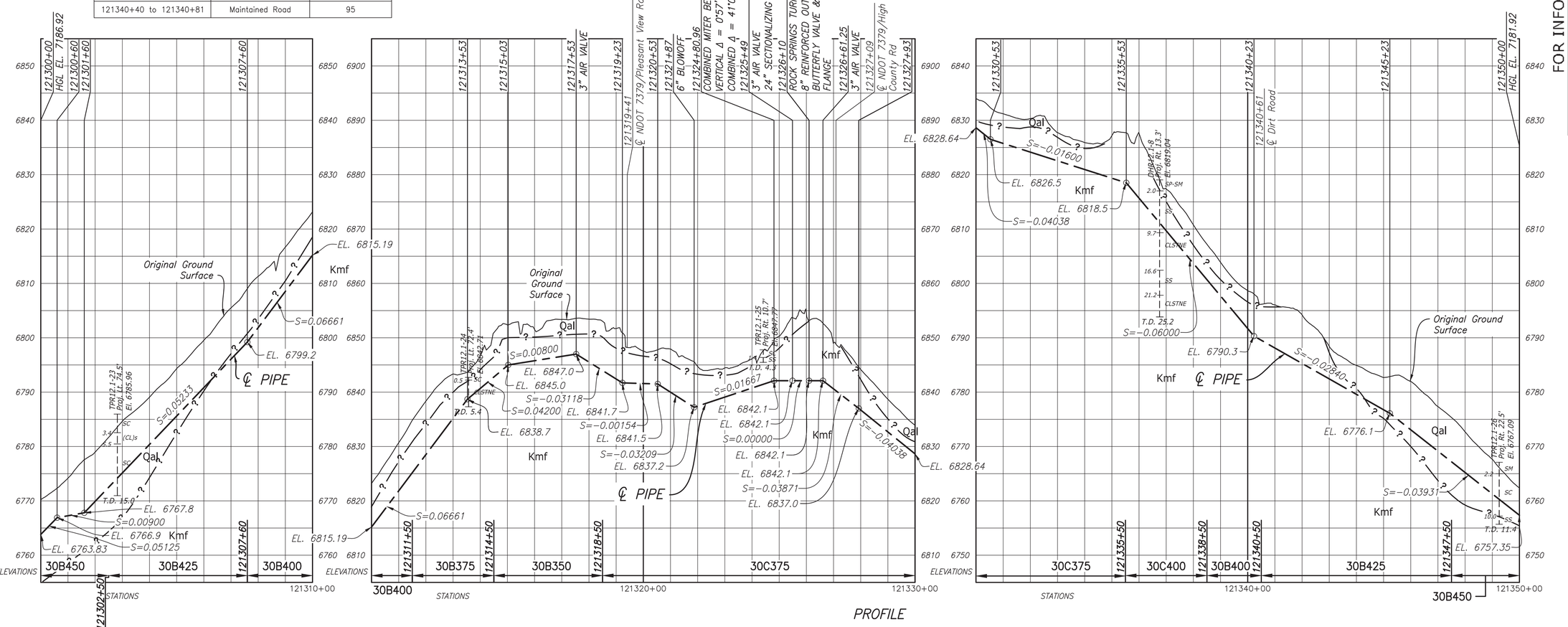
SHEET 2



COMPACTION AT CROSSINGS		
Station	Crossing Type	% Compaction
121319+21 to 121319+61	Maintained Road	95
121326+87 to 121327+30	Maintained Road	95
121340+40 to 121340+81	Maintained Road	95

NOTE

1. For General Notes, Abbreviations, Symbols and Pipe Classifications see, (60113).



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MARCH 25, 2026 11:58
PLOTTED BY
KFELZER

CAD SYSTEM
AutoCAD Rel. 25.0s
CAD FILENAME
1695-D-60119-RD-7.DWG

GEOLOGY
Date 8-23-17
TECHNICAL APPROVAL JUSTIN GILBERT
SUBMITTED DEBBY THORNBERG
APPROVED CHRIS BEYER

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NEW MEXICO

SAN JUAN LATERAL – REACH 12.1
NAVAJO CODE TALKERS SUBLATERAL
LINE PIPE

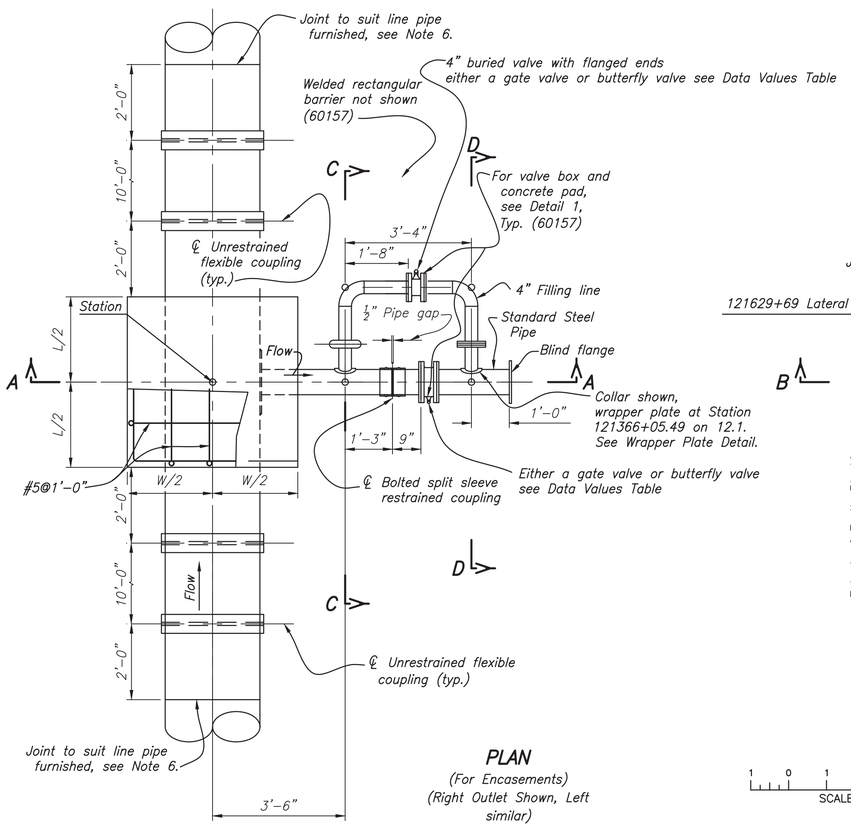
REV NO	DATE	BY	
4	2023-04-10	Jerry R. Waugh, P.E.	D Profile changed to account for the relocation of the Rock Springs Turnout
3	2021-05-03	Jerry R. Waugh, P.E.	D Deleted temporary construction easement table. Added NDOT 7379 Pleasant View Rd and NDOT 7379/High County Rd.
2	2020-09-14	Jerry R. Waugh, P.E.	D Eliminated station equation.
1	2018-12-21	Jerry R. Waugh, P.E.	D Updated title block.

D.L. GESUNDHEIT
DESIGNED
K. FINK
DRAWN
JERRY R. WAUGH, P.E.
CHECKED
JERRY R. WAUGH, P.E.
TECH. APPR.
LINDA M. BOWLES, P.E.
APPROVED
PEER REVIEWER – CIVIL ENGINEER
DENVER, CO 2018-05-18

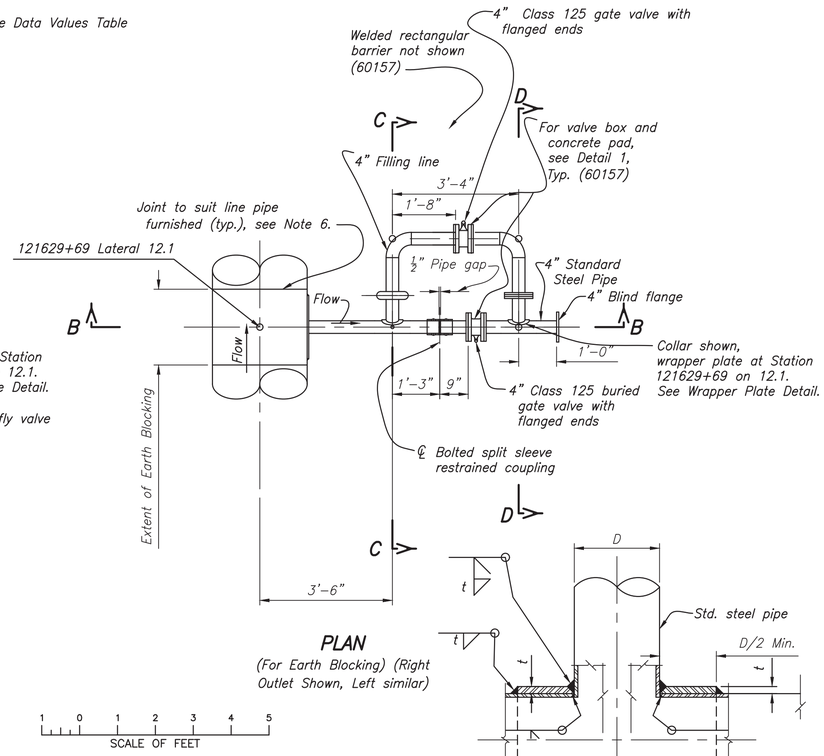
STA. 121300+00
TO
STA. 121350+00

PLAN AND PROFILE

1695-D-60119
SHEET 7 OF 13



PLAN
(For Encasements)
(Right Outlet Shown, Left similar)



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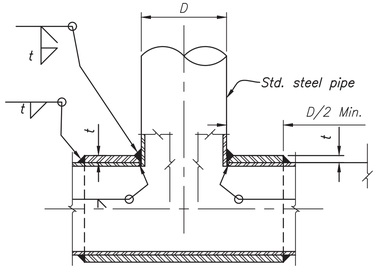
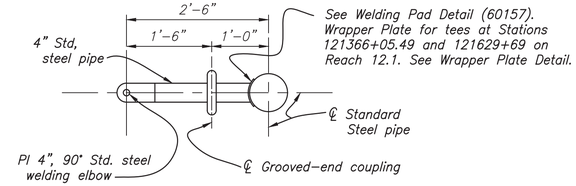
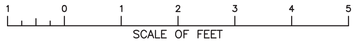
(For Earth Blocking) (Right Outlet Shown, Left similar)

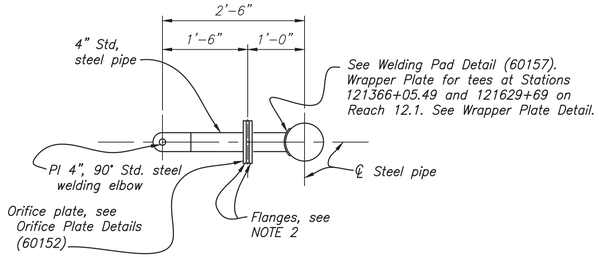


WRAPPER PLATE DETAIL
Design According to AWWA M11
(Not to Scale)

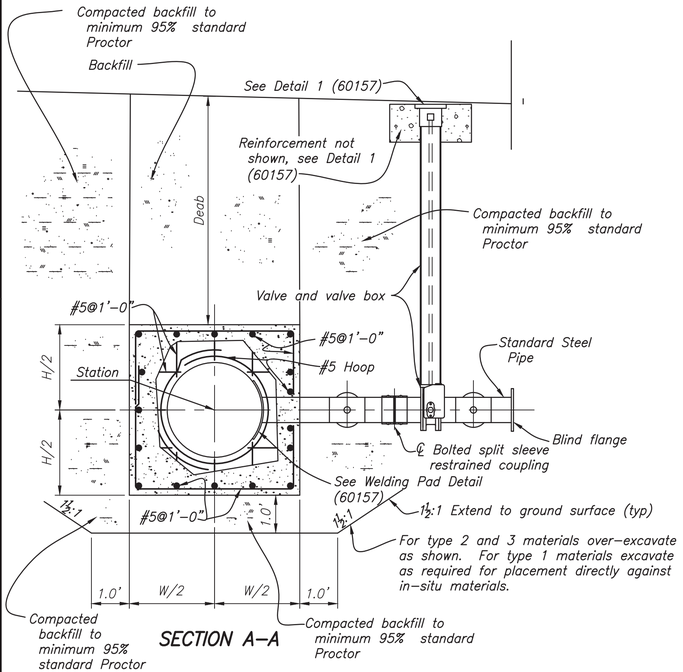


SECTION C–C

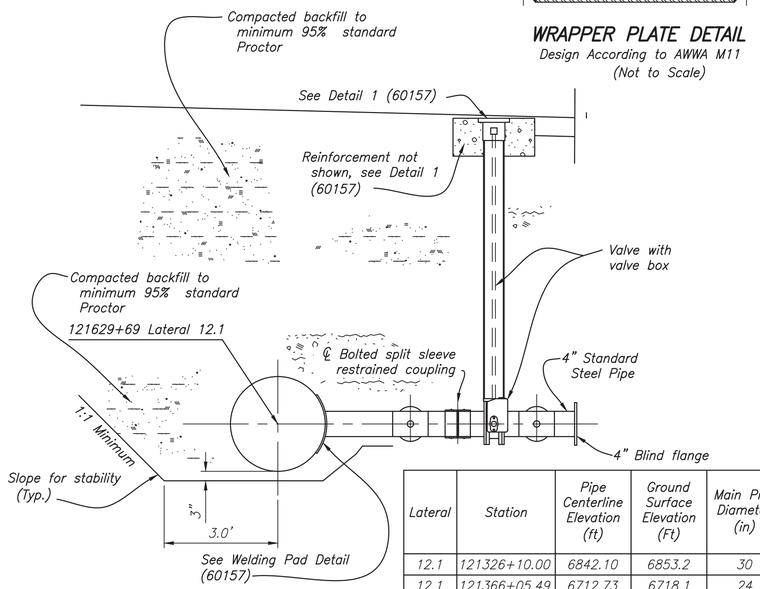




SECTION D–D



SECTION A–A



SECTION B–B

OPERATING CRITERIA

Bypass butterfly valve shall be fully open when operated.

NOTES

1. *For Plan and Profile, see (60120) and (60128).*
2. *Flange class shall meet pressure requirement designated by adjacent symbol line pipe. Flange drilling shall match mating flange.*
3. *For general notes and minimum requirements for detailing reinforcement, see 40–D–60004. Concrete compressive strength, (f'c), is 4,500 psi and specified yield point for reinforcement, (fy), is 60,000 psi.*
4. *For general concrete outline notes, see 40–D–60003.*
5. *The valves are either gate valves or butterfly valves for class of valve see Data Values Table. Compacted backfill to minimum 95% standard Proctor to be installed within 1 foot around the valve. CLSM is not allowed.*
6. *Provide dielectric joint for dissimilar metallic pipes.*
7. *Direction when looking downstream.*

DATA VALUES

Lateral	Station	Pipe Centerline Elevation (ft)	Ground Surface Elevation (Ft)	Main Pipe Diameter (in)	Outlet Diameter (in)	Direction (Note 7)	Head Class (ft)	Type of Valve	Class of Valve	Flange Type	Number of Holes in Orifice Plate	Concrete Encasement or Earth Blocking	W	L	H	Deab (ft)
12.1	121326+10.00	6842.10	6853.2	30	8	Right	375	Butterfly	250B	Class E	7	Encasement	4'–6"	4'–6"	4'–6"	8.8
12.1	121366+05.49	6712.73	6718.1	24	4	Right	500	Gate	250	Class E	6	Encasement	4'–0"	4'–0"	4'–0"	3.4
12.1	121629+69.00	7061.10	7071.4	24	4	Right	100	Gate	125	Class D	15	Earth Blocking	N/A	N/A	N/A	N/A
12.2	122111+00.00	6897.85	6903.2	24	6	Left	300	Butterfly	150B	Class D	8	Encasement	4'–0"	4'–0"	4'–0"	3.4
12.2	122176+00.00	6828.51	6833.5	24	6	Right	425	Butterfly	250B	Class E	7	Encasement	4'–0"	4'–0"	4'–0"	3.0
12.2	122206+00.00	6801.30	6813.0	24	8	Right	475	Butterfly	250B	Class E	7	Encasement	4'–0"	4'–0"	4'–0"	4.8

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL – REACH 12.1
NAVAJO CODE TALKERS SUBLATERAL
ROCK SPRINGS TURNOUT

REV. NO			
6	D	2023-03-27 Jerry R. Waugh, P.E.	Changed Station of Rock Springs Turnout on Reach 12.1 from Sta. 121353+53.11 to Sta. 121326+10.
5	D	2021-08-03 Jerry R. Waugh, P.E.	Revisions to data items for tee at Station 122206+00 on 12.2.
4	D	2021-07-08 Jerry R. Waugh, P.E.	Removed gravel surfacing.
3	D	2021-05-03 Jerry R. Waugh, P.E.	Added Notes 6 and 7 and revised callout for tee. Added tee details.
2	D	2020-09-17 Jerry R. Waugh, P.E.	Rock Springs Turnout shortened to valve and flange and revised earthwork notes.
1	D	2018-12-21 Jerry R. Waugh, P.E.	Updated title block. Changed flow meter and flow meter vault and updated Notes as revised.

D.L. GESUNDHEIT
DESIGNED
D.L. GESUNDHEIT
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JERRY R. WAUGH, P.E.
CHECKED
JERRY R. WAUGH, P.E.
TECH. APPR.
LINDA M. BOWLES, P.E.
APPROVED
PEER REVIEWER – CIVIL ENGINEER

DENVER, CO 2018–05–18

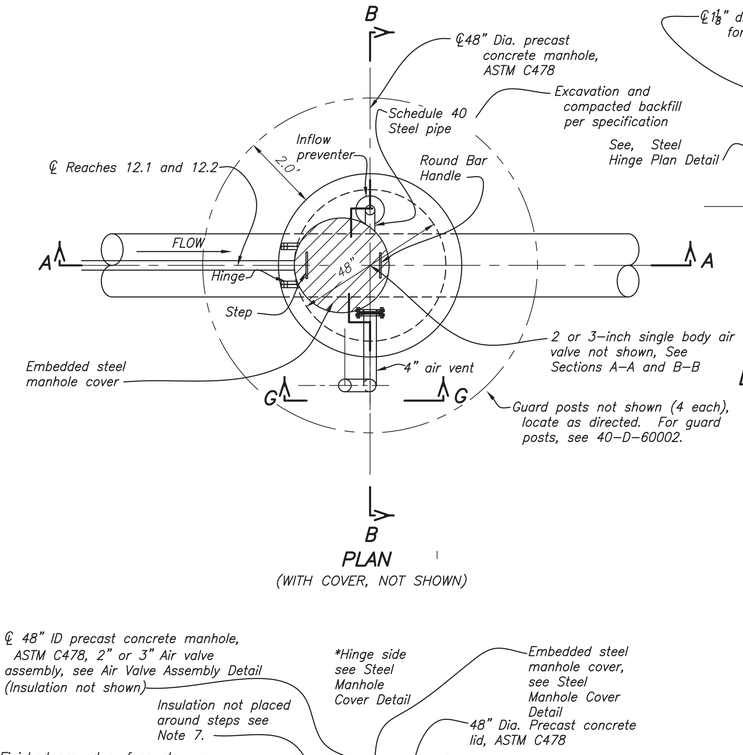
PLAN SECTIONS, AND DETAILS

OUTLETS

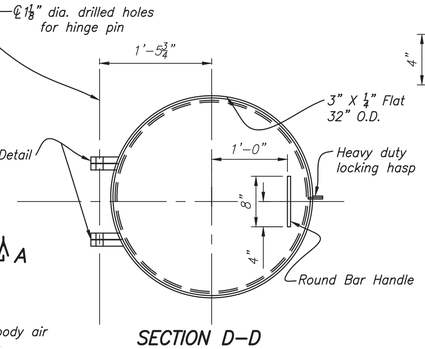
1695–D–60129

SHEET 1 OF 2

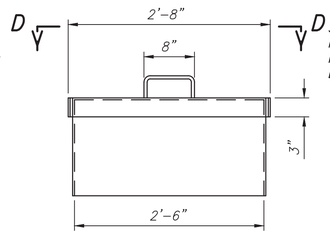
DATE AND TIME PLOTTED MARCH 24, 2026 17:17
PLOTTED BY KFELZER
CAD SYSTEM AutoCAD Rel. 25.0s
CAD FILENAME 1695-D-60129 ROCKSPRINGSTURNOUT.DWG



PLAN
(WITH COVER, NOT SHOWN)



SECTION D–D



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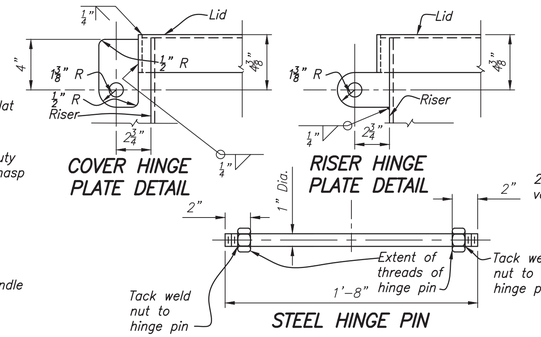
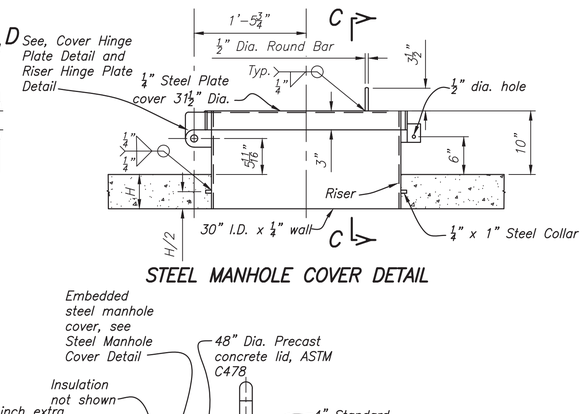
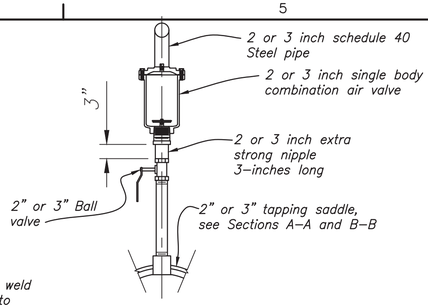


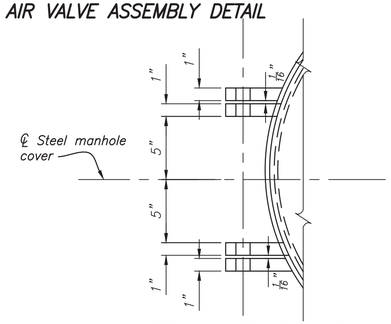
STEEL HINGE PIN



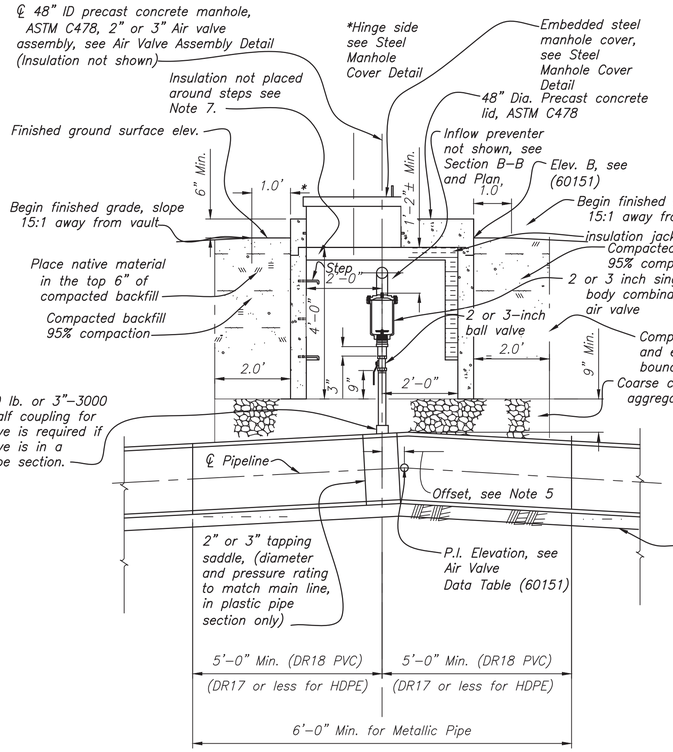
STEEL MANHOLE COVER DETAIL



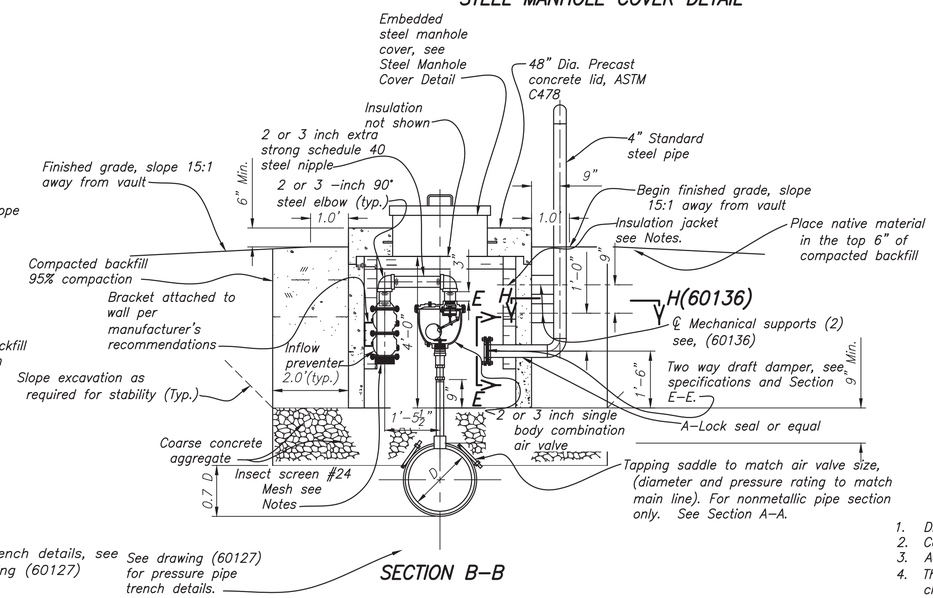
AIR VALVE ASSEMBLY DETAIL



STEEL HINGE PLAN DETAIL

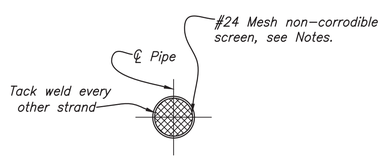


SECTION A–A



SECTION B–B

SECTION G–G



DETAIL A

SECTION E–E

SECTION F–F

NOTES

1. Drawing not to scale.
2. Construct air valve vertical.
3. Actual trench slopes will depend on soil stability.
4. The height of 48" precast manhole shall be high enough to allow clearance between the top of the air valve and the bottom the concrete lid of the precast vault per the air valve manufacturer's recommendations.
5. Locate air valve on the side of the P.I. with the flatter slope and as close to the P.I. as is practical. For air valve configuration where air valve occurs at a concrete encased miter bend, see (60251) and (60148).
6. For air valve locations and sizes, see (60151).
7. Extend insulation jacket a minimum of 36" below finish grade insulation. Provide 1'–0" clearance of vault insulation jacket from centerline of steps on either side. Insulate bottom of embedded steel and precast concrete lids. Keep insulation jacket from support brackets of the inflow preventer.
8. Sections A–A and B–B are for configurations without a pressure transducer. For the air valve upstream of the McKinley Hill Tank at Station 122069+50 on Reach 12.2, see Air Valve Assembly Detail.
9. The #24 non–corrodible screen uses a wire of 0.014 inches which produces an opening of 0.0277 inches.
10. Mount inflow preventer at highest point in concrete manhole where possible.
11. Pipe fittings to be coated and lined according to the specifications.

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NEW MEXICO

SAN JUAN LATERAL
NAVAJO CODE TALKERS SUBLATERAL
REACHES 12.1 & 12.2

REV NO	DATE	BY	DESCRIPTION
6	2022-02-11	Jerry R. Waugh, P.E.	Added inflow preventer, two way draft damper and insect screen.
5	2021-08-11	Jerry R. Waugh, P.E.	Revisions to hinge details.
4	2021-07-26	Jerry R. Waugh, P.E.	Removed air valve insulation. Changed access lid for air valve.
3	2021-05-03	Jerry R. Waugh, P.E.	Changed spray-on insulation to insulation jacket.
2	2020-09-17	Jerry R. Waugh, P.E.	Eliminated Surge Tank Pressure Transmitter
1	2018-12-21	Jerry R. Waugh, P.E.	Updated title block. Increased manholes to 30-inches. Deleted fiberglass pipe material option. Deleted steel manhole cover details.

D.L. GESUNDHEIT
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D.L. GESUNDHEIT
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JERRY R. WAUGH, P.E.
CHECKED
JERRY R. WAUGH, P.E.
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LINDA M. BOWLES, P.E.
APPROVED
PEER REVIEWER – CIVIL ENGINEER

DENVER, CO 2018–05–18

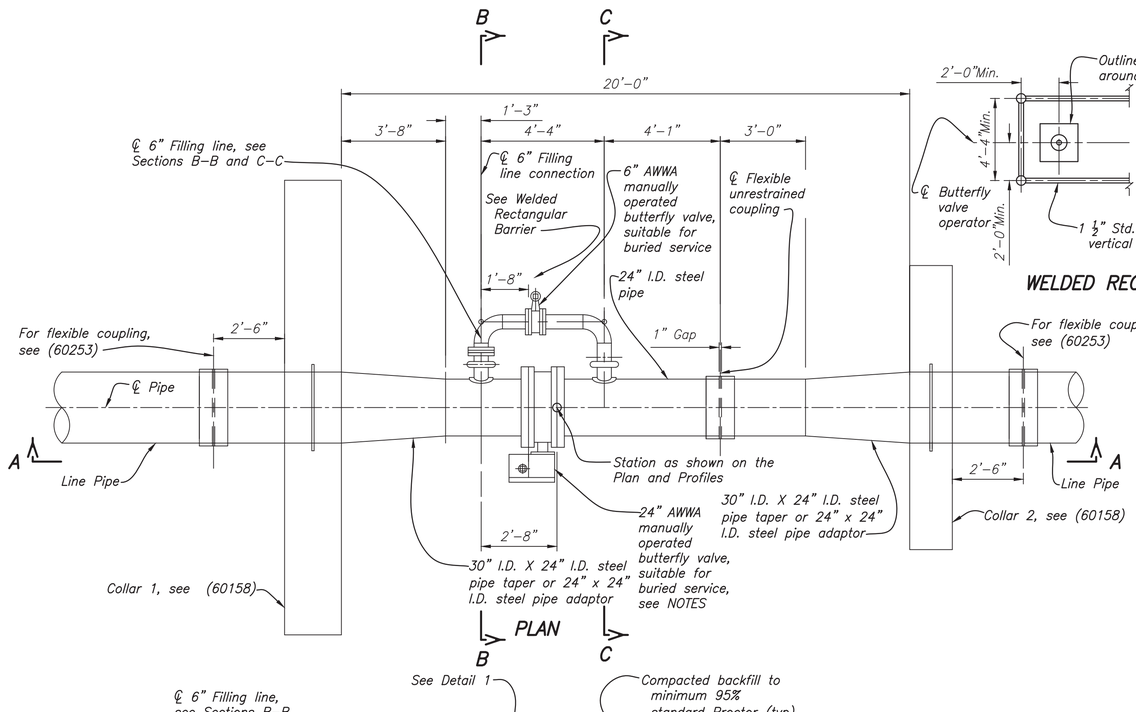
2 OR 3–INCH AIR VALVE

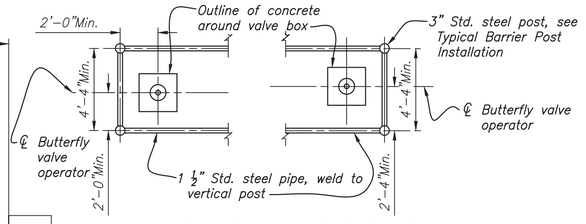
PLAN, SECTIONS, AND DETAILS

1695–D–60150

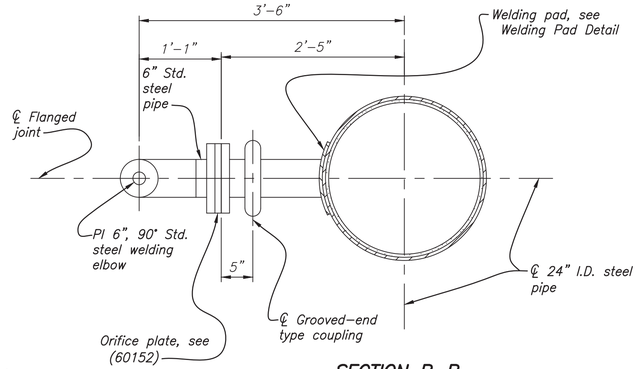
SHEET 1 OF 2

CAD SYSTEM AutoCAD Rel. 25.0s
CAD FILENAME 1695–D–60150 2–INCH AIR VALVE INSTALLATION PLAN SECTIONS AND DETAILS POST AMENDMENT 2
DATE AND TIME PLOTTED MARCH 24, 2026 17:21
PLOTTED BY DANIEL GESUNDHEIT

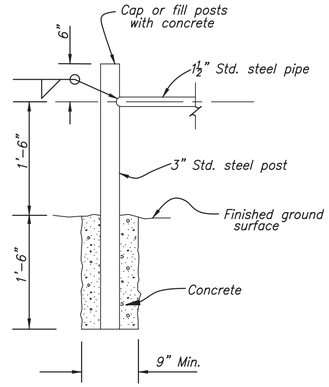




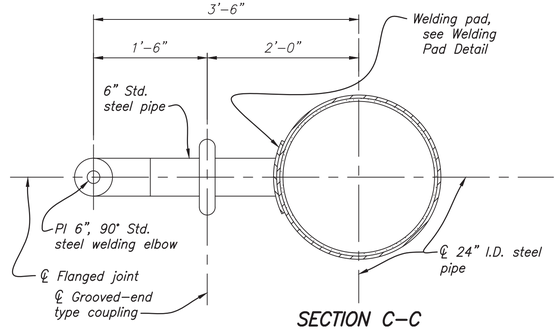
WELDED RECTANGULAR BARRIER



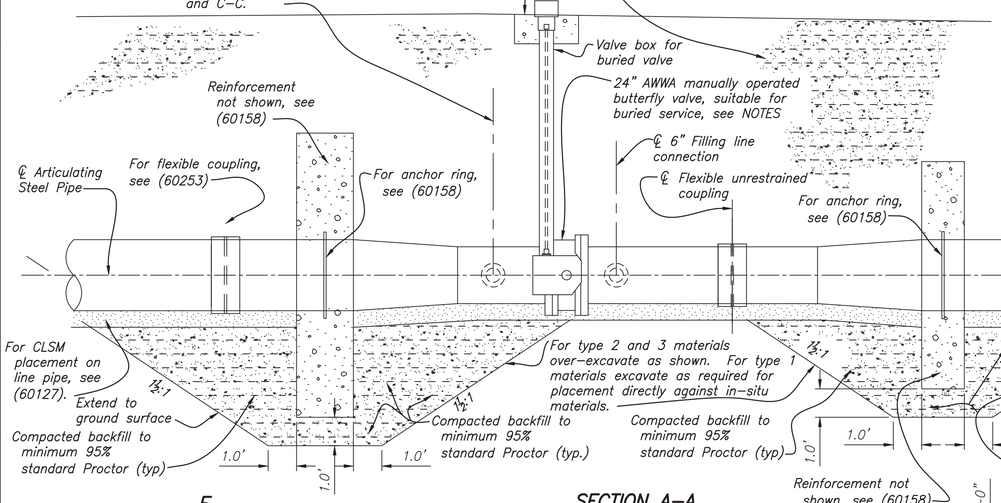
SECTION B-B



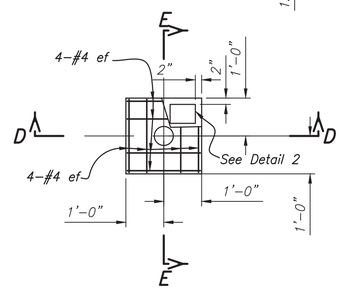
TYPICAL BARRIER POST INSTALLATION DETAIL



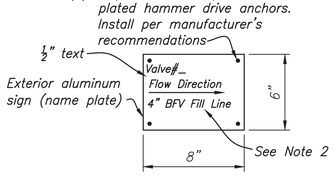
SECTION C-C



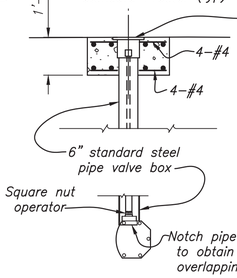
SECTION A-A



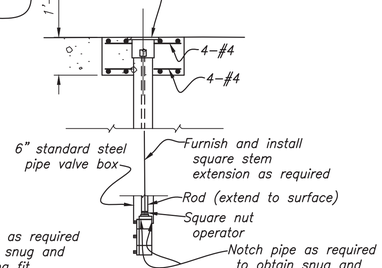
DETAIL 1



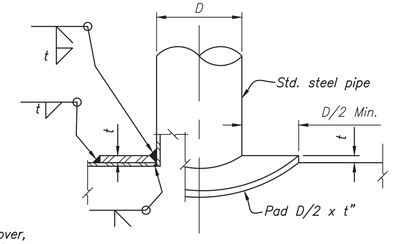
DETAIL 2



SECTION D-D



SECTION E-E



WELDING PAD DETAIL
Design According to AWWA M11
(Not to Scale)

OPERATING CRITERIA

1. High performance bypass butterfly valve shall be fully open when operated.
2. Excavation around blocking shall not occur when the pipeline is under hydrostatic pressure.

NOTES

1. Drawing not to scale.
2. Sectionalizing valve will require a bypass.
3. For pipe trench earthwork details, see (60127).
4. The minimum steel plate thickness shall be the thickness of the steel line pipe. For the steel line pipe thickness see specifications.
5. Piping next to butterfly valves may require 1" oversizing so that the applied interior lining will not reduce the inside dimension to a smaller size which could prevent the butterfly valve from opening. Valve bypass piping shall be epoxy lined.
6. For details of collars, see (60158).
7. Flange class shall meet pressure requirement designated by adjacent symbol line pipe. Flange drilling shall match mating flange.
8. Contractor may submit alternate designs (eg. other types of pipe and fittings) for approval.
9. The inside diameter of the steel pipe is to the inside of the cement mortar lining.
10. For sectionalizing valve installation layout, see (60253).
11. Provide 300 psi minimum rated manually-operated butterfly valve.

SECTIONALIZING VALVE TABLE REACH 12.1

Valve Number	STATION	PIPE CENTERLINE ELEVATION	HEAD CLASS (FT)	BUTTERFLY VALVE TYPE	NUMBER OF HOLES IN THE ORIFICE PLATE
1	121125+20	6576.09	650	See Note11	11
2	121217+80	6661.07	550	250B	12
3	121325+49	6842.10	375	250B	16
4	121491+49	6708.44	450	250B	13

SECTIONALIZING VALVE TABLE REACH 12.2

Valve Number	STATION	PIPE CENTERLINE ELEVATION	HEAD CLASS (FT)	BUTTERFLY VALVE TYPE	NUMBER OF HOLES IN THE ORIFICE PLATE
5	122205+50	6801.35	475	250B	15
6	122296+60	6774.25	500	250B	14

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SAN JUAN LATERAL

NAVAJO CODE TALKERS SUBLATERAL
REACHES 12.1 & 12.2

REV NO	DATE	BY	DESCRIPTION
D	2023-03-27	Jerry R. Waugh, P.E.	Changed location of Sectionalizing Valve No. 12.1 from Sta. 121352+93 to Sta.121323+45
6	2023-02-09	Jerry R. Waugh, P.E.	To avoid utility conflict, the elevation for the sectionalizing valve at Sta. 122296+60 was adjusted from El. 6780.73 to El. 6774.25.
5	2021-08-02	Jerry R. Waugh P.E.	Changed location of Sectionalizing valve on 12.2.
4	2021-05-03	Jerry R. Waugh, P.E.	Added sectionalizing valve on Reach 12.2. Changed callouts for flanges and butterfly valves. Added Table 12.2.
3	2020-09-16	Jerry R. Waugh, P.E.	Earthwork lines have changed.
2	2018-12-21	Jerry R. Waugh, P.E.	Updated title block. Updated stations after relocation ring. Deleted sectionalizing valve for Rock Springs Turnout. Changed earthwork.
1			

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LINDA M. BOWLES, P.E.
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PEER REVIEWER – CIVIL ENGINEER

DENVER, CO 2018-05-18

SECTIONALIZING VALVE INSTALLATION

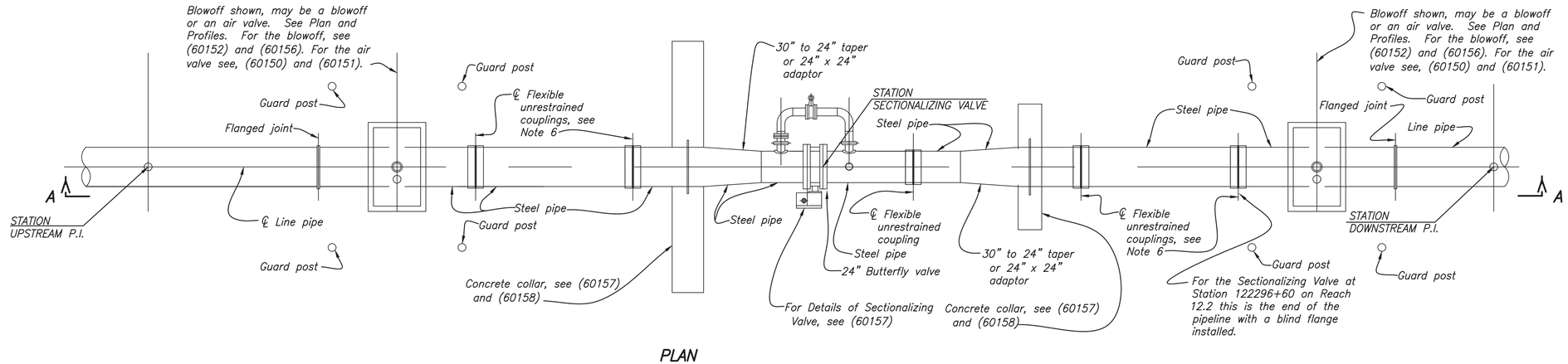
PLAN, SECTIONS, AND TABLE

1695-D-60157

SHEET 2 OF 3

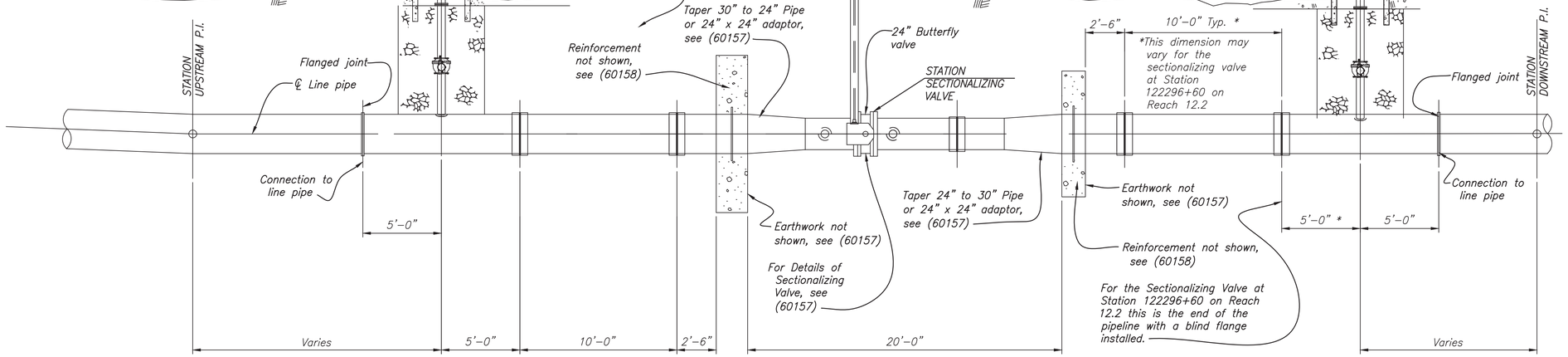
DATE AND TIME PLOTTED
MARCH 25, 2026 12:17
PLOTTED BY
KFELZER

CAD SYSTEM
AutoCAD Rel. 25.0s
CAD FILENAME
1695-D-60157-RD-1.DWG



PLAN

1 0 5 10 15
SCALE OF FEET



SECTION A–A

1 0 5 10 15
SCALE OF FEET

SECTIONALIZING VALVE FEATURE DATA TABLE FOR REACH 12.1				
SECTIONALIZING VALVE STATION	SECTIONALIZING VALVE PIPE CENTERLINE ELEVATION	HEAD CLASS (FT)	UPSTEAM P.I. STATION	DOWNSTREAM P.I. STATION
121125+20	6576.09	650	121123+75	121125+80
121217+80	6661.07	550	121217+30	121219+60
121325+49*	6842.10	375	121324+80.96	121326+61.25
121491+49	6708.44	450	121489+21.02	121491+89

* The downstream air valve at this location is at Station 121326+61.25, see (60119).

SECTIONALIZING VALVE FEATURE DATA TABLE FOR REACH 12.2				
SECTIONALIZING VALVE STATION	SECTIONALIZING VALVE PIPE CENTERLINE ELEVATION	HEAD CLASS (FT)	UPSTEAM P.I. STATION	DOWNSTREAM P.I. STATION
122205+50	6801.35	475	122205+29.43	122206+39.95
122296+60	6774.25	500	122294+99	122296+87.43**

** This is the end of Reach 12.2,

NOTES

1. For details of blowoff, see (60152).
2. For details of sectionalizing valve, see (60157).
3. For pipe trench earthwork, see (60127).
4. Locate guard posts as shown on (60152). For guard posts detail, see 40–D–60002.
5. P.I. is point of intersection.
6. Use flexible restrained couplings when line pipe is HDPE. For the end of Reach 12.2 (Station 122296+87.43) this is a blind flange.
7. For Plan and Profiles, see (60113–60125) and (60254–60260).

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SAN JUAN LATERAL
NAVAJO CODE TALKERS SUBLATERAL
REACHES 12.1 & 12.2

REV NO		
6	D	Changed location of Sectionalizing Valve in 12.1 from Sta. 121352+93 to 121325+49.
5	D	To avoid utility conflict, the elevation for the sectionalizing valve at Sta. 122296+60 was adjusted from El. 6780.73 to El. 6774.25
4	D	Revised location of sectionalizing valve on 12.2.
3	D	Option for Air Valve upstream shown. Added Sectionalizing valve on Reach 12.2.
2	D	Updated Stations after restationing
1	D	Updated title block.

REV NO	2023-03-27	Jerry R. Waugh, P.E.
5	2023-02-09	Jerry R. Waugh, P.E.
4	2021-08-03	Jerry R. Waugh, P.E.
3	2020-11-09	Jerry R. Waugh, P.E.
2	2020-09-17	Jerry R. Waugh, P.E.
1	2018-12-21	Jerry R. Waugh, P.E.

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JERRY R. WAUGH, P.E.
CHECKED
JERRY R. WAUGH, P.E.
TECH. APPR.
LINDA M. BOWLES, P.E.
APPROVED
PEER REVIEWER – CIVIL ENGINEER
DENVER, CO 2018–05–23

SECTIONALIZING VALVE INSTALLATIONS LAYOUT

PLAN, SECTION, AND TABLES

1695–D–60253

SHEET 1 OF 3

DATE AND TIME PLOTTED
MARCH 25, 2026 12:13
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KPELZER

CAD SYSTEM
AutoCAD Rel. 25.0s
CAD FILENAME
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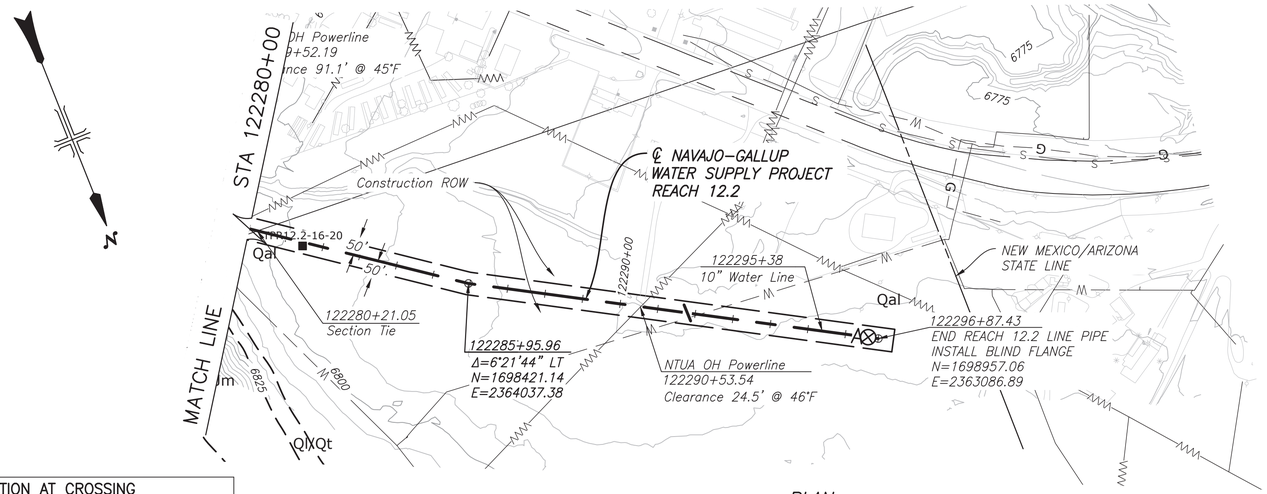
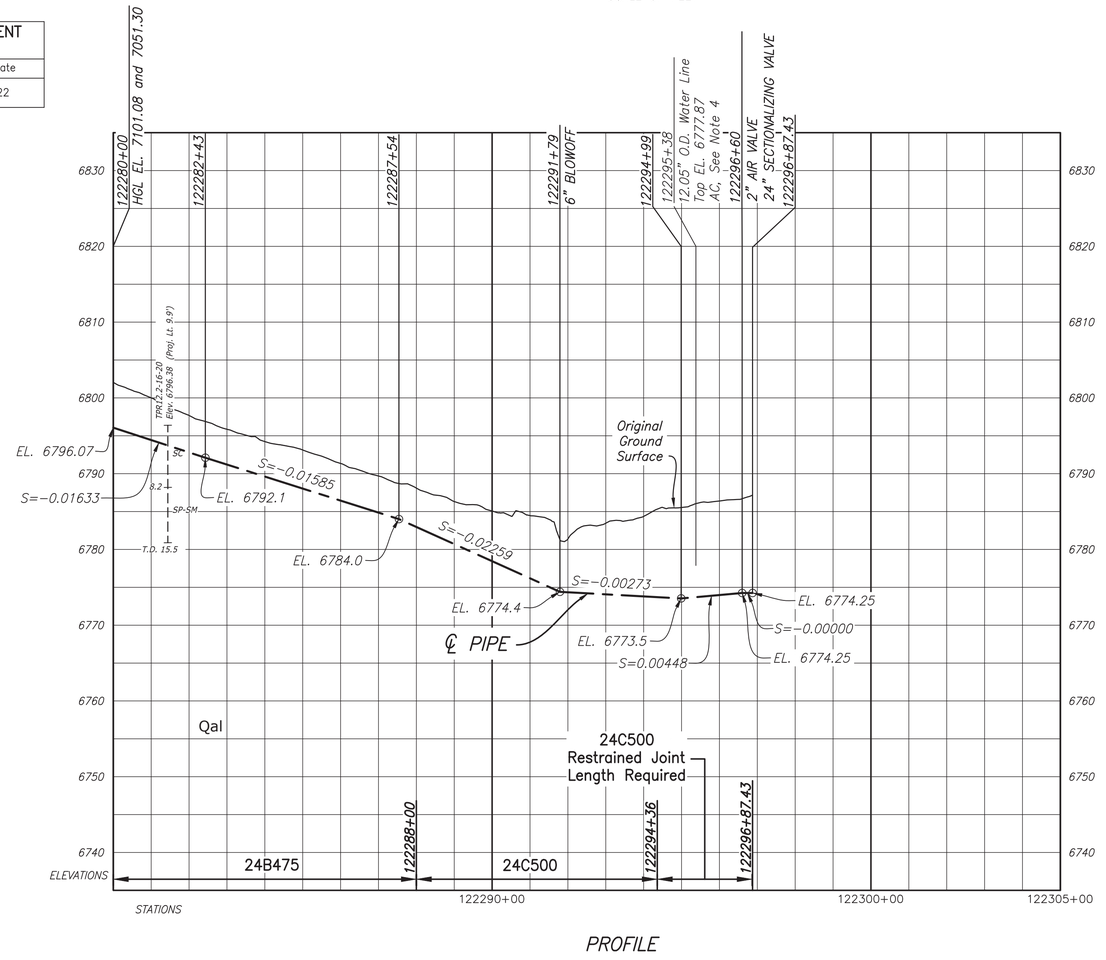


PLAN

200 0 200 400 600
SCALE OF FEET

COMPACTION AT CROSSING		
Station	Crossing Type	% Compaction
122292+71 to 122295+71	Erosion Control	85

TEMPORARY CONSTRUCTION EASEMENT EXPIRATION	
Station	Expiration Date
122277+30.62 to 122296+87.42	09/25/2022



PROFILE

NOTES

1. For General Notes, Abbreviations, Symbols and Pipe Classifications, see (60113).
2. Blind flange at Station 122296+87.43 rated for 250 psi.
3. Provide 2" tap and ball valve in blind flange for pipeline testing. Locate tap at pipe crown level.
4. Asbestos Cement (AC) pipe description provided by S.J. Louis.

CONSTRUCTION CONSIDERATIONS

1. Line pipe must have at least 3.5 feet of cover between Station 122294+36 and the end of the line pipe when pressure testing is performed.
2. PVC pipe may not be used between Stations 122294+36 and the end of the line pipe unless concrete collar restraint is provided in lieu of restrained line pipe.
3. Concrete collar restraint may be submitted for approval in lieu of restrained line pipe.

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NEW MEXICO

SAN JUAN LATERAL – REACH 12.2
NAVAJO CODE TALKERS SUBLATERAL
LINE PIPE

REV NO		
4	2023-01-26	D Modified end of profile to avoid utility conflict at Sta. 122295+38
3	2021-05-03 Jerry R. Waugh, P.E.	D Updated Note 2, added Air Valve and Sectionalizing Valve at Sta. 122296+60.
2	2020-09-17 Jerry R. Waugh, P.E.	D Updated stationing and note 3.
1	2018-12-21 Jerry R. Waugh, P.E.	D Updated title block. Added Note 4 & Section Tie. Updated CH clearances & alignment to end at Sta. 122310.464. Deleted Air Valve at Sta. 122296+65.

D.L. GESUNDHEIT
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K. FINK
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JERRY R. WAUGH, P.E.
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JERRY R. WAUGH, P.E.
TECH. APPR.
LINDA M. BOWLES, P.E.
APPROVED
PEER REVIEWER – CIVIL ENGINEER
DENVER, CO 2018–05–18

GEOLOGY
Date 8-23-17

TECHNICAL APPROVAL JUSTIN GILBERT
SUBMITTED DEBBY THORNBERG
APPROVED CHRIS BEYER

STA. 122280+00
TO
END

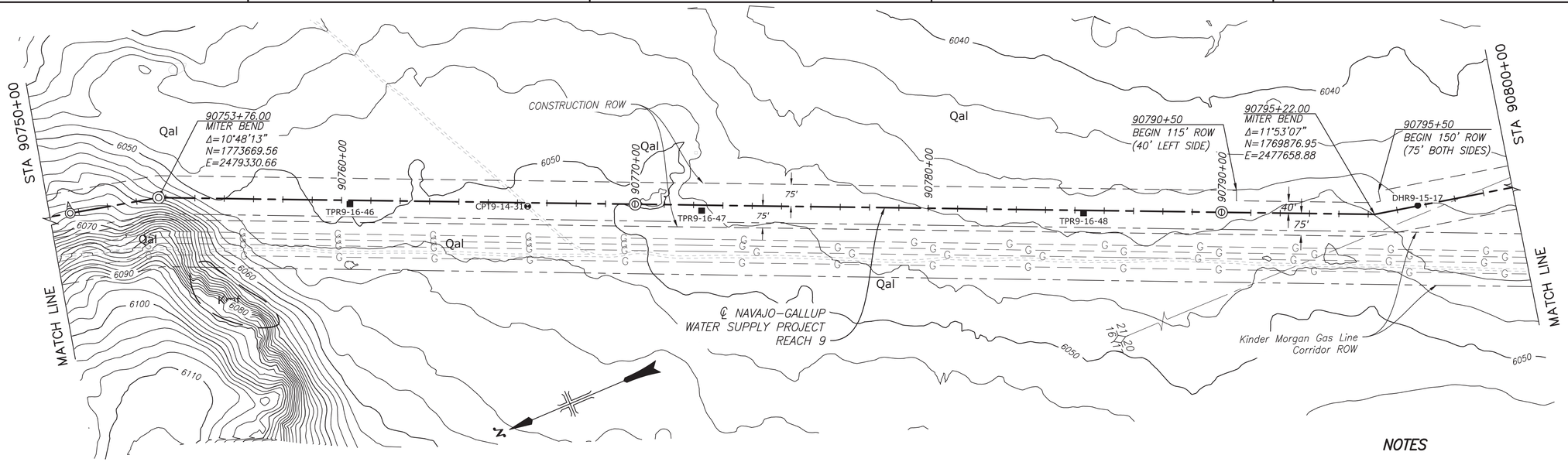
PLAN AND PROFILE

1695–D–60260

SHEET 7 OF 7

DATE AND TIME PLOTTED
MARCH 25, 2026 12:09
PLOTTED BY
KPELZER

CAD SYSTEM
AutoCAD Rel. 25.0s
CAD FILENAME
1695-D-60260-R0-1.DWG



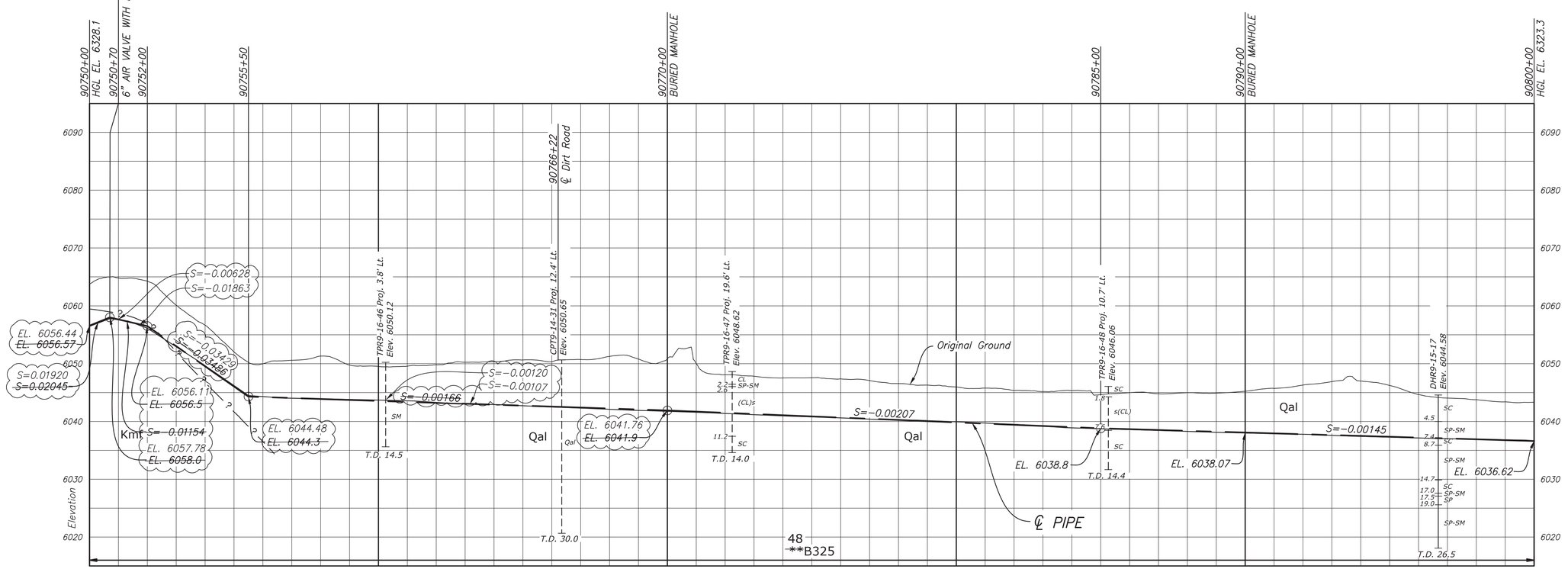
PLAN

200 0 200 400 600
SCALE OF FEET

OPEN–CUT CROSSINGS		
Station	Road / Wash	% Compaction
90766+10 TO 90766+35	Unmaintained Road	95

NOTES

1. For General Notes, Symbols, Abbreviations, and Pipe Classification, see (60281).
2. Diameters with ** designation are dependent on pipe material type, see Line Pipe Material and Diameter Table (60306).



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**B325

PROFILE

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SAN JUAN LATERAL – REACH 9

LINE PIPE

REV NO		
2	2023-08-04	D As-builts.
	Gladden, Ryan L P.E.	
1	2018-08-15	D Revised Construction ROW between Station 90790+150 and Station 90795+50.
	Bowles, Linda M P.E.	

Andrew Marner
DESIGNED
Andrew Marner
DRAWN
D. L. Gesundheit
CHECKED
Bowles, Linda M P.E.
TECH. APPR.
Duke, Wylie C (Chris) P.E.
APPROVED Manager - Water Conveyance Group
PEER REVIEWER
DENVER, CO 2017-03-30

GEOLOGY

TECH. APPR.	JUSTIN GILBERT
SUBMITTED	PAUL GARDNER
APPROVED	JUSTIN GILBERT Supervisory Geologist

PLAN AND PROFILE

STA. 90750+00
TO
STA. 90800+00

1695-D-60297

SHEET 17 OF 26

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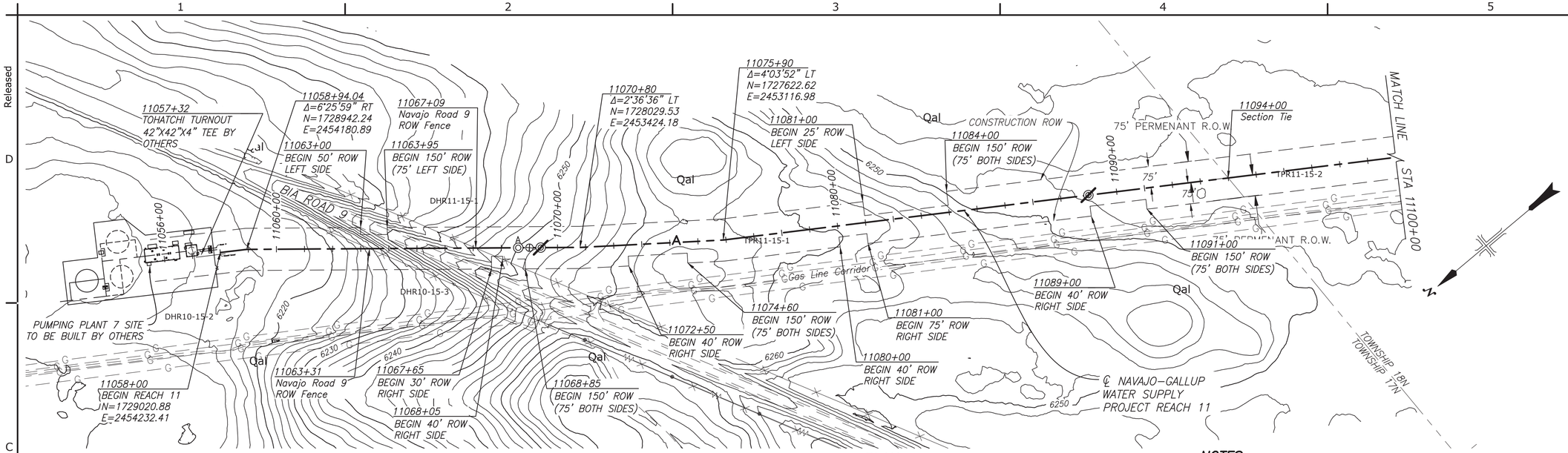
MICROSTATION (LMS TECH)
CAD FILENAME
1695-D-60297-RD-17 OF 26.DWG

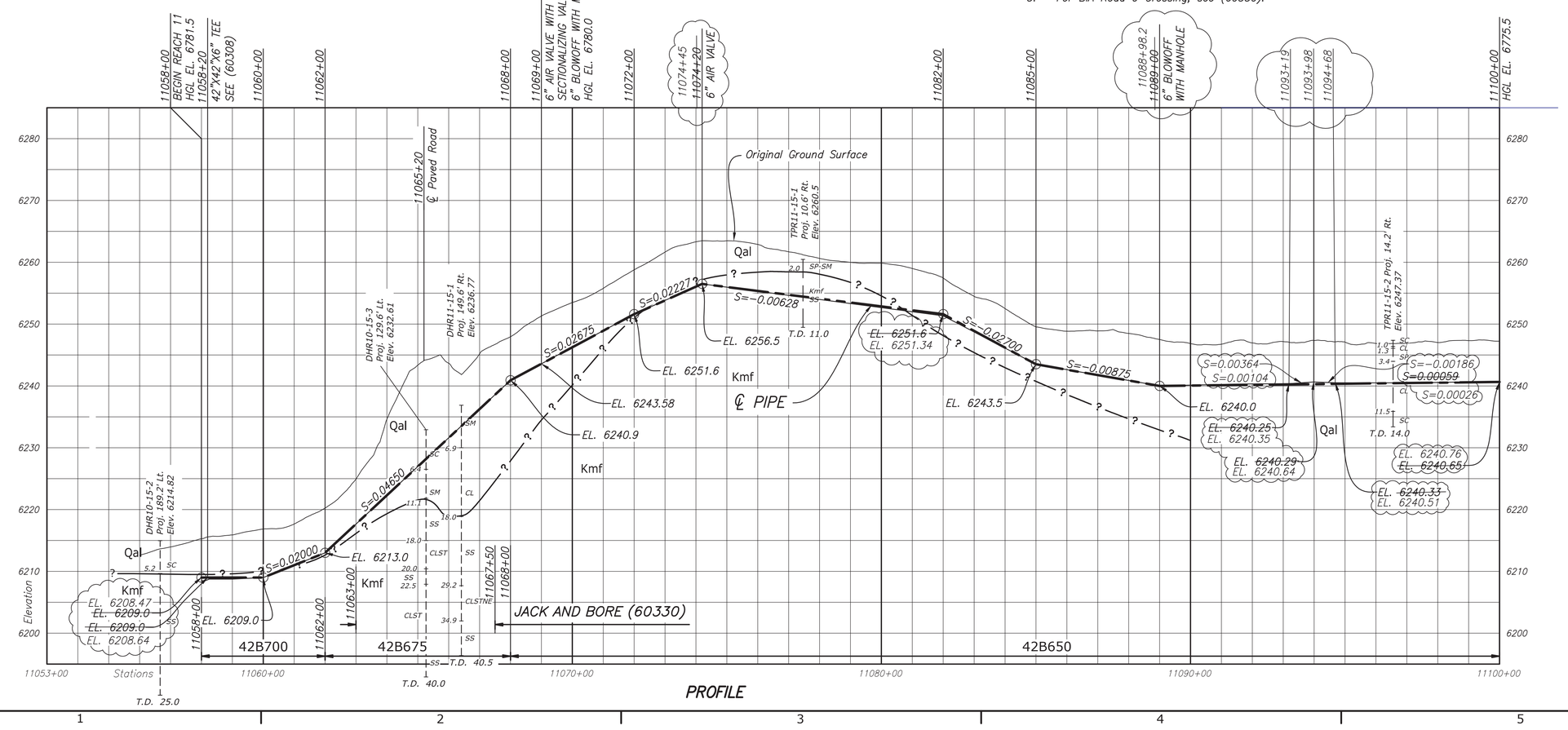


PLAN

200 0 200 400 600
SCALE OF FEET

NOTES

1. For General Notes, Symbols, Abbreviations, and Pipe Classification, see (60281).
2. For Temporary 6" Construction Water Line, see (60308).
3. For BIA Road 9 Crossing, see (60330).



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NEW MEXICO

SAN JUAN LATERAL - REACH 11

LINE PIPE

REV NO	DATE	BY	DESCRIPTION
4.2	2023-08-04	Gladden, Ryan L P.E.	D As-builts
3	2018-02-23	Bowless, Linda M P.E.	D Adjusted Construction ROW widths to 100' total from Sta. 11061+00 to Sta. 11064+00.
2	2018-02-23	Bowles, Linda M P.E.	D Reduced Construction ROW widths to 25' left from Sta. 11081+50 to Sta. 11083+25.
1	2017-05-19	Bowles, Linda M P.E.	D Revised Construction ROW widths.

Andrew Marner
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Andrew Marner
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Bowles, Linda M P.E.
TECH. APPR.
Duke, Wylie C (Chris) P.E.
APPROVED Manager - Water Conveyance Group
ADMIN APPROVAL
DENVER, CO 2017-03-30

GEOLOGY

TECH. APPR. JUSTIN GILBERT
SUBMITTED PAUL GARDNER
APPROVED JUSTIN GILBERT Supervisory Geologist

PLAN AND PROFILE

BEGINNING
TO
STA. 11100+00

1695-D-60309

SHEET 1 OF 5

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AutoCAD Civil 3D (LMS TECH)
CAD FILENAME
1695-D-60309-RD-1 OF 5.DWG

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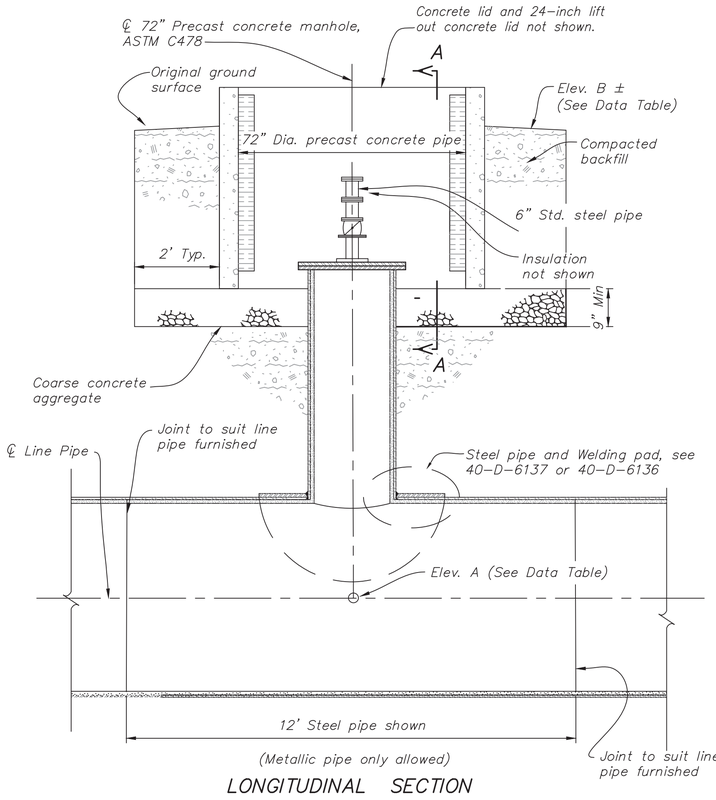
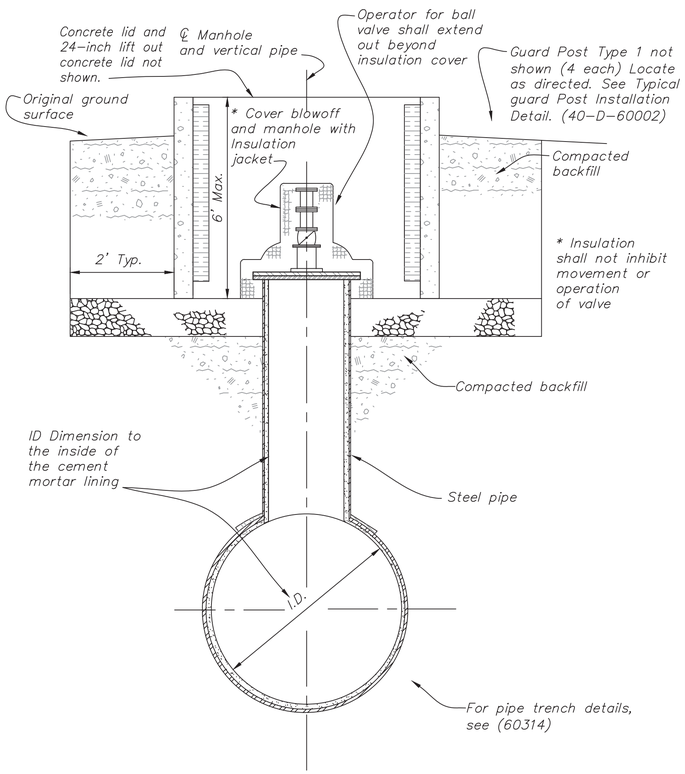
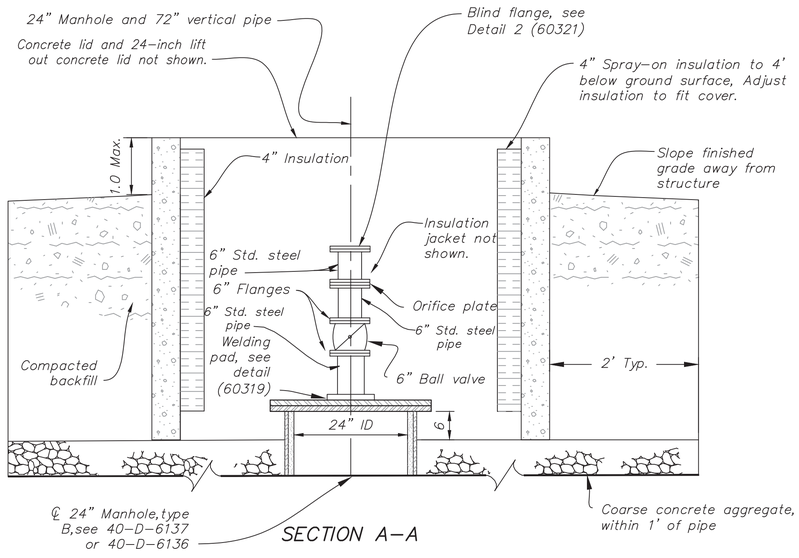


LONGITUDINAL SECTION



TRANSVERSE SECTION



SECTION A-A

NOTES

1. For blowoff locations, see Plan and Profiles (60281 to 60313).
2. For coating and lining requirements for std. steel pipe and fittings, see specification paragraphs.
3. For 6" flange ratings, see Table. Match all mating flanges with highest rated flange.
4. For orifice plate, see Blowoff tables (60321).
5. Any welding to be completed prior to CML application.

BLOWOFF DATA TABLE–REACH 9

STATION	HEAD CLASS (FT)	LINE PIPE DIAMETER (IN)	BALL VALVE TYPE	ORIFICE PLATE NO. HOLES	PIPE CL ELEV A (FT)	GROUND SURFACE ELEV B (FT)	DEPTH OF COVER +/– (FT)	MANHOLE?
90000+11.25	600	48	CLASS 300	NR	5863.83	5873.1	7.3	YES
90023+90	625	48	CLASS 300	38	5827.20	5846.5	17.3	YES
90040+73	625	48	CLASS 300	35	5833.00	5840.0	5.0	NO
90059+00	650	48	CLASS 300	31	5820.20	5827.4	5.2	YES
90059+00	650	48	CLASS 300	NR	5820.20	5827.4	5.2	YES
90078+90	650	48	CLASS 300	NR	5814.80	5830.6	13.8	YES
90098+00	650	48	CLASS 300	NR	5810.60	5826.5	13.9	YES
90105+50	625	48	CLASS 300	NR	5819.75	5828.3	6.6	NO
90115+00	625	48	CLASS 300	33	5830.80	5838.0	5.2	YES
90145+00	575	48	CLASS 300	NR	5864.70	5871.6	4.9	NO
90167+20	575	48	CLASS 300	25	5875.10	5884.7	7.6	YES
90207+50	500	48	CLASS 300	NR	5948.50	5960.6	10.1	NO
90262+00	500	48	CLASS 300	46	5923.80	5938.8	13.0	YES
90275+50	525	48	CLASS 300	38	5921.80	5930.2	6.3	NO
90285+30	525	48	CLASS 300	48	5923.81	5931.4	5.6	YES
90320+50	525	48	CLASS 300	43	5907.50	5920.0	10.5	NO
90330+00	525	48	CLASS 300	25	5912.80	5920.3	5.5	YES
90368+50	500	48	CLASS 300	29	5932.40	5941.0	6.6	NO
90389+50	475	48	CLASS 300	NR	5947.20	5963.0	13.8	NO
90402+00	500	48	CLASS 300	38	5925.60	5938.3	10.7	YES
90440+00	500	48	CLASS 300	38	5928.00	5938.7	8.7	YES
90470+00	450	48	CLASS 300	26	5963.50	5970.9	5.4	YES
90499+50	400	48	CLASS 300	39	6005.40	6016.1	8.7	YES
90547+00	375	48	CLASS 300	NR	6015.00	6022.2	5.2	YES
90566+00	375	48	CLASS 300	48	6007.00	6015.8	6.8	YES
90596+00	375	**	CLASS 300	NR	6008.30	6016.1	5.8	YES
90624+00	400	**	CLASS 300	44	5991.44	6000.0	6.6	YES
90637+00	400	**	CLASS 300	33	5975.80	5981.6	3.8	NO
90660+00	400	**	CLASS 300	36	5982.00	5989.1	5.1	YES
90681+50	375	**	CLASS 300	29	5998.15	6007.3	7.2	YES
90703+00	350	**	CLASS 300	33	6015.70	6022.3	4.6	YES
90725+00	350	**	CLASS 300	46	6035.40	6042.0	4.6	YES
90739+50	350	**	CLASS 300	41	6031.00	6037.4	4.4	YES
90739+50	350	**	CLASS 300	41	6031.00	6037.4	4.4	YES
90810+00	325	**	CLASS 150	44	6034.40	6041.4	5.0	YES
90810+00	325	**	CLASS 150	48	6034.40	6041.4	5.0	YES
90832+50	325	**	CLASS 150	48	6029.66	6042.3	10.6	YES
90879+00	300	42	CLASS 150	48	6052.19	6058.7	4.8	YES
90921+50	275	42	CLASS 150	36	6064.30	6073.3	7.3	YES
90980+00	275	42	CLASS 150	37	6067.90	6075.0	5.4	YES
90980+00	275	42	CLASS 150	48	6067.90	6075.0	5.4	YES
91037+00	250	42	CLASS 150	35	6081.80	6088.7	5.1	YES
91088+00	200	42	CLASS 150	48	6120.50	6127.5	5.3	YES
91142+00	175	42	CLASS 150	NR	6133.70	6141.1	5.7	YES
91142+00	175	42	CLASS 150	41	6133.70	6141.1	5.7	YES
91188+00	125	42	CLASS 150	NR	6161.50	6168.6	5.4	YES

** Denotes where diameter is determined by Pipe Material Type, see (60306)
NR: Not Required

BLOWOFF DATA TABLE–REACH 10

STATION	HEAD CLASS (FT)	LINE PIPE DIAMETER (IN)	BALL VALVE TYPE	ORIFICE PLATE NO. HOLES	PIPE CL ELEV A (FT)	GROUND SURFACE ELEV B (FT)	DEPTH OF COVER +/– (FT)	MANHOLE?
10015+00	100	42	CLASS 150	35	6164.7	6173.5	7.1	YES

BLOWOFF DATA TABLE–REACH 11

STATION	HEAD CLASS (FT)	LINE PIPE DIAMETER (IN)	BALL VALVE TYPE	ORIFICE PLATE NO. HOLES	PIPE CL ELEV A (FT)	GROUND SURFACE ELEV B (FT)	DEPTH OF COVER +/– (FT)	MANHOLE?
11069+00	650	42	CLASS 300	35	6243.58	6251.3	6.0	YES
11089+00	650	42	CLASS 300	33	6240.00	6247.5	5.8	YES
11168+80	600	42	CLASS 300	23	6283.13	6290.4	5.5	YES
11210+00	575	42	CLASS 300	23	6285.10	6292.3	5.4	NO
11241+50	525	42	CLASS 300	30	6321.00	6333.4	10.6	NO
11273+00	475	42	CLASS 300	48	6366.30	6372.3	4.3	YES

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

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BLOCK 9-11 - LINE PIPE

FOR INFORMATION ONLY

REV NO		
3	2018-11-13 Marner, Andrew (Andy) J P.E.	Revised blowoff location from 90262+50 to 90262+60.
2	2017-12-06 Bowles, Linda M P.E.	D Revised Blowoff location from 90000+00 to 90000+11.25 and 90272+00 to 90275+50.
1	2017-05-22 Marner, Andrew (Andy) J P.E.	D Corrected the title block text.

Andrew Marner — DESIGNED
Derek Kosakowski — DRAWN
D.L. Gesundheit — CHECKED
Bowles, Linda M P.E. — TECH. APPR.
Duke, Wylie C (Chris) P.E. — APPROVED
PEER REVIEWER - Manager - Water Conveyance Group

DENVER, CO 2017-03-30

MANHOLE WITH BLOWOFF

SECTIONS, DETAILS, AND TABLES

1695-D-60322

SHEET 1

LAST SAVED DATE 2018-11-13
LAST SAVED BY AMARNER

CAD SYSTEM AutoCAD Civil 3D
CAD FILENAME 1695-D-60322-RD-1.DWG

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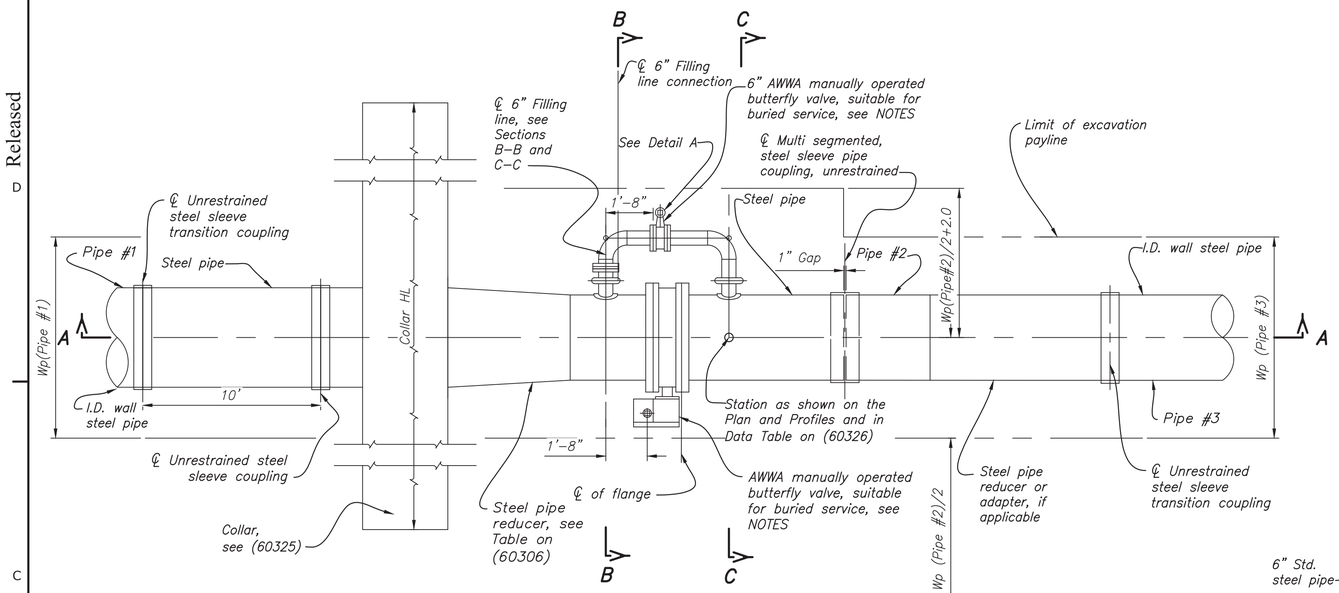
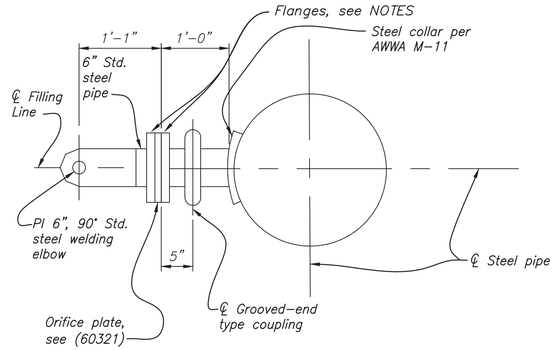


PLAN



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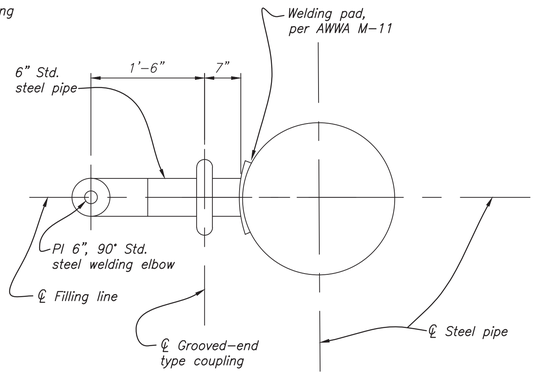
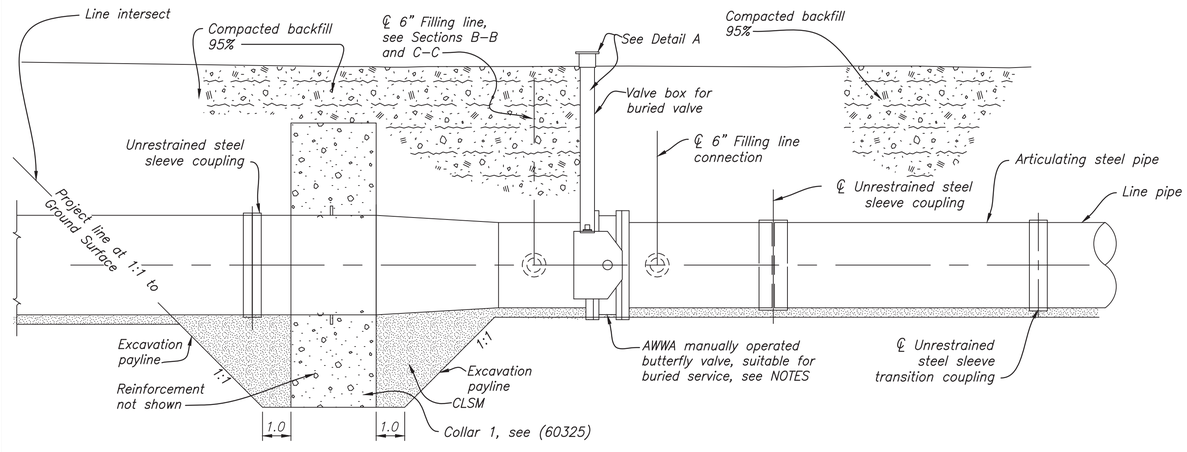


SECTION C–C



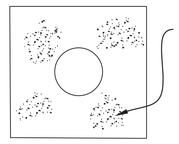
SECTION A–A

OPERATING CRITERIA

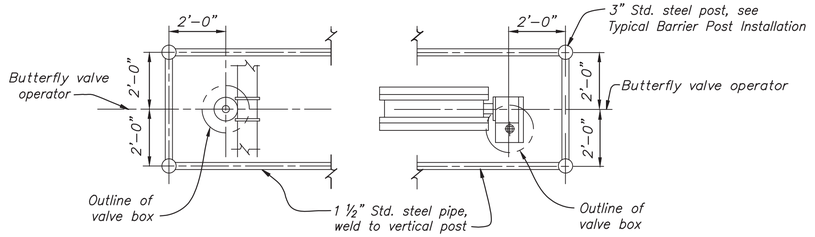
1. High performance Bypass butterfly valve shall be fully open when operated.
2. Excavation around blocking shall not occur when the pipeline is under hydrostatic pressure.

NOTES

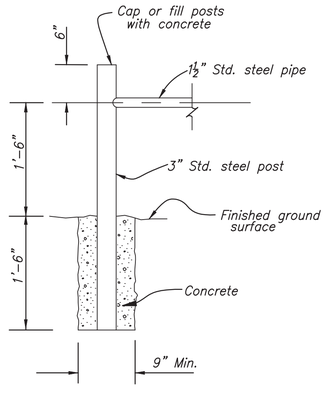
1. Drawing not to scale.
2. All sectionalizing valves will require a bypass.
3. For Wp and pipe trench earthwork details, see (60314).
4. The minimum steel cylinder thickness for the section of line pipe with the valve outlet shall be the thickness of the steel line pipe. For the steel line pipe thickness see the specification.
5. Piping next to butterfly valves may require 1" oversizing so that the applied interior lining will not reduce the inside dimension to a smaller size which could prevent the butterfly valve from opening. Valve bypass piping shall be epoxy lined.
6. For details of tapers, adaptors and collars, see (60325).
7. Flange class shall meet pressure requirement designated by adjacent line pipe. Flange drilling shall match mating flange.
8. For Flange and Valve ratings, see notes (60319).



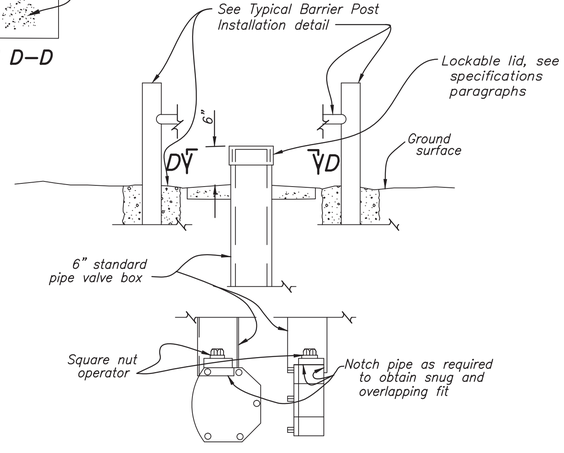
SECTION D–D



WELDED RECTANGULAR BARRIER



TYPICAL BARRIER POST INSTALLATION DETAIL



DETAIL A
(PROFILE VIEW)

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BLOCK 9-11 - LINE PIPE

REV NO	DATE	BY	DESCRIPTION
1	2017-06-09	Bowles, Linda M P.E.	Corrected distances shown on Section B-B and Section C-C. Revised limits of backfill compacting.

Andrew Marner — DESIGNED
Derek Kosakowski — DRAWN
David Gesundheit — CHECKED
Bowles, Linda M P.E. — TECH. APPR.
Duke, Wylie C (Chris) P.E. — APPROVED
PEER REVIEWER - Manager - Water Conveyance Group
DENVER, CO 2017-03-30

SECTIONALIZING VALVE INSTALLATION

PLAN, SECTIONS AND DETAILS

1695-D-60324

SHEET 1

DATE AND TIME PLOTTED
3/28/2017 5:43 PM
PLOTTED BY
svc-Corp-EDM

CAD SYSTEM
AutoCAD Rel. 20.0s
CAD FILENAME
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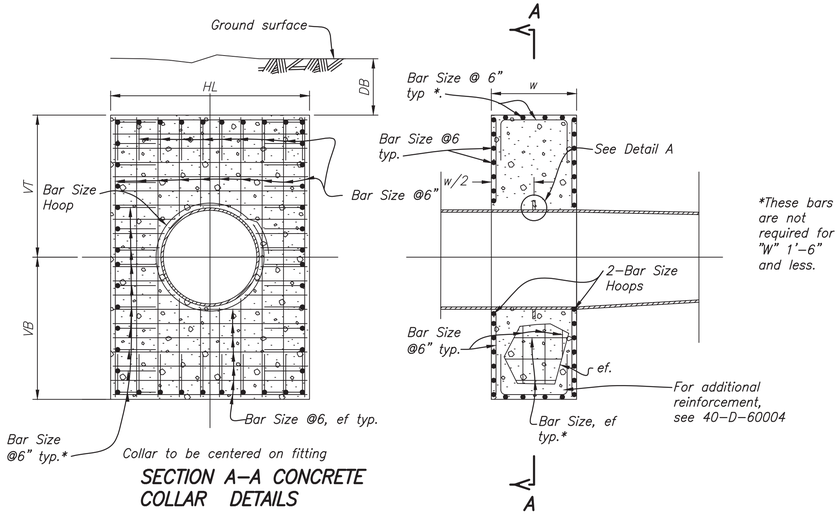
Released



SECTION A–A CONCRETE COLLAR DETAILS

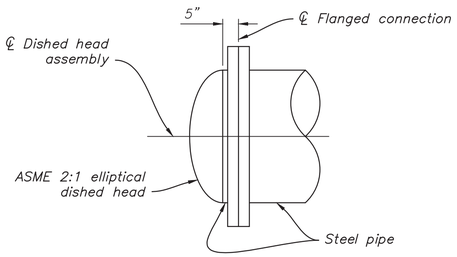
Locate collars as shown on (60326) for valve installations.
For Collar reinforcement and thickness, see Collar Table.
Earthwork not shown, see (60324).

COLLAR TABLE REACH 9

SECTIONALIZING VALVE STATION	DESIGN HEAD (FT)	PIPE DIAMETER (IN)	HL	VT	VB	DB (FT)	W	BAR SIZE
90000+11.25	375	48	25'–0"	5'–0"	9'–0"	4.1	2'–6"	7
90059+00	418	48	25'–0"	5'–0"	10'–0"	2.1	2'–9"	7
90115+00	408	48	25'–0"	5'–0"	9'–0"	2.1	2'–9"	7
90167+20	363	48	25'–0"	6'–6"	8'–0"	3.1	2'–6"	7
90224+60	282	48	20'–0"	7'–0"	8'–0"	0.1	2'–3"	7
90285+30	315	48	25'–0"	6'–0"	7'–0"	1.6	2'–3"	7
90330+00	326	48	25'–0"	5'–6"	8'–0"	2.1	2'–3"	7
90384+00	276	48	25'–0"	5'–6"	7'–0"	1.6	2'–0"	8
90419+40	301	48	25'–0"	5'–6"	7'–0"	1.6	2'–3"	8
90470+00	275	48	25'–0"	5'–0"	6'–3"	2.1	2'–0"	8
90523+00	212	48	20'–0"	5'–6"	6'–0"	1.6	1'–9"	7
90577+00	223	48	20'–0"	5'–6"	7'–0"	2.6	1'–9"	7
90624+00	247	48	20'–0"	6'–0"	7'–0"	2.1	2'–0"	6
90681+50	240	48	20'–0"	5'–0"	6'–0"	4.1	2'–0"	7
90739+50	207	48	20'–0"	4'–0"	7'–0"	2.1	1'–9"	7
90810+00	204	48	20'–0"	5'–0"	6'–0"	2.1	1'–9"	7
90879+00	186	48	20'–0"	5'–0"	6'–6"	1.1	1'–9"	7
90921+50	174	42	15'–0"	5'–0"	5'–0"	3.8	1'–6"	6
90980+00	171	42	15'–0"	5'–0"	6'–0"	1.8	1'–6"	6
91037+00	157	42	15'–0"	5'–0"	5'–0"	1.8	1'–6"	6
91088+00	118	42	10'–0"	5'–0"	6'–0"	1.8	1'–3"	5
91142+00	105	42	10'–0"	5'–0"	5'–0"	2.3	1'–3"	5
91188+00	77	42	10'–0"	4'–0"	4'–0"	2.8	1'–0"	5

COLLAR TABLE REACH 11

SECTIONALIZING VALVE STATION	DESIGN HEAD (FT)	PIPE DIAMETER (IN)	HL	VT	VB	DB (FT)	W	BAR SIZE
11069+00	482	42	29'–0"	5'–6"	7'–0"	1.8	2'–9"	8
11168+00	442	42	25'–0"	5'–6"	8'–0"	1.8	2'–6"	7



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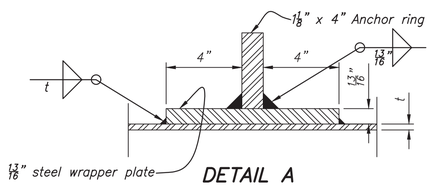

DETAIL A

OPERATING CRITERIA

1. Excavation around the blocking shall not occur when the pipeline is under hydrostatic pressure.

NOTES

1. Drawing not to scale.
2. Flange class shall meet pressure requirement designated by adjacent symbol line pipe. Flange drilling shall match mating flange. See Notes on (60319).
3. Steel plate minimum thickness, t, shall not be less than specification requirements.
4. Concrete compressive strength, f'c, is 4,500 psi and specified yield point for reinforcement, fy, is 60,000 psi.
5. For general concrete outline notes and for minimum requirements for detailing reinforcement, see 40–D–60003, and 40–D–60004.
6. For coating and lining options, see specification paragraphs.

DESIGNED

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BLOCK 9-11 - LINE PIPE

REV NO	DATE	BY	DESCRIPTION
0	2017-12-11	Bowles, Linda M P.E.	Sectionalizing valve at 90000+00 changed to 90000+11.25.
1			

Andrew Marner
DESIGNED
Derek Kosakowski
DRAWN
D.L. Gesundheit
CHECKED
Bowles, Linda M, P.E.
TECH. APPR.
Duke, Wylie (Chris), P.E.
APPROVED
PEER REVIEWER - Manager - Water Conveyance Group

DENVER, CO 2017-03-30

SECTIONALIZING VALVE INSTALLATION

TAPERS, ADAPTORS, AND COLLARS

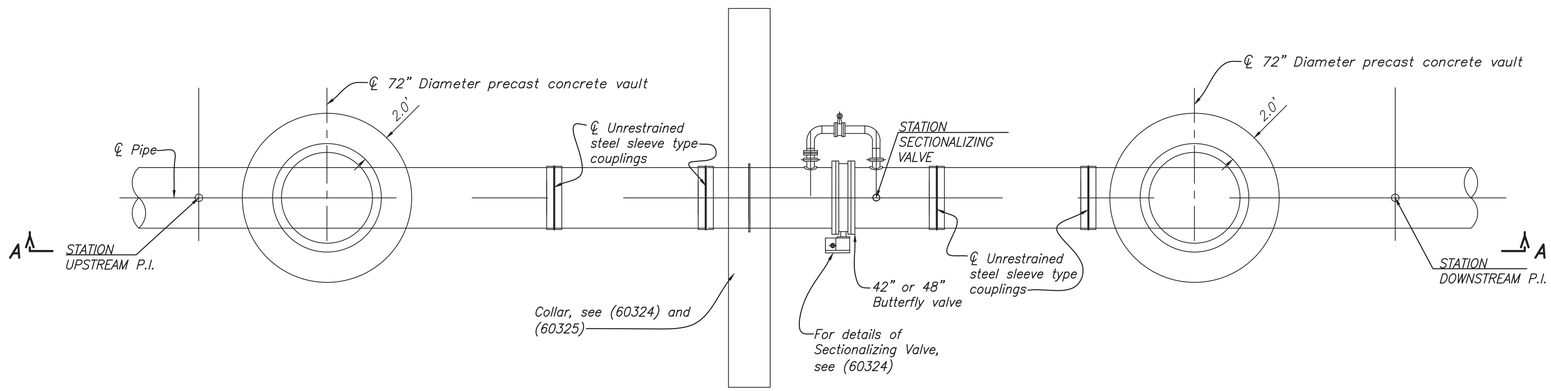
1695-D-60325

SHEET 1

DATE AND TIME PLOTTED
3/28/2017 5:24 PM
PLOTTED BY
svc-Corp-EDM

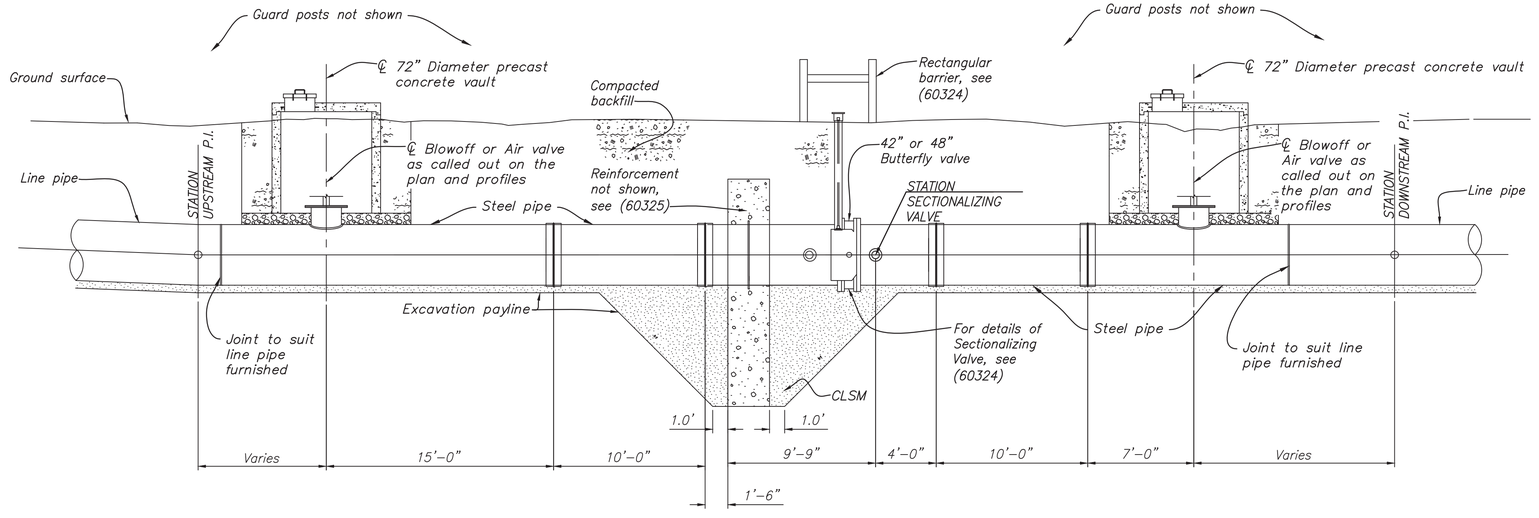
CAD SYSTEM
AutoCAD Rel. 20.0s
CAD FILENAME
1695-D-60325-RD-1.dwg

Released



PLAN

SECTIONALIZING VALVE DATA TABLE–REACH 9			
SECTIONALIZING VALVE STATION	VALVE HEAD CLASS (FT)	ORIFICE DESIGN HEAD (FT)	NUMBER OF ORIFICE HOLES
90000+11.25	600	381	13
90059+00	650	425	12
90115+00	625	414	12
90167+20	575	370	13
90224+60	500	289	15
90285+30	525	321	14
90330+00	525	332	14
90384+00	450	282	15
90419+40	475	308	14
90470+00	450	282	15
90523+00	375	219	16
90577+00	375	229	16
90624+00	400	254	15
90681+50	375	247	16
90739+50	350	214	17
90810+00	325	211	17
90879+00	300	193	17
90921+50	275	181	18
90980+00	275	177	18
91037+00	250	163	19
91088+00	200	125	21
91142+00	175	111	22
91188+00	125	84	25
91220+00	100	73	27



SECTION A–A

SECTIONALIZING VALVE DATA TABLE–REACH 11			
SECTIONALIZING VALVE STATION	VALVE HEAD CLASS (FT)	ORIFICE DESIGN HEAD (FT)	NUMBER OF ORIFICE HOLES
11069+00	650	496	11
11168+80	600	443	12

NOTES

1. Drawing not to scale.
2. CLSM – Controlled Low Strength Material.
3. For Sectionalizing Valve installation, see (60324).
4. For Tapers, Adaptor and Collars, see (60325).

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BLOCK 9-11 - LINE PIPE

REV NO	DATE	DESCRIPTION
1	2017-12-08	Sectionalizing valve at 90000+00 changed to 90000+11.25
	Bowles, Linda M P.E.	

Andrew Marner
DESIGNED
Derek Kosakowski
DRAWN
D.L. Gesundheit
CHECKED
Bowles, Linda M P.E.
TECH. APPR.
Duke, Wylie C (Chris) P.E.
APPROVED
PEER REVIEWER - Manager - Water Conveyance Group

DENVER, CO 2017-03-30

SECTIONALIZING VALVE INSTALLATION

TYPICAL LAYOUT

1695-D-60326

SHEET 1

DATE AND TIME PLOTTED
12/11/2017 10:11 AM
PLOTTED BY
svc Corp-EDM

CAD SYSTEM
AutoCAD Rel. 20.0s
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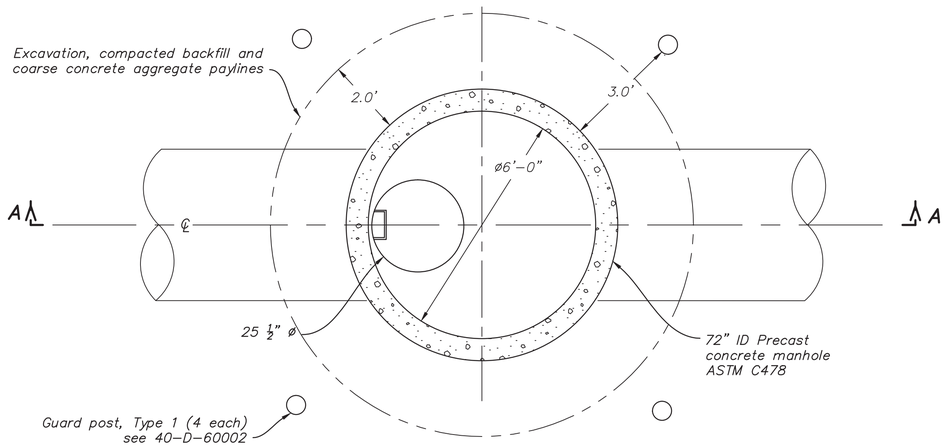
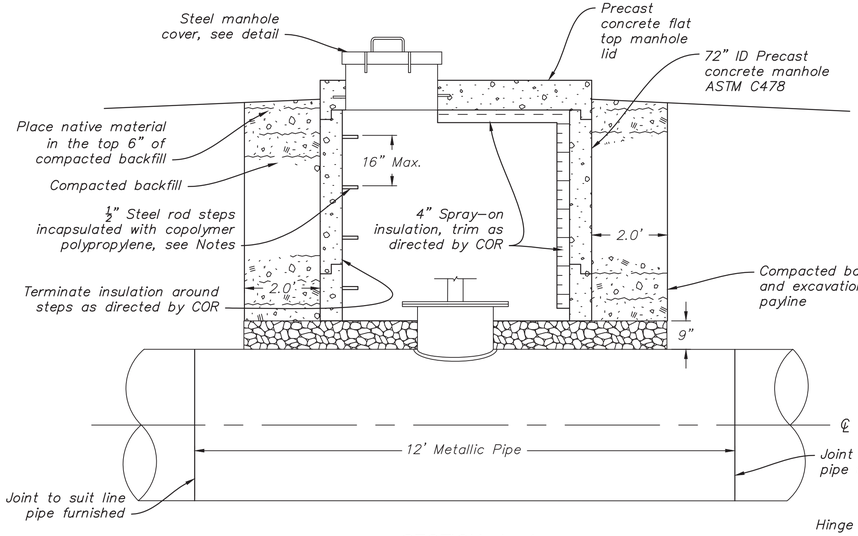
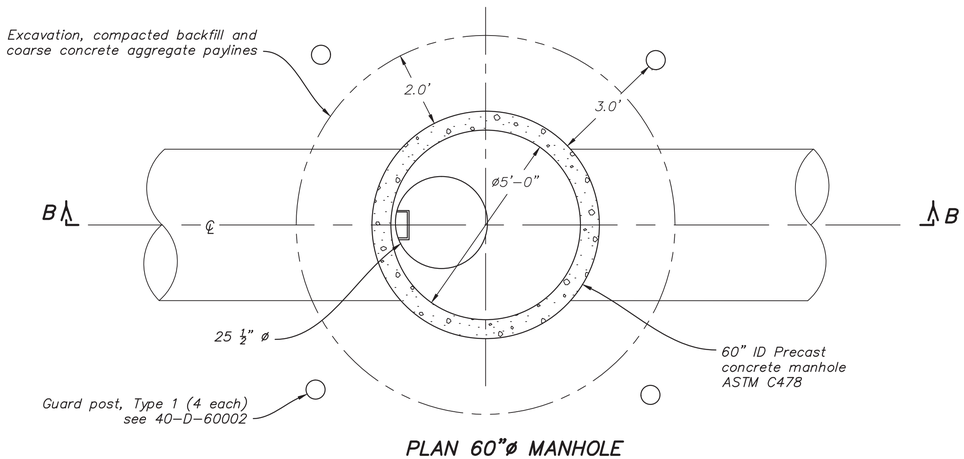


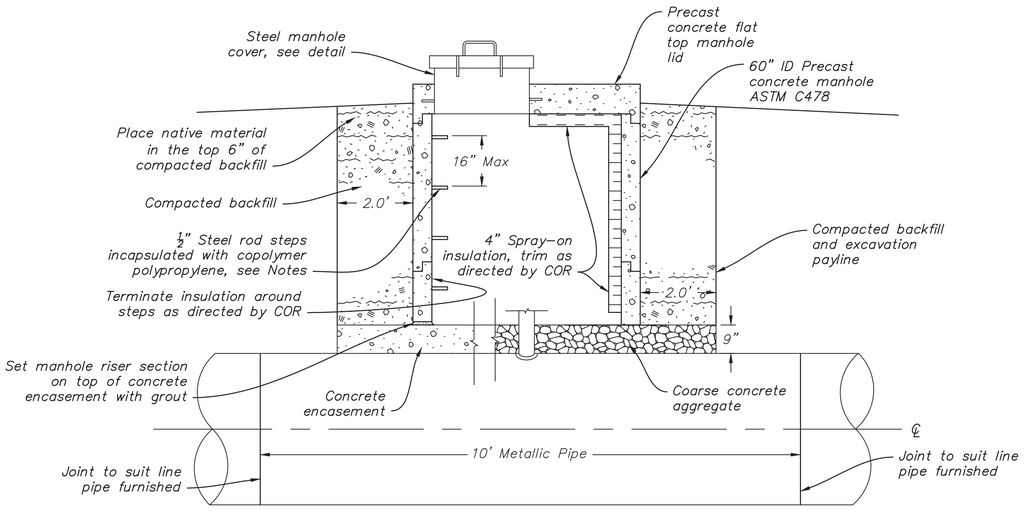
PLAN 72"ø MANHOLE



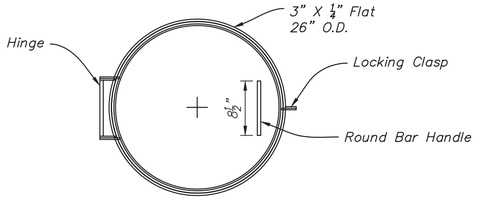
SECTION A–A

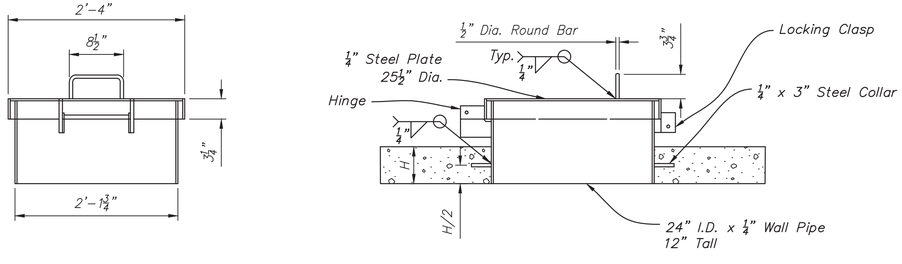


PLAN 60"ø MANHOLE



SECTION B–B





STEEL MANHOLE COVER DETAIL

NOTES

1. Manhole steps shall be made of ½" steel rod encapsulated with copolymer polypropylene or approved equal and shall conform to ASTM C478. Steps shall have minimum projections of 4–inches, spaced no more than 16" apart, minimum overall widths of 14 inches and anchored into the walls at least 3½ inches deep.
2. Steel manhole cover shall be provided with a lockable lid.
3. Drawing not to scale.

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BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

BLOCK 9-11 - LINE PIPE

REV NO	DATE	DESCRIPTION
1	2017-07-14 Bowles, Linda M P.E.	D Added note about lockable lids for steel manhole cover detail.

Andrew Marner — DESIGNED
Derek Kosakowski — DRAWN
D.L. Gesundheit — CHECKED
Bowles, Linda M P.E. — TECH. APPR.
Duke, Wylie C (Chris) P.E. — APPROVED
PEER REVIEWER - Manager - Water Conveyance Group
DENVER, CO 2017-03-30

PRECAST CONCRETE MANHOLES

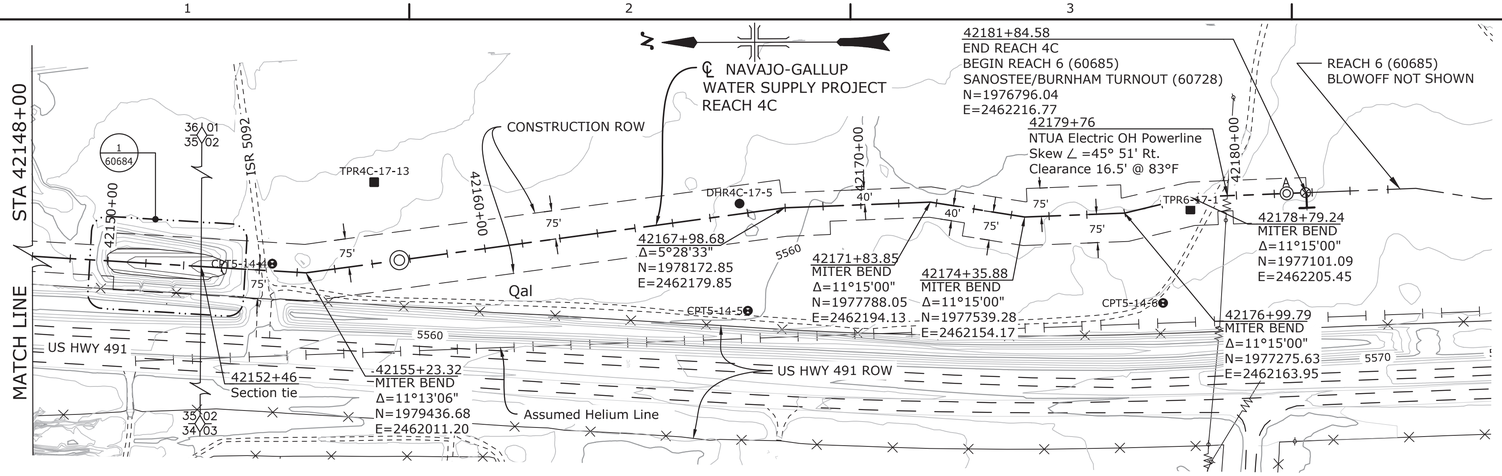
DETAILS

1695-D-60327

SHEET 1

DATE AND TIME PLOTTED
3/27/2017 7:36 PM
PLOTTED BY
svc-Corp-EDM

CAD SYSTEM
AutoCAD Rel. 20.0s
CAD FILENAME
1695-D-60327-AD-1.dwg

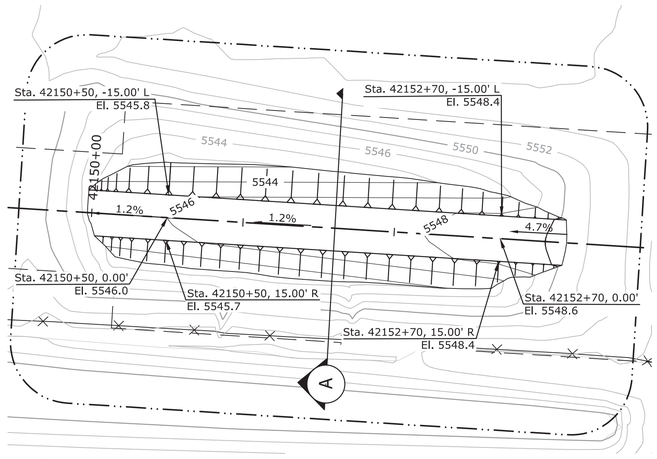


PLAN

0 200' 400'

OPEN-CUT ROAD CROSSINGS	
Station	Surfacing
42154+05 to 42154+20	Maintained Road
42179+39 to 42179+59	Unmaintained Road

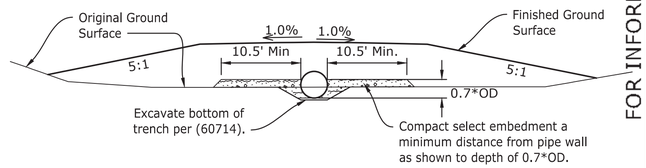
CONSTRUCTION ROW			
Station		Width (Ft.)	
Start	End	Left	Right
42148+00	42168+00	75	75
42168+00	42169+19	40	75
42169+19	42172+84	40	40
42172+84	42174+36	40	75
42174+36	42177+00	75	75
42177+00	End	40	75



1 GRADING PLAN

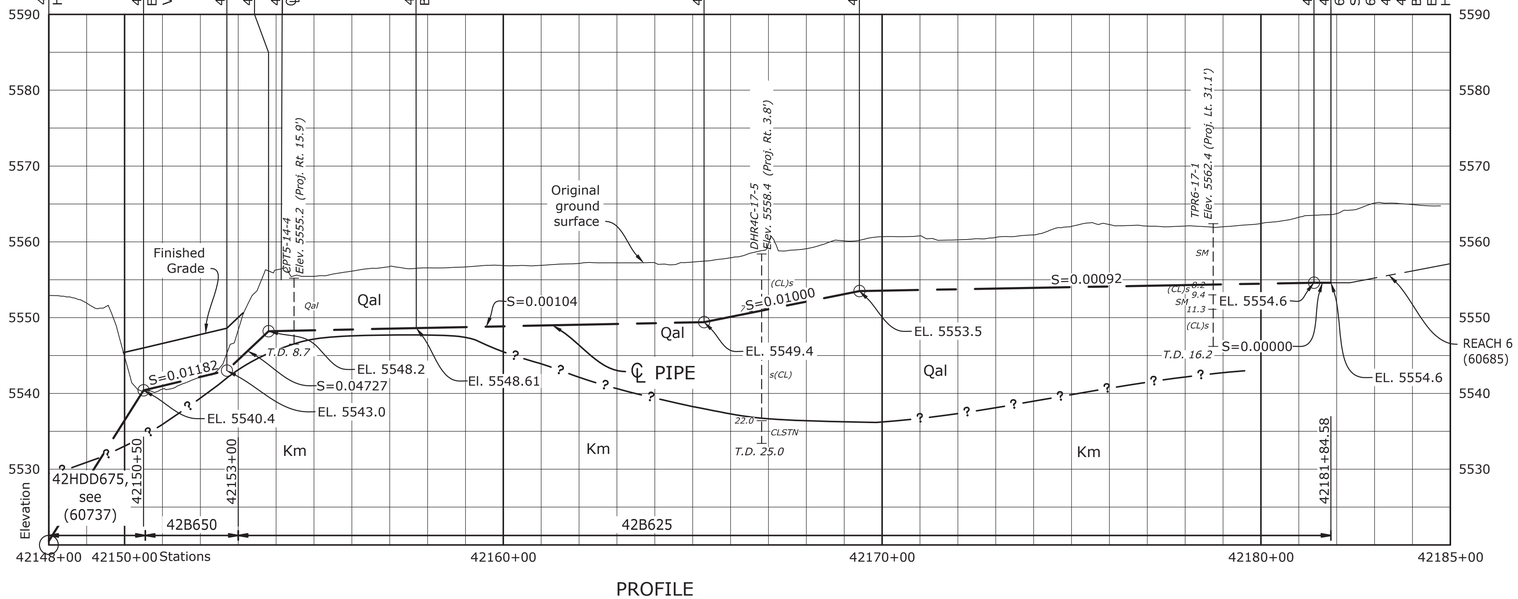
STA. 42150+00 TO STA. 42153+00

0 50' 100'



A SECTION

0 10' 20'



PROFILE

NOTES

1. For General Notes, Symbols, Abbreviations, Pipe Classification and Geology Note, see (60677).
2. The 14-inch ID turnout pipe is based on a phase 2 peak flow of 2.27 cubic feet per second.
3. Slope finished ground surface away from the pipe as directed by OGR for positive drainage between stations 42149+60 and 42153+20.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

REACH 4C LINE PIPE

REV NO		
1	D	Added geology explorations to profile.
	2019-05-20	Clough, Nicholas W P.E.

M. Norris
DESIGNED
D. Kosakowski
DRAWN
A. Marner
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer

GEOLOGY

TECH. APPR. Chris Beyer
SUBMITTED Debby Thornburg
APPROVED Paul Gardner Supervisory Geologist

DENVER, CO 2019-03-13

PLAN AND PROFILE

STA. 42150+00
TO
END

1695-D-60684

SHEET 5 OF 5

LAST SAVED DATE
2019-05-20
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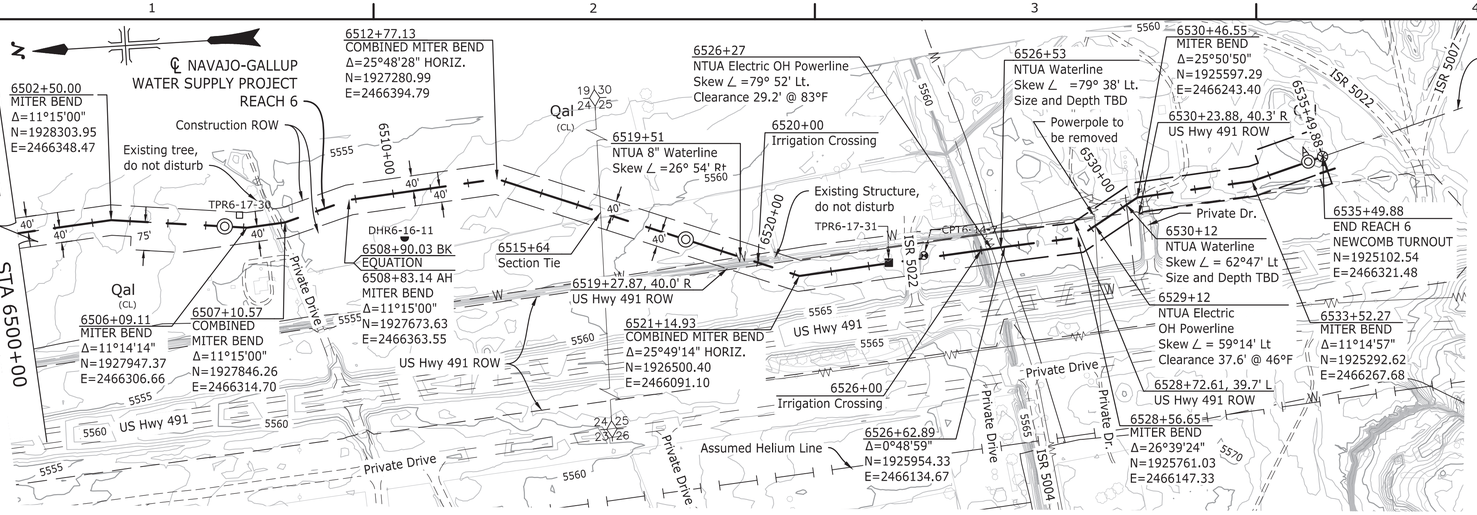
AMARNER

CAD SYSTEM
AutoCAD 22.0s
CAD FILENAME
1695-D-60684-RD-5 OF 5.DWG

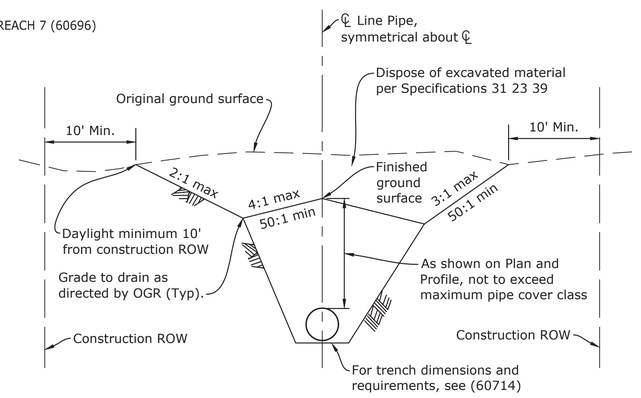


PLAN

0 200' 400'

OPEN-CUT ROAD CROSSINGS	
Station	Surfacing
6506+84 to 6506+95	Unmaintained Road
6524+03 to 6524+19	Maintained Road
6530+61 to 6530+69	Maintained Road

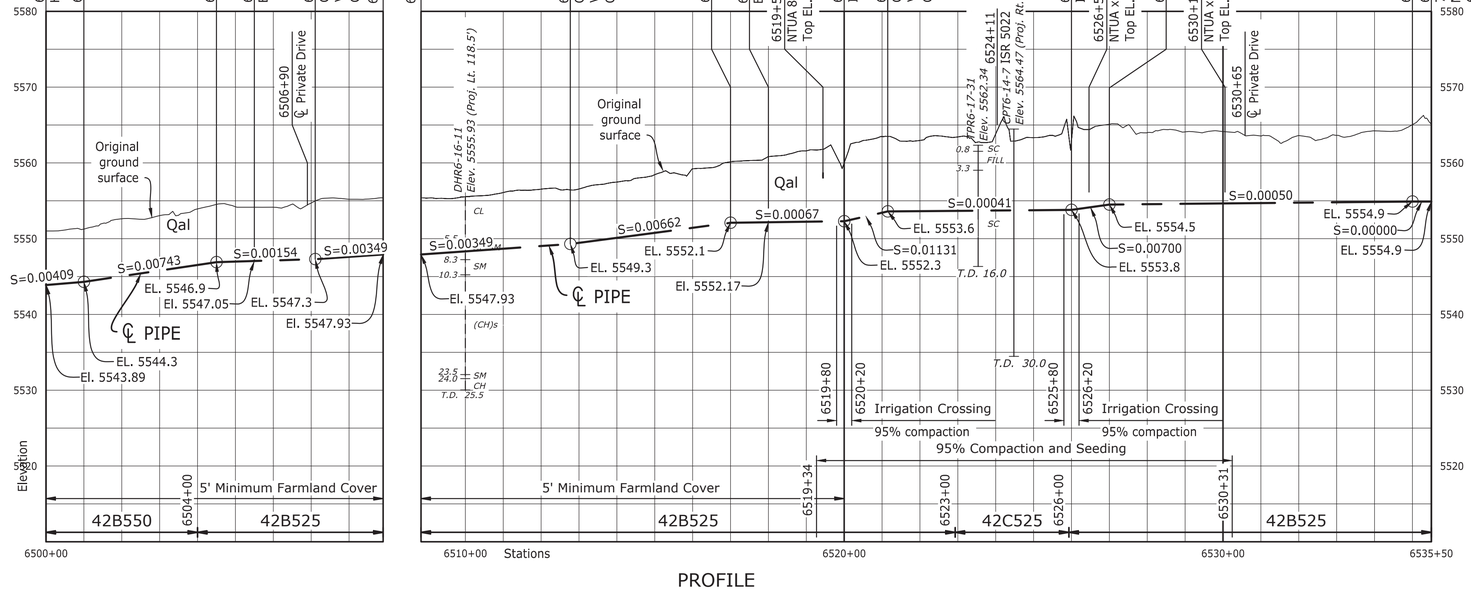
CONSTRUCTION ROW			
Station		Width (Ft.)	
Start	End	Left	Right
6500+00.0	6503+04.9	40	40
6503+04.9	6504+36.8	40	75
6504+36.8	6506+09.1	40	varies
6506+09.1	6533+83.4	40	40
6533+83.4	6535+24.3	75	40
6535+24.3	6535+49.87	75	75



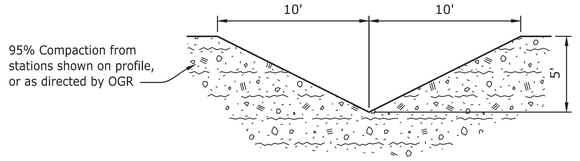
TYPICAL FINISHED GROUND DETAIL
NO SCALE

NOTES

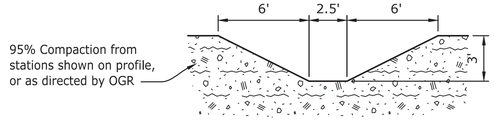
1. For General Notes, Symbols, Abbreviations, Pipe Classification, and Geology Note, see (60677).
2. The Newcomb turnout pipe has a peak flow of 1.52 cubic feet per second.
3. Irrigation crossings at Sta. 6520+00 and Sta. 6526+00 dimensions shall be verified and constructed per OGR.



PROFILE



EARTHEN IRRIGATION CROSSING DETAIL
STA. 6520+00



EARTHEN IRRIGATION CROSSING DETAIL
STA. 6526+00

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SAN JUAN LATERAL - BLOCK 4C - 8

REACH 6 LINE PIPE

REV NO		
2	2019-09-19	Clough, Nicholas W P.E.
1	2019-05-17	Nicholas W. Clough PE

D REALIGNMENT, ADDED STATION EQUATION.
D ADDED EARTHEN IRRIGATION CROSSING DETAILS. MINOR REVISIONS.

M. LEAVITT
DESIGNED
M. LEAVITT
DRAWN
D. L. GESUNDHEIT
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer

GEOLOGY
DENVER, CO 2019-03-11

TECH. APPR. Chris Beyer
SUBMITTED Debby Thornburg
APPROVED Paul Gardner Supervisory Geologist

PLAN AND PROFILE

STA. 6500+00
TO
END

1695-D-60695
SHEET 11 OF 11

Released

LAST SAVED DATE
2019-09-19
LAST SAVED BY
MLEAVITT

CAD SYSTEM
AutoCAD 23.0s
CAD FILENAME
1695-D-60695-RD-11 OF 11.DWG

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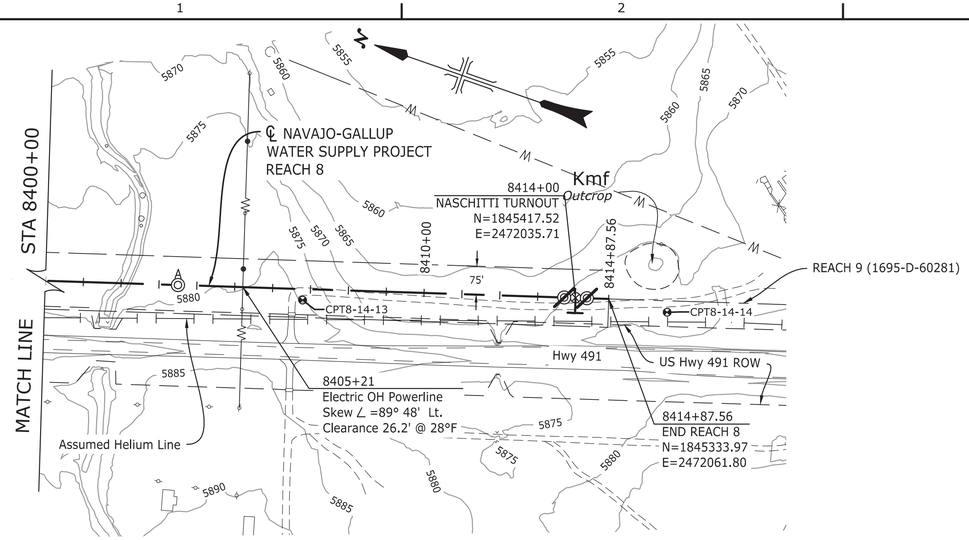
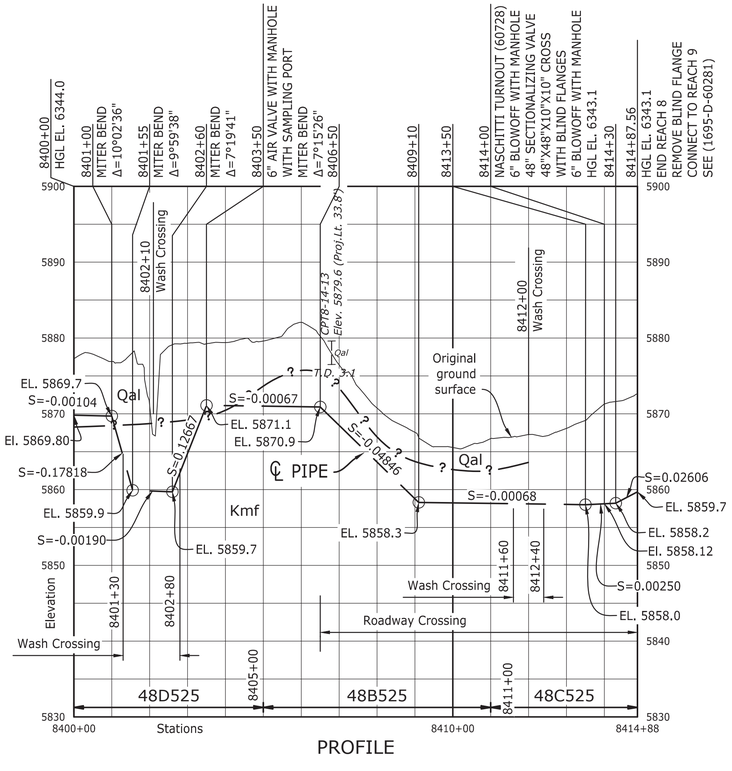


PLAN

0 200' 400'

OPEN-CUT ROAD CROSSINGS	
Station	Surfacing
8406+50 to 8414+87.56	Maintained Road

CONSTRUCTION ROW			
Station		Width (Ft.)	
Start	End	Left	Right
8400+00.0	8414+88.45	75	75



PROFILE

NOTES

1. For General Notes, Symbols, Abbreviations, Pipe Classification, and Geology Note, see (60677).
2. The Naschitti turnout pipe has a peak flow of 1.77 cubic feet per second.
3. Field verify Reach 9 blind flange elevation and location.
4. For wash crossings, see (60736).

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NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

REACH 8 LINE PIPE

REV NO			
4	D	2020-05-15	Manner, Andrew (Andy) J P.E.
		Adjusted profile for wash crossing at Sta. 8402+10.	
3	D	2020-03-10	Manner, Andrew (Andy) J P.E.
		Moved Naschitti Turnout to 8414+00 from 8398+20.	
2	D	2019-07-18	Clough, Nicholas W P.E.
		MINOR REVISIONS.	
1	D	2019-05-17	Clough, Nicholas W P.E.
		MINOR REVISIONS.	

M. LEAVITT
DESIGNED
M. LEAVITT
DRAWN
D. L. GESUNDHEIT
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer

DENVER, CO 2019-03-13

GEOLOGY
TECH. APPR. Chris Beyer
SUBMITTED Debby Thornburg
APPROVED Paul Gardner, Supervisory Geologist

PLAN AND PROFILE

STA. 8400+00
TO
END

1695-D-60713

SHEET 9 OF 9

LAST SAVED DATE
2020-05-15
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CAD SYSTEM
AutoCAD 2018
CAD FILENAME
1695-D-60713-RD-9 OF 9.DWG

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AIR VALVE DATA TABLE - REACH 4C								
STATION	HEAD CLASS (FT)	AIR VALVE SEAT RATING (PSI)	PI ELEVATION (FT)	GROUND SURFACE ELEVATION (FT)	DEPTH OF COVER '+/- (FT)	MANHOLE?	SAMPLING PORT	MITER?
42003+10.00	625	270.8	5,589.20	5,597.1	6.1	NO	NO	NO
42029+00.00*	625	270.8	5,575.73	5,582.0	4.4	YES	NO	NO
42089+60.00	625	270.8	5,574.70	5,582.5	6.0	YES	YES	YES
42137+06.17*	650	281.7	5,548.70	5,555.7	5.1	YES	NO	YES
42181+84.58	625	270.8	5,554.60	5,563.6	7.1	YES	YES	NO

AIR VALVE DATA TABLE - REACH 6								
STATION	HEAD CLASS (FT)	AIR VALVE SEAT RATING (PSI)	PI ELEVATION (FT)	GROUND SURFACE ELEVATION (FT)	DEPTH OF COVER '+/- (FT)	MANHOLE?	SAMPLING PORT	MITER?
6041+10.00*	600	260.0	5,569.20	5,575.8	4.7	YES	NO	YES
6061+00.00	600	260.0	5,570.16	5,578.2	6.2	YES	NO	NO
6090+90.00	550	238.3	5,604.50	5,613.8	7.5	NO	NO	YES
6117+00.00	575	249.2	5,567.99	5,575.9	6.1	YES	YES	NO
6202+30.00	475	205.8	5,657.55	5,665.8	6.4	YES	NO	NO
6202+30.00	475	205.8	5,657.55	5,665.8	6.4	YES	NO	NO
6280+60.00	550	238.3	5,589.70	5,597.8	6.2	YES	YES	NO
6300+60.00	525	227.5	5,598.30	5,608.6	8.5	NO	NO	NO
6370+00.00	500	216.7	5,616.90	5,624.8	6.0	YES	NO	NO
6399+00.00	550	238.3	5,560.76	5,568.7	6.1	YES	YES	NO
6535+49.88	525	227.5	5,554.90	5,565.1	8.3	YES	YES	NO

AIR VALVE DATA TABLE - REACH 7								
STATION	HEAD CLASS (FT)	AIR VALVE SEAT RATING (PSI)	PI ELEVATION (FT)	GROUND SURFACE ELEVATION (FT)	DEPTH OF COVER '+/- (FT)	MANHOLE?	SAMPLING PORT	MITER?
7009+86.97	525	227.5	5,554.90	5,565.1	8.3	YES	YES	NO
7022+60.00	550	238.3	5,547.40	5,554.5	5.2	NO	NO	YES
7101+60.00	475	205.8	5,581.80	5,589.1	5.4	NO	NO	NO
7148+00.00	450	195.0	5,580.45	5,588.3	6.0	YES	YES	NO
7213+50.00	400	173.3	5,607.62	5,614.9	5.4	YES	NO	NO
7224+00.00	375	162.5	5,623.80	5,632.4	6.8	NO	NO	NO
7277+00.00	325	140.8	5,651.65	5,659.7	6.2	YES	YES	NO
7312+20.00	300	130.0	5,680.12	5,689.1	7.1	YES	NO	NO
7394+50.00	125	54.2	5,795.50	5,802.3	5.0	YES	NO	NO

AIR VALVE DATA TABLE - REACH 8								
STATION	HEAD CLASS (FT)	AIR VALVE SEAT RATING (PSI)	PI ELEVATION (FT)	GROUND SURFACE ELEVATION (FT)	DEPTH OF COVER '+/- (FT)	MANHOLE?	SAMPLING PORT	MITER?
8001+00.00*	625	270.8	5,859.00	5,865.30	4.4	NO	NO	NO
8029+20.00	600	260.0	5,818.40	5,826.60	6.1	YES	NO	YES
8052+00.00	575	249.2	5,851.70	5,859.80	6.0	YES	NO	NO
8061+00.00	575	249.2	5,830.95	5,840.50	7.4	YES	NO	NO
8100+00.00	550	238.3	5,861.30	5,869.40	6.0	YES	NO	NO
8124+00.00*	550	238.3	5,868.70	5,876.20	5.4	YES	NO	YES
8132+20.00	550	238.3	5,870.22	5,878.40	6.0	YES	YES	NO
8158+00.00	475	205.8	5,941.40	5,949.10	5.6	NO	NO	YES
8172+30.00	525	227.5	5,882.40	5,891.10	6.6	YES	NO	YES
8211+00.00	475	205.8	5,919.29	5,928.30	6.9	YES	NO	NO
8231+69.78	450	195.0	5,957.80	5,965.90	6.0	YES	NO	NO
8260+00.00	500	216.7	5,895.43	5,903.60	6.1	YES	YES	NO
8264+50.00	525	227.5	5,892.00	5,901.10	7.0	YES	NO	YES
8285+30.00	575	249.2	5,866.10	5,874.10	5.9	YES	NO	YES
8315+60.00	525	227.5	5,866.38	5,874.50	6.0	YES	NO	NO
8315+60.00	525	227.5	5,866.38	5,874.50	6.0	YES	NO	NO
8343+00.00*	550	238.3	5,851.70	5,859.60	5.7	YES	NO	YES
8344+60.00*	550	238.3	5,852.80	5,859.30	4.4	NO	NO	YES
8385+90.00	525	227.5	5,885.80	5,894.50	6.5	YES	NO	NO
8403+50.00	525	227.5	5,871.10	5,879.20	6.0	YES	YES	YES

* Earthwork may be mounded at the air valves at these stations.

NOTE
For air valve details see (60721) and (60722)

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

LINE PIPE

REV NO			
6	2021-05-18	DENVER, CO	Removed Air Valve at Sta. 41976+00 from relocation of pumping plant at beginning of Reach 4C.
5	Marner, Andrew (Andy) J P.E.		
4	2021-01-27	DENVER, CO	Revised Table Values relocation of pumping plant at beginning of Reach 4C.
3	Marner, Andrew (Andy) J P.E.	D	
2	2020-05-15	D	Revised Table Values for Wash Crossing changes.
1	Clough, Nicholas W P.E.		
	2020-03-10	D	Revised Table Values for Naschitti Turnout relocation.
	Marner, Andrew (Andy) J P.E.		
	2019-05-17	D	Revised Table Values
	Clough, Nicholas W P.E.		
	2019-05-17	D	Revised Table Values
	Clough, Nicholas W P.E.		

D. L. Gesundheit
DESIGNED
K. Whitesell
DRAWN
M. Gemperline
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer

DENVER, CO 2019-03-13

AIR VALVES

DATA TABLES

1695-D-60720

SHEET 1

LAST SAVED DATE
2021-05-18
LAST SAVED BY
AMARNER

CAD SYSTEM
AutoCAD 24.0s
CAD FILENAME
1695-D-60720-RD-1.DWG

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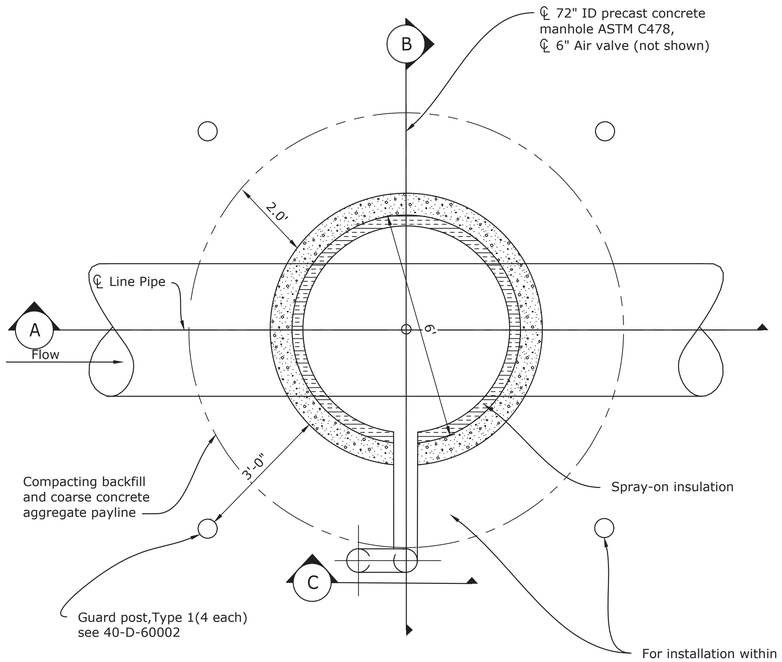
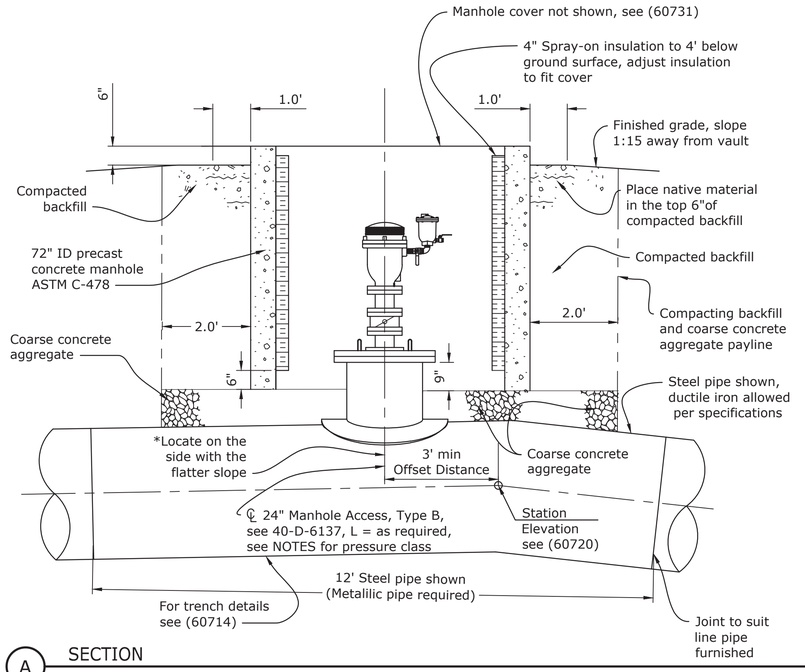


PLAN

WITH COVER REMOVED



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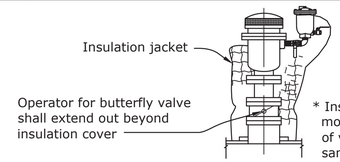
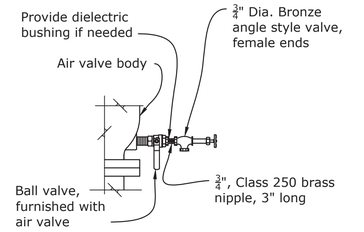
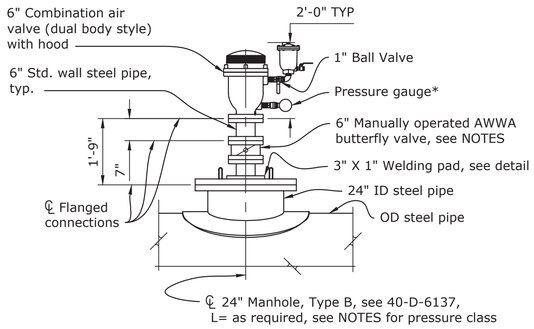


1 AIR VALVE INSULATION DETAIL



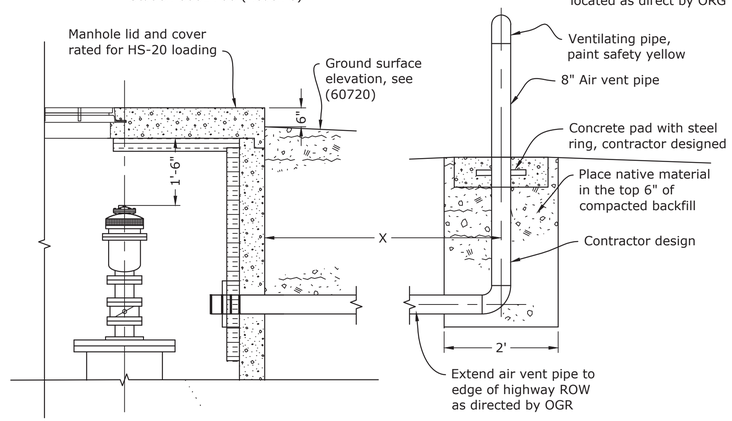
WATER QUALITY SAMPLING PORT DETAIL

For locations, see Air Valve Data Tables (60720)



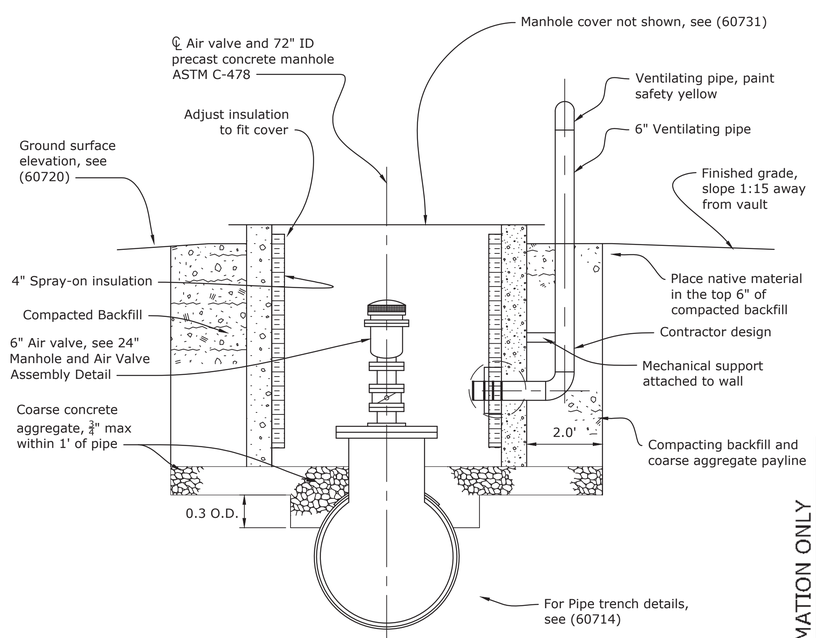
24" MANHOLE AND AIR VALVE ASSEMBLY

* Pressure gauge to be installed at air valve assembly for the following locations only:
Station 42003+10 (Reach 4C)
Station 6042+80 (Reach 6)
Station 7022+60 (Reach 7)
Station 8001+00 (Reach 8)



HIGHWAY ROW AIR VALVE DETAIL

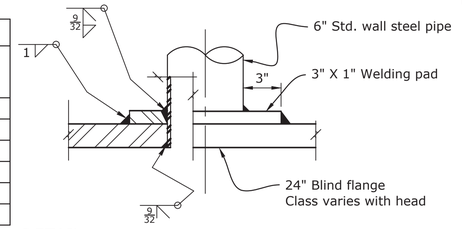
See Note 11



B SECTION

HIGHWAY ROW AIR VALVES		
STATION	"X" OFFSET LENGTH (FT) (Approximate)	MANHOLE
8260+00	40	YES
8264+50	60	YES
8315+60	30	YES
8315+60	30	YES
8343+00	30	YES
8344+60	30	NO

Offset distances are approximate and perpendicular to the line pipe. Vent pipe located as direct by ORG



DETAIL

3" X 1" WELDING PAD

NOTES

1. For Air valve data table, see (60720)
2. Flanges should be AWWA class D when the head class is less than or equal to 346 ft, AWWA class F when the head class is greater than 346 ft and is less than or equal to 692 ft.
3. The butterfly valve shall be AWWA class 150B when the head class is less than or equal to 346 ft, AWWA class 250B when the head class is less than or equal to 577 ft. See specifications for butterfly valves with head classes greater than 577 ft.
4. The air valve inlet flange shall be ANSI B16.1 class 125 or ANSI B16.5 class 150 when the head class of the pipe is less than 346 ft and ANSI B16.1 class 250 or ANSI B16.5 class 350 for a head class of pipe greater than or equal to 692 ft.
5. The Manhole class shall be Class 150 when the head class of the pipe is less than 346 feet and Class 300 for a head class of pipe less than or equal to 692 ft.
6. Drawing not to scale.
7. Construct air valve vertical.
8. Earthwork compacted backfill and excavation minimums are shown. Actual slope will depend on soil conditions.
9. The height of 72" vertical pipe shall be high enough to allow enough clearance between the top of the air valve and the bottom of the insulation on the cover as per the air valve manufacturer's recommendations.
10. Match all mating flanges with highest rated flanges.
11. For manholes installed within NMDOT ROW, do not install guard posts. Design to withstand HS-20 loading.

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SAN JUAN LATERAL - BLOCK 4C - 8

LINE PIPE

1	2021-05-18	Changed pressure gauge location in Reach 4C due to Pumping Plant 3 relocation.
	DENVER, CO	Marrier, Andrew (Andy) J P.E.

K. Whitesell
DESIGNED
K. Whitesell
DRAWN
D. Kosakowski
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer

DENVER, CO 2019-03-11

AIR VALVES WITH MANHOLES

SECTIONS AND DETAILS

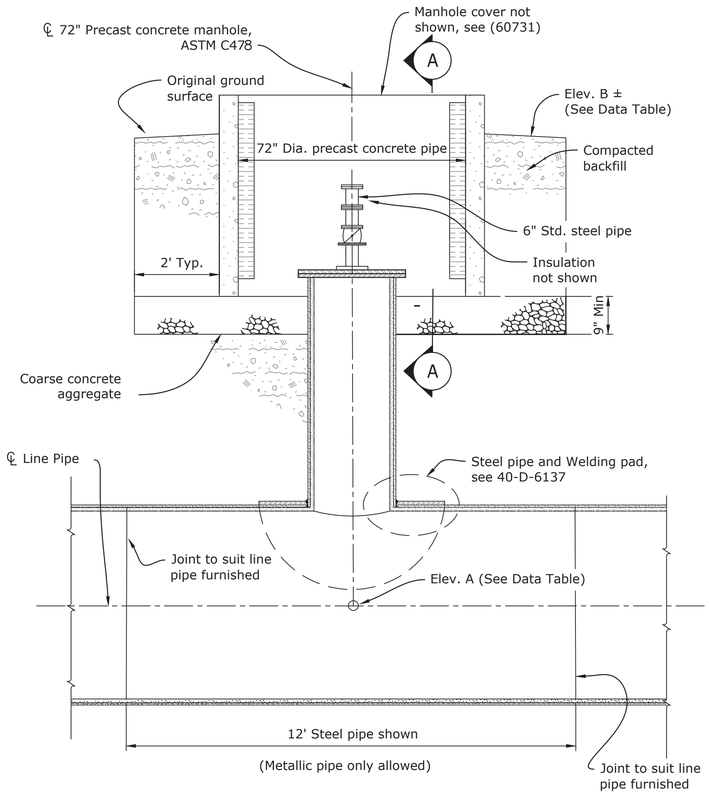
1695-D-60722

SHEET 1

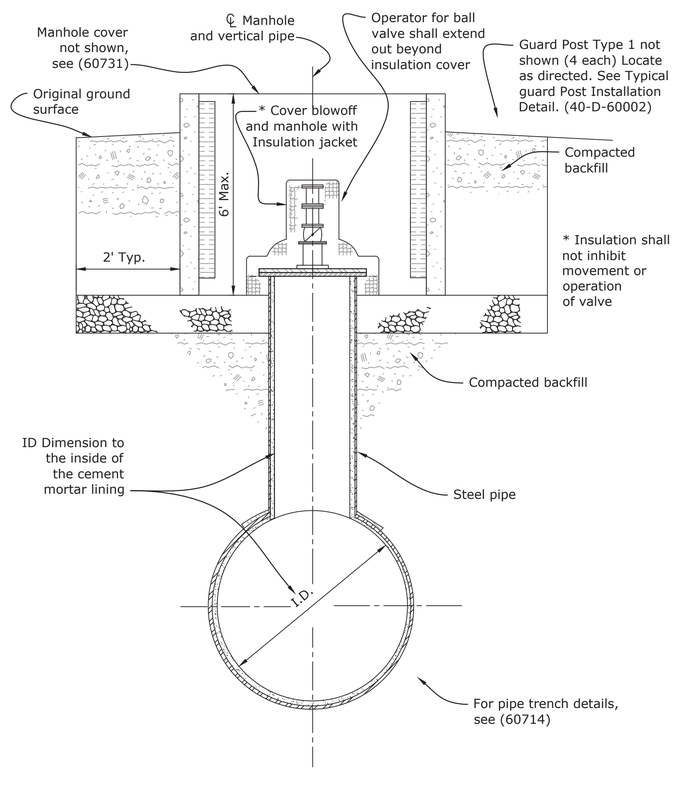
LAST SAVED DATE
2021-05-18
LAST SAVED BY
AMARNER

CAD SYSTEM
AutoCAD Civil 3D
CAD FILENAME
1695-D-60722-RD-1.DWG

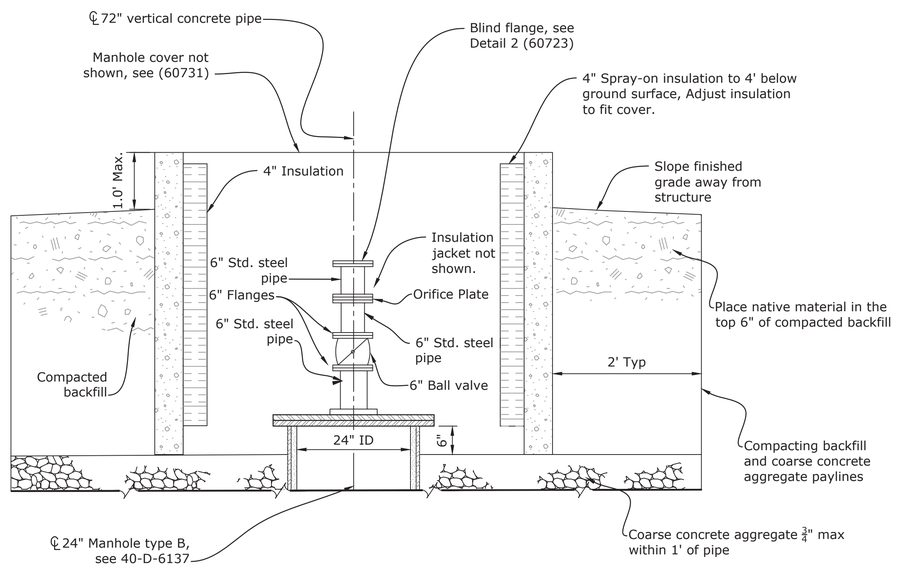
Released



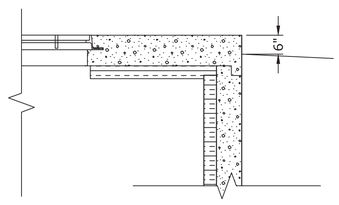
LONGITUDINAL SECTION



TRANSVERSE SECTION



A SECTION



HIGHWAY ROW BLOWOFF DETAIL

See Note 6

NOTES

1. For blowoff locations, see Plan and Profiles (60679 to 60713).
2. For coating and lining requirements for std. steel pipe and fittings, see specification paragraphs.
3. For 6" flange ratings, see Table. Match all mating flanges with highest rated flange.
4. For orifice plate, see Blowoff Sections and Details (60723).
5. Any welding to be completed prior to cement mortar lining application.
6. For manholes installed within NMDOT ROW at Sta. 8260+00 and Sta. 8294+20, do not install guard posts. Design to withstand HS-20 loading.

BLOWOFF DATA TABLE-REACH 4C

STATION	HEAD CLASS (FT)	LINE PIPE DIAMETER (IN)	BALL VALVE TYPE	ORIFICE PLATE NO. HOLES	PIPE CL ELEV A (FT)	GROUND SURFACE ELEV B (FT)	DEPTH OF COVER +/- (FT)	MANHOLE?
42029+00	625	42	CLASS 300	N/A	5575.73	5582.0	4.4	YES
42080+00	650	42	CLASS 300	36	5536.28	5544.1	6.0	YES
42080+00	650	42	CLASS 300	36	5536.28	5544.1	6.0	YES

BLOWOFF DATA TABLE-REACH 6

STATION	HEAD CLASS (FT)	LINE PIPE DIAMETER (IN)	BALL VALVE TYPE	ORIFICE PLATE NO. HOLES	PIPE CL ELEV A (FT)	GROUND SURFACE ELEV B (FT)	DEPTH OF COVER +/- (FT)	MANHOLE?
6004+05	625	42	CLASS 300	42	5554.6	5563.6	7.1	YES
6061+00	600	42	CLASS 300	37	5570.2	5578.2	6.2	YES
6117+00	575	42	CLASS 300	36	5568.0	5575.9	6.1	YES
6129+20	600	42	CLASS 300	23	5552.6	5560.5	6.0	YES
6251+20	600	42	CLASS 300	22	5543.3	5551.1	5.9	YES
6251+20	600	42	CLASS 300	30	5543.3	5551.1	5.9	YES
6289+70	550	42	CLASS 300	42	5579.7	5587.7	6.1	YES
6324+40	575	42	CLASS 300	36	5558.3	5568.6	8.5	YES
6324+40	575	42	CLASS 300	30	5558.3	5568.6	8.5	YES
6399+00	550	42	CLASS 300	30	5560.8	5568.7	6.1	YES
6459+40	550	42	CLASS 300	42	5534.7	5542.8	6.3	YES
6459+40	550	42	CLASS 300	42	5534.7	5542.8	6.3	YES

BLOWOFF DATA TABLE-REACH 7

STATION	HEAD CLASS (FT)	LINE PIPE DIAMETER (IN)	BALL VALVE TYPE	ORIFICE PLATE NO. HOLES	PIPE CL ELEV A (FT)	GROUND SURFACE ELEV B (FT)	DEPTH OF COVER +/- (FT)	MANHOLE?
7071+50	525	42	CLASS 300	42	5534.06	5542.2	6.2	YES
7071+50	525	42	CLASS 300	32	5534.06	5542.2	6.2	YES
7138+00	475	42	CLASS 300	42	5561.70	5569.5	6.0	YES
7148+00	450	42	CLASS 300	42	5580.45	5588.3	6.0	YES
7213+50*	400	42	CLASS 300	33	5607.62	5614.9	5.4	YES
7230+50	375	42	CLASS 300	37	5617.00	5624.8	6.0	YES
7277+00	325	42	CLASS 150	41	5651.65	5659.7	6.2	YES
7312+20	300	42	CLASS 150	22	5680.12	5689.1	7.1	YES
7394+50*	125	42	CLASS 150	N/A	5795.50	5802.3	5.0	YES

BLOWOFF DATA TABLE-REACH 8

STATION	HEAD CLASS (FT)	LINE PIPE DIAMETER (IN)	BALL VALVE TYPE	ORIFICE PLATE NO. HOLES	PIPE CL ELEV A (FT)	GROUND SURFACE ELEV B (FT)	DEPTH OF COVER +/- (FT)	MANHOLE?
8004+90	625	42	CLASS 300	30	5800.70	5809.7	7.1	YES
8061+00	575	48	CLASS 300	42	5830.95	5840.5	7.4	YES
8071+90*	600	48	CLASS 300	34	5827.60	5835.3	5.6	NO
8106+60	550	48	CLASS 300	N/A	5854.80	5869.9	13.0	YES
8132+20	550	48	CLASS 300	27	5870.22	5878.4	6.0	YES
8186+55	525	48	CLASS 300	29	5874.80	5887.9	11.0	NO
8211+00	475	48	CLASS 300	36	5919.29	5928.3	6.9	YES
8260+00	500	48	CLASS 300	28	5895.43	5903.6	6.1	YES
8297+90*	575	48	CLASS 300	38	5863.30	5869.6	4.1	YES
8331+60	550	48	CLASS 300	36	5845.80	5855.5	7.6	NO
8376+50	550	48	CLASS 300	35	5843.00	5853.0	7.9	NO
8414+00	525	48	CLASS 300	42	5858.13	5871.3	11.0	YES
8414+00	525	48	CLASS 300	N/A	5858.13	5871.3	11.0	YES

N/A No orifice plate

* Earthwork may be mounded at the blowoffs at these stations.

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SAN JUAN LATERAL - BLOCK 4C - 8

LINE PIPE

REV NO	DATE		
6	2021-05-18	DENVER, CO	Removed Blowoff at Sta. 4+98.1+70 in Table due to pumping plant relocation at beginning of Reach 4C.
5	2021-01-27	DENVER, CO	Revised Table values due to pumping plant relocation at beginning of Reach 4C.
4	2020-05-15	D	Revised Table values due to Wash Crossing changes.
3	2020-03-10	D	Revised Table values due to Naschitti Turnout relocation.
2	2019-09-19	D	Revised Table values
1	2019-05-17	D	Revised Table values

Marmer, Andrew (Andy) J P.E. — Clough, Nicholas W P.E.

K. Whitesell — DESIGNED
K. Whitesell — DRAWN
D. Gesundheit — CHECKED
Clough, Nicholas W P.E. — TECH. APPR.
Bowles, Linda M P.E. — APPROVED
PEER REVIEWER - Peer Reviewer

DENVER, CO 2019-03-13

MANHOLE WITH BLOWOFF

SECTIONS, DETAILS, AND TABLES

1695-D-60724

SHEET 1

LAST SAVED DATE 2021-05-18
LAST SAVED BY AMARMER

CAD SYSTEM AutoCAD 2018
CAD FILENAME 1695-D-60724-RD-1.DWG

Released

SECTIONALIZING VALVE COLLAR TABLE REACH 4C

SECTIONALIZING VALVE STATION	DESIGN HEAD (FT)	PIPE DIAMETER (IN)	HL (FT)	VT (FT)	VB (FT)	DB (FT)	W	VERTICAL REINFORCEMENT AND SPACING	HORIZONTAL REINFORCEMENT AND SPACING
42029+00	318	42	30	5.5	5.5	0.5	2'-1''	#5 @1'-0''	#9 @6''
42080+00	357	42	27	5.5	5.5	2	2'-3''	#5 @1'-0''	#8 @6''
42181+84.58	339	42	22	5.5	5.5	3.5	2'-3''	#5 @1'-0''	#7 @6''

SECTIONALIZING VALVE COLLAR TABLE REACH 6

SECTIONALIZING VALVE STATION	DESIGN HEAD (FT)	PIPE DIAMETER (IN)	HL (FT)	VT (FT)	VB (FT)	DB (FT)	W	VERTICAL REINFORCEMENT AND SPACING	HORIZONTAL REINFORCEMENT AND SPACING
6061+00	323	42	25	5.5	5.5	2.5	2'-3''	#5 @1'-0''	#7 @6''
6117+00	326	42	25	5.5	5.5	2	2'-3''	#5 @1'-0''	#7 @6''
6202+30	236	42	18	5.5	5.5	2.5	1'-9''	#5 @1'-0''	#6 @6''
6251+20	350	42	28	5.5	5.5	2	2'-3''	#5 @1'-0''	#8 @6''
6324+40	335	42	21	5.5	5.5	4.5	2'-3''	#5 @1'-0''	#7 @6''
6399+00	333	42	26	5.5	5.5	2	2'-3''	#5 @1'-0''	#8 @6''
6459+40	359	42	26	5.5	5.5	2.5	2'-3''	#5 @1'-0''	#8 @6''
6535+49.88	339	42	21	5.5	5.5	4.5	2'-3''	#5 @1'-0''	#7 @6''

SECTIONALIZING VALVE COLLAR TABLE REACH 7

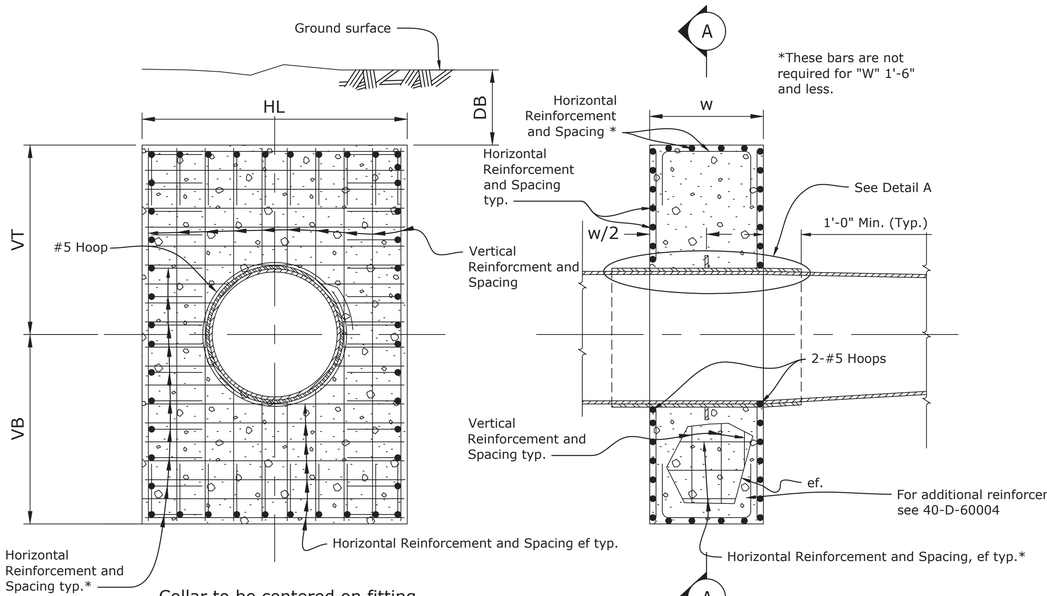
SECTIONALIZING VALVE STATION	DESIGN HEAD (FT)	PIPE DIAMETER (IN)	HL (FT)	VT (FT)	VB (FT)	DB (FT)	W	VERTICAL REINFORCEMENT AND SPACING	HORIZONTAL REINFORCEMENT AND SPACING
7071+50	359	42	24	6	6	2	2'-3''	#5 @1'-0''	#7 @6''
7148+00	313	42	22	6	6	1.5	2'-3''	#5 @1'-0''	#7 @6''
7213+50	286	42	24	5.5	5.5	1.5	2'-0''	#5 @1'-0''	#7 @6''
7277+00	242	42	17	6	6	2	1'-9''	#5 @1'-0''	#6 @6''
7312+20	213	42	14	6	6	2.5	1'-9''	#5 @1'-0''	#5 @6''
7394+50	98	42	11	5	5	1.5	1'-3''	#5 @1'-0''	#5 @1'-0''

SECTIONALIZING VALVE COLLAR TABLE REACH 8

SECTIONALIZING VALVE STATION	DESIGN HEAD (FT)	PIPE DIAMETER (IN)	HL (FT)	VT (FT)	VB (FT)	DB (FT)	W	VERTICAL REINFORCEMENT AND SPACING	HORIZONTAL REINFORCEMENT AND SPACING
8061+00	414	48	29	6	6	3.5	2'-9''	#5 @1'-0''	#8 @6''
8132+20	375	48	37	5	5	3	2'-9''	#5 @1'-0''	#10 @6''
8211+00	326	48	25	6	6	3	2'-6''	#5 @1'-0''	#7 @6''
8260+00	350	48	41	4	4	4	3'-0''	#5 @1'-0''	#10 @6''
8315+60	379	48	37	5	5	3	2'-9''	#5 @1'-0''	#10 @6''
8414+00	387	48	24	5	5	8	2'-9''	#5 @1'-0''	#7 @6''

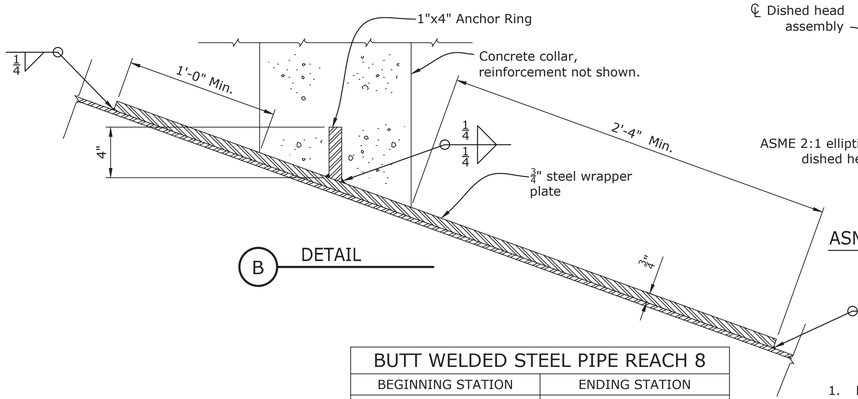
TAPER COLLAR TABLE REACH 8

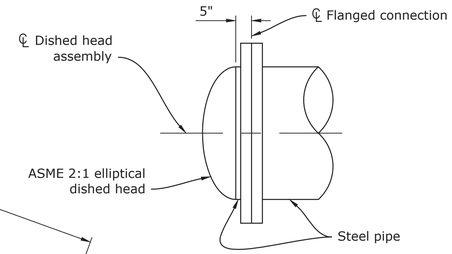
COLLAR STATION	DESIGN HEAD (FT)	TAPER DIMENSIONS	HL (FT)	VT (FT)	VB (FT)	DB (FT)	W	VERTICAL REINFORCEMENT AND SPACING	HORIZONTAL REINFORCEMENT AND SPACING
8006+00	625	42''X48''	21	3.5	3.5	5.5	1'-9''	#5 @1'-0''	#7 @6''



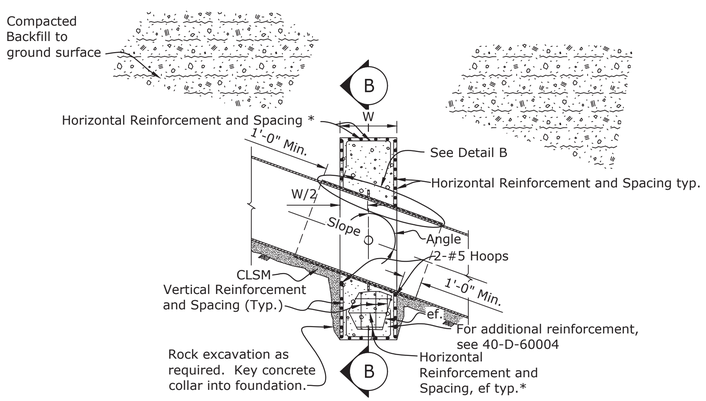
Locate collars as shown on (60725) for valve installations. For Collar reinforcement and thickness, see Collar Table. Earthwork not shown, see (60724).

SECTIONALIZING VALVE COLLAR SECTION





ASME ELLIPTICAL DISHED HEAD ASSEMBLY



STEEP PIPE ANCHOR BLOCK SECTION

BUTT WELDED STEEL PIPE REACH 8	
BEGINNING STATION	ENDING STATION
8001+00.00	8003+30.00

STEEP ANCHOR BLOCK DATA REACH 8

STATION	PIPE CENTERLINE ELEVATION	PIPE SLOPE	PIPE DIAMETER (IN)	ANGLE	HL (FT)	VT (FT)	VB (FT)	DB (FT)	W	VERTICAL REINFORCEMENT AND SPACING	HORIZONTAL REINFORCEMENT AND SPACING
8002+03	5839.57	-0.36656	42	110.1°	11'-0"	4'-0"	4'-0"	13.2	1'-9"	#5@1'-0"	#6@1'-0"

OPERATING CRITERIA

1. Excavation around the blocking shall not occur when the pipeline is under hydrostatic pressure.

NOTES

1. Drawing not to scale.
2. Flange class shall meet pressure requirement designated by adjacent symbol line pipe. Flange drilling shall match mating flange. See Notes on (60722).
3. Steel plate minimum thickness, t, shall not be less than specification requirements.
4. Concrete compressive strength, f'c, is 4,500 psi and specified yield point for reinforcement, fy, is 60,000 psi.
5. For general concrete outline notes and for minimum requirements for detailing reinforcement, see 40-D-60003, and 40-D-60004.
6. For coating and lining options, see specification paragraphs.

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

LINE PIPE

REV NO		
4	2021-01-27	Marner, Andrew (Andy) J P.E.
3	2020-03-10	Marner, Andrew (Andy) J P.E.
2	2019-09-19	Clough, Nicholas W P.E.
1	2019-05-17	Clough, Nicholas W P.E.

D Revised Table for relocation of pumping plant at beginning of Reach 4C.
D Revised Table for relocation of Naschitti Turnout.
D Revised Table Values
D Revised Table Values

Doug Richards
DESIGNED
Doug Richards
DRAWN
D. Gesundheit
CHECKED
Marner, Andrew (Andy) J P.E.
TECH. APPR.
Duke, Wylie C (Chris)
APPROVED
PEER REVIEWER - Water Conveyance Manager

DENVER, CO 2019-03-13

TAPERS, ADAPTORS, AND COLLARS

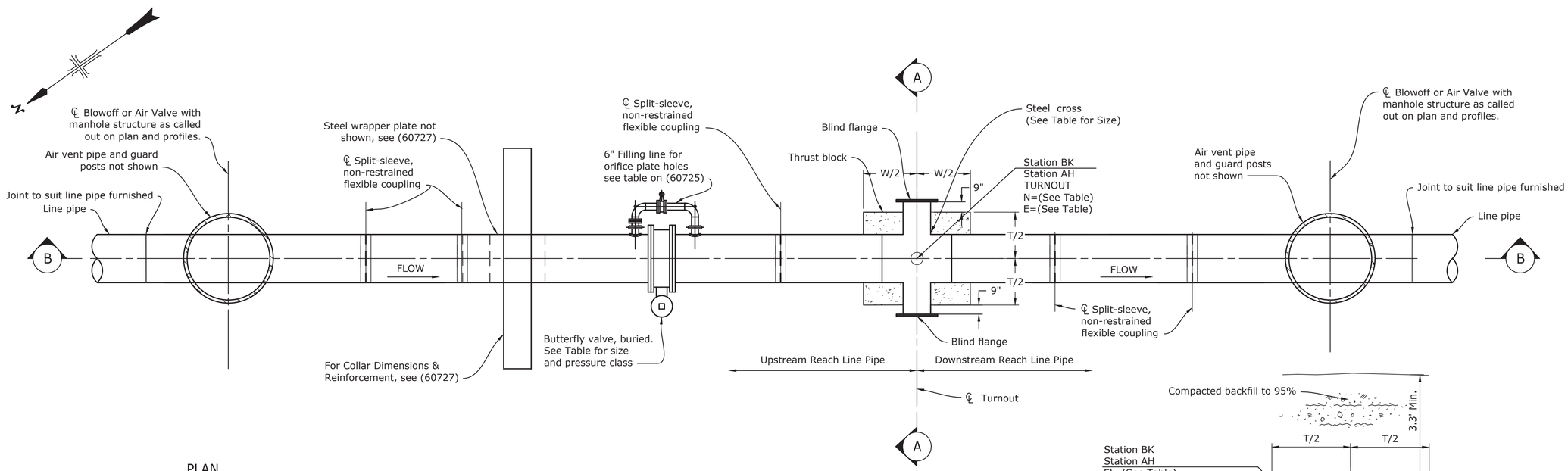
DETAILS, SECTIONS AND DATA TABLES

1695-D-60727

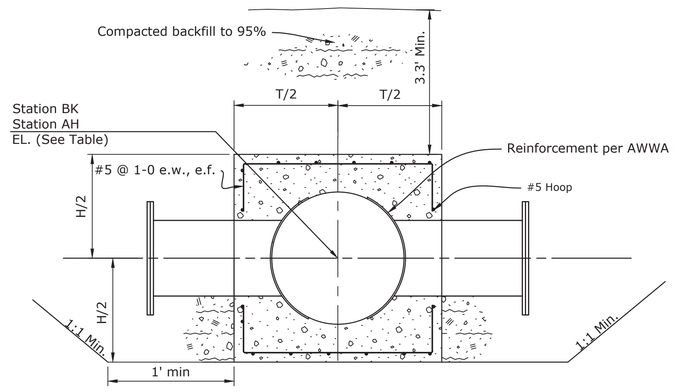
SHEET 1

LAST SAVED DATE
2021-01-27
LAST SAVED BY
DGESUNDHEIT

CAD SYSTEM
MicroStation V8i
CAD FILENAME
1695-D-60727-RD-1.DWG



PLAN
NOT TO SCALE

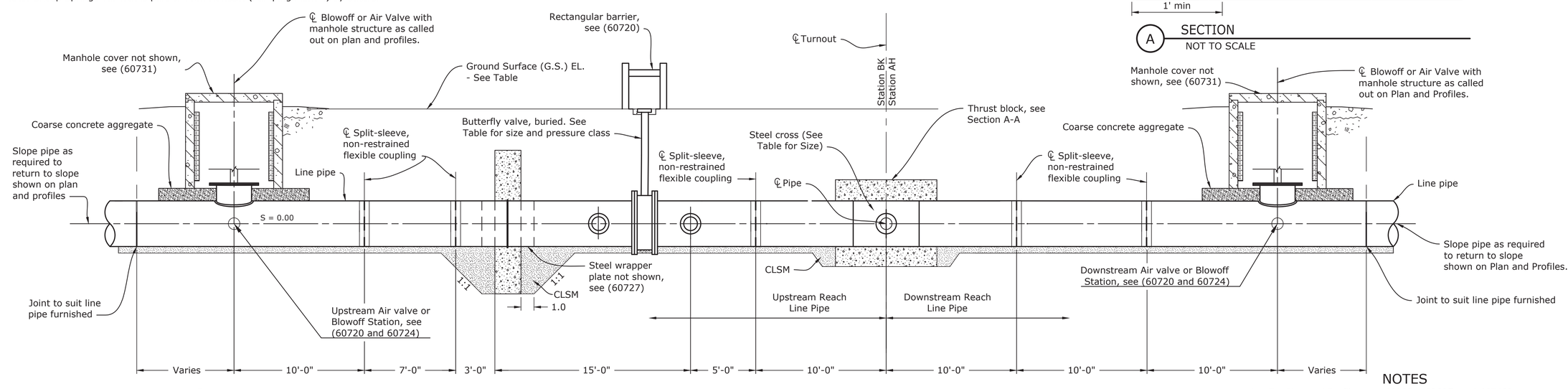


A SECTION
NOT TO SCALE

4C-8 Turnout Summary

Reach	Turnout Name	Flow (cfs)	Max. Transient Pressure (ft)	Cross (inch)	Valve Size (inch)	N	E	Upstream Reach Sta. BK	Downstream Reach Sta. AH	G.S. Elev.	CenterLine Elev.	H (ft)	T (ft)	W (ft)
4C	Sanonstee/Burnham	2.27	625	42x42x14	42	1,976,796.04	2,462,216.77	42181+84.58	6004+05.34	5,564.0	5,554.6	5'-9"	5'-9"	8'-6"
6	Newcomb	1.52	525	42x42x8	42	1,925,102.54	2,466,321.48	6535+49.88	7009+86.97	5,565.0	5,554.9	5'-9"	5'-9"	5'-9"
8	Naschitti	1.77	525	48x48x10	48	1,845,417.52	2,472,035.71	8414+00.00	8414+00.00	5,871.3	5,858.1	6'-3"	6'-3"	6'-3''

Note: Sheep Springs Turnout is part of Tooh Haltsooi (Pumping Plant 4) by others.



B SECTION
NOT TO SCALE

NOTES

1. For Plan and Profiles, see (60684, 60695, and 60712).
2. Concrete compressive strength, f'c, is 4,500 psi and specified yield point for reinforcement fy, is 60,000 psi.
3. For general concrete outline notes and minimum requirements for detailing reinforcement, see 40-D-60003 and 40-D-60004.
4. Outlet pipes to be blind flanged.
5. Turnout assembly piping is metallic.

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SAN JUAN LATERAL - BLOCK 4C - 8

LINE PIPE

REV NO	DATE		
3	2020-03-10	Ramer, Andrew (Andy) J P.E.	Revised table for Naschitti Turnout relocation.
2	2019-09-19	Clough, Nicholas W P.E.	Added coupling. Minor revisions.
1	2019-05-17	Clough, Nicholas W P.E.	D Revised Table Values

Doug Richards
DESIGNED
Doug Richards
DRAWN
D. Kosakowski
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Duke, Wylie C (Chris)
APPROVED
PEER REVIEWER - Water Conveyance Manager
DENVER, CO 2019-03-13

TYPICAL TURNOUT

PLAN, SECTIONS AND DETAILS

1695-D-60728

SHEET 1

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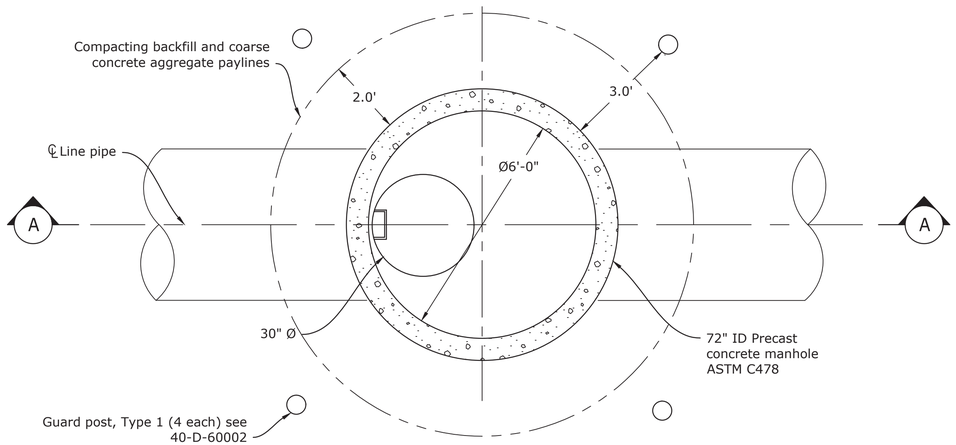
LAST SAVED DATE
2020-03-10
LAST SAVED BY
DGESUNDHEIT

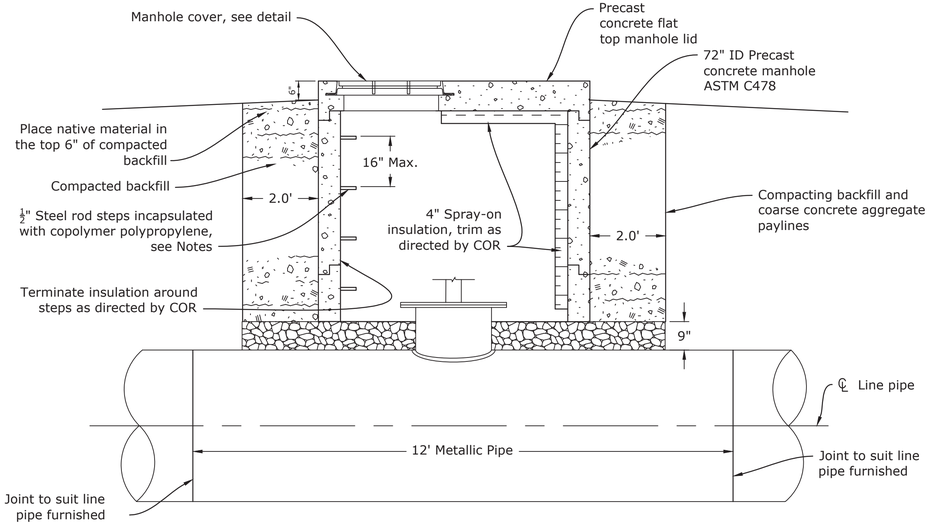
CAD SYSTEM
AutoCAD 23.0s
CAD FILENAME
1695-D-60728-RD-1.DWG

Released



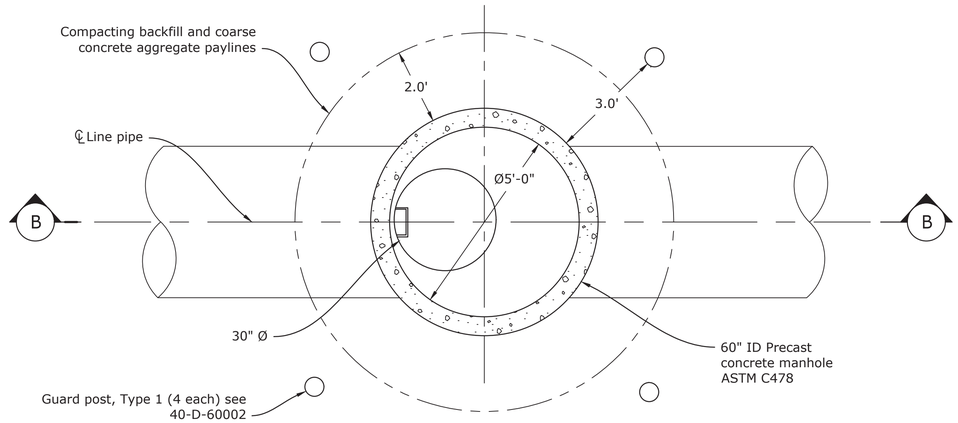
PLAN - 72" Ø MANHOLE

NOT TO SCALE



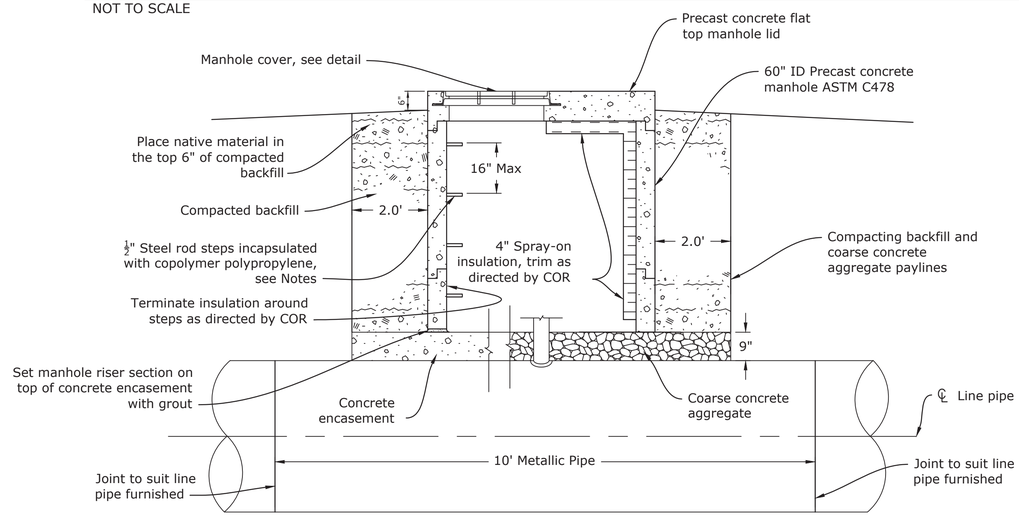
A SECTION

NOT TO SCALE



PLAN - 60" Ø MANHOLE

NOT TO SCALE



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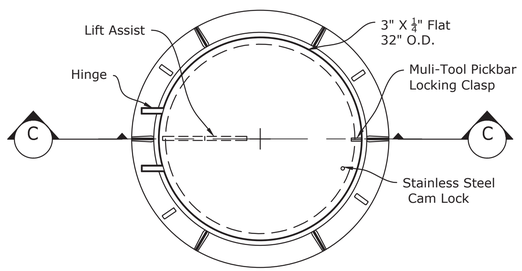
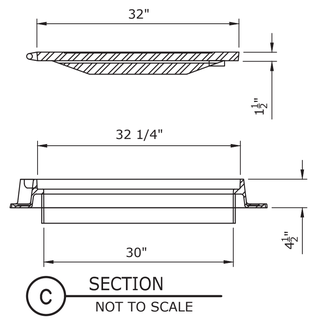
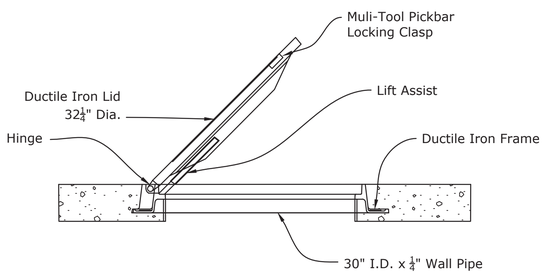




C SECTION

NOT TO SCALE



MANHOLE COVER DETAILS

NOT TO SCALE

NOTES

1. Manhole steps shall be made of ½-inch steel rod encapsulated with copolymer polypropylene or approved equal and shall conform to ASTM C478. Steps shall have minimum projections of 4-inches, spaced no more than 16" apart, minimum overall widths of 14 inches and anchored into the walls at least 3½ inches deep.
2. Manhole cover shall be provided with a lockable lid.
3. Drawing not to scale.

RECLAMATION
Managing Water in the West

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL - BLOCK 4C - 8

LINE PIPE

K. Whitesell
DESIGNED
K. Whitesell
DRAWN
D. Kosakowski
CHECKED
Clough, Nicholas W P.E.
TECH. APPR.
Bowles, Linda M P.E.
APPROVED
PEER REVIEWER - Peer Reviewer

DENVER, CO 2019-03-11

PRECAST CONCRETE MANHOLES

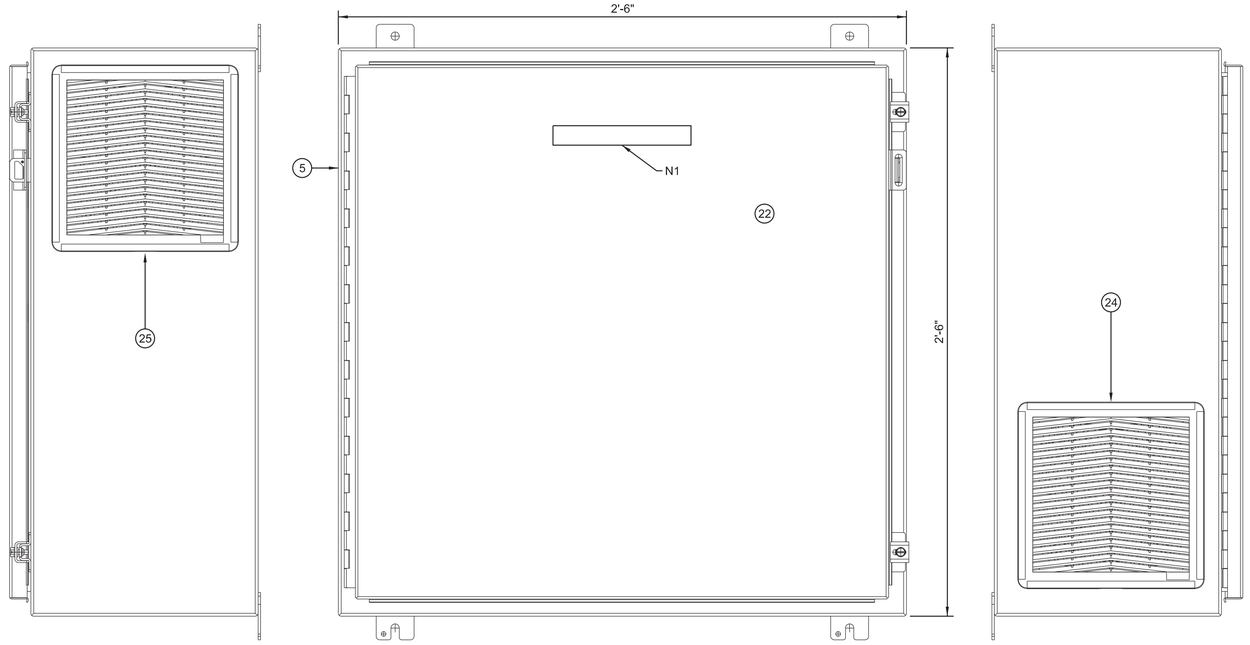
PLANS, SECTIONS AND DETAILS

1695-D-60731

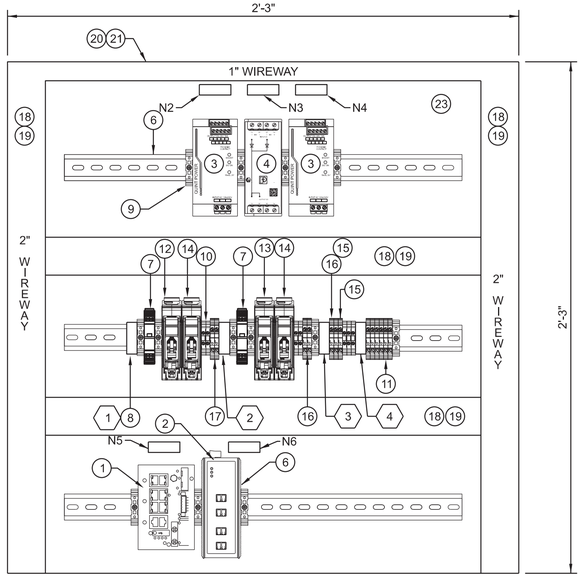
SHEET 1

LAST SAVED DATE
2019-03-11
LAST SAVED BY
DKOSAKOWSKI

CAD SYSTEM
AutoCAD Civil 3D
CAD FILENAME
1695-D-60731-RD-1.DWG



ENCLOSURE ELEVATION
3"=1'-0"



PANEL INTERNAL ELEVATION
3"-1'-0"

GENERAL SHEET NOTES

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2. GROUND BACKPANEL TO ENCLOSURE.
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4. FRONT OF PANEL ENCLOSURE SHALL HAVE A WARNING LABEL STATING IT IS POWERED FROM TWO SOURCES.
5. REFER TO 000-G-0093 FOR ADDITIONAL NAMEPLATE CONSTRUCTION STANDARDS.
6. REFERENCE DETAIL 4091-384 FOR WALL MOUNTING THE PANEL. FINAL ANCHORAGE DETAILS TO BE DETERMINED BY INSTALLING CONTRACTOR.

SHEET KEY NOTES

1. LABEL SHALL READ UTIL
2. LABEL SHALL READ UPS
3. LABEL SHALL READ 24VDC-1
4. LABEL SHALL READ 24VDC-2

BILL OF MATERIALS

ITEM	QTY	MANUFACTURER	PART NUMBER	DESCRIPTION	DIMENSIONS IN INCHES			NOTES
					HEIGHT	WIDTH	DEPTH	
1	1	FORTINET	FGR-70F-BDL-950-36	FORTIGATE RUGGED FIREWALL	4.800	3.200	4.400	NA
2	1	MOXA	OBU-102-SS-LC	2-CHANNEL OPTICAL FIBER BYPASS UNIT	5.310	1.800	4.130	NA
3	2	PHOENIX CONTACT	2866763	QUINT-PS/1AC/24DC/10 - POWER SUPPLY UNIT	5.118	2.362	4.921	OR EQUAL
4	1	PHOENIX CONTACT	2907719	QUINT4-DIODE/12-24DC/2X20/1X40	5.118	1.969	4.921	OR EQUAL
5	1	HOFFMAN, nVent	A303012LP	CONTINUOUS HINGE ENCLOSURE WITH CLAMPS LP TYPE 4, 30x30x12, GRAY, MILD STEEL	30.000	30.000	12.000	OR EQUAL
6	1	Black Box	JPM397A	JPM39 Series DIN Rail Mini Fiber Enclosure - 2 Port	2.010	5.310	5.510	OR EQUAL
7	1	Black Box	JPM467C-R2	High-Density Single mode Fiber Adapter Panel - Ceramic Sleeve, (12) LC Duplex, Blue	3.930	5.310	5.510	OR EQUAL
6	AS REQ'D	PHOENIX CONTACT	801733	NS 35/ 7,5 PERF 2000MM - DIN RAIL PERFORATED	1.378	AS REQ'D	0.295	OR EQUAL
7	2	PHOENIX CONTACT	2907918	SURGE PROTECTIVE DEVICE W/BASE	3.674	0.071	2.930	OR EQUAL
8	4	PHOENIX CONTACT	800310	UBE - MARKER CARRIERS	0.787	1.685	1.720	OR EQUAL
9	13	PHOENIX CONTACT	800886	E/NS 35 N - END BRACKET	1.913	0.374	1.291	OR EQUAL
10	7	PHOENIX CONTACT	3007327	UK 5 N WH - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
11	3	PHOENIX CONTACT	3044102	UT 4 - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
12	1	ALLEN BRADLEY	1489-M1C100	UL489 2 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
13	1	ALLEN BRADLEY	1489-M1C005	UL489 5 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
14	2	ALLEN BRADLEY	1489-M1C150	UL489 15 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
15	6	PHOENIX CONTACT	3046126	UT-4HESILED 60 (5X20) - FUSE MODULAR TERMINAL BLOCK	2.276	0.244	2.976	OR EQUAL
16	3	LITTLEFUSE	215002P	215 SERIES 5X20MM, TIME-LAG FUSE, 2A	0.197	0.197	0.787	OR EQUAL
17	4	PHOENIX CONTACT	441504	USLKG 5 - GROUND TERMINAL	1.673	0.244	1.850	OR EQUAL
18	AS REQ'D	THOMAS & BETTS	TYD2X4WPG6	2" X 4" WIRING DUCT, GREY	AS REQ'D	2.000	4.000	OR EQUAL
19	AS REQ'D	THOMAS & BETTS	TYD2CPG6	2" WIDE WIRING DUCT COVER, GREY	AS REQ'D	2.000	4.000	OR EQUAL
20	AS REQ'D	THOMAS & BETTS	TYD1X4WPG6	1" X 4" WIRING DUCT, GREY	1.000	AS REQ'D	4.000	OR EQUAL
21	AS REQ'D	THOMAS & BETTS	TYD1CPG6	1" WIDE WIRING DUCT COVER, GREY	1.000	AS REQ'D	4.000	OR EQUAL
22	1	HOFFMAN, nVent	ALFSWD	REMOTE DOOR SWITCH, 120/250VAC 10AMP, GALVANIZED, STEEL	1.260	0.670	0.870	OR EQUAL
23	1	HOFFMAN, nVent	A30P30	PANEL FOR TYPE 3R 4 4X 12 13 ENCLOSURE, FITS 30x30, WHITE, STEEL	27.000	27.000	0.750	OR EQUAL
24	1	HOFFMAN, nVent	HN0916513	HN 3R SIDE-MOUNT FILTER FANS 115V, 9.84X9.84X4.44, BLACK, PLASTIC	9.840	9.840	4.440	OR EQUAL
25	1	HOFFMAN, nVent	HN0900503	HN 3R FILTER FAN EXHAUST GRILLE, 9.84x9.84X1.33,BLACK, PLASTIC	9.840	9.840	0.970	OR EQUAL

NAMEPLATE SCHEDULE

ITEM	TAG	NAMEPLATE SIZE (W X L)	LETTER SIZE
N1	SAN-NIP-01	1" X 4"	1/4" NORMAL
N2	SAN-PS24-01	1/2" X 3"	5/32" NORMAL
N3	SAN-DIO24-01	1/2" X 3"	5/32" CONDENSED
N4	SAN-PS24-02	1/2" X 3"	5/32" NORMAL
N5	SAN-FW-01	1/2" X 3"	3/16" CONDENSED
N6	SAN-FBU-01	1/2" X 3"	3/16" CONDENSED

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BUREAU OF RECLAMATION

Jacobs

A DOMINGOS — DESIGNED
J BOSS — DRAWN
K BROWN — CHECKED
S MENK — TECH. APPR.
J PETERSON — APPROVED

VERIFY SCALE
BAR IS ONE INCH ON ORIGINAL DRAWING.
0 1"

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
SANOSTEE
TURNOUT
NIP LAYOUT

DBSJLWTP

085-N-1110
SHEET 273

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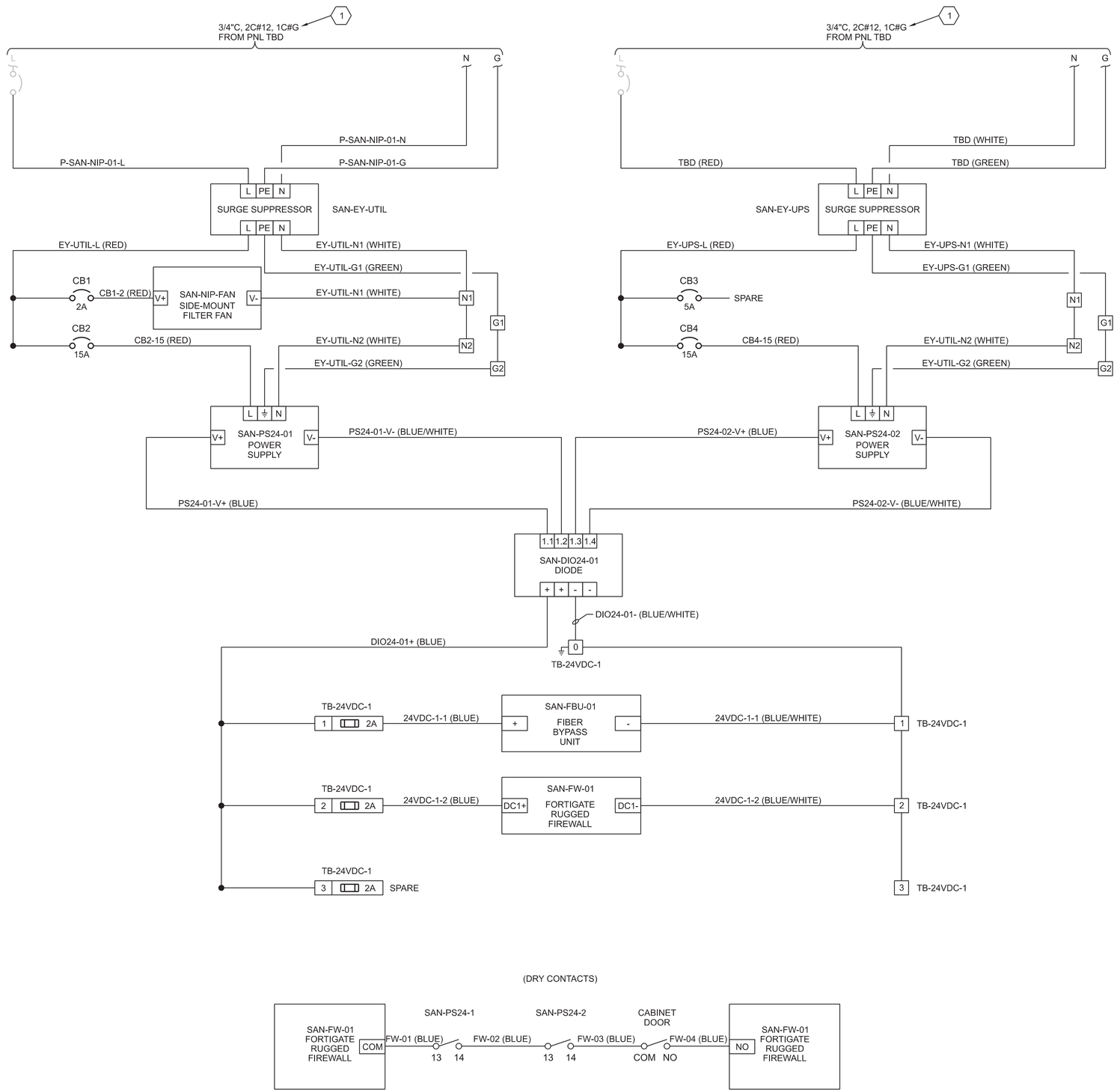
GENERAL SHEET NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY PER SPECIFICATION DIVISIONS 26 AND 27.

SHEET KEY NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY FROM POWER SOURCE



(DRY CONTACTS)

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DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

KIEL W. BROWN
NEW MEXICO
31004
PROFESSIONAL ENGINEER

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SANOSTEE TURNOUT NIP POWER DISTRIBUTION

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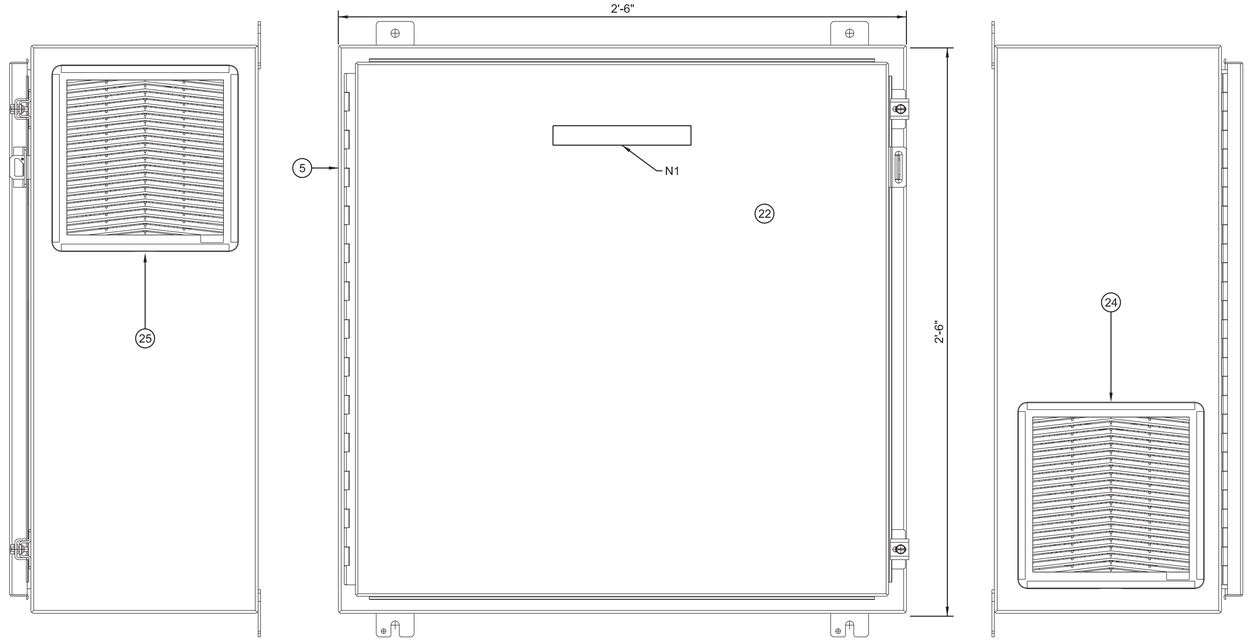

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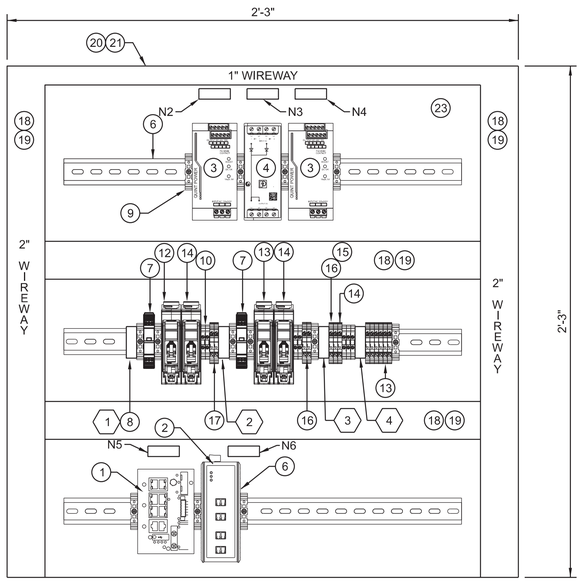

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BUREAU OF RECLAMATION

Jacobs

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J BOSS — DRAWN
K BROWN — CHECKED
S MENK — TECH. APPR.
J PETERSON — APPROVED

VERIFY SCALE

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0 1"

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
NEWCOMB
TURNOUT
NIP LAYOUT

DBSJLWTP

085-N-1115

SHEET 275

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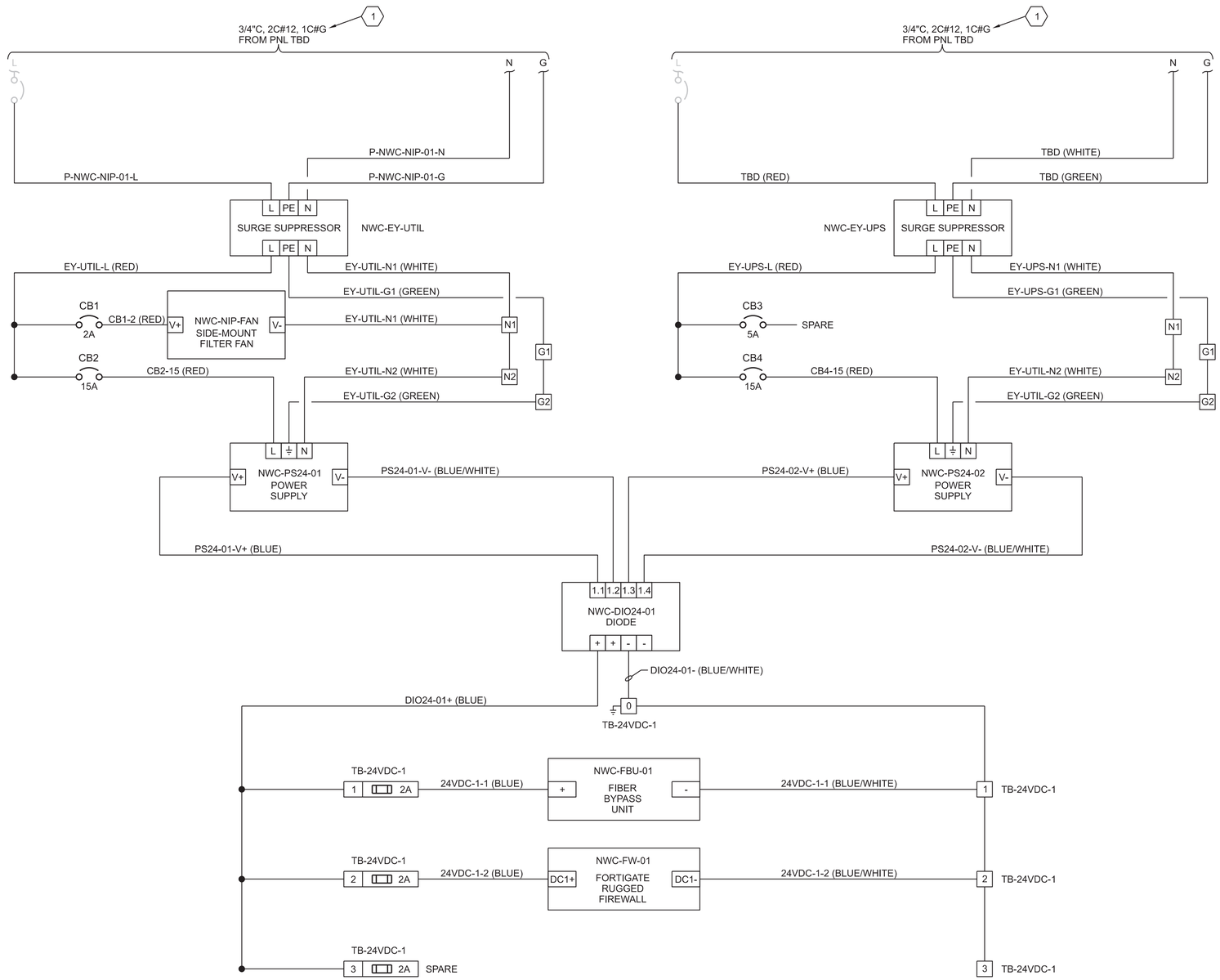

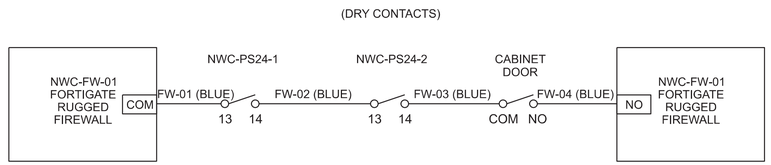

— BUREAU OF — RECLAMATION

Jacobs

DESIGNED	A DOMINGOS
DRAWN	J BOSS
CHECKED	K BROWN
TECH. APPR.	S MENK
APPROVED	J PETERSON

VERIFY SCALE

BAR IS ONE INCH ON ORIGINAL DRAWING.
0 1"

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SAN JUAN LATERAL

WATER TREATMENT PLANT

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FARMINGTON, NM FEBRUARY 2026

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TURNOUT
NIP POWER
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085-N-1116
SHEET 276

NIEL W. BROWN
NEW MEXICO
21004
PROFESSIONAL ENGINEER

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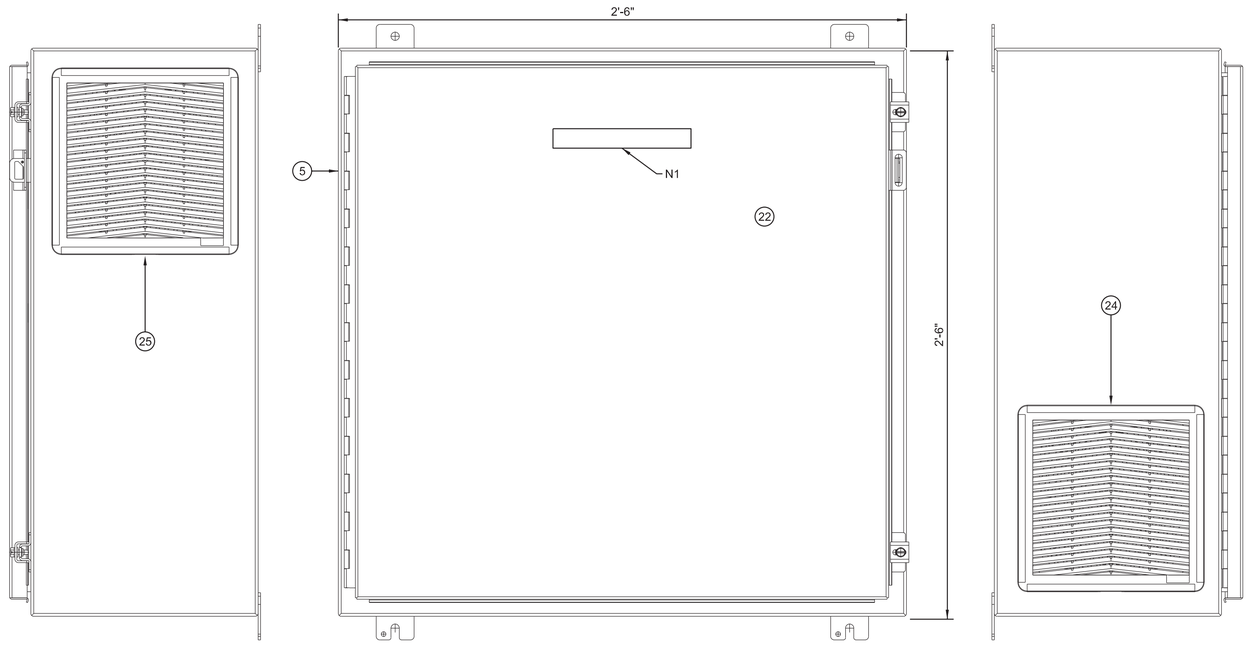
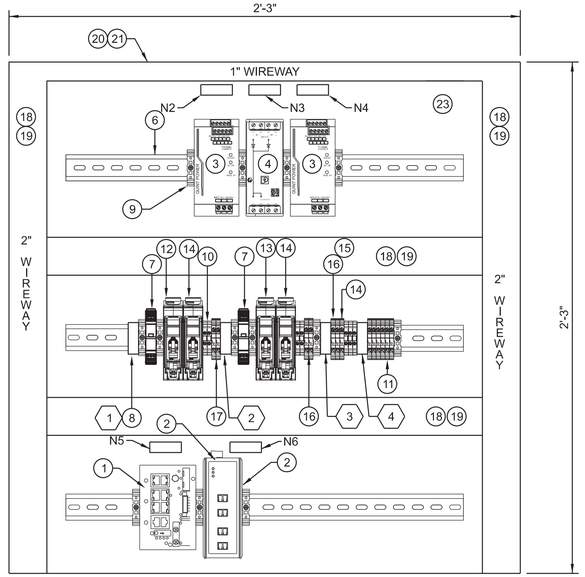


ENCLOSURE ELEVATION

3"=1'-0"



PANEL INTERNAL ELEVATION

3"-1'-0"

GENERAL SHEET NOTES

1. CABINET LAYOUT DESIGNED USING FIRST NAMED MANUFACTURER LISTED IN THE BOM. IT IS THE SUBCONTRACTORS RESPONSIBILITY TO UPDATE THE BOM AND CABINET LAYOUT USING THE SELECTED ITEM DIMENSIONS.
2. GROUND BACKPANEL TO ENCLOSURE.
3. BOM ITEM 23 SHALL BE INSTALLED SUCH THAT THE DOOR SWITCH CONTACTS OPEN WHEN THE ENCLOSURE IS OPENED. TO BE INSTALLED ALONG THE TOP ELEVATION OF THE DOOR/ENCLOSURE.
4. FRONT OF PANEL ENCLOSURE SHALL HAVE A WARNING LABEL STATING IT IS POWERED FROM TWO SOURCES.
5. REFER TO 000-G-0093 FOR ADDITIONAL NAMEPLATE CONSTRUCTION STANDARDS.
6. REFERENCE DETAIL 4091-384 FOR WALL MOUNTING THE PANEL. FINAL ANCHORAGE DETAILS TO BE DETERMINED BY INSTALLING CONTRACTOR.

SHEET KEY NOTES

1. LABEL SHALL READ UTIL
2. LABEL SHALL READ UPS
3. LABEL SHALL READ 24VDC-1
4. LABEL SHALL READ 24VDC-2

BILL OF MATERIALS

ITEM	QTY	MANUFACTURER	PART NUMBER	DESCRIPTION	DIMENSIONS IN INCHES HEIGHT	WIDTH	DEPTH	NOTES
1	1	FORTINET	FGR-70F-BDL-950-36	FORTIGATE RUGGED FIREWALL	4.800	3.200	4.400	NA
2	1	MOXA	OBU-102-SS-LC	2-CHANNEL OPTICAL FIBER BYPASS UNIT	5.310	1.800	4.130	NA
3	2	PHOENIX CONTACT	2866763	QUINT-PS/1AC/24DC/10 - POWER SUPPLY UNIT	5.118	2.362	4.921	OR EQUAL
4	1	PHOENIX CONTACT	2907719	QUINT4-DIODE/12-24DC/2X20/1X40	5.118	1.969	4.921	OR EQUAL
5	1	HOFFMAN, nVent	A303012LP	CONTINUOUS HINGE ENCLOSURE WITH CLAMPS LP TYPE 4, 30x30x12, GRAY, MILD STEEL	30.000	30.000	12.000	OR EQUAL
6	1	Black Box	JPM397A	JPM39 Series DIN Rail Mini Fiber Enclosure - 2 Port	2.010	5.310	5.510	OR EQUAL
7	1	Black Box	JPM467C-R2	High-Density Single mode Fiber Adapter Panel - Ceramic Sleeve, (12) LC Duplex, Blue	3.930	5.310	5.510	OR EQUAL
6	AS REQ'D	PHOENIX CONTACT	801733	NS 35/ 7,5 PERF 2000MM - DIN RAIL PERFORATED	1.378	AS REQ'D	0.295	OR EQUAL
7	2	PHOENIX CONTACT	2907918	SURGE PROTECTIVE DEVICE W/BASE	3.674	0.071	2.930	OR EQUAL
8	4	PHOENIX CONTACT	800310	UBE - MARKER CARRIERS	0.787	1.685	1.720	OR EQUAL
9	13	PHOENIX CONTACT	800886	E/NS 35 N - END BRACKET	1.913	0.374	1.291	OR EQUAL
10	7	PHOENIX CONTACT	3007327	UK 5 N WH - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
11	3	PHOENIX CONTACT	3044102	UT 4 - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
12	1	ALLEN BRADLEY	1489-M1C100	UL489 2 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
13	1	ALLEN BRADLEY	1489-M1C005	UL489 5 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
14	2	ALLEN BRADLEY	1489-M1C150	UL489 15 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
15	6	PHOENIX CONTACT	3046126	UT-4HESILED 60 (5X20) - FUSE MODULAR TERMINAL BLOCK	2.276	0.244	2.976	OR EQUAL
16	3	LITTLEFUSE	215002P	215 SERIES 5X20MM, TIME-LAG FUSE, 2A	0.197	0.197	0.787	OR EQUAL
17	4	PHOENIX CONTACT	441504	USLKG 5 - GROUND TERMINAL	1.673	0.244	1.850	OR EQUAL
18	AS REQ'D	THOMAS & BETTS	TYD2X4WPG6	2" X 4" WIRING DUCT, GREY	AS REQ'D	2.000	4.000	OR EQUAL
19	AS REQ'D	THOMAS & BETTS	TYD2CPG6	2" WIDE WIRING DUCT COVER, GREY	AS REQ'D	2.000	4.000	OR EQUAL
20	AS REQ'D	THOMAS & BETTS	TYD1X4WPG6	1" X 4" WIRING DUCT, GREY	1.000	AS REQ'D	4.000	OR EQUAL
21	AS REQ'D	THOMAS & BETTS	TYD1CPG6	1" WIDE WIRING DUCT COVER, GREY	1.000	AS REQ'D	4.000	OR EQUAL
22	1	HOFFMAN, nVent	ALFSWD	REMOTE DOOR SWITCH, 120/250VAC 10AMP, GALVANIZED, STEEL	1.260	0.670	0.870	OR EQUAL
23	1	HOFFMAN, nVent	A30P30	PANEL FOR TYPE 3R 4 4X 12 13 ENCLOSURE, FITS 30x30, WHITE, STEEL	27.000	27.000	0.750	OR EQUAL
24	1	HOFFMAN, nVent	HN0916513	HN 3R SIDE-MOUNT FILTER FANS 115V, 9.84X9.84X4.44, BLACK, PLASTIC	9.840	9.840	4.440	OR EQUAL
25	1	HOFFMAN, nVent	HN0900503	HN 3R FILTER FAN EXHAUST GRILLE, 9.84x9.84X1.33,BLACK, PLASTIC	9.840	9.840	0.970	OR EQUAL

NAMEPLATE SCHEDULE

ITEM	TAG	NAMEPLATE SIZE (W X L)	LETTER SIZE
N1	NAS-NIP-01	1 X 4"	1/4" NORMAL
N2	NAS-PS24-01	1/2" X 3"	5/32" NORMAL
N3	NAS-DIO24-01	1/2" X 3"	5/32" CONDENSED
N4	NAS-PS24-02	1/2" X 3"	5/32" NORMAL
N5	NAS-FW-01	3/4" X 3"	3/16" CONDENSED
N6	NAS-FBU-01	3/4" X 3"	3/16" CONDENSED

BUREAU OF RECLAMATION

Jacobs

A DOMINGOS — DESIGNED
J BOSS — DRAWN
K BROWN — CHECKED
S MENK — TECH. APPR.
J PETERSON — APPROVED

VERIFY SCALE
BAR IS ONE INCH ON ORIGINAL DRAWING.
0 1"

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ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED
FARMINGTON, NM FEBRUARY 2026

KIEL W. BROWN
NEW MEXICO
31004
PROFESSIONAL ENGINEER

DIGITALLY STAMPED ON 2/11/2026

I&C
NASCHITTI
TURNOUT
NIP LAYOUT

DBSJLWTP

085-N-1120
SHEET 277

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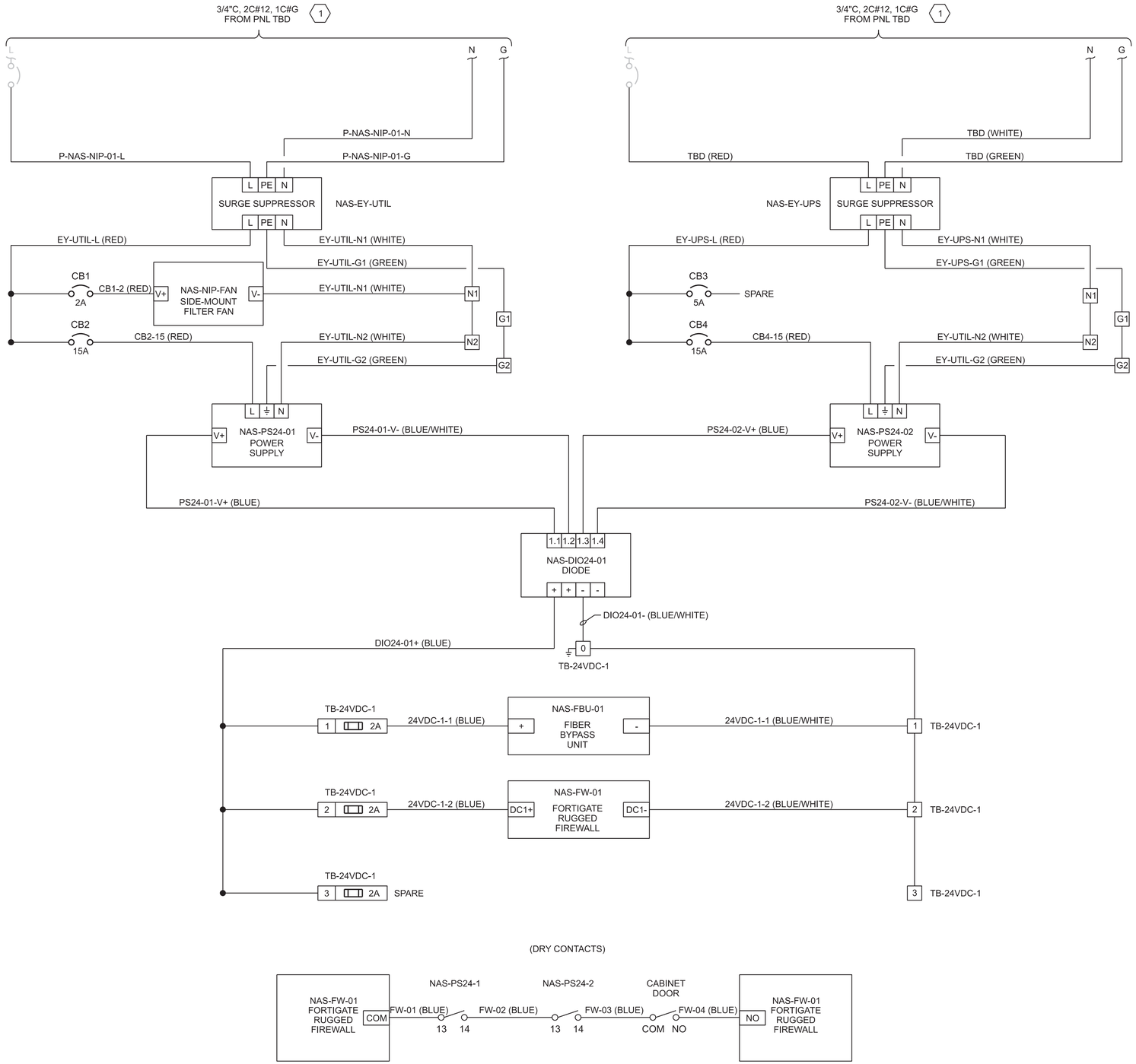
GENERAL SHEET NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY PER SPECIFICATION DIVISIONS 26 AND 27.

SHEET KEY NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY FROM POWER SOURCE



BUREAU OF RECLAMATION

Jacobs

DESIGNED	A DOMINGOS
DRAWN	J BOSS
CHECKED	K BROWN
TECH. APPR.	S MENK
APPROVED	J PETERSON

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NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
NASCHITTI TURNOUT
NIP POWER DISTRIBUTION
DBSJLWTP

085-N-1121
SHEET 278

DIGITALLY STAMPED ON 2/11/2026

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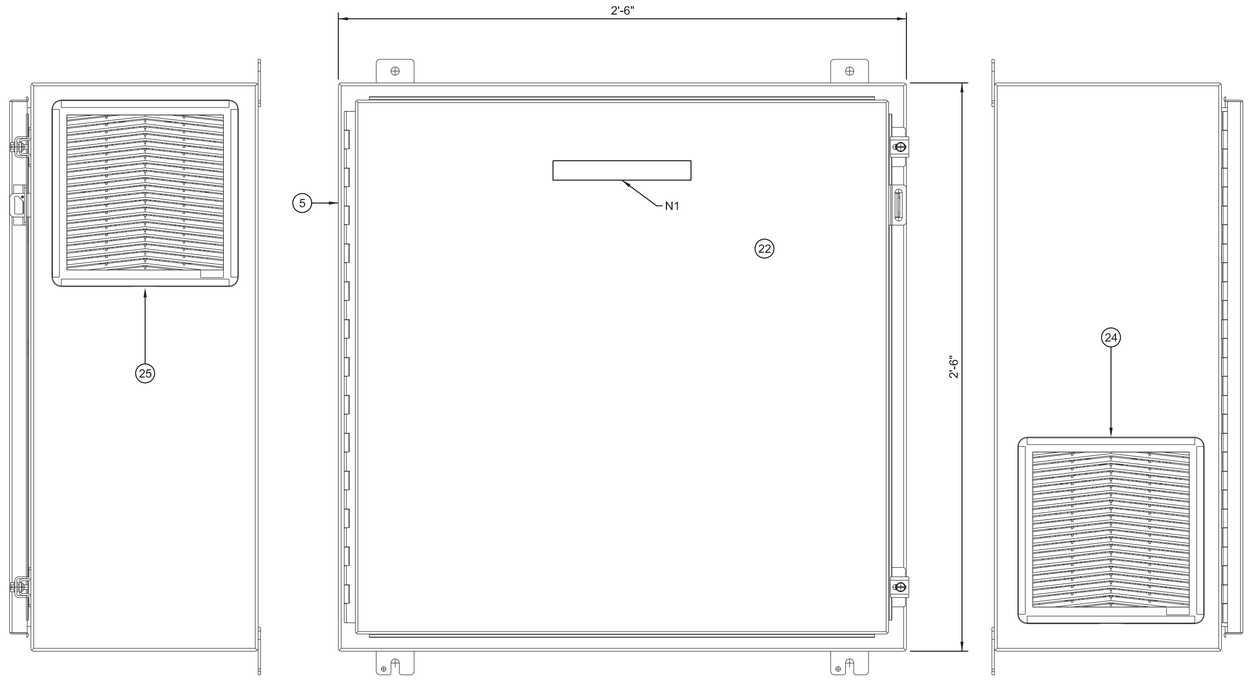
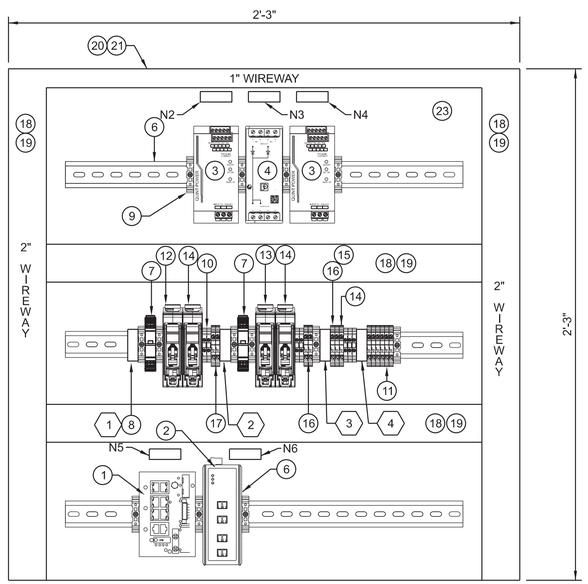


ENCLOSURE ELEVATION
3"=1'-0"



PANEL INTERNAL ELEVATION
3"=1'-0"

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4. FRONT OF PANEL ENCLOSURE SHALL HAVE A WARNING LABEL STATING IT IS POWERED FROM TWO SOURCES.
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SHEET KEY NOTES

1. LABEL SHALL READ UTIL
2. LABEL SHALL READ UPS
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BUREAU OF RECLAMATION

Jacobs

A DOMINGOS — DESIGNED
J BOSS — DRAWN
K BROWN — CHECKED
S MENK — TECH. APPR.
J PETERSON — APPROVED

VERIFY SCALE
BAR IS ONE INCH ON ORIGINAL DRAWING.
0 1"

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
BBN9 TURNOUT
NIP
LAYOUT

DBSJLWTP

085-N-1125
SHEET 279

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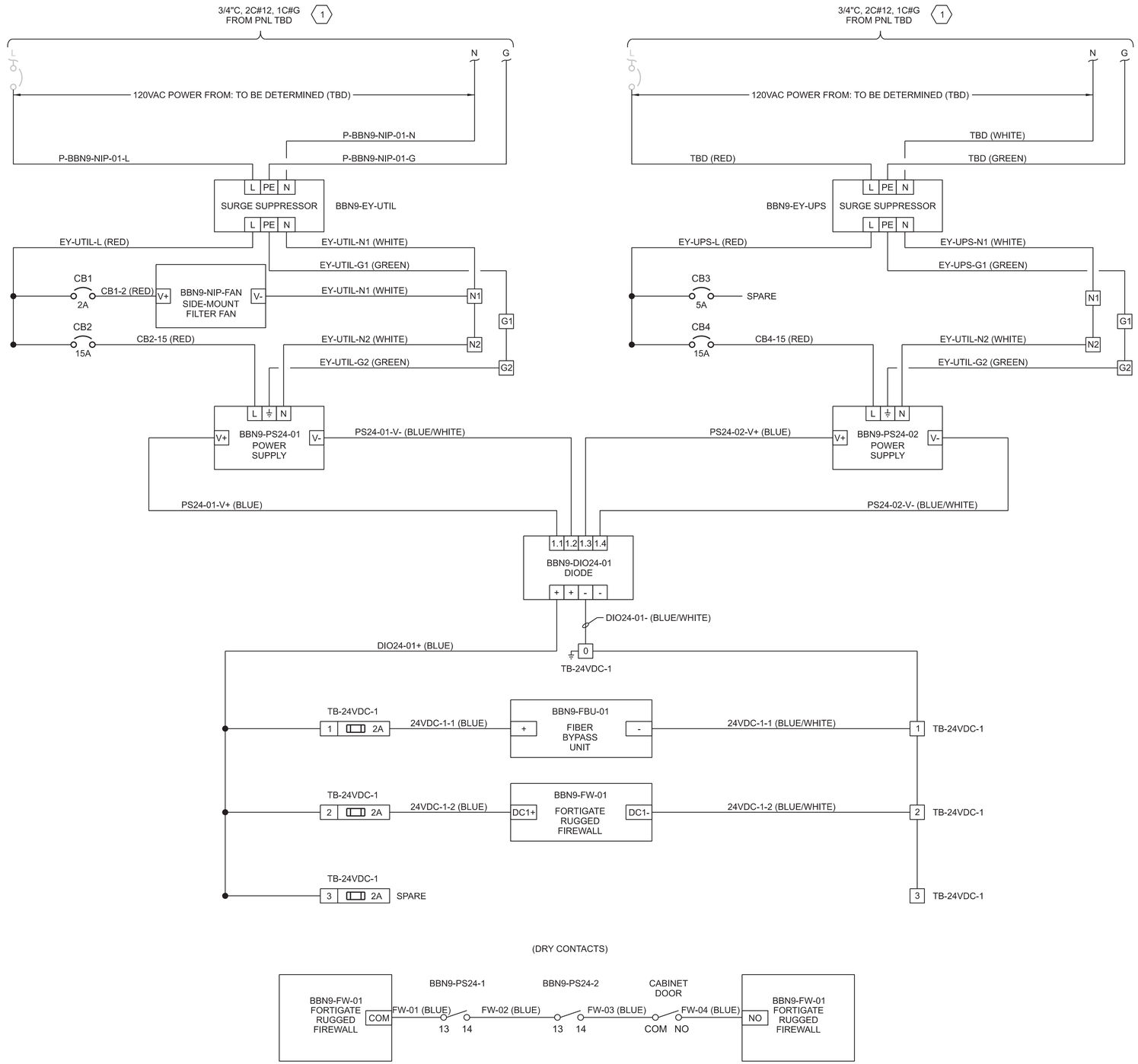

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SAN JUAN LATERAL

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NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C

BBN9
TURNOUT
NIP POWER
DISTRIBUTION

DBSJLWTP

085-N-1126

SHEET 280

KIEL W. BROWN
NEW MEXICO
21004
PROFESSIONAL ENGINEER

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MICROSTATION CONNECT
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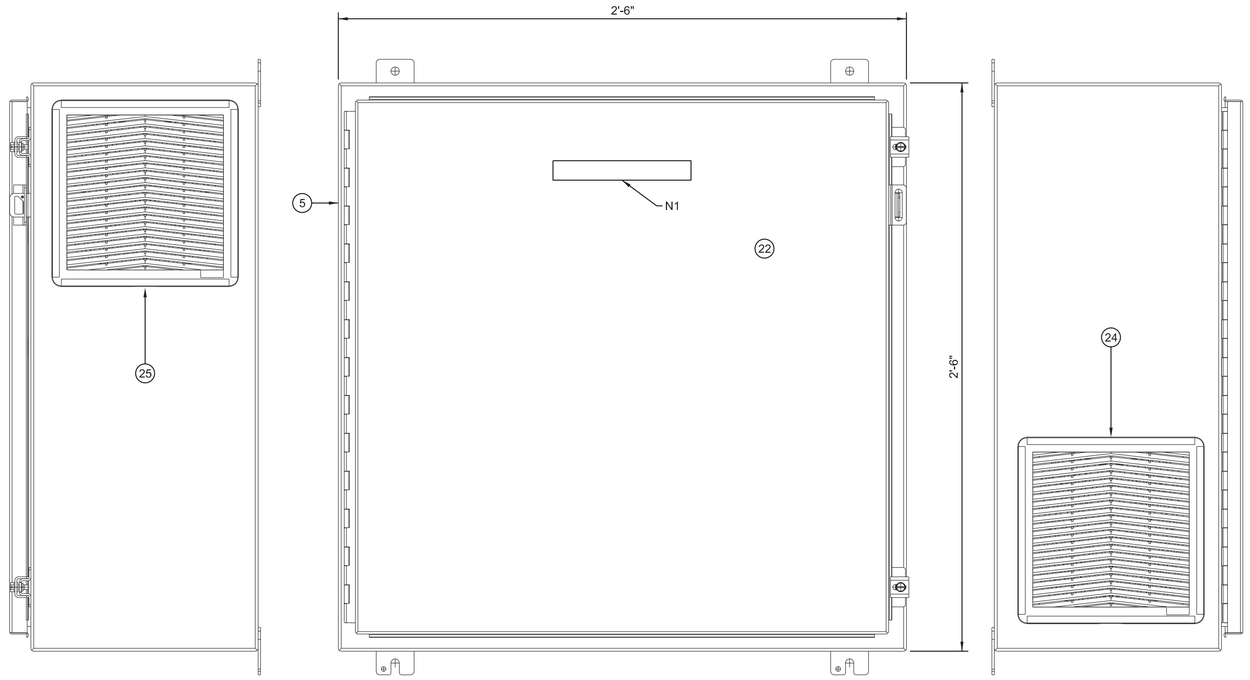

ENCLOSURE ELEVATION
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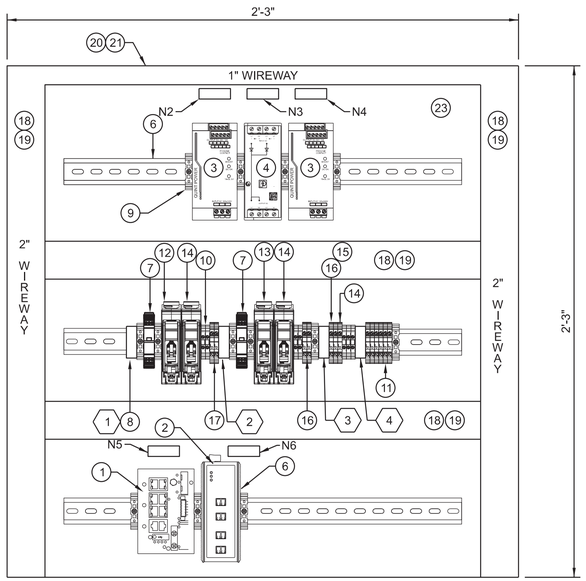

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9	13	PHOENIX CONTACT	800886	E/NS 35 N - END BRACKET	1.913	0.374	1.291	OR EQUAL
10	7	PHOENIX CONTACT	3007327	UK 5 N WH - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
11	3	PHOENIX CONTACT	3044102	UT 4 - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
12	1	ALLEN BRADLEY	1489-M1C100	UL489 2 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
13	1	ALLEN BRADLEY	1489-M1C005	UL489 5 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
14	2	ALLEN BRADLEY	1489-M1C150	UL489 15 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
15	6	PHOENIX CONTACT	3046126	UT-4HESILED 60 (5X20) - FUSE MODULAR TERMINAL BLOCK	2.276	0.244	2.976	OR EQUAL
16	3	LITTLEFUSE	215002P	215 SERIES 5X20MM, TIME-LAG FUSE, 2A	0.197	0.197	0.787	OR EQUAL
17	4	PHOENIX CONTACT	441504	USLKG 5 - GROUND TERMINAL	1.673	0.244	1.850	OR EQUAL
18	AS REQ'D	THOMAS & BETTS	TYD2X4WPG6	2" X 4" WIRING DUCT, GREY	AS REQ'D	2.000	4.000	OR EQUAL
19	AS REQ'D	THOMAS & BETTS	TYD2CPG6	2" WIDE WIRING DUCT COVER, GREY	AS REQ'D	2.000	4.000	OR EQUAL
20	AS REQ'D	THOMAS & BETTS	TYD1X4WPG6	1" X 4" WIRING DUCT, GREY	1.000	AS REQ'D	4.000	OR EQUAL
21	AS REQ'D	THOMAS & BETTS	TYD1CPG6	1" WIDE WIRING DUCT COVER, GREY	1.000	AS REQ'D	4.000	OR EQUAL
22	1	HOFFMAN, nVent	ALFSWD	REMOTE DOOR SWITCH, 120/250VAC 10AMP, GALVANIZED, STEEL	1.260	0.670	0.870	OR EQUAL
23	1	HOFFMAN, nVent	A30P30	PANEL FOR TYPE 3R 4 4X 12 13 ENCLOSURE, FITS 30x30, WHITE, STEEL	27.000	27.000	0.750	OR EQUAL
24	1	HOFFMAN, nVent	HN0916513	HN 3R SIDE-MOUNT FILTER FANS 115V, 9.84X9.84X4.44, BLACK, PLASTIC	9.840	9.840	4.440	OR EQUAL
25	1	HOFFMAN, nVent	HN0900503	HN 3R FILTER FAN EXHAUST GRILLE, 9.84x9.84X1.33,BLACK, PLASTIC	9.840	9.840	0.970	OR EQUAL

NAMEPLATE SCHEDULE

ITEM	TAG	NAMEPLATE SIZE (W X L)	LETTER SIZE
N1	NAS-NIP-01	1 X 4"	1/4" NORMAL
N2	NAS-PS24-01	1/2" X 3"	5/32" NORMAL
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N4	NAS-PS24-02	1/2" X 3"	5/32" NORMAL
N5	NAS-FW-01	3/4" X 3"	3/16" CONDENSED
N6	NAS-FBU-01	3/4" X 3"	3/16" CONDENSED

BUREAU OF RECLAMATION

Jacobs

A DOMINGOS — DESIGNED
J BOSS — DRAWN
K BROWN — CHECKED
S MENK — TECH. APPR.
J PETERSON — APPROVED

VERIFY SCALE
BAR IS ONE INCH ON ORIGINAL DRAWING.
0 1"

FOR INFORMATION ONLY

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
COYOTE CANYON
TURNOUT PP
NIP LAYOUT

DBSJLWTP

085-N-1130
SHEET 281

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TIME PLOTTED
01:24

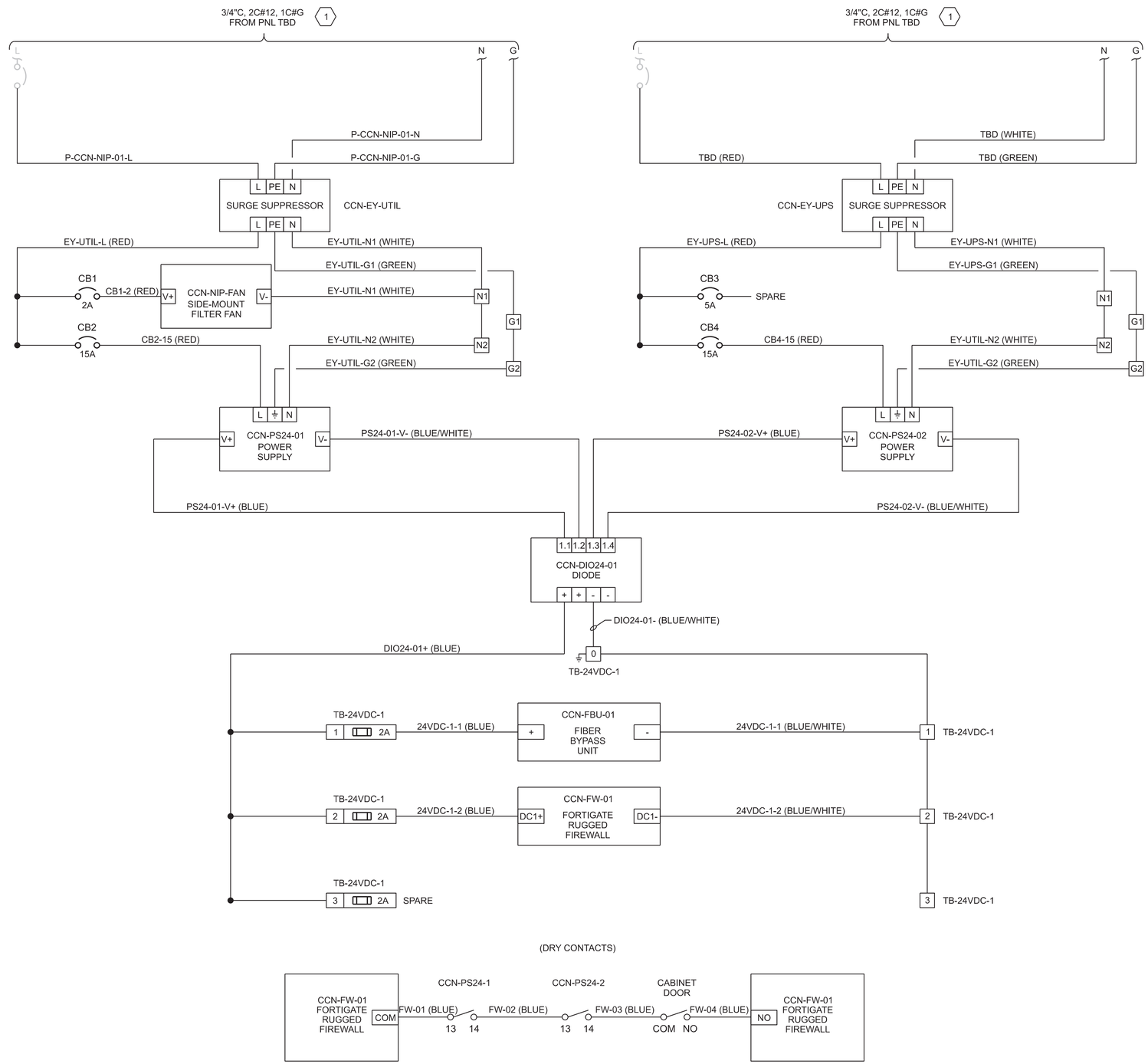
MICROSTATION CONNECT
$PWPATHS.

GENERAL SHEET NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY PER SPECIFICATION DIVISIONS 26 AND 27.

SHEET KEY NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY FROM POWER SOURCE



BUREAU OF RECLAMATION

Jacobs

DESIGNED	A DOMINGOS
DRAWN	J BOSS
CHECKED	K BROWN
TECH. APPR.	S MENK
APPROVED	J PETERSON

VERIFY SCALE

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0 1"

FOR INFORMATION ONLY

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
COYOTE CANYON
TURNOUT
NIP POWER
DISTRIBUTION
DBSJLWTP

085-N-1131
SHEET 282

NIEL W. BROWN
NEW MEXICO
31004
PROFESSIONAL ENGINEER

DIGITALLY STAMPED ON 2/11/2026

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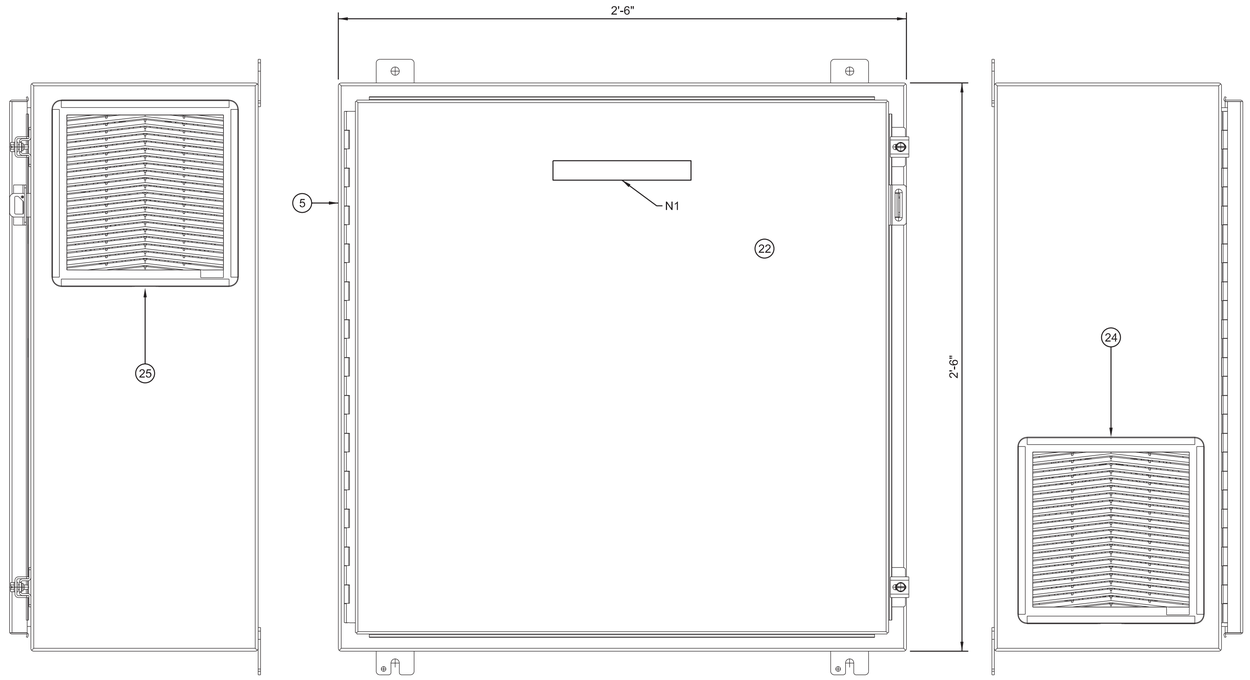
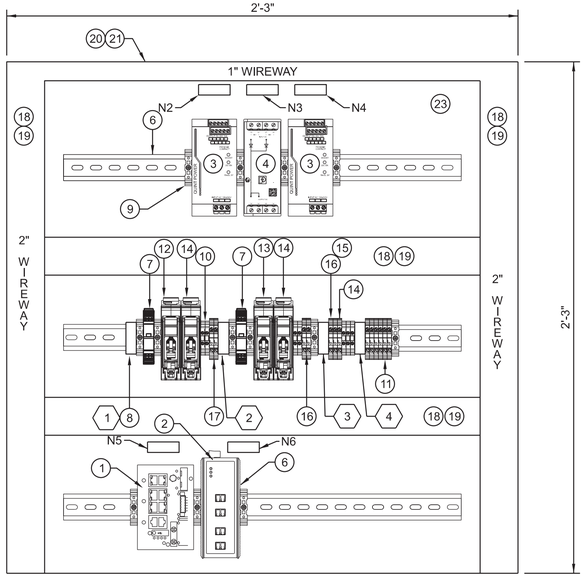
MICROSTATION CONNECT
$PWPATHS;



ENCLOSURE ELEVATION
3"=1'-0"



PANEL INTERNAL ELEVATION
3"=1'-0"

BILL OF MATERIALS

ITEM	QTY	MANUFACTURER	PART NUMBER	DESCRIPTION	DIMENSIONS IN INCHES			NOTES
					HEIGHT	WIDTH	DEPTH	
1	1	FORTINET	FGR-70F-BDL-950-36	FORTIGATE RUGGED FIREWALL	4.800	3.200	4.400	NA
2	1	MOXA	OBU-102-SS-LC	2-CHANNEL OPTICAL FIBER BYPASS UNIT	5.310	1.800	4.130	NA
3	2	PHOENIX CONTACT	2866763	QUINT-PS/1AC/24DC/10 - POWER SUPPLY UNIT	5.118	2.362	4.921	OR EQUAL
4	1	PHOENIX CONTACT	2907719	QUINT4-DIODE/12-24DC/2X20/1X40	5.118	1.969	4.921	OR EQUAL
5	1	HOFFMAN, nVent	A303012LP	CONTINUOUS HINGE ENCLOSURE WITH CLAMPS LP TYPE 4, 30x30x12, GRAY, MILD STEEL	30.000	30.000	12.000	OR EQUAL
6	1	Black Box	JPM397A	JPM39 Series DIN Rail Mini Fiber Enclosure - 2 Port	2.010	5.310	5.510	OR EQUAL
7	1	Black Box	JPM467C-R2	High-Density Single mode Fiber Adapter Panel - Ceramic Sleeve, (12) LC Duplex, Blue	3.930	5.310	5.510	OR EQUAL
6	AS REQ'D	PHOENIX CONTACT	801733	NS 35/ 7,5 PERF 2000MM - DIN RAIL PERFORATED	1.378	AS REQ'D	0.295	OR EQUAL
7	2	PHOENIX CONTACT	2907918	SURGE PROTECTIVE DEVICE W/BASE	3.674	0.071	2.930	OR EQUAL
8	4	PHOENIX CONTACT	800310	UBE - MARKER CARRIERS	0.787	1.685	1.720	OR EQUAL
9	13	PHOENIX CONTACT	800886	E/NS 35 N - END BRACKET	1.913	0.374	1.291	OR EQUAL
10	7	PHOENIX CONTACT	3007327	UK 5 N WH - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
11	3	PHOENIX CONTACT	3044102	UT 4 - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
12	1	ALLEN BRADLEY	1489-M1C100	UL489 2 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
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18	AS REQ'D	THOMAS & BETTS	TYD2X4WPG6	2" X 4" WIRING DUCT, GREY	AS REQ'D	2.000	4.000	OR EQUAL
19	AS REQ'D	THOMAS & BETTS	TYD2CPG6	2" WIDE WIRING DUCT COVER, GREY	AS REQ'D	2.000	4.000	OR EQUAL
20	AS REQ'D	THOMAS & BETTS	TYD1X4WPG6	1" X 4" WIRING DUCT, GREY	1.000	AS REQ'D	4.000	OR EQUAL
21	AS REQ'D	THOMAS & BETTS	TYD1CPG6	1" WIDE WIRING DUCT COVER, GREY	1.000	AS REQ'D	4.000	OR EQUAL
22	1	HOFFMAN, nVent	ALFSWD	REMOTE DOOR SWITCH, 120/250VAC 10AMP, GALVANIZED, STEEL	1.260	0.670	0.870	OR EQUAL
23	1	HOFFMAN, nVent	A30P30	PANEL FOR TYPE 3R 4 4X 12 13 ENCLOSURE, FITS 30x30, WHITE, STEEL	27.000	27.000	0.750	OR EQUAL
24	1	HOFFMAN, nVent	HN0916513	HN 3R SIDE-MOUNT FILTER FANS 115V, 9.84X9.84X4.44, BLACK, PLASTIC	9.840	9.840	4.440	OR EQUAL
25	1	HOFFMAN, nVent	HN0900503	HN 3R FILTER FAN EXHAUST GRILLE, 9.84x9.84X1.33,BLACK, PLASTIC	9.840	9.840	0.970	OR EQUAL

NAMEPLATE SCHEDULE

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GENERAL SHEET NOTES

1. CABINET LAYOUT DESIGNED USING FIRST NAMED MANUFACTURER LISTED IN THE BOM. IT IS THE SUBCONTRACTORS RESPONSIBILITY TO UPDATE THE BOM AND CABINET LAYOUT USING THE SELECTED ITEM DIMENSIONS.
2. GROUND BACKPANEL TO ENCLOSURE.
3. BOM ITEM 23 SHALL BE INSTALLED SUCH THAT THE DOOR SWITCH CONTACTS OPEN WHEN THE ENCLOSURE IS OPENED. TO BE INSTALLED ALONG THE TOP ELEVATION OF THE DOOR/ENCLOSURE.
4. FRONT OF PANEL ENCLOSURE SHALL HAVE A WARNING LABEL STATING IT IS POWERED FROM TWO SOURCES.
5. REFER TO 000-G-0093 FOR ADDITIONAL NAMEPLATE CONSTRUCTION STANDARDS.
6. REFERENCE DETAIL 4091-384 FOR WALL MOUNTING THE PANEL. FINAL ANCHORAGE DETAILS TO BE DETERMINED BY INSTALLING CONTRACTOR.

SHEET KEY NOTES

1. LABEL SHALL READ UTIL
2. LABEL SHALL READ UPS
3. LABEL SHALL READ 24VDC-1
4. LABEL SHALL READ 24VDC-2

BUREAU OF RECLAMATION

Jacobs

A DOMINGOS — DESIGNED
J BOSS — DRAWN
K BROWN — CHECKED
S MENK — TECH. APPR.
J PETERSON — APPROVED

VERIFY SCALE
BAR IS ONE INCH ON ORIGINAL DRAWING.
0 1"

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
ROCK SPRINGS
TURNOUT
NIP LAYOUT

DBSJLWTP

085-N-1135
SHEET 283

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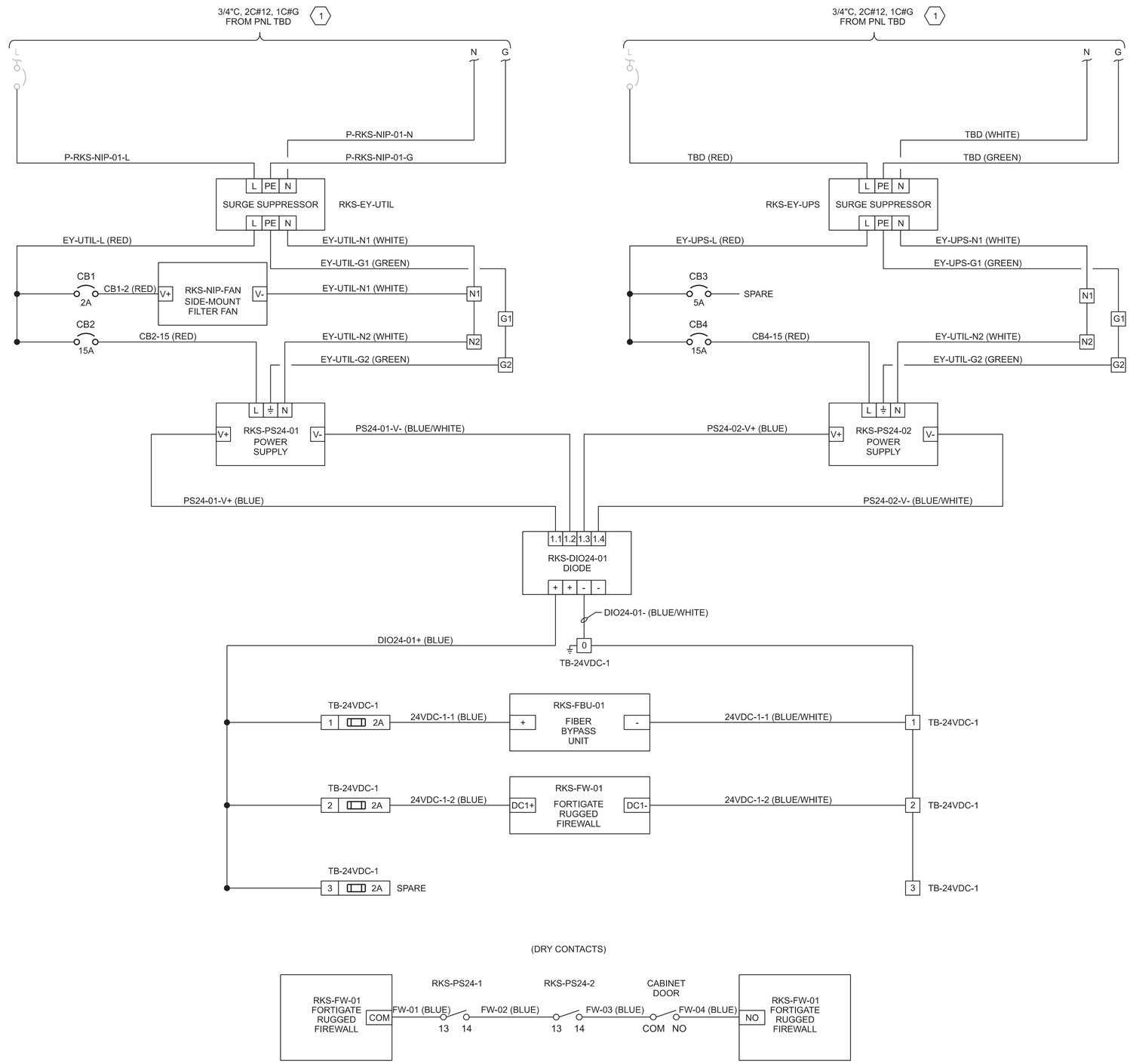
MICROSTATION CONNECT
$PWPATHS.

GENERAL SHEET NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY PER SPECIFICATION DIVISIONS 26 AND 27.

SHEET KEY NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY FROM POWER SOURCE



BUREAU OF RECLAMATION

Jacobs

DESIGNED	A DOMINGOS
DRAWN	J BOSS
CHECKED	K BROWN
TECH. APPR.	S MENK
APPROVED	J PETERSON

VERIFY SCALE

BAR IS ONE INCH ON ORIGINAL DRAWING.
0 1"

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ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C

ROCK SPRINGS
TURNOUT
NIP POWER
DISTRIBUTION

DBSJLWTP

085-N-1136

SHEET 284

KIEL W. BROWN
NEW MEXICO
21004
PROFESSIONAL ENGINEER

DIGITALLY STAMPED ON 2/11/2026

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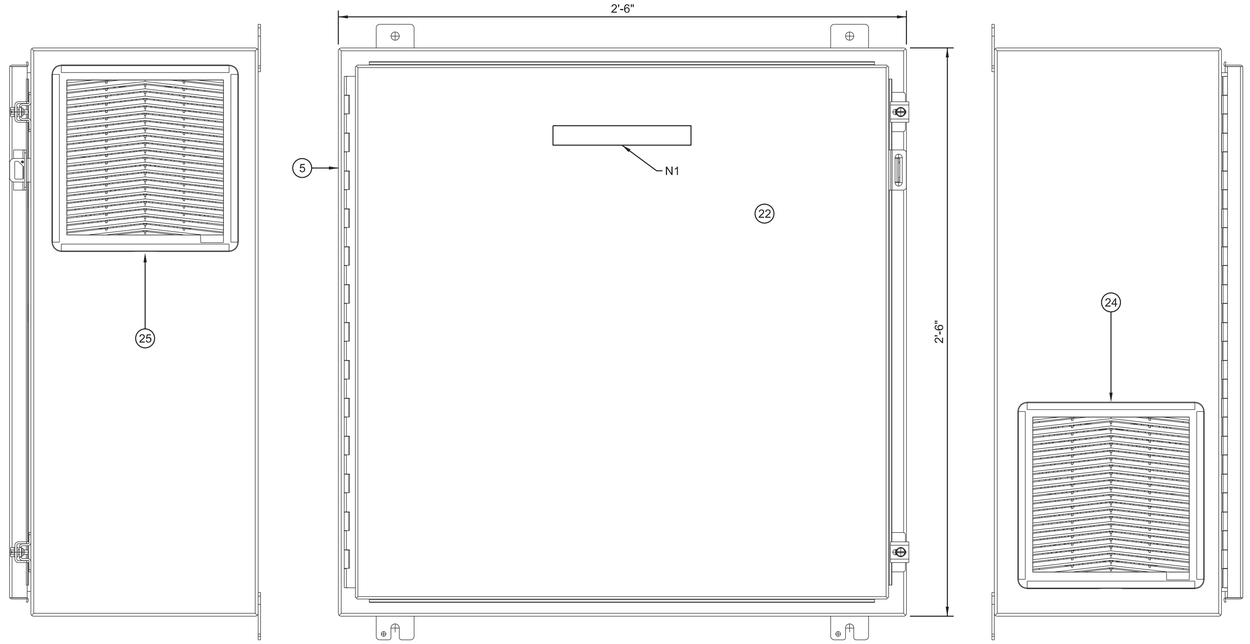
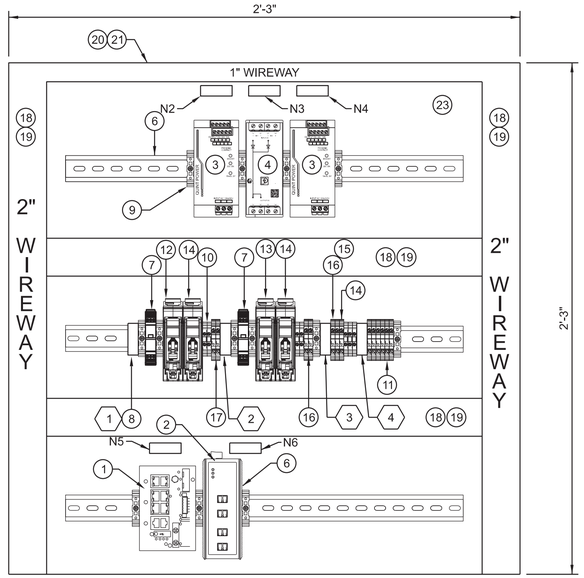
MICROSTATION CONNECT
$PWPATHS.



ENCLOSURE ELEVATION

3"=1'-0"



PANEL INTERNAL ELEVATION

3"-1'-0"

GENERAL SHEET NOTES

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3. LABEL SHALL READ 24VDC-1
4. LABEL SHALL READ 24VDC-2

BILL OF MATERIALS

ITEM	QTY	MANUFACTUER	PART NUMBER	DESCRIPTION	DIMENSIONS IN INCHES HEIGHT	WIDTH	DEPTH	NOTES
1	1	FORTINET	FGR-70F-BDL-950-36	FORTIGATE RUGGED FIREWALL	4.800	3.200	4.400	NA
2	1	MOXA	OBU-102-SS-LC	2-CHANNEL OPTICAL FIBER BYPASS UNIT	5.310	1.800	4.130	NA
3	2	PHOENIX CONTACT	2866763	QUINT-PS/1AC/24DC/10 - POWER SUPPLY UNIT	5.118	2.362	4.921	OR EQUAL
4	1	PHOENIX CONTACT	2907719	QUINT4-DIODE/12-24DC/2X20/1X40	5.118	1.969	4.921	OR EQUAL
5	1	HOFFMAN, nVent	A303012LP	CONTINUOUS HINGE ENCLOSURE WITH CLAMPS LP TYPE 4, 30x30x12, GRAY, MILD STEEL	30.000	30.000	12.000	OR EQUAL
6	AS REQ'D	PHOENIX CONTACT	801733	NS 35/ 7,5 PERF 2000MM - DIN RAIL PERFORATED	1.378	AS REQ'D	0.295	OR EQUAL
7	2	PHOENIX CONTACT	2907918	SURGE PROTECTIVE DEVICE W/BASE	3.674	0.071	2.930	OR EQUAL
8	4	PHOENIX CONTACT	800310	UBE - MARKER CARRIERS	0.787	1.685	1.720	OR EQUAL
9	13	PHOENIX CONTACT	800886	E/NS 35 N - END BRACKET	1.913	0.374	1.291	OR EQUAL
10	7	PHOENIX CONTACT	3007327	UK 5 N WH - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
11	3	PHOENIX CONTACT	3044102	UT 4 - FEED-THROUGH TERMINAL BLOCK	1.673	0.244	1.328	OR EQUAL
12	1	ALLEN BRADLEY	1489-M1C100	UL489 2 A MINATURE CIRCUIT BREAKER	4.370	0.693	2.854	OR EQUAL
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17	3	PHOENIX CONTACT	441504	USLKG 5 - GROUND TERMINAL	1.673	0.244	1.850	OR EQUAL
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19	AS REQ'D	THOMAS & BETTS	TYD2CPG6	2" WIDE WIRING DUCT COVER, GREY	AS REQ'D	2.000	4.000	OR EQUAL
20	AS REQ'D	THOMAS & BETTS	TYD1X4WPG6	1" X 4" WIRING DUCT, GREY	1.000	AS REQ'D	4.000	OR EQUAL
21	AS REQ'D	THOMAS & BETTS	TYD1CPG6	1" WIDE WIRING DUCT COVER, GREY	1.000	AS REQ'D	4.000	OR EQUAL
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23	1	HOFFMAN, nVent	A30P30	PANEL FOR TYPE 3R 4 4X 12 13 ENCLOSURE, FITS 30x30, WHITE, STEEL	27.000	27.000	0.750	OR EQUAL
24	1	HOFFMAN, nVent	HN0916513	HN 3R SIDE-MOUNT FILTER FANS 115V, 9.84X9.84X4.44, BLACK, PLASTIC	9.840	9.840	4.440	OR EQUAL
25	1	HOFFMAN, nVent	HN0900503	HN 3R FILTER FAN EXHAUST GRILLE, 9.84x9.84X1.33,BLACK, PLASTIC	9.840	9.840	0.970	OR EQUAL

NAMEPLATE SCHEDULE

ITEM	TAG	NAMEPLATE SIZE (W X L)	LETTER SIZE
N1	WDR-NIP-01	1" X 4"	1/4" NORMAL
N2	WDR-PS24-01	1/2" X 3"	5/32" NORMAL
N3	WDR-DIO24-01	1/2" X 3"	5/32" CONDENSED
N4	WDR-PS24-02	1/2" X 3"	5/32" NORMAL
N5	WDR-FW-01	1/2" X 3"	3/16" CONDENSED
N6	WDR-FBU-01	1/2" X 3"	3/16" CONDENSED

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BUREAU OF RECLAMATION

Jacobs

DESIGNED	A DOMINGOS
DRAWN	J BOSS
CHECKED	K BROWN
TECH. APPR.	S MENK
APPROVED	J PETERSON

VERIFY SCALE

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0 1"

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
WINDOW ROCK
TURNOUT
NIP LAYOUT

DBSJLWTP

085-N-1150
SHEET 289

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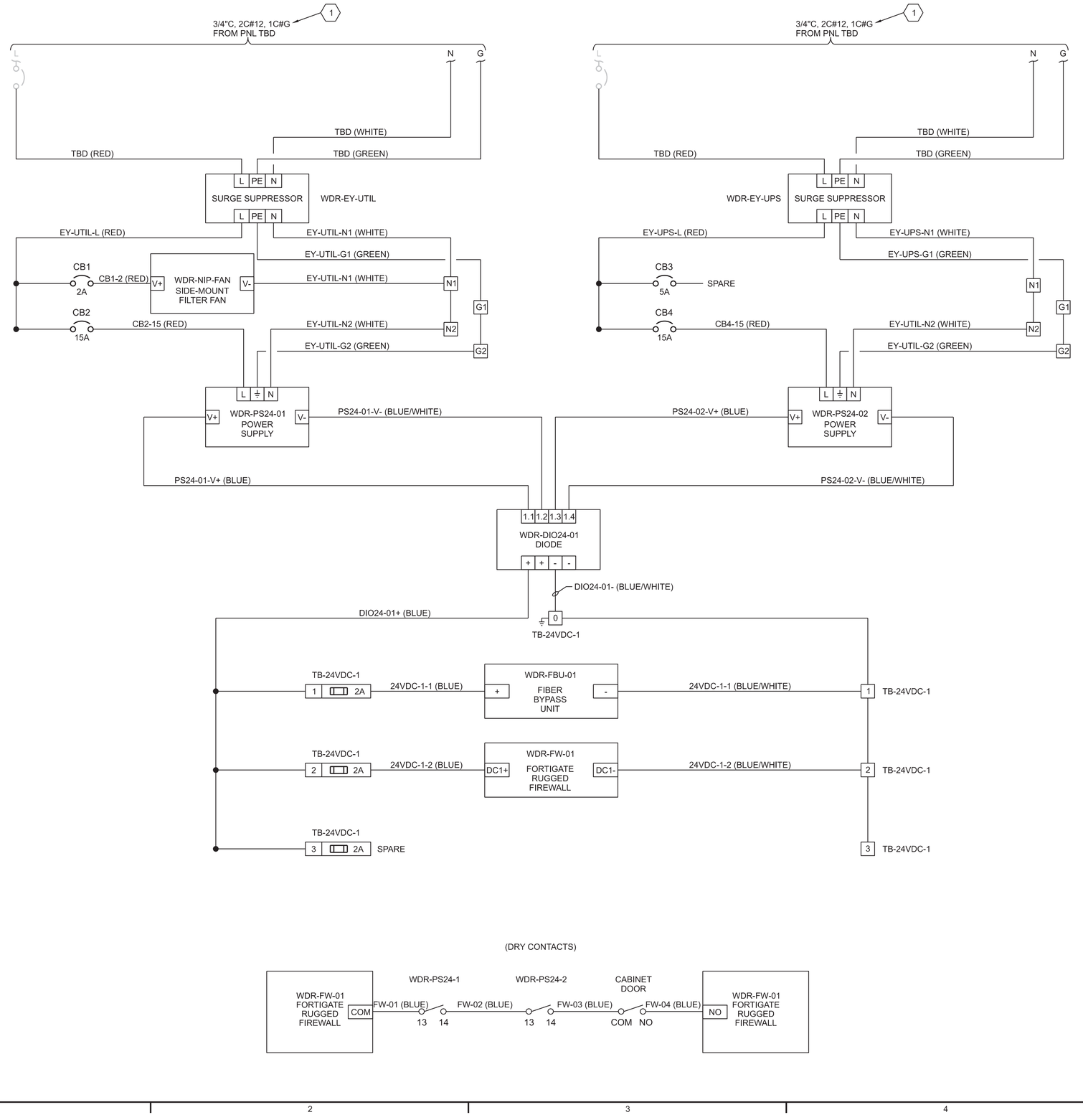
MICROSTATION CONNECT
$PWPATHS;



GENERAL SHEET NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY PER SPECIFICATION DIVISIONS 26 AND 27.

SHEET KEY NOTES

1. PROVIDE CABLES, CONDUCTORS, AND RACEWAY FROM POWER SOURCE

BUREAU OF RECLAMATION

Jacobs

A DOMINGOS — DESIGNED
J BOSS — DRAWN
K BROWN — CHECKED
S MENK — TECH. APPR.
J PETERSON — APPROVED

VERIFY SCALE
BAR IS ONE INCH ON ORIGINAL DRAWING.
0 1"

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

NAVAJO-GALLUP WATER SUPPLY PROJECT
NEW MEXICO

SAN JUAN LATERAL

WATER TREATMENT PLANT

NO.	DATE	REVISION	BY	APVD

CONTRACTOR
DRAWN

ACCEPTED

FARMINGTON, NM FEBRUARY 2026

I&C
WINDOW ROCK
TURNOUT
NIP POWER
DISTRIBUTION
DBSJLWTP

085-N-1151
SHEET 290

NIEL W. BROWN
NEW MEXICO
31004
PROFESSIONAL ENGINEER
DIGITALLY STAMPED ON 2/11/2026

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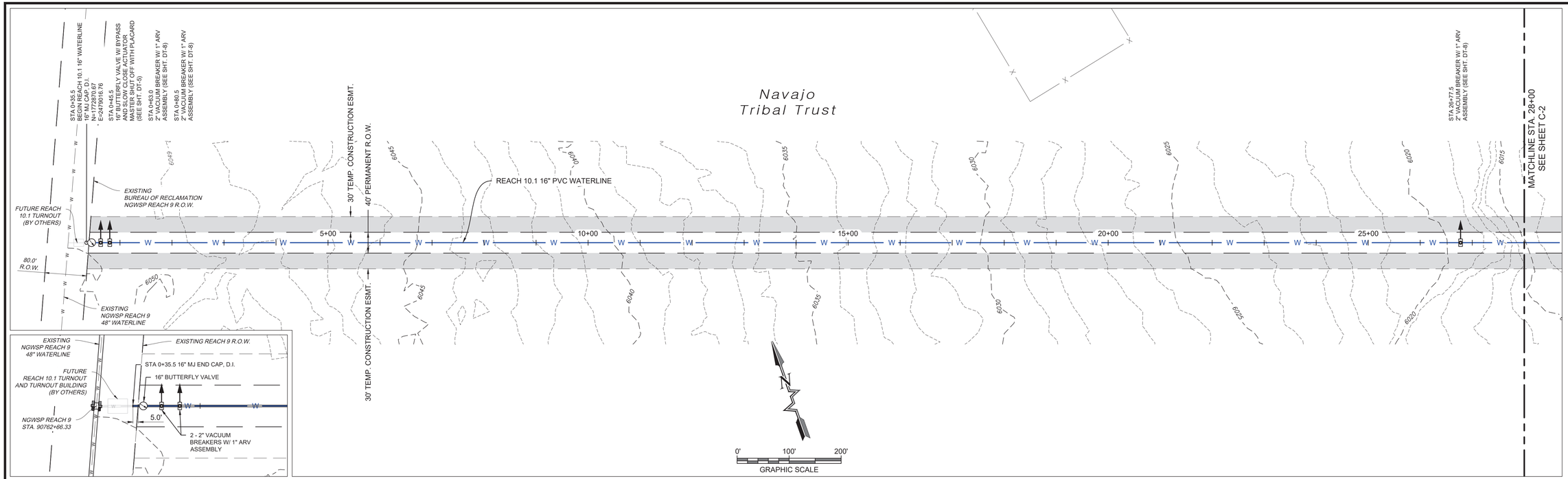
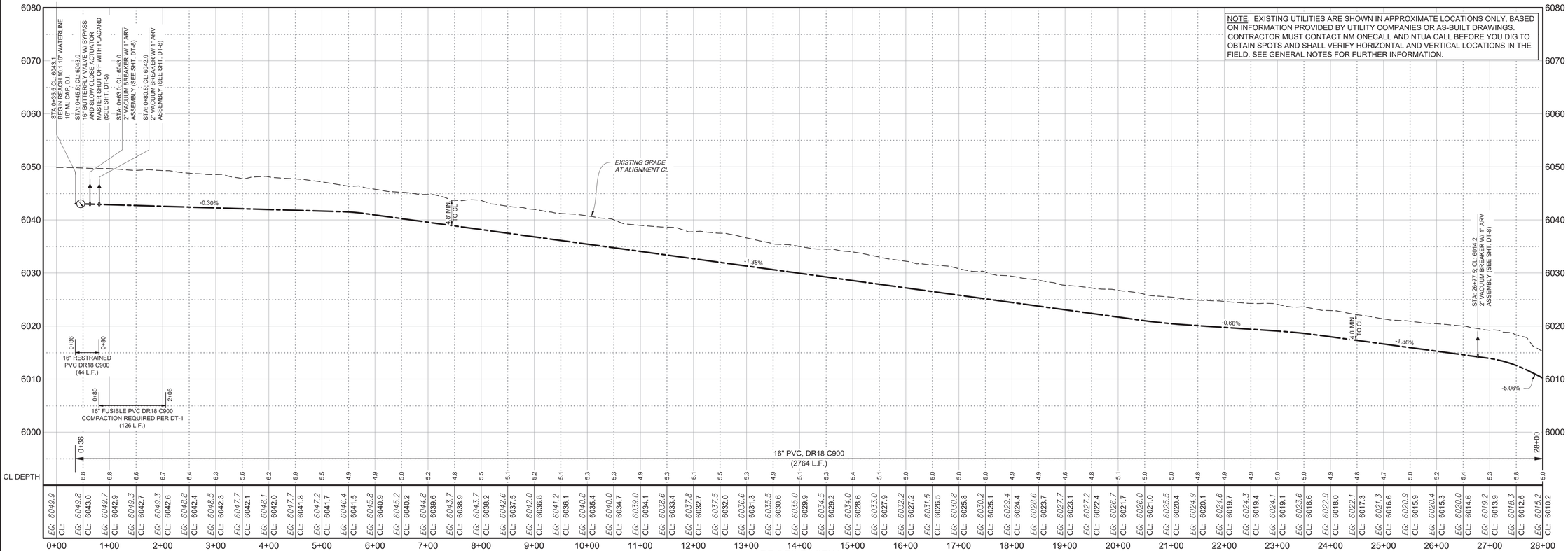


REACH 10.1 TURNOUT DETAIL
SCALE: 1'=50'

PLAN VIEW
16" WATERLINE



PROFILE VIEW
16" WATERLINE

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Revision	Date	Description	By	Chk'd

Client: The Navajo Nation
Location: Coyote Canyon, NM

PLAN AND PROFILE STA 0+35.5 To STA 28+00
NAVAJO GALLUP WATER SUPPLY PROJECT
BEACON BISTTN9 - REACH 10.1
McKINLEY COUNTY, NEW MEXICO
SECTION 21, T. 19 N., R. 17 W.

TORY TADANO
NEW MEXICO
21073
15/12/25
LICENSED PROFESSIONAL ENGINEER

Designed	Drawn	Checked
EAP	MvH/AaV	TTT

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Date: May 2025
Scale: Horiz: 1" = 200'
Vert: 1" = 20'
Project No: 6922268
Sheet: C-1

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P:\6-Beacon Bist N9 Water Line (6922268)\CAD\Civil\Reach 10.1 Plan Set\Construction Plans\Reach 10.1 - (1).dwg
5/27/2025 8:38:06 AM aav

REACH 10.1.1 TURNOUT DETAIL
SCALE: 1"=40'

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P:\6-Beacon Bist N9 Water Line (6922268)\CAD\Civil\Reach 10.1.1 Plan Set\Record\Construction Plans\Reach 10.1.1 BOR Tie In Location.dwg
5/19/2026 9:12:00 AM aav

Revision	Date	Description	By	Chkd

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TURNOUT AS BUILT
NAVAJO GALLUP WATER SUPPLY PROJECT
BEACON BISTI N9 - REACH 10.1.1
McKINLEY COUNTY, NEW MEXICO
SECTION 34, T. 18 N., R. 18 W.

AS BUILT

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Date: May 2026
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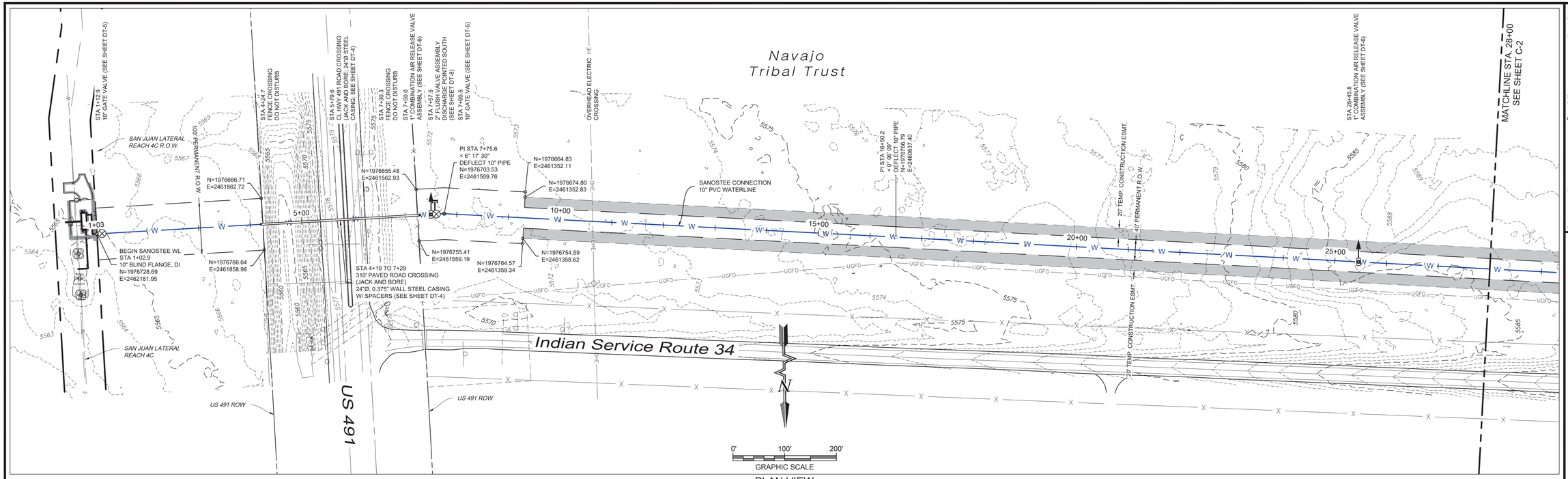
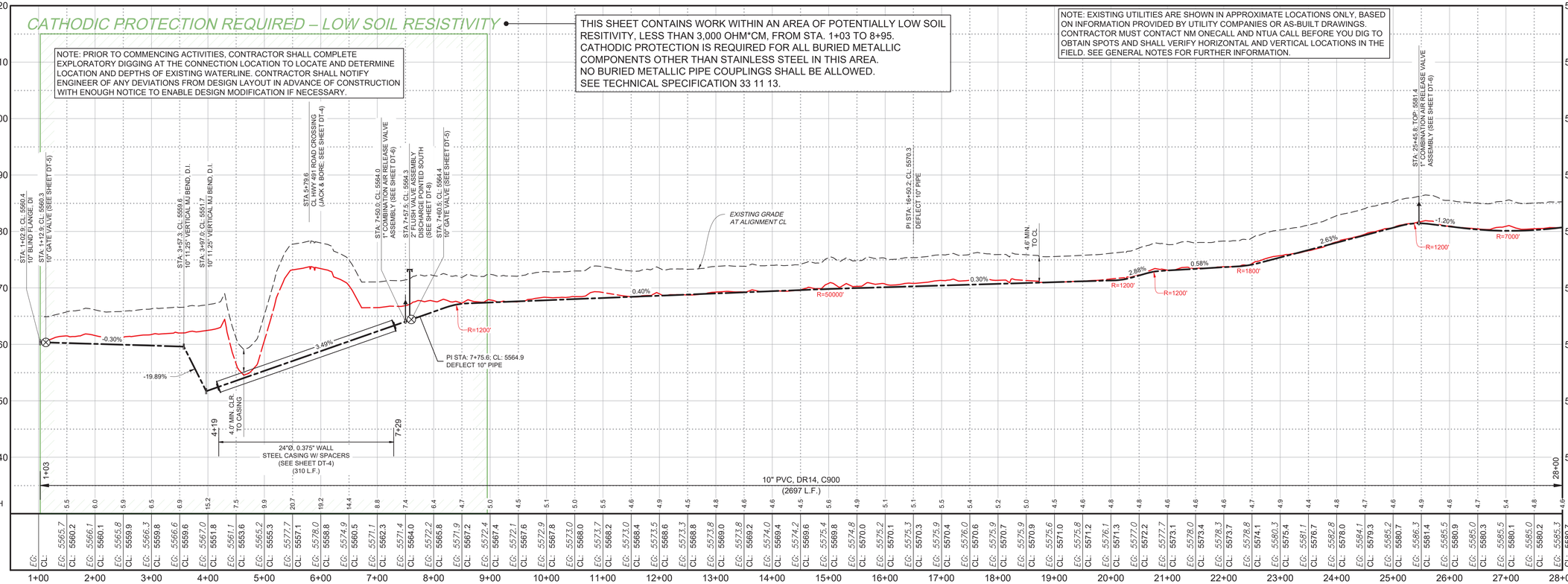
Vert: N/A
Project No: 6922268
Sheet: FIG 1



PLAN VIEW
10" WATERLINE



PROFILE VIEW
10" WATERLINE

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Revision	Date	Description	By	Chk'd

Location: Sanostee, NM
Client: The Navajo Nation

PLAN AND PROFILE STA 1+03 TO STA 28+00
SANOSTEE CHAPTER
SAN JUAN LATERAL CONNECTION
SAN JUAN COUNTY, NEW MEXICO
SECTION 2, T. 25 N., R. 18 W.

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Date: May 2026
Scale: Horiz: 1" = 200'
Vert: 1" = 20'
Project No: 6928904
Sheet: C-1

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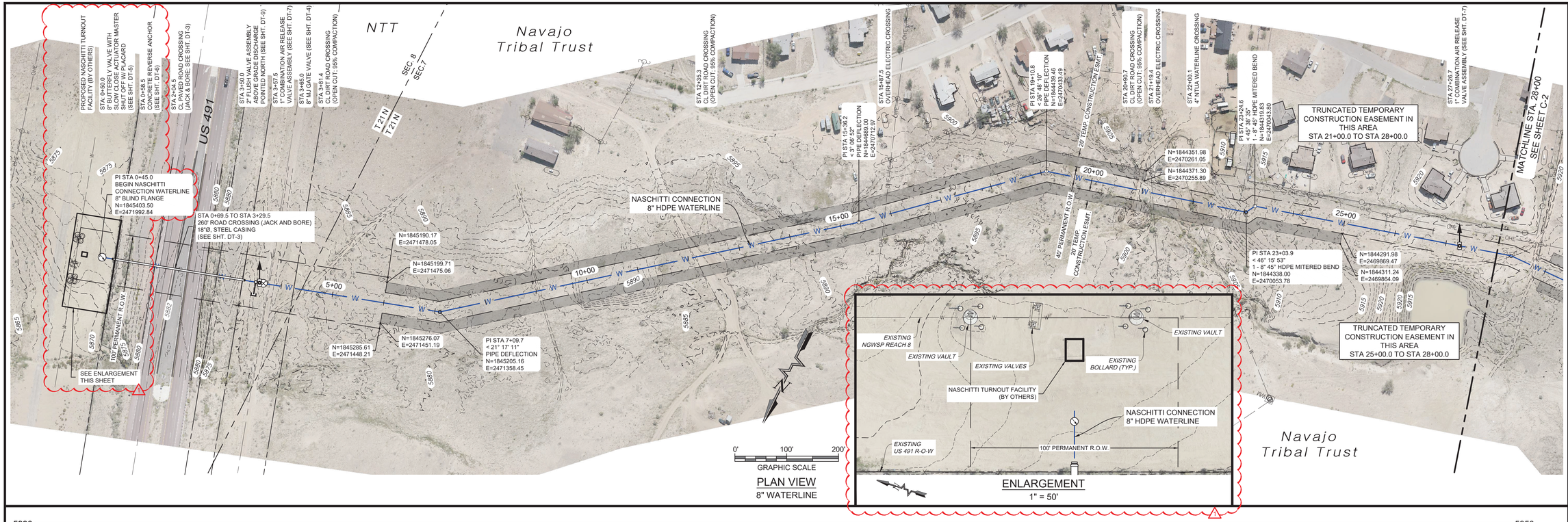
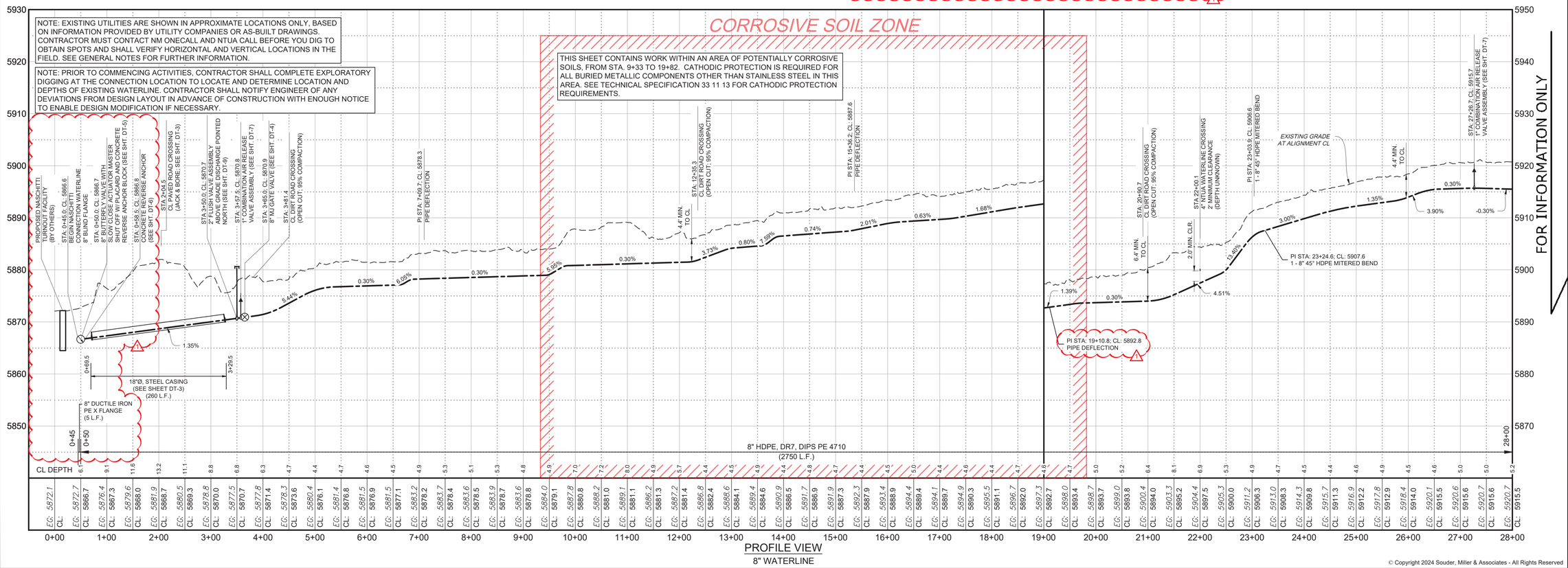
5/16/2026 12:57:14 PM aav





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Revision	Date	Description	By	Chk'd
1	5-14-2024	REVISED ALIGNMENT BEGINNING, PIPE BEND	AAV	TBT

Location: Naschitti, NM
Client: The Navajo Nation

PLAN AND PROFILE STA 0+00 To STA 28+00
NASCHITTI CHAPTER
SAN JUAN LATERAL CONNECTION
SAN JUAN COUNTY, NEW MEXICO
SECTION 7 AND 8, T. 21 N., R. 17 W.

JAY B. TSINNIJINNIE — NEW MEXICO — 25520 — LICENSED PROFESSIONAL ENGINEER — 5/14/24

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TBT	MvH/AaV	TTT

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Project No: 6928997
Sheet: C-1

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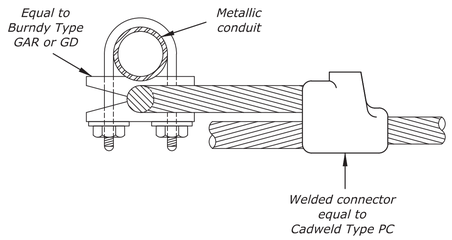
P:\6-NGWSP Naschitti Chapter Connection (6928997)\CAD\Civil\Construction Plans\Naschitti Align - (2).dwg
5/14/2024 12:23:13 PM aav

Released

FIGURE 1

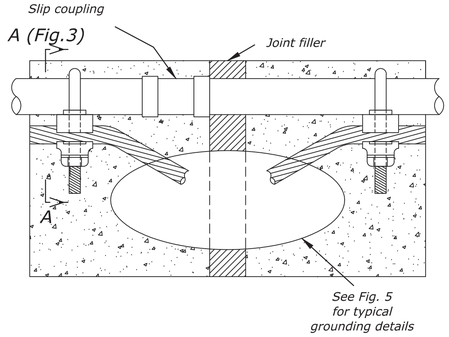
GROUNDING METAL CONDUITS



METAL CONDUIT GROUNDING TABLE			
CONDUIT SIZE	MINIMUM SIZE OF GROUND CABLE	NUMBER OF TAPS	CLAMP SIMILAR TO BURNDY TYPE
¡" to 1þ"	No.4 A.W.G.	1	GAR
2" to 3"	No.1 A.W.G.	1	GAR
Above 3"	No.1 A.W.G.	2	GD

FIGURE 2

BONDING CONDUITS AT EXPANSION JOINTS

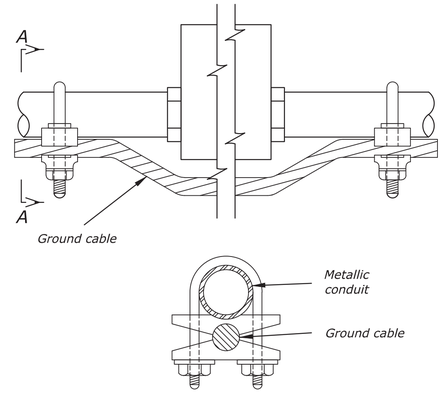


NOTE

CLAMPS SHALL BE EQUAL TO BURNDY TYPE GAR OR GD AND BARE COPPER CABLE IN ACCORDANCE WITH THE TABLE IN FIGURE 1 OR A BONDING JUMPER EQUAL TO O.Z. TYPE BJ (WITH BRONZE FITTINGS) MAY BE USED.

FIGURE 3

BONDING CONDUITS AROUND JUNCTION BOXES



SECTION A-A

NOTE

CLAMPS SHALL BE EQUAL TO BURNDY TYPE GAR OR GD AND BARE COPPER CABLE IN ACCORDANCE WITH THE TABLE IN FIGURE 1 OR A BONDING JUMPER EQUAL TO O.Z. TYPE BJ (WITH BRONZE FITTINGS) MAY BE USED.

NOTES

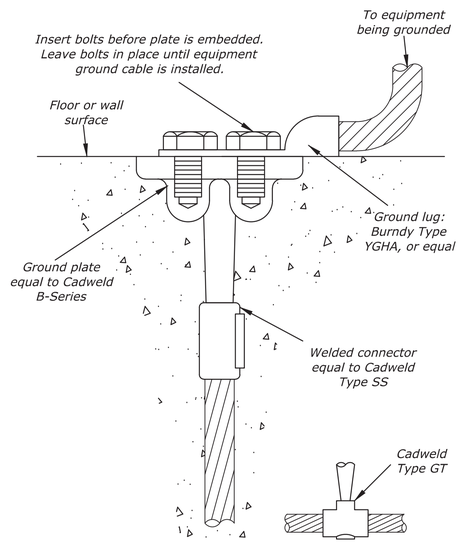
1. Ground cables for connection to equipment shall be of sufficient length to bolt to equipment when ultimately installed. Terminations as shown in Figure 4 Dwg 40-D-4334 shall be used in lieu of stubouts. Approximate locations only are shown on drawings. Exact locations to be determined in field.
2. Refer to conduit drawings or general arrangement drawings whenever available, for more exact location of equipment.
3. Metal conduits shall be grounded at both ends. If a break occurs in a run of metal conduit, the conduit shall be bonded across the break. In case the above is not feasible, the each section shall be grounded separately. Each contractor shall ground each metal conduit run, or portion therof, that he installs. Junction or pull boxes not containing protective equipment in which the conduits are installed with a locknut and bondnut shall not be considered a break when the voltage is below 240 volts to ground, but shall be so considered for higher voltages. All junction or pull boxes in runs of non-metallic conduit shall be individually grounded. All boxes containing overload protective devices shall be individually grounded.
4. Ground cable shall be No.4 A.W.G. or larger.
5. All paint, enamel, and scale shall be removed from point of contact on metal surfaces before applying ground connections.
6. All bolts, machine screws, nuts and washers used in making grounding connections shall be silicon bronze.
7. Cable shields and one conductor of multiconductor cable shall be grounded at both ends.
8. Thermoweld, Cadweld, or an approved equivalent process shall be used in making connections between cables where embedded in concrete, direct buried, or under plant superstructure.
9. Where Thermoweld, Cadweld, or an approved equivalent process is used it shall be performed with heavy duty welding equipment in accordance with manufacturer's instructions and the material shall be new fresh stock.
10. All ground connectors shall be UL listed. All below grade connectors to be in accordance with IEEE 837.

REFERENCE DRAWINGS

TYPICAL GROUNDING DETAILS (SHT 2 OF 2)	40-D-4335
SWITCHYARDS & SUBSTATIONS - GROUNDING DETAILS	40-D-4753
PLATFORM GROUNDING DETAILS	40-D-4874
CABLE TRENCH GROUNDING	40-D-6182
GROUNDING SUPPORTS ON TUBULAR STEEL STRUCTURES	40-D-6309
FENCE GROUNDING DETAILS	40-D-6376

FIGURE 4

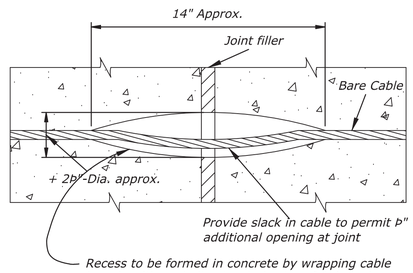
EMBEDDED GROUND CONNECTOR WITH EQUIPMENT GROUND CABLE



SIDE VIEWS

FIGURE 5

CROSSING EXPANSION AND CONTRACTION JOINTS WITH GROUNDING CABLE



Recess to be formed in concrete by wrapping cable loosely with asphalt saturated cotton fabric and given a heavy coating of asphaltum.

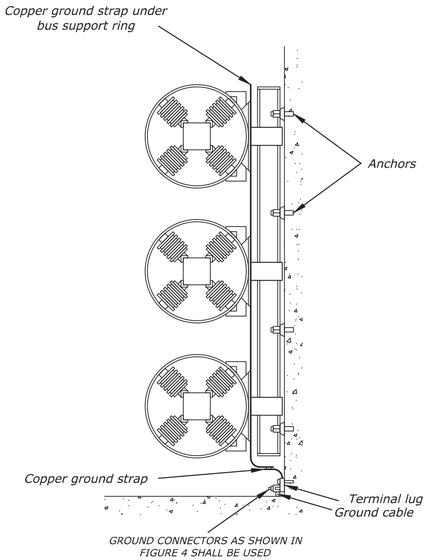
+ This dimension is approximate for No. 4/0 cable only. For larger cables, increase diameter of wrapping accordingly

NOTE

THE CROSSING OF EXPANSION JOINTS WITH EMBEDDED GROUND CABLE SHOULD BE AVOIDED WHENEVER POSSIBLE. THE EMBEDDED GROUNDING SYSTEM FOR EACH UNIT OR BLOCK SHOULD BE SELF CONTAINED.

FIGURE 6

GROUNDING OF THREE PHASE BUS STRUCTURE



GROUND CONNECTORS AS SHOWN IN FIGURE 4 SHALL BE USED

REV NO		
3	2004-07-06 D - J. R. Z.	CHANGED NOTE 10, ADDED FIG.4 CONNECTOR REF., ADDED REF. DRAWINGS, OTHER MINOR REVISIONS.
	3 - 10 - 00 D - L.G.	REDRAWN IN AUTOCAD R.14 FORMAT
	12 - 2 -85 D - G.D.O.	REVISED TO SHOW CURRENT PRODUCTS OR PRACTICES.

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IN PART DRAWING 40-D-4336, 01-14-2003

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STANDARD DESIGNS

ELECTRICAL INSTALLATION

TYPICAL GROUNDING DETAILS

DRAWN R.Y.N. - S.J.S. SUBMITTED D. CLARK

TRACED J. M. H. RECOMMENDED H. H. PLUMB

CHECKED V.N.P - R.F.H. - T.J.L APPROVED L. N. McCLELLAN
ACTING CHIEF ENGINEER

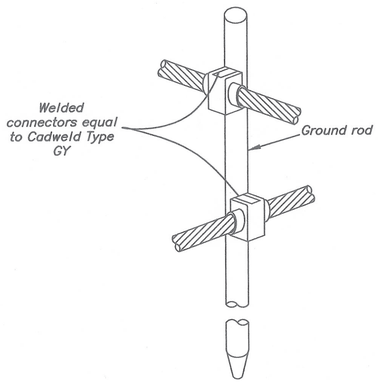
DENVER, COLORADO MAY 1, 1947
SHEET 1 OF 2

40-D-4334

CAD SYSTEM AutoCAD Rel. 16.0s
CAD FILENAME 40-D-4334.DWG
DATE AND TIME PLOTTED FEBRUARY 27, 2008 09:48
PLOTTED BY RNOI

FIGURE 1

GROUND MATS



GROUND ROD DETAILS

FIGURE 2

GROUND MATS

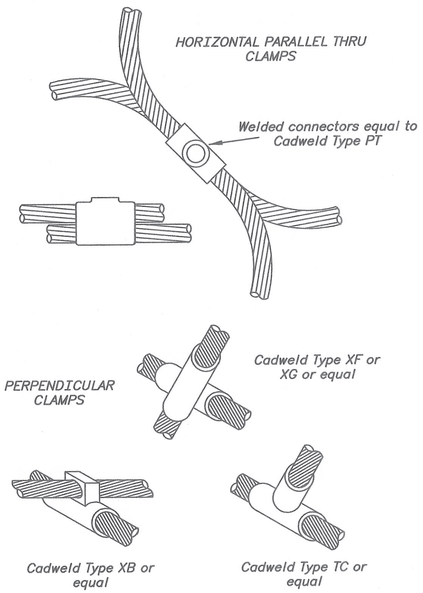
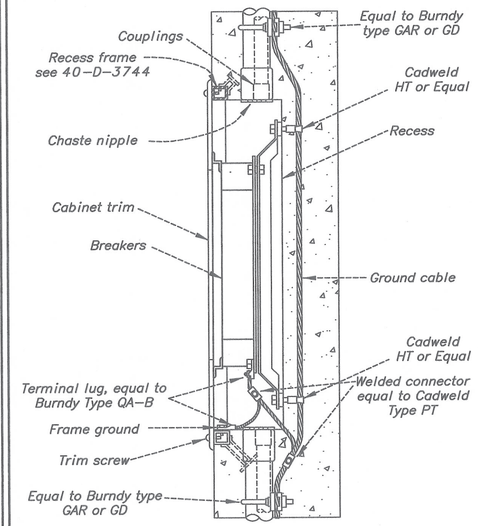


FIGURE 3

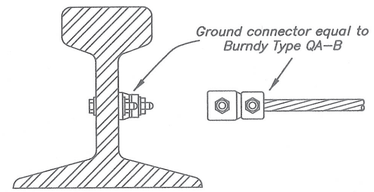
LIGHTING OR MISC POWER CABINETS



SECTIONAL ELEVATION

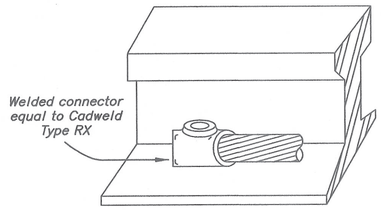
FIGURE 4

RAILS



EXPOSED INSTALLATION

CONNECTION TO RAIL



EMBEDDED INSTALLATION

CONNECTION TO RAIL

NOTES

1. See drawing 40-D-4334 for notes and list of reference drawings.

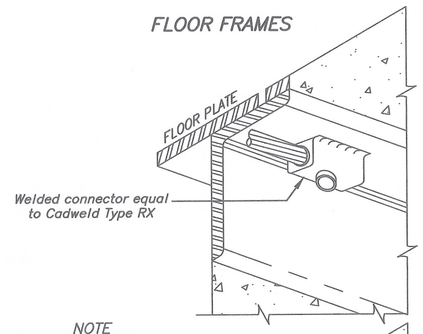
GROUND SYMBOLS

(Used on Plan Drawings)

- Indicates ground cable embedded in concrete
- Indicates ground cable concealed but not embedded in concrete
- Indicates ground cable exposed
- Indicates ground rod
- Indicates ground connection
- Indicates ground cable riser
- Indicates embedded ground plate

FIGURE 5

FLOOR FRAMES



NOTE
WELD CONNECTOR WITH ANGLE INVERTED PRIOR TO PLACING FRAME IN FINAL POSITION.

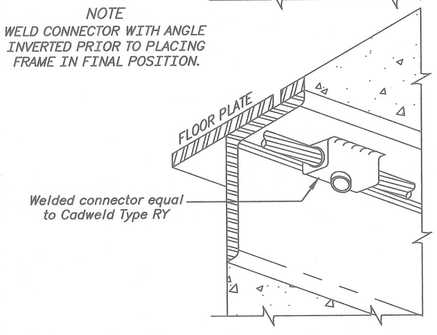
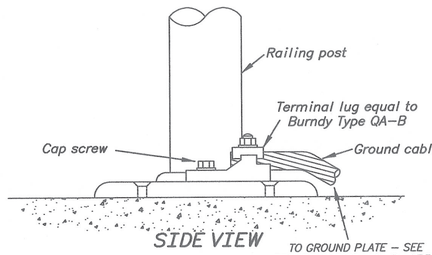
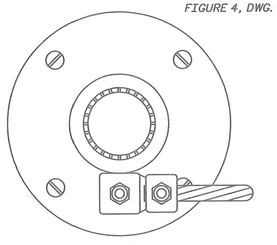


FIGURE 6

RAILING POST BASES AND HANDRAIL SUPPORTS



SIDE VIEW



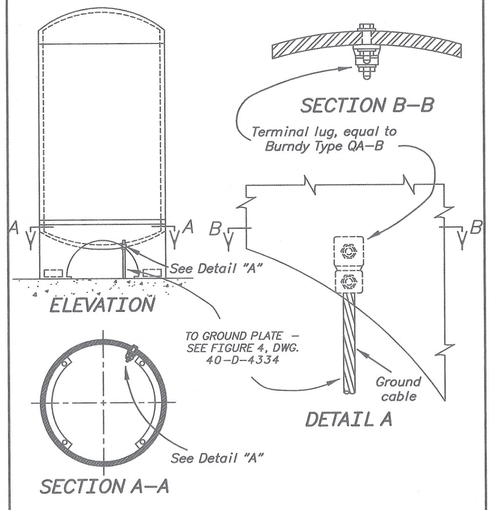
PLAN VIEW

CONNECTION TO RAILING POST BASE

(DETAILS FOR GROUNDING HANDRAIL SUPPORTS ARE SIMILAR)

FIGURE 7

OIL TANK GROUNDING DETAILS



ELEVATION

SECTION A-A

SECTION B-B

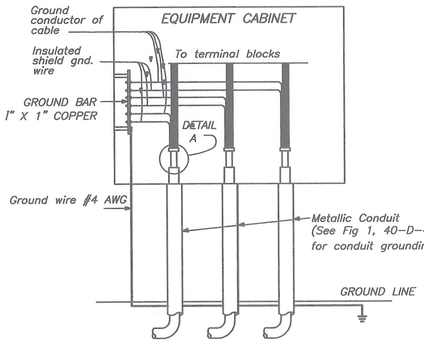
Terminal lug, equal to Burndy Type QA-B

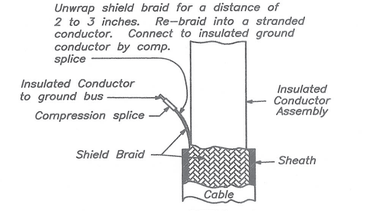
Ground cable

DETAIL A

FIGURE 8

CONTROL CABLE SHIELD





DETAIL A

LONGITUDINAL CABLE SECTION

REFERENCE DRAWINGS

ELECTRICAL INSTALLATION - TYPICAL GROUNDING DETAILS (SHT 1 OF 2) 40-D-4334

REV NO		
3	2004-07-06 D - NRS	MINOR REVISIONS; ADDED FIG.'S 2 AND 8.
	3 - 10 - 00 D - L.G.	REDRAWN IN AUTOCAD R.14 FORMAT
	12 - 2 -85 D - G.D.O.	REVISED TO SHOW CURRENT PRODUCTS OR PRACTICES.

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TYPICAL GROUNDING DETAILS

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TRACED J. M. H. RECOMMENDED H. H. PLUMB

CHECKED V.N.P. - R.F.H. - T.J.L APPROVED L. N. McCLELLAN
ACTING CHIEF ENGINEER

DENVER, COLORADO APRIL 24, 1947
SHEET 2 OF 2

40-D-4335

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DATE AND TIME PLOTTED: JUNE 30, 2004 16:31
PLOTTED BY:
DRAWING#:

MAR 01 2004

FIGURE 1

GROUNDING TO FLAT SURFACE

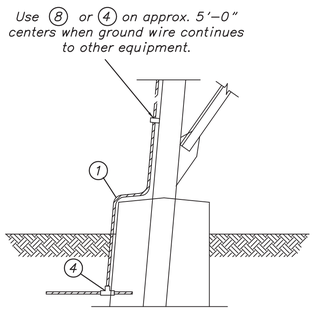
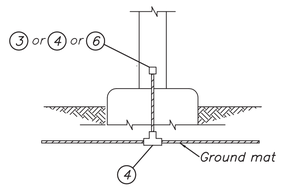
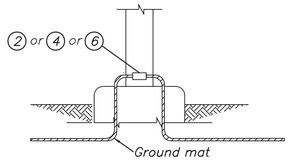


FIGURE 2

GROUNDING TO FENCE POST OR PIPE-TYPE SWITCHYARD LIGHTING POLE



OPTION A



OPTION B

FIGURE 3

GROUNDING TO CABLE TRENCH OR RACK

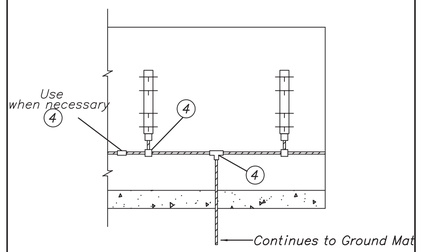
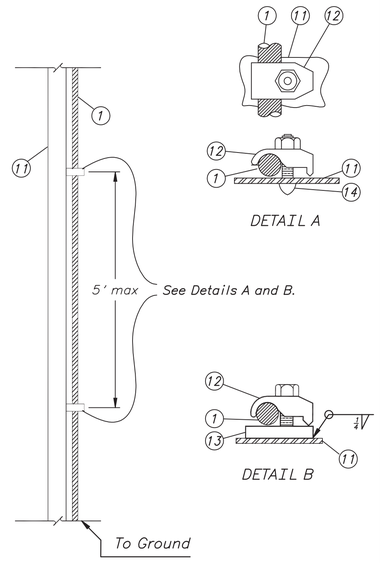


FIGURE 4

GROUNDING TO TUBULAR STEEL STRUCTURE



LIST OF MATERIALS

ITEM NO.	DESCRIPTION
1	Copper cable, bare, 4/0 AWG minimum
2	Bolted ground connector similar to Burndy type GB
3	Bolted ground clamp similar to Penn-Union type GWL
4	Exothermic welded ground connector similar to Cadweld or ThermOweld
5	Bolt: ½" silicon-bronze with nut and lock washer. Length as required for Item 8.
6	Bolted ground connector similar to Burndy type GAR
7	Compression-type ground connector
8	Cable to flat bar connector similar to Burndy QA or GB
9	Ground rod, 10'x3/4", copper-clad steel
10	Copper-clad steel stranded cable, bare, 7 No. 5
11	Tubular steel structure
12	Bolted ground connector similar to Burndy type GBM
13	Steel plate, 1-inch thick (Notes 5, 7)
14	3/8" stud, power driven (Note 6)
15	Copper-clad steel stranded cable, bare, 7 No. 9

NOTES

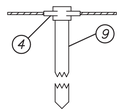
1. See drawing 40-D-4334 for general notes and list of reference drawings.
2. See drawing 40-D-4335 for typical ground system symbols.
3. All ground cables to be stubbed out approx. where indicated on plan drawings and be of sufficient length to bolt to equipment.
4. Ground all metal gates or each half of all double gates to the gate post with flexible metal braid installed across gate hinges.
5. Steel plate size to be larger than projected area of clamp.
6. Power driven stud equal to Ramset Fastening System as manufactured by Illinois Tool Works, Wood Dale, IL.
7. Plate to be welded to structure and be drilled and tapped for a ⅜" US standard bolt.
8. All grounding connections shall be as close as possible to the fence post.
9. Weave ground cable thru fencing and attach to fence with at least three (3) connectors.

REFERENCE DRAWINGS

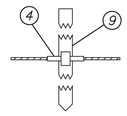
TYPICAL GROUNDING DETAILS (SHT 1 OF 2) . . 40-D-4334
TYPICAL GROUNDING DETAILS (SHT 2 OF 2) . . 40-D-4335

FIGURE 5

GROUND ROD CONNECTIONS



OPTION A



OPTION B

FIGURE 6

GROUNDING TO CABLE TUNNEL CABLE TRAYS

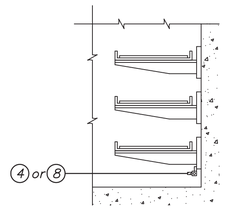
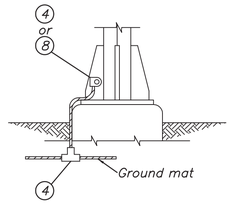
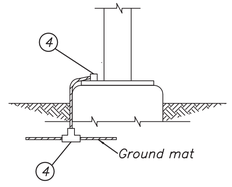


FIGURE 7

GROUNDING TO STRUCTURE BASES



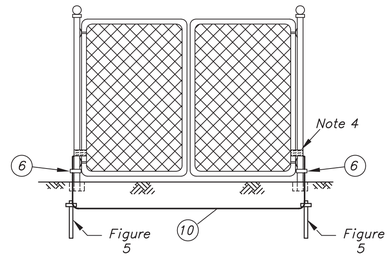
TYPICAL PIPE BUS SUPPORT OR PIPE TIE DOWN CONNECTION



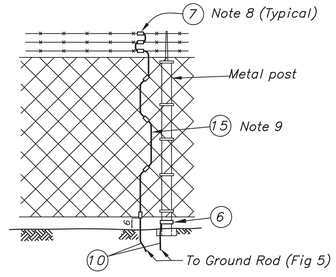
TYPICAL GROUNDING TO TUBULAR STRUCTURE

FIGURE 8

FENCE GROUNDING



GATE AND FENCE POSTS



CHAIN LINK FENCING AND POST

THIS DRAWING SUPERSEDES DRAWING 40-D-6309, 2004-09-30 IN PART

REV NO		
4	2004-09-30 D - J. R. Z.	ADDED FIGURES 4 AND 8; RENUMBERED FIGURES ACCORDINGLY; ADDED NOTES 8 & 9; ADDED ITEMS 7 & 15.
3	2004-07-06 D - J. R. Z.	ADDED ITEMS 2 & 3 TO FIG. 2 AND GROUND CABLE SIZE (ITEM 1); REPLACED 'DETAIL' VIEWS w/ 'FIGURES'; OTHER MINOR REVISIONS.
	3 - 8 - 00 D - L. G.	REDRAWN IN AUTOCAD R.14 FORMAT
	04-20-70 D - G. W. C.	ADDED DETAIL H

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UNITED STATES
DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

STANDARD DESIGNS

ELECTRICAL INSTALLATION

TYPICAL GROUNDING DETAILS FOR SWITCHYARDS AND SUBSTATIONS

DRAWN - J.L.L. - P.J.S. CHECKED - H.H. PLUMB

TRACED - L.W.G. RECOMMENDED - O.L. RICE

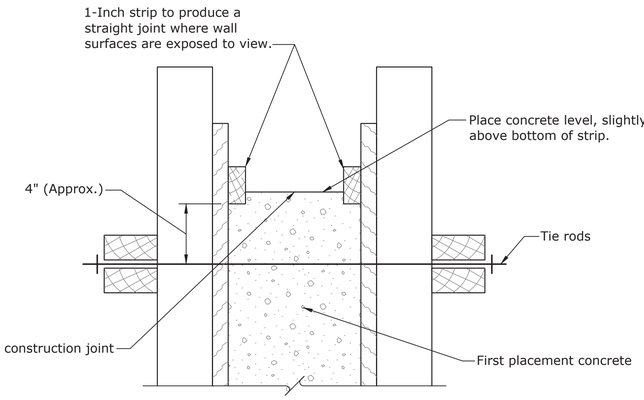
CHECKED - J.A.W. - D.C.M. - M.H.K. APPROVED - L.N. McCLELLAN
CHIEF ENGINEER

DENVER, COLORADO APRIL 5, 1949
SHEET 1 OF 1

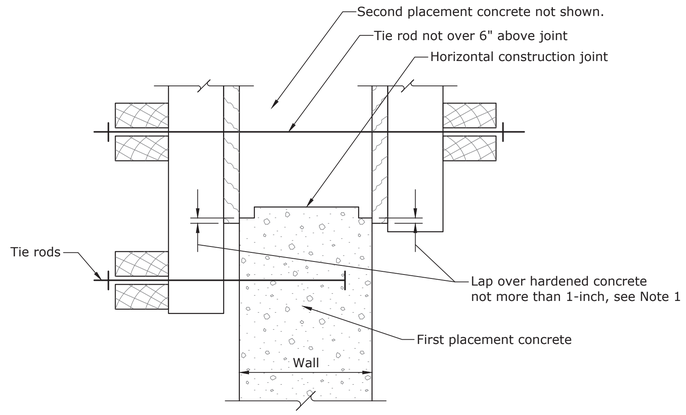
40-D-4753

CAD SYSTEM
AutoCAD Rel. 16.0s
CAD FILENAME
40-D-4753.DWG
DATE AND TIME PLOTTED
OCTOBER 14, 2004 1:30:06
PLOTTED BY
DCBWFOR

Released



FORMS IN PLACE - FIRST PLACEMENT
NOT TO SCALE



FORMS IN PLACE - SECOND PLACEMENT
NOT TO SCALE

NOTES

1. A 1-inch overlap produces a far tighter contact than a much larger overlap, and results in prevention of form spread with consequent offset at the joint.
2. For General Notes for Concrete Outline and Reinforcement, see 40-D-60003.

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BUREAU OF RECLAMATION
STANDARD DRAWINGS

STANDARD AND TYPICAL DESIGNS

CONCRETE STRUCTURES

REV NO		
3	D	2023-04-14 Bartlet, Rodney J P.E. — UPDATED DRAWING TEXT STYLE, BORDER AND TITLE BLOCK TO CURRENT DRAWING RECORDS SYSTEM. REMOVED SUBMITTED SIGNATURE. MINOR REVISIONS.
2	D	2006-08-15 J.W. — REDRAWN IN AUTOCAD.
1	D	1985-05-28 K.S.K. — REDRAWN AND DELETED V-GROOVE.

DESIGNED
C.J. MELSON
DRAWN
Dethlefsen, Scott R
CHECKED
Christopher G
TECH. APPR.
G.A. ERICKSON
APPROVED
PEER REVIEWER - CHIEF, STRUCT. AND ARCH. BRANCH

DENVER, CO 1985-05-27

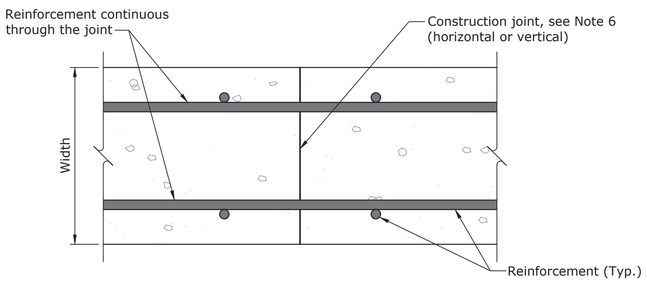
BUILDINGS AND OTHER STRUCTURES

METHOD OF BUILDING FORMS FOR SUCCESSIVE LIFTS,DETAIL

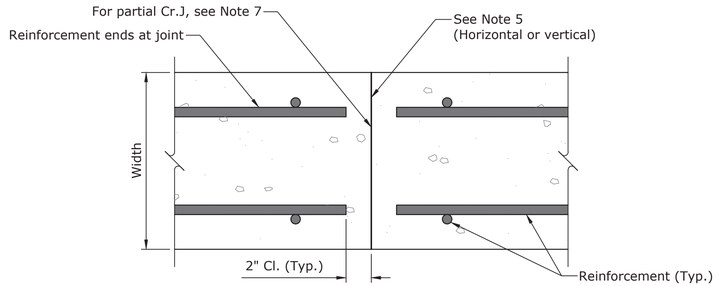
40-D-5246

SHEET 1

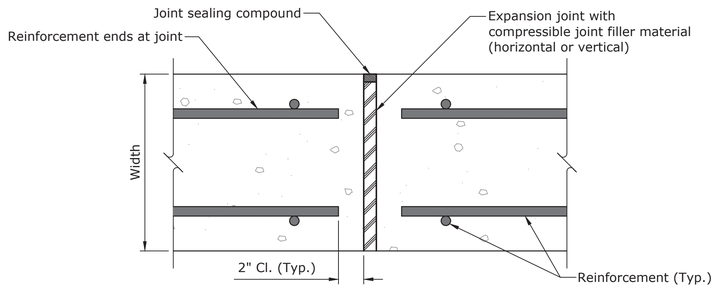
Released



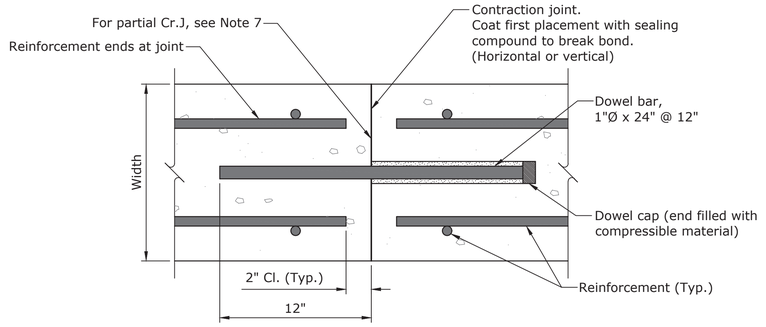
CONSTRUCTION JOINT (CJ)
NOT TO SCALE



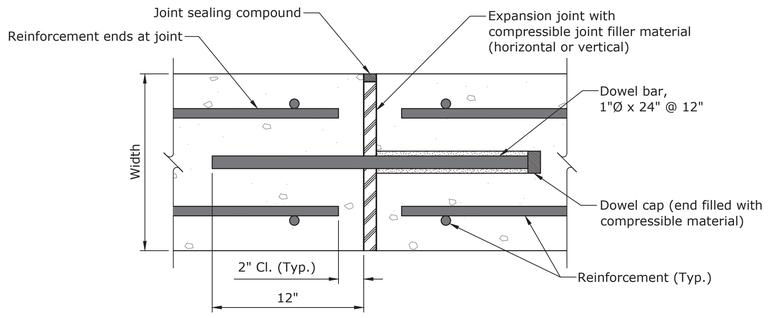
CONTRACTION JOINT (Cr.J)
NOT TO SCALE



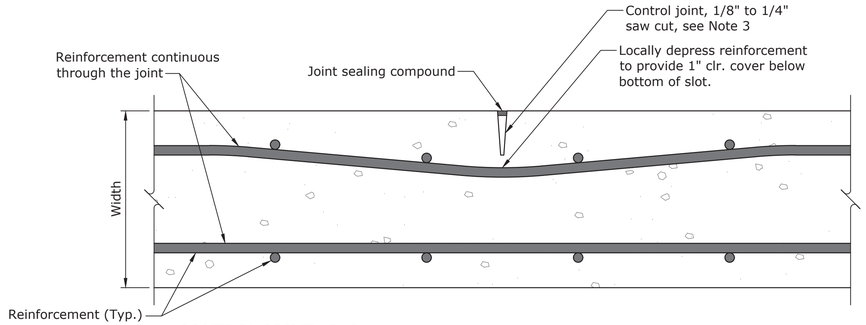
EXPANSION JOINT (EJ)
NOT TO SCALE



CONTRACTION JOINT (Cr.J) WITH DOWEL
NOT TO SCALE



EXPANSION JOINT (EJ) WITH DOWEL
NOT TO SCALE



CONTROL JOINT (Ct.J)
NOT TO SCALE

NOTES

1. Construction joint keys and waterstops required if shown on construction drawings.
2. For General Notes for Concrete Outline and Reinforcement, see 40-D-60003.
3. Minimum depth of saw cuts shall be 1/4 of the depth of concrete unless noted otherwise on construction drawing. Saw cut joints shall be completed within 24 hours following concrete placement.
4. For joint type definitions, see construction documents (i.e., see Specifications Section 03 30 00 - "Cast-in-Place Concrete").
5. Coat surface of concrete first placed, with curing compound that prevents bond, before placing concrete on other side of joint.
6. Clean, roughen, and surface dry surfaces of CJ to be covered with fresh concrete. Do not use a mortar layer on CJ.
7. If noted on construction drawings, at partial contraction joints, discontinue every other reinforcement bar perpendicular to the joint (i.e., 1/2 of perpendicular reinforcement shall cross the joint).

FORMERLY 103-D-219

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STANDARD DRAWINGS

STANDARD AND TYPICAL DESIGNS

CONCRETE STRUCTURES

REV NO		
3	D	UPDATED DRAWING TEXT STYLE, BORDER AND TITLE BLOCK TO CURRENT DRAWING RECORDS SYSTEM. ADDED JOINT TYPES. REMOVED SUBMITTED SIGNATURE.
2	D	REDRAWN IN AUTOCAD.
1	D	DELETED "CORK" AND REFERENCE TO SEALS.

2023-04-12	Barthel, Rodney J P.E.
2006-08-28	J.W.
1971-02-11	H.O.B.

DESIGNED	H.H.B.
DRAWN	C.A.R.
CHECKED	N.A.A. - H.B.
TECH. APPR.	K.B. KEENER
APPROVED	W.E. BLOMGREN
	PEER REVIEWER - ACT. CHIEF ENGINEER

DENVER, CO 1952-12-04

BUILDINGS AND OTHER STRUCTURES

JOINTS IN CONCRETE STRUCTURES

40-D-5247

SHEET 1

SAMPLE NAMEPLATES

STRAINER DRAIN

EXTENDED TYPE

STRAINER
DRAIN

COMPACT TYPE

FIRE PROTECTION
WATER SUPPLY
CAUTION
2 1/2" OUTLET FOR FIRE
DEPARTMENT USE ONLY

INSCRIPTION AND CLOSE RANGE
INFORMATIONAL LETTERING

GOTHIC LETTERING CHART

1/4" NORMAL (Avg. width .271")
ABCDEFGHIJKLMN
ABCDEFGHIJKLMNOP
CONDENSED (Avg. width .234")

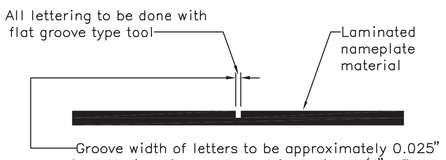
3/16" NORMAL (Avg. width .203")
ABCDEFGHIJKLMNOPQR
ABCDEFGHIJKLMNOPQRSTU
CONDENSED (Avg. width .175")

5/32" NORMAL (Avg. width .163")
ABCDEFGHIJKLMNOPQRSTUV
ABCDEFGHIJKLMNOPQRSTUVWXYZ
CONDENSED (Avg. width .140")

1/8" NORMAL (Avg. width .135")
ABCDEFGHIJKLMNOPQRSTUVWXYZ
ABCDEFGHIJKLMNOPQRSTUVWXYZ
CONDENSED (Avg. width .116")

3/32" NORMAL (Avg. width .103")
ABCDEFGHIJKLMNOPQRSTUVWXYZ
ABCDEFGHIJKLMNOPQRSTUVWXYZ
CONDENSED (Avg. width .086")

*Average widths listed include spacing between letters.



LETTERING DETAIL

SIZES AND LETTER SPACING OF NAMEPLATES

FOR DIMENSIONS SEE TABLES 1 & 2

TABLE 1
WIDTH
TABLE FOR ALL LETTERING THE SAME SIZE

GOTHIC LETTER SIZE	NO. OF LINES 1 W	1 B	2 W	2 S	2 B	3 W	3 S	3 B
1/4	1.00	.375	1.25	.15	.30	1.75	.15	.35
3/16	.75	.28	1.00	.11	.26	1.25	.11	.23
5/32	.50	.17	.75	.08	.18	1.00	.09	.18
1/8	.50	.19	.75	.08	.21	1.00	.08	.23
3/32	.50	.20	.50	.07	.12	.75	.08	.15

For other lettering sizes, more than three lines, or various size lettering on same nameplate, arrange width as follows:
Space(S)=Average of letter sizes on each side of space x 0.6.
Border(B)=Largest lettering size on name-plate x 1.2 + half of odd amounts to give total width(W) to even 1/4".

Dim. in inches

TABLE 3
STANDARD NAMEPLATES
FOR ELECTRICAL EQUIPMENT ONLY

NAMEPLATE SIZE NO.	NAMEPLATE DIMENSIONS L	W	B	LETTERING SIZE
1	1 1/8	7/16	1/16	3/32
2	1 1/2	9/16	1/8	3/32
3	1 3/4	5/8	1/8	1/8
4	2 1/4	3/4	1/8	3/16
5	3	1	7/32	3/16
6	4	1 1/4	1/4	1/4
7	6	1 3/4	1/4	1/2

Above nameplates shall be used for electrical equipment unless otherwise specified.

Dim. in inches

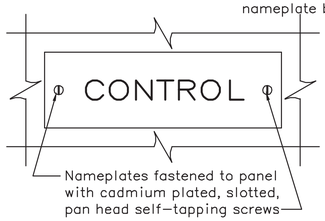
TABLE 2
LENGTH

NO.OF LETTERS AND SPACES	LETTERING SIZE ON LONGEST LINE (NORM=NORMAL; COND=CONDENSED) 1/4 NORM	1/4 COND	3/16 NORM	3/16 COND	5/32 NORM	5/32 COND	1/8 NORM	1/8 COND	3/32 NORM	3/32 COND
1										
2										
3										
4	3	3	3	3	3	3	3	3	3	3
5										
6										
7	3 1/2									
8		3 1/2								
9	4		3 1/2							
10		4		3 1/2						
11	4 1/2				3 1/2					
12	5	4 1/2	4	3 1/2		3 1/2				
13										
14	5 1/2	5	4 1/2	4		4	3 1/2	3 1/2		
15										
16	6									
17		5 1/2	5	4 1/2						
18	6 1/2				4 1/2	4				
19		6	5 1/2	5			4	3 1/2		
20	7				5					
21		6 1/2				4 1/2				
22	7 1/2		6	5 1/2			4 1/2	4		
23		7								
24	8				5					
25	8 1/2	7 1/2	6 1/2	6	5 1/2	5	5	4 1/2	4	3 1/2
26										

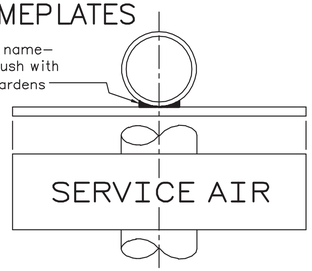
Dim. in inches

For other lettering sizes or longer nameplates, use the following formula:
L = A(N+n) + 2M (Nameplate length cut to nearest 1/2").
L – Length of nameplate.
A – Average letter width on longest line.
N – Number of letters on longest line.
n – Number of spaces on longest line
M – Margin=Average letter width plus 5/8".

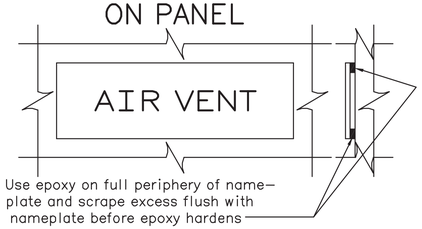
MOUNTING OF NAMEPLATES



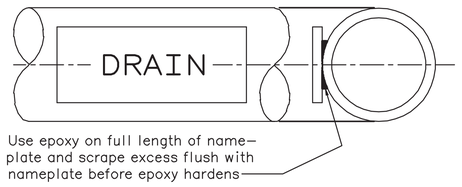
TYPICAL MOUNTING ON PANEL



TYPICAL MOUNTING ON VERTICAL PIPE

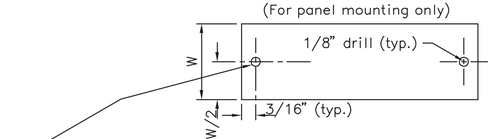


TYPICAL MOUNTING ON CONCRETE WALL

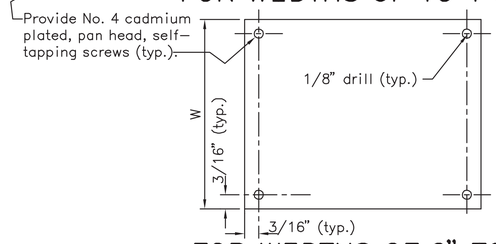


TYPICAL MOUNTING ON HORIZONTAL PIPE

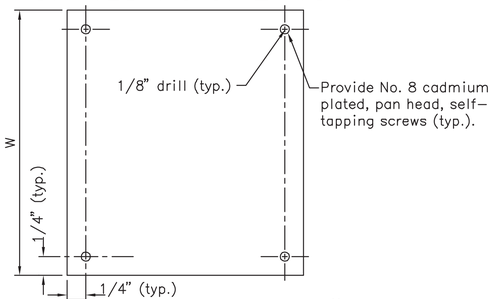
DRILLING OF NAMEPLATES
(For panel mounting only)



FOR WIDTHS UP TO 1 3/4"



FOR WIDTHS OF 2" TO 3"



FOR WIDTHS OVER 3"

NAMEPLATE MATERIAL

NOTE – All nameplate material shall be suitable for exposure to direct sunlight without detrimental fading.
TYPE A – Material shall be laminated phenolic resin sheets 1/16" or 3/32" thick (as specified) opaque black surface, white layer adjacent to surface layer, buffed or satin finish. Sheets to be cut to size with square edges. Engraving to cut through black surface to white layer. Shall conform to Military Specification MIL – P – 78A and Federal Specification L–P–509a.
TYPE B – Same as Type A except Chinese red surface or equivalent.
TYPE C – Same as Type A except 1/8", 3/16", or 1/4" thick (as specified).
TYPE D – Same as Type A except Chinese red surface or equivalent and 1/8", 3/16", or 1/4" thick (as specified).

NAMEPLATE USES

TYPE A – For panel mounting with self-tapping screws. For concrete wall or other flat mounting to identify valves, equipment, etc. Mount with epoxy cement-indoors or outdoors.
TYPE B – Red nameplates are restricted to uses such as fire protection equipment, warning inscriptions, and emergency devices. Methods of mounting shall be the same as for Type A.
TYPE C – For mounting on non-flat surface with epoxy cement.
TYPE D – Restricted to uses such as fire protection, equipment warning inscriptions, and emergency devices. For mounting on non-flat surfaces with epoxy cement.

LETTERING

Use normal or condensed letter types shown on the Gothic Lettering Chart. The normal size of lettering shall be not less than 3/16" high (1/4" preferred), except that 5/32" and smaller may be used for information and instructions that may be read at close range. Larger size lettering may be specified for out-of-doors installations or where necessary.

NOTES

This drawing supersedes Drawings No. 40–D–5173, 106–D–341, and 106–D–342.
This drawing and Drawings No. 40–D–6124, –6130, and –6187 supersede Drawing No. 40–D–2567.
Table 3 for ELECTRICAL EQUIPMENT ONLY, Tables 1 and 2 for all other equipment.

10 – 13 – 00 D – L.G.	Redrawn in AutoCAD
12 – 11 – 79 D – L.H.	Redrawn
11 – 6 – 70 D – D.A.W.	Add Table 3 and Note

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UNITED STATES
DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

STANDARD DESIGNS

STANDARD NAMEPLATES

DESIGNED D.A.W. TECHNICAL APPROVAL C.G.Bates
DRAWN J.M.F. SUBMITTED Jack W. Hilf
CHECKED D.M.B.–S.K.K.–F.M.J. APPROVED B.P.Bellport
CHIEF ENGINEER

CADD SYSTEM	CADD FILENAME	DATE AND TIME PLOTTED
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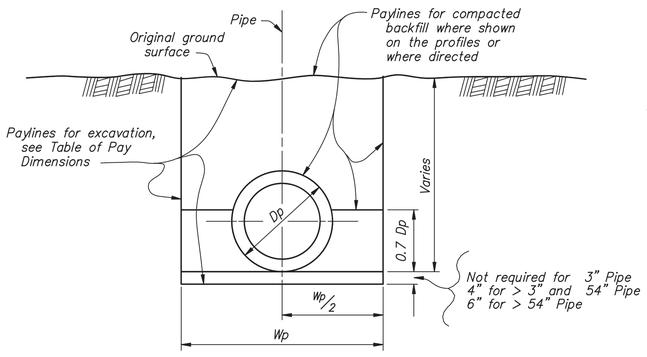
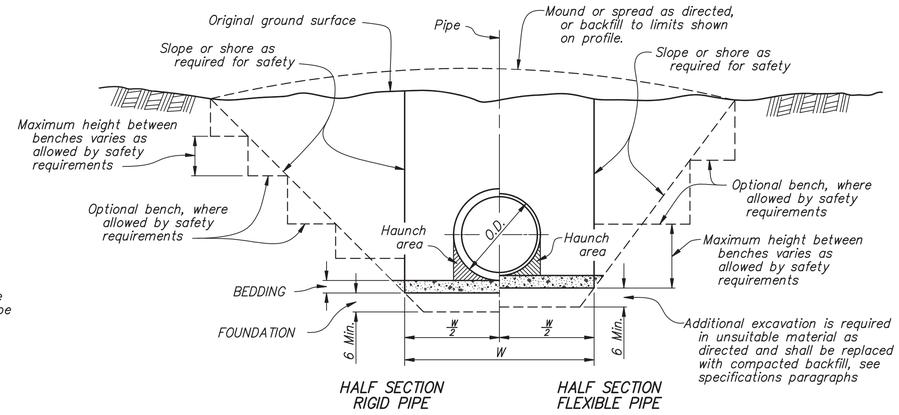
DENVER, COLORADO	MAY 1970	40–D–6234

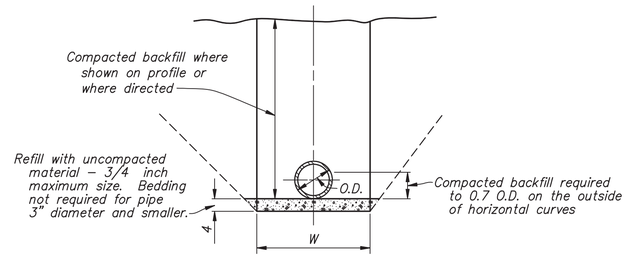
Released



TRENCH FOR PAYLINES ONLY
ALL TYPES OF PIPE



TYPICAL TRENCH DETAILS



PIPE 10 INCH DIAMETER AND SMALLER

PVC
STEEL
DUCTILE IRON

TABLE OF PAY DIMENSIONS

Pipe I.D. (Inches)	Dp (Inches)	Wp (Feet)
6 and less	I.D. + 2	2.0
Over 6 thru 18	I.D. + 4	$\frac{1}{12}$ (I.D. + 24)
Over 18 thru 24	I.D. + 4	$\frac{1}{12}$ (I.D. + 40)
Over 24	1.167 I.D.	$\frac{1}{12}$ (1.167 I.D. + 36)

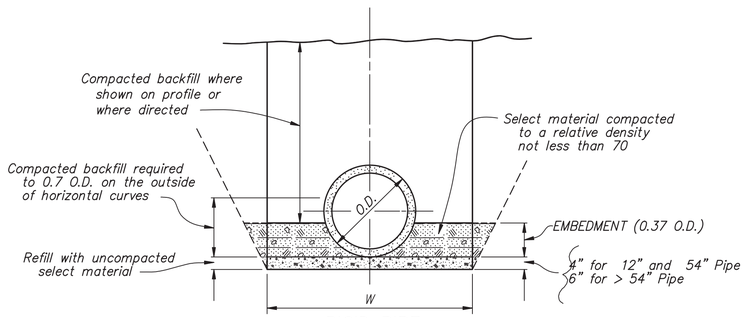
MINIMUM INSTALLATION WIDTH

PIPE I.D. (INCHES)	W (FEET)
6 and less	2.0
Over 6 thru 18	$\frac{1}{12}$ (O.D. + 20)
Over 18	$\frac{1}{12}$ (O.D. + 36)

GRADATION LIMITS FOR SELECT MATERIAL

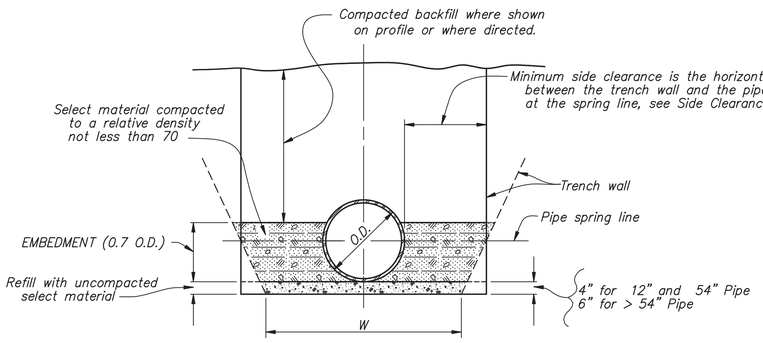
SIZE *	PERCENT BY WEIGHT
Passing No. 200 sieve	5 or less
Passing No. 50 sieve	25 or less

* Maximum size shall not exceed 3/4 inch.



RIGID PIPE

DUCTILE IRON_____12" – 20"
REINFORCED CONCRETE_____12" AND LARGER
REINFORCED CONCRETE CYLINDER_____48" AND LARGER



FLEXIBLE PIPE

PVC_____12" – 36"
PRETENSIONED CONCRETE CYLINDER_____12" – 66"
STEEL_____12" AND LARGER
FIBERGLASS_____12" AND LARGER
DUCTILE IRON_____24" AND LARGER

SIDE CLEARANCE TABLE

TRENCH TYPE	MINIMUM SIDE CLEARANCE [W1](INCHES)
1	10 INCHES FOR 12" THRU 18" I.D. 18 INCHES FOR OVER 18" I.D.
2	ONE O.D.
3	TWO O.D.

For location of Trench Types, see Specifications.

NOTES

Dp and Wp are used for calculating pay quantities for all pipe and trench types. Calculations are based on vertical walls.
Paylines for backfill will be the paylines for excavation, except the volume of the pipe, based on the diameter Dp will be deducted, and except where the depth of backfill is limited as shown on profiles.
W is minimum width of excavation in feet at bottom of bedding. The minimum side clearance for flexible pipe may require a wider trench bottom than dimension W.
Pipe diameters shown are the nominal inside diameter (I.D.) of the pipe in inches unless otherwise indicated.
O.D. is outside diameter in inches of the pipe actually installed.
Where pipe slope exceeds 0.3, see specifications paragraphs for backfill in pipe trenches.
Sloping, shoring, and benching shall be in accordance with Reclamation Construction Safety Standards.
Other installation methods with comparable pipe designs may be submitted for approval.

REV. NO		
3	2006-12-11 D – R.S.D.	REDRAWN IN AUTOCAD.
	11/27/91 D– L.A.K.	REMOVED NOTES ABOUT EQUAL LIFTS UNDER "RIGID PIPE" AND "FLEXIBLE PIPE".
	10/17/91 D– L.A.K.	REDRAWN WITH MINOR REVISIONS.

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UNITED STATES
DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION

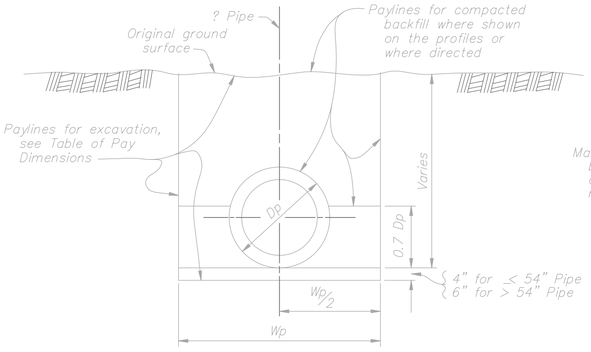
STANDARD DESIGNS

PRESSURE PIPE TRENCH INSTALLATION SELECT MATERIAL

DESIGNED Richard P. Fuerst — TECHNICAL APPROVAL Leo A. Kinney, Jr.
DRAWN Bob Scully — SUBMITTED Douglas H. Wegener
CHECKED Leo A. Kinney, Jr. — APPROVED W. L. Long
CHIEF, WATER CONVEYANCE BRANCH

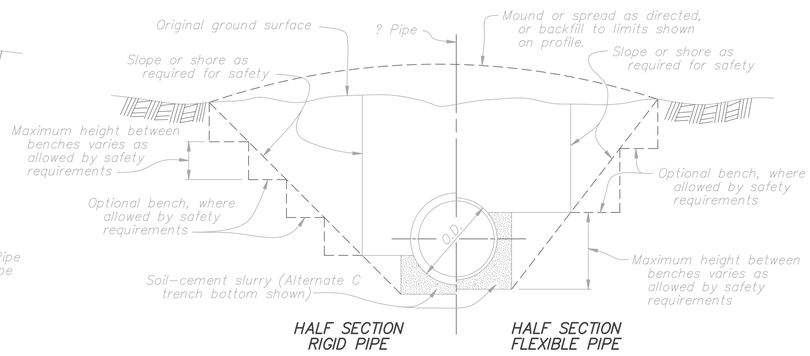
CAD SYSTEM: ARMSOFT R186.0s (LMS Tech) | CADD FILENAME: 40-D-6551.DWG | CADD PLT FILENAME: 40-D-6551.PLT | DATE : DEC. 11, 2006
DENVER, COLORADO — APRIL 1, 1991 — 40-D-6551

DATE AND TIME PLOTTED
DECEMBER 11, 2006 11:27
PLOTTED BY
RDUNBAR

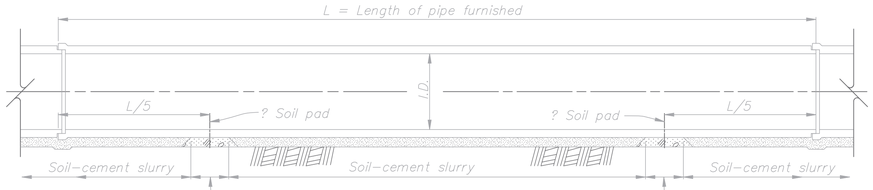


TRENCH FOR PAYLINES ONLY

ALL TYPES OF PIPE



TYPICAL TRENCH DETAILS



INSTALLATION WITH SOIL PADS

TYPICAL FOR ALL PIPE TYPES

SOIL PAD TABLE

PIPE I.D.	X
12" – 27"	.05L
30" – 66"	.10L
69" – 114"	.15L

NOTE: Soil pads allow observation of the soil-cement slurry flowing from one side of the pipe to the other.

TABLE OF PAY DIMENSIONS

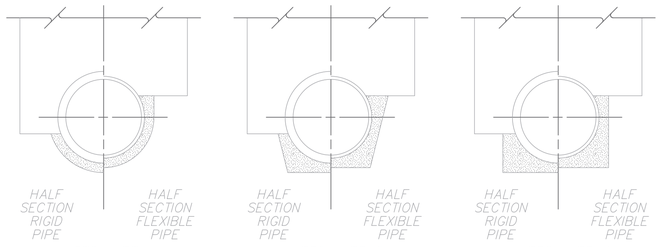
Pipe I.D. (Inches)	Dp (Inches)	Wp (Feet)
12 thru 18	I.D. + 4	$\frac{1}{12}$ (I.D. + 24)
Over 18 thru 24	I.D. + 4	$\frac{1}{12}$ (I.D. + 40)
Over 24	1.167 I.D.	$\frac{1}{12}$ (1.167 I.D. + 36)

GRADATION LIMITS FOR SELECT MATERIAL

SIZE *	PERCENT BY WEIGHT
Passing No. 200 sieve	12 or less **
Passing No. 50 sieve	25 or less

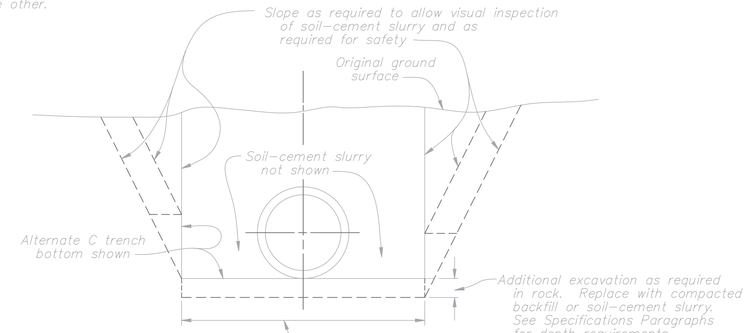
* Maximum particle size depends on pipe type and coating supplied, see Specifications paragraphs.

** When saturation and internal vibration method is used for compaction, percent by weight shall be 5 or less.



ALTERNATE A ALTERNATE B ALTERNATE C

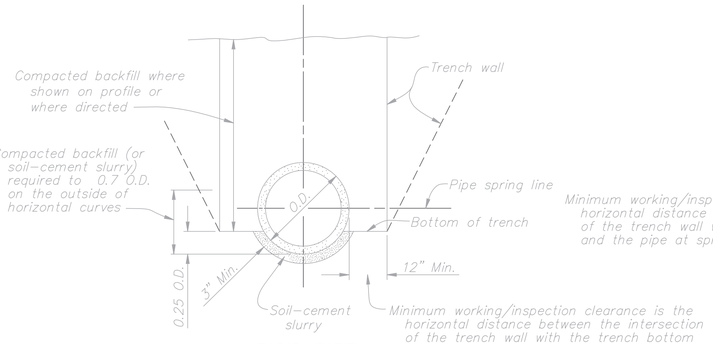
ACCEPTABLE TRENCH BOTTOMS FOR SOIL-CEMENT SLURRY



NOTE: Trench dimensions and slopes shall be submitted for approval.

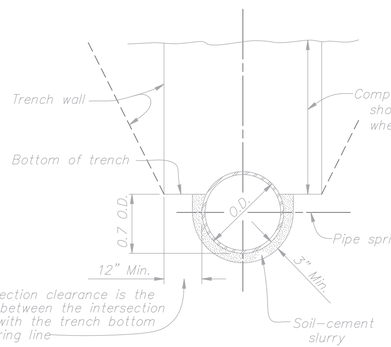
OPTIONAL INSTALLATION ON TRENCH BOTTOM

REQUIRED FOR PIPE > 114"

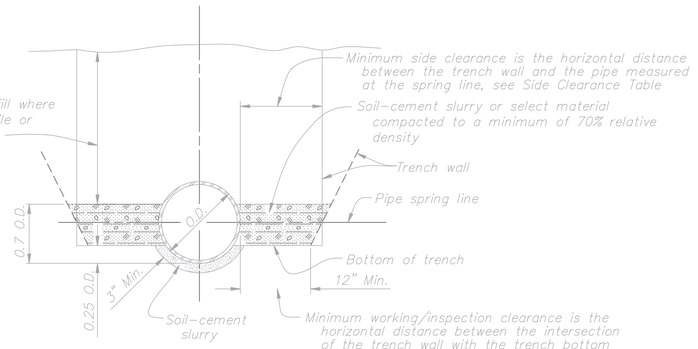


RIGID PIPE

DUCTILE IRON	12" – 20"
REINFORCED CONCRETE	12" AND LARGER
REINFORCED CONCRETE CYLINDER	48" AND LARGER



TRENCH TYPE 1



TRENCH TYPE 2

MAY BE USED FOR TRENCH TYPE 1

FLEXIBLE PIPE

PVC	12" – 36"
PRETENSIONED CONCRETE CYLINDER	12" – 66"
STEEL	12" AND LARGER
DUCTILE IRON	24" AND LARGER
FIBERGLASS	12" AND LARGER

SIDE CLEARANCE TABLE

TRENCH TYPE	MINIMUM SIDE CLEARANCE (INCHES)
1	12
2	One O.D.

For location of Trench Types, see Specifications.

NOTES

Dp and Wp are used for calculating pay quantities for all pipe and trench types. Calculations are based on vertical walls.

Paylines for backfill will be the paylines for excavation, except the volume of the pipe, based on the diameter Dp will be deducted, and except where the depth of backfill is limited as shown on profiles.

Pipe diameters shown are the nominal inside diameter (I.D.) of the pipe in inches unless otherwise indicated.

O.D. is the outside diameter, in inches, of the pipe actually installed.

Sloping, shoring, and benching shall be in accordance with Reclamation Construction Safety Standards.

Other installation methods with comparable pipe designs may be submitted for approval.

2	2006-12-11 D - R.S.D.	REDRAWN IN AUTOCAD.
	2-13-92 D - R.P.F.	CHANGE BOTTOM WIDTH NOTE IN "OPTIONAL INSTALLATION ON TRENCH BOTTOM".

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UNITED STATES
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BUREAU OF RECLAMATION

STANDARD DESIGNS

PRESSURE PIPE TRENCH INSTALLATION

SOIL- CEMENT SLURRY

12-INCH OR LARGER PIPE

DESIGNED	R. P. FUERST	TECHNICAL APPROVAL	L. A. KINNEY JR.
DRAWN	B. SCULLY	SUBMITTED	D. H. WEGENER
CHECKED	L. A. KINNEY JR.	APPROVED	K. L. LONG, CHIEF, WATER CONVEYANCE BRANCH

CADD SYSTEM AutoCAD Rel. 16.0s	CADD FILENAME 40-D-6552.DWG
DENVER, COLORADO	APRIL 1,1997

40-D-6552

FILE PATH: H:\CASSO SHARE\DB012\Drawing Files\Standard Drawings\dwgs_DONE\40-D\
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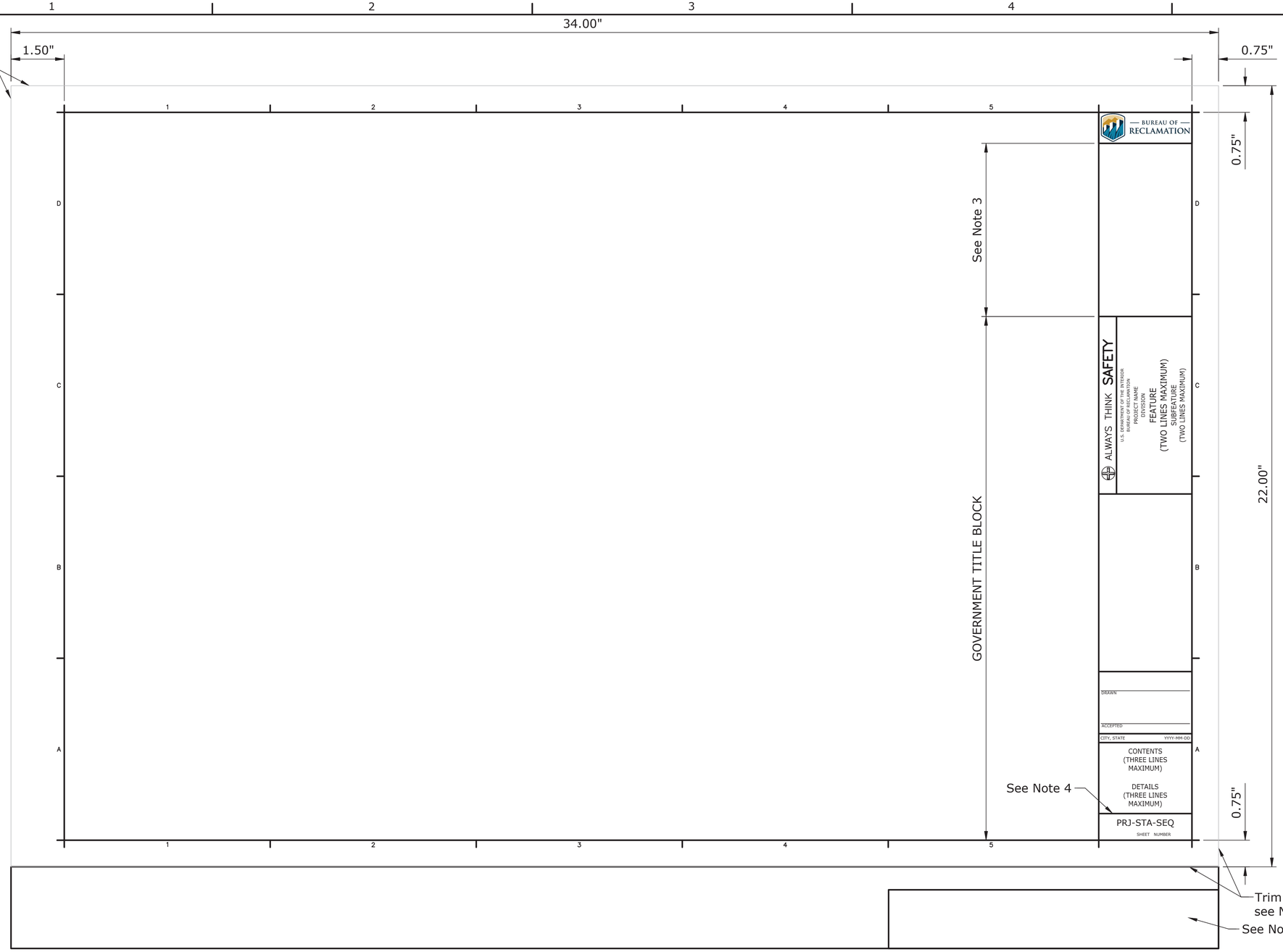
DECEMBER 11, 2006 13:14

Released



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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
YAKIMA PROJECT
KENNEWICK DIVISION

CHANDLER PUMPING PLANT

SWITCHYARD - STEEL STRUCTURES

SAMPLE TITLE

DRAWING BORDER

MANUFACTURER OR CONTRACTOR
DRAWN
ACCEPTED
TITLE

SAMPLE SIGNATURE LINES

REFERENCE

1. Information Management Handbook (IMH - 2020), Chapter 2, Section 9.A.4 - "Sheet Border" and Figures G2, G3, and G4.

NOTES

1. Government will provide contractor with an electronic AutoCAD template to be used by manufacturer for submittal of manufacturer's/contractors drawing in Government format.
2. The manufacturer's title block and drawing revisions as a result of the submittal process, may be located anywhere outside of the trim lines. The location shown on this drawing is the Government's preferred area.
3. The manufacturer shall show the Government solicitation/specification number, the manufacturer's name, drawing and sheet number in this area.
4. Government drawing number(s) will be issued at the end of the submittal process and drawing will be entered into the Government's electronic drawing system by the Government.
5. Trim lines shall be shown on all submitted drawings.

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U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
STANDARD DRAWINGS

STANDARD AND TYPICAL DESIGNS

GOVERNMENT DRAWING FORMAT

REV NO		
3	2023-04-20 Barthel, Rodney J P.E.	D UPDATED DRAWING TO CURRENT DRAWING RECORDS SYSTEM. MINOR REVISIONS.
2	2015-03-25 Barthel, Rodney J P.E.	D UPDATED TITLE BLOCK AND SIGNATURE LINES.
1	2014-12-17 E. Mendin	D UPDATED TITLE BLOCK AND SIGNATURE LINES.

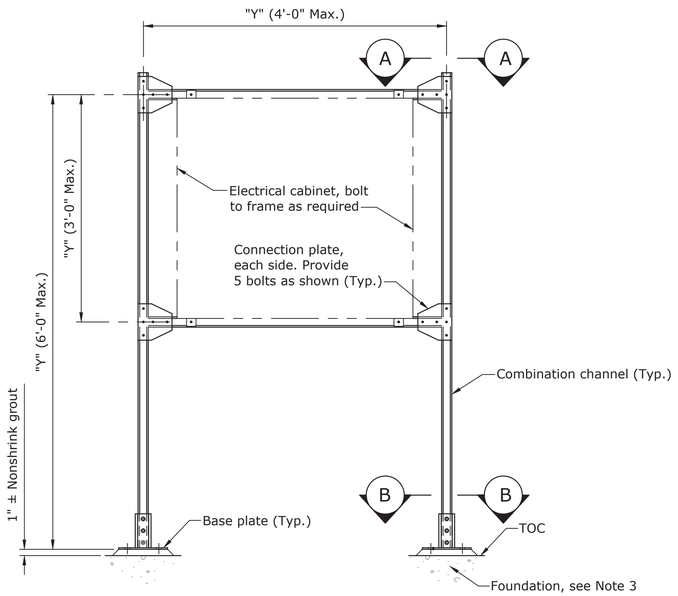
S. NITTA
DESIGNED
S. NITTA
DRAWN
D. CRAWFORD
CHECKED
R. BARTHEL, P.E.
TECH. APPR.
R.W. LAFOND, P.E.
APPROVED
ADMIN APPROVAL -

DENVER, CO

DRAWING BORDER, SAMPLE TITLE BLOCK, AND SIGNATURE LINES

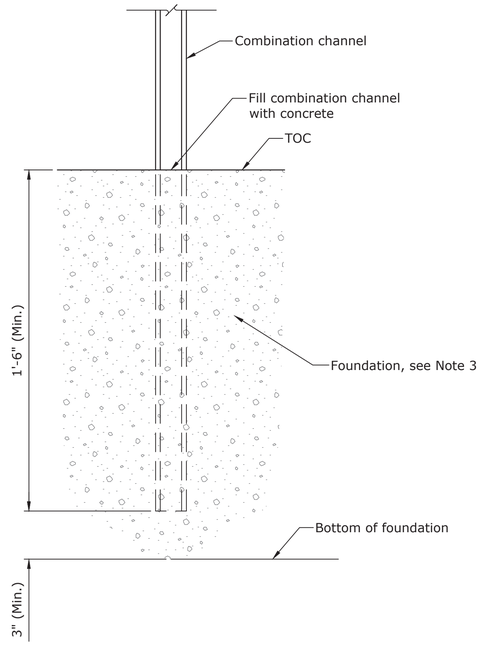
40-D-7102

SHEET 1

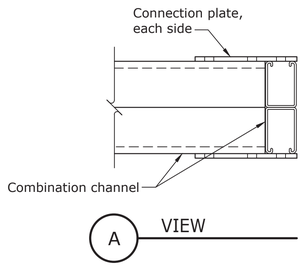


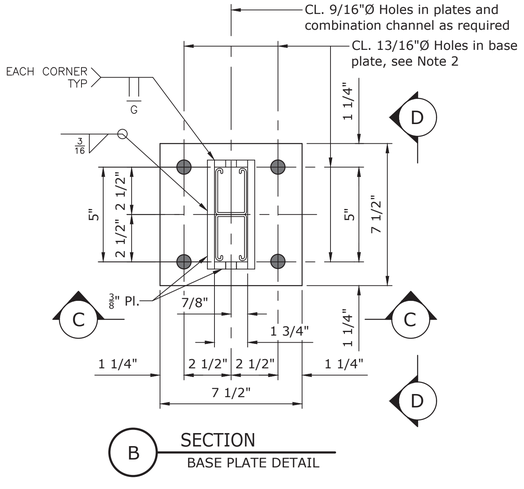
ELECTRICAL EQUIPMENT SUPPORT, TYPE I

(Est. Wt. 135 lbs.)



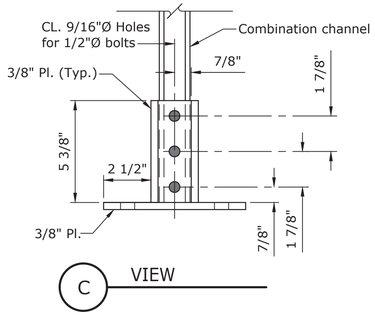
DETAIL - ALTERNATE ANCHORAGE

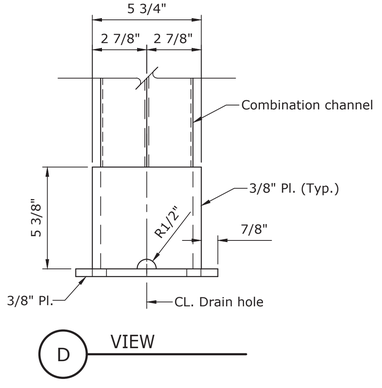




B SECTION

BASE PLATE DETAIL





DESIGN CRITERIA

Electrical equipment support is designed to withstand a three-second gust basic wind speed of 90 mph in accordance with ASCE 7-05. It is assumed that no electrical cabinets will be installed in lower portion of frame.

GENERAL NOTES

ASCE denotes American Society of Civil Engineers

ASTM denotes American Society for Testing and Materials

AWS denotes American Welding Society

COR denotes Contracting Officer's Representative

Max. denotes maximum

Min. denotes minimum

Pl. denotes plate

TOC denotes top of concrete

Typ. denotes typical

All welds, sections and details are typical for all similar conditions.

Electro-galvanized finish combination channels.

Unless noted otherwise, galvanize base plate and plates after fabrication per ASTM A385.

Unless noted otherwise, galvanize hardware per ASTM A123.

Metal framing combination channels shall be as manufactured by Globe Strut, United Gypsum Corporation, 101 South Wacker Drive, Chicago, IL 60606; Unistrut Corporation, 35660 Clinton Street, Wayne, MI 48184; or Cooper B-Line, 509 West Monroe Street, Highland, IL 62249.

Combination Channels: Globe Strut No. G7612A, B-Line B12A, or Unistrut P5501; or equal, having the following essential characteristics: No. 12 gauge steel and channel 1 5/8-inch wide by 4 7/8-inch deep with 7/8-inch openings.

Connection Plates:
With five holes per Globe Strut G2057, B-Line B532, Unistrut P1726; or equal.
With seven holes per Globe Strut G2068, B-Line B334, Unistrut P1950; or equal.

Connection hardware shall be suitable for combination channel strut attachments.

Fabricated Metalwork:
Plates and bars_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ ASTM A36

Length of bolt threads per ASME B18.2.1 and thread class 2 free-fit, American National coarse-thread series.

Unless noted otherwise, make all welds continuous and in the shop.

Start and stop all welds at center of member or tubing.

Unless noted otherwise, fabricator shall determine edge preparation and root opening for groove welds. Shop detail drawing shall show edge preparation.

Arc-Welding Electrodes: Filler metal and required shielding gases or fluxes shall be per AWS D1.1 with filler metal for steel having a minimum tensile strength of 70,000 psi.

Unless noted otherwise, all bolted connection bolt holes are standard 1/16" oversize.

"Y" dimensions are the responsibility of the Contractor and are based on the electrical cabinet furnished.

NOTES

1. Contractor determines final size of electrical equipment support based on electrical cabinet furnished.
2. Unless noted otherwise, or as directed by the foundation design engineer, install 5/8"Ø post-installed anchors with a minimum embedment of 6" or 5/8"Ø cast-in-place threaded anchors with a minimum embedment of 1'-6" having a minimum projection of 2".
3. Foundation design engineer shall verify foundation stability, concrete and reinforcement strength requirements, and anchorage requirements.
4. Install electrical equipment supports at locations shown on specifications drawings and/or as directed by COR.
5. Estimated electrical equipment support weight shown is based on maximum dimensions and five hole connection plates.

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BUREAU OF RECLAMATION
STANDARD DRAWINGS

STANDARD AND TYPICAL DESIGNS

MISCELLANEOUS METALWORK

REV NO		
2	2023-04-25	D UPDATED DRAWING TEXT STYLE TO CURRENT DRAWING RECORDS SYSTEM. MINOR REVISIONS.
	Putnam, Sara K P.E.	
1	2015-04-14	D MINOR MODIFICATIONS.
	BARTHEL, RODNEY J. P.E.	

Bernard Lajimodiere
DESIGNED
RBarthel
DRAWN
Olaff G. Huerta
CHECKED
Barthel, Rodney J P.E.
TECH. APPR.
Bernstein, Alfred I P.E.
APPROVED Plant Structures Group
ADMIN APPROVAL

DENVER, CO 2015-03-23

ELECTRICAL EQUIPMENT SUPPORT, TYPE I

SECTIONS AND DETAIL

40-D-60001

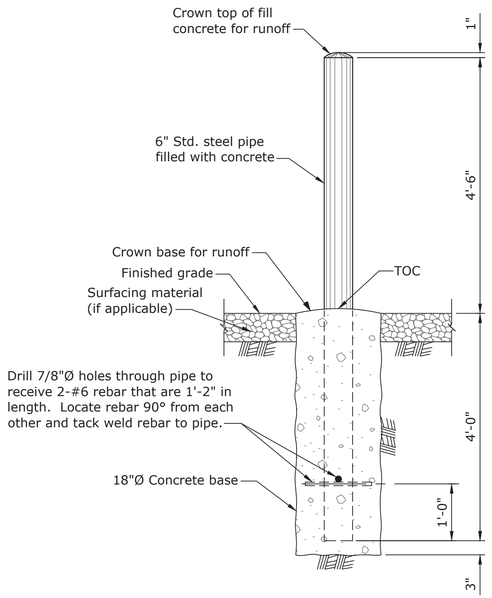
SHEET 1

Released

LAST SAVED DATE
2023-04-25
LAST SAVED BY
TRYAN

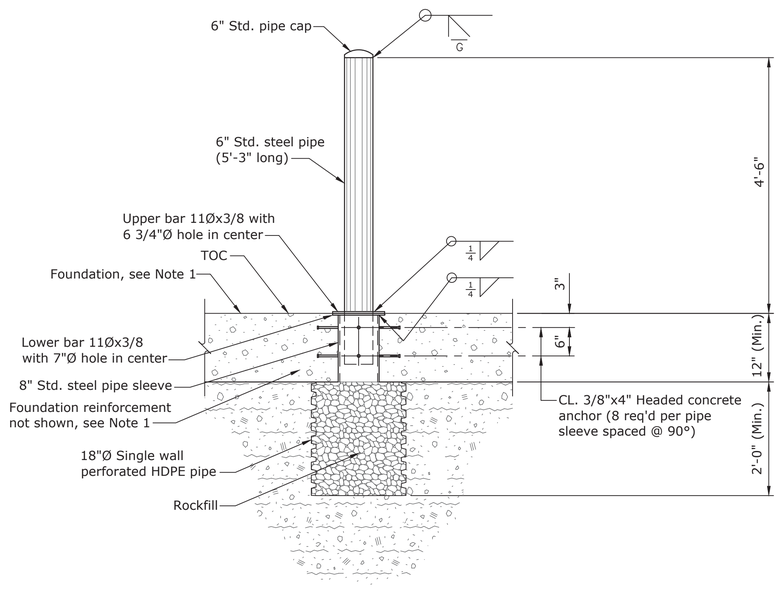
CAD SYSTEM
AUTOCAD LT 2018
CAD FILENAME
40-D-60001-RD-1.DWG

Released



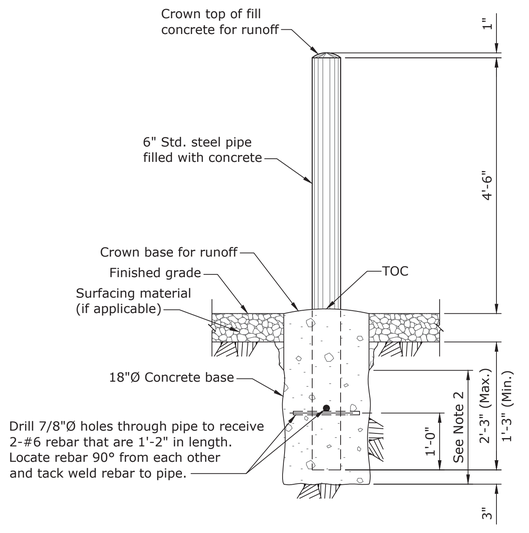
GUARD POST, TYPE I

FOR FOUNDATION EMBEDDED IN COMMON SOIL CONDITION
DO NOT GALVANIZE STEEL PIPE



GUARD POST, TYPE III

REMOVABLE GUARD POST AND SLEEVE FOR SLAB-ON-GRADE CONDITION
GALVANIZE PIPE SLEEVE
DO NOT GALVANIZE STEEL PIPE



GUARD POST, TYPE II

FOR FOUNDATION EMBEDDED IN ROCK CONDITION
DO NOT GALVANIZE STEEL PIPE

GENERAL NOTES

ASTM denotes American Society for Testing and Materials

AWS denotes American Welding Society

AWWA denotes American Water Works Association

COR denotes Contracting Officer's Representative

HDPE denotes high density polyethylene

Max. denotes maximum

Min. denotes minimum

psi denotes pounds per square inch

Rebar denotes concrete reinforcement

Std. denotes standard

TOC denotes top of concrete

Typ. denotes typical

All welds, sections and details are typical for all similar conditions.

Coat steel guard post and upper bar per specifications or as directed by COR.

Unless noted otherwise, galvanize pipe sleeve and lower bar after fabrication per ASTM A385.

Fabricated Metalwork:
Bars _____ ASTM A36
Std. Steel Pipe _____ ASTM A53, Grade B
Rebar _____ ASTM A615

HDPE Pressure Pipe: ASTM F405.

Unless noted otherwise, make all welds continuous and in the shop.

Unless noted otherwise, fabricator shall determine edge preparation and root opening for groove welds.

Arc-Welding Electrodes: Filler metal and required shielding gases or fluxes shall be per AWS D1.1 with filler metal for steel having a minimum tensile strength of 70,000 psi.

Rockfill per ASTM C33, Coarse Aggregate, Size No. 57.

Unless noted otherwise, concrete structural design is based on concrete with a minimum compressive strength of 2,500 psi at 28 days and reinforcement with a minimum yield strength of 60,000 psi.

NOTES

1. Foundation design engineer shall verify foundation stability and the concrete and reinforcement strength requirements.
2. 12" Embedment in moderately weathered or less weathered rock and 24" embedment in intensely-to-moderately weathered or more weathered rock. Rock weathering classifications are defined within drawing 40-D-7022.
3. Install guard posts at locations shown on specifications drawings and/or as directed by COR.

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STANDARD DRAWINGS

STANDARD AND TYPICAL DESIGNS

MISCELLANEOUS METALWORK

REV NO		
2	2023-04-25	D UPDATED DRAWING TEXT STYLE TO CURRENT DRAWING RECORDS SYSTEM. MINOR REVISIONS.
	Putnam, Sara K P.E.	
1	2015-04-14	D MINOR MODIFICATIONS.
	BARTHEL, RODNEY J, P.E.	

Bernard Lajimodiere
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Barthel, Rodney J P.E.
TECH. APPR. Plant Structures Group
Bernstein, Alfred I P.E.
APPROVED
ADMIN APPROVAL - Plant Structures Group
DENVER, CO 2015-03-23

TYPICAL GUARD POSTS

NON-SECURITY APPLICATION

40-D-60002

SHEET 1

LAST SAVED DATE
2023-04-25
LAST SAVED BY
RRODRIGUEZ

CAD SYSTEM
MICROSTATION (LMS TECH)
CAD FILENAME
40-D-60002-RD-1.DWG

Released

ABBREVIATIONS

add'l = additional
bf = bottom face
bl = bottom layer
br = bottom row
CJ = Construction joint
℄ = Centerline
CL. = Centerline
clr. = clear
Cr.J = Contraction joint
Ct.J = Control joint
ctr. = center or centers
d_b = nominal diameter of reinforcing bar
Dwg. = Drawing
ec = each corner
ef = each face
EJ = Expansion joint
El. = Elevation
eq. = equal or equally
eq. spc. = equally spaced, equal spaces
er = each row
es = each side
ew = each way
ff = far face
fr = far row
fs = far side
HP = High point
if = inside face
ir = inside row
ℓ_d = development length
ℓ_{dh} = development length of standard hook
LP = Low point
ml = middle layer
mr = middle row
nf = near face
nr = near row
ns = near side
OCJ = Optional construction joint
of = outside face
or = outside row
spc. = spaced or spaces
Std. = Standard
tf = top face
tl = top layer
TOC = Top of concrete
tr = top row
TW = Tailwater
Typ. = Typical
uv = uniformly varying lengths of bars between lengths shown
VCJ = Vertical construction joint
WP = Working point
WS = Water surface, Waterstop

CONCRETE SYMBOLS

The different concrete placements are indicated by the following symbols:

Concrete - First stage
Concrete - Second stage
Concrete - Blockout
Concrete - Existing or in adjacent structures
Concrete - Precast
Grout

NOTES TO DESIGNERS AND DETAILERS

1. Splice lengths shown in TABLES 3.0-60 thru 5.0-60 on 40-D-60004 are for Class B tension lap splices in accordance with ACI 318 and ACI 350. Assumed conditions for these tables in addition to the requirements shown on this drawing are uncoated reinforcement, normal weight concrete, reinforcement yield strength fy = 60,000 psi, and the transverse reinforcement index (K_{tr}) equal to zero. Splices or development lengths other than those shown in the tables must be detailed on the reinforcement specifications drawings.
2. Some factors which require additional consideration are: Beams or columns with ties, lightweight aggregate concrete, epoxy-coated reinforcement, excess reinforcement, bars in compression, bundled bars, and special seismic provisions.
3. The minimum centerline to centerline bar spacing requirements shown in TABLES 3.0-60 thru 5.0-60 on 40-D-60004 may not be applicable when using the provisions of ACI 350 Provision 10.6 for distribution of flexural reinforcement to control flexural cracking.
4. The minimum centerline to centerline bar spacing requirements shown in TABLES 3.0-60 thru 5.0-60 on 40-D-60004 is based on using 1.5 inch aggregate to calculate the spacings.
5. Recalculation of lap splice lengths and development lengths by the designer may be required for changes in combinations of controlling variables per ACI 318 Provisions 25.4 and 25.5, or ACI 350 Provisions 12.8 and 12.9.
6. Lap splices and development lengths for horizontal bars in walls are designated as top bar splices and development lengths unless otherwise noted or detailed on the reinforcement specifications drawings.
7. Development lengths for deformed bars in tension terminating in a standard hook shall be calculated by the designer in accordance with ACI 318 Provision 25.4.3.1 or ACI 350 Provision 12.8.5. In cases where a restricted member thickness does not allow for the full development length of a standard hook, the designer shall use a reduced tensile capacity.

CONCRETE OUTLINE GENERAL NOTES

CHAMFER :
Chamfer edges of permanently exposed concrete surfaces, except slabs and top edges of curbs, with a 45° bevel, $\frac{3}{4}$" x $\frac{3}{4}$", unless otherwise shown on the specifications drawings.

CONCRETE PLACEMENT :
Before placing concrete, see specifications and manufacturer's drawings, for all embedded material which is required in the placement.

DIMENSIONS :
All dimensions to a joint are to the centerline of the joint unless otherwise shown.
Dimensions to beams, columns, and walls are from reference lines or other control points.
Beam and slab depths shall be measured from the top of the structural slab.
Dimensions given for the depth of recesses are from the surface of the structural concrete.
Thicknesses shown for walls and slabs placed against soil or rock are minimum dimensions.

EMBEDDED MATERIALS :
Before placing concrete, care shall be taken that all embedded items are in position and securely fastened in place. Anchor bolt locations and settings are shown on the structural steel, miscellaneous metal, concrete, and equipment drawings. Conduits and grounding cables are shown on the manufacturer's drawings and the electrical drawings. All projecting items, including waterstops, shall be supported and protected from damage and exposure.

FINISHES :
For location and classes of finishes for formed and unformed concrete surfaces, see the specifications drawings and/or paragraphs.

JOINTS :
All construction joints, identified as CJ, shall be provided where shown on the drawings. All optional construction joints, identified as OCJ, are to reduce the volume of placements or to facilitate construction. If optional joints are provided they shall be only at the locations shown. Relocation, addition, or elimination of construction joints will be subject to approval.

TOOLED EDGES :
Tool exposed edges of slabs and top edges of curbs to a radius of $\frac{1}{4}$", unless otherwise shown on the specifications drawings.

REINFORCEMENT GENERAL NOTES (See Note 4)

ACCESSORIES :
Bar supports, spacers, and other accessories are not shown on the design drawings. The recommendations of the ACI Detailing Manual-2020 (MNL-66), or other approved supporting systems shall be used.

CODE AND DETAILING REFERENCES :
ACI Building Code Requirements for Structural Concrete (ACI 318-19).
ACI Detailing Manual-2020 (MNL-66).
ACI Code Requirements for Environmental Engineering Concrete Structures (ACI 350-20).

COVER :
Place the reinforcement so that the clear distance between face of concrete and nearest reinforcement is 2" for #8 bars and smaller, and 3" for #9 bars thru #11 bars, unless otherwise shown on specifications drawings. Provide 3" clear distance from face of concrete for all bars when concrete exposure is cast against and permanently in contact with the ground. For corner bars cover, see ADDITIONAL REINFORCEMENT AROUND OPENINGS AND RECESSES, SECTION A-A on 40-D-60004. Clear distance is to the design dimension line.
Reinforcement parallel to construction joints shall have a minimum of 2" clear cover.

DIMENSIONS :
Dimensions are to the centerline of the bars except for embedment of hooks, which are dimensioned to the outside of the bar.
Clear cover dimensions are marked "clr." and are dimensioned to the outside of the bar.

PLACING :
Reinforcement at small openings (Max. 1'-6") in walls and slabs may be spread apart not more than 1.50 times the bar spacing.
Unless otherwise shown on the specifications drawings, reinforcement shall be adjusted laterally to maintain a clear distance of at least 1" between the reinforcement and keys, waterstops, anchor bolts, form ties, conduits, and other embedded materials.
In heavily reinforced areas, relocation of the embedded material or cutting of reinforcement must be approved.
When bars are bent due to offsets less than 3" and recesses less than 3" deep, the slope of the inclined portion must not exceed 6 to 1.
Reinforcement parallel to anchor bolts or other embedded material shall be placed to maintain a clear distance of at least 1.33 times the maximum size aggregate.

PLAIN DOWELS :
Plain dowels across contraction joints shall be smooth bars uniformly coated with a film of oil before concrete placement.

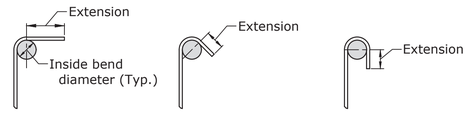
REINFORCEMENT DOWELS :
Unless otherwise shown, dowels indicated on the drawing, such as #8(d), shall be embedded a length equal to l_d and shall have a projection equal to that required for lap splicing to a bar of the same diameter.

SPACING :
The first and last bars in walls and slabs, stirrups in beams, and ties in columns are to start and end at a maximum of one half of the adjacent bar spacing. The minimum edge spacing shall be $2.5d_b$. For #9 thru #11 bars where minimum centerline bar spacing is 6" and the lap splice lengths requiring staggered placement shown in TABLES 3.0-60 thru 5.0-60 on 40-D-60004 are used, the minimum edge spacing shall be 3".

SPLICES :
The minimum length of lap for splicing parallel bars in tension shall be as given in TABLES 3.0-60 thru 5.0-60 on 40-D-60004.
Staggered lap splices shall be separated to give 12 inches clear between ends of adjacent splices.
Bars spliced by noncontact lap splices shall not be spaced transversely farther apart than one-fifth the required lap splice length, nor 6" on centers.
When reinforcing bars of different size are to be spliced, splice length shall be the larger of l_d of larger bar and tension lap splice length of smaller bar.
Splices are to be made so that the required clear distances to face of concrete will be maintained.

STANDARD HOOKS FOR DEVELOPMENT OF DEFORMED BARS IN TENSION :
90-degree bend plus $12d_b$ extension at free end of the bar.
180-degree bend plus $4d_b$ extension, but not less than $2\frac{1}{2}$", at the free end of the bar.

STANDARD HOOKS FOR STIRRUPS, TIES, AND HOOPS :
#5 bar and smaller, 90-degree bend plus $6d_b$ extension, but not less than 3", at the free end of the bar.
#6, #7, and #8 bars, 90-degree bend plus $12d_b$ extension at the free end of the bar.
#8 bars and smaller, 135-degree bend plus $6d_b$ extension, but not less than 3", at the free end of the bar.
#8 bars and smaller, 180-degree bend plus $4d_b$ extension, but not less than 2.5", at the free end of the bar.

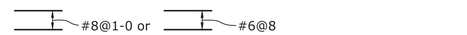


90-degree 135-degree 180-degree

TABLE OF INSIDE BEND DIAMETERS (INCHES)

BAR NO.	3	4	5	6	7	8	9	10	11
Standard	$2\frac{1}{4}$	3	$3\frac{3}{4}$	$4\frac{1}{2}$	$5\frac{1}{4}$	6	$9\frac{1}{2}$	$10\frac{3}{4}$	12
Stirrups, ties, and hoops	$1\frac{1}{2}$	2	$2\frac{1}{2}$	$4\frac{1}{2}$	$5\frac{1}{4}$	6			

SYMBOLS :
Bars shown thus:



Indicate a group of the same size bars equally spaced.

An open circle at the end of a bar indicates a bend with the bar turned away from the observer.
A closed circle at the end of a bar indicates a bend with the bar turned towards the observer.

Splices shown thus indicate a lap splice, not a bend in the bar.

Mechanical splice
Headed deformed bar

NOTES

1. This drawing shall be used in conjunction with 40-D-60004.
2. For Minimum Requirements For Detailing Reinforcement, see 40-D-60004.
3. Unless otherwise shown on the concrete outline and/or reinforcement specifications drawings, the notes shown are minimum requirements and typical for all drawings that refer to this drawing.
4. Unless otherwise shown on the reinforcement specifications drawings or 40-D-60004, follow the recommendations established by the ACI Detailing Manual - 2020 (MNL-66).
5. # Denotes reinforcement bar number.
6. Numerals in parentheses ( ) following notes, details, and section letters or numbers indicate the number of the drawing upon which the section or detail is indicated or shown. For example (60524) denotes drawing XXX-D-60524. The letters XXX shown here will be numbers denoting the project.

THIS DRAWING SUPERCEDES DRAWINGS 40-D-6263 AND 40-D-7012

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STANDARD AND TYPICAL DESIGNS

CONCRETE OUTLINE AND REINFORCEMENT

REV NO	DATE	BY	DESCRIPTION
2	2022-09-21	Barthel, Rodney J P.E.	D REVISED TO CONFORM TO ACI 350-20 AND MINOR REVISIONS.
1	2020-05-20	Barthel, Rodney, P.E.	D REVISED TO CONFORM TO ACI 318-19 AND MINOR REVISIONS.

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J. WAGNER
CHECKED
McGovern, Robert K P.E.
TECH. APPR.
Bernstein, Alfred I
APPROVED
PEER REVIEWER - Plant Structures Group

DENVER, CO 2016-09-23

GENERAL NOTES FOR CONCRETE OUTLINE AND REINFORCEMENT

40-D-60003

SHEET 1 OF 2

DATE AND TIME PLOTTED
MAY 27, 2020 13:55
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RRODRIGUEZ

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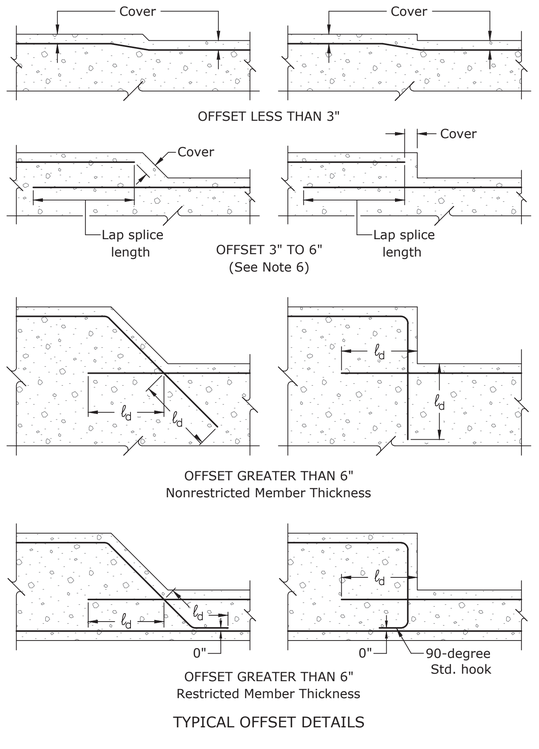
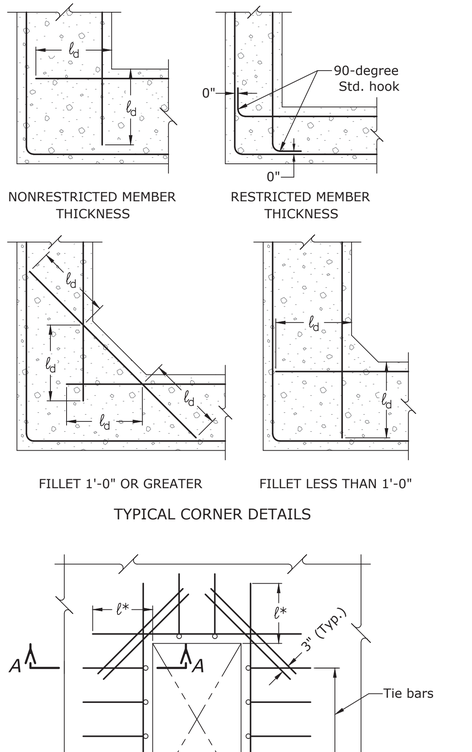
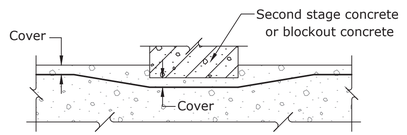
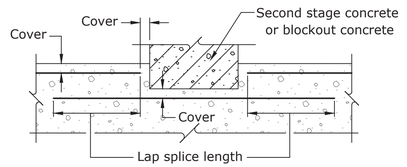


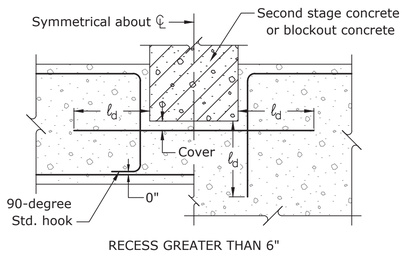
TYPICAL OFFSET DETAILS



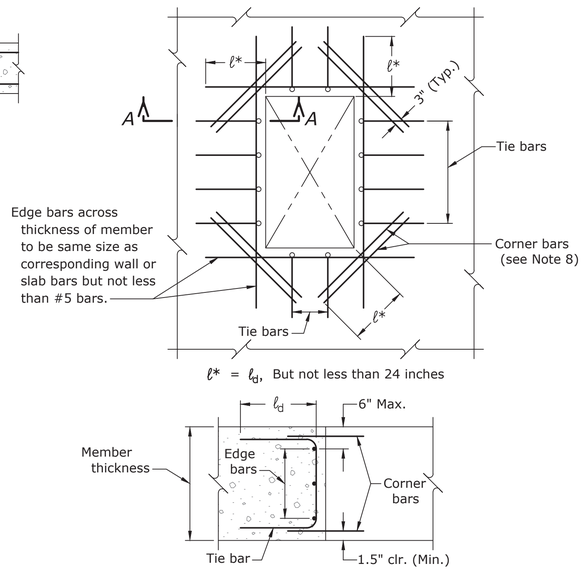
TYPICAL CORNER DETAILS







TYPICAL BLOCKOUT RECESS DETAILS
(Second stage concrete shown)



OPENINGS :
TABLE FOR ADDITIONAL REINFORCEMENT

MEMBER THICKNESS	TIE BARS	EDGE BARS	CORNER BARS
Less than 9"	None	1 - ctr.	2 - #4 ctr.
9" thru 1'-5"	None	2 - (1 ef)	4 - #4 (2 ef)
1'-6" thru 3'-0"	#4@1'-0"	3 - eq. spc.	4 - #4 (2 ef)
Over 3'-0"	#6@1'-0"	Spc. @ 1'-0"	4 - #4 (2 ef)

Omit edge and tie bars along sides of openings where opening dimension is less than 1'-6".
Omit corner bars at sides of openings adjacent to floors, walls, or beams.
Omit corner bars if both dimensions of opening are less than 1'-6".

RECESSES :
Use corner bars in face of recesses deeper than 4" if either dimension of recess is equal to or greater than 1'-6".

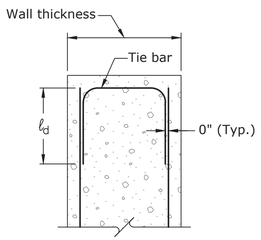
ADDITIONAL REINFORCEMENT AROUND OPENINGS AND RECESSES

f'c = 3,000 psi TABLE 3.0-60 fy = 60,000 psi

Bar size No. *** IP (SM)	MINIMUM ℄ to ℄ BAR SPACING (INCHES)	LAP SPLICE LENGTH (INCHES)****		DEVELOPMENT LENGTH ℓ_d (INCHES)	
		TOP BARS *	OTHER BARS	TOP BARS *	OTHER BARS
3 (10)	3	17	13	13	12
4 (13)	3	23	18	18	14
5 (16)	3.5	28	22	22	17
6 (19)	4	34	26	26	20
7 (22)	4.5	49	38	38	29
8 (25)	5	56	43	43	33
9 (29)	6	63	49	49	38
10 (32)	6.5	71	55	55	42
11 (36)	7.5	79	61	61	47
10 (32)	6	75**	58**	58	45
11 (36)	6	93**	71**	71	55

f'c = 4,000 psi TABLE 4.0-60 fy = 60,000 psi

Bar size No. *** IP (SM)	MINIMUM ℄ to ℄ BAR SPACING (INCHES)	LAP SPLICE LENGTH (INCHES)****		DEVELOPMENT LENGTH ℓ_d (INCHES)	
		TOP BARS *	OTHER BARS	TOP BARS *	OTHER BARS
3 (10)	3	15	12	12	12
4 (13)	3	20	15	15	12
5 (16)	3.5	25	19	19	15
6 (19)	4	29	23	23	18
7 (22)	4.5	43	33	33	25
8 (25)	5	49	37	37	29
9 (29)	6	55	42	42	33
10 (32)	6.5	62	47	47	37
11 (36)	7.5	68	53	53	41
10 (32)	6	65**	50**	50	39
11 (36)	6	80**	62**	62	48



TYPICAL TOP OF WALL DETAIL

For wall thickness less than 1'-6", tie bars not required.
For wall thickness 1'-6" and larger, tie bars to be #6@1'-0" unless otherwise shown on reinforcement specifications drawings.

f'c = 3,500 psi TABLE 3.5-60 fy = 60,000 psi

Bar size No. *** IP (SM)	MINIMUM ℄ to ℄ BAR SPACING (INCHES)	LAP SPLICE LENGTH (INCHES)****		DEVELOPMENT LENGTH ℓ_d (INCHES)	
		TOP BARS *	OTHER BARS	TOP BARS *	OTHER BARS
3 (10)	3	16	12	12	12
4 (13)	3	21	16	16	13
5 (16)	3.5	26	20	20	16
6 (19)	4	31	24	24	19
7 (22)	4.5	45	35	35	27
8 (25)	5	52	40	40	31
9 (29)	6	59	45	45	35
10 (32)	6.5	66	51	51	39
11 (36)	7.5	73	56	56	43
10 (32)	6	70**	54**	54	41
11 (36)	6	86**	66**	66	51

f'c = 4,500 psi TABLE 4.5-60 fy = 60,000 psi

Bar size No. *** IP (SM)	MINIMUM ℄ to ℄ BAR SPACING (INCHES)	LAP SPLICE LENGTH (INCHES)****		DEVELOPMENT LENGTH ℓ_d (INCHES)	
		TOP BARS *	OTHER BARS	TOP BARS *	OTHER BARS
3 (10)	3	14	12	12	12
4 (13)	3	19	14	14	12
5 (16)	3.5	23	18	18	14
6 (19)	4	28	21	21	17
7 (22)	4.5	40	31	31	24
8 (25)	5	46	35	35	27
9 (29)	6	52	40	40	31
10 (32)	6.5	58	45	45	35
11 (36)	7.5	64	50	50	38
10 (32)	6	61**	47**	47	37
11 (36)	6	76**	58**	58	45

f'c = 5,000 psi TABLE 5.0-60 fy = 60,000 psi

Bar size No. *** IP (SM)	MINIMUM ℄ to ℄ BAR SPACING (INCHES)	LAP SPLICE LENGTH (INCHES)****		DEVELOPMENT LENGTH ℓ_d (INCHES)	
		TOP BARS *	OTHER BARS	TOP BARS *	OTHER BARS
3 (10)	3	13	12	12	12
4 (13)	3	18	14	14	12
5 (16)	3.5	22	17	17	13
6 (19)	4	26	20	20	16
7 (22)	4.5	38	29	29	23
8 (25)	5	44	34	34	26
9 (29)	6	49	38	38	29
10 (32)	6.5	55	43	43	33
11 (36)	7.5	61	47	47	36
10 (32)	6	58**	45**	45	35
11 (36)	6	72**	55**	55	43

* Top bars are defined as horizontal reinforcement with a casting position of more than 12 inches of fresh concrete placed below the bar (see NOTES TO DESIGNERS AND DETAILERS, Note 5 on 40-D-60003).
** Splices must be staggered.
*** IP = Inch-pound bar size designation.
SM = Soft metric bar size designation.
**** Splice lengths are based on Class B tension lap splices.

NOTES

1. This drawing shall be used in conjunction with 40-D-60003.
2. For Concrete Outline General Notes and Reinforcement General Notes, see 40-D-60003.
3. For ABBREVIATIONS and CONCRETE SYMBOLS, see 40-D-60003.
4. For NOTES TO DESIGNERS AND DETAILERS, see 40-D-60003.
5. Unless otherwise shown on the reinforcement specifications drawings, the details shown are minimum requirements and typical for all reinforcement drawings that refer to this drawing.
6. This detail may not be appropriate for tension areas of shallow structural members. If in doubt, use detail for recess greater than 6".
7. TABLES 3.0-60 thru 5.0-60 generated using the general development length equations ACI 318 Eq. 25.4.2.4a and ACI 350 Eq. 12-1.
8. Development of diagonal corner bars in close proximity to another opening, wall or slab, will require bends or hooks.

THIS DRAWING SUPERCEDES DRAWING 40-D-6263

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STANDARD DRAWINGS

STANDARD AND TYPICAL DESIGNS

CONCRETE OUTLINE AND REINFORCEMENT

REV NO			
3	D	2022-09-21 Barthel, Rodney J P.E.	MINOR REVISIONS.
2	D	2020-05-20 Barthel, Rodney, P.E.	REVISED TO CONFORM TO ACI 318-19 AND MINOR REVISIONS.
1	D	2016-12-08 McGovern, Robert K P.E.	MINOR REVISION

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Bernstein, Alfred I
APPROVED
PEER REVIEWER - Plant Structures Group

DENVER, CO *2016-09-23*

MINIMUM REQUIREMENTS FOR DETAILING REINFORCEMENT

40-D-60004

SHEET 2 OF 2

DATE AND TIME PLOTTED
MAY 27, 2020 13:59
PLOTTED BY
RRODRIGUEZ

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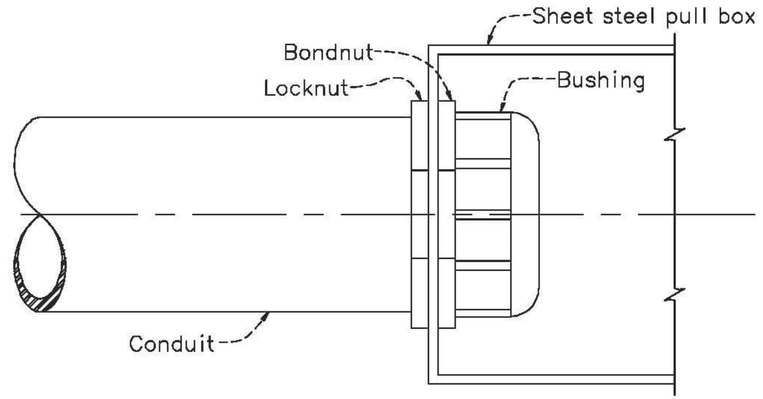
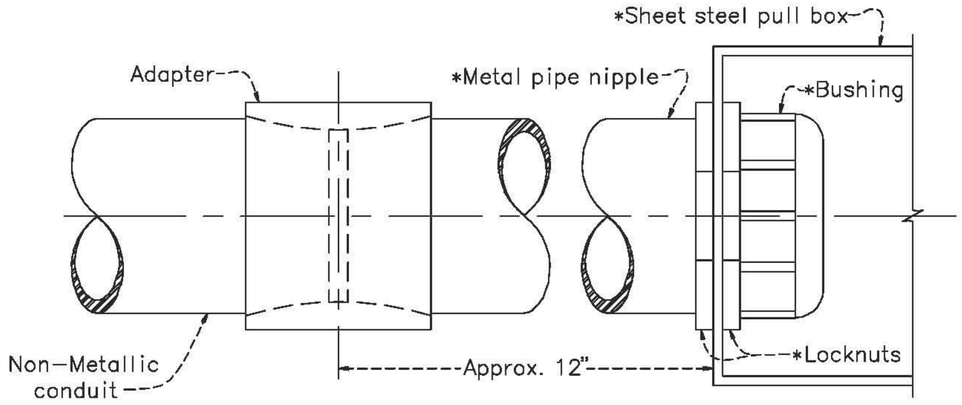
CAD FILENAME
40-D-60004-RD-2.DWG

BOX TERMINATION



DETAIL "C"

TERMINATING METAL CONDUIT IN SHEET STEEL PULL BOXES



DETAIL "D"

TERMINATING NON–METALLIC CONDUIT IN SHEET STEEL PULL BOXES

*Use aluminum or brass and slot steel box between knockouts for single phase.

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FORMERLY X–D–1523–7

REV. 3–25–46
REV. 4–1–42

DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
DENVER OFFICE
ELECTRICAL STANDARDS

DENVER, COLO. 6–14–38 | 104–D–254

MISCELLANEOUS DESIGNATIONS

ALT _____ Alternator
A _____ Suffix designation for auxilliary control circuit
ASC _____ Adjustable speed controller
BFV _____ Butterfly valve
C or CAP _____ Capacitor
CB_____ Circuit breaker
CLF _____ Current limiting fuse
CNT _____ Counter
DV_____ Discharge valve
EXC _____ Exciter
FU_____ Fuse
GFCI _____ Ground fault circuit interrupter
GRD _____ Ground
GRS _____ Galvanized rigid steel conduit
GV_____ Gate or guard valve
HR_____ Hand reset (Use as suffix)
I _____ Input
IL_____ Indicating lamp, add prefix for color
IMC _____ Intermediate steel conduit
MCE _____ Motor control equipment (2500 - 5000 volts)
M _____ Motor
MCC _____ Motor control center (600 - volts and lower)
NP_____ Nameplate
NC_____ Normally closed
NO_____ Normally opened
NSPB _____ Nonsegregated phase bus
O _____ Output
PB _____ Pushbutton (Momentary contact type)
PBM _____ Pushbutton (Maintaining contact type)
PC_____ Programmable controller
PR_____ Probe operated relay (Refer to FS for suffix)
REC _____ Rectifier
R/I _____ Resistance to current transducer
SF_____ Service factor
SO_____ Solenoid oiler
SV_____ Solenoid operated valve
TE_____ Time delay on energization
TD_____ Time delay on deenergization
TT _____ Thermal switch
V/I _____ Voltage to current transducer
VLV _____ Valve
WL _____ Water level contact on telemeter receiver (Refer to FS for suffix)

WRM_____ Wound rotor motor
X,Y,X_____ Suffix for auxiliary relay, switch, or contactor (ALX,CSX,FLX, ect.)

-1,-2,-3, etc._____ Suffix for unit number for pumping units, gates, valves, ect.

1,2,3, etc. _____ Suffix for electrical device numbering, such as TR1, TR2, ect.

* _____ Estimated rating

RELAY DESIGNATIONS

AL _____ Alarm relay
BG_____ Bearing temperature relay
CR_____ Control relay
FL _____ Field loss relay
FR_____ Field application relay
GP_____ Ground protective relay
IS_____ Incomplete sequence relay
LO_____ Lockout relay
OC_____ Overcurrent relay
OL_____ Overload relay
OV_____ Overvoltage relay
PO_____ Pullout (Loss of synchronism relay)
RC_____ Remote control relay
RSR _____ Remote sensing relay
SC_____ Squirrel cage (damper winding(relay
SR_____ Shift register
TR _____ Time delay relay
UF_____ Under frequency relay
UV-OPR _____ Undervoltage single and reverse phase relay
WT_____ Winding temperature relay

CONTACTOR DESIGNATIONS

FC_____ Field contactor
LC_____ Lighting contactor
M _____ Main contactor
MA_____ Air compressor motor contactor
MF_____ Vent fan motor contactor
MG* _____ Gate motor contactor
MO* _____ Oil pump motor contactor
MV* _____ Valve motor contactor
*Use suffix as application
-C Closing function
-L Lowering function
-O Opening function
-R Rasing function
S _____ Starting contractor
1S_____ Start contactor
2S_____ Start transition contactor } Reduced voltage starting
R _____ Run contactor

SWITCH DESIGNATIONS

CS_____ Control switch
FCD _____ Foreign circuit disconnect switch
FS_____ Float switch (Use suffix letter_ _ C-canal, P-pipe, R-reservoir, S-sump, T-tank)
LS_____ Limit switch
PS_____ Pressure switch
SS_____ Selector switch
TQ_____ Torque switch
TSW _____ Transfer switch

TRANSFORMER DESIGNATIONS

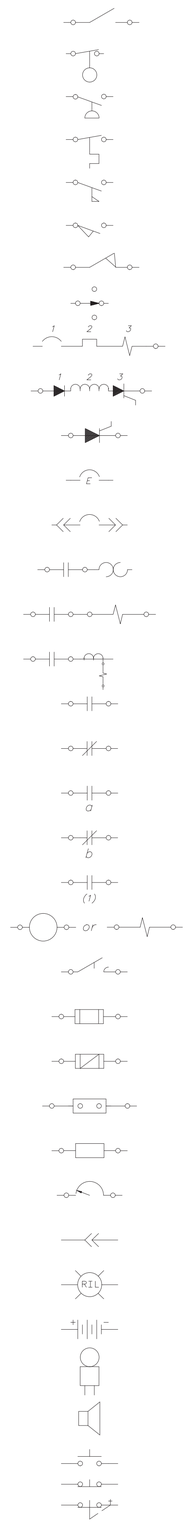
CCT _____ Control circuit transformer
CT_____ Current transformer
PT_____ Potential transformer
T _____ Transformer
TH_____ Transformer oil temperature
TL_____ Transformer oil level
TP_____ Transformer pressure relief
SST _____ Station service transformer

INSTRUMENT AND METER DESIGNATIONS

AM_____ Ammeter
AS_____ Ammeter transfer switch
CNT _____ Start counter
INT _____ Integrating instrument
PF_____ Power-factor meter
PST _____ Phase shifting transformer
RDM _____ Recording demand meter
TM_____ Time meter
V _____ Voltmeter
VAR _____ Varmeter
VARH_____ Varhour meter
VS_____ Voltmeter transfer switch
W _____ Wattmeter
WH_____ Watthour meter
WHD _____ Watthour demand meter
(R*) _____ Indicates recording instruments
The asterisk is not part of the symbol.
Replace the asterisk with letter or letters depending on the function of the instrument.

DEVICE SYMBOLS

CS — Control switch
AS — Ammeter transfer switch
VS — Voltmeter tansfer switch
Surge arrester
Capacitor
Reactor
Potential transformer
Current transformer
Power or distribution transformer
3Ø, Wye grounded connection
3Ø, Delta connection
3Ø, Broken delta connection
3Ø, Open delta connection (corner grounded)



Disconnecting switch
Liquid level switch (Normally open)
Vacuum and pressure switch (Normally closed)
Temperature actuated switch (Normally open)
Flow actuated switch (Normally open)
Limit switch-direct actuated (Normally open)
Torque switch (Normally closed)
3-Position selector switch
Air circuit breaker thermomagetic trip 1 Frame size, 2 Inverse time trip, 3 Instantaneous trip
1 Rectifier, 2 Reactor, 3 Inverter
Silicon controlled rectifier (Solid state DC switch)
Air circuit breaker, electrically operated
Air circuit breaker, withdraw type
Contactor with thermal trip
Contactor with magnetic trip
Contactor with CT and magnetic trip relay
Contactor (Normally open)
Contactor (Normally closed)
Auxiliary contactor (Closes when main device is closed)
Auxiliary contactor (Open when main device is closed)
Relay contact number
Relay coil, contactor coil
Horn-gap disconnectiong switch
Fuse
Fuse, current limiting type
Instrument shunt
Resistor
Rheostat, or potentiometer
Separable connector
Indicating lamp W-white, R-red, G-green, Y-yellow, A-amber, B-blue
Battery
Bell
Horn
Pushbutton (Normally open, momentary contact)
Pushbutton (Normally closed, momentary contact)
Maintained contact pushbutton

Probe, water level detector
Load interrupter switch
Motor
Triac (Solid state AC switch)
Wound rotor motor

WIRING SYMBOLS

Ground connection
Interconnection between separately owned systems
Duplex single-gang plug receptacle
Single weatherproof plug receptacle
Watertight three-phase power receptacle
Single pole switch
Three way switch
Luminaire, letter indicates type, number indicates wattage
Represents connection from external equipment. Does not necessarily represent a terminal block.
High voltage cable termination
Represents connection from external equipment via-multi-pair control (signal) cable.

GROUND SYMBOLS

Cable exposed
Cable embedded in concrete
Cable concealed but not embedded
Cable buried directly in earth
Ground rod
Welded or bolted connection
Cable riser
Indicates ground when required for clarity
Ground connector per 40-D-4334, Fig. 4
Ground ring

CONDUIT SYMBOLS

Exposed
Embedded in concrete
Concealed but not embedded
Buried directly in earth
Bending toward observer
Bending away from observer
Capped
Pull box or junction box
Designation number
Liquid tight flexible metal conduit
Sealing bushing, conduit to cable

NOTES

1. If designations and symbols other than the ones shown on this drawing used, they shall be shown on the approprate drawing.
2. For additional designations and symbols refer to NEMA Standards Pub, ICS-1970.
3. Contacts in control circuits are shown in the deenergized position. Liquid level switches are shown with liquid container empty. Vaccum and pressure switches are shown at ambient pressure. Temperature switches are shown at ambient temperature.

3 - 24 - 00 D - L.G.	REVISED TITLE BLOCK AND REDRAWN IN AUTOCAD 14.
3 - 11 - 81 D - D.G.	MINOR REVISIONS

ALWAYS THINK SAFETY

UNITED STATES
DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
STANDARD DESIGNS

PUMPING PLANT
ELECTRICAL INSTALLATIONS
DESIGNATIONS AND SYMBOLS

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CADD SYSTEM	CADD FILENAME	DATE AND TIME PLOTTED
AutoCAD Rel. 16.0s	757.DWG	APRIL 30, 2004 13:14
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Released

IEEE STANDARD CONTROL & PROTECTION DEVICE FUNCTION NUMBERS

(SEE NOTES 1, 2, AND 3)

01___Master Element
02___Time-Delay Starting or Closing Relay (TDPU)
03___Checking or Interlocking Relay
04___Master Contactor
05___Stopping Device
06___Starting Circuit Breaker
07___Rate-of-Change Relay
08___Control Power Disconnecting Device
09___Reversing Device
10___Unit Sequence Switch
11___ Multifunction Device
12___Overspeed Device
13___Synchronous-speed Device
14___Underspeed Device
15___Speed or Frequency Matching Device
16___Reserved for future application
17___Shunting or Discharge Switch
18___Accelerating or Decelerating Device
19___Starting-to-Running Transition Contactor
20___Electrically Operated Valve
21___Distance Relay
22___Equalizer Circuit Breaker
23___Temperature Control Device
24___Volts Per Hertz Relay
25___Synchronizing or Synchronism-Check Device
26___Apparatus Thermal Device
27___Undervoltage Relay
28___Flame Detector
29___Isolating Contactor
30___Annunciator Relay
31___Separate Excitation Device
32___Directional Power Relay
33___Position Switch
34___Master Sequence Device
35___Brush-Operating or Slip-Ring Short-Circuiting Device
36___Polarity or Polarizing Voltage Device
37___Undercurrent or Underpower Relay
38___Bearing Protective Device
39___Mechanical Condition Monitor
40___Field Relay
41___Field Circuit Breaker
42___Running Circuit Breaker
43___Manual Transfer or Selector Device
44___Unit Sequence Starting Relay
45___Atmospheric Condition Monitor
46___Reverse-Phase or Phase-Balance Current Relay
47___Phase-Sequence Voltage Relay
48___Incomplete Sequence Relay
49___Machine or Transformer Thermal Relay
50___Instantaneous Overcurrent or Rate-of-Rise Relay
51___AC Time Overcurrent Relay
52___AC Circuit Breaker
53___Exciter or DC Generator Relay
54___Turning Gear Engaging Device
55___Power Factor Relay
56___Field Application Relay
57___Short-Circuiting or Grounding Device
58___Rectification Failure Relay
59___Overvoltage Relay
60___Voltage or Current Balance Relay
61___Density Switch or Sensor
62___Time-Delay Stopping or Opening Relay (TDDO)
63___Pressure Switch
64___Ground Detector Relay
65___Governor
66___Notching or Jogging Device
67___AC Directional Overcurrent Relay
68___Blocking Relay
69___Permissive Control Device
70___Rheostat
71___Level Switch
72___DC Circuit Breaker
73___Load-Resistor Contactor
74___Alarm Relay
75___Position Changing Mechanism
76___DC Overcurrent Relay
77___Pulse Transmitter
78___Phase-Angle Measuring or Out-of-Step Protective Relay
79___AC Reclosing Relay
80___Flow Switch
81___Frequency Relay
82___DC Reclosing Relay
83___Automatic Selective Control or Transfer Relay
84___Operating Mechanism
85___Carrier or Pilot-Wire Receiver Relay
86___Lockout Relay
87___Differential Protective Relay
88___Auxiliary Motor or Motor Generator
89___Line Switch
90___Regulating Device
91___Voltage Directional Relay
92___Voltage and Power Directional Relay
93___Field-Changing Contactor
94___Tripping or Trip-Free Relay
95___Reserved for future application
96___Reserved for future application
97___Reserved for future application
98___Creep Detector Device
99___Reserved for future application

FIRST LETTER SUFFIX OF THE DEVICE DESIGNATION

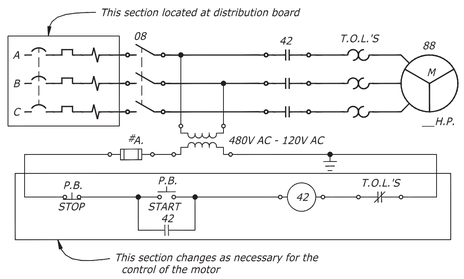
(SEE NOTES 1, 2, AND 3)

A____Governor System (or Actuator Systems--gates)
B____Battery Charging & Monitoring System
C____High-Voltage Cable System
D____Data Acquisition System
E____Excitation System including Transformer and Regulator but not Main Field
F____Fire and CO2 System
G____Main Generator including Auxiliary Systems
G/M___Generator Motor including Auxiliary Systems in Pumped Storage Applications
H____Turbine or Main Pump Including Auxiliary Systems
I____Isolated and Other Power Bus Systems (not High Voltage Cable)
J____Power Circuit Breaker including Auxiliary Systems
K____Power Transformer including Auxiliary Systems
L____Annunciator System, Security System
M____Main Pump Motor including Auxiliary Systems and Variable Speed Drive
N____Air (Pneumatic) System
O____Not used
P____Penstock or Discharge Line System
Q____Oil Storage, Handling, Purification System
R____Field Flashing System or Phase Reversal Switch including Auxiliary Systems
S____Station Service Substation System including Engine/Generator System
T____Tone and Transfer Trip System
U____Unit Control Circuit System or Uninterruptible Power Supply System
V____Intake and/or Discharge Valve System
W____Water Systems including Intake/Outlet Works and Plant Water and Sump Systems
X____Defined for systems unique to a facility
Y____Defined for systems unique to a facility
Z____Defined for systems unique to a facility

SECOND AND SUBSEQUENT LETTER SUFFIXES OF THE DEVICE DESIGNATION

(SEE NOTES 1, 2, AND 3)

A___Abnormal, A.C., Acceleration, Admission, Alarm, Amperes, Automatic, Auxiliaries, Phase A, etc.
B___Backup, Bearing, Block, Blower, Booster, Brakes, Bus, Button, Bypass, Phase B, etc.
C___Cable, Carrier, Charger, Check, Chlorination, Close, Collector, Common, Compensator, Compressor, Control, Cooling, Current, Cycle, Cylinder, Phase C, Conveyor, etc.
D___D.C., Deceleration, Delay, Depress, Detector, Dielectric, Differential, Discharge, Disconnect, Discordance, Domestic, Down, Downstream, Draft Tube, Drain, etc.
E___Ejector, Elevator, Emergency, Explosive, etc.
F___Failure, Fan, Fault, Feeder, Field, Filter, Fire, Flame, Flow, Follower, Forward, Frequency, Full, Fumes, Fuse, etc.
G___Gas, Gate, Gating (SCR), Generate, Ground, Guide Bearing, etc.
H___Halon, Hand, Heat, Heaters, High, Hoist, Horn, Hot, Housing, Hydropneumatic Tank, etc.
I___Indication, Initial, Inlet, Input, Instantaneous, Intake, Interface, Interlock, Interrupt, Inverter, Ionization, etc.
J___Jacking, Jet, etc.
K___Key, Transformer
L___Lamps, Left, Level, Lights, Limit, Line, Liquid, Local, Logic, Loss, Louvers, Low, Lower, Lubrication, etc.
M___Main, Malfunction, Manual, Meter, Metering, Motor, etc.
N___Negative, Network, Neutral, Normal, etc.
O___Open, Outlet, Output, etc.
P___Packing Box, Parallel, Parameter, Penstock, Phase, Phaseback, Pilot, Pit, Position, Positive, Potential, Pothead, Power, Pressure, Primary, Protection, Pulse, Pump, Purification, Push, etc.
Q___Oil, etc.
R___Raise, Reactor, Reclose, Record, Rectifier, Reed, Refrigeration, Regulate, Relay, Release, Relief, Remote, Reservoir, Reset, Resistor, Right, Rotation, Rotor, Runner, etc.
S___Seals, Secondary, Selector, Sewage, Shorting, Shutdown, Signal, Skimmer, Sludge, Smoke, Solenoid, Speed, Spiral or Scroll Case, Splice, Stabilizer, Standby, Starting, Stator, Stepping, Storage, Strainer, Suction, Sump, Supply, Switch, Synchronizing, etc.
T___Tank, Temperature, Test, Thermal, Thrust Bearing, Thyratron, Tie, Time, Transducer, Transfer, Transmitter, Trip, Trouble, Trashrake, etc.
U___Unit, Unloader, Unwatering, Up, Upper, Upstream, etc.
V___Valve, Vars, Vibration, Voltage, etc.
W___Water, Watts, Winding, etc.
X___Auxiliary Device, etc.
Y___Auxiliary to Device X, Antipump Relay. etc.
Z___Auxiliary to Device Y



BASIC MOTOR CONTACTOR SCHEMATIC

DEVICE SYMBOLS

(SEE NOTES 5 AND 6)

- RELAY COIL, SOLENOID COIL OR CONTACTOR COIL
- INDICATING LAMP
- INDICATING METER
- RECORDING METER
- INTEGRATING METER
- TRANSFER SWITCH
- TRANSDUCER
- TELEMETRY
- TELEMETRY FOR REMOTE INDICATION
- (REC.-LOG.) TELEMETRY FOR REMOTE LOGGING AND/OR RECORDING
- (REC.-AGC.-LOG.) TELEMETRY FOR REMOTE RECORDING AND/OR LOGGING ASSOCIATED WITH AUTOMATIC GENERATION CONTROL
- MICROWAVE, TRANSMITTER-RECEIVER, FOR RELAYING (SEE NOTE 7)
- CHANNEL NO. ___ MICROWAVE, TRANSMITTER-RECEIVER, FOR SUPERVISORY CONTROL (SEE NOTE 7)
- CHANNEL NO. ___ MICROWAVE, TRANSMITTER-RECEIVER FOR TELEMETERING (SEE NOTE 7)
- REMOVABLE LINK
- DEVICE TERMINAL POINT (SEE NOTE 11)
- CONTACT, NORMALLY OPEN (SEE NOTE 8)
- CONTACT, NORMALLY CLOSED (SEE NOTE 8)
- PUSHBUTTON, NORMALLY CLOSED
- PUSHBUTTON, NORMALLY OPEN
- PUSHBUTTON, MAINTAINED
- CONTACTOR WITH THERMAL OVERLOAD TRIP ELEMENT
- CONTACTOR WITH MAGNETIC TRIP ELEMENT
- THERMAL OVERLOAD DEVICE
- MAGNETIC OVERLOAD DEVICE
- THERMOMAGNETIC OVERLOAD DEVICE
- TIME DELAY MAGNETIC OVERLOAD DEVICE
- FUSE
- FUSE, CURRENT LIMITING TYPE
- RESISTOR
- RESISTOR, VARIABLE
- RESISTOR, TAPPED
- VARISTOR
- THYRISTOR OR TRIAC
- DIODE
- ZENER DIODE
- SEMI-CONDUCTOR CONTROLLED RECTIFIER (SCR)
- RESISTANCE TEMPERATURE DETECTOR (RTD)
- INSTRUMENT SHUNT
- CAPACITOR

- REACTOR
- THERMOCOUPLE
- LIGHTNING OR SURGE ARRESTER
- PROTECTIVE GAP
- HORN GAP
- CABLE TERMINATION
- BELL
- HORN
- BUZZER
- CARRIER FREQUENCY WAVE TRAP
- MECHANICAL INTERLOCK
- KEY INTERLOCK
- ELECTRICAL INTERLOCK
- REPRESENTS CONNECTION FROM EXTERNAL EQUIPMENT, DOES NOT NECESSARILY REPRESENT A TERMINAL POINT
- SYNCHRO (SELF-SYNCHRONOUS) DEVICE FOR POSITION INDICATION OR CONTROL
- THREE-PHASE, 3-WIRE, DELTA
- THREE-PHASE, BROKEN DELTA
- THREE-PHASE, OPEN DELTA
- THREE-PHASE, Y (OR STAR), UNGROUNDED
- THREE-PHASE, Y (OR STAR), WITH NEUTRAL BROUGHT OUT AND GROUNDED
- THREE-PHASE, ZIG-ZAG, UNGROUNDED

--- SEE NOTE 9 ---

- A.C. GENERATOR, MOTOR OR CONDENSER, BASIC SYMBOL
- SQUIRREL CAGE INDUCTION MOTOR
- SLIP-RING OR WOUND ROTOR INDUCTION MOTOR
- SYNCHRONOUS GENERATOR MOTOR OR CONDENSER WITH DIRECT CURRENT FIELD EXCITATION
- SYNCHRONOUS CONVERTER
- D.C. GENERATOR, MOTOR OR CONDENSER, BASIC SYMBOL
- D.C. GENERATOR, MOTOR OR CONDENSER WITH SHUNT AND SERIES FIELD
- SPLIT FIELD MOTOR
- DIRECT CONNECTED UNITS. USE BASIC AC OR DC GENERATOR OR MOTOR SYMBOLS
- USE MANUFACTURERS DIAGRAM ROTATING AMPLIFIER
- PERMANENT MAGNET GENERATOR
- SINGLE-PHASE TWO-WINDING TRANSFORMER
- POLYPHASE TRANSFORMER
- THREE-WINDING TRANSFORMER
- AUTOTRANSFORMER OR COMPENSATOR
- CONSTANT-CURRENT TRANSFORMER

- LOAD-RATIO CONTROL TRANSFORMER WITH TAPS
- STEP-VOLTAGE REGULATOR OR LOAD-RATIO CONTROL AUTOTRANSFORMER
- CURRENT TRANSFORMER (C.T.), WOUND TYPE
- CURRENT TRANSFORMER (C.T.), BUSHING TYPE
- POTENTIAL TRANSFORMER (P.T.)
- CAPACITOR-BUSHING POTENTIAL DEVICE
- CARRIER-COUPLING CAPACITOR POTENTIAL DEVICE
- TO CARRIER EQUIPMENT
- DISCONNECTING SWITCH, HOOK-STICK-OPERATED
- DISCONNECTING SWITCH, MANUALLY-GANG-OPERATED
- AIR-BREAK SWITCH, HORN GAP, MANUALLY-GANG-OPERATED
- DISCONNECTING SWITCH, MOTOR-OPERATED
- INTERRUPTER SWITCH
- MANUALLY OPENED, ELECTRICALLY TRIPPED GROUNDING SWITCH
- FUSED DISCONNECT SWITCH
- POWER CIRCUIT BREAKER
- POWER CIRCUIT BREAKER, DRAWOUT TYPE
- AIR CIRCUIT BREAKER
- AIR CIRCUIT BREAKER, DRAWOUT TYPE
- AIR CIRCUIT BREAKER WITH ELECTRICAL TRIP
- 3-POLE CIRCUIT BREAKER WITH THERMAL-OVERLOAD DEVICE OR INVERSE-TIME-TRIP ELEMENT IN ALL 3 POLES
- 3-POLE CIRCUIT BREAKER WITH MAGNETIC-OVERLOAD DEVICE OR INSTANTANEOUS-TRIP ELEMENT IN ALL 3 POLES

NOTES

1. Select the most appropriate ANSI device function number which identifies the device used by function and the most appropriate suffix letters which further identify the device by system, subsystem, quantity or condition monitored or controlled.

   Examples: 127UC....Unit 1 Unit Control Bus Undervoltage Relay
   251EF....Unit 2 Exciter Field Time Delay Overcurrent Relay
   71WSH....Plant Sump High Water Level Switch

2. Device numbers 1 through 99 are used for the overall plant and switchyard operation. Device numbers 101 through 199, 201 through 299, 301 through 399, etc. are used for Units 1, 2, 3, etc. Device numbers prefixed by letters RE indicates devices controlled by a supervisory system.

3. No device designation shall be duplicated in the plant or switchyard. When two or more devices perform exactly the same function, the designations will be followed by a number, starting at number 1, to differentiate between the devices.

   Examples: 127UCX1, 2.....Auxiliary Relays to Device 127UC
   71WSHX1, 2, 3...Auxiliary Relays to Device 71WSH
   104UC1, 2.....Unit 1 Master Control Starting Relays

4. All schematic diagrams shall have an explanation list for devices shown on that diagram. This explanation list shall consist of a device designation with a concise description for each device and be located above the title block (e.g. See the examples in Notes 1 and 3 above).

5. This list of device symbols shows the preferred symbol to be used on Bureau of Reclamation single-line and schematic diagrams.

6. The following symbols are not part of the device symbols they are shown with, always replace these symbols by the following:

   *...Show device function numbers and letter suffixes within circle.

   ▽...Show color of lamp by the following: A-Amber, B-Blue, C-Clear, G-Green, O-Orange, R-Red, W-White.

   ■...Substitute symbol from the following list:

   A - Ammeter
   AH - Ampere-hour Meter
   CMA - Contact-making Ammeter
   CMC - Contact-making Clock
   CMV - Contact-making Voltmeter
   DM - Demand Meter
   ETM - Elapsed Time Meter
   F - Frequency Meter
   G - Galvanometer
   GD - Ground Detector
   OHM - Ohmmeter
   OSC - Oscillograph
   PF - Power Factor Meter
   PH - Phase Meter
   PI - Position Indication
   RF - Reactive Factor Meter
   RPM - Speed Indicator
   SY - Synchroscope
   T - Temperature Meter
   TS - Time Switch
   V - Voltmeter
   VAR - Varmeter
   VH - Varhour Meter
   W - Watt Meter
   WH - Watthour Meter
   WHDM - Watthour-demand Meter

   Use "R" in front of letters for a recording type meter.

   □....Show location to which data or information is being telemetered.

   ⬬...Show device function numbers and letter suffixes.

   ⬭...Show a descriptive note, letter, abbreviation or word to indicate when the contact opens or closes. For latching relays, selector or control switches, use the normally open contact with a letter or word indicating when the contact closes. For time delay relays use TDC or TDO for contacts. All other relay contacts do not require any descriptive note.

   △....Show TR for torque-synchro receiver or TX for torque-synchro transmitter.

   ┼....Show C, M, or G within symbol. Internal connections for A.C. machines should be shown, e.g. three phase, single phase, etc.

   ⬡.....Show S.T. for shunt trip, R.C.T. for reverse current trip, U.V. for undervoltage trip, etc.

7. Channels are via microwave or leased facilities, not via powerline carriers.

8. These contact symbols apply to the following devices: control, flow, level, limit, position, pressure, speed, and temperature switches; relays; contactors; auxiliary contacts; etc. For selector switches, always use a normally-open contact with the position(s) the contact is closed listed below contact.

9. The symbols shown before this point are used on single-line and schematic diagrams. The symbols shown after this point are used as follows: the left-hand symbols are used on single-line diagrams; the right-hand symbols are used on schematic diagrams.

10. All devices shall be shown in the nonoperated and deenergized positions, e.g. breakers shall be open; latching relays shall be in the reset position; selector switches shall be in the off position; level switches shall be in the empty container state, pressure switches shall be in the ambient pressure state; temperature switches shall be in the ambient temperature state; valves, gates, and doors shall be shown in the closed position; etc.

11. All devices, shown on schematic diagrams, shall be shown with their device terminal points numbered and/or lettered for identification of internals for wiring purposes.

12. Whenever this drawing and drawing number 104-D-757 are both referenced, the designations and symbols shown on this drawing shall take precedence over those shown on 104-D-757.

This drawing supersedes drawings 104-D-200 through 104-D-214.

REV NO		
4	2017-10-24 Patterson, James B P.E.	SHADED IN TOP OVAL FOR NOTE 6. ("SHOW DEVICE FUNCTION NUMBERS...") CLARIFIED NOTE 8 FOR SELECTOR SWITCHES
6-24-2003 D- L.G.	REVISED STANDARDS REFERENCES.	
4-4-00 D- L.G.	REDRAWN IN AUTOCAD.	
6 - 1 - 90 D- J.W.R.	ADDED LIST OF SPERCEDED DRAWINGS, NEW TITLE BLOCK AND DRAWING NUMBER IN UPPER LEFT CORNER.	

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UNITED STATES
DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
STANDARD DESIGNS

MAJOR POWER/PUMPING PLANTS
ELECTRICAL INSTALLATION
DEVICE DESIGNATIONS AND SYMBOLS TO BE USED ON SINGLE-LINE AND SCHEMATIC DIAGRAMS

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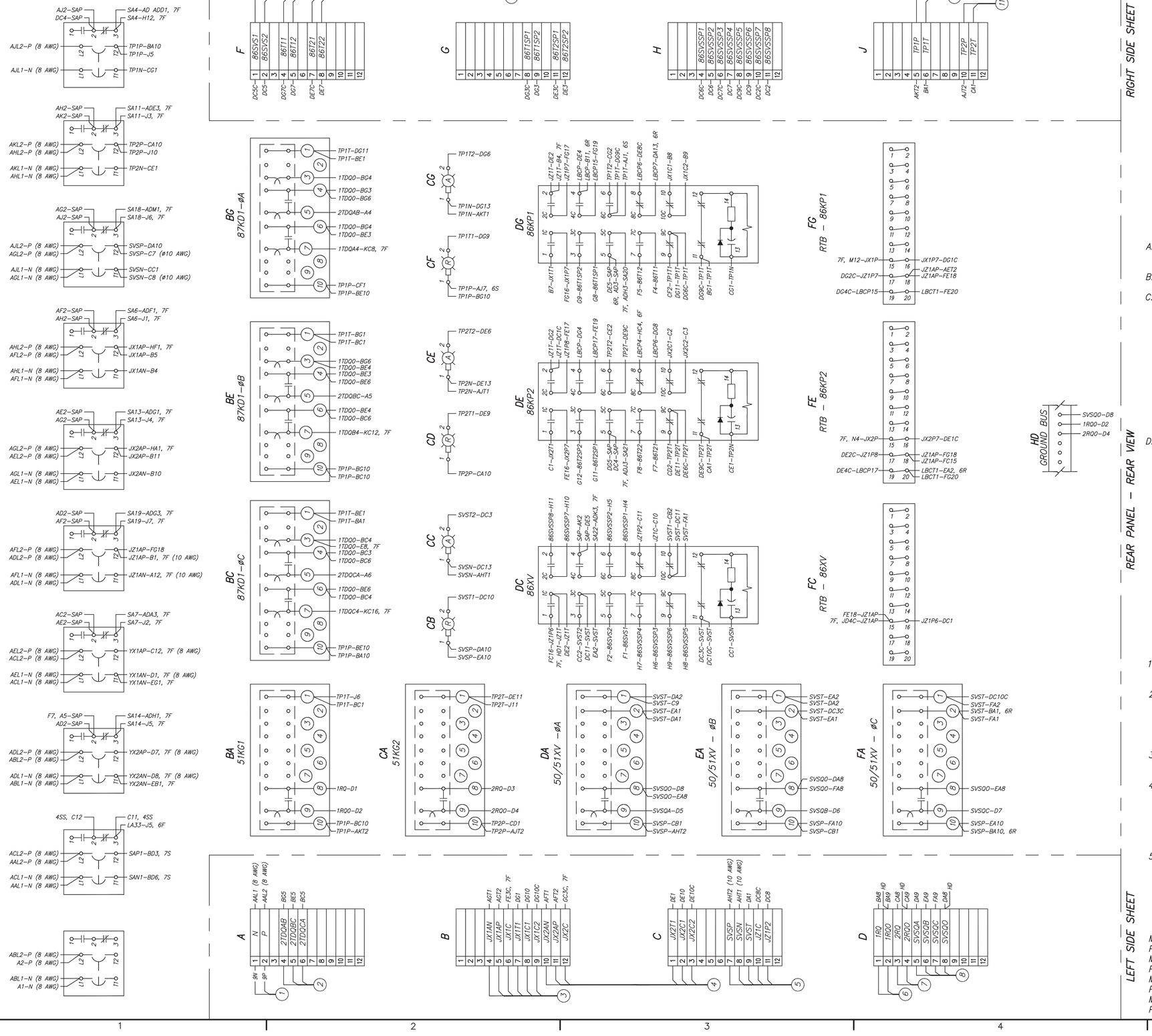
CADD SYSTEM	CADD FILENAME	DATE AND TIME PLOTTED
AutoCAD Rel. 21.0s	104-D-1150-RD-1.DWG	OCTOBER 24, 2017 09:33
DENVER, COLORADO	OCTOBER 21,1983	104-D-1150

Released



CABLE LIST

CABLE NUMBER	CABLE DESIGNATION	DESTINATION REFERENCE DRAWING	CABLE SIZE
1	9BSA–CSA7R	(6300)	2–1/C–8
2	1JX1A–CSA7R	(6121)	1–5/C–10
3	1CSA7R–JX1A	(6121)	1–9/C–10
4	2CSA7R–JX2A	(6230)	1–9/C–10
5	3CSA7R–JZ1A	(6249)	1–12/C–10
6	1KX1A–CSA7R	(6097)	1–3/C–10
7	1KX2A–CSA7R	(6201)	1–3/C–10
8	1JX1A–CSA7R	(6249)	1–5/C–10
9	4CSA7R–RTU	(6599)	1–12/C–16
10	5CSA7R–KX1A	(6097)	1–5/C–10
11	6CSA7R–KX2A	(6201)	1–5/C–10

EXPLANATION

POINT–TO–POINT WIRING DIAGRAMS

A. Double letters in bold face, such as AA, indicate the location of each device on the wiring diagram except for terminal blocks.

B. Single letters in bold face, such as A, indicate the location of each terminal block on the wiring diagram.

C. A control board is made of many sections which in turn are made of different panels. The sections are designated numerically 1 through "n" with "n" being the last section of the control board. The different panels are designated alphabetically as follows:
1. F – Indicates the front panel.
2. R – Indicates the rear panel.
3. S – Indicates the swing panel when there is only one swing panel per section.
4. FS and RS – Indicate the front and the rear swing panels, respectively, when there are two swing panels per section.
5. SS – Indicates the shipping split terminal block panel/section location.

D. All point–to–point wiring diagrams shall be drawn as follows:
1. All devices shown on the drawing shall represent the relative physical location of the devices.
2. Circuit (wire) designations shall be located next to the device stud (pin).
3. The circuit (wire) termination point shall be placed second from the device stud with a hyphen between the circuit designation and the circuit termination point.
4. The panel destination (number) of the wire shall be placed in the last position from the device stud with a comma between the circuit termination point and the panel destination. All panel wiring within the same panel will not require a panel destination.
5. Therefore, an example of point–to–point wiring is "LBCP4–HC4, 6F" which is located at device, DE, stud 8 on this drawing. The circuit (wire) designation is LBCP4. The circuit (wire) termination point is at device HC, stud (pin) number 4, located on the front panel of section 6.

NOTES

1. All cable shields and spare conductors shall be connected to the ground bus.

2. An internal wiring diagram of each type of protective relay, except lockout relays, shall be shown on a separate drawing. All other devices, including lockout relays, shall have their internal wiring diagrams shown on their external point–to–point wiring diagrams.

3. Do not edit the lineweights or text styles in this drawing. Only use text styles that are used in this drawing.

4. Do not explode or burst the titleblock. Contractor to edit titleblock information for each drawing based on information provided by Government. The titleblock information includes Subfeature, Contents, Details, and Drawing Number. Obtain latest version of Government titleblock and standards prior to commencing with Government Format Drawings.

5. The space below the Reclamation Logo at the top right side of the titleblock is reserved for manufacturer's information. At a minimum, include the following information in the space provided: Manufacturer's name, address, phone number, date, and solicitation number. Use Government provided block.

REFERENCE DRAWINGS

MAIN CONTROL BOARD CSA – PANEL 6R WIRING DIAGRAM . . . . . XXX–X–XXXXX
MAIN CONTROL BOARD CSA – PANEL 6F WIRING DIAGRAM . . . . . XXX–X–XXXXX
MAIN CONTROL BOARD CSA – PANEL 7F WIRING DIAGRAM . . . . . XXX–X–XXXXX
MAIN CONTROL BOARD CSA – PANEL 7S WIRING DIAGRAM . . . . . XXX–X–XXXXX

RIGHT SIDE SHEET

REAR PANEL – REAR VIEW

LEFT SIDE SHEET

RECLAMATION
Managing Water in the West

Manufacturer Information
Name:
Address:
Phone:
Date:
Solicitation Number:
Other Info:

ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
DESIGN, DATA (ELECTRICAL)
COLORADO

DESIGN DATA TYPICAL POWERPLANT_PUMPING PLANT
ELECTRICAL INSTALLATION

REV. NO		
3	2016–04–01	Mendlin, Eric T P.E.
2	2000–04–03	LG
1	1991–03–01	JWR

D REVISED TO NEW DRAFTING STANDARDS.
D REDRAWN IN AUTOCAD.
D REVISED TO NEW DRAFTING STANDARDS.

R.W MC AFEE — DESIGNED
M.F. JOJOLA — DRAWN
R.W MC AFEE — CHECKED
G.D. OSBURN — TECH. APPR.
F.W. COOK, SR — APPROVED
ADMIN APPROVAL – CHIEF, ELECTRICAL BRANCH
D 1990–12–20

MAIN CONTROL BOARD CSA – PANEL 7R

WIRING DIAGRAM

104–D–1165

SHEET 1

DATE AND TIME PLOTTED
Not Plotted
PLOTTED BY
Not Plotted

CAD SYSTEM
AUTOCAD
CAD FILENAME
UNKNOWN